U.S. Fish and Wildlife Service Species' Draft Biological Assessment for the 2001 Roadless Area Conservation Rule Rescission

USDA Forest Service

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List of preparers:

John Morse, Nichole Panico, Barbara Adams, Jody Tucker, Helen Chmura, Dan Saenz, Rebecca Kirby, Rebeca Rothenberg, Tanya Skurski, Tova Spector

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INTRODUCTION

The U.S. Department of Agriculture proposes to rescind the 2001 Roadless Area Conservation Rule (2001 Roadless Rule) (66 FR 3244, 36 CFR Subpart B (2001)), which prohibits road construction, road reconstruction, and timber harvesting in inventoried roadless areas on National Forest System lands, with limited exceptions. Rescinding the 2001 Roadless Rule is intended to return decision making for these activities to local officials, in conjunction with Forest-level land management planning. The 2001 Roadless Rule applies to approximately 44.7 million acres of National Forest System (NFS) lands, including in Alaska.

Accompanying the Proposed Rule, which is published in the Federal Register, and an environmental impact statement, the Forest Service has prepared this draft 'framework programmatic' biological assessment in compliance with the Endangered Species Act. This draft biological assessment discloses the potential effects to ESA-listed resources under the U.S. Fish and Wildlife Service jurisdiction that would result from rescinding the proposed 2001 Roadless Rule. An additional biological assessment has been prepared to assess effects to ESA-listed resources under National Oceanic and Atmospheric Administration, National Marine Fisheries Services (NFMS) jurisdiction. The Forest Service is requesting framework consultation and conferencing and commits to consulting on any future projects in the action area that may affect ESA-listed species or designated critical habitats.

All documents, including the proposed rule and other supporting information, may be found on the following website: https://www.fs.usda.gov/managing-land/planning/roadless.

Roadless Rule History

Throughout its history, the Forest Service has managed undeveloped areas to conserve their unique characteristics. After passage of the Wilderness Act of 1964 (Pub. L. 88-577), management of these areas, commonly known as "roadless areas," became the focus of national attention when the Forest Service initiated a review to determine their suitability for inclusion in the National Wilderness Preservation System, known as the Roadless Area Review and Evaluation (RARE) I (1973) and RARE II (1977). In the decades following the inventory, roadless area management continued to be controversial in land management planning and project proposals to harvest timber, build roads, or otherwise develop these areas. In 1999, President Clinton directed the Forest Service to undertake a public process to "provide appropriate long-term protection for most or all of these currently inventoried 'roadless' areas...", initiating the rule-making process. In 2001, the Forest Service issued the Roadless Area Conservation Rule establishing a national-level rule to maintain the values of inventoried roadless areas, limit their development, and address the history of controversy.

The 2001 Roadless Rule prohibited road construction, road reconstruction, and timber harvesting on inventoried roadless areas (IRAs) on 58.5 million acres, or 31 percent, of National Forest System lands (defined in 36 CFR 294.11 (2001) identified in the set of maps contained in the 2001 Final EIS). The Rule provided for certain limited exceptions, including for forest stewardship purposes, public health and safety, reserved or outstanding rights, existing or

renewed mineral leases, and realignment to prevent resource damage or address safety hazards (36 CFR 294.12 and 294.13 (2001)).

In the promulgation of the 2001 Roadless Rule, the Forest Service found that the Rule's prohibitions on road construction and timber harvest were not likely to adversely affect threatened or endangered species or adversely modify designated or critical habitat; were not likely to jeopardize proposed species or adversely modify proposed critical habitat; and may beneficially affect threatened, endangered, and proposed species and critical habitat (USFS 2000c). The Services concurred with the determination (66 FR 9 3271).

Inventoried Roadless Areas

Today, inventoried roadless areas total approximately 58.2 million[1] acres, or about 30 percent of all National Forest System lands. The 2001 Roadless Rule applies to approximately 44.7 million acres of National Forest System lands, including about 9.3 million acres within the Tongass National Forest. The 44.7 million acres exclude approximately 9.3 million acres in Idaho and 4.2 million acres in Colorado, which are under state specific roadless rules. Current areas are reflected on national maps at the Forest Service Roadless Area website (https://www.fs.usda.gov/managing-land/planning/roadless)[2].

More than 95 percent of IRAs managed under the 2001 Roadless Rule (outside of Colorado and Idaho) are in ten Western states (from largest inventoried roadless area acreage to smallest): Alaska, Montana, California, Utah, Wyoming, Nevada, Washington, Oregon, New Mexico, and Arizona. In the West, IRAs were typically 5,000 acres or greater. Some roadless areas are less than 5,000 acres—either because they are adjacent to Congressionally designated wilderness or in the eastern U.S. (east of the 100th meridian). While they are referred to as "roadless", because of the history of their inventory and designation, there are thousands of miles of existing roads within lands designated under the 2001 Roadless Rule, including National Forest System roads as well as roads of other ownerships. Additionally decommissioned roads and unauthorized routes that are not part of the official system or records may exist.

PURPOSE AND NEED

The U.S. Department of Agriculture (USDA or Department) is proposing to rescind the 2001 Roadless Area Conservation Rule (2001 Roadless Rule), which established broad prohibitions on road construction, road reconstruction, and timber harvesting within inventoried roadless areas on National Forest System lands. The intent of this proposed action is to reduce regulatory burden and return decision-making for the management of inventoried roadless areas to the land management planning process at the individual national forest level. Rescission of the national-level prohibitions provides responsible officials with flexibility to better guide management of National Forest System lands and respond to changing local resource conditions. The Forest and

[1] While the original rule listed 58.5 million acres, mapping modernization and minor corrections reduced that to 58.2 million acres.

[2] To view the national maps in an interactive map viewer select 'Map Service for Inventoried Roadless Areas designated by the 2001 Roadless Rule', then select ArcGIS Online Map Viewer.

Rangeland Renewable Resources Planning Act of 1974, as amended by the National Forest Management Act of 1976, the 2012 National Forest System Land Management Planning Rule (2012 Planning Rule), and the associated land management planning processes and plans are the appropriate and effective mechanisms to guide sustainable, integrated management of the resources within the plan areas in the context of the broader landscape, giving due consideration to the relative values of the various resources in particular areas. For additional discussion on the purpose and need please refer to the Final Rule.

CONSULTATION HISTORY

July 8, 2025 – The ESA compliance team is stood up for the Roadless Rule Rescission.

August 21, 2025 – The ESA compliance team decided to conduct a programmatic analysis for the rule rescission despite all future activities being required to conduct project-level ESA consultation.

August 29, 2025 – The Forest Service confirms a full analysis is preferred for this action given the history of the 2001 Roadless Rule.

November 21, 2025 – The analytical data needed to conduct exposure analyses is completed.

December 24, 2025 – Preliminary draft BA submitted to the Services to begin technical assistance.

DESCRIPTION OF THE ACTION

The proposed action described below is a "framework programmatic action" as defined in 50 CFR 402.02, where the framework programmatic action only establishes a framework for the development of specific future action(s) but does not authorize any future action(s). The effects to listed species and designated critical habitat of future actions that are subsequently authorized, funded, or carried out would be addressed in subsequent section 7 consultations. While this is a 'framework programmatic action', it does not provide a new framework. Instead, the proposed rule would remove the Roadless Area Conservation Rule regulation/prohibitions from the broader existing legal framework in which the Forest Service plans, authorizes, and implements activities. The programmatic effects are evaluated based on the legal framework that would remain in place, focusing on the effects of the removal of the prohibitions on roads construction and timber harvest (see Legal Framework, below).

Under the proposed rulemaking, the 2001 Roadless Area Conservation Rule prohibiting activities in inventoried roadless areas would be rescinded (also referred to as 'repealed'). This action would eliminate the national prohibitions set forth in the 2001 Roadless Rule – road construction, reconstruction, and timber harvest may be authorized within inventoried roadless areas, where they are consistent with land management plans, or other statutory and regulatory

requirements. In addition, the proposed rule would remove the national "inventoried roadless area" designation from the 44.7 acres million acres of National Forest System lands currently designated as inventoried roadless areas under the 2001 Roadless Rule[3].

Management of lands formerly managed as inventoried roadless areas would be governed by applicable land management plan direction alongside other relevant law, regulation, and policy. Some land management plans include components that restrict timber harvest or road building to meet other management objectives or legal requirements, such as to protect sensitive areas, to provide for primitive recreation opportunities, or to provide wildlife habitat. The proposed rescission of the Roadless Rule would not alter this existing land management plan direction.

Rescinding the Roadless Rule does not authorize, fund, or carry out future forest management activities that could occur in these areas. All future site-specific project decisions, including those to authorize road construction, reconstruction, and timber harvesting, must be consistent with land management plans. Any proposed actions must undergo environmental review, including National Environmental Policy Act compliance prior to a decision. Project decisions must comply with applicable Federal and State laws, such as the Clean Water Act, the Clean Air Act, Endangered Species Act, National Historic Preservation Act, and the Wilderness Act, as well as provisions for government-to-government consultation with Tribes and Alaska Native Corporations.

In addition, the proposed rule would not compel the amendment or revision of any land management plan. While the rulemaking would not require such changes, plans may be amended in any manner whatsoever and are revised periodically (16 U.S.C. 1604(f)). Local responsible officials would continue to have discretion to initiate land management plan amendment or revise, as deemed necessary. A determination to amend or revise a land management plan is based on a variety of factors. Any future administrative changes, amendments, or revisions to land management plans would comply with NFMA and planning regulations (36 CFR 219 and the planning directives at FSM 1920 and FSH 1909.12), which includes substantive requirements to provide for species biodiversity, recovery of threatened and endangered species, and ecological conditions to provide viable populations of species of conservation concern. To the degree that a land management plan could change in the future, the effects of such a change would be attributed to and analyzed as part of the decision-making for that action.

Rescinding the Roadless Rule does not constrain or alter the statutory requirement of sections 7(a)(1) through section 7(a)(4) of the Endangered Species Act. All future activities in currently inventoried roadless areas will require compliance with the ESA prior to implementation.

[3] The 9.3 million acres of inventoried roadless area within the Tongass National Forest would be included in the 44.7 million acres no longer managed according to the 2001 Roadless Rule. Rescinding the 2001 Roadless Rule does not affect state-level roadless rules that exist in Colorado and Idaho, nor does it preclude the development of state-led roadless efforts in the future.

ANALYTICAL METHODS

Action Area Determination

There are 44.7 million acres of IRAs subject to the 2001 Roadless Rule, although only ~40 million acres are included in the action area where the proposed rulemaking could result in changes to future management activities. In this assessment, the term "action area" and "affected environment" are used interchangeably.

The action area for the proposed Roadless Rule Rescission was defined by removing non-National Forest System lands, wilderness, wilderness study areas, and wild portions of wild and scenic rivers from Inventoried Roadless Areas managed under the 2001 Roadless Rule (resulting in ~40 million acres of action area) as potential future management of these areas would not be affected by the proposed rulemaking.

More specifically, review of the geospatial data found that 44.3 million acres of these are on National Forest System lands, meaning about 0.4 million acres of lands have had land ownership change since the 2001 Roadless Rule or there may be vertical integration problems (mapping errors) between the inventoried roadless data and official land ownership boundary data[4].

In addition, over 4 million acres are within Congressionally-designated wilderness[5], wilderness study areas[6], and wild segments of wild and scenic rivers[7]. These statutory designations have more restrictive and more permanent mandates than the 2001 Roadless Rule. Therefore, changes in management activities in these areas under any alternative are highly unlikely and not reasonably certain to occur. The Forest Service did not arbitrarily buffer the action area to account for the possibility that some stressors such as noise and dust may extend beyond the IRA boundary. At the national scale, it is not reasonably certain that a majority of stressors associated with a single action may extend beyond the boundary of the IRAs, with the exception of sedimentation being carried downstream. Stressors such as noise and dust are expected to be minimally impactful where they occur, whereas sedimentation from proposed activities is

[4] Vertical integration is a term used in the Geographic Information System (GIS) community to describe the process of aligning datasets from different sources. This includes the process of aligning the 2001 IRA boundary lines with National Forest System (NFS) boundary lines. NFS boundaries include, but are not limited to NFS property boundaries, national forest/national grassland boundaries, and special designated area boundaries (such as wilderness). Misalignments occur over time due to new surveys, improved technologies, and datum shirts (USDA Forest Service 2024 – Memo Procedure for Aligning and Making Technical Corrections (https://usdagcc.sharepoint.com/sites/fs-wo-emcarlh/NEPA-EADM/Roadless/Forms/AllItems.aspx?id=%2Fsites%2Ffs%2Dwo%2Demcarlh%2FNEPA%2DEADM%2FRoadless%2FProcedure%20for%20Aligning%20and%20Making%20Technical%20Corrections%20to%202001%20IRA%20Boundary%20Lines%20%28eSign%29%5FCombined%2Epdf&parent=%2Fsites%2Ffs%2Dwo%2Demcarlh%2FNEPA%2DEADM%2FRoadless). Very slight misalignment of the coverages can result in polygon slivers between the coverages, which can produce acreage differences initially. These differences can amount to tens or hundreds of acres or more, especially because of the large area under analysis.

[5] 1.3 million acres have been designated wilderness. Congress has included approximately 1.3 million acres (or 2.8 percent) of inventoried roadless areas managed under the 2001 Roadless Rule in the National Wilderness Preservation System.

[6] 2.8 million acres are Congressionally-designated wilderness study areas. Agency policy requires that "[a]ny area recommended for wilderness or wilderness study designation is not available for any use or activity that may reduce the wilderness potential of an area" (FSM 1923.03(3)).

[7] 85,000 acres are Wild segments of designated Wild and Scenic Rivers. Agency policy states that "[g]enerally, a wild river is accessible only by trail" (FSM 2354.42g) and that we must "[e]nsure trees are not sold, cut, or otherwise harvested in a designated wild river area except under emergency conditions, such as for insect or disease control, fire, natural catastrophe, disaster, public safety, or under specified conditions on valid mining claims" (FSM 2354.42d).

considered a major stressor to the aquatic environment. Outside of the action area (i.e., outside IRAs) there are no prohibitions to road construction or timber harvest such that these activities could have already been occurring, suggesting a buffer to include these areas may not be prudent. The fourth circuit court has defended that the action area for a programmatic action need not include areas outside of the relevant administrative boundary because the programmatic action does not authorize any specific activity and extension beyond those borders would be largely speculative. (Case 1:24-cv-00118-MR-WC).

For impacts considerations for aquatic species, downstream portions of aquatic habitats were included in the action area for up to 2 miles, to capture the most likely locations where sedimentation may be an issue. Instream work has been shown to cause turbidity that is generally not detectable more than half a mile downstream (see the Sedimentation section of the Fish discussion for more details). While activities on land can lead to increased sedimentation entering surface waters, best management practices will reduce these sediment loads. Although sediment transport is dependent on stream flow and particle size, a 2-mile downstream buffer is expected to capture a majority of the reasonably certain effects from sedimentation that would potentially occur with the future actions (indirect effects). Although storm events may cause sediment plumes that travel more than 2 miles downstream, these events are often short in duration, unpredictable, and not a result of forest management. This downstream buffer distance is similar or longer than most aquatic effect analyses have required from consultations on timber and road construction activities in the west. National grasslands and forests in Idaho and Colorado are not included in the action area because roadless areas in these states are managed under state-specific roadless rules, which are not affected by rescission of the national roadless rule.

Legal Framework for Future Projects[8]

As stated under the description of the action, the proposed rulemaking does not authorize any site-specific actions, however, it does remove prohibitions on certain actions such that they are reasonably certain to be proposed in some of these areas in the future (see also “Scope” below). Under the proposed rulemaking, all future projects that authorize road construction and timber harvest within the action area must still be consistent with all federal laws, including, Endangered Species Act (ESA), National Forest Management Act (NFMA) of 1976, National Environmental Policy Act, and, Clean Water Act, and must comply with the relevant land management plan. To ensure compliance with these laws and plans, during project-level environmental review and decision-making, the Forest Service includes project design measures that minimize[9] adverse effects to species and their habitats.

[8] Not all inclusive.
[9] Apply pre-project planning, best available technology, management practices, and scientific knowledge to reduce the magnitude, extent and/or duration of impacts.

Endangered Species Act

The Endangered Species Act of 1973 (ESA) the primary U.S. law protecting imperiled species and their habitats from extinction. Under Section 7 of the ESA, federal agencies must ensure actions they authorize, fund, or carry out, such as permits, construction, or resource management, do not jeopardize listed species or adversely modify critical habitats. Agencies must consult with the U.S. Fish and Wildlife Service (USFWS) or National Marine Fisheries Service (NMFS) if a project "may affect" protected species. Typically, ESA consultations include design measures that minimize adverse effects to ESA listed species and their designated critical habitats. Per the ESA, federal agencies are not allowed to jeopardize the continued existence of ESA listed species and destroy or adversely modify critical habitat. Because there is no way to predict where projects will occur and all future projects that occur in the affected environment will undergo ESA Section 7 consultation, this analysis will be high level effects that may be expected for actions proposed under this Environmental Assessment. In some cases, previous consultation effects to species and critical habitats for similar actions were utilized to determine effects.

National Forest Management Act

NFMA requires "guidelines for land management plans developed to achieve the goals of the Program which…provide for diversity of plant and animal communities based on the suitability and capability of the specific land area in order to meet overall multiple-use objectives, and within the multiple-use objectives of a land management plan adopted pursuant to this section, provide, where appropriate, to the degree practicable, for steps to be taken to preserve the diversity of tree species similar to that existing in the region controlled by the plan." NFMA is implemented through either the 2012 planning rule or the 1982 planning rule.

A primary purpose of the 2012 Planning Rule (planning rule) is to promote the ecological integrity of national forests and grasslands and other National Forest System administrative units. Ecological integrity – a substantive requirement of the planning rule – is designed to support Ecological Sustainability (36 CFR 219.8), Diversity of Plant and Animal Communities (36 CFR 219.9), and Multiple Uses (36 CFR 219.10). Similarly, the 1982 planning rule dictates "Fish and wildlife habitat shall be managed to maintain viable populations of existing native and desired non-native vertebrate species in the planning area. For planning purposes, a viable population shall be regarded as one which has the estimated numbers and distribution of reproductive individuals to insure its continued existence is well distributed in the planning area. In order to insure that viable populations will be maintained, habitat must be provided to support, at least, a minimum number of reproductive individuals and that habitat must be well distributed so that those individuals can interact with others in the planning area" and that "Forest planning shall provide for diversity of plant and animal communities and tree species consistent with the overall multiple-use objectives of the planning area. Such diversity shall be considered throughout the planning process. Inventories shall include quantitative data making possible the evaluation of diversity in terms of its prior and present condition. For each planning alternative, the interdisciplinary team shall consider how diversity will be affected by various mixes of resource outputs and uses, including proposed management practices. (Refer to Sec. 219.27(g).)"

All forest and grassland management plans, regardless of which planning rule they follow, must be consulted on under the Endangered Species Act.

Aquatic Species Conservation Strategies

Specific to aquatic species habitat, NFMA mandates that U.S. Forest Service protect streams, streambanks, shorelines, lakes, and wetlands from detrimental changes in water temperature, sediment deposits, and blocked watercourses during timber harvesting and management. It requires, through land management plans, that soil and water resources are not irreversibly damaged, often by establishing riparian management areas and utilizing Best Management Practices (BMPs). Further, individual land management plans contain standards and guides for aquatic management zones, some in the form of Aquatic Conservation Strategies (ACS) (such as the Northwest Forest Plan (USFS and USBLM 1994b), PacFish (USFS 1995a), and InFish (USFS 1995b)). Aquatic conservation strategies and land management plans standards and guides are designed to move the aquatic management zones to desired ecological conditions.

Aquatic conservation strategies use an ecosystem approach to management of riparian and aquatic habitats and were designed to (1) protect watersheds that had good-quality habitat and strong fish populations at the time they were drafted, and (2) halt further declines in watershed condition and restore ecological processes that create and maintain favorable conditions in aquatic ecosystems in degraded ecosystems (FEMAT 1993). The long-term goal (100+ years) is to develop a network of functioning watersheds that supports populations of fish and other aquatic and riparian-dependent organisms (such as the Northwest Forest Plan (USFS and USBLM 1994b), PacFish (USFS 1995a), and InFish (USFS 1995b)). Aquatic conservation strategies are based on preserving and restoring key ecological processes, including the natural disturbance regimes (USFS and USBLM 1994b) that create and maintain habitat for native aquatic and riparian-dependent organisms, and it recognizes that periodic natural disturbances may be required to sustain ecological productivity. As a result, aquatic conservation strategies do not expect all watersheds to be in favorable condition (highly productive for the same aquatic organisms) at any point in time, nor does it expect that any watershed will remain in a certain condition through time. The primary objective of aquatic conservation strategies is to maintain and restore the distribution, diversity, and complexity of watershed-level features and processes to which aquatic and riparian species are uniquely adapted. Aquatic conservation strategies were also designed to develop a strategy to guide management of aquatic ecosystems on federal lands, where amended, that would meet potential ESA requirements (Reeves et al. 2018). Aquatic conservation strategies were expected to make significant contributions to the recovery of the ESA-listed fish by increasing the quantity and quality of freshwater habitat for Pacific salmon and protecting and enhancing habitats of other species (Reeves et al. 2006, FEMAT 1993).

Best Management Practices (Clean Water Act)

Forest Service National Best Management Practices for Water Quality Management on National Forest System Lands (USFS 2012a) was developed to improve agency performance and accountability in managing water quality consistent with the Federal Clean Water Act (CWA) and State water quality programs. Current Forest Service policy directs compliance with required CWA permits and State regulations and requires the use of BMPs to control nonpoint source pollution to meet applicable water quality standards and other CWA requirements. The National

BMP program document provides consistency among Forest Service administrative units to efficiently administer the program and demonstrate improvements in performance and accountability at multiple scales. The National BMP Program consists of four main components: (1) a set of National Core BMPs, (2) a set of standardized monitoring protocols to evaluate implementation and effectiveness of those BMPs, (3) a data management and reporting structure, and (4) corresponding national direction.

Relationship Between the 2001 Roadless Rule and Land Management Planning

Administration of each national forest and grassland is directed by a land management plan, along with numerous land management laws, regulations, policies, and agency directives. A land management plan provides a framework for future site-specific project analysis and decision making. Land management plans are developed, amended, and revised under the National Forest Management Act and the 2012 Planning Rule at 36 CFR 219. All land management plans must conform to existing laws and regulations. Laws and regulations take precedence over management direction in the land management plans. When new or changed laws or regulations create conflicts with land management plan content, plans are administratively changed as needed to conform to new statutory and regulatory requirements (36 CFR 219.1(f) and 219.13(c)).

The 2001 Roadless Rule did not amend, revise, or compel amendment or revision of any land management plans (36 CFR 294.14(b) (2001); 66 FR 3273); however, the Roadless Rule supersedes individual land management plans where a provision of the rule is more restrictive than a plan. Similarly, for some areas, the plans may contain plan components that are more restrictive than the 2001 Roadless Rule. Under the 2012 Planning Rule, inventoried roadless areas are treated as a type of designated area,[10] with the option to also identify these areas as unique geographic areas or management areas in the plans. Plans may also include additional plan components for inventoried roadless areas if they are compatible with the restrictions of the applicable roadless rule. Within land management plans, inventoried roadless areas currently overlap other management areas or geographic areas, such as core habitats, research natural areas, alpine areas, or areas managed for backcountry recreation where there are expected to be low levels of development, including timber harvest and road construction.

Current land management-plans often include plan direction and components that identify where timber management and road construction may or may not be allowed. Example management

[10] 36 CFR 219.7(d) Management areas or geographic areas. Every plan must have management areas or geographic areas or both. The plan may identify designated or recommended designated areas as management areas or geographic areas.
36 CFR 219.9 Designated area. An area or feature identified and managed to maintain its unique special character or purpose. Some categories of designated areas may be designated only by statute and some categories may be established administratively in the land management planning process or by other administrative processes of the Federal executive branch. Examples of statutorily designated areas are national heritage areas, national recreational areas, national scenic trails, wild and scenic rivers, wilderness areas, and wilderness study areas. Examples of administratively designated areas are experimental forests, research natural areas, scenic byways, botanical areas, and significant caves.
Geographic area. A spatially contiguous land area identified within the planning area. A geographic area may overlap with a management area.
Management area. A land area identified within the planning area that has the same set of applicable plan components. A management area does not have to be spatially contiguous

areas where timber management or road construction may be limited include recommended wilderness, research natural areas, or backcountry management areas. Some plans include plan components with direction specific to IRAs.

A national review of plan components in management areas overlapping the potentially affected environment was completed to identify where plans do or do not further restrict timber harvest and road construction activities. There is much variation between individual land management plans and Regions reflecting differences in local conditions, age of the plans, and changes in planning policies over time. Overall, within potentially affected IRAs, land management plan components:

- Further restrict (do not allow or generally do not allow) road construction on 21.8 million acres (54.5 percent of the potentially affected IRAs), and
- Further restrict (do not allow or generally do not allow) timber harvest on 26.7 million acres (66.5 percent of the potentially affected IRAs).

Note that this information is intended for programmatic analysis purposes, actual implementation of any activities would be subject to local interpretation of the land management plans based on unique local conditions, which would also include numerous considerations of forest-wide or habitat related forest plan direction also to likely be limiting on where timber management and road construction would be allowed or proposed.

Consideration of Existing Protections

Species-specific conservation measures and design criteria were considered on an individual basis for certain species. Conservation measures largely come from forest plans but might also originate from regional strategies and previous consultations. These measures were reviewed and analyzed to determine if they affected the standard determinations made based on the procedures described above. Species-specific measures associated with forest plans typically apply to any location on that forest that is inhabited by the species unless otherwise stated. Species-specific measures from forest plans and other approved strategy documents are mandatory in nature unless otherwise stated.

Scope

The following describes how the agency drew a reasonable and manageable line in consideration of effects of the proposed action that are reasonably certain to occur, and those that may extend outside the geographical territory of the proposal or might materialize later in time.

In general, it is reasonably likely activities with potential for beneficial or adverse environmental effects could occur within or near inventoried roadless areas following the proposed rulemaking. However, the rulemaking would not compel, require, or authorize any site-specific activities. Therefore, site-specific locations and amounts of activities that could occur in the future under the alternatives are uncertain. Any future decisions on these activities are greatly influenced by factors beyond the proposed rulemaking such as local decision-making through land management plans and projects, local industry capacity, agency budget allocations and resources,

and other regulations and policies. Site-specific effects can be meaningfully measured and evaluated at the project and activity stage and are evaluated during the project-level ESA Section 7 consultation process. In addition, some of the potential adverse effects described could be reduced or mitigated at the project level, through complying with policy, following land management plan direction, implementing best management practices, or developing project-specific design criteria. Protections/requirements that reduce impacts are noted throughout this analysis.

The analysis focuses on the difference where (what locations) activities, particularly road construction and timber harvest, are more or less reasonably likely to occur within inventoried roadless areas, and the resulting effects to listed species. Estimates of areas where activities could occur, were prepared for effects analysis purposes – they do not reflect proposed or planned activity areas. The stressors associated with these activities were assumed to potentially occur when species would be present and affected as the timing of future activities is not known.

The geographic extent of the environmental effects analysis focuses on activities with a close causal relationship to the rulemaking. Changes to regulations remain in place until additional rulemaking or Congressional action. Short- and long-term reasonably likely to occur environmental impacts of the rulemaking are considered.

The short-term potential for road construction and timber harvest or related activities and associated effects is more predictable and reasonably likely to occur. Within the potentially affected environment the following areas are more likely to have activities or impacts, in the short term (5 years):

- areas where activities are allowed under existing land management plans (see environmental baseline below);
- areas with a greater management priority or 'need' (ex. high risk firesheds and areas affected by insect and disease or other disturbances);
- areas that are more feasible or economical for equipment and workers to access (more operable); and,
- areas with high product potential/economic viability for timber production to support management objectives.

In the longer term (up to 20 years and beyond), the extent of activities is more uncertain. Still, it is reasonable to expect a greater proportion of the potentially effected environment could be impacted by activities over time. Factors affecting long term change include natural events such as storms and wildfires, local decision-making through projects and land management plan amendments/revisions, national policy changes, technological changes, industry and economic changes, etc. Areas that are least likely for development in the short term (currently restricted, less developed, etc.) could be impacted by change, in the longer term. However, given the long causal chain and distance from the action proposed, predicting the specific amounts is too speculative.

The Roadless Rule prohibited road construction and timber harvest (with exemptions) but did not directly restrict other activities. Rescinding the Roadless Rule increases the likelihood of timber and road construction activities occurring within the boundary of previously designated IRAs. However, removing the prohibitions on road construction in IRAs may increase accessibility and therefore increase use of the land for purposes other than timber harvest. These activities are speculative, not reasonably certain to occur, and would require a lengthy causal chain to tie to the proposed rescission of the Roadless Rule. As such, only timber and road construction activities are considered in this analysis.

Action Deconstruction

Although rescinding the Roadless Rule does not dictate or prescribe forest management in previously inventoried roadless areas, it does remove the current prohibitions on timber harvest and road construction. Therefore, it is reasonably certain that additional active forest management and road construction activities could occur in these areas under the proposed roadless rule rescission (authorized under future decisions) for a variety of purposes. A deconstruction of potential future road construction and timber harvest activities is provided in Table 1 and the analysis assumptions for these activities that follow. The deconstruction table below does not include the impacts (stressors) associated with each tool or the expected response of listed species as those differed between aquatic and terrestrial organisms. Each taxonomic section has a deconstruction table that includes the stressors and responses relevant to those species.

Table 1. Deconstruction of the Roadless Rule Rescission.

Activity	*Management Element*	*Task/Method*	*Tool*
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment
		Tree and Unit Marking	Walking
			ATV
		Landings - Vegetation Clearing and Removal	Heavy Equipment
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment
		Timber Haul	Heavy Equipment
	Temporary Access Control	Road Gates Installation	Heavy Equipment
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment

Activity	*Management Element*	*Task/Method*	*Tool*
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment
		Temporary stream flow divergence; Bypass	Heavy Equipment; Pumps; Screens
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment/Hand Tools
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment, ATV
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bank full	Heavy Equipment
	Snow	Plowing	Heavy Equipment
Permanent and Temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools
		Snags - Hand Felling	Motorized Hand Tools
		Snags - Mechanical Felling	Heavy Equipment
		Tree Removal - Log Decks	Heavy Equipment
		Tree Removal - Hauling	Heavy Equipment
	Roadway template	Soil Excavation	Heavy Equipment, Blasting
		Soil Removal	Heavy Equipment
		Fill Placement	Heavy Equipment
	Drainage	Ditch lines, Relief Culverts, Water bars, Riprap	Heavy Equipment
		Stream Crossings - Culvert Installation	Heavy Equipment
	Surfacing	Hauling of surfacing materials	Heavy Equipment
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment

Activity	*Management Element*	*Task/Method*	*Tool*
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment
	scarifying	Material removal	Heavy equipment
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment

Road Construction

Road construction[11] or road reconstruction[12] or road maintenance[13] may be associated with timber harvest or other forest vegetation management (fuels reduction, thinning, etc.), private or special uses, existing rights, minerals and energy exploration[14], watershed restoration, habitat improvement, and recreation development. Aside from timber harvest, these activities are currently allowed under the 2001 Roadless Rule (see environmental baseline section).

The objective of the NFS transportation system is to provide the minimal mileage road system needed for safe and efficient travel, and for administration, utilization and protection of NFS lands. Any new roads would be designed and maintained in a way that would minimize negative effects on water quality, soil stability, and wildlife habitat. The decision to construct a new, permanent road that would become part of the NFS road system would be made as part of transportation planning associated with planned resource management activities and use of NFS lands. Proposals for new roads related to development and the associated environmental effects would be considered through project-level planning. If future decisions authorize it, modifications to the transportation network would follow law, regulation and policy found at 16 USC 1608, FSM/FSH 7700, relevant State and Federal statutes such as the Clean Water Act, NEPA, and Endangered Species Act, and national forest land management plans. Implementation of these policies can minimize, but not eliminate, some adverse environmental effects.

Timber harvest activities would likely generate the highest need for permanent and temporary roads on the landscape. New roads will not be required for some timber activities. For example, it's assumed timber related activities within 0.5 miles of existing roads where terrain slopes are 40 percent or less, would less likely require the development of new, permanent roads (11.3 million acres or 28.3 percent of the affected environment). Conversely, where timber activities are occurring beyond 0.5 miles of existing roads, some construction of new temporary and permanent roads would be more likely[15]. New projects would not automatically be approved as

[11] Activity that results in the addition of forest classified or temporary road miles.
[12] Road reconstruction. Activity that results in improvement or realignment of an existing classified road defined as follows:
(1) Road improvement. Activity that results in an increase of an existing road's traffic service level, expansion of its capacity, or a change in its original design function.
(2) Road realignment. Activity that results in a new location of an existing road or portions of an existing road, and treatment of the old roadway.
[13] Road maintenance. The ongoing upkeep of a road necessary to retain or restore the road to the approved road management objective.
[14] General Mining Act of 1872 and the 2001 Roadless Rule's exceptions (36 CFR 294.12(b)(3); 36 CFR 294.13(b)(2))
[15] A recent review of timber related projects in Forest Service Northern Region (1) revealed that approximately 25 percent of roads constructed in connection with timber contracts or other permits/leases were considered temporary and would be

Land Management Plan consistency and other environmental factors would need to be in alignment with the proposal. Any new permanent road construction[16] would be limited to areas where it is currently allowed by land management plans (18.2 million acres or 45.5 percent of the affected environment).

Timber Harvest

Vegetation and Forest Management include commercial timber harvest[17] and non-commercial timber treatments[18] and are completed for multiple resource objectives, including but not limited to timber production, fuels reduction, public health and safety, forest health improvement, watershed improvement, and wildlife habitat improvement. Land management plans must ensure: (1) Timber harvest is only allowed in areas that will not be irreversibly damaged, where cutting units can be visually blended into the terrain, where streams, wetlands, riparian areas, and water bodies are protected; (2) Timber harvest does not exceed sustained yield and is not designated solely to maximize outputs; and, (3) Harvested areas must be regenerated within 5 years. Any project activity, including timber harvest, must be consistent with land management plans. The 2012 Planning Rule provides further direction for forest management activities including timber harvest and non-commercial treatments. These plans include plan components (desired conditions, goals, objectives, standards, and guidelines) and identify areas suitable and not suitable for timber production to maintain sustained yield levels, identify maximum harvest levels on suitable lands (allowable sale quantity or sustained yield limit), and allow, prohibit, or limit timber harvest to protect resources.

Timber harvest effects are the same and not dependent on the resource objective; therefore, the deconstruction table (Table 1) only includes a basic timber harvest activity. Silvicultural prescriptions are utilized to meet direction within individual land management plans. Timber sales and other removals of forest products support agency strategic objectives to foster resilient, adaptive ecosystems, mitigate wildfire risk, and strengthen communities (USFS 2015). Table 1 was developed to cover all types of timber harvest and silvicultural prescriptions covered by the Roadless Rule Rescission. On average per region, NFS harvests less than 1 percent of operable areas annually (Forest Service Cut and Sold Reports 2014-2024). About 4.8 million acres (16

decommissioned within 10 years of contract, permit, or lease termination (16 USC 1608). For example, the Sawmill-Petty Project on the Lolo National Forest in Montana estimated road construction needs of approximately 19 miles of permanent road and 9 miles of temporary road to treat just under 17,000 acres. Another project, the Cyclone Bill Project on the Flathead National Forest estimated road construction needs of approximately 12 miles of permanent road and 2.5 miles of temporary road to treat approximately 12,000 acres.

[16] New road construction would be accordance with 16 U.S.C. 1608 which states "Unless the necessity for a permanent road is set forth in the forest development road system plan, any road constructed on land of the National Forest System in connection with a timber contract or other permit or lease shall be designed with the goal of reestablishing vegetative cover on the roadway and areas where the vegetative cover has been disturbed by the construction of the road, within ten years after the termination of the contract, permit, or lease either through artificial or natural means. Such action shall be taken unless it is later determined that the road is needed for use as a part of the National Forest Transportation System."

[17] Timber Harvest – Forest management activities that include the removal of sawtimber. Timber sales are often used as a least-cost method (revenue is returned to the Federal treasury to offset the costs of preparing and carrying out the timber harvest while portions of receipts are used for post-sale activities) of managing vegetation to meet multiple resource objectives (Forest Service Manual 2400, Forest Service Handbook 2409.14).

[18] Non-commercial forest management – Forest management activities that do not include the removal of sawtimber, non-commercial uses of timber, or any forest products removed for personal or incidental use. These activities may be completed with mechanized systems or by hand. Non-commercial forest management are higher cost options for managing vegetation to meet multiple resource objectives than timber harvest (Forest Service Manual 2460, Forest Service Handbook 2409.18).

percent of forested areas in the potentially affected IRAs) are in areas where current land management plans allow timber harvest and are more likely to be operable.

Other Activities Potentially Involving Road Construction or Tree Cutting

Access to National Forest System Lands and Special Use Authorizations

Access to the National Forest System (NFS) lands come in many forms from general usage by members of the public that do not require permits in most cases, to lands and special use authorization uses that are allowed through Land Management Plans and require a special use permit or similar legal instrument. Special use authorizations are needed for activities on or occupancy of National Forest System Lands including water transmission/storage, utilities, research, military training, communication uses, agriculture, pipelines, organized recreation events, ski areas, road access and much more. Pre-existing uses and those related to reserved or outstanding rights for access, occupancy, and use of NFS lands were not affected by the 2001 Roadless Rule. The continuation, re-issuance, or renewal of existing special use authorizations for activities in inventoried roadless areas is not prohibited by the 2001 Roadless Rule. Any future or additional special use authorizations would be reactive in nature to respond to public and customer requests. Under the proposed rulemaking, it is reasonably certain to occur that there would be an increase in new proposals for uses involving road construction or reconstruction that benefit the public or provide access to non-Federal lands within or adjacent to IRAs, as well as requests for expansion of existing uses (for example, additional access routes or improved road quality). An authorization is issued only after being evaluated and having passed through screening criteria (36 CFR 251.54), including compliance with the applicable land management plan and environmental analysis processes. Under the proposed action, new special use proposals would be evaluated on their own merits and no longer be rejected outright due to prohibitions for road construction or reconstruction. Requests for and approval of road construction-related authorizations are most likely to occur in areas that are 'operable' and where current land management plans allow these activities. For the purposes of this assessment, road construction is considered but future activities requiring a special use permit are outside the scope of this analysis and will be consulted on at the time of permit application.

Minerals Exploration and Development

Mineral and energy exploration and development occur within inventoried roadless areas (though to a lesser degree than other NFS lands) and would continue where there are existing rights or permits/leases/authorizations. Road construction, reconstruction, and tree cutting for locatable (gold, silver, copper, lead, zinc, etc.) mineral exploration and development is considered a right of access as provided by the General Mining Law and the 2001 Roadless Rule's exceptions (36 CFR 294.12(b)(3); 36 CFR 294.13(b)(2)). Changes may occur for leasable minerals exploration and development (primarily oil and gas and coal). Any new, modified or renewed leases would continue to be managed under existing laws and regulations governing oil and gas and coal, including project specific environmental review documents and compliance with other relevant law, regulation, and policy prior to leasing. For new leases or modifications of leases, construction or reconstruction of roads that are reasonable and necessary for development of existing energy or mineral leases, for access to new or existing energy or mineral

leases, and for access to associated product conveyance lines could be allowed as necessary in accordance with the terms of the lease, where they are consistent with land management plans or other laws and policy requirements. In addition, the 2001 Roadless Rule and IRA boundary would not apply or be considered in any future analyses for the modification of existing leases. However, in a site-specific leasing analysis, a no surface occupancy stipulation may still be applied to protect surface resources. For example, to prevent excessive erosion on steep slopes, a no surface occupancy may still be applied to the Federal lease. The Forest Service is not proposing to extend this consultation to include future minerals exploration and development. Future mineral exploration and development will require standalone consultation to comply with the Endangered Species Act.

Species Lists

Threatened, endangered, and proposed species and their designated or proposed critical habitat were compiled from existing consultation documents for species under NMFS's jurisdiction. A shapefile of IRA boundaries by administrative unit were imported into IPaC to compile the list of species under FWS's jurisdiction. Both species lists were reviewed by regional FS staff and species that do not actually occur on NFS lands were removed. Species lists were sent to the Services in September and October 2025 for review. Where appropriate, species were grouped taxonomically to facilitate discussing the effects to a large number of species and critical habitats. Species of high public interest and forest species known to be severely impacted by timber harvest and road construction are often discussed separately from taxonomic groupings.

Data Collection and Use

We acquired species occurrence data from the Global Biodiversity Information Facility (GBIF) and Natural Resource Manager (NRM) using R and ArcGIS Pro, respectively. The databases do not differentiate between DPSes of vertebrate species listed under the ESA, which causes potential inflation of exposure to these entities. The data is managed in such a way that occasionally species occurrences occur in locations where the entity is not listed or protected by the ESA; however, the FS does not believe these discrepancies have altered the conclusions provided.

A custom exposure analysis is performed for some species known to be impacted by roads and timber activities and where national data could support such analyses. Where custom analyses are performed, they will be documented in either the administrative record or in the taxonomic sections below. While the potential for inflated exposure is possible the FS does not believe it would significantly affect the determinations made. These data discrepancies will also be addressed during project-level consultation. We processed GBIF data using three Python scripts to unzip the downloaded GBIF folders and join relevant data (e.g., NLCD, basic ownership, Roadless Rule Affected Environments), filter to ESA sensitive species, and summarize the data into region-wide csv files. For Forest Service Alaska Region (Region 10), we separated the GBIF data from the conterminous United States (CONUS) data and ran a separate Python script due to available NLCD data (2024 Annual NLCD data for CONUS and the 2016 NLCD Alaska data for Region 10). We designated NRM data features using ArcGIS Pro's ArcPy as either inside or outside of Roadless Rule Affected Environments in National Forest land. We then determined

each feature's NLCD value and class using pixel count majorities in R, joined relevant data (e.g., basic ownership and Roadless Rule Affected Environments), created Forest Service region-wide summaries, and exported them as an excel workbook. Species occurrence data are opportunistic and do not represent the results of exhaustive structured species surveys nor do they include negative occurrence data. Additionally, for mobile species, an observation of an individual in a specific location does not mean that that location is occupied by resident individuals- some of these observations could have been of transient individuals.

Exposure was determined by comparing species occurrence data with roadless area operability (as defined in the Environmental Baseline section below) and categorized as high, moderate, and low likelihood of exposure. Low exposure was determined for species that had no occurrences in IRAs. Medium exposure was determined for species with less than 50% of occurrences in IRAs. High exposure was determined for species that had greater than 50% of occurrences in IRAs. Where there was no occurrence data the exposure was classified as unknown. Exposure estimates were performed in a manner to err on the side of likely exposure. Existing protections afforded by land management plans and other documents/decisions were also reviewed to determine how they may affect the determinations for each species, particularly if effects to the species were determined to be highly impactful. The response of species was derived from published scientific literature and previous consultations. Together, exposure and response were used to make the determinations: No effect (NE), Not likely to adversely affect (NLAA), and likely to adversely affect (LAA).

	Exposure		
	Low or unknown	Medium or unknown	High
Response is impactful	NLAA or LAA	LAA	LAA
Response is minimal	NE or NLAA	NLAA or LAA	LAA

Where available information indicated a less impactful determination was warranted compared to the general determinations a less impactful determination was made. Where information is lacking the Forest Service erred on the side of making the more impactful determination.

The exposure of critical habitat to stressors associated with road construction and timber activities was performed by comparing the overlap of critical habitat with the action area categorized by operability. Where overlap occurred, the physical and biological features of the critical habitat were assessed to determine whether the stressors associated with road construction and timber harvest could be expected to adversely affect those PBFs and the functionality of the critical habitat. Determinations for critical habitat were either no effect, not likely to adversely affect, or likely to adversely affect. Where information was lacking the Forest Service erred on the side of making the more impactful determination.

ENVIRONMENTAL BASELINE

This section briefly describes the status of inventoried roadless areas that may be affected (Roadless Affected Environment or Environmental Baseline). There are approximately 40

million acres of IRAs managed under 2001 Roadless Rule[19] within the action area. Additional information on the status of the action area is available in the environmental impact statement for the roadless rule rescission.

Land Management Plans and Operability

A national review of plan components in management areas overlapping the potentially affected environment was completed to identify where plans do or do not further restrict timber harvest and road construction activities[20]. There is much variation between individual land management plans and Regions reflecting differences in local conditions, age of the plans, and changes in planning policies over time. Current land management-plans include plan direction and components that identify where timber management and road construction uses may or may not be allowed (Table 3). Example management area types where timber management or road construction may be limited include recommended wilderness, research natural areas, or backcountry management areas are administrative land management planning designations that limit active management.

- Across the Roadless Affected Environment, land management plan components generally allow timber harvest across 13.4 million acres (33.4% of NFS land within the Roadless Affected Environment). [36% not allowed, 26.8 generally not allowed (with some exceptions)].

- Across the Roadless Affected Environment, land management plan components generally allow road construction across 18.2 million acres (45.5% of NFS land within the Roadless Affected Environment).

This analysis is directed at actions that are reasonably likely to occur in the remaining acres of inventoried roadless areas where road construction and reconstruction or timber harvest would be most likely to occur under current land management plans. A nationwide geospatial analysis was used to estimate potential operability, or the capability of mechanized treatment or activities, in each area based on slope, distance from roads, site productivity, and statutory restrictions (Table 2):

- Likely Operable: Less than 0.5 miles from a road, on slopes less than 40%, are productive for restocking (greater than 20 cubic feet/acre/year), are potentially forested, and have no statutory restrictions from timber harvest.

- Likely Operable but Complex: Less than 0.5 miles from a road and on slopes between 40% and 70% or between 0.5 miles and one mile from a road and on slopes less than 70%. These areas have the same productivity for restocking as likely operable areas, are potentially forested, and no statutory restrictions from timber harvest.

[19] Current areas are reflected on national maps at the Forest Service Roadless Area website: https://www.fs.usda.gov/managing-land/planning/roadless

[20] Note that this national level review is currently undergoing quality control and Regional/local vetting and the data presented will be updated in the final BA.

- Likely Not Operable: Include slopes over 70%, further than one mile from a road or areas over one mile from a road and less than 70% percent slope. These areas have low productivity, are potentially forested, and no statutory restrictions from timber harvest.

The affected environment's overlap with hydrologic unit codes (HUC) is presented in Table 4. Within the Tongass National Forest 9.3 million acres of inventoried roadless area managed under the 2001 Roadless Rule, or about 56 percent of the Forest. Approximately 9.2 million acres of these are in the affected environment (on National Forest System lands and outside of wilderness, wilderness study area and wild river segments). Operability analysis excluded portions of the action area that were not forested.

Table 2. Operability by region within the action area (affected environment).

Region	Roadless Affected Environment Acres	Likely Operable		Likely Operable, but Complex		Likely Not Operable	
		Acres	*Percent of Affected Environment*	*Acres*	*Percent of Affected Environment*	*Acres*	*Percent of Affected Environment*
1	5,951,089	990,944	16.7%	2,020,381	33.9%	2,218,591	37.3%
2	1,733,284	411,357	23.7%	452,877	26.1%	505,389	29.2%
3	2,528,791	156,219	6.2%	262,988	10.4%	1,689,352	66.8%
4	8,474,301	1,030,112	12.2%	1,295,380	15.3%	4,219,049	49.8%
5	3,634,708	553,856	15.2%	1,216,690	33.5%	1,278,172	35.2%
6	3,729,946	787,299	21.1%	1,512,164	40.5%	1,182,516	31.7%
8	839,339	337,357	40.2%	384,545	45.8%	103,114	12.3%
9	605,812	270,002	44.6%	193,661	32.0%	137,154	22.6%
10	12,565,353	396,547	3.2%	651,833	5.2%	5,574,698	44.4%
Total	**40,062,623**	**4,933,693**	**12.3%**	**7,990,519**	**19.9%**	**16,908,035**	**42.2%**

Table 3. Acres of the affected environment (action area) where land management plans currently allow road construction and timber harvest.

Region	Total National Forest System Lands (thousands of acres)	2001 Roadless Rule IRA not including Co or ID (thousands of acres)	Potentially affected environment (thousands of acres)	Potentially affected environment where current LMPs allow/generally allow road construction (thousands of acres)	Potentially affected environment where current LMPs allow/generally allow timber harvest (thousands of acres)
1	25,624	6,707	5,951	3,635	1,253
2	22,059	1,761	1,733	109	443
3	20,539	2,724	2,529	1,085	1,216
4	31,856	8,946	8,474	6,187	5,621
5	20,253	4,200	3,635	1,085	1,578
6	25,012	3,967	3,730	617	611
8	13,457	954	839	113	158
9	12,180	664	606	264	271
10	22,132	14,779	12,565	5,125	2,255
Total	**193,112**	**44,701**	**40,063**	**18,221**	**13,406**

Table 4. HUC4s within the affected environment (action area) and their operability.

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
0104	Androscoggin	2,257,874	27,166	1.2%	13,740	4,682	8,575
0106	Saco	3,105,641	38,808	1.2%	17,898	6,344	14,315
0107	Merrimack	3,203,363	59,281	1.9%	29,321	9,699	19,819
0108	Connecticut	7,208,942	90,744	1.3%	41,564	19,026	29,739
0202	Upper Hudson	8,105,456	977	0.01%		692	221
0207	Potomac	9,394,592	159,330	1.7%	23,653	49,566	85,465
0208	Lower Chesapeake	13,237,535	160,810	1.2%	20,666	49,374	89,928
0301	Chowan-Roanoke	12,199,819	237	0.002%	0	200	29
0302	Neuse-Pamlico	9,139,348	20,742	0.2%	11,702	3,823	5,125
0305	Edisto-Santee	15,166,690	37,403	0.2%	10,388	8,533	18,238
0306	Ogeechee-Savannah	10,913,203	18,757	0.2%	614	8,522	9,489
0307	Altamaha-St. Marys	13,260,203	4,187	0.0%	525	1,675	1,969
0308	St. Johns	7,798,794	4,820	0.1%	0	4,171	268
0311	Suwannee	9,035,995	13,888	0.2%	1,324	7,420	5,076
0312	Ochlockonee	2,430,245	3,907	0.2%	0	3,617	221
0313	Apalachicola	13,203,758	20,578	0.2%	41	13,977	6,467
0315	Alabama	14,555,603	33,302	0.2%	2,764	14,981	15,352
0316	Mobile-Tombigbee	13,970,558	621	0.004%	0	598	17
0401	Western Lake Superior	6,225,236	56,712	0.9%	5,164	35,322	14,911
0402	Southern Lake Superior	4,606,993	4,336	0.1%	8	3,335	905
0403	Northwestern Lake Michigan	13,259,458	17,503	0.1%	0	16,393	876
0406	Northeastern Lake Michigan	5,903,166	3,854	0.1%	0	3,184	502
0420	Northern Lake Huron	8,713,390	7,615	0.1%	19	4,526	2,406
0424	Lake Huron	16,085,424	223	0.001%	4	115	94
0430	Middle Saint Lawrence River	8,593,027	14,115	0.2%	3,525	4,535	5,875

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0501	Allegheny	7,506,369	24,767	0.3%	4,572	10,512	9,534
0502	Monongahela	4,720,782	45,408	1.0%	4,502	21,059	19,527
0505	Kanawha	7,838,989	139,350	1.8%	9,602	70,451	58,414
0507	Big Sandy-Guyandotte	3,822,713	4,744	0.1%	851	892	2,976
0510	Kentucky-Licking	6,829,672	2,847	0.0%	185	1,548	1,084
0514	Lower Ohio	8,063,733	14,833	0.2%	0	14,575	121
0601	Upper Tennessee	11,067,332	157,075	1.4%	25,561	43,597	87,009
0602	Middle Tennessee-Hiwassee	3,302,727	38,239	1.2%	6,571	9,579	21,855
0701	Mississippi Headwaters	12,870,704	8	0.0%	0	1	5
0703	St. Croix	4,940,765	948	0.0%		842	99
0705	Chippewa	6,105,019	35,519	0.6%	563	28,152	5,706
0707	Wisconsin	7,636,945	282	0.004%	0	257	0
0714	Upper Mississippi-Kaskaskia-Meramec	10,938,157	3,747	0.03%	3	2,764	958
0802	Lower Mississippi-St. Francis	10,351,746	2,676	0.03%	87	1,330	1,240
0804	Lower Red-Ouachita	13,124,880	8,087	0.1%	1,567	2,779	3,611
0806	Lower Mississippi-Big Black	4,530,136	2,620	0.1%	0	2,592	0
0902	Red	30,126,196	46,051	0.2%	2	3,073	250
0903	Rainy	17,195,988	17,931	0.1%	789	10,798	5,879
1002	Missouri Headwaters	8,982,371	1,652,068	18.4%	579,631	343,656	460,122
1003	Missouri-Marias	12,706,687	819,632	6.5%	364,078	129,948	258,897
1004	Missouri-Musselshell	15,031,372	232,225	1.5%	89,065	47,002	78,355
1007	Upper Yellowstone	9,254,806	568,818	6.1%	233,379	68,768	169,390
1008	Big Horn	14,641,079	805,533	5.5%	270,407	167,482	207,159
1009	Powder-Tongue	12,033,003	332,777	2.8%	94,538	100,594	91,394
1010	Lower Yellowstone	8,891,206	25,101	0.3%	2	447	46
1011	Missouri-Little Missouri	11,029,150	195,615	1.8%	2,310	31,277	13,407

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1012	Cheyenne	15,501,512	116,830	0.8%	1,278	8,274	7,357
1013	Missouri-Oahe	23,734,929	14,165	0.1%	0	24	16
1018	North Platte	19,885,105	291,618	1.5%	74,813	92,947	92,862
1101	Upper White	14,302,523	25,260	0.2%	1,004	18,617	5,407
1108	Upper Canadian	7,983,248	11,869	0.1%	3,955	2,550	3,455
1111	Lower Arkansas	10,112,696	87,361	0.9%	981	72,048	13,945
1114	Red-Sulphur	17,697,863	16,835	0.1%	6	13,301	3,357
1203	Trinity	11,486,295	1,339	0.0%	0	1,323	10
1204	Galveston Bay-San Jacinto	5,323,577	2,754	0.1%	0	2,670	49
1301	Rio Grande Headwaters	4,871,558	7,575	0.2%	366	3,260	2,739
1302	Rio Grande-Elephant Butte	17,358,505	508,029	2.9%	309,272	40,678	68,261
1303	Rio Grande-Mimbres	13,916,028	200,847	1.4%	181,513	5,608	10,448
1305	Rio Grande Closed Basins	11,246,551	108,924	1.0%	59,917	2,433	4,520
1306	Upper Pecos	15,058,643	88,505	0.6%	40,071	14,108	22,133
1403	Upper Colorado-Dolores	5,352,777	77,699	1.5%	62,734	2,835	3,364
1404	Great Divide-Upper Green	13,283,979	700,232	5.3%	231,148	189,288	150,924
1405	White-Yampa	8,486,235	68,265	0.8%	7,287	33,741	22,634
1406	Lower Green	9,316,233	897,863	9.6%	296,324	197,988	218,344
1407	Upper Colorado-Dirty Devil	8,714,077	546,145	6.3%	305,736	73,368	64,362
1408	San Juan	15,955,361	49,434	0.3%	42,319	219	843
1501	Lower Colorado-Lake Mead	19,373,477	271,873	1.4%	199,422	15,403	16,340
1502	Little Colorado	17,316,510	57,712	0.3%	36,597	6,938	6,514
1503	Lower Colorado	13,280,094	45,500	0.3%	41,583	389	876
1504	Upper Gila	9,722,929	865,873	8.9%	650,032	53,057	78,942
1505	Middle Gila	11,508,412	235,202	2.0%	126,359	7,181	32,979
1506	Salt	8,624,188	256,342	3.0%	138,469	10,256	18,130

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1507	Lower Gila	9,648,506	54,314	0.6%	39,299	2,444	5,650
1508	Sonora	15,386,447	42,206	0.3%	22,314	1,817	5,707
1601	Bear	4,809,334	355,039	7.4%	134,408	77,651	99,687
1602	Great Salt Lake	18,358,652	794,991	4.3%	434,506	108,199	171,935
1603	Escalante Desert-Sevier Lake	10,543,993	996,473	9.5%	664,234	94,409	133,741
1604	Black Rock Desert-Humboldt	18,329,433	475,723	2.6%	160,320	26,199	51,779
1605	Central Lahontan	8,115,069	574,982	7.1%	279,617	58,824	77,923
1606	Central Nevada Desert Basins	30,543,216	1,649,766	5.4%	1,006,726	17,294	33,295
1701	Kootenai-Pend Oreille-Spokane	33,298,571	2,610,131	7.8%	1,037,377	386,911	1,101,449
1702	Upper Columbia	29,592,139	827,984	2.8%	371,475	107,337	288,076
1703	Yakima	3,935,933	204,042	5.2%	57,517	46,419	91,620
1704	Upper Snake	22,958,517	944,925	4.1%	394,121	147,133	252,815
1705	Middle Snake	23,671,623	364,179	1.5%	74,047	68,957	100,804
1706	Lower Snake	22,428,309	425,288	1.9%	114,622	71,907	160,448
1707	Middle Columbia	19,138,642	553,310	2.9%	120,331	159,731	232,331
1708	Lower Columbia	3,985,028	180,740	4.5%	49,310	41,127	87,870
1709	Willamette	7,336,249	175,545	2.4%	31,119	56,721	85,324
1710	Oregon-Washington Coastal	16,121,566	490,054	3.0%	107,803	124,677	241,186
1711	Puget Sound	13,085,895	467,199	3.6%	237,227	42,819	164,847
1712	Oregon Closed Basins	11,155,153	86,239	0.8%	14,855	38,158	25,541
1801	Klamath-Northern California Coastal	16,744,271	774,800	4.6%	239,956	123,855	324,032
1802	Sacramento	17,794,824	715,873	4.0%	219,382	155,657	286,475
1803	Tulare-Buena Vista Lakes	10,501,344	493,058	4.7%	229,044	52,982	154,083
1804	San Joaquin	10,127,905	231,575	2.3%	75,488	59,843	75,879
1806	Central California Coastal	8,472,485	408,848	4.8%	130,027	61,749	145,265
1807	Southern California Coastal	8,861,368	437,036	4.9%	141,749	76,724	187,761

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1808	North Lahontan	2,913,621	36,488	1.3%	7,118	9,919	14,513
1809	Northern Mojave-Mono Lake	18,105,223	559,848	3.1%	225,083	25,081	32,199
1810	Southern Mojave-Salton Sea	10,934,158	23,140	0.2%	8,025	4,320	5,116
1901	Southeast Alaska	69,591,689	9,473,879	13.6%	4,626,703	357,067	589,753
1902	South Central Alaska	65,799,951	3,089,520	4.7%	942,450	36,276	61,108
1907	Upper Yukon River	76,785,649	1,897	0.002%	13	0	0
2101	Puerto Rico	3,373,641	12,607	0.4%	n/a	n/a	n/a

General Characteristics of the Affected Environment

Roadless Rule prohibitions do not mean no activities occur in Inventoried Roadless Areas. Activities that are ongoing and permissible, if not restricted by land management plans or other law, regulation, and policy include, but are not limited to:

- Access for reserved or outstanding rights/existing rights for minerals,
- Continuation of existing land use authorizations (ex. utilities, recreation special uses, private access, easements, etc.),
- Road maintenance on existing roads,
- Motorized and non-motorized trail construction or maintenance,
- Hunting, fishing, hiking, camping, and other recreational uses,
- Use of a motorized vehicle on a trail open to motorized use,
- Mountain biking on a trail open to mechanized use,
- Prescribed burning, and
- Livestock grazing.

The 2001 Roadless Rule prohibition on road construction and reconstruction in IRAs did not affect already existing access, occupancy, and use rights. These rights include those that exist by law, treaty, or by other authority. They include, but are not limited to, the right to provide reasonable access across NFS lands to private property, mining claims for locatable minerals under the 1872 Mining Law, and land uses protected by Native American treaty rights. These rights include, but are not limited to, rights of access provided in the ANILCA (Public Law 96-487), and recognized highway rights-of-way granted over NFS lands under Revised Statute 2477 (Public Law 94-579). The most common type of access pursued in conjunction with these two prominent statues is motorized road access. Land use authorizations and rights-of-way existing prior to the 2001 Roadless Rule are ongoing and part of baseline conditions.

While they are referred to as “roadless”, because of the decades between inventory and designation, there are thousands of miles of existing roads within lands designated under the 2001 Roadless Rule, including National Forest System roads as well as roads of other ownerships. Additionally decommissioned roads and unauthorized routes that are not part of the official system or records may exist.

Between 1984 and 2024, wildfire burned 20.5% of potentially affected IRAs (or 29.9% of potentially affected IRAs outside of Alaska). Of these, 4.2% (1.7 million acres) were high severity wildfire and 5.2% (2.1 million acres) were moderate severity wildfire. Nationally, wildfire hazards are commonly characterized using the Wildfire Hazard Potential (WHP) index, which integrates burn probability and potential fire intensity. Across potentially affected IRAs outside of Alaska, 42 percent of lands are classified as high or very high Wildfire Hazard

Potential, indicating a high potential for intense wildfire under severe weather conditions. Regional values range from 5 percent in the Eastern Region to 60 percent in the Pacific Southwest Region. Within potentially affected IRAs 9.8 million acres (24 percent) overlap with Healthy Forests Restoration Act Wildland Urban Interface. Within the potentially affected environment 5.9 million acres (14.8 percent) overlap with Community Wildfire Risk Reduction Zones (CWRRZ). Nearly all CWRRZ within the potentially affected environment occurs within the transition zone (within the 1.5-mile buffer surrounding building clusters).

2001 Roadless Area Conservation Rule Implementation

The 2001 Roadless Rule provided that the Responsible Official may grant certain limited exceptions to the road construction and timber harvest prohibitions, including for forest stewardship purposes, public health and safety, reserved or outstanding rights, existing or renewed mineral leases, and realignment of roads to prevent resource damage or address safety hazards (36 CFR 294.12(b) and 294.13(b) (2001)). It also provided exceptions to the timber harvest prohibitions for generally small diameter tree cutting to improve habitat for threatened, endangered, or sensitive species or to maintain or restore ecosystem characteristics and where it maintains or enhances one or more of the roadless characteristics, or when the activity is incidental to activities not otherwise prohibited (36 CFR 294.13(b) (2001)). The rule states that use of the timber harvest exceptions would be expected to be infrequent. Amid legal challenges to the 2001 Roadless Rule and to provide for consistent application, exceptions for specific activities in IRAs were reviewed and authorized by the Secretary and Forest Service Chief until 2012. In 2012, certain exceptions were delegated to the applicable regional forester. In 2018 all exceptions were delegated to the regional foresters, and some exceptions for public health and safety or small diameter tree cutting[21] may be further delegated to responsible officials. Approval of exceptions generally requires review documentation, including maps and justification.

Regional forester approval of exceptions under the 2001 Roadless Rule has been limited, and the majority have been for stewardship purposes. There is no national-level tracking of the review, approval, or implementation of 2001 Roadless Rule exceptions, though the Washington Office has recommended that Responsible Officials document reviews and approvals. Regional level tracking of review and approvals between 2018 and 2025 shows about 188 documented projects were granted exceptions. About 10 percent involved road related activities and 96 percent involved tree cutting (some involved both road and tree cutting activities). Since 2018, the majority were related to maintaining and restoring ecosystem characteristics. Eight exceptions approved tree cutting for the explicit purpose of "improving threatened, endangered, or sensitive species habitat," although some of the exceptions outside of this category may have served multiple purposes, such as providing benefits to species habitat. This information does not include responsible official reviews or approvals below the regional forester level[22], so it is not possible to list all exceptions that were granted with existing data sources.

[21] Regional foresters and deputy regional foresters must retain sole authority for exceptions under §294.12(b)(3)-(7) and §294.13(b)(4) (2001).
[22] Similarly, the Agency does not track when requests for exceptions were abandoned or not approved.

The best available information on activities that have been implemented within IRAs over the last decade comes from the Forest Service Activity Tracking System (FACTS). This data shows an average of 500 acres per year of timber harvest and a total of 1.8 million acres of hazardous fuels reduction within IRAs managed under the 2001 Roadless Rule. More than half of the hazardous fuels reduction was from wildfire use, which is reported when acres burned by wildfire are determined to meet land management planning objectives in the post-fire assessment. An additional 25 percent of the hazardous fuels reduction was from prescribed burning activities for reducing fuel hazards and as resource management treatments. Best available records of permanent road construction indicate that approximately 60 miles of roads were built within inventoried roadless areas after 2013.

Use of the available exceptions has been limited in scope and inconsistent. This is due in part to rigorous project-level analysis requirements and associated reviews and approvals needed to demonstrate consistency with Roadless Rule requirements, as well the rule's intent for infrequent use.

STATUS OF THE SPECIES AND CRITICAL HABITAT

Given the near-national scope of the project and large number of ESA-listed resources it is not prudent to include a description of the status of every species and critical habitat in the action area. Given the large geographic scope of the action the status of the species in the action area is essentially the same as the status of the species overall, which is managed and published by the Fish and Wildlife Service online or in a number of species review documents held by the Services. During early coordination the FWS indicated development of a species status section for this project would not be prudent or efficient. As such, no species status section is included. For information on the current status of each species see the Fish and Wildlife Service's ECOS website.

EFFECTS ANALYSES

A discussion of effects on taxonomic groups is below. Where necessary for species of high interest, effects discussions are provided at the species level.

Mammals

The potential impacts of the proposed action are diverse as listed mammalian species in the action area include large carnivores, small herbivores, ungulates, and marine mammals in disparate portions of the United States and its territories. As such, we have provided evaluations for each species separately, with narrow exceptions for subspecies whose evaluations depend on similar literature. The mammal group includes 32 total species including 22 endangered, 8 threatened, and 2 experimental populations of mammals (Table 5).

Table 5. Threatened (T), endangered (E), proposed threatened (PT), and experimental population (EXP) mammal species in the action area. The presence of critical habitat (and it's percent overlap with the action area) is also provided. A summary of exposure (Low, Moderate, High, Unknown) and the effects determination (No Effect (NE), NLAA (Not Likely to Adversely Affect), LAA (Likely to Adversely Affect), NA (Not Applicable)) are also shown.

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment *	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Black-footed ferret	*Mustela nigripes*	E	NA	Low	LAA	NA
Black-footed ferret (Experimental)	*Mustela nigripes*	EXP	NA	Low	LAA	NA
Canada lynx	*Lynx canadensis*	T	Yes ~12.32%	High	LAA	LAA
Carolina northern flying squirrel	*Glaucomys sabrinus coloratus*	E	NA	High	LAA	NA
Fisher	*Pekania pennanti*	E	Yes ~10.72%	High	LAA	LAA
Gray bat	*Myotis grisescens*	E	NA	Low	LAA	NA
Gray wolf (MN)	*Canis lupus*	T	Yes ~1.04%	Low	LAA	NLAA
Gray wolf (44-entity)	*Canis lupus*	E	NA	High	LAA	NA
Grizzly bear	*Ursus arctos horribilis*	T	NA	High	LAA	NA
Indiana bat	*Myotis sodalis*	E	Yes ~0.31%	High	LAA	NLAA
Jaguar	*Panthera onca*	E	Yes ~9.92%	Unknown	LAA	LAA
Mexican long-nosed bat	*Leptonycteris nivalis*	E	NA	Low	LAA	NA
Mexican wolf	*Canis lupus baileyi*	E	NA	High	LAA	NA
Mexican wolf (Experimental)	*Canis lupus baileyi*	EXP	NA	High	LAA	NA
Mount Graham red squirrel	*Tamiasciurus fremonti grahamensis*	E	Yes ~41.22%	High	LAA	LAA
New Mexico meadow jumping mouse	*Zapus hudsonius luteus*	E	Yes ~0.79%	High	LAA	LAA
North American Wolverine	*Gulo gulo luscus*	T	NA	High	LAA	NA
Northern long-eared bat	*Myotis septentrionalis*	E	NA	High	LAA	NA
Ocelot	*Leopardus (=Felis) pardalis*	E	NA	Low	LAA	NA
Ozark big-eared bat	*Corynorhinus townsendii ingens*	E	NA	High	LAA	NA
Pacific marten, Coastal DPS	*Martes caurina*	T	Yes ~17.64%	High	LAA	LAA
Penasco least chipmunk	*Tamias minimus atristriatus*	E	NA	Unknown	NLAA	NE
Peninsular bighorn sheep	*Ovis canadensis nelsoni*	E	NA	Low	LAA	NE
Preble's meadow jumping mouse	*Zapus hudsonius preblei*	T	NA	Low	NLAA	NA

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment *	Exposure	Species Effects Determination	Critical Habitat Effects Determination
San Bernardino Merriam's kangaroo rat	*Dipodomys merriami parvus*	E	Yes ~6.24%	High	LAA	LAA
San Joaquin kit fox	*Vulpes macrotis mutica*	E	NA	High	LAA	NA
Sierra Nevada bighorn sheep	*Ovis canadensis sierrae*	E	Yes ~2.57%	Moderate	LAA	NLAA
Sierra Nevada red fox	*Vulpes vulpes necator*	E	NA	High	LAA	NA
Southern mountain caribou DPS	*Rangifer tarandus ssp. caribou*	E	Yes ~4.44%	High	LAA	LAA
Stephens' kangaroo rat	*Dipodomys stephensi (incl. D. cascus)*	T	NA	Low	NLAA	NA
Tricolored bat	*Perimyotis subflavus*	PE	NA	High	LAA	NA
Utah prairie dog	*Cynomys parvidens*	T	NA	High	LAA	NA
Virginia big-eared bat	*Corynorhinus townsendii virginianus*	E	Yes 2%	High	LAA	LAA
West Indian Manatee	*Trichechus manatus latirostris, Trichechus manatus manatus*	T	No	Low	NLAA (NE in Puerto Rico)	NE

Marine Mammals

West Indian Manatee

The West Indian Manatee, which is a general name for two distinct subspecies, the Florida manatee (*Trichechus manatus latirostris*) and the Antillean manatee (*Trichechus manatus manatus*), has a complex listing history beginning with the listing of the Florida manatee (32 FR 4001) as endangered under the Endangered Species Preservation Act of 1966. An amendment added the Antillean manatee to the listing (35 FR 18319). These subspecies were subsequently grandfathered into listing under the ESA as the West Indian manatee. In 2017, the West Indian manatee was reclassified as threatened (82 FR 16668), however there is currently a proposed rule to list the two entities separately with the Florida manatee receiving threatened status and the Antillean manatee receiving endangered status (90 FR 3131).

Trends for the Florida manatee were stable or increasing in all 4 management units until recent unusual mortality events (UME) that affect the Atlantic Management Unit. The demographic impact of the Atlantic mortality unit that began in December 2020 was not known as of the 2024 SSA (USFWS 2024a). The Antillean manatee in Puerto Rico is thought to be stable, although survey data is a few years old and deaths caused by watercraft are thought to be increasing (USFWS 2024b).

Exposure to the Proposed Action Area and Proposed Actions

The listed entity, West Indian manatee is found in both Florida and Puerto Rico. Manatees inhabit coastal waters and lakes and spend time in both fresh and salt water. Manatees in Florida are reliant on springs and other sites where water temperatures exceed a specific threshold. Both species are dependent on seagrass food sources. Critical habitat for the Florida manatee was designated in 1976 (41 FR 41914). In 2024, revisions to critical habitat for the Florida manatee and designation of critical habitat for the Antillean manatee were proposed (89 FR 78134). None of the current West Indian manatee critical habitat overlaps with an inventoried roadless area affected by the proposed action. In the NRM and GBIF databases, there are no observations of West Indian Manatee in inventoried roadless areas affected by the proposed action, and only 36 observations of them at all on USFS lands. It is difficult to draw conclusions from this data, so we conducted a supplementary analysis overlapping the USFWS range for the species (USFWS 2026a) with inventoried roadless areas affected by the proposed action. There were no points of intersection. Florida Manatees are known to occur in waterbodies in Ocala National Forest (1999 land management plan, available at USFS 2025a), and there are two inventoried roadless areas within this forest one of which (Alexander Springs Creek) abuts a stream below a spring site. The true impacts of any potential future activities in that roadless area would be more accurately assessed by surveys of that creek. Proposed Antillean Manatee critical habitat includes coastal habitat in Puerto Rico, some of which is downstream from waterbodies originating in El Yunque National Park, which contains multiple inventoried roadless areas which could be affected by the proposed action.

Effects of Roads

Sedimentation from roads can affect seagrass growth, an important food source for manatees (USFWS 2024b). The consequences of seagrass quality on manatee forage and health are detailed in multiple sources (USFWS 2024b, Hieb et al. 2021). Manatees frequent waterbodies used by human recreationists, if road development led to increased recreation this could cause increased disturbance to manatees from human presence (USFWS 2024a). This is more of a concern for the Florida manatee which has an inventoried roadless area right next to a spring site visited by recreationists.

Effects of Timber/ Vegetation Management

Timber harvest can affect sedimentation of nearby waterbodies, thus affecting water quality in manatee habitat. Sedimentation can also affect the growth of seagrasses, a critical food source for manatees (USFWS 2024b). The consequences of seagrass quality on manatee forage and health are detailed in multiple sources (USFWS 2024bean, Hiep et al. 2021).

Existing Protections

The 1999 Land Management Plan for Forests in Florida provides two guidelines with respect to manatees. They are to 1) “Restrict the construction of boating facilities to areas where their construction and subsequent use will not adversely affect manatees” and 2) “Restrict activities that degrade manatee habitat.” There is no mention of the manatees in the El Yunque Forest Plan.

Critical Habitat

The physical and biological features identified as critical for Florida manatee habitat in the 2024 proposed critical habitat are as follows:

"(1) Areas of water warmed by natural processes (e.g., spring discharges, passive thermal basins) that have either:

(a) Reliable thermal quality throughout the winter (i.e., having at least a medium thermal quality as defined by the Florida Manatee WWHAP (Valade et al. 2020, pp. 25–32)), which consists of water temperatures that stay at or above:

(i) 72 degrees Fahrenheit (°F) (22 degrees Celsius (°C)) during mild weather,
(ii) 68 °F (20 °C) during cold weather, and
(iii) 64 °F (18 °C) during severe cold weather; or

(b) Established manatee use throughout the winter each year (see the Florida Manatee WWHAP (Valade et al. 2020, pp. 25–32)).

(2) Areas supporting submerged, emergent, or floating aquatic vegetation within 18.6 miles (30 kilometers) of:

(a) The natural warm-water sources described in paragraph (1), above; or

(b) Other established winter manatee aggregation areas (i.e., power plants with established manatee use." (89 FR 78134)

None of the 2024 proposed Florida manatee critical habitat overlaps with an inventoried roadless area affected by the proposed action. There is one small roadless area that is upstream of manatee critical habitat in the Ocala National Forest.

Although forest streams occur upstream of the critical habitat designated in Florida, the downstream distance suggests effects from sedimentation will be greatly reduced as sediment settles out of the stream as it progresses toward the ocean. Further, very little of the action area is upstream of designated critical habitat in Florida. Inland activities are not expected to produce so much sediment that the critical habitat will be permanently impacted. A such, the FS has determined West Indian manatee critical habitat in Florida will not be affected.

For the Antillean manatee elements of critical habitat were defined as "nearshore marine waters with at least two of the following resources within a 3-mile (5-kilometer) radius:

(1) Freshwater sources, such as streams and wastewater outfalls;

(2) Seagrass in waters less than 43 ft (13 m) deep; and

(3) Calm waters, such as shallow bays and coves, with water depths less than 9.8 ft (3 m) and wave heights less than 0.98 ft (0.3 m)." (89 FR 78134)

None of the 2024 proposed Antillean manatee critical habitat overlaps with an inventoried roadless area affected by the proposed action, however there are large tracts of inventoried roadless areas that are upstream of critical manatee habitat in El Yunque National Forest. It is possible that sedimentation in these streams from actions that become more likely with the rescission of the roadless rule could affect seagrass in critical manatee habitat.

Although forest streams occur upstream of the critical habitat designated off the coast of Puerto Rico, the distance is greater than 5 stream miles, suggesting effects from sedimentation will be

greatly reduced as sediment settles out of the stream as it progresses toward the ocean. Any sediment reaching the ocean will settle and be redistributed over time. Inland activities are not expected to produce so much sediment that the coastal environment will be permanently impacted. A such, the FS has determined West Indian manatee critical habitat off the coast of Puerto Rico will not be affected.

Effects Determination

Given the low likelihood of exposure to activities that may occur in the action area following rescission of the 2001 Roadless Rule and that the Forests of Florida land management plan contains direction to restrict activities that degrade manatee habitat, the Forest Service has determined rescission of the 2001 Roadless Rule is not likely to adversely affect the West Indian Manatee because the likelihood of effects to this species are so low as to be discountable. Effects to the subpopulation in the coastal waters of Puerto Rico are far enough removed from the inland waters of El Yunque NF that the Forest Service has made a separate determination of No Effect for that population. Should the Caribbean population of West Indian Manatee be listed as a separate entity in the future, this determination will apply. The Forest Service has determined the proposed activity will have No Effect on West Indian manatee critical habitat.

Bats

Ozark big-eared bat

The Ozark big-eared bat (*Corynorhinus townsendii ingens*), a subspecies of Townsend's big-eared-bat, is an insectivorous bat that was listed as endangered under the Endangered Species Act in 1979 (44 FR 69206). Current conservation actions are informed by a recovery plan (USFWS 1995a) and a 5-year review (USFWS 2020a). As of 2020, population trends in seven maternity colonies and nine hibernacula were mixed, with most displaying stable trends, several declining, and a few increasing (USFWS 2020a). However, the methods used for both maternity counts and hibernacula surveys are variable and contribute to substantial uncertainty in trend estimates (USFWS 2020a). Ozark big-eared bats are at risk of extinction due to their small population size, limited distribution, and vulnerability to human disturbance (USFWS 1995a, USFWS 2020a).

Exposure to the Proposed Action Area and Proposed Actions

Ozark big-eared bats are currently found in Oklahoma and Arkansas, and historically they also inhabited Missouri. They hibernate and roost year-round in caves and karst habitats that are surrounded by oak-hickory hardwood forests (USFWS 1995a). They forage for insects, and especially moths, near trees and in stream- and forest-edge habitats (USFWS 1995a, Clark et al. 1993). The greatest threat to Ozark big-eared bats is human disturbance of caves, especially during the maternity and hibernation periods (USFWS 1995a). No designated critical habitat currently exists for Ozark big-eared bats (44 FR 69206), but many stakeholders engage in habitat conservation efforts on both public and private land.

There are several inventoried roadless areas (IRAs) on the Ozark-St. Francis National Forest. Although road construction or reconstruction is allowed on more than half of this IRA acreage, there are also some areas where it is not currently allowed (USFS 2000a). As such, rescinding

the 2001 Roadless Rule could eventually lead to more road and timber development in areas near Ozark big-eared bat habitat. There is only one observation of the species in inventoried roadless areas and it occurred on lands classified as operable, suggesting this species has potentially high likelihood of exposure to at least a few individuals, although it is extremely difficult to draw conclusions from such a limited sample size. As such we conducted a supplementary analysis intersecting the USFWS range with inventoried roadless areas affected by the proposed action and found that approximately 0.004% of the current range is in the proposed action area.

Additional bat habitat is located on both public and private land. The Ozark Plateau National Wildlife Refuge was created to conserve habitat for threatened and endangered species including the Ozark big-eared bat, and it is accessible by permit only to protect the bats (USFWS 2025a). Key conservation easements have also been implemented on private land, including two separate 40-acre tracts purchased by the Cherokee Nation for bat cave habitat protection (USFWS 1995a).

Effects of Roads

The specific effects of roads on Ozark big-eared bat population trends have not been explicitly quantified to date. This may be due in part to the fact that big-eared bats have relatively low-intensity echolocation calls which are often not well recorded by detectors which makes researching them challenging (Loeb et al. 2011). However, given that human disturbance to caves is a major threat to population persistence (USFWS 1995a, USFWS 2020a), the increased access that roads can provide to caves is a concern. For example, in another bat species, a maternity colony decline was thought to be the result of increased recreation by cavers after a road was paved (USFWS 1995a).

If new roads must be constructed near Ozark big-eared bat habitat, then installing gates and signage that deter cave access without impeding bats is an effective measure that can help mitigate negative effects (USFWS 2020a). Many stakeholders have contributed to successful cave gating efforts for Ozark big-eared bat protection on both public and private lands (USFWS 2020a). On the Ozark Plateau National Wildlife Refuge, which was created in part to protect cave habitat, proposed management alternatives included avoiding new road development, closing caves to the public, building and repairing cave gates, and posting signs prohibiting entry (78 FR 9410).

Although published studies are unavailable, it is reasonable to assume that roads could lead to increases in direct mortalities from vehicle collisions. Road construction could also impact bat habitat including nearby caves, which is a documented concern for another *Corynorhinus* bat (U.S. Fish and Wildlife Service 2019f). Although not specific to roads, human disturbance is known to be a major threat to Ozark big-eared bat populations (USFWS 2020a).

Effects of Timber/ Vegetation Management

Research on the effects of timber production activities on Ozark big-eared bat populations has been limited to date, so impacts remain uncertain. The 1995 Revised Recovery Plan noted that there had been a loss of caves and foraging habitat used by Ozark big-eared bats due to various types of development, including timber development (USFWS 1995a). This Plan also noted that since human disturbance is the main threat to the species, protecting caves and foraging habitat

from all types of development is a key recovery strategy (USFWS 1995a). However, the 2020 Five-Year Review acknowledged that timber thinning, planting, and prescribed burns can be important parts of habitat enhancement and restoration (USFWS 2020a).

More recently, a study conducted in Arkansas' Ozark National Forest investigated the effects of prescribed fire and mechanical thinning treatments on Ozark big-eared bat foraging behavior and prey availability (Marsh 2023). This research suggested that prescribed burning and mechanical thinning may benefit Ozark big-eared bat prey availability and foraging because bat prey was higher in areas with open canopies and more groundcover (Marsh 2023). However, the authors noted that study design challenges and forest complexity made it difficult to draw strong conclusions about the bats' relative use of burned and thinned habitats (Marsh 2023).

In contrast, a previous review of *Corynorhinus* (the genus encompassing Ozark big-eared bats) foraging behavior found that preferred moth prey were typically captured in areas of mature timber and avoided open habitats, but that *Corynorhinus* bats made decisions about where to forage based on forest structure, not solely based on prey availability (Loeb et al. 2011). This review showed that the bats selected forest corridors and edges which may make it easier for them to glean prey (Loeb et al. 2011) which may have different implications for forest management than if maximizing prey availability was the primary management concern. Additional research found that moths were more abundant in poletimber stands, riparian areas, and upland forests, and suggested that using forest management to maintain habitat heterogeneity around roosts may benefit bats (Dodd et al. 2008).

Although the specific effects of timber production on Ozark big-eared bat populations remains uncertain, the 2005 Ozark-St. Francis Forest Plan limits the cutting of overstory vegetation, road construction, and prescribed burning within 200 feet of colony sites (USFS 2025a). Given the bats' sensitivity to human disturbance, it is reasonable to assume that any timber harvest or forest management activities would typically use design features to limit short-term disturbance while offering compelling evidence of future benefits to habitat.

Existing Protections

The USDA Forest Service manages important habitat for these bats on the Ozark-St. Francis National Forest in Arkansas (2005 land management plan, USFS 2025a) and there are some existing protections for these bats in the Forest's Plan. For example, the Ozark-St. Francis Forest Plan specifies a forest-wide standard designed to protect habitat which states, "A 200-foot buffer of undisturbed forest will be maintained around […] Ozark big-eared bat maternity sites, bachelor sites, or winter colony sites. Prohibited activities within this buffer include cutting of overstory vegetation; construction of roads, trails, or wildlife openings or development of pastures; and prescribed burning…" (2005 land management plan, USFS 2025a).

Effects Determination

Given the low degree of overlap of the proposed action with the species' range, we expect the potential for exposure to be low. Although the Ozark-St. Francis NF forest plan provides protections for this species, given the high degree of sensitivity of the species to human

disturbance, we have determined the proposed action is likely to adversely affect Ozark big-eared bats.

Virginia big-eared bat

The Virginia big-eared bat (*Corynorhinus townsendii virginianus*), a subspecies of Townsend's big-eared bat, is an insectivorous bat that was listed as endangered under the Endangered Species Act in 1979 (44 FR 69206). Current conservation actions are informed by a recovery plan (USFWS 2024c) and a 5-year review (USFWS 2019a). As of 2019, the range-wide population of Virginia big-eared bats was thought to have increased by 28-30 percent since 2008, with an estimated 19,574 bats in hibernacula and 11,778 in maternity sites (USFWS 2019a). However, some populations and regions experienced declines (USFWS 2019a). There are four geographically distinct management units of this subspecies that do not intermix and that each contribute to the genetic diversity, adaptive capacity, and distribution of the subspecies (USFWS 2024c). As such, maintaining healthy populations across all four of these management units is important. However, Virginia big-eared bats are very sensitive to human disturbance so their small population size and low genetic variability are causes for concern (USFWS 2019a).

Exposure to the proposed action area and proposed actions

Virginia big-eared bats are currently found in West Virginia, Virginia, Kentucky, North Carolina, and Tennessee; historically, they were also found in Pennsylvania (USFWS 2024c). They hibernate, roost, breed, and raise their young in caves and cave-like areas such as abandoned mines and rock crevices (USFWS 2024c). They forage primarily for moths in forested habitats that are interspersed with open fields, cliff lines, rock outcroppings, riparian areas, and water features (USFWS 2024c).

Critical habitat for Virginia big-eared bats was designated at the time of initial listing under the Endangered Species Act and encompasses five caves in West Virginia (44 FR 69206). These caves were designated as critical habitat because they contain precise conditions that support Virginia big-eared bat populations, and any actions that affect these caves' temperature, structure, humidity, air flow, or disturbance are specified as actions that could adversely affect the critical habitat (44 FR 69206). Other important caves have since been identified, of which some are considered protected, and some of which are not (USFWS 2019a).

The USDA Forest Service manages federal land across the subspecies' range that provides important habitat for Virginia big-eared bats. The Monongahela, George Washington, Jefferson, Cherokee, Daniel Boone, Nantahala and Pisgah National Forests all contain critical habitat for the subspecies. There are currently inventoried roadless areas on all of these Forests except for the Daniel Boone National Forest (USFS 2000a).

There are only 12 acres of critical habitat in inventoried roadless areas, which is approximately 2% of total critical habitat. Of these 12 acres, all are classified in habitat that is considered operable. There are no observations of the species in inventoried roadless areas, which could be due to low sampling effort (there are only 38 observations of the species in the NRM and GBIF database), the rarity of the species, or other factors. As such, no conclusions about the relative vulnerability of the species to potential development from the proposed action can be drawn

from this occurrence data. Given this we ran a supplemental analysis that intersected the USFWS range for this species (USFWS 2026a) with inventoried roadless affected by the proposed action. The result was approximately 1.7%.

Effects of Roads

The effects of roads on Virginia big-eared bat populations are generally negative (USFWS 2019a, USFWS 2024c). Based on concern about the increasing threats posed by barriers to movement, the latest recovery plan amendment identified a new recovery action: "Within each [recovery unit], avoid and/or minimize barriers to movement and sources of direct mortality to [Virginia big-eared bats] (e.g., roads,…)" (USFWS 2024c). Since roosts in caves are in separate places from foraging habitats, there is concern that roads cause fragmentation by separating discrete habitat areas (USFWS 2019a). Further contributing to this concern, evidence suggests that Virginia big-eared bats avoid crossing major roads when foraging (Weber et al. 2016).

Roads can also damage Virginia big-eared bat habitat. In West Virginia and North Carolina, road development and widening was proposed within the Virginia big-eared bat's range, and there was concern that this activity would lead to a loss or alteration of important cave and mine habitat or reduce foraging success (USFWS 2019a, Weber et al. 2016). Additionally, since human development is increasing in North Carolina, there is concern that associated road widening, road development, human access, and increased predation from domesticated cats could be detrimental (USFWS 2019a). Although it is plausible that larger roads could have a larger effect on bat habitat, even small roads have caused direct mortalities of Virginia big-eared bats (USFWS 2019a). There is also concern that roads could contribute to reduced reproductive success which could contribute to population declines (USFWS 2019a).

Effects of Timber/ Vegetation Management

The specific effects of timber production on Virginia big-eared bat populations are uncertain, but there are likely tradeoffs between the potential negative effects of disturbance and the potential benefits of certain forest management activities on bat habitat. Biologists have long voiced concerns about the effects of timber harvesting on Virginia big-eared bat habitat (Adam et al. 1994), but some research suggests that Virginia big-eared bats populations may be able to tolerate some timber harvest if a diversity and abundance of local plants are retained to ensure adequate levels of prey (Loeb et al. 2011). A review of *Corynorhinus* (the genus encompassing Virginia big-eared bats) foraging behavior found that prey were typically captured in areas of mature timber and avoided open habitats (Loeb et al. 2011). These bats select forest corridors and edges (Loeb et al. 2011), and they do not forage in areas that have been clearcut (USFWS 2019a). As such, selective logging may be a useful strategy for maintaining Lepidopteran moth prey levels (USFWS 2019a).

Existing Protections

Land management plans for the Monongahela (2011), George Washington (2014), Jefferson (2004), Cherokee (2004), and Daniel Boone (2004) National Forests all contain standards that protect Virginia big-eared bat habitat within a short distance of known bat colonies (typically either within 200 feet or within ¼ mile; USFS 2025a). The 2023 Nantahala and Pisgah National

Forests land management plan mentions that the presence (or absence) of Virgina big-eared bats needs to be documented on the forest (USFS 2025a).

Critical Habitat

"The Service has concluded that five caves in West Virginia should be designated as Critical Habitat for the Virginia big-eared bat. Because of precise conditions of physical structure, temperature, and humidity, these caves are suitable for use by the species as sites for hibernation and reproduction. The species has a limited range and is highly susceptible to changes in its habitat. Even minor disturbance or physical changes in the caves occupied may result in extinction. Therefore, these caves are essential for its conservation. The physical and biological features of its habitat are such as to require special management considerations and protection." (44 FR 69206)

There are only 12 acres of critical habitat in inventoried roadless areas, which is approximately 2% of total critical habitat. Of these 12 acres, all are classified in habitat that is considered operable. Despite minimal overlap, given minor disturbances to habitat can have a profound effect on the species we have determined the proposed activity is likely to adversely affect critical habitat for this species.

Effects Determination

Given the available data for assessing occurrence of the species, we are assigning the species an exposure level of high". Given the species' sensitivity to human disturbance, we have determined the proposed activity is likely to adversely affect this species and its designated critical habitat.

Mexican long-nosed bat

The Mexican long-nosed bat (*Leptonycteris nivalis*, also known as the greater long-nosed bat) is a migratory, mostly-nectivorous bat that was listed as endangered under the Endangered Species Act in 1988 (53 FR 38456). Conservation actions are informed by a recovery implementation strategy (USFWS 2024d), a recovery plan (USFWS 2024eand a species status assessment (USFWS 2018a).

The Mexican long-nosed bat is considered a single population (Pourshoushtari and Ammerman 2020) that is composed of several subpopulations associated with different roosting sites (USFWS 2018a). The current population size of Mexican long-nosed bats is uncertain, in part due to the species' migratory behavior (USFWS 2018a). However, most of the 11 recently-monitored roosting sites (of ~30 known roosts) support fewer than 2,000 individuals, and the two known roosts in the United States support 500-3,000 individuals (Texas) and 500-7,000 individuals (New Mexico) (USFWS 2018a). The greatest threats facing Mexican long-nosed bats are the reduced availability of roosting sites, a lack of connectivity among their foraging habitat, and human disturbances to their roosting sites (USFWS 2018a, USFWS 2024e).

Exposure to the Proposed Action Area and Proposed Actions

In the United States portion of their range, Mexican long-nosed bats are primarily located in New Mexico and Texas (USFWS 2018a), but they were also recently documented in Arizona (Bat

Conservation International 2025). They hibernate and roost in caves, abandoned mines, and cavities on cliff faces (USFWS 2018a). They forage in agave- and desert-scrub woodlands where they primarily feed on agave, cactus pollen, and insects (USFWS 2024e). These bats rely heavily on a "nectar corridor" (Burke et al. 2019) created by flowering plants, especially agave, during their migrations (USFWS 2018a).

At present, there is no designated critical habitat for this species, but the known roosts in the United States portion of their range are on protected land that is part of Big Bend National Park (NPS 2004) and a Bureau of Land Management Wilderness Protected Area (USFWS 2018a). Potential connectivity habitat may also exist on National Forests and Grasslands in New Mexico and Texas, although agave is only mentioned in the land management plans for the Cibola (2022) and Coronado (2018) National Forests (USFS 2025a) and the Mexican long-nosed bat is only listed as present in the 2018 land management plan for the Coronado National Forest (USFS 2025a). There are three occurrence records for the species in inventoried roadless areas in NRM and GBIF databases and none of them occur in operable areas, however given the low sample size and the opportunistic nature of many detections in these data sources the relative exposure of the species to the proposed action cannot be determined using these data. We ran a supplementary analysis overlaying the USFWS range for the species with inventoried roadless and found that approximately 0.005% of the species' range occurs in the proposed action area.

Additionally, other stakeholders including non-governmental organizations and states are engaged in improving habitat for Mexican long-nosed bats. For example, in New Mexico where these bats are a "species of greatest conservation need" (NMDW 2025a), partners are monitoring and improving a large patch of agave to serve as habitat for long-nosed bats on a state-managed Wildlife Management Area that is adjacent to the Gila National Forest (NMDW 2025b).

Effects of Roads

The effects of roads on Mexican long-nosed bat populations are unknown. However, due to this bat species' long-distance migratory movements and the existing challenges posed by fragmentation of important food resources such as agave (USFWS 2018a), it is feasible that new road construction could further fragment habitat and cause additional challenges for the species. At present, there are some inventoried roadless areas that do not allow road construction or reconstruction on the Coronado National Forest and close to the southern border with Mexico (USFS 2000a). It is uncertain whether the proposed action would lead to an increase in road construction in this area, but both Mexican long-nosed bats and important agave species are documented on the Coronado National Forest, meaning that some impacts to bat habitat could be possible (2018 land management plan, available at USFS 2025a).

Similarly, since Mexican long-nosed bats are sensitive to human disturbance (USFWS 2018a), roads could negatively affect the species by increasing recreation levels around important caves, which could exacerbate the existing challenge posed by all-terrain vehicle disturbance (USFWS 2018a). Since these bats are sensitive to noise (USFWS 2024e), it would be important to ensure that road construction did not contribute to noise at roosts. Lastly, it is theoretically possible that roads could cause mortalities from collisions as well since this has been documented in other bat species.

Effects of Timber/ Vegetation Management

The effects of timber production on Mexican long-nosed bat populations are unknown. It is also unclear how much of the species' range on National Forests would be suitable for timber production. However, it is worth noting that deforestation around caves was thought to be enough of a concern for the species that it merited monitoring around at least one cave in Mexico (USFWS 2018a). If timber harvest were to impact agave connectivity or areas around important caves and bat roosts, then it is reasonable to assume that this might have a negative effect on the species.

Existing protections

Plans for National Forests in New Mexico and Texas (with the exception of the Lincoln and Gila National Forests) contain standards and guidelines designed to protect general bat habitat, such as caves (1986 Lincoln NF, 1986 Gila NF, 1996 Texas NFs, 2020 Coronado NF, 2022 Carson NF, and 2022 Cibola NF land management plans, available at USFS 2025a).

Effects Determination

Given the low overlap between the species' range and inventoried roadless areas affected by the proposed action we are assigning an exposure level of low. However, given its sensitivity to human disturbance we have determined the proposed action is likely to adversely affect this species**.**

Gray bat

The gray bat (*Myotis grisescens*) is an insectivorous bat that was listed as endangered under the Endangered Species Act in 1976 (41 FR 17736). Conservation actions are informed by a recovery plan (USFWS 1982) and a five-year review (USFWS 2025b). As of 2021, the range-wide gray bat population was considered stable (USFWS 2025b). The 2021 winter survey estimated approximately 4.9 million gray bats present within 15 major hibernacula as compared to 5.3 million bats estimated during the 2019 winter survey (USFWS 2025b). Experts determined that these counts are within the margin of error due to the challenges of counting hundreds of thousands of bats at a time (USFWS 2025b).

At present, the major threats to gray bats include human disturbance and destruction of roosting sites, cave flooding due to natural or human causes, pesticide contamination and siltation of waterways near bat habitat (which can affect their insect prey), deforestation along waterways where the bats forage, wind turbine interactions, and white-nose syndrome (USFWS 1982, USFWS 2025b).

Exposure to the Proposed Action Area and Proposed Actions

Gray bats are currently found in Alabama, Arkansas, Kentucky, Missouri, and Tennessee; they were also documented in Florida as recently as 2011 (USFWS 2025b). They may rarely occur in parts of Georgia, Kansas, Indiana, Illinois, Oklahoma, Mississippi, Virginia, and North Carolina (USFWS 1982). Year-round, they hibernate and roost in limestone karst habitats such as caves, sinkholes, and springs, and their habitat is very closely tied to areas around these caves (USFWS 1982). Many bats migrate seasonally between maternity caves and hibernacula (USFWS 1982).

They tend to forage over and around bodies of water where they rely in part on access to mayflies, and they prefer deep, vertical caves that occur near rivers (USFWS 1982, USFWS 2025b). Approximately 98% of the species occurs in only 15 hibernacula (USFWS 2025b), putting them at an elevated risk of extirpation. Although there is not a specific subpopulation structure of gray bats, there is some genetic variation between eastern and western groups which should be considered as part of species and habitat management (Holliday et al. 2023).

There is not currently any federally designated critical habitat for gray bats, but some key habitat elements are protected in other ways. For example, three National Wildlife Refuges in Alabama were specifically established to protect gray bat habitat (USFWS 2025c, USFWS 2025d, USFWS 2025e), and there are some habitat protections on National Forests across their range (see below). In the NRM and GBIF databases, there are 3 occurrences of the species in potentially affected IRAs out of 250 records on NFS lands. All of these occurrence records occur in areas that are operable. Given the potential bias in opportunistic datasets like this, we also conducted a supplementary analysis overlaying the species' USFWS range with inventoried roadless areas affected by the proposed action. The overlap was approximately 0.004%. Given the importance of a limited number of hibernacula to the species, we visually inspected the locations of 15 caves highlighted in the 2021 SSA as being important to the species relative to inventoried roadless areas. This visual inspection did not identify areas of potential overlap. Project level analyses should also conduct evaluations of potential impacts to maternity and other colony locations.

Effects of Roads

The specific effects of roads on gray bat populations are unknown, but evidence suggests that the effects of roads on this species are probably mixed, with some benefits and some negative effects. Gray bats use roads during flight, so the development of rural roads may serve as flight corridors to foraging areas (Jackson et al. 2023). Gray bats have also been documented roosting under bridges when caves are unavailable (in North Carolina, Etchison & Weber 2020) as well as during migration (in Arkansas, Sasse 2019), meaning that road infrastructure could be an important alternate source of roosting habitat.

However, roads likely also have negative implications for gray bats. Roads can provide access to caves that are otherwise inaccessible, increasing human disturbance, although roads and caves can be gated to prevent human entry (USFWS 2025b). In theory, road construction could also impact roosting and foraging habitat or otherwise disturb bats, as well as cause mortalities from vehicle collisions. Recent research mapped particularly high-use areas of gray bat habitat as occurring in north-central Arkansas, central Missouri, north-central and north-eastern Tennessee, south-central Kentucky, and northeastern Alabama (Nagel et al. 2024), indicating that new road development in these areas may need special consideration to mitigate any negative impacts on gray bats and their habitat.

Effects of Timber/ Vegetation Management

The specific effects of timber production on gray bat populations are unknown, but based on gray bat behavior and the impacts of timber production on other similar species, it is plausible that the effects of timber harvest might be negative. Gray bats – and particularly young bats –

rely on forests for cover, and deforestation in areas near caves and waterways used for foraging is listed as a threat to the species (USFWS 1982, USFWS 2025b). Gray bats have also been documented using daytime tree roosts during migration periods (Samoray et al. 2020), but more research is needed to understand gray bat reliance on forests since trees were not considered to be roosting habitat previously (USFWS 2025b). Experts suggest that any activities that could affect foraging habitat (which includes forested areas near waterways) within 25 kilometers of gray bat caves should be carefully considered to protect bat habitat, particularly given the context that "substantial areas of habitat have been destroyed or degraded by clearing…" (USFWS 1982).

Existing Protections

Gray bats inhabit several National Forests that are managed by the USDA Forest Service where they benefit from some habitat protections. On the Ozark-St. Francis National Forest in Arkansas, a 200-foot buffer of undisturbed forest must be maintained around gray bat maternity and hibernation colonies where overstory cutting, road construction, pasture development, and prescribed burning are prohibited (2005 land management plan, available at USFS 2025a). On the Shawnee National Forest in Illinois, seismic testing cannot be conducted within 2.5 miles of gray bat hibernacula during the hibernation season, smoke effects on gray bat hibernacula must be considered when responding to wildland fire or conducting prescribed fire, and no prescribed burns may be conducted in upland or bottomland forests during specific spring-summer seasons to protect maternity roosts (2006 land management plan, available at USFS 2025a). On the Hoosier National Forest in Indiana, known gray bat hibernacula are surrounded by a ¼ mile buffer where new construction is prohibited, caves containing gray bats may be considered for access restrictions, and at least 70% of forest canopy cover should be maintained within a one-mile radius of gray bat hibernacula. Additionally, "timber harvest should be restricted to single-tree and group selection within this zone," but vegetation management can be implemented to maintain or improve bat habitat (2006 land management plan, available at USFS 2025a).

On the Cherokee National Forest in Tennessee and Georgia, consultation with the U.S. Fish and Wildlife Service is conducted for any project that is proposed within 20 miles of known gray bat maternity sites if the project is anticipated to affect canopy cover in riparian corridors or along forested lakeshores (2004 land management plan, available at USFS 2025a). Additionally, a ¼ mile buffer of undisturbed forest is maintained around maternity and hibernation colony sites where overstory cutting, prescribed burning, road construction, and prescribed burning are prohibited (2006 land management plan, available at USFS 2025a). Although gray bats are not currently documented in Texas, National Forests and Grasslands in this state restrict the use of aerially-applied herbicides within 300 feet of gray bat habitat and ground-applied herbicides within 60 feet of gray bat habitat (1996 land management plan, available at USFS 2025a).

Effects Determination

Given the proportion of the species' range that overlaps with inventoried roadless areas and the absence of an indication that major hibernacula occur in inventoried roadless, the FS has determined that the exposure of the species to the proposed action is low. This could be revised should better spatial evaluations of maternity and other colony locations suggest overlap with the

proposed action area. Despite existing protections, given the sensitivity of the species to human disturbance the FS has determined the proposed action is likely to adversely affect this species.

Northern long-eared bat

The northern long-eared bat (*Myotis septentrionalis*) is an insectivorous bat that was listed as endangered under the Endangered Species Act in 2023 (88 FR 4908) after initially being listed as threatened in 2015 (80 FR 17974). Conservation actions are informed by a Species Status Assessment (USFWS 2022a).

As of 2023, the northern long-eared bat has been experiencing precipitous declines across most of its range, due in large part to white-nose syndrome (88 FR 4908). There are no range-wide population estimates for this species, but both winter and summer surveys conducted across the range show evidence of dramatic decline over the last two decades (88 FR 4908). At present, the major threat to northern long-eared bats is the spread of white-nose syndrome, but other stressors include habitat loss, wind energy development, and changes in climate patterns (88 FR 4908, USFWS 2022a).

Exposure to the Proposed Action Area and Proposed Actions

There is no federally designated critical habitat for the northern long-eared bat (88 FR 4908). These bats currently inhabit 37 states, ranging from Montana and Wyoming in the West to Maine in the northeast to Georgia, Alabama, Mississippi, and Louisiana in the southeast (88 FR 4908). In the eastern portion of the range, these bats were widely distributed prior to the arrival of white-nose syndrome, but since then, they are generally rare with documented extinctions of hibernacula across the range (88 FR 4908).

In the winter, northern long-eared bats primarily hibernate in caves and mines with stable, cool temperatures, but they have also been documented hibernating in tunnels and bunkers, or more rarely in sewers and aqueducts or other similar areas when caves and mines are not present (88 FR 4908). In the summer, northern long-eared bats roost in tree cavities, tree crevices, and under the bark of both live and dead trees, as well as occasionally in caves, mines, and human structures like barns and utility poles (88 FR 4908). Although these bats do not seem to select particular tree species, in some areas, they frequently choose shade-tolerant, deciduous hardwoods, particularly for maternity colonies (as opposed to males roosting alone) (88 FR 4908). In other areas, they may opt for a higher proportion of dead conifers or live trees with dead branches and other visible signs of decay (88 FR 4908). Females may roost in more open areas than males which may provide solar radiation or aid bats learning to fly, and roost tree diameter is highly variable (88 FR 4908). Northern long-eared bats forage in forests, typically between the understory and the canopy, although they occasionally forage near water, along roads, or in small clearings (88 FR 4908). They generally do not use larger forest openings or clearings (88 FR 4908).

Given the broad range of northern long-eared bats, they have been documented on many National Forests and Grasslands across the country. In the bats' listing decision, 18 different National Forests and Grasslands were mentioned by name as being places that northern long-eared bats were found (88 FR 4908). As such, it is likely that public lands managed by the

USDA Forest Service are important sources of habitat and monitoring effort for these bats. In the GBIF and NRM databases there are 66 occurrences of the species in inventoried roadless areas potentially affected by the proposed action, all but one of which occur in operable or operable but complex areas. This leads to an assessment of "high" exposure to the proposed action given the analytical framework defined for this BA. However, given the sources of biases inherent in opportunistic datasets, we also conducted an analysis overlaying the total USFWS species range with inventoried roadless areas affected by the proposed action and found an overlap of 0.004%.

Effects of Roads

The specific effects of roads on northern long-eared bats are likely mixed. These bats are known to forage over small roads, and some research has suggested that they use road corridors more than expected, making use of skid trails for travel and foraging habitat within areas that experienced timber harvest (Owen et al. 2003). However, other research has suggested the opposite, indicating that northern long-eared bats are less likely to be present in corridors and edge habitats and that their occupancy is positively associated with the distance from the nearest road (Hyzy et al. 2020). In general, there is consensus that these bats prefer intact forests of mixed types that contain small gaps (i.e., forest trails, small roads) over clear cuts (USFWS 2022a), indicating that bigger roads with more open space likely contribute to habitat fragmentation. Like other bat species, roads also present the risk of increased disturbance to bats (particularly via increased access to caves and hibernacula) as well as mortality from collisions.

Effects of Timber/Vegetation Management

The specific effects of timber production on northern long-eared bat populations and habitat are likely mixed, but given these bats' reliance on trees for both roosting and foraging habitat, forest management likely has both benefits and risks depending on how it is conducted. According to the species' listing decision, the benefits of timber harvest could include maintaining or increasing suitable roosting and foraging habitat, whereas the negative impacts could include bat death (88 FR 4908).

Forest conversion to other landcover types is listed as a cause of habitat destruction in the listing decision for northern long-eared bats (88 FR 4908). Forest conversion can make it more difficult for bats to find roosting and foraging habitats, and it may require them to travel farther between roosting and foraging habitats due to the fragmentation of travel corridors (88 FR 4908). This is of particular concern in the areas of the bat's range where forest is already fragmented (88 FR 4908).

Evidence suggests that this bat species prefers intact forest over fragmented forest patches (88 FR 4908), including mixed-type forests with small gaps (i.e., forest trails, small roads) rather than clearcuts (USFWS 2022a). One study suggested that these bats selected water, areas that were historically thinned, and patch cuts over other habitats, and they generally avoided larger openings such as clearcuts (Divoll et al. 2022). Similarly, another study suggested that partial timber harvests that leave a closed canopy could improve foraging habitat (Owen et al. 2003).

At the same time, some research has indicated that northern long-eared bats continue to roost in areas that have been harvested or thinned (88 FR 4908), and that timber harvest openings did not

negatively impact northern long-eared bats' use of roosts (Bergeson et al. 2021). Other authors found that silvicultural practices could support northern long-eared bat habitat and stated, "use of patch cuts or selective logging practices could allow for the retention of large-diameter snags, while also providing regenerating forest (Lacki and Schwierjohann 2001)."

Ultimately, as stated in the listing decision, "Impacts to northern long-eared bats from forest management would be expected to vary depending on the timing of removal, location (within or outside northern long-eared bat home range), and extent of removal (88 FR 4908)."

Existing Protections

Many states and National Forests protect habitat for various bat species, including other tree-roosting bats that have similar habitat needs as the northern long-eared bat, such as the Indiana bat (87 FR 56381). For example, some caves are restricted or gated to protect Indiana bats and other bat habitat, and all known hibernacula within USDA Forest Service's Rocky Mountain Region (Colorado, Kansas, Nebraska, Wyoming, South Dakota) are closed during the winter to reduce the risk of transmitting white-nose syndrome (87 FR 56381).

Since the northern long-eared bat was listed under the Endangered Species Act relatively recently, it is not mentioned by name in many Forest Plans for National Forests and Grasslands that are managed by the USDA Service. However, since the northern long-eared bat's range overlaps with the Indiana bat in several southeastern states, it benefits from the habitat protections offered by several National Forests and Grasslands (e.g., 2004 Daniel Boone NF, 2005 Ouachita NF, 2005 Ozark-St. Francis NF, 2006 Hoosier NF, 2006 Shawnee NF, 2011 Monongahela NF, and 2023 Nantahala and Pisgah NFs land management plans, available at USFS 2025a). Despite these protections, it is possible that northern long-eared bats could benefit from greater protections in the more northern portions of its range, although most forest plans do have general protections to safeguard cave habitats for sensitive bats in general.

Effects Determination

In the GBIF and NRM databases there are 66 occurrences of the species in inventoried roadless areas potentially affected by the proposed action, all but one of which occur in operable or operable but complex areas. This leads to an assessment of "high" exposure to the proposed action given the analytical framework defined for this BA. However, we note that inventoried roadless areas constitute only 0.004% of the species range. Given this exposure and the sensitivity of the species to human disturbance the FS has determined the proposed action is likely to adversely affect this species**.**

Indiana bat

The Indiana bat (*Myotis sodalis*) is an insectivorous bat that was originally listed as being at risk of extinction under the Endangered Species Preservation Act in 1966 (32 FR 4001) and is now listed as endangered under the Endangered Species Act. Critical habitat was designated for Indiana bats in 1976 and 1977 (41 FR 41914, 42 FR 47840) and was upheld after a petition to expand it in 2007 was found to be not warranted (72 FR 9917). Current conservation actions are informed by a recovery plan (USFWS 2007a) and a five-year review (USFWS 2019b). Indiana bat conservation benefits from a greater depth of research and monitoring information than many

other bat species, including a database containing records of all known hibernacula and a biennial, range-wide population estimation system (USFWS 2019b).

The species experienced a substantial decline from the 1980s until 2003, after which there was a four-year reprieve followed by another decline that has extended to the present day, due in part to the range-wide effects of white-nose syndrome (USFWS 2019b). As of 2019, the range-wide population is estimated to be 537,297 bats (USFWS 2019b). Historical records indicate that the range used to be broader and abundance used to be higher (USFWS 2019b).

Historically, the main range-wide threats to Indiana bats were habitat loss and degradation, forest fragmentation, human disturbance, and environmental contaminants. More recently, additional threats have included white-nose syndrome, invasive species, and wind turbines (USFWS 2019b).

Exposure to the Proposed Action Area and Proposed Actions

Indiana bats currently occur in 19 states in the eastern United States, ranging from Arkansas and Alabama in the south to Oklahoma, Illinois, Missouri in the Midwest, and to Connecticut and Vermont in the northeast (USFWS 2019b). Designated critical habitat for Indiana bats encompasses one mine in Illinois, two caves in Indiana, two caves in Kentucky, five caves and one mine in Missouri, one cave in Tennessee, and one cave in West Virginia (41 FR 41914). These sites were designated as critical habitat because at the time of the rulemaking, 75% of the known Indiana bat population hibernated in these caves and mines (41 FR 41914).

Indiana bats hibernate over the winter in caves and mines, and in the summer, they forage in forested areas (USFWS 2019b). The bats may either stay close to their hibernacula (especially males and nonreproductive females) or migrate hundreds of miles to summer habitat (especially reproductive females) (USFWS 2007a). In the summer, reproductive females often roost in tree crevices and under the bark of both live and dead trees, choosing primarily dead trees with dried bark or occasionally live trees with naturally peeling bark like shagbark and shellbark hickory (USFWS 2007a). Maternity colonies are formed in roost trees as well, and there is some (albeit mixed) evidence that roost trees tend to be larger-diameter trees that receive some sunlight (USFWS 2007a).

In 2005, around 90% of the species hibernated in five states (Indiana, Missouri, Kentucky, Illinois, and New York) and 72% of the species hibernated in just 10 hibernacula (USFWS 2007a).

Of the critical habitat designated for this species, 0.31% of it occurs in inventoried roadless areas potentially affected by the proposed action. 74% of such habitat is classified as operable or operable but complex. Of occurrence records in NRM and GBIF, there are 15 observations of bats in inventoried roadless areas potentially affected by the proposed action with all of them occurring in operable or operable but complex areas. Given the small sample size and known potential for bias in opportunistic sampling, we conducted a supplemental analysis overlaying the USFWS species' range with inventoried roadless and found that approximately 0.003% of the range is in inventoried roadless areas affected by the proposed action.

Effects of Roads

Roads pose risks to Indiana bats and their habitats, but they may offer some benefits as well. Caves (which serve as hibernacula for bats) are very sensitive to changes on the land's surface, especially from a drainage perspective (USFWS 2007a). Road construction and urban development can contribute to increased stormwater runoff and siltation which can enter caves, increase flood risk, and lead to contamination from pesticides and other chemicals (USFWS 2007a). For this reason, experts suggest that forest buffers should be maintained between roads and hibernacula to stabilize soils and reduce erosion and sedimentation issues within these caves (USFWS 2007a).

Roads can also contribute to mortality risk and increased human disturbance. At least one record exists of an Indiana bat being killed by a vehicle collision, and while this may not pose a major threat to the population, it may be worth considering if new roads are being developed. (USFWS 2007a). There is also concern that roads can provide recreational access to caves that would otherwise be inaccessible, although this concern can be mitigated by gating, fencing, blocking, or removing roads and caves to restrict access (USFWS 2007a).

Despite these risks, roads may provide some habitat for Indiana bats that would otherwise be unavailable. These bats have been documented roosting in bridges (USFWS 2019b) and hibernating in a railroad tunnel (USFWS 2007a). In areas without waterways, Indiana bats may be attracted to road ruts that fill with water, although water features are plentiful throughout much of their range (USFWS 2007a).

In general, Indiana bats forage and roost in forest openings created by downfall, riparian areas, or road and trail edges (USFWS 2007a). For this reason, corridors produced by roads may be useful if they are wooded. However, Indiana bats avoid large open areas, meaning that larger roads may fragment their habitat (although they have been documented crossing highways) (USFWS 2007a).

Effects of Timber/ Vegetation Management

Timber production can pose risks to Indiana bats and their habitat, but it is likely that silvicultural treatments and forest management can also be used to benefit their habitat. According to the Indiana bat recovery plan (which contains detailed information about the impacts of forest management), "Quality habitat around hibernacula is very important to roosting and foraging Indiana bats […] adjacent and nearby forests typically need some special management to ensure that a continual supply of roost trees and foraging habitat are maintained and to avoid the potential for take occurring during timber harvests or prescribed burns (USFWS 2007a)." In general, retaining buffers of intact forest around hibernacula can reduce the risk of flooding, contaminant runoff, stormwater runoff, siltation, and temperature or humidity change that can occur during logging, land clearing and other land uses. (USFWS 2007a, USFWS 2019b).

If timber production and forest management is designed to maintain small, forested openings, this may benefit Indiana bats. When foraging, Indiana bats use small regeneration openings over a large area (Divoll et al. 2022), and they may be most active at harvest edges and in adjacent

forests (Caldwell et al. 2019). As such, maintaining a heterogenous forest structure may support these bats along with other species (Divoll et al. 2022). Importantly, Indiana bats avoid traveling through large open areas, meaning that large-scale clearing could fragment their habitat. Deforestation was one of the initial identified causes of decline (USFWS 2019b), and the loss of forest habitat connectivity was added as a threat later (USFWS 2007a). Maintaining wooded areas and small forest openings is important to reduce fragmentation for this species.

It is also important for forest management activities to protect roost trees. Indiana bat deaths and injuries have been recorded because of the inadvertent felling of roost trees, but this can be minimized by retaining snags, preserving potential roost trees, and avoiding cutting trees when bats are occupying their summer range (USFWS 2007a). Similarly, restricting the cutting of firewood to downed trees (rather than standing dead trees) near hibernacula may help to protect potential Indiana bat habitat. Other work suggests that managing some forest elements to develop older-growth characteristics, developing a plan to ensure that live trees can develop into snags, and using two-age shelterwood cuts that retain plenty of live trees and snags may be beneficial (USFWS 2007a).

Existing Protections

Indiana bats inhabit several National Forests that are managed by the USDA Forest Service where they benefit from habitat protections. On some Forests, protections are minimal to moderate. For example, on the Shawnee National Forest in Illinois, seismic testing cannot be conducted within 2.5 miles of Indiana bat hibernacula during the hibernation season, smoke effects on gray bat hibernacula must be considered when responding to wildland fire or conducting prescribed fire, and no prescribed burns may be conducted in upland or bottomland forests during specific spring-summer seasons to protect maternity roosts (2006 land management plan, available at USFS 2025a). On the Ouachita National Forest in Arkansas and Oklahoma, a cave gate must be maintained, ground-disturbing projects and prescribed burns proposed within 2 miles of a hibernaculum must be evaluated to determine their effects on bats, and bat surveys must be conducted prior to roof, bridge, mine, or well modifications (2005 land management plan, available at USFS 2025a). On the Nantahala and Pisgah National Forests in North Carolina, hibernation, foraging, roosting, and maternity habitat must be maintained (2023 land management plan, available at USFS 2025a). Additionally, despite Indiana bats' presumed absence in Texas, National Forests and Grasslands in the state must restrict the use of aerially applied herbicides within 300 feet of Indiana bat habitat and ground-applied herbicides within 60 feet of Indiana bat habitat (1996 land management plan, available at USFS 2025a).

In contrast, four National Forests in Indiana, Arkansas, Kentucky, and West Virginia have more extensive management direction related to Indiana bat habitat. The land management plan for the Hoosier National Forest in Indiana contains multiple pages of guidelines that specify hibernaculum protection zones, canopy cover goals during timber harvest, prescribed fire restrictions, forest composition goals in roosting habitat that specific tree DBH, species, and density, and firewood and snag cutting restrictions, among others (2006 land management plan, available at USFS 2025a). The land management plan for the Ozark-St. Francis National Forest in Arkansas contains a similar level of detail and provides guidance on tree thinning, improving roosting and foraging conditions, retaining potential roost trees of specific DBHs and densities

during salvage logging, maintaining specific overstory densities around hibernacula, and restricting road construction, certain agricultural allotments, and pest management near hibernacula (2005 land management plan, available at USFS 2025a). Similarly, the land management plans for the Monongahela National Forest in West Virginia (2011 land management plan, available at USFS 2025a) and the Daniel Boone National Forest in Kentucky (2004 land management plan, available at USFS 2025a) contain multiple standards and guidelines related to habitat maintenance, silvicultural prescriptions, and mining.

Critical Habitat

Of the critical habitat designated for this species, 0.31% of it occurs in inventoried roadless areas potentially affected by the proposed action. 74% of such habitat is classified as operable or operable but complex. There are no PCEs for this species. Given that small amount of overlap of critical habitat with the action area (<1%) and the existing protections discussed above, the FS has determined the proposed action is not likely to adversely affect designated critical habitat for this species.

Effects Determination

While there are only 15 occurrence records of the species in inventoried roadless areas, all of them occur in operable or operable but complex. Additionally, even though only a small portion of the total range intersects with inventoried roadless, there is inventoried roadless in the species' critical habitat which encompasses sensitive hibernacula. Based on these data, the FS has assigned an exposure level of high to the proposed action. Given this exposure and the sensitivity of the species to human disturbance the FS has determined the proposed action is likely to adversely affect this species. Given that small amount of overlap of critical habitat with the action area (<1%) and the existing protections discussed above, the FS has determined the proposed action is not likely to adversely affect designated critical habitat for this species.

Tricolored bat

The tricolored bat (*Perimyotis subflavus*) is a broadly distributed, insectivorous bat that was proposed for listing as endangered under the Endangered Species Act in 2022 (87 FR 56381). The tricolored bat is also listed as endangered under Canada's Species At Risk Act (COSEWIC 2013) and has additional protections in several U.S. states (87 FR 56381). Conservation actions in the United States are informed by a species Status Assessment (USFWS 2021a).

As of 2021, the tricolored bat population is in danger of extinction throughout its range (87 FR 56381). Both the species' abundance and range have contracted, and these reductions have likely impacted genetic diversity, exacerbating its vulnerability (USFWS 2021a). The major threat to the tricolored bat is white-nose syndrome which has caused populations declines of 90-100% across 59% of the species' range (87 FR 56381). Other threats include changes to climate patterns, mortality from wind turbines, human disturbance during hibernation, and habitat loss (87 FR 56381).

Exposure to the Proposed Action Area and Proposed Actions

The tricolored bat is a wide-ranging bat that is found across 39 states (87 FR 56381) spanning the midwestern and eastern United States, extending as far west as Colorado and Wyoming and

ranging into parts of central America to the south and Canada to the north. Tricolored bats hibernate over winters in caves, abandoned mines, and road-associated culverts (in areas where caves and mines are unavailable, 87 FR 56381). They disperse between winter hibernacula and summer roosting habitat, and they engage in migrations of up to 150 miles (USFWS 2021a).

During spring, summer, and fall, tricolored bats forage near water and in forested habitats where they roost in trees (USFWS 2021a). They tend to select tree roosts in the live and dead foliage of deciduous hardwoods, but they also roost in Spanish moss, lichen, pine needles, and human structures (e.g., barns, roofs, bridges, tunnels, and road-associated culverts) (87 FR 56381). There is no federally designated critical habitat for this species (87 FR 56381).

Of records in the NRM and GBIF databases, there are 13 records of the species in inventoried roadless areas, of which 69% occurred in areas that were operable or operable but complex. Given the wide range of this species, we also conducted an analysis overlaying the USFWS range map (USFWS 2026a) with inventoried roadless areas affected by the proposed action. The result was approximately a 0.002% overlap.

Effects of Roads

The specific effects of roads on tricolored bat populations and habitat are unknown, but it is likely that roads can create both benefits and risks. Like many bats, tricolored bats are sensitive to disturbance, meaning that the increased human access to caves that is possible from roads could be detrimental. However, gating caves can be an effective way of reducing disturbance to hibernating bats (USFWS 2021a). Additionally, it is possible that roads can contribute to mortalities from vehicle collisions as documented in other bats.

That said, roads offer important habitat elements to bats in some portions of their range. In the southern United States where climate conditions are subtropical and there is an absence of cold winters, many tricolored bats hibernate in road-associated culverts when caves and mines are sparse (USFWS 2021a). In fact, road-associated culverts comprise the majority of winter hibernacula in Mississippi, Georgia, and Louisiana (USFWS 2021a). These culverts are likely an important habitat element for tricolored bats, particularly because there is some evidence suggesting that bats overwintering in culverts may be less susceptible to white-nose syndrome because they experience shorter torpor bouts and forage intermittently year-round (USFWS 2021a).

Effects of Timber/ Vegetation Management

The specific effects of timber production on tricolored bat populations and habitat are mixed and likely depend on the circumstance. However, given tricolored bats' reliance on trees for both roosting and foraging habitat, forest management activities likely have both benefits and risks for bats and their habitats depending on how they are conducted.

Forest removal and conversion is listed as a cause of habitat loss in the proposed listing decision for tricolored bats, and there is concern that this could become a greater issue in the future (87 FR 56381). Forest removal can make it more difficult for bats to find roosting and foraging habitats, and it may require them to travel farther between roosting and foraging habitats due to the fragmentation of travel corridors (87 FR 56381, USFWS 2021a). The loss of roosts, foraging

habitat, the need to expend more energy can all reduce survival and reproductive success (USFWS 2021a). Forest removal is particularly risky in areas where forested areas are already fragmented because there is a higher chance of losing connectivity or removing roost trees (USFWS 2021a).

If conducted with consideration for tricolored bat habitat needs, forest management can likely benefit the bats by improving roosting and foraging habitat. Tricolored bats likely benefit from forest management that prioritizes retaining mature forests and well-connected corridors (USFWS 2021a) along with large-diameter snags and allowing for roost-tree regeneration (87 FR 56381). Making forests more heterogeneous in terms of forest type, age, and structure may also benefit tree-roosting bats (87 FR 56381). Studies have also shown that tricolored bats were more active in areas that were harvested 3 years prior (Silvis et al. 2016) and at harvest edges (Caldwell et al. 2019). The timing of timber harvest and forest management activities is important to consider as well. Due to their reliance on trees for roosting, tree removal when bats are active can cause injuries or death (87 FR 56381). For this reason, some states conduct forest management in the winter within bat home ranges to protect them (USFWS 2021a).

Existing Protections

Many states and National Forests protect habitat for various bat species, including other tree-roosting bats that have similar habitat needs as the tricolored bat, such as the Indiana bat (87 FR 56381). For example, some caves are restricted or gated to protect Indiana bats and other bat habitat, and all known hibernacula within USDA Forest Service's Rocky Mountain Region (Colorado, Kansas, Nebraska, Wyoming, South Dakota) are closed during the winter to reduce the risk of transmitting white-nose syndrome (87 FR 56381).

Since the tricolored bat was proposed for listing under the Endangered Species Act relatively recently, it is not mentioned by name in many Forest Plans for National Forests and Grasslands that are managed by the USDA Service. However, since the tricolored bat's range overlaps with the Indiana bat in several southeastern states, it benefits from the habitat protections offered by several National Forests and Grasslands (e.g., 2004 Daniel Boone NF, 2005 Ouachita NF, 2005 Ozark-St. Francis NF, 2006 Hoosier NF, 2006 Shawnee NF, 2011 Monongahela NF, and 2023 Nantahala and Pisgah NFs land management plans, available at USFS 2025a). Despite these protections, it is possible that tricolored bats could benefit from greater protections in other regions since the tricolored bat's range is much greater than the Indiana bat's range.

Effects Determination

While there are only 13 records of the species in inventoried roadless areas in the NRM and GBIF databases, 69% of them occur in areas that are operable or operable but complex. Given this, we are assigning them an exposure level of "high." However, note that only approximately 0.002% of the range occurs in inventoried roadless areas affected by the proposed action. Given this exposure and the sensitivity of the species to human disturbance the FS has determined the proposed action is likely to adversely affect this species.

Ungulates

Peninsular bighorn sheep and Sierra Nevada bighorn sheep

The Peninsular bighorn sheep (*Ovis canadensis nelsonii)* and Sierra Nevada bighorn sheep *(Ovis canadensis sierrae)* are both listed as endangered.

The Peninsular bighorn sheep was listed in 1998 (63 FR 13134). Since 2008 it has been recognized as a distinct population segment of desert bighorn sheep (74 FR 17288). The most recent status review for this DPS was completed in 2020. It found that the DPS has been generally increasing since listing, but that sparse data since 2011 limits knowledge of the species currently (USFWS 2020b). In 2024 the population was estimated at 742 [95% CI: 656-829] individuals- there may have been a decline in the population between 2016 and 2022 associated with drought (CADFW 2025).

The Sierra Nevada bighorn sheep was listed in 2000 (65 FR 20–30). It is recognized as a subspecies of bighorn sheep (73 FR 45534). In 2023 the total population was estimated to be about 350 individuals (USFWS 2024f). This represents a steep decline after approximately 50% of adults were lost in the harsh winter of 2022-2023. This population also faced a steep decline after a harsh winter in 2016-2017 (USFWS 2024f). Small population size is one of the threats that contributes to the species' current status (USFWS 2024f)- it also faces the challenge of having fragmented subpopulations across its range.

Exposure to the Proposed Action

Both the Peninsular bighorn sheep and the Sierra Nevada bighorn sheep live in the mountains of California.

The Peninsular bighorn sheep DPS lives in the Colorado Desert of Southern California. They can be found in the Peninsular Mountain Ranges from the San Jacinto Mountains south to the border with Mexico (USFWS 2011a) and additional range for the subspecies extend into Mexico. They typically inhabit relatively low elevation habitat for bighorn sheep <4,600 ft. While sheep habitat is typically steep slopes with low levels of vegetation they will also use more gentle terrain (USFWS 2011a). Critical habitat was designated for the DPS in 2009 (74 FR 17288), but no critical habitat overlaps with IRAs in the proposed action area. While critical habitat does not overlap with IRAs in the proposed action, the San Bernadino National Forest Plan (2005 land management plan Part B, available at USFS 2025a) reports that there over 5,000 acres of key peninsular bighorn sheep habitat in and near the Pyramid Peak Inventoried Roadless Area. Bighorn sheep habitat is also found in the Cactus Springs A and B Inventoried Roadless Areas (2005 land management plan Part B, available at USFS 2025a) which abut the Santa Rosa Wilderness. The NRM and GBIF databases had very limited observations of this DPS, so we used the USFWS range map (USFWS 2026a) to evaluate the exposure of the species to the proposed action. Approximately 1% of the DPS's range overlaps with inventoried roadless areas in the proposed action area.

The Sierra Nevada bighorn sheep lives in Eastern California in the Sierras within Tuolumne, Mono, Fresno, Inyo, and Tulare counties (73 FR 45534). Best sheep habitat is open, for visibility, and on steep rocky slopes (USFWS 2008a). While they avoid dense forest, they will

use open woodlands (USFWS 2008a). They can be found across a wide range of elevations, typically selecting high alpine habitat >10,000 ft in the summer, but using areas down to ~4,000 ft in the winter (73 FR 45534). Critical habitat for the species was designated in 2008 (73 FR 45534) and approximately 2.6% of it is within an IRA affected by the proposed action, with 15% of that habitat in areas that are designated as operable or operable but complex. There are only 13 observations of Sierra Nevada bighorn sheep in IRAs in the NRM and GBIF database. As such it is not possible to use these data to draw conclusions about their relative exposure areas that may be more likely to be altered by the proposed action. We ran a supplemental analysis intersecting the USFWS range map (USFWS 2026a) for this species with inventoried roadless areas in the proposed action area. The overlap was approximately 8.2%. The majority of herd units have some portion of their habitat overlapping or immediately adjacent to IRAs.

Effects of Roads

Vehicle strikes on roads have led to the deaths of both Sierra Nevada and Peninsular Bighorn sheep. While there are not extensive road networks in Sierra Nevada bighorn sheep habitat, several mortalities have been reported on roads in Lee Vining Canyon on several occasions (listing decision). Road networks are more extensive in Peninsular sheep habitat and both highways and county roads have led to mortality from strikes (USFWS 2011a).

There are some data from Peninsular Bighorn sheep to suggest that road avoidance behavior may fragment bighorn sheep habitat. Roads or traffic on roads may make sheep, especially ewes (Rubin et al. 1998) reluctant to cross roads. There are several sheep subpopulations whose home ranges are delineated by roads (USFWS 2011a). There are also fairly detailed accounts of sheep behavior near the Dunn Road within the Santa Rosa Mountains. While this road, which was built illegally, was actively used by humans, sheep appeared to stay 1.5-3 km east of it. Because of this avoidance behavior, authorities closed the road and sheep have since started using habitat to the west of it (USFWS 2011a). Another study suggested that on days of high recreational vehicle traffic in Anza-Borrego Desert State Park, sheep reduced use of waterholes by as much as 50% (Jorgenson 1974 as cited in USFWS 2011a). Similarly, a study reported that noise from motorcycles also deterred sheep from using waterholes Olech (1979). The effect of roads on sheep may be sex specific, a study in Peninsular sheep showed that a greater proportion of males were more likely to cross roads than females, but that this difference in propensity occurred during periods of segregation between the sexes and not aggregation (Bleich et al. 2016). The same study found that on an annual basis, sheep showed no difference in their likelihood to cross a two-lane paved road with ~100 vehicle daily traffic volume versus a dirt road that received ~ 1 vehicle a day (Bleich et al. 2016). While habituation to roads may occur as shown in a study in Desert Sheep in Utah, most sheep avoid road corridors and road corridors cause a loss in habitat (Papouchis et al. 2001).

Given that vehicle noise appears to disturb bighorn sheep, it is perhaps not surprising that bighorn sheep may also be impacted by human recreation, which could increase in IRAs affected by rule rescission. There are several examples reported in the Peninsular Bighorn sheep species status assessment that suggests sheep avoid trails when being used by humans and that closing trails to humans leads sheep to return to habitat near trails (USFWS 2011a). The recovery plan calls for prevention of infrastructure (including roads) and activities (increased use of offroad

vehicle areas) that may serve as barriers to sheep movement. Data from Desert bighorn sheep in Utah suggests that sheep have a particularly strong behavioral response to hikers, fleeing well over half the time (Papouchis et al. 2001). Another study from Utah found that Desert bighorn sheep spent less time grazing and were more vigilant in an area of high human use compared to low human use (Sproat et al. 2020).There are fewer data specifically addressing the response of Sierra Nevada Bighorn sheep to recreation, although the critical habitat designation notes the need to better understand this relationship. The sensitivity of other bighorn sheep subspecies to human recreation suggests there is a similar potential for disturbance during exposure.

Increased risk of wildfire caused by human ignitions could also affect Bighorn sheep. Severe wildfires, which may be more likely due to long term fire suppression policies, are thought to degrade sheep habitat, at least temporarily. This may be especially true in habitat types used by sheep that are not well adapted to fire disturbance, such as the Sonoran scrub used by the Peninsular sheep (USFWS 2000a). For Peninsular sheep, catastrophic fire is thought to be a higher risk at the lower elevation sites where invasive plant cover has increased (USFWS 2011a).

Effects of Timber/ Vegetation Management

Commercial timber harvest is not discussed in listing decisions and status assessments for either sheep species. However, other vegetation management and removal activities do have the potential to affect them. Evaluations for both sheep species suggest that fire suppression has degraded sheep habitat. Specifically expanding dense vegetative cover is thought to reduce visibility for sheep and provide cover for mountain lions, which are a predator of sheep and have a big impact on population numbers (74 FR 17288, Gammons et al. 2025). The 2011 Species Status assessment for Peninsular Sheep identified several areas where habitat conversion due to fire suppression was thought to have reduced bighorn sheep habitat (USFWS 2011a). Therefore, vegetation management activities that reduce dense cover and provide greater visibility for sheep may benefit these two species if it reduces predation by mountain lions (74 FR 17288). Fire mitigation activities, such as prescribed fire or other vegetation management practices, may create habitat that are used by desert bighorn sheep (Holl et al. 2012: Bleich et al. 2019). For example, a population in the San Rafael mountains used recently burned chapparal habitat for forage (Holl et al. 2012;). A recent review suggested that prescribed fire has the potential to be especially beneficial for sheep habitat may be especially important for chaparral habitat in the Southwestern United States (Whiting et al. 2023).

The literature on Sierra Nevada bighorn sheep is more limited but suggests a similar pattern. Greene et al. 2012 documented an improvement in winter sheep forage after a natural fire. Mountain lion predation is also a primary cause of mortality for Sierra Nevada sheep (Gammons et al. 2025) and improved visibility in burned areas may also be attractive to this subspecies. The authors recommend more research is needed to evaluate whether prescribed fire or other vegetation management activities confer similar benefits to natural fire. In study of Rocky Mountain bighorn sheep, authors found that the timing of vegetation recovery after prescribed fire, which can be affected by weather conditions, had a strong impact on whether fire benefitted forage (Clapp and Beck 2016).

Existing Protections

The San Bernadino Forest Plan has some specific directives for Peninsular Bighorn sheep. Specifically, the desired conditions in the Santa Rosa and San Jacinto Mountains National Monument region of the forest include the enhancement of all habitat for TES species. The "program" for this region states that "Enhancement of wildlife habitat for threatened, endangered, proposed, candidate, and sensitive species (such as the Peninsular Ranges bighorn sheep) will be emphasized in all management activities" and specifically emphases that "The habitat linkage from the Santa Rosa Mountains into the Anza Borrego State Park for Peninsular Range bighorn sheep will be maintained." The Inyo National Forest has desired conditions specific to Sierra Nevada Bighorn sheep, and the guideline that the forest should "manage recreation, or other disturbances, where research has found it to cause Sierra Nevada bighorn sheep to avoid important habitat as described in the Sierra Nevada Bighorn Sheep Recovery Plan or other guidance from the U.S. Fish and Wildlife Service" (USFS 2025a).

Critical Habitat

For the Peninsular Bighorn sheep, critical habitat was designated for the DPS in 2009 (74 FR 17288), but no critical habitat overlaps with IRAs in the proposed action area. As such we do not discuss PCEs and we have determined the proposed action will have no effect on critical habitat for this species**.**

For the Sierra Nevada Bighorn sheep, critical habitat was designated in 2008 (73 FR 45534) and approximately 2.6% of it is within an IRA affected by the proposed action, with 15% of that habitat in areas that are designated as operable or operable but complex. The PCEs for this species habitat are:

- "(1) Non-forested habitats or forest openings within the Sierra Nevada from 4,000 ft (1,219 m) to 14,500 ft (4,420 m) in elevation with steep (greater than or equal to 60 percent slope), rocky slopes that provide for foraging, mating, lambing, predator avoidance, and bedding and allow for seasonal elevational movements between these areas.
- (2) Presence of a variety of forage plants as indicated by the presence of grasses (e.g., *Achnanthera* spp.; *Elymus* spp.) and browse (e.g., *Ribes* spp.; *Artemisia* spp., *Purshia* spp.) in winter, and grasses, browse, sedges (e.g., *Carex* spp.) and forbs (e.g., *Eriogonum* spp.) in summer.
- (3) Presence of granite rock outcroppings containing minerals such as sodium, calcium, iron, and phosphorus that could be used as salt licks/mineral licks in order to meet nutritional needs."

Given that there is critical habitat in inventoried roadless areas, even though it is a small amount, we have defined the exposure level as medium. Given that the PCEs for the species are unlikely to be modified by roads and sheep habitat is high elevation and steep we have determined the proposed action is not likely to adversely affect critical habitat for this species because the effects are discountable.

Effects Determination

The exposure of the Peninsular Bighorn sheep to the proposed action is low. However, given that the species is sensitive to roads and human activities associated with roads, we have determined the proposed action is likely to adversely affect this species.

The exposure of the Sierra Nevada Bighorn sheep to the proposed action is medium because even though a decent portion of their range overlaps with the proposed action it is not a landscape that is conducive to road construction. Given that this species is sensitive to roads and human activities associated with roads, we have determined the proposed action is likely to adversely affect this species.

Southern mountain caribou DPS

The southern mountain caribou distinct population segment (DPS) of woodland caribou *(Rangifer tarandus caribou)* is a transboundary (United States and Canada) caribou population that is listed as endangered by both the Endangered Species Act (ESA) (84 FR 52598) and the Committee on the Status of Endangered Wildlife in Canada (COSEWIC) (COSEWIC 2014). For the portion of the range in the United States, critical habitat for this DPS was originally designated in 2012 (77 FR 71042) and then reaffirmed in the 2019 listing decision (84 FR 52598), and a recovery plan outline was published in 2019 (USFWS 2019c). During that same year, the last female southern mountain caribou known to be present in the United States was captured in Idaho and moved to a captive breeding facility in Canada (USFWS 2023a), marking the extirpation of the subpopulation that inhabited remote areas of northeast Washington and northwest Idaho (WDFW 2025).

The southern mountain caribou DPS was estimated to encompass 1,356 adults in 2014, reflecting a decline of 27% since 2002 and a decline of 45% in the previous three generations (COSEWIC 2014). The DPS consists of 17 subpopulations, of which at least two have been extirpated, and most of the other extant populations in Canada are very small (COSEWIC 2014). As of 2019, the southern mountain caribou DPS is thought to be in an accelerating state of decline (84 FR 52598). A main driver of southern mountain caribou DPS decline is predation by gray wolves, bears, and mountain lions resulting from habitat fragmentation (COSEWIC 2014) due to roads and human disturbance. When old-growth caribou habitat is fragmented, the early-seral vegetation that grows after disturbance supports moose and deer that were not previously present, in turn supporting a greater number of predators that then use caribou as secondary prey (Wittmer et al. 2007, Seip 1992). For this reason, one of the stated goals of the Mountain Caribou Recovery Implementation Plan is to "protect [5,436,320 ac] of caribou habitat from logging and road building" (84 FR 52598).

Exposure to the Proposed Action Area and Proposed Actions

Southern mountain caribou are strongly associated with old-growth forests (Apps et al. 2001). In particular, they inhabit high-elevation Engelmann spruce and subalpine fir forests and western red cedar and western hemlock at low elevations (Apps et al. 2001). They use different habitat types seasonally, including lower elevations with green vegetation (spring), high-elevation areas (during calving for females), valley bottoms and low slopes (early winter), and upper slopes and ridgetops (late winter) (Apps et al. 2001, 84 FR 52598). This caribou DPS is adapted to living in

areas with deep snow, and their winter diet is composed almost entirely of arboreal lichens, meaning that they require large intact areas of old forest (which support hair lichens in the overstory, Goward et al. 2024) and steep mountainous terrain to achieve their nutritional needs. However, maintaining intact areas of habitat in the areas surrounding these steep old-forest areas is also very important for population abundance (Lochhead et al. 2022) because caribou benefit from access to large areas of mature forest so they can space out from other competing ungulates and their shared predators (Seip 1992).

Critical habitat for this species encompasses 30,010 acres in the northeast corner of Washington and the northwest corner of Idaho (77 FR 71042). The entirety of this designated critical habitat occurs on federal land that is managed by the USDA Forest Service (77 FR 71042). The critical habitat is located on the Idaho Panhandle National Forest (IPNF), which encompasses portions of Idaho, Montana, and Washington, and the Colville National Forest (CNF) in Washington (77 FR 71042). Both the IPNF and the CNF contain inventoried roadless areas (USFS 2000a). Approximately 4.4% of critical habitat for the species is in a potentially affected IRA, and of that habitat 82% is classified as being in areas that are operable or operable but complex. There are no observations of the species in NRM or GBIF, making these datasets unsuitable for drawing conclusions about the relative exposure of the species to the proposed action. We ran a supplemental analysis overlaying the USFWS range map (USFWS 2026a) for this species with inventoried roadless areas affected by the proposed action to further evaluate exposure and the overlap was approximately 7.8%.

Effects of Roads

A large body of research has demonstrated that roads and other linear features can cause caribou, their competitors (moose, deer, and elk), and their shared predators to modify their space use (Dickie et al. 2023) with ramifications for caribou demographic rates (Serrouya et al. 2017) and population declines (Wittmer et al. 2007). Competitor ungulates and predators can use roads to access parts of caribou range that were previously inaccessible to them (Dickie et al. 2020). This can lead to increased predation on caribou populations and depressed population growth.

Additionally, a recent study provided evidence that roads in both currently used (“core”) and surrounding (“matrix”) habitat were negatively associated with caribou abundance. The research predicts that for every 1% increase in roads in the matrix habitat, there will be a 42% decline in southern mountain caribou abundance (Lochhead et al. 2022). This is concerning for a DPS that is already characterized by small, isolated subpopulations with limited interbreeding (COSEWIC 2014, 84 FR 52598).

Roads also create opportunities for more human disturbance in areas that were previously less accessible (84 FR 52598). Human recreation activities, especially winter activities like backcountry skiing, heli-skiing, cat-skiing, and snowmobiling, are known to affect caribou behavior (84 FR 52598), and many of these activities make use of existing roads and require road maintenance.

Finally, roads are a source of direct mortality for caribou. Highways have led to several fatal collisions (84 FR 52598) which can have a major effect on already-small subpopulations. To address these concerns, the Forest Service has been working to mitigate threats from roads

within caribou critical habitat by avoiding road construction through mature forest when possible and by emphasizing road closure and habitat mitigation (USFWS 2019c).

Effects of Timber/ Vegetation Management

In general, timber harvest is associated with negative effects on southern mountain caribou. Caribou avoid areas where timber was recently harvested, and they abandon areas that experienced high levels of harvest (Smith et al. 2000, Stevenson et al. 2025). For example, timber harvest and other anthropogenic disturbances in low-elevation portions of seasonal habitat were associated with a shift from migratory to resident behavior in one population of southern mountain caribou which ultimately corresponded with a decrease in survival rates (Williams et al. 2021). Further, a study that compared the effects of timber harvest on caribou to the effects of wildfire found that caribou are far "more likely to shift, abandon, or contract their range in response to timber harvest than in response to wildfire" and that caribou population declines are stronger in response to timber harvest than wildfire as well, possibly due to timber-associated roadbuilding (Stevenson et al. 2025).

Timber production contributes to habitat fragmentation and loss via conversion to younger forest that leads to increased ungulate presence, increased predation, and then population decline (COSEWIC 2014, 84 FR 52598, Wittmer et al. 2007, Lochhead et al. 2022). To address this concern, recommendations for forest management in mountain caribou habitat include retaining upper-elevation mature forest at specific age levels, using partial cutting and uneven-aged silvicultural methods, and retaining mature forest corridors and mature forest remnants within cutblocks (Seip 1998).

Negative effects of salvage logging after insect outbreaks have been documented in some caribou populations, although results depend on the population and the season (Labadie et al. 2021, Griffin et al. 2025). One population of central mountain caribou whose habitat experienced a pine beetle outbreak avoided areas that experienced higher timber harvest and fire density (Griffin et al. 2025). These caribou also increased their avoidance of timber harvest as it became more prevalent on the landscape (Griffin et al. 2025). Similarly, a population of boreal caribou experienced increased mortality after a spruce budworm outbreak led to more deciduous vegetation, more moose, and more wolf predation that was eventually exacerbated by salvage logging (Labadie et al. 2021).

Existing Protections

Both the IPNF and CNF Forest Plans contain provisions to protect woodland caribou habitat. The IPNF Forest Plan emphasizes connectivity with habitat in Canada, low levels of disturbance, reducing disturbance in seasonal habitats during key times, and management activities in seasonal habitat that are aimed to steer vegetation toward target stand conditions (2015 IPNF land management plan, available at USFS 2025a). This plan also incorporates a court-ordered snowmobile closure that restricts access and trail grooming and incorporates a protected caribou travel corridor (84 FR 52598). Similarly, the CNF Forest Plan restricts over-the-snow vehicle use to designated routes within the caribou recovery area and emphasizes disturbance reduction and vegetation management to ensure the availability of preferred lichens for forage (2019 CNF land

management plan, available at USFS 2025a). Both the IPNF and the CNF contain inventoried roadless areas (USFS 2000a).

Critical Habitat

Designated critical habitat for the United States portion of the southern mountain caribou DPS' range includes five elements: 1) mature hemlock/western red cedar and subalpine fir/Engelmann spruce forests above 5,000 feet in elevation, 2) ridgetops and basins above 6,000 feet in elevation, 3) the presence of arboreal hair lichens, 4) high-elevation areas with shallow slopes, meadows, and flowering plants, and 5) corridors connecting the habitats described above (77 FR 71042). This designated critical habitat encompasses 30,010 acres in the northeast corner of Washington and the northwest corner of Idaho (77 FR 71042). Approximately 4.4% of critical habitat for the species is in a potentially affected IRA, and of that habitat 82% is classified as being in areas that are operable or operable but complex. Given that 82% of critical habitat in inventoried roadless areas is in operable or operable but complex environments we are assigning an exposure level of "high" for critical habitat. Given this exposure and that a number of the PBFs for this species' critical habitat may be affected by road and timber activities, the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species.

Effects Determination

The exposure level of the Southern Mountain Caribou DPS to the proposed action is high given the overlap of the proposed action with its range. Given the importance of mature forest to the species, the FS has determined the proposed action is likely to adversely affect this species. Given exposure and the requirement for mature forest, the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species.

Small Rodents

Utah Prairie Dog

The Utah Prairie dog *(Cynomys parvidens)* was listed as threatened in 1973 (38 FR 14678) and was uplisted to endangered (76 FR 36053) in 2011. There is considerable variability in the estimated population size, and although the overall long-term trend is for stability or potentially an increase in population numbers, the range-wide count of animals was noticeably lower in 2020 than it was in 2016. These are the data available in the most recent species status assessment (USFWS 2021b). Plague is one of the most important threats to the species (USFWS 2021b).

Exposure to the Proposed Action Area and Proposed Actions

Currently the Utah Prairie Dog only occurs in southwestern Utah (USFWS 2021b) where it lives in semiarid shrub-steppe and grassland habitats and build burrows in well-drained soils (76 FR 36053). This distribution represents a retraction from its historical range which extended further north (USFWS 2021b). It is thought to be present on the Dixie, Fishlake, and Manti-la Sal National Forests. There are 28 observations of the Utah Prairie dog in inventoried roadless areas that may be affected by the proposed actions in the NRM and GBIF databases. Of these, only 4% are in areas that are potentially operable or operable but complex. Given the opportunistic nature

of sampling from these data sources, we also conducted an analysis overlaying the USFWS range for this species (USFWS 2026a) with inventoried roadless areas affected by the proposed action to evaluate the consistency of these two approaches. The overlap is approximately 12.3%. There is no designated critical habitat for the Utah prairie dog.

Effects of Roads

Utah prairie dogs can be killed by vehicle strikes on roads (76 FR 36053). Additionally, the use of off-highway vehicles and the creation of trails by these vehicles by recreationists can cause habitat loss for the species and potentially extirpation of colonies (76 FR 36053). At the time of the 2011 listing decision, the USFWS concluded that "Road construction, off-highway vehicle (OHV) use, and recreation may have effects to individuals or colonies that occur adjacent to the roadways, trails, or play areas; however, these are localized areas and do not result in population level effects" (76 FR 36053).

Effects of Timber/ Vegetation Management

The 2011 listing decision and the 2021 species status assessment do not address the impacts of timber harvest on this species, which is perhaps not surprising given the species' habitat association.

Existing Protections

The Dixie National Forest Plan has the goal of maintaining or enhancing prairie dog habitat (1986 land management plan, available at USFS 2025a). Similarly, the Fishlake National Forest Plan states that "it will also meet the habitat requirements of the Utah prairie dog recovery plan" (1986 land management plan, available at USFS 2025aa). The Manti-La Sal National Forest plan does not contain specific direction for the species (USFS 2025a).

Effects Determination

Given that 12.3% of the species' range overlaps with inventoried roadless areas, the FS has determined that the exposure of the species to the proposed action is high. There is potential for individual impacts to survival due to road construction and road usage. Therefore, the FS has determined that the proposed action is likely to adversely affect this species.

Carolina Northern Flying Squirrel

The Carolina northern flying squirrel *(Glaucomys sabrinus coloratus)* was listed as endangered in 1985 (50 FR 26666). The 2022 Species Status Assessment noted that some data in a preliminary occupancy analysis suggests that populations in 7/10 management units for the subspecies were stable, but that a number of methodological limitations limited the strength of conclusions from this data (USFWS 2022b). The recent assessment notes that many of the populations of this subspecies are isolated and that some have only a little high-quality habitat (USFWS 2022b).

Exposure to the Proposed Action Area and the Proposed Actions

The Carolina northern flying squirrel is a subspecies occurring at the southern extent of the range of the northern flying squirrel in the Southern Appalachians in North Carolina, Tennessee and

Virginia. They are present on the Cherokee, George Washington and Jefferson, Nantahala, and Pisgah National Forests and all of these forests have inventoried roadless areas that could be affected by the proposed action. Spruce fir forests are thought to be especially important for foraging (USFWS 2013a) and hardwood stands with yellow birch are thought to be important for denning (USFWS 2013a). The subspecies is thought to primarily occur above 4,500 ft, although more recent work has demonstrated use in the 4,000-4,500 ft elevational band (USFWS 2022b). Analyses of vegetation on the Cherokee, Nantahala, and Pisgah National Forests show that large portions of the forest that used to be spruce-fir are now uncharacteristic vegetation types (USFWS 2022b). Habitat fragmentation from logging is thought to have played a role in their decline (50 FR 26999).

There is no critical habitat for this subspecies. There are no records of this subspecies in the NRM and GBIF databases. As such, it is not possible to use these data for an assessment of the potential impact of the proposed action on this species. We conducted a supplementary analysis overlapping the USFWS range map (USFWS 2026a) for this species with inventoried roadless affected by the proposed action. Approximately 6% of the species' range is in the proposed action area.

Effects of Roads

Roads can create a barrier to movement for this subspecies. For example, the Cherohala Skyway was a barrier to dispersal within the Unicoi Mountains GRA when it was constructed, even though it was a two-lane rural scenic byway with relatively low traffic volume (Kelly et al. 2013). Researchers suggest that even roads of this size create a large enough gap to impede traverse by gliding, and that squirrels were averse to crossing the road by foot as evidenced by the absence of significant roadkill data for the subspecies across its range (Weigl et al. 2002 in Kelly et al. 2013). In this specific instance, red spruce have been planted and artificial crossing structures have been installed (USFWS 2022b; Kelly et al. 2013) to improve connectivity, but the example illustrates how road construction can affect movement behavior and gene flow within the subspecies.

Effects of Timber/ Vegetation Management

According to the 2022 5-year review, "significant areas [of squirrel habitat] are still suffering from legacy effects of past disturbances and land use (e.g., industrial logging in the late 1800s/early 1900s), and work is needed to properly restore and manage lands to help recover the species." Several national forests where the species occurs currently have uncharacteristic vegetation types. Restoration of spruce-fir habitat is identified as a need for this species (USFWS 2022b). The 5-year review also notes that "while there is the potential for timber harvesting in Carolina northern flying squirrel habitat, these areas are largely avoided, and any harvest would likely be small in scale and undertaken for the purpose of restoration" (USFWS 2022b). Vegetation management activities that restore this spruce-fir type, when properly designed, could be beneficial for the species.

Having "sufficient ecological data and timber management data ... to assure future protection and management" of the Carolina northern flying squirrel is a requirement for recovery (USFWS

2022b). As of the assessment in 2022, studies explicitly evaluating the impact of different timber management practices on the subspecies had not been conducted (USFWS 2022b).

Existing Protections

The Cherokee National Forest plan lists the management objective of "Provide high quality northern hardwood/spruce fir habitats; improve habitat connectivity where possible" for this subspecies (2004 land management plan, available at USFS 2025a). The Nantahala- Pisgah National Forest Plan includes the desired condition that "habitat components at finer scales provide for wildlife occupancy, are present in sufficient amounts, and distributed across all ecozones. For example...smaller live or dead trees with crevices provide critical nesting and roosting habitat for flying squirrels and bats" (2024 land management plan, available at USFS 2025a). Additionally, the forest should "Maintain species presence within currently occupied habitat" on the forest and offers the goal of "Emphasize restoration of spruce-fir habitat for the Carolina northern flying squirrel and maintain the health and resiliency of this forest type in the face of climate change" (2024 land management plan, available at USFS 2025a). The Jefferson National Forest Plan also has objectives specific to the recovery of the subspecies (2004 land management plan, available at USFS 2025a).

The assessment of the 2013 5-year review was that while it "appears" that "sufficient habitat for population maintenance is maintained" but it is "unclear if sufficient habitat is being managed for expansion or to allow for habitat corridors that connect GRAS" (USFWS 2013a).

Effects Determination

Given that 6% of this species' range is affected by the proposed action is affected by the proposed action we have assigned an exposure level of high. Additionally, given the species' dependence on specific forest types, the negative impacts of habitat loss, and the negative impact of roads on connectivity the FS has determined the proposed action is likely to adversely affect this species.

San Bernardino Merriam's kangaroo rat

San Bernardino Merriam's kangaroo rat (*Dipodomys merriami parvus*) is a granivorous subspecies of the Merriam's kangaroo rat that was listed as endangered under the Endangered Species Act in 1998 (63 FR 51005). At the time of listing, the subspecies' historical range had contracted by 95% due to human development including agricultural conversion, urbanization, and industrial development (63 FR 51005).

At present, range-wide estimates of population trends and occupancy are not available (USFWS 2023b). However, ongoing human-caused threats to the subspecies include habitat destruction and loss, fragmentation from gravel mining operations, urbanization, off-highway vehicle use, and flood control projects (63 FR 51005). Some populations of San Bernardino kangaroo rats are also threatened by changes in stream flow patterns and related vegetation changes (63 FR 51005).

Exposure to the Proposed Action Area and Proposed Actions

Critical habitat for the San Bernardino kangaroo rat was originally designated in 2002 and then revised in 2008 (73 FR 61936). Approximately 7,779 acres across five units are designated as critical habitat in San Bernardino and Riverside counties of California, including multiple locations (specific creeks and washes) on the San Bernardino National Forest (73 FR 61936; 2005 land management plan, available at USFS 2025a). Approximately 6% of critical habitat is in inventoried roadless areas, and, of that, 35% of it is within areas that are operable or operable but complex.

San Bernardino kangaroo rats inhabit alluvial habitats and adjacent upland areas with low vegetation cover and sandy soils that enable them to dig shallow burrows, such as flood plains and alluvial fans (63 FR 51005). This type of area typically contains gravel, rock, sand, silt, and clay that is deposited by moving water (73 FR 61936). Areas near rivers that experience flooding every few years to maintain lower vegetation cover tend to support higher densities of San Bernardino kangaroo rats (63 FR 51005). Due to the changing vegetation and soil patterns, not all alluvial scrub habitat is appropriate for this subspecies at any given time since they use more open areas with low shrub canopy cover (63 FR 51005). Alluvial sage scrub is a "distinct and rare plant community that dominates major outwash fans at the mouths of canyons along the coastal side of the San Gabriel, San Bernardino, and San Jacinto mountains and some smaller floodplain and riverine areas of southern California" (73 FR 61936). Maintaining habitat connectivity between the stream and the upland alluvial terraces is important for supporting San Bernardino kangaroo rat populations (USFWS 2024g, USFWS 2024h).

San Bernardino kangaroo rats have occasionally been documented in areas that have occasionally been altered by humans such as the edges of orchards, vineyards that are inactive, and areas of wildland-urban interface that are near occupied alluvial habitat (73 FR 61936). Although these are not areas of high-quality habitat, these individuals may serve an important role in recolonizing after flooding or other dynamic events within the primary habitat (73 FR 61936).

There are no records of the species in the NRM or GBIF databases, making it impossible to use these data sources to evaluate the exposure of the species to the proposed action. As such we ran a supplementary analysis overlaying the USFWS range map (USFWS 2026a) for the species with inventoried roadless areas affected by the proposed action. The overlap was approximately 7.2%.

Effects of Roads

The specific effects of roads on San Bernardino kangaroo rats are generally negative. This species cannot use areas of compacted soil due to its weak forelegs and the need for looser soil that enables burrowing (73 FR 61936), meaning that road construction fragments existing habitat. Urbanization, development, and associated road construction is a major threat across the entirety of the subspecies' range due to the increasing human population in the surrounding areas (73 FR 61936). Sand and gravel mining is a particular concern for this subspecies' habitat due to the way it alters soil composition (73 FR 61936), and mining access roads could also pose

challenges. Additionally, off-road vehicle use has been noted as a concern due to the way it can damage kangaroo rat burrows, soil structure, and the surrounding vegetation (73 FR 61936).

Effects of Timber/ Vegetation Management

The specific effects of timber production on San Bernardino kangaroo rats are likely negligible. The species inhabits alluvial fans with low shrub canopy cover, and especially areas that flood periodically (73 FR 61936). It is unlikely that timber harvest would occur within the subspecies' habitat, and neither timber harvesting nor forest management are mentioned in the Species Status Assessment (USFWS 2023b).

Existing Protections

San Bernardino kangaroo rats and some of their designated critical habitat are located on the USDA Forest Service's San Bernardino National Forest (2005 land management plan, available at USFS 2025a). The land and resource management plan for the San Bernardino National Forest was revised in 2005 and incorporates multiple provisions intended to benefit this subspecies' habitat (73 FR 61936; 2005 land management plan, available at USFS 2025a). However, the U.S. Fish and Wildlife Service opted to designate critical habitat on these lands in addition to the provisions outlined in this plan since the habitat contains elements that are "essential to [San Bernardino kangaroo rat] conservation" (73 FR 61936).

Critical Habitat Determination

The primary constituent elements of critical habitat for this species are as follows:

- "(1) Alluvial fans, washes, and associated floodplain areas containing soils consisting predominately of sand, loamy sand, sandy loam, and loam, which provide burrowing habitat necessary for sheltering and rearing offspring, storing food in surface caches, and movement between occupied patches;
- (2) Upland areas adjacent to alluvial fans, washes, and associated floodplain areas containing alluvial sage scrub habitat and associated vegetation, such as coastal sage scrub and chamise chaparral, with up to approximately 50 percent canopy cover providing protection from predators, while leaving bare ground and open areas necessary for foraging and movement of this subspecies; and
- (3) Upland areas adjacent to alluvial fans, washes, and associated floodplain areas, which may include marginal habitat such as alluvial sage scrub with greater than 50 percent canopy cover with patches of suitable soils (PCE 1) that support individuals for re-population of wash areas following flood events. These areas may include agricultural lands, areas of inactive aggregate mining activities, and urban/wildland interfaces." (73 FR 61936)

Given that approximately 6% of critical habitat is in inventoried roadless areas, and of that 35% of it is within areas that are operable or operable but complex the exposure of the habitat to the proposed action is high. Given that roads, soil compaction, and vegetation trampling or removal

may adversely affect critical habitat the FS has determined the proposed activity is likely to adversely affect critical habitat for this species**.**

Effects Determination

Given that 7.2% of the species' range intersects with inventoried roadless areas affected by the proposed action exposure is high. Given the potential for roads and OHV use to damage soils and vegetation the FS has determined the proposed action is likely to adversely affect this species. Given that roads, soil compaction, and vegetation trampling or removal may adversely affect critical habitat the FS has determined the proposed activity is likely to adversely affect critical habitat for this species**.**

Stephens' kangaroo rat

Stephens' kangaroo rat (*Dipodomys stephensi*) is a nocturnal mammal that was listed as endangered under the Endangered Species Act in 1988 after experiencing extensive habitat loss and degradation that yielded small, isolated populations (53 FR 38465). More recently, the species was downlisted to threatened in 2022 due to a reduction in extinction risk, and a "4(d) rule" that prohibits intentional take and specifies exceptions was published (87 FR 8967). Current habitat loss, fragmentation, and population isolation were noted as reasons that the species should be reclassified as threatened, as well as the ongoing effects of past habitat fragmentation (87 FR 8967). The current range-wide population trend and density is unknown due to incomplete surveys, challenges of detecting the animals, and natural year-to-year population fluctuations (87 FR 8967).

As of the time of downlisting, experts concluded that "the Stephens' kangaroo rat continues to occur in suitable habitat in seemly stable populations across its range" (87 FR 8967). Conservation actions are informed by a draft recovery plan (USFWS 1997) and a five-year review (USFWS 2025f).

Exposure to the Proposed Action Area and Proposed Actions

Critical habitat has not been designated for Stephens' kangaroo rats (53 FR 38465). Populations occur in three geographic regions of southern California (53 FR 38465). Historically, they likely inhabited a broader range, but they were slowly confined to a smaller area as urban and agricultural development in the region expanded (53 FR 38465). They inhabit open scrublands and grasslands with herbaceous annuals that are native to California (53 FR 38465, 87 FR 8967). They are found in areas with slightly disturbed or plowed soil and flat or rolling terrain (53 FR 38465). They live in underground burrows and come to the surface to forage at night (53 FR 38465), occupying patchy areas of habitat containing burrow entrances and connecting paths (87 FR 8967). They may prefer plant communities that are maintained by disturbances like fire, grazing, and agriculture that do not have dense grass or high cover of non-native grasses (USFWS 2025f).

There are only 6 records of the species in the NRM and GBIF databases, none of which occur in inventoried roadless areas. Given the paucity of records for this species and the potential for sampling bias, the relative exposure of this species to the proposed action cannot be assessed with these data. We ran a custom analysis overlaying habitat for the species (Conservation

Biology Institute 2019) with inventoried roadless areas. Less than 0.001% of modelled habitat overlapped with the proposed action area.

Effects of Roads

The specific effects of roads on Stephens' kangaroo rats are uncertain, but it is plausible that they could have detrimental effects. This species does not use areas of compacted soil (53 FR 38465), meaning that road construction could fragment existing habitat. Additionally, off-road vehicle use has been noted as a concern due to the risk of vegetation destruction, soil and burrow compaction, and collisions (53 FR 38465, USFWS 1997). For this reason, the recovery plan notes that habitat blocks without roads and that are inaccessible to people are more appropriate for Stephens' kangaroo rat conservation efforts (USFWS 1997).

However, under certain circumstances, the species may be able to colonize areas that have been modified for agriculture, grazing, clearing, and roads (USFWS 1997). Stephens' kangaroo rats are frequently found near disturbed areas including dirt roads (USFWS 1997).

Effects of Timber/ Vegetation Management

The specific effects of timber production on Stephens' kangaroo rats are negligible. The species inhabits a small area of habitat that is dominated by grasslands, scrub, and some shrublands. It is unlikely that timber harvest would occur within the species' habitat.

Existing Protections

The majority of Stephens' kangaroo rat's range occurs on private land (53 FR 38465). However, the USDA Forest Service manages the Cleveland National Forest which contains suitable habitat for Stephens' kangaroo rat (2005 land management plan, available at USFS 2025a) and which may provide connectivity habitat to other areas (USFWS 1997). The Cleveland National Forest Plan does not contain specific standards and guidelines related to this species; rather, it tiers to the Recovery Plan (2005 land management plan, available at USFS 2025a). Other federally managed land is also an important component of the species' habitat, including several military bases and areas under the Bureau of Land Management's jurisdiction (53 FR 38465). At least two major habitat conservation efforts have been implemented within the species' range, and the military installations within the range have also developed natural resources management plans and are engaged in habitat conservation efforts (87 FR 8967).

Species Effects Determination

Given that the overlap of modelled habitat for this species and inventoried roadless areas is extremely low the FS has determined the proposed action is not likely to adversely affect this species because the effects are so unlikely as to be discountable.

Peñasco Least Chipmunk

The Peñasco Least Chipmunk (*Neotamias minimus atristriatus*) was listed as Endangered effective as of January 9. 2025. The Peñasco least chipmunk is a disjunct subspecies at the southernmost extent of least chipmunks, historically found in the Sacramento and White Mountains of New Mexico. In the Sacramento Mountains, it occupied canyon-bottom meadows and open ponderosa pine forests between 2,103–2,438 m (6,900–8,000 ft), while in the White

Mountains it occurred in high-elevation (>3,109 m/10,200 ft) Thurber's fescue grasslands with rock outcrops.

The Peñasco least chipmunk population is currently restricted to a small portion of the White Mountains within Lincoln National Forest, New Mexico. The 2024 Species Status Assessment (SSA) prepared by the U.S. Fish and Wildlife Service concluded that the subspecies persists at extremely low abundance, with the best available data indicating fewer than 50 individuals remain (USFWS 2024i). The population is confined to a single, isolated habitat patch, representing a significant contraction from its historical range. Genetic diversity is presumed to be severely reduced, increasing the risk of inbreeding depression and limiting adaptive potential. The SSA determined that the population has experienced a long-term decline with no evidence of recovery or expansion. Surveys and monitoring efforts have consistently documented low detection rates, suggesting that the subspecies is at imminent risk of extirpation. The decline is attributed to multiple interacting stressors, including habitat degradation, disturbance and fragmentation, predation and alteration of weather.

Exposure to the Proposed Action Area and the Proposed Action

The Peñasco least chipmunk (*Neotamias minimus atristriatus*) had critical habitat designated simultaneously with the Endangered listing date effective January 2025 (89 FR 99656). The U.S. Fish and Wildlife Service listed the subspecies as endangered and concurrently designated approximately 4,386 acres (1,774 hectares) of critical habitat in Lincoln County, New Mexico, within the Lincoln National Forest. This designation encompasses the chipmunk's remaining occupied range and is intended to protect native meadow and ponderosa pine forest habitats essential for survival. One of the smallest chipmunks, the Peñasco least chipmunk uses mature conifers to provide shade and protection from overhead predators as it forages on seeds and flowers in meadows and other openings in subalpine forests. All of the designated critical habitat for the Peñasco least chipmunk (100%) is on federal lands, specifically within the Lincoln National Forest managed by the U.S. Forest Service. However, there are no roadless areas in the affected environment in this critical habitat.

Effects of Roads

The 2024 Species Status Assessment did not identify roads as a primary driver of decline compared to grazing, ski resort operations, feral hogs, drought, and wildfire (USFWS 2024i). There are no documented cases of direct mortality from vehicle strikes in this species however, roads may contribute to habitat fragmentation, disturbance, and reduced connectivity in the chipmunk's already restricted range. Road construction and maintenance can degrade native meadow and ponderosa pine understory habitats, increase edge effects, and facilitate human access that elevates predation and competition pressures. Roads also act as barriers to dispersal, further isolating the small population and limiting gene flow. Studies of small mammals in western forests have shown that road networks reduce movement across landscapes, increase mortality risk from vehicles, and alter predator–prey dynamics by providing travel corridors for predators. Research on chipmunks and other sciurids indicates sensitivity to habitat fragmentation, with reduced occupancy in areas near roads and disturbed corridors.

Effects of Timber/ Vegetation Management

The 2024 Species Status Assessment did not identify timber harvest or logging as a primary driver of the species decline (USFWS 2024i). However, timber harvest activities can negatively affect the chipmunk's habitat by reducing old growth ponderosa pine stands and associated native grassland understory that historically supported the subspecies in the Sacramento Mountains. Logging can fragment meadows and forest openings, alter canopy structure, and diminish the parklike understory conditions that provide food and cover. Increased human access associated with timber harvest may further exacerbate disturbance and predation risks. While not highlighted as a major threat, timber harvest can contribute to habitat degradation and reduced resiliency, compounding the species' vulnerability given its already small population size and restricted distribution.

Existing Protections

Because all designated habitat occurs on U.S. Forest Service lands, habitat will also be subject to management guidelines under the Lincoln National Forest Land Management plan (1986) and amendments (USFS 2025a). At the state level, the New Mexico Department of Game and Fish recognizes the subspecies as a Species of Greatest Conservation Need, providing additional monitoring and coordination with federal agencies.

Critical Habitat Determination

Given that no critical habitat overlaps with the proposed action area the Forest Service has determined there will be No Effect to designated critical habitat for this species.

Effects Determination

Exposure data on species occurrence are insufficient for assessment so we will assume potential exposure. However, given the limited spatial extent of the population and that area does not overlap with the affected environment exposure is likely low. As the impacts of roads or timber on this species are mostly indirect and not identified as major stressors, combined with low exposure to the affected environment, we determine that the proposed action is not likely to adversely affect this species.

Mount Graham Red Squirrel

The Mount Graham red squirrel *(Tamiasciurus fremonti grahamensis)* was listed as endangered subspecies in 1987 (52 FR 20994). The results from a census in 2023 suggest that there were 144 individuals (USFWS 2024j). This is a decline from the 200-300 squirrels that were documented consistently from 2000-2016, but an improvement from the 35 individuals that were estimated following the 2017 Frye Fire, which burned much of the squirrel's habitat (Merrick et al. 2021; USFWS 2024j). Midden surveys suggest that the species had previously experienced a decline in the mid-1990s to early 2000s associated with loss of spruce-fir habitat from wildfires and fire suppression activities (USFWS 2011b). The Mount Graham squirrel is genetically distinct and isolated from other red squirrels and the dramatic decline in population after the Frye fire suggests a recent population bottleneck (USFWS 2024j).

Exposure to the Proposed Action and the Proposed Action Area

The Mount Graham squirrel lives in a very restricted habitat in the Pinaleño Mountains of southeastern Arizona (52 FR 20994) in the Safford Ranger District of the Coronado National Forest. It occurs in mature mixed conifer and spruce-fir forest above 8,000 feet and requires full forest canopy cover (USFWS 2011b). The 2024 SSA reported that due to widespread habitat loss there are currently about 6,400 acres of squirrel habitat remaining, 34 of which is spruce-fir forest. Critical habitat for the squirrel was designated in 1990 (55 FR 425) and approximately 41% of it occurs in IRAs and of that critical habitat, 76% of it is classified as operable or operable but complex.

Our initial evaluation of squirrel occurrence records (in NRM and GBIF) was not informative because there are only 2 observations of the species in this dataset. However, given the extensive overlap of designated CH there are likely more individuals present within the action area. We ran a custom analysis overlaying squirrel habitat (USGS 2018) with the footprint of inventoried roadless areas and the Frye Fire to evaluate potential exposure. The proportion of total squirrel habitat (pre-fire) that overlapped with the proposed action was ~ 16.3%. However, approximately 77% of modeled squirrel habitat burned in the Frye Fire. If roadless areas were to be affected by the proposed action, the total proportion of squirrel habitat affected by the Frye Fire or inventoried roadless areas is ~82.2%. Of the habitat that was not in the Frye Fire footprint, approximately 22.8% is inventoried roadless. This analysis does not consider burn severity and the potential for squirrel habitat to remain within the Frye Fire footprint.

Effects of Roads

Roads have the potential to impact Mount Graham red squirrels through direct mortality, loss of habitat, and behavioral avoidance. There is no systematic data collection of direct mortality, however 14 road-killed squirrels have been reported incidentally between 1987 and the early 2020s (USFWS 2024j). Openings created by roads may also increase vulnerability of squirrels to avian predators (Chen and Koprowski 2016a).

Squirrels avoid crossing roads (Chen and Koprowski 2016a) and move more quickly while crossing than immediately before or after (Chen and Koprowski 2016b). Interestingly probability of crossing was not related to traffic volume, but gaps in tree cover (<50%) created by roads, may lead to crossing avoidance (Chen & Koprowski 2016a). Squirrel occupancy of middens exposed to higher levels of traffic noise was also lower than middens in quieter environments (Chen and Koprowski 2015). This avoidance may be particularly impactful during reproduction as squirrels cross roads more often during the mating season and when there are larger number of mates on the opposite side of the road (Chen and Koprowski 2016a). Squirrels may also be negatively impacted if roads lead to increased human caused wildfires that cause habitat loss.

Effects of Timber/ Vegetation Management

While Mount Graham red squirrels are negatively affected by tree removal, they are also negatively affected by loss of habitat from high severity wildfire. Tree removal from selective thinning, harvest, or prescribed burning can result in loss of habitat, food resources, and protection from aerial predators (USFWS 2024j).

However, high severity fires can also severely impact red squirrels. In the Nuttall-Gibson Complex Fire 35% of marked squirrels within the burn perimeter were never relocated and 75% of middens within the perimeter were destroyed (Koprowski et al. 2006). 31% of surviving squirrels within the burn perimeter abandoned middens within 2 weeks of the fire- all were within areas with intensive crown fire (Koprowski et al. 2006). In the days following the Frye Fire, squirrel survivorship was lower than expected, potentially due to increased avian predation (Merrick et al. 2021). 31% of squirrel habitat was lost in the 2017 Frye Fire, and substantial amounts of habitat was lost from other disturbance events prior to that (Merrick et al. 2021).

These data suggest that Mount Graham red squirrels may benefit from carefully planned forest management approaches that reduce the risk of high intensity wildfire, while maintaining sufficient canopy cover.

Existing Protections

The Arizona-Idaho Conservation Act of 1989 and the Coronado National Forest land management plan are important vehicles for the conservation of Mount Graham red squirrels. The conservation act established the Mount Graham Astrophysical and Biological Research Area, which is ~2,900 acres of land encompassing an astrological observatory and the Mount Graham Red Squirrel Refugium, which focuses on research for red squirrels and spruce-fir.

The Coronado National Forest land management plan has several components that address conservation of Mount Graham Red Squirrels (available at USFS 2025a). It identifies that a desired condition for the biological research area is habitat for the red squirrel and that "management activities do not degrade these special habitats." The standard for this area is that "within habitat for the Mount Graham red squirrel, no new recreational residence or developed recreation areas will be established" and guidelines advise that red squirrel habitat needs "should supersede the needs of all other species of plants and animals", that hiking levels should not negatively impact individual squirrels or their habitat, and that vegetation treatments "should be designed and implemented to avoid disturbance of ... squirrel middens." Another standard for a forest treatment objective states that "vegetation treatments should be designed to create stand conditions that enhance cone production of white fir, corkbark fir, Engelmann spruce, and Douglas-fir in order to provide a reliable Mount Graham red squirrel food source."

Critical Habitat Determination

Critical habitat for the squirrel was designated in 1990 (55 FR 425). When critical habitat was designated, it was determined that selected areas "contain major concentrations of the Mount Graham red squirrel, and the habitat necessary to its survival, including cover, food sources, nest sites, and midden sites. The winter survival of the red squirrel depends primarily on the availability of seeds of cones stored in middens. Therefore, an environment in which the midden-cached cones will stay cool and moist, and be prevented from opening and losing their seeds, is of critical importance. Such an environment is most often found in dense, shady forest above 10,000 feet (3,048 meters) and at lower elevations on north-facing slopes or in protected pockets and small basins (Spicer et al. 1985)."

Approximately 41% of squirrel critical habitat occurs in IRAs and of that critical habitat, 76% of it is classified as operable or operable but complex. Given that a large portion of the habitat is exposed to the proposed action, the exposure level is high. Given the importance of cones and shady forest for critical habitat, the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species**.**

Effects Determination

A considerable proportion of pre-fire and post-fire squirrel habitat that is inventoried roadless. Given the impacts of roads on squirrels and the importance of forested vegetation, the Forest Service has determined the proposed action is likely to adversely affect this species. Given overlap and the importance of cones and shady forest for critical habitat, the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species**.**

New Mexico meadow jumping mouse

The New Mexico meadow jumping mouse *(Zapus hudsonius luteus)* is a subspecies of jumping mice that was listed as endangered in 2014 (79 FR 33119). The approximate population size has not been measured directly (USFWS 2020c). Habitat loss is the major threat facing the subspecies. As its habitat is currently quite fragmented and is likely to become more so, lack of connectivity is a threat to their persistence. The viability of the subspecies was described as compromised in the near term at the time of listing (79 FR 33119).

Exposure to the Proposed Action Area and Proposed Actions

This subspecies is found in Arizona, New Mexico and small portions of Colorado. It has specific habitat requirements, thriving in tall and dense herbaceous vegetation in riparian habitat corridors with perennial or seasonal flowing water (USFWS 2020c). It also requires nearby upland habitat for foraging and other activities (USFWS 2020c).

The species is thought to occur on the Apache-Sitgreaves, Gila, Lincoln, Rio Grande, San Juan, and Santa Fe National Forests. It might occur on the Comanche National Grassland and Pike and San Isabel National Forest though confirmation is needed.

Critical habitat was designated for the subspecies in 2016 (81 FR 14263). Approximately 0.79% of critical habitat for the subspecies occurs in inventoried roadless areas, of that 28% is considered operable or operable but complex. In the NRM and GBIF databases there are eight occurrences of the subspecies in inventoried roadless areas, of which five occur in operable or operable but complex environments. Given the opportunistic nature of most occurrence data, we conducted a supplementary analysis overlaying the USFWS species range with inventoried roadless affected by the proposed action and found that 1.8% of the species' range occurred in these areas.

Effects of Roads

Roads can have a number of direct and indirect effects on the subspecies. In addition to direct loss of habitat during construction, soil erosion, mowing, and flooding from inappropriately size culverts could all cause habitat loss (USFWS 2020c). Debris flows from small landslides during heavy rain events also have the potential to modify water flow in jumping mouse habitat

(USFWS 2020c). Habitat can also become fragmented due to road construction creating a barrier to movement, although there is at least one individual jumping mouse that was observed to cross a gravel road (USFWS 2020c). The potential for sedimentation from dirt roads to influence New Mexico meadow jumping mouse was noted in the listing decision (79 FR 33119), however at the time the Services noted that they did not have data to evaluate this risk.

If human use of riparian habitat increased due to increased road access, this could increase the negative effects of dispersed camping, OHV use, and dispersed hiking on jumping mouse habitat (USFWS 2020c).

Effects of Timber/ Vegetation Management

Timber harvest upstream has the potential to affect the riparian habitats used by this subspecies by modifying waterflows to downstream habitat (USFWS 2020c). Historically this has affected the subspecies in portions of its range (USFWS 2020c).

Fire is considered a threat to the species and the listing decision notes that "following fires, we found that, depending on fire intensity and the subsequent ash and debris flow within stream reaches, New Mexico meadow jumping mouse populations can be significantly affected and likely extirpated, even when 15 km (9 mi) of continuous suitable habitat existed prior to the fire" (79 FR 33119). The 2020 SSA notes that activities to reduce the risk of high severity wildfire in headwaters to jumping mouse habitat may reduce the risk of fire on the species (USFWS 2020c). At the same time prescribed fire, in the wrong place, can be detrimental to the species. At the time of listing, the Services noted that "unauthorized modification or manipulation of riparian habitat, including mowing or prescribed burning of occupied habitats, especially during the active season" would constitute a violation of Section 9 of the ESA (79 FR 33119).

Existing Protections

The jumping mouse is mentioned multiple times in the Apache Sitgreaves National Forest Plan and there are components of the forest plan that protect the species. For example, one forest guideline is that "the needs of localized species (e.g., New Mexico meadow jumping mouse, Bebb willow, White Mountains paintbrush) should be considered and provided for during project activities to ensure their limited or specialized habitats are not lost or degraded" (2015 Apache-Sitgreaves NFs land management plan, available at USFS 2025a). Another desired condition that applies to the whole forest states that "floodplains and adjacent upland areas provide diverse habitat components (e.g., vegetation, debris, logs) as necessary for migration, hibernation, and brumation (extended inactivity) specific to the needs of riparian-obligate species (e.g., New Mexico meadow jumping mouse, Arizona montane vole, narrow-headed gartersnake)" (2015 Apache-Sitgreaves NFs land management plan, available at USFS 2025a). There are no forest plan components specific to the jumping mouse in the Santa Fe Forest Plan (2022 Santa Fe NF land management plan, available at USFS 2025a). This illustrates the variety of existing protection for the jumping mouse in different jurisdictions.

Critical Habitat Determination

The primary constituent elements of jumping mouse habitat identified in the critical habitat designation are:

- "1) Riparian communities along rivers and streams, springs and wetlands, or canals and ditches that contain:
    - a) Persistent emergent herbaceous wetlands especially characterized by presence of primarily forbs and sedges (*Carex* spp. or *Schoenoplectus pungens*); or
    - b) Scrub-shrub riparian areas that are composed of willows (Salix spp.) or alders (Alnus spp.) with an understory of primarily forbs and sedges; and
- 2) Flowing water that provides saturated soils throughout the New Mexico meadow jumping mouse's active season that supports tall (average stubble height of herbaceous vegetation of at least 61 cm (24 in)) and dense herbaceous riparian vegetation composed primarily of sedges (*Carex* spp. or *Schoenoplectus pungens*) and forbs, including, but not limited to one or more of the following associated species: spikerush (*Eleocharis macrostachya*), beaked sedge (*Carex rostrata*), rushes (*Juncus* spp. and *Scirpus* spp.), and numerous species of grasses such as bluegrass (*Poa* spp.), slender wheatgrass (*Elymus trachycaulus*), brome (*Bromus* spp.), foxtail barley (*Hordeum jubatum*), or Japanese brome (*Bromus japonicas*), and forbs such as water hemlock (*Circuta douglasii*), field mint (*Mentha arvense*), asters (*Aster* spp.), or cutleaf coneflower (*Rudbeckia laciniata*); and
- 3) Sufficient areas of 9 to 24 kilometers (5.6 to 15 miles) along a stream, ditch, or canal that contains suitable or restorable habitat to support movements of individual New Mexico meadow jumping mice; and
- 4) Adjacent floodplain and upland areas extending approximately 100 m (330 ft) outward from the boundary between the active water channel and the floodplain (as defined by the bankfull stage of streams) or from the top edge of the ditch or canal." (81 FR 14264)

Approximately 0.79% of critical habitat for the subspecies occurs in inventoried roadless areas, and of that 28% is considered operable or operable but complex. Given the sensitivity of habitat to any action that alter waterflow, even actions that occur at a distance from the critical habitat itself, the FS has determined the proposed activity is likely to adversely affect designated critical habitat for this species, however we do not expect those effects to be widespread**.**

Effects Determination

Given that 1.8% of the species' range occurs in areas affected by the proposed action, the FS has determined that exposure is high. Given the susceptibility of the subspecies to activities that modify waterflow to riparian areas, the FS has determined the proposed activity is likely to adversely affect this species. Given the sensitivity of habitat to any action that alter waterflow, even actions that occur at a distance from the critical habitat itself, the FS has determined the proposed activity is likely to adversely affect designated critical habitat for this species, however we do not expect those effects to be widespread**.**

Preble's meadow jumping mouse

The Preble's meadow jumping mouse (*Zapus hudsonius preblei*) ("Preble's mouse") was listed as threatened under the Endangered Species Act in 1998 (63 FR 26517). In 2001, a rule was published to clarify some circumstances when "take" might be acceptable (e.g., certain agricultural practices, rodent control, structure modifications, and water rights situations, 66 FR 28125). Critical habitat was finalized in 2010 (75 FR 78430), and current conservation actions are informed by a Recovery Plan (USFWS 2018b).

Current population trends are uncertain in Wyoming, and at the time the Recovery Plan was written, it was not clear whether existing survey data were adequate to determine mouse occupancy and distribution in Wyoming (USFWS 2018b). However, the Recovery Plan describes clear population targets within two recovery units and outlines future research and information that would support delisting efforts (USFWS 2018b). The primary threat to the Preble's mouse is habitat loss and degradation resulting from residential development, commercial development, and agricultural conversion (63 FR 26517).

Exposure to the Proposed Action Area and Proposed Actions

The Preble's mouse inhabits densely vegetated riparian areas of the high plains spanning southeastern Wyoming to south-central Colorado (U.S. Fish and Wildlife Service 2018). This mouse depends on wet grasslands, riparian shrublands, and near-stream habitats (63 FR 26517), inhabiting a relatively restricted area of the landscape that is very closely tied to riparian areas (USFWS 2018b). Areas used by the Preble's mouse usually have multi-layer canopies of woodland and herbaceous vegetation and high volumes of litter that the mice can use for burrowing and nesting (63 FR 26517). These mice have also been found in irrigation ditches, road ditches, and wetlands near highways, and it is possible that these types of adjacent habitats provide individuals with dispersal opportunities (63 FR 26517).

At the time of listing, the Preble's mouse inhabited two counties in Wyoming and seven counties in Colorado, but historical records show that its range was previously broader in both states (63 FR 26517). Critical habitat was designated in Colorado in in 2010 (75 FR 78430), but no critical habitat has been designated in Wyoming (USFWS 2018b). The designated critical habitat encompasses 34,935 acres including 411 miles of rivers and streams in seven Colorado counties (75 FR 78430), including on the Arapaho-Roosevelt and the Pike San Isabel National Forests (75 FR 78430).

The Preble's mouse digs to create nests belowground, is mostly crepuscular or nocturnal, and hibernates in riparian areas from October through May (63 FR 26517, USFWS 2018b). Their specific diet is uncertain, but it likely eats seeds, fruits, fungus, and insects (63 FR 26517). The Preble's mouse may need an open water source for drinking which could explain its close ties to streams and rivers, and at least one study has indicated that trapping success decreased after a seasonal stream stopped flowing (63 FR 26517).

There are only nine observations of the species in the NRM and GBIF databases in Wyoming, making these data insufficient to draw conclusions about the relative exposure of the species to the proposed action. We ran a supplementary analysis overlaying the USFWS range of the

species with inventoried roadless areas affected by the proposed action. The overlap is less than 0.005%.

Effects of Roads

The effects of roads on the Preble's mouse are likely negative. The initial listing decision stated, "Roads, trails, or other linear development through Preble's [mouse] habitat may act as barriers to movement," and emphasized that maintaining dispersal corridors between habitat patches is likely important (63 FR 26517). The Recovery Plan also noted that vehicle accidents in riparian areas can pose a risk of fuel and chemical spills that could contaminate important areas of riparian habitat (USFWS 2018b).

Effects of Timber/ Vegetation Management

The effects of timber production on the Preble's mouse would likely be negative if they were to occur within areas surrounding important habitat areas. There is at least one example of commercial timber production being restricted from an area to protect Preble's mouse habitat (75 FR 78430), but timber production is not listed as a concern in the Recovery Plan (USFWS 2018b). However, given that the Preble's mouse uses areas with multi-layered vegetation including trees, changes to vegetation structure and ground disturbance in riparian areas are expected to negatively impact Preble's mouse habitat.

Existing Protections

The proposed action overlaps with the Medicine Bow National Forest, which contains standards and guidelines specific to the Preble's meadow jumping mouse (2003 land management plan, available at USFS 2025a). These include guidelines specific to road construction, such as "In suitable habitat within the range of the Preble's meadow jumping mouse, avoid placing new recreation sites, trails or roads within the riparian zone. Existing roads in designated critical habitat will be reviewed for possible closure or Relocation" (2003 land management plan, available at USFS 2025a). There are also guidelines about the way that vegetation management actions, such as prescribed fire, can be implemented in the species' habitat and for surveys for listed species during project planning.

Critical Habitat Determination

There is no critical habitat in the proposed action area; therefore, the FS has determined there would be No Effect.

Effects Determination

Given that less than 0.005% of the species' range that intersects with IRAs in the proposed action area, the exposure of the species to the action is low. While there are some potential impacts of roads and timber on this species and its habitat, the Medicine Bow National Forest plan has standards and guidelines that shape how these activities can occur. Given this, the FS has determined the proposed activity is not likely to adversely affect this species.

Carnivores

Gray wolf

The gray wolf was first listed throughout the contiguous United States under the Endangered Species Preservation Act of 1969 (38 FR 14678), a predecessor to the ESA. The listing status of gray wolves in the U.S. is complex, with numerous revisions to listable units and their listing status (see Ward 2020 for a review of wolf regulatory history in the U.S.). In February 2024, FWS published a 12-month finding that concluded (1) the Northern Rocky Mountain DPS no longer constitutes a valid listable entity, and (2) that the Western United States is a valid DPS but does not meet the definition of threatened or endangered (89 FR 8391). In August 2025, the finding was partially vacated by a federal judge and FWS was ordered to reconsider the relisting petition for the delisted Northern Rocky Mountain region (NRMR). Currently, gray wolves in the United States (excepting the Mexican wolf subspecies, considered separately below) have one of the four listing statuses depending on geographic region:

1. Delisted due to recovery in the Northern Rocky Mountain region (Idaho, Montana, Wyoming, eastern Washington and Oregon, and a small portion of north-central Utah (74 FR 15123)
2. Nonessential and experimental in Colorado (88 FR 77014)
3. Threatened in Minnesota (43 FR 9607)
4. Endangered in the remainder contiguous states, sometimes referred to as the "44-entity"

Wolves in the western U.S. have generally increased in abundance and expanded in distribution since their reintroduction in the 1990s (USFWS 2023c). According to state wildlife agencies, the per state abundance estimates for delisted populations in 2022 were as follows: Idaho (958), Montana (1,087), Wyoming (338), Washington (inside the NRMR = 159; outside the NRMR 57), and Oregon (inside the NRMR = 140, outside the NRMR 38), and California (18). In the winter of 2021, approximately 2,700 wolves were estimated to be present in the state of Minnesota (MNDNR 2022) and 1,126 in Wisconsin (WIDNR 2023). Abundance estimates are collected by state agencies using varying methods with different biases and degrees of precision (USFWS 2023c).

Connectivity is considered important to the ability of wolves to recolonize vacant but suitable habitat and maintain demographic resilience. Excepting the Mexican wolf subspecies and the isolated population in Isle Royale National Park, North American gray wolves generally exhibit high levels of genetic diversity and low levels of population differentiation (USFWS 2023c). Range-wide genetic analyses support relatively high rates of effective dispersal and gene flow, suggesting metapopulation dynamics are being maintained under current conditions (USFWS 2023c).

The FWS deems the primary stressor to wolf populations as human-caused mortality, which includes regulated harvest in the NRMR, illegal take outside of the NRMR, and lethal control of wolves depredating livestock (USFWS 2023c). Human-caused mortality was the main factor

responsible for decline that led to initial listing and accounts for 60 to 80 percent of total mortalities in the coterminous U.S now (USFWS 2023c, Hill et al. 2022, Morales-González et al. 2025). Determination of how much human-caused mortality can be tolerated without affecting population viability is complex and long-contested in the scientific literature, depending not only on the total mortality rate but also the degree to which mortality is compensatory or additive in nature (see USFWS 2023c for literature review of compensatory dynamics). Other potential stressors to population resiliency include episodic disease outbreaks, inbreeding depression, disease in prey species, and other sources of habitat modification, such as human development and wildfire (USFWS 2023c).

Exposure to the Proposed Action Area and Proposed Actions

Gray wolves are habitat generalists that successfully occupy a wide range of environments. The two principal factors limiting their distribution in the U.S. are density of large ungulate prey and human-caused mortality (USFWS 2023c). Accordingly, gray wolf presence is positively correlated with Federal land ownership and forest cover and is negatively correlated with the density of roads and humans (Mladenoff et al. 1995, Carroll et al. 2006, Oakleaf et al. 2006). National forests accounted for approximately 52% of the occupied range for wolves of the western U.S. in 2023 (USFWS 2023c). Large areas of modeled wolf habitat exist that are currently unoccupied or in early stages of recolonization exist that could theoretically support wolves and contribute to range-wide recovery (Carroll et al. 2001, Carroll et al. 2003, Mladenoff and Sickley 1998, van den Bosch et al. 2022). Critical habitat was designated in Minnesota and Isle Royale, Michigan in 1978 (43 FR 9607) but no critical habitat has been designated in the western states. Approximately 1% of the Minnesota critical habitat overlaps with potentially affected IRAs, and of that approximately 89% is in areas classified as operable or operable but complex. Of gray wolf occurrence data in the NRM and GBIF databases, 571 occurrence records occur in IRAs, 59% of them in regions that are operable or operable but complex. Since there are both threatened and endangered wolves, which are not differentiated in the occurrence analysis, we ran analyses overlaying the threatened and endangered ranges with inventoried roadless areas affected by the proposed action. Approximately 0.002% of the Minnesota threatened gray wolf range is affected by the proposed action, whereas approximately 4.2% of the range of the “44-entity”(the endangered population) is affected by the proposed action.

Effects of Roads

Roads have mixed effects on gray wolves at the individual level. Roads negatively affect wolves by increasing mortality and degrading habitat quality and cause positive effects by increasing hunting efficiency for ungulate prey. At the population level, roads have the potential to inhibit population growth and recolonization of new areas.

Mortality

Roads expose wolves to human-caused mortality directly through vehicle collisions and indirectly by increasing risk of illegal killings due to more frequent human encounters.

Vehicle strikes contribute to wolf mortality but rank well below other sources of anthropogenic mortality, such as lethal control and illegal killings (Tables S2,S3; Morales-González et al. 2025). In a study of 711 collared wolves (1982-2004) in the NRMR prior to de-listing, 18 wolves

(2.5% of all wolves, 5.0% of known mortalities) died from vehicle strikes (Murray et al. 2010). A study of wolves across four National Parks found that vehicle strikes accounted for 5.9% of total mortalities and 9% of human-caused mortalities (Cassidy et al. 2023). In Wisconsin, between 1979 and 2012, vehicle strikes were estimated to account for 13% of mortalities (Treves et al. 2017a). A meta-analysis of mortality data from studies across Europe and North America found that mortalities increase as the density of roads increase (Morales-González et al. 2025). However, none of these mortality studies distinguished between strikes that occurred on paved, high traffic roads and unpaved, secondary or forest roads.

Illegal killing is one of the highest sources of mortality for gray wolves and has the potential to significantly alter population dynamics. In the NRMR prior to delisting, 24% of known mortalities were attributed to illegal killings (Smith et al. 2010). Other studies have found illegal killings accounted for between 5 and 12 percent of the known wolf population in the NRM (Smith et al. 2010, Murray et al. 2010, Ausband et al. 2017, Stenglein et al. 2018). The true rate of illegal killing is almost certainly higher, after also accounting for collared wolves with unknown fates or unknown causes of mortality (Treves et al. 2017b, Liberg et al. 2012, Liberg et al. 2020).

Roads increase susceptibility to illegal killings by acting as points of access for humans that increase their encounter rates with wolves. Multiple studies have found lower rates of illegal killings in areas associated with low road density and low human presence (Fritts et al. 2003, Smith et al. 2010, O'Neil et al. 2017, Simpson et al. 2023) but generally have been challenged to tease apart road access from other confounding factors, such as proximity to livestock, private land, and human settlements. However, a wolf mortality study centered on the Superior National Forest from 1968 to 2018 found that wolves in wilderness areas had fewer mortalities from illegal killings than non-wilderness wolves, providing strong evidence that increased human access from trails and roads was responsible for the difference (Barber-Meyer et al. 2021). In contrast, a meta-analysis of mortality patterns from 21 studies in North America (12 in the conterminous U.S.) found no significant effect of road density on mortality due to illegal killing (Hill et al. 2022). Authors were surprised and suggested this may in part be explained by varying conditions and context-specific effects, such as differing spatial scales and regional densities of roads. As an example of the latter, a study of wolves on the Tongass National Forest found that the rate of wolf harvest (in this case, pooled legal and illegal mortality) increased with density of roads when overall road density was low (<0.9 km/km2) but had no additional effect on harvest rates when road density was higher (Person and Russel 2008).

Displacement and Inhibition of Recolonization

Wolves can clearly tolerate living in areas with high road densities and dense human settlement, as evidenced by recolonization of densely populated regions in Europe. Less understood is the degree that long-term occupancy in areas with higher rates of human encounters is sustained by inputs from remote or largely inaccessible regions. Like designated wilderness, roadless areas may act as refugia from human mortality, helping stabilize demography in areas with more development and foster recolonization of suitable but vacant habitat (Mech 1989, Haight et al. 1998, Mladenoff et al. 1999; Person and Russel 2008, Barber-Meyer et al. 2021).

Most studies conducted at broader spatial scales (e.g., home range and larger) agree that wolf occupancy is negatively correlated with road density (Thiel 1985, Mech et al. 1988, Mladenoff et al. 1995). Construction of new roads could negatively affect gray wolf recovery by slowing or inhibiting recolonization of vacant but suitable areas. A global meta-analysis of data from throughout Eurasia and North America found that reproductive dens and rendezvous sites are located significantly farther from roads than random points (Saztornil et al. 2016). However, this meta-analysis lumped forest roads with highways and other paved roads; when only main roads were considered, the avoidance was stronger, suggesting paved roads were driving much of the global signal. Telemetry studies of recolonizing wolves have shown that new territories are preferentially established in areas with low road density, but that affinity for low road density relaxes as wolves proliferate. For example, analysis of territory locations for 56 recolonizing packs in Idaho, Montana, and Wyoming in the early 2000s found that both overall territories as well as core use areas had a significantly lower density of 2-wheel and 4-wheel drive roads relative to the surrounding environment (Oakleaf et al. 2006). Similarly, in northern Wisconsin and upper Michigan, a telemetry study of 14 packs found that territories mostly consisted of areas with a density of paved and unpaved roads <0.45 km/km2 and no portion of any territory area had a road density >1.0 km/km2 (Mladenoff et al. 1995). Habitat models parameterized with this relationship correctly predicted the location of early re-colonizing packs in Wisconsin (18 of 23; Mladenoff et al. 1999) but failed to predict pack locations in later years when wolves occupied a broader range of habitat type (Mech 2006). Additional analyses encompassing the first 28 years of re-establishment in Wisconsin showed that wolves in marginal habitat—defined as regions with higher road density, lower amounts of federal land, and higher proportions of agricultural land—had lower reproductive output and higher human-caused mortality than habitat with low road density, high proportions of federal land, and low proportions of agriculture (Simpson et al. 2023). The study, however, did not distinguish the effects of roads relative to land ownership and human use categories.

Habitat degradation

Wolves, like all wildlife, presumably select for habitat types and landscapes that maximize their fitness. In theory, evaluating how wolves select or avoid roads offers insight into how the presence of roads affects habitat quality for wolves. In practice, evidence shows that the behavioral response of gray wolves to roads is complex and highly contextual, depending on attributes of the road, its surrounding environment, the individual wolf, time of day and year, and the physical scale of inference.

Studies generally agree that roads and other linear features facilitate wolf movements and allow wolves to travel at faster speeds, but the interpretation of this response has varied by context (McKenzie et al. 2012, Zimmerman et al. 2014, Dickie et al. 2017, Dickie et al. 2020). In the boreal forests of Canada, where there is a dense network of logging roads and seismic lines, greater speed allows for longer distances traveled per day (McKenzie et al. 2012, Zimmerman et al. 2014, Dickie et al. 2017, Dickie et al. 2020), which in turn increases search and encounter rates for ungulate prey (Whittington et al. 2011). In this scenario unpaved forest roads may confer a slight fitness advantage to wolves. Yet in areas with a more extensive human footprint, such as near towns or in areas with significant recreation, faster travel has instead been interpreted as a response to increased risk from humans (Dickie et al. 2017, Whittington et al.

2022). In this case, the loss of important "slow state" movements such as resting, foraging, and reproduction, may represent a fitness loss and degradation of habitat quality.

Many studies have demonstrated that wolves use roads frequently (e.g., St-Pierre et al. 2022, Dickie et al. 2017, Dickie et al. 2020, Zimmerman et al. 2014, Muhly et al. 2019). However, the direction and strength of selection for roads within a home range varies widely depending on road characteristics. First, wolves vary their response to roads based on the amount of human use of a given road (Thurber et al. 1994, Bojarska et al. 2020, Dennehy et al. 2021) as well as of the broader surrounding region (Hebblewhite and Merrill 2008). For example, a 3-year study in Alberta used remote cameras to show that use of temporary and gravel roads by wolves increased with low rates of human traffic (<5 vehicles/day) but declined at higher traffic rates (Keim et al. 2019). A study in the vicinity of Kenai National Wildlife Refuge similarly found wolves used secondary roads more than expected by chance but avoided oilfield access roads that received regular traffic and were open year-round (Thurber et al. 1994). Other studies have found wolves temporally adjust their use of roads to avoid peak periods of human activity (Anton et al. 2020, Bojarska et al. 2020, Dennehy et al. 2021).

The behavioral response to roads differs depending on the surrounding road density, with the nature of this functional response also varying by study. Analysis of GPS-locations in Quebec found wolves strongly selected for forest roads in territories where roads occurred in low density, but the strength of positive selection declined as road density increased. (Houle et al. 2010). In contrast, breeding wolves in Scandinavia showed the opposite pattern, in which wolves used gravel roads more often as road density within their home range increased (Zimmerman et al. 2014). In a large study of 172 GPS-collared wolves across multiple Canadian provinces, wolves selected for areas within their home range that had high road density in provinces where the average regional road density was high (for example, >0.075 km/km^2), but preferentially chose sites with low road density in provinces where the average regional road density is low (Muhly et al. 2019). Another analysis of 20-years of GPS data from wolves collared in and around Banff National Park found that wolves negatively selected for roads in all contexts, but the avoidance was greatest when road densities were high (Whittington et al. 2022).

Finally, some studies have found no uniform response at the population level, but rather a wide range of individual- or pack-specific responses (Gregovich et al. 2025, Ferreiro-Arias et al. 2024, Anton et al. 2020, Zimmermann et al. 2014). Such lack of uniformity may reflect a wide degree of behavioral plasticity or, alternatively, could be a manifestation of roads exerting opposing effects in different contexts. Regardless, the range of variation observed within and across studies suggests behavioral effects of road construction is likely to be mixed and/or location specific.

Timber/ Vegetation Management

The effects of timber harvest on wolves are predominantly mediated through its influence on ungulate prey and, to a lesser extent, on the human risk landscape. Logging creates early seral forests that provide high-quality forage that attracts deer, elk, and moose and concentrates their distribution (McKay and Finnegan 2023). Simultaneously, the removal of overcanopy can increase sight lines, which along with logging roads, may facilitate prey searching efficiency for

wolves (Boucher et al. 2022). Accordingly, many studies in boreal coniferous forests found that wolves select for early regenerating cutblocks, especially those logged within <5 years (Houle et al. 2010, Muhly et al. 2019, Boucher et al. 2022, Johnson-Bice et al. 2023). Other studies have found either no significant habitat selection for regenerating timber harvested areas (Gregovich et al. 2025, Kuzyk et al. 2004) or mixed responses that were mediated by proximity to other types of human disturbance and activity (Lesmerises et al. 2012, Boczulak et al. 2023, Person and Russell 2008). We note that most studies were conducted in the context of large clearcuts in Alaskan or Canadian boreal forest, with considerably less information available on the effects on wolves of other timber harvest practices.

Existing Protections

Gray wolves in the United States are protected through a combination of federal, state, Tribal, and land management measures. State regulations for wolf management is varied across the range. For example, in Minnesota, wolves are managed under a state plan that prohibits recreational hunting and trapping, while allowing limited depredation control to protect livestock. In the Northern Rocky Mountains distinct population segment (Idaho, Montana, Wyoming), wolves are managed under state authority following delisting and harvest is permitted subject to established state quotas, but conservation safeguards remain in place such as conflict mitigation strategies such as range riders and carcass removal to reduce livestock conflicts. Conversely, outside of the delisted Northern Rockies DPS in coastal states state regulations emphasize wolf conservation. For example, in Washington state wolves are listed as endangered under state law, and the Washington Department of Fish and Wildlife prohibits hunting while implementing a Wolf Conservation and Management Plan that emphasizes recovery zones and nonlethal conflict deterrence. Tribal governments exercise sovereign authority on Tribal lands, such as the Confederated Salish and Kootenai Tribes in Montana, which prohibit wolf hunting on Tribal lands and contribute to regional monitoring. On federal lands, project level decisions must consider wolf impacts under ESA consultation, while state and private lands employ conflict response programs, compensation for verified depredations, and nonlethal deterrents. On Forest Service land, numerous individual forest Land Management Plans also provide additional guidance and protections for gray wolves and their habitat. Collectively, these overlapping protections — ranging from ESA take prohibitions to state population floors and Tribal hunting bans — aim to maintain sustainable wolf populations, reduce human–wildlife conflict, and prevent the need for future federal relisting.

Critical Habitat

One percent (1%) of critical habitat is in inventoried roadless areas and of that 89% is in areas that are operable or operable but complex. There are no PBFs for gray wolves; however, given the generalist nature of gray wolves and the primary threat on NFS lands being human conflicts, we determined the proposed activity is not likely to adversely affect designated critical habitat for this species.

Effects Determination

A high proportion of gray wolf occurrence records in the NRM and GBIF databases occur in inventoried roadless areas that are operable or operable but complex. However, the exposure of

the wolf range to the proposed action differs between the threatened and endangered entities. Due to this difference, the FS has determined that the exposure of the threatened entity to the proposed action is low, whereas the exposure of the endangered entity to the proposed action is high. Despite this difference, there are risks posed to the species by human conflict with increased human access, so the FS has determined the proposed activity is likely to adversely affect both the threatened and endangered entities for this species.

Mexican Wolf

The Mexican wolf is the southernmost subspecies of gray wolf in North America. It was originally listed in 1976 (41 FR 17736), with reestablishment in the wild in the U.S. in 1998. In 2015 FWS issued a rule to list the Mexican wolf as endangered separately from the gray wolf listing (80 FR 2488). Mexican wolves exist in the wild in the United States in a single introduced population in Arizona and New Mexico south of Interstate 40, designated as the Mexican Wolf Experimental Population Area (MWEPA). Mexican wolves in the MWEPA are accordingly managed as a nonessential experimental population (80 FR 2512, 87 FR 39348) whereas outside the MWEPA they are considered endangered. Mexican wolf conservation strategy is guided by the 2022 Mexican Wolf Recovery Plan, Second Revision (USFWS 2022c).

The experimental population of Mexican wolves is small but steadily growing (USFWS 2017b, USFWS 2024k). In 2023, there were 257 individuals inside the MWEPA (USFWS 2024k). All Mexican wolves today descend from only seven captive founders, resulting in high mean kinship and low genetic diversity, particularly in the wild population where managers have less control over breeding.

The most important biological threats to recovery are 1) excessive human-caused mortality, 2) demographic stochasticity associated with small population size, and 3) loss of genetic diversity and risk of inbreeding depression (USFWS 2017b).

Exposure to the Proposed Action Area and the Proposed Actions

Highly suitable habitat for Mexican wolves is forested, montane terrain with adequate densities of ungulate prey (elk, white-tailed deer, mule deer) and low density of livestock and human (USFWS 2017b, Martinez-Meyer et al. 2021). National forests make up only 15.5% of the total MWEPA but comprise a disproportionate share of suitable forested habitat: approximately 70% of forest vegetation types in the MWEPA fall on national forest lands (USFWS 2017b, Martinez-Meyer et al. 2021). The Service identifies National Forest lands as the most suitable lands for reintroduction and recovery of the Mexican wolf, predominantly due to lower levels of human development and disturbance.

As an experimental population, Mexican wolves do not have designated critical habitat. There are only two records of the species on USFS lands in the NRM and GBIF databases, neither of which are in potentially affected IRAs. Given this low sample size, no meaningful conclusions about the exposure of this species to the action can be drawn based on these data. We conducted a supplemental analysis overlaying the USFWS ranges of the endangered and experimental populations with inventoried roadless areas affected by the proposed action. The overlap for the

endangered population was approximately 1.8% and the overlap for the experimental population was approximately 5.1%.

Effects of Roads

The individual effects of roads are expected to be similar to other gray wolves (see above) but carry greater consequences at the level of the population. Low road density is considered a key component of suitable Mexican wolf habitat (USFWS 2017b, Martinez-Meyer et al. 2021). Roads are associated with increased human-caused mortality, either directly from vehicle strikes or indirectly because they act as access points for humans (see gray wolf section). Monitoring data and population viability modeling indicate that adult mortality is a major factor driving population dynamics in the MWEPA, with the potential to slow population growth or cause declines in years when adult mortality is high (USFWS 2017b, Miller 2017, USFWS 2024k). Like other gray wolves, the biggest source of mortality for the Mexican wolf mortality is human caused, predominantly by illegal killings and vehicle collisions. Of 133 mortalities documented between 1998 to 2016, 77 (58%) were attributed to illegal killings and 16 were caused by vehicle strikes (12%) (USFWS 2017b). Human-caused mortalities in the 5 years following 2017 were stable or slightly declining (USFWS 2024k), suggesting management actions to minimize human mortalities outlined in the recovery plan have been effective (USFWS 2024k).

Effects of Timber/ Vegetation Management

Individual effects of logging are expected to be mixed and highly similar to other gray wolves (see above). Effects that increase ungulate abundance and concentrate their distribution are likely to be positive, whereas effects that increase exposure to human encounters are likely to be negative.

Existing Protections

There are some protections for the Mexican wolf in some existing forest plans. The Apache-Sitgreaves National Forest plan references its commitment to the ESA and a memorandum of understanding for conservation, recovery and persistence of the species (Apache-Sitgreaves, 2015). The Cibola National Forest, where the wolf is known to forage but not den, states that it will work with the USFWS to aid recovery for listed species like the Mexican wolf in both the experimental and non-experimental range (2022 Cibola NF land management plan, available at USFS 2025a). There are no mentions of the Mexican wolf in the Coconino National Forest Plan (2018) or Kaibab National Forest Plan (2014) or Gila National Forest Plan (1986; available at USFS 2025a.

Effects Determination

Given the overlap of both population's ranges with inventoried roadless areas, we have assigned an exposure level of high to both experimental and endangered populations. Given the potential for roads to bring humans and Mexican wolves into greater conflict, the FS has determined the proposed activity is likely to adversely affect Mexican wolves. This determination is made for both experimental and threatened populations.

North American wolverine

The North American wolverine (*Gulo gulo luscus*) was listed as threatened in 2023 (Federal Register 83726) and a recovery plan outline was published that same year. The true population size of wolverines in the contiguous United States is not known, but estimates based on habitat modeling place it at approximately 300 individuals (Inman et al. 2013). Trends in this population are also unknown, although work in a wolverine population in southwestern Canada suggests a decline of 41% between 2011 and 2020 (Barrueto et al. 2022). The impacts of this decline on wolverines in the contiguous United States are not known (USFWS 2023d), although the 2023 wolverine listing decision notes that "connectivity with Canada is essential to the long-term viability of North American wolverines in the contiguous United States" citing Cegelski et al. 2006.

Exposure to the Proposed Action Area and Proposed Action

Wolverines in the contiguous United States inhabit small, isolated high-altitude habitats separated from each other by low valleys of less hospitable terrain. Wolverines are known to reproduce in the Northern Rockies of Idaho, Montana, and Wyoming and the Cascade Mountains of Washington (88 FR 83726). Dispersing individuals have been detected in California, Oregon, Utah and Colorado (88 FR 83726) and North Dakota (Rogers 2016). Federal land management decisions are important for wolverine populations in the contiguous United States. Analyses estimate that 96% of wolverine primary habitat identified in Inman et al., 2013 is owned or managed by federal agencies (USFWS 2018c). Critical habitat has not been proposed for this species. Approximately 41% of primary wolverine habitat in the contiguous U.S. is managed as wilderness (USFWS 2018c) and is unaffected by the proposed action.

Of the 137 wolverine detections in inventoried roadless areas, 69 of them were in operable areas or in areas that were classified as operable, but complex in the NRM and GBIF databases. This analysis places the wolverine in the highest vulnerability category to the proposed action. This is surprising given that the 2023 listing decision notes that most wolverine reproductive habitat in the contiguous U.S. is rugged and not appropriate for timber harvest. To evaluate whether this analysis was biased due to opportunistic sampling or sampling close to existing infrastructure, we evaluated the proportion of wolverine habitat in the current USFWS range map for the species (USFWS 2026a) that intersected with roadless areas. We found that approximately 10.2% of the range intersects with an inventoried roadless area affected by the proposed action. Given that the spatial extent of the range includes areas that are not wolverine habitat, we would expect that the impact to actual habitat to be higher.

Effects of Roads

The 2023 listing decision for wolverines determined that, in isolation, roads could have negative impacts on individuals, but in combination with other stressors, roads and high road densities could negatively impact population "resilience, distribution and gene flow" (Federal Register 83726). Previous research has found that roads are a mortality risk for wolverines (summarized in USFWS 2018c; Scrafford et al. 2024). Between 2018 and 2023, the USFWS became aware of 4 wolverine mortalities near roads in the contiguous United States (USFWS 2023d)

encompassing both interstates and smaller state roads. Mortalities may be biased towards young dispersing males (USFWS 2023d).

Even though roads are not an absolute barrier to wolverine movement, roads can lead to behavioral avoidance and changes in habitat usage by wolverines. Work by Scrafford et al. 2018 in Alberta, suggests that in summer wolverines "avoid roads, increase speed near roads, and increase speed more near roads with greater traffic volumes." Several studies have suggested that there is a negative relationship between roads and wolverine distribution or occupancy, although the mechanism behind this relationship is unknown (Rowland et al. 2003, Bowman et al. 2010; Kortello et al. 2019; Mowat et al. 2020; Chow-Fraser et al. 2022; Schepens et al. 2023). Degradation of wolverine habitat quality can occur with the presence of roads even when they have low traffic volume (Scrafford et al. 2018). Wolverine den sites may be located away from roads, but it is unclear whether this is due to available denning habitat or avoidance of roads (USFWS 2023d). Hausleitner et al. 2024 reviewed the literature and found that the distance between recorded wolverine dens and roads varied across studies with some reporting distances as low as 1km from smaller industrial roads or as high as 17 km from major highways.

Overall wolverines are characterized as having "low" tolerance of human disturbance (88 FR 83726) and the construction of new roads may affect wolverines indirectly by increasing access by human recreationists. Wolverines appear to be sensitive to both motorized and non-motorized human recreation (Heinemeyer et al. 2019, Barrueto et al. 2022; Kortello et al. 2019). In some instances, these interactions may be mediated by roads, as one study found that wolverine distribution is negatively correlated with forest roads used by winter snowmachines (Kortello et al. 2019). Females may be especially sensitive to recreational disturbance (Heinemeyer et al. 2019). While lethal trapping of wolverine is prohibited in the contiguous United States, increased road access may affect incidental bycatch of wolverine from trappers legally trapping wolves in Montana (USFWS 2023d), although the likelihood of this is unknown. From 2012-2023, 10 wolverines have been trapped in Montana incidentally and reported, with 3 events resulting in mortality (88 FR 83726).

Increased access by humans may also lead to increased risk of human-caused wildfires. While wildfires are considered a low threat to wolverines (USFWS 2018c; 88 FR 83726), they are projected to increase in the western United States including in wolverine habitat (USFWS 2018c). In addition to the risk of injury or mortality from fire, wolverines may be displaced by wildfire and the habitat modifications that result (88 FR 83726). A study in Idaho found at least occasional use of recently burned areas, and in one instance, even an area still smoldering from fire, by wolverine (Copeland 1996). Overall, however, the literature on this relationship is sparse. In the 2018 SSA, authors indicated that they were not aware of any studies investigating the direct effects of fire on wolverines and the recent listing decision noted that indirect effects of wildfire on wolverines are not well-characterized (USFWS 2018c; 88 FR 83726).

Changes in the density of linear features like roads could also affect wolverines by increasing interactions with predators and competitors. Linear features like roads may affect predation of wolverines by wolves (Scrafford et al. 2024). Additionally, increases in human infrastructure like roads may facilitate use of previously occupied wolverine habitat by coyotes (Chow-Fraser et al. 2022), which authors postulate could displace wolverines and cause range contraction.

As a result of these impacts, the 2023 listing decision notes that wolverine may benefit from "limitations on development and road construction" and" limitations on road densities in certain areas" (88 FR 83726). Multiple federal listing documents have determined that "the fragmented nature and distribution of wolverine habitat in the contiguous United States results in a population that is highly vulnerable to extirpation because of lack of connectivity between subpopulations, and this also makes them more vulnerable to external threats" (75 FR 78030; December 14, 2010 cited in 2023 listing decision). Fragmentation of wolverine habitat by major highways has already been shown to influence genetic connectivity, with gene flow being especially restricted in females (Sawaya et al. 2019). A broad scale analysis of wolverine connectivity also found that human disturbance negatively influenced genetic connectivity (Day et al. 2024).

Effects of Timber/ Vegetation Management

Several studies in North America suggest that wolverines avoid timber harvested areas. Given the "low" tolerance of wolverines to human disturbance (88 FR 83726) avoidance could occur during the act of timber harvesting, although one study did find usage of timber areas during active logging (Scrafford et al. 2017). The long-term relationship between wolverines and landscapes altered by timber harvest may be nuanced and sensitive to seasonality and time since harvest. Bowman et al. 2010 found that wolverines in Ontario were associated with mature, but not logged, forest. Female wolverines in British Columbia avoided recently logged areas at some sites and during some seasons (Krebs et al. 2007). A study in Alberta found that both males and females avoided the interior of cutblocks aged 11-25 years but used cutblock edges in a sex and season-dependent manner (Scrafford et al. 2017). Franklin et al. 2019 suggest that harvest regimes could affect wolverine habitat usage. They found that logged areas (15-18 yr post-harvest) with higher levels of stand retention were used by wolverines more than areas with lower levels of stand retention. Even when wolverines use timber harvested areas, their behavior may be modified by reduced cover. A study in Montana found that wolverines crossed cutblocks, but made directed linear movements at a quick pace when traveling within them (Hornocker and Hash 1981), unlike the slower movements associated with areas without harvest. Recently work by Day et al. 2024 suggested that forest cover is associated with wolverine connectivity. Incidental take of wolverine during vegetation management activities to reduce the risk or severity of wildfire was expressly exempted from the section 9 prohibitions in the wolverine listing determination (88 FR 83726) because these activities were determined to not result in jeopardizing the continued existence of the species.

Existing Protections

Some Forest Plans contain management direction for wolverine and/ or roadless areas. For example, the Flathead National Forest land management plan (2018) identifies specific areas within the forest where the desired condition is that it provides connectivity for wide ranging species like wolverine (available at USFS 2025a). Some inventoried roadless areas (e.g. Bear-Marshall-Scapegoat-Swan Inventoried Roadless Area #1485) in the Flathead National Forest are identified as "Recommended Wilderness" and maternal denning habitat for wolverine and connectivity are listed as one component for why these lands should be considered for Wilderness designation. A forest-wide objective for the Beaverhead-Deerlodge is to "maintain...

winter habitat integrity for wolverine." A number of additional management areas within the Beaverhead-Deerlodge National Forest have the stated management objective to "maintain or enhance roadless characteristics and naturalness" (2009 land management plan, available at USFS 2025a). The 2015 Kootenai-National land management plan commits to maintaining 16% of IRAs as having "wilderness potential" and that these areas of recommended wilderness contribute "habitats for species with large home ranges such as wide-ranging carnivores... and species found primarily in these habitats" as well as conditions that "contribute to wildlife movement within and across the Forest" (available at USFS 2025a). While this is a small sample of applicable forest plans, this illustrates that at least some plans provide protection beyond the federal designation of IRAs. However, these examples also illustrate that there is tremendous variety in what that protection can look like.

Effects Determination

Based on the overlap of the wolverine range with the proposed action area, we have determined that the exposure is high. Given that wolverines have high potential to be exposed to the proposed action and, that if exposed, individuals may exhibit a number of significant responses, the FS has determined the proposed activity is likely to adversely affect this species.

Ocelot

The ocelot was listed as endangered in 1972 under the authority of the Endangered Species Conservation Act of 1969 (37 FR 6476). Following passage of the ESA in 1973, the ocelot was included on the January 4, 1974, list of "Endangered Foreign Wildlife" that "grandfathered" species from the lists under the 1969 Endangered Species Conservation Act. Endangered status was extended to the U.S. portion of the ocelot's range for the first time with a final rule published July 21, 1982 (47 FR 31670).

Currently, the ocelot ranges from extreme southern Texas southern Arizona through the coastal lowlands of Mexico to Central America, Ecuador and northern Argentina. Currently there are two known breeding populations in southern Texas, located 30km (19 mi) from one another near the Gulf coast. Together, both populations may potentially total less than 100 individuals (Lombardi et al. 2022). These populations are isolated from populations in Mexico (Janecka et al. 2014). The quality and quantity of optimal habitat in Texas is on a downward trend, with the Lower Rio Grande Valley's agricultural lands rapidly developing (Wilkens et al. 2000) and much of the area that could be restored to suitable habitat occurs on private lands. Ocelot populations in Arizona are extremely rare, consisting of only a few individuals occasionally dispersing from Mexico, with perhaps two currently known in the state (as of late 2024), but no established breeding population.

Due to its small population, size ocelots are vulnerable to demographic and genetic risks that elevate extinction risk. Demographically, isolated populations are more sensitive to environmental variability and random events, making them less resilient to disturbance. Barriers to connectivity further exacerbate these risks by preventing dispersal, which would otherwise supplement declining populations and allow recolonization of vacant habitat. Consequently, preservation and restoration of habitat connectivity is of critical importance for ocelot recovery.

The vast majority of the U.S. ocelot range is not within National Forest System lands. The remaining U.S. population is almost exclusively confined to small, fragmented populations in southern Texas and some areas of Arizona, primarily on private lands and National Wildlife Refuges, with minimal overlap with National Forests (USFWS 2016a, USFWS 2025g).

Exposure to the Proposed Action Area and Proposed Actions

Critical habitat has not been designated for ocelot to date. In the United States, ocelots require dense thorn scrub and brushland habitats with high canopy cover, abundant understory vegetation, and structural complexity that provides concealment, denning sites, and connectivity between patches. These habitats are concentrated in South Texas and southeastern Arizona, where fragmentation from roads, agriculture, and development poses major threats. Most ocelot habitat in the United States occurs on private lands, but federal lands do contain important habitat areas. The Laguna Atascosa National Wildlife Refuge in Texas contains the largest block of federally protected thorn scrub, while Forest Service lands in the Coronado National Forest in Arizona offer smaller but strategically important habitat corridors that connect U.S. populations to Mexico. There are no designated roadless areas that overlap occupied ocelot habitat in Texas. In southern Arizona, roadless areas occur throughout potential ocelot habitat on the Coronado National Forest that may be important areas for movement and dispersal (USFWS 2016a). There are no observations of ocelot in inventoried roadless areas in the NRM and GBIF databases, making these data sources insufficient for assessing the relative exposure of the species to the proposed action.

Effects of Roads

Roads have been identified as one of the most significant ongoing threats to ocelot populations in the United States. Vehicle collisions are the leading cause of mortality for the small Texas populations, accounting for 45% of mortalities, in a study of 80 radio-tagged ocelots. Vehicle collisions have also been documented in Arizona and Mexico. In addition to direct mortality, roads fragment dense thornscrub and brushland, reducing habitat connectivity and limiting dispersal between patches. This isolation restricts gene flow, prevents recolonization of vacant habitat, and increases extinction risk for small populations. Additionally, some otherwise suitable habitats may not be occupied by ocelots or may serve as a population sink due to the risks posed by vehicles or avoidance of habitat adjacent to roads as has been reported in other species (USFWS 2016a, USFWS 2025g).

Effects of Timber/ Vegetation Management

Timber harvest poses a secondary but relevant threat to ocelot habitat in the United States. While large-scale commercial forestry does not occur within the thornscrub landscapes of South Texas, where ocelots are most abundant, timber-related activities in southeastern Arizona can reduce habitat quality and connectivity. Removal of canopy and understory vegetation diminishes the dense cover and structural complexity that ocelots require for concealment, denning, and hunting. Timber harvest and its associated infrastructure can further fragment habitat, creating openings that ocelots are reluctant to cross and increasing isolation of small populations. Both the 2016 Recovery Plan and the 2025 Status Review emphasize that maintaining dense woody cover—particularly thornscrub with canopy closure exceeding 75 percent—is critical for ocelot

persistence, and that timber operations which reduce this cover or increase fragmentation may undermine recovery objectives (USFWS 2016a, USFWS 2025g).

Existing Protections

Existing protections for ocelots in the United States are provided in the 2010 Conservation Plan for the Laguna Atascosa National Wildlife Refuge (USFWS 2010) in South Texas which provides the largest block of federally protected thornscrub. In Arizona, lands managed by the Coronado National Forest and other federal units contribute to connectivity with Mexican populations, with management guidance provided in the 2018 Coranado National Forest Land Management Plan (available at USFS 2025a).

In addition to federal protections, ocelots are safeguarded by state-level conservation frameworks. In Arizona and Texas, the species is recognized as a Species of Greatest Conservation Need, included in Arizona's *Comprehensive Wildlife Conservation Strategy: 2005–2015* (AGFD 2006) and the *Texas Conservation Action Plan* (TPWD 2012). These strategies provide policy guidance to agencies and the public but do not confer direct regulatory protection. In Texas, ocelots are legally protected under state law and formally listed as endangered (TPWD 2012).

Effects Determination

Given that Arizona is the only location where the ocelot could occur on inventoried roadless, we are assigning the exposure of the species to the proposed action as low. However, given that roads and timber harvest may degrade habitat that is important for ocelot movement the FS has determined the proposed activity is likely to adversely affect this species.

Canada Lynx

Canada lynx (*Lynx canadensis*) were listed as threatened in the contiguous United States in 2000 (65 FR 16053). The status of this species is currently evaluated for 6 geographic units. In the 2023 Species Status Assessment (USFWS 2023e), population sizes for each unit were estimated based on expert opinion and published estimates of carrying capacity as follows: Northeast (750 – 1,000), Midwest (100 – 200), Northern Rockies (200 – 300), Northern Cascades (30-35), Greater Yellowstone (0-10), and Southern Rockies (75-150). Based on the best available information at the time of the assessment, it was determined that there is no reliable evidence that the distribution or abundance of lynx is reduced from historical levels at the level of the DPS. The status of each of these 6 units is thought to be largely independent from the other units (USFWS 2023e). Connectivity of these units to Canada is important and immigration of lynx from Canada into the contiguous United States is thought to contribute to population persistence (McKelvey et al. 1999a; Schwartz et al. 2002), although the degree to which this is important is unknown (USFWS 2023e).

Exposure to the Proposed Action

Lynx in the contiguous United States inhabit the patchy southern edge of the boreal forest in areas with high densities of snowshoe hares (USFWS 2017a). This includes subalpine coniferous forest (Douglas-fir, Spruce-fir, and hemlock-fir) in western mountains, boreal forest in the Great Lakes, and mixed coniferous and deciduous forest and high tundra vegetation types in the east

(Aubry et al. 1999). There is evidence that, currently, lynx both occur and regularly reproduce within Colorado, Maine, Minnesota, Montana, and Washington (USFWS 2023e). While lynx have been detected in North Idaho, there is only one documented reproductive event in the state in recent decades (USFWS 2017a). There is also evidence suggesting recent lynx reproduction in New Hampshire (2023 SSA) and Vermont (USFWS 2017a).

While lynx have been detected recently in other states, there is no evidence to suggest contemporary resident breeding populations. While a number of lynx reintroduced into Colorado dispersed into northern New Mexico, they did not establish a resident breeding population (USFWS 2023e). Lynx dispersing from the Colorado reintroduction were also briefly detected in Nevada, Utah, Arizona, South Dakota, Nebraska and Kansas (Devineau et al. 2010). Validated records show lynx present in Wyoming as far back as the 1850s (Reeve et al. 1986) and lynx reproduction was verified in Northwest Wyoming in the early 2000s, however more recent detections have been sporadic (USFWS 2023e). Some recent federal survey efforts in the state have failed to detect lynx (USFWS 2023e), although a lynx was photographed in the Wyoming Range in 2024 by a member of the public (Koshmrl 2024). Verified lynx detections have occurred in Michigan, and Wisconsin (USFWS 2023e). There was a reintroduction effort for lynx in the Adirondacks of New York between 1989-1991, however these animals have not been detected since 1993 and there was no evidence that they bred successfully (McKelvey et al. 1999a).

Federal land management decisions have the strong potential to influence lynx populations and lynx habitat. Overall, 64% of lynx habitat in the 6 geographic units are federally managed (USFWS 2017a) and 87% of this land is managed by the US Forest Service (USFWS 2017a), although this percentage varies by geographic unit. In 2000, the inadequacy of Forest Service and Bureau of Land Management land management plans for lynx was central to their listing (65 FR 16053), however the contemporary regulatory landscape has changed considerably from that time (USFWS 2017a; see discussion below). Critical habitat for lynx was designated in 2006 and revised in 2009 and 2014 (79 FR 54782). A proposed rule on revised critical habitat was published in 2024 (89 FR 94656), but this is pending as of 12/1/2025.

The 2017 Species Status Assessment considered the role of roadless areas in its evaluation of factors potentially influencing future conditions for the species. Specifically for the Northern Rockies geographic unit, the 2017 SSA noted that "we anticipate that future Federal management direction will include continued management of national parks, designated wilderness and roadless areas, and other areas with non-developmental land-use allocations to maintain natural ecological processes, which should maintain natural disturbance regimes and landscape-level habitat mosaics to which lynx are adapted" (USFWS 2017a). Of the 160 lynx detections in inventoried roadless areas, 121 of them (~76%) occurred in operable or operable but complex environments. To evaluate whether sampling bias near roads affected this assessment, we conducted a secondary analysis evaluating the overlap of roadless areas with lynx habitat in the Northern and Southern Rockies. Our analysis overlapping with inventoried roadless areas affected by the proposed action with lynx habitat in the Greater Yellowstone Area of Wyoming, Montana, and Washington as modeled by Olson et al. 2021 estimated that approximately 26.6% of lynx habitat in this region occurs in IRAs. While most of the southern Rockies lynx habitat is

excluded from the proposed action, a similar proportion of lynx habitat in this part of their range is administered by the U. S. Forest Service as roadless areas (Squires et al. 2025). We also evaluated the overlap between the current USFWS range for the species and inventoried roadless areas and found that approximately 6.8% of the lynx range occurs in these areas. This is likely because the proportion of roadless areas in the midwest and east is lower in lynx habitat than it is in the west. Altogether this shows that, the management of current roadless areas in ways that enhance the conservation of lynx is important at the southern-most range periphery in the continental United States. Approximately 12% of lynx designated critical habitat occurs within the area affected by the proposed action and 46% of that critical habitat within IRAs is on lands designated as operable or operable but complex.

Effects of Roads

Roads have direct and indirect impacts on Canada lynx. While overall the negative impacts of roads in lynx habitat are expected to outweigh any benefits (USFWS 2017a), as of the 2017 SSA, negative impacts of roads to the lynx DPS population as a population-level stressor had not been documented.

Direct mortality from vehicle strikes is a threat to individual lynx. The 2017 SSA reported that between 2010 and 2015 lynx were struck and killed 54 times in Maine, 9 times in Minnesota, 5 times in Montana and 1 time in Idaho. Importantly, mortalities occurred on both paved and unpaved roads. For example, 22 out of 54 lynx road mortalities in Maine occurred on low traffic, low speed, dirt forestry roads and 2 out of 9 mortalities in Minnesota occurred on unpaved forest roads (USFWS 2017a). Translocated individuals in re-introduction efforts may be especially vulnerable to vehicle strikes (Devineau et al. 2010).

While lynx have been documented crossing roads successfully, including highways (Squires and Oakleaf 2005; Squires et al. 2013), several studies suggest that lynx avoid crossing highways or that their behavior is modified while crossing these roads. Baigas et al. 2017 showed that lynx regularly crossed two-lane highways present in their home-ranges in Colorado, although they typically crossed in proximity to vegetative cover and at times of day when traffic volume was low. Thirty-nine percent of lynx in this study crossed roads at lower rates than expected. Work by Apps (2007) suggest that the likelihood of lynx crossing a highway decreased with increasing traffic volume.

Road construction may fragment lynx habitat although studies on avoidance behavior near small and low traffic roads are mixed. Fuller et al. 2007 found that while lynx in Maine would travel along unplowed forest roads in winter, they selected against road and road-edge habitat within their winter home range. This may be mediated by prey availability, as lynx killed fewer hares near roads and road edges than in preferred habitat types (Fuller and Harrison 2010). Another study in Maine found that lynx selected against logging roads at the homerange scale, but that logging roads had no effect on third order selection or selection “within core areas” (Vashon et al. 2008). In contrast, one study in central Ontario found no impact of road density on lynx detection in the study area (Walpole et al. 2012), and authors speculate it may have been due to the low traffic volume of area roads during winter. A small study in Washington found no evidence of road avoidance for lynx crossing small forestry roads or evidence of selection

against roads (McKelvey et al. 1999b). In the Superior National Forest, lynx were detected within 200m or roads and trails while making long distance movements (Moen et al. 2010). In the same forest, selection for roads did not differ from random expectation for lynx movement within home ranges, but authors note that the density of roads within the study area was homogenous (Terwilliger and Moen 2012). Similarly, a study in the Northern Rockies showed no effect of low traffic forest roads on lynx habitat use (Squires et al. 2010). In Colorado, a study found that lynx and a neutral to slightly positive selection for snow-covered forest roads (Squires et al. 2019). The effects of roads on lynx may be mediated by presence of competitors as well as prey. One study in an area of logging and energy development in Alberta found that higher densities of roads negatively predicted lynx occupancy in places where coyotes were present (Bayne et al. 2008). However, a winter study in Montana did not find evidence that coyotes were eating lots of hares, which would be indicative exploitative competition with lynx (Kolbe et al. 2007). Almost all studies on the impact of roads focused on occupancy and movement behavior, however one study evaluated the impact of roads on denning. Squires et al. 2008 found that lynx den further from forest roads than random expectation, but that this wasn't driven by avoidance of forest roads open to traffic, specifically.

Road construction also has the potential to fragment lynx habitat, however, attributing the impacts of road construction to habitat fragmentation, as opposed to habitat loss, can be difficult. One study in southern Ontario, also at the southern margin of lynx habitat, evaluated metrics of habitat availability and fragmentation and their ability to predict lynx occupancy (Hornseth et al. 2014). They found that in landscapes with large amounts of suitable habitat, lynx occurred in areas with high proportions of suitable habitat and that occupancy was not related to fragmentation metrics. In landscapes with less suitable habitat, occupancy was unrelated to habitat suitability and fragmentation metrics (Hornseth et al., 2014), suggesting flexibility in habitat use in marginal areas. The authors suggest that habitat loss, rather than habitat fragmentation, may be more important for lynx. However, in the Northern Rockies of Montana, the proclivity of female lynx to produce kittens did relate to connected patches of mature forests in home ranges (Kosterman et al 2018, Holbrook et al 2019)

A few recent studies shed light on the net effects of habitat loss and habitat fragmentation on lynx connectivity. Several studies suggest that Canada lynx can travel through a wide range of habitat types between core areas (reviewed in Vanbianchi et al. 2017 and Squires et al., 2013). Furthermore, a recent study on dispersing lynx during a crash in snowshoe hare abundance found that dispersing lynx were more tolerant of roads than resident lynx, but that both residents and dispersers were less likely to cross roads as compared to random steps (Arnold et al. 2025). Other work suggests that movement corridors for Canada lynx, particularly between the US and Canada in places like the Rocky Mountains, may be facilitated by a limited number of putative corridors (Squires et al. 2013). To the extent that new forest roads could affect the propensity of Canada lynx to use these corridors, it is theoretically possible that habitat fragmentation from roads could affect lynx connectivity. Nonetheless it is difficult to evaluate how likely this is given that lynx do cross roads (see above), albeit in a context dependent way, and genetic evidence to date does not suggest genetic isolation from core populations in northern North America (Schwartz et al. 2002). However, given the importance of gene flow between southern

peripheral populations and Canada (Schwartz et al. 2002), maintenance of connectivity is important for lynx management.

Indirectly, road construction may facilitate access of humans to lynx habitat resulting in mortality from incidental trapping. In Minnesota between 2010 and 2016 11 of 26 lynx trapped incidentally by trappers targeting other species were killed (USFWS 2017a). Increased road networks may also increase access of other recreationists to lynx habitat. Work in Southern Colorado suggests that in areas with high intensity dispersed recreation (backcountry skiing, snowboarding, snowmobiling, and packed trail skiing) lynx became more active at night and less active during the day (Olson et al. 2018). The same study found some evidence that lynx avoided areas with motorized recreation (Olson et al. 2018). Another analysis from the same study system found that lynx avoided areas used by snowmachines and selected areas used by backcountry skiers (Squires et al. 2019).

Increased road networks may lead to increased incidence of human-caused wildfires. Wildfires have the potential to alter lynx habitat, at least temporarily. Large wildfires have reduced the carrying capacity of lynx habitat in the Cascades (Lyons et al. 2023) and the population in this unit has likely declined since listing (USFWS 2023e). Early studies suggested that lynx avoid recently burned habitat (Koehler et al. 2008). In a more recent study, lynx briefly entered fire perimeters only one-year post-fire in Washington, but they primarily used unburned areas with mature trees suitable for snowshoe hares (Vanbianchi et al. 2017). Another recent study in the Northern Rockies focusing on recent burns (<3 years old) found that lynx avoided burned areas within their home-ranges and that they increasingly avoided burned areas as they became more available within the home-range (Squires et al. 2026). Lynx in this study also preferred unburned and low fire severity areas relative to high fire severity areas (Squires et al. 2026). Patterns of habitat use change as time since fire increases. For example, lynx tended to use all areas of older (20+ year) burns in a study in the North Cascades (Vanbianchi et al. 2017). One study in the Yukon has documented lynx use of downed woody debris for denning habitat approximately 30 years postfire (Slough 1999). A study in the Northern Rockies showed lynx using burned areas within 25 years, but use differed depending upon forest management approach (see below) (Olson et al. 2023). The length of time for forest conditions to become favorable to lynx and snowshoe hare again after a burn may also vary across the lynx range (USFWS 2017a).

Effects of Timber/ Vegetation Management

The initial listing for Canada lynx identified that timber harvest and vegetation management can be "benign, beneficial, or detrimental to lynx depending on harvest methods, spatial and temporal specifications, and the inherent vegetation potential of the site" (65 FR 16053). Recent peer reviewed literature on lynx reflect this heterogeneity.

"Intensive" forest harvesting such as "large scale clearcuts" can be detrimental to lynx by eliminating habitat associated with prey and lynx denning (65 FR 16053). Although lynx generally avoid crossing clearcuts during winter (Squires et al 2010), they either select or avoid harvested landscapes depending on mosaics of forest structure across time (Holbrook et al. 2018). Lynx selected mature forests at the landscape scale, but preferentially used 20-30 yr old advanced generating forests from forest management within home ranges. (Holbrook et al.

2017). In addition, lynx in the Northern Rockies increasingly avoided forests 5 years or less post-disturbance as it became more available in home ranges (Holbrook et al. 2017). Work in Maine shows similar results with lynx avoiding recent clearcuts in winter (Fuller et al. 2007). Regardless of silvicultural treatment, lynx may avoid recently harvested landscapes. Work in the Northern Rockies showed that lynx use of areas subjected to group selection cuts, liberation cuts, improvement cuts, pre-commercial thinning, regeneration cuts with natural regeneration, and regeneration cuts with planting was low for the first 10 years post-action regardless of treatment (Holbrook et al. 2018). After that, it took from approximately 20 yrs (thinning) and 34-40 yrs (selection, regeneration cuts) following tree harvest for vegetation to sufficiently recover as to support 50% of lynx-use (Holbrook et al. 2018)

The time-dependent nature of lynx habitat use post-disturbance is seen in the literature on the value of regeneration cuts to lynx. A study in Western Canada suggested that large patches (>1000 hectares) of regenerating forest (15-40 years post-harvest) had sufficient hare densities to support lynx reproduction (Reid et al. 2024). Similarly, a study in the Northern Rockies showed that advanced regenerating forests (20-80 years old) had hare densities at least 2.8 times that of other structural classes (Holbrook et al. 2019). Related studies found that small-diameter regenerating forest contributed to models that predicted successful reproduction, but within a matrix of connected mature forest (Kosterman et al. 2018, Holbrook et al 2019). An early study in Maine found evidence of lynx selecting regenerating forest (Hoving et al. 2004). Several additional studies in the region have further explored this relationship. Lynx in Maine selected for tall regenerating clearcuts (10+ years post-harvest) and established partially harvested stands (11-21 years post-harvest) (Fuller et al. 2007). Another study in Maine found that both male and female lynx preferred conifer dominated sapling stands (mid-regeneration) at multiple spatial scales (Vashon et al. 2008). A subsequent study suggested that lynx may capitalize on edge habitat, as at the landscape scale, lynx used areas straddling both regenerating forests with high numbers of hares and mature coniferous forest efficient for travel (Simons-Legaard et al. 2013). In contrast, young forests (<5 years post-harvest) are not selected for by Canada lynx. For example, lynx intensity of use of forests in the Rocky Mountains is low for approximately 10 years post silvicultural treatment (Holbrook et al. 2018) and resource selection analyses suggest that lynx use forests in stand initiation phases (<5 years post-disturbance), less than older forests and sparsely forested areas (Holbrook et al. 2017).

There is less literature on the effects of retention harvest on lynx. Retention harvest of different levels did not affect lynx use of habitat in a study by Franklin et al. 2019 in Alberta. During winter, lynx in a study area in Maine selected against recent partially harvested stands (Fuller et al. 2007).

The significance of mature forest may vary across the lynx range. In the Northern Rockies lynx select for mature forest and experience fitness benefits from this habitat type. One study showed that across seasons Canada lynx in the Northern Rockies used mature (50-200 year old) spruce-fir forest most (Holbrook et al. 2017). During winter, lynx foraged primarily in areas with mature large diameter trees and high horizontal cover (Squires et al. 2010). Females in this study region were more likely to produce kittens in areas of mature well-connected forest, primarily comprised of spruce-fir (Kosterman et al. 2018; Holbrook et al. 2019).

The significance of mature forest is less clear for lynx in Maine. During winter, lynx in a study in Maine selected against mature forest (40+ years post disturbance) (Fuller et al. 2007). However, another study suggested that lynx may select for mature forest when it borders high quality hare habitat (Simons-Legaard et al. 2013) and a third study found no selection for or against mature forest (Hoving et al. 2004). We are unaware of a study evaluating the impact of mature forest on lynx productivity in the East.

A few studies have focused on the effects of logging associated with fire mitigation and post-fire salvage on lynx. A study in the Northern Rockies showed that lynx habitat use varied with time post-fire and silvicultural treatment, with lynx using forest with regeneration cuts ~25 years post-disturbance the most (Olson et al. 2023). While pre-fire thinning and salvage reduced high intensity fire within forest stands, lynx only used these stands at a low-level post-fire, albeit consistently and at qualitatively higher levels than stands that did not receive pre-fire management (Olson et al. 2023). The temporal change in use of post-fire habitat by snowshoe hares may be especially important for Canada lynx. Kelly and Hodges (2020) showed that snowshoe hare density decreased immediately following wildfire, but that densities started to rebound by 8-9 years post-fire in areas that were not-salvage logged. In contrast, snowshoe hares were largely absent in salvaged logged areas in Washington even 8-9 years post-fire (Kelly and Hodges 2020). A subsequent study showed that lynx avoided salvage logged areas at the 50 and 150m scales at sites approximately 5-15 years post-fire (Volkman and Hodges 2024).

In sum, the effect of forest management and timber harvest on Canada lynx is nuanced and time-dependent. Several studies recommend that managers maintain a mosaic of forest types and/or a minimum percentage of high-quality hare habitat (Simons-Legaard et al. 2013; Holbrook et al. 2018; Holbrook et al. 2019). Overall, management activities that reduce horizontal cover, such as thinning, are thought to degrade lynx habitat (Squires et al. 2010).

Existing Protections

The impact of timber and vegetation management on lynx is now informed by additional land-management guidance specific to lynx. The Northern Rockies Lynx Management Direction (USFS 2007) and Southern Rockies Lynx Amendment (USFS 2008) direct vegetation management in occupied lynx habitat for most of the intermountain west through incorporation into land management plans. These documents follow the Lynx Conservation Assessment Strategy (LCAS), which identifies risks to lynx and recommends conservation measures for actions that could increase lynx risk through habitat alteration. The LCAS recommends that ≤30% of a lynx analysis unit (LAU) should be currently unsuitable (in stand initiation structural stage) for lynx, that non-salvage timber harvest should not change more than 15% of lynx habitat in an LAU to currently unsuitable over the course of a decade, that no precommercial thinning that reduces hare habitat takes place in stand initiation structure classes until stands no longer provide winter snowshoe hare habitat, and that no vegetation management takes place in multistory forests that reduces winter snowshoe hare habitat. The SRLA provides for some minimal allowances for precommercial treatments in regenerating stands with winter snowshoe hare habitat. The SRLA also provides for limited uneven-aged vegetation management activities in mature multi-story stands with snowshoe hare habitat. These limits apply to all projects, except for fuels management activities within the WUI. These management directions also offer

guidelines to forests on minimizing the impact of roads on lynx. Outside of the Rockies, all forest plans have been formally amended to incorporate guidance from the LCAS except for the Okanogan-Wenatchee. The forest manages landscapes according to the LCAS (USFWS 2023e) and the Canada Lynx Conservation Agreement between the USFS and USFWS, which commits the USFS to implement the LCAS until all forest plans are amended to incorporate it (USFS & USFWS 2000).

Critical Habitat

The physical and biological features of Canada lynx critical habitat are:

- "(1) Boreal forest landscapes supporting a mosaic of differing successional forest stages and containing: (a) Presence of snowshoe hares and their preferred habitat conditions, which include dense understories of young trees, shrubs or overhanging boughs that protrude above the snow, and mature multistoried stands with conifer boughs touching the snow surface; (b) Winter conditions that provide and maintain deep fluffy snow for extended periods of time; (c) Sites for denning that have abundant coarse woody debris, such as downed trees and root wads; and (d) Matrix habitat (e.g., hardwood forest, dry forest, non-forest, or other habitat types that do not support snowshoe hares) that occurs between patches of boreal forest in close juxtaposition (at the scale of a lynx home range) such that lynx are likely to travel through such habitat while accessing patches of boreal forest within a home range." (65 FR 54782)

Approximately 12% of lynx designated critical habitat occurs within the area affected by the proposed action and 46% of that critical habitat within IRAs is on land designated as operable or operable but complex. Given that much of this habitat is operable or operable but complex terrain and given the importance of vegetation features for the habitat of this species the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species.

Effects Determination

Given that at least 25% of the habitat for the species in the Northern Rockies is in the proposed action area and given that lynx could be subject to mortality and habitat degradation with the proposed action, the FS has determined the proposed activity is likely to adversely affect Canada lynx**.** Given that much of this habitat is operable or operable but complex terrain and given the importance of vegetation features for the habitat of this species the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species.

Coastal DPS of Pacific Marten

The coastal DPS of Pacific marten (*Martes caurina humboldtensis*), also referred to as the Humboldt marten, was proposed as Federally Threatened in 2018 (83 FR 50574) and then listed as Federally Threatened in 2020 under a 4(d) rule (85 FR 63806) with critical habitat designated in 2024 (89 FR 46576).

The coastal marten had a historical range throughout coastal Oregon and coastal northern California (85 FR 63806). Recently, the species has not been detected throughout much of its

historical range. The coastal marten DPS is currently persisting in approximately 7% of its historical range in four small populations with fewer than 100 individuals each (USFWS 2023f). Given the small population size and fragmentation of the remaining habitat, connectivity between the remaining populations is a major concern for species persistence (USFWS 2026b). Both Oregon populations are functionally isolated without connectivity to any other population, and the two California populations only have connectivity to each other (USFWS 2020d). Consequently, if either of these two Oregon populations were eliminated, natural recolonization from other populations would not be feasible (USFWS 2020d).

Exposure to the Proposed Action and the Proposed Action Area

Critical habitat was designated in 2024 and consists of mature, structurally complex, late-successional forests primarily in northwestern California and southwestern Oregon, encompassing roughly 1.2 million acres. The vast majority, 92% (1.1. million acres) of critical habitat, is located on federal lands with 88% located on Forest Service lands and 17.6% of total critical habitat is within the affected environment. The essential physical and biological features necessary for the marten's survival and recovery include a dense understory for cover and foraging habitat, a complex forest canopy, and abundant coarse woody debris like large snags and downed logs for denning sites and prey habitat. Coastal marten are found in older, conifer–dominated forests, serpentine habitats, and shore pine and transitional shore pine/Douglas–fir–hemlock forests at the Oregon Dunes National Recreation Area (Eriksson et al. 2019). However, the common denominator with serpentine and older forest habitats is the presence of dense, spatially extensive ericaceous shrub understories and diverse and abundant prey.

Martens are more selective of habitat conditions at den sites than at resting sites and can be a limiting factor for the population. Suitable denning habitat requires both denning structures as well as suitable foraging habitat near den sites that supports energy requirements of female during lactation and minimizes time spent away from kits. Research has found most coastal marten den sites are in cavities of live and dead trees that were larger than average diameter compared to the surrounding stand with structural features such as complex crowns, large limbs, broken tops, hollow bases, or multiple cavities. Coastal marten frequently reuse denning structures with 50 percent of reproductive females monitored in multiple breeding seasons reusing a den from a previous season, and 60 percent reusing a site within the same season.

We conducted multiple analyses to evaluate the exposure of Coastal marten to the proposed action. Analyses were not conducted on occurrence data because these occurrence records did not distinguish the coastal marten DPS from other *Martes caurina.* We overlayed the USFWS Coastal marten range map (USFWS 2026a) with the proposed action area and found that 8.9% of the range overlapped with IRAs affected by the proposed action. We also conducted an analysis using a data product from the USFWS[23] to evaluate the amount of core habitat and linkage corridors for this species that overlap with the proposed action area. Approximately 14% of core habitat and 7.3% of linkage corridor areas overlapped with IRAs affected by the proposed action.

[23] Used Primary Model Outputs from the USFWS Coastal Marten Analysis available at: https://www.fws.gov/project/coastal-marten-analysis#PrimaryModelOutputs (accessed 1/9/2026).

Effects of Roads

Direct Effects

Roads exert several direct impacts on coastal marten populations. Roads can present a direct mortality risk, with individuals vulnerable to vehicle collisions when attempting to cross. Since 1980, nineteen vehicle-caused mortalities have been documented for coastal marten and all were in Oregon. Of the four marten population areas, the central coastal Oregon population appears substantially more vulnerable to roadkill mortalities than the other three, which has been attributed to the increased traffic volume and closer proximity of habitat to busy highways, along with an increased human presence making the reporting of roadkill animals more likely. Roads also have the potential to increase habitat fragmentation, as road construction and associated clearings break up continuous forest into smaller, isolated patches. While research concerning road impacts on coastal marten movement is lacking, studies of populations of American marten have found that roads can reduce habitat quality, finding reduced activity near logged roads in Ontario and reduced marten detection rates with increasing road density in Idaho. This reduces the ability of martens to move freely across their range and can restrict opportunities for dispersal and genetic exchange. In addition, traffic noise and human activity along roads can displace martens from otherwise suitable habitats, forcing them into suboptimal areas. Even relatively small or low-use roads can act as barriers to movement, effectively restricting access to critical resources and diminishing the functional connectivity of the landscape.

Indirect Effects

Beyond these immediate impacts, roads also generate a suite of indirect effects that further compromise coastal marten viability. Road corridors often alter predator-prey dynamics by creating edge habitats that favor generalist predators such as coyotes or bobcats, thereby increasing predation risk for martens. Displacement from road-adjacent areas can lead to crowding in remaining habitat, intensifying competition, disease transmission, and stress. Human disturbance from off-road vehicles and recreation in dune or forest habitats that overlaps with critical breeding areas may potentially disrupt maternal care and juvenile survival. Finally, roads contribute to long-term connectivity loss, limiting movement between suitable habitat patches and further isolating already small and vulnerable populations (Schrott and Shinn 2020). High–traffic, multi–lane highways have been shown to be significant barriers to movement of American martens although some types of culverts can be used by martens to pass under such highways.

Exposure to rodenticides has potential lethal and sub-lethal effects on coastal martens. Illegal marijuana cultivation sites on public, Tribal, and private forest lands are implicated as the likely source of these rodenticides. In fishers, a similar carnivore species, 85% of carcasses tested were exposed to rodenticides, with the cause of death attributed to exposure in 13% (83 FR 50574). While there is less data on rodenticide exposure in coastal marten they use similar habitats that also have high potential for illegal marijuana cultivation and 3 of the 9 coastal marten tested in recent work were positive for anticoagulant rodenticide exposure (83 FR 50574). As illegal cultivation has been associated with proximity to roads (Wengert et al. 2021) removing roadless area protections could potentially increase use of these areas for illegal cultivation and expand and increase the exposure of coastal marten to toxicants used at these sites.

Effects of Timber/ Vegetation Management

Timber harvest has been identified as a primary threat to coastal marten (83 FR 50574). Marten habitat loss due to logging of older forests is considered a likely cause of population declines and low population levels in northwest California since trapping has been banned since 1946 (USFWS 2023f). Historically, older forests dominated coastal California, the Klamath–Siskiyou, and the Oregon Coast Range, but as a result of past timber harvest there are only a~ 5%, 20%, and 38% of those forests remaining, respectively. Timber harvest can negatively impact coastal marten habitat by directly reducing the structural complexity of forests that martens require and can result in the removal of mature conifers and dense understory vegetation diminishing the availability of resting, and denning sites, which are often located in large diameter logs, snags, or complex root systems. Clearcutting and intensive thinning eliminate canopy cover, exposing martens to increased predation risk while also increasing preferred habitat their primary predator the bobcat, which have been shown to prefer regenerating harvested stands <30 years old. Harvest operations also disturb martens through noise, machinery, and human presence, which can displace individuals from otherwise suitable habitat. Detection rates of coastal martens rise with the size of late-successional forest patches, indicating that timber harvest practices that reduce patch size likely negatively impact marten populations. Studies have shown that marten occupancy declines sharply in areas subjected to intensive harvest, reflecting the species' reliance on late-successional forest conditions.

Coastal martens are highly sensitive to habitat fragmentation, which reduces the availability of mature and old forest stands and creates large expanses of unsuitable habitat with little protective cover. Timber harvest and associated forest management practices have been shown to lower marten population density when late-successional forest cover falls below 70 percent of the landscape. Because martens have a small body size, high metabolic rates, and spatial requirements several times larger than similar-sized carnivores, they are particularly vulnerable to habitat loss. Increased fragmentation forces martens to travel longer distances through regenerating forests, which has been linked to reduced body condition. Clearcutting and heavy thinning have been shown to be especially detrimental for marten creating openings that martens consistently avoid. Poor body condition can impair predator avoidance, foraging efficiency, reproduction, and resistance to disease, further compounding the effects of timber-driven habitat degradation.

Existing Protections

Conservation and management by a number of federal guidelines. The FWS published a Recovery Outline in 2020 that serves as an interim strategy to guide recovery efforts until a draft recovery plan has been completed. Then in early 2026, a Draft Recovery Plan (USFWS 2026c), an updated Species Status Assessment (USFWS 2026b), a Recovery Implementation Strategy (USFWS 2026d), and a Five-Year Review (USFWS 2026e) were also published.

Management guidance and protections are also provided in the National Forest Land Management Plans for National Forests within the range of coastal marten. Furthermore, coastal marten are managed under the Northwest Forest Plan which provides critical management guidance by conserving late-successional and old-growth forests, limiting clearcutting, and

maintaining structural complexity that are critical elements of coastal marten habitat. These measures reduce fragmentation and preserve habitat quality for marten populations.

Critical Habitat Determination

Of the total critical habitat designated, 17.6% is within the affected environment. The essential physical and biological features necessary for the marten's survival and recovery include a dense understory for cover and foraging habitat, a complex forest canopy, and abundant coarse woody debris like large snags and downed logs for denning sites and prey habitat. Given this, we have determined that the exposure of critical habitat to the proposed action is high and that the proposed action is likely to adversely affect designated critical habitat.

Effects Determination

The exposure of the coastal marten to the proposed action is high. Given this exposure level, and the potential impacts of roads and timber harvest on the DPS, the FS has determined the proposed action is likely to adversely affect this species. We have also determined that the proposed action is likely to adversely affect designated critical habitat.

Black Footed Ferret

The black footed ferret *(Mustela nigripes)* was initially listed as endangered 1967 under the Endangered Species Preservation Act of 1966 (32 FR 4001) and subsequently incorporated into listing under the ESA (39 FR 1175). Currently the species is listed as endangered and most reintroduced populations are listed as nonessential experimental. Black footed ferrets on Buffalo Gap National Grassland are designated experimental.

The 2019 species status assessment indicates that the ferret population size declined dramatically after 2010. The sylvatic plague is thought to be a main contributing factor to these losses (USFWS 2019d).

Exposure to the Proposed Action Area and the Proposed Actions

Black-footed ferrets have a close association with prairie dogs occurring in grasslands and other prairie dog habitat in the Great Plains (USFWS 2013b). They rely on prairie dogs for food and use their burrows for shelter (USFWS 2013b). There are two national grasslands where the ferret is thought to occur in South Dakota.

There is no critical habitat for black-footed ferret. There is one observation of a ferret in inventoried roadless areas out of more than two hundred for the management units and it is in an area that is not operable, however this low sample size makes it difficult to draw conclusions from these data. We conducted a supplementary analysis overlaying range maps for the endangered population of ferrets and experimental populations of ferrets (USFWS 2026a) with IRAs. There is less than 1 square kilometer of overlap of the endangered ferret range with inventoried roadless on the Custer Gallatin National Forest in Montana. Less than 0.001% of the experimental population range overlaps with inventoried roadless areas, but it all occurs in ~94 square kilometers of the range in Buffalo Gap National Grassland.

Effects of Roads

Roads and road construction could lead to direct mortality of black footed ferrets from vehicle strikes and from predation by raptors using associated infrastructure (USFWS 2019d). Roads may also affect ferrets indirectly through habitat fragmentation and soil compaction (USFWS 2019d).

Effects of Timber/ Vegetation Management

Given that ferrets are largely associated with grasslands, we do not discuss the effects of timber.

Existing Protections

The Nebraska Grasslands Management Plan includes standards and guidelines to benefit the black-footed ferret (available at USFS 2025a). They are:

- "18. In prairie dog colonies known or thought to be occupied by black-footed ferrets, limit oil and gas development to one location per 80 acres to help maintain suitable ferret habitat.
- 19. To help provide suitable habitat for black-footed ferrets and their young during the breeding and whelping seasons, prohibit the following activities within 1/8 mile of prairie dog colonies, or those portions of larger colonies, occupied or thought to be occupied by ferrets from March 1 through August 31:
    - Construction (e.g., roads, water impoundments, oil and gas facilities),
    - Reclamation,
    - Gravel mining operations,
    - Drilling of water wells,
    - Oil and gas drilling.
- 20. To help provide suitable habitat for black-footed ferrets and their young during the breeding and whelping seasons, do not authorize the following activities within 1/8 mile of prairie dog colonies, or those portions of larger colonies, occupied or thought to be occupied by ferrets from March 1 through August 31:
    - Construction (e.g., pipelines, utilities, fencing),
    - Seismic exploration,
    - Permitted recreation events involving large groups of people. Guideline...
- [#21 was stricken by an amendment]
- 22. For routine maintenance, access to oil and gas facilities in prairie dog colonies occupied or thought to be occupied by black-footed ferrets shall be limited to daylight hours. This does not apply to emergency repairs."

In addition to these forest and grassland-wide directions, there is detailed guidance on the management of black-footed ferret in reintroduction habitat.

Effects Determination

Given the extremely small amount of overlap between the endangered population range and the proposed action, the exposure of the ferret to the proposed activity is low.

Given that there is overlap between inventoried roadless areas and the experimental population range, but it is a small portion of the range, the exposure of the ferret to the proposed activity is low.

For both populations, given the low likelihood of exposure to road and timber activities, existing protections, and the limited direct impact roads would have on black-footed ferrets even if exposed, the FS has determined the proposed activity is not likely to adversely affect this species because effects are discountable.

Jaguar

Jaguar (*Panthera onca*) in the United States was listed as federally endangered in 1997 (62 FR 39147). Jaguars once occupied a range stretching from the southern United States to central Argentina. Today, their distribution extends from the southwestern U.S. down to northern Argentina, with populations present in every country except El Salvador and Uruguay. Current estimates suggest they inhabit roughly 51% of their former range. The IUCN Red List assessment indicates a declining global trend, with a suspected 20–25% reduction over the past three generations (21 years) in occupied area, geographic extent, and habitat quality, leading to a comparable 20–25% decrease in mature individuals during that period (Quigley et al. 2017).

In the United States historically, jaguars were found in California, Arizona, New Mexico, Texas, and possibly Louisiana, but were largely extirpated from these states, with only occasional sightings since the early 20th century. Between 1996 and July 2017, only 7-8 male jaguar sightings have been confirmed in the U.S. Because females have been absent for decades, evidence indicates that jaguars in the U.S. represent extensions of populations based primarily in Mexico, which is supported by genetic studies that confirm connectivity between individuals in Arizona and Sonora. Consequently, maintaining connectivity with jaguar populations in Mexico is critical to recovering U.S. populations.

Exposure to the Proposed Action and the Proposed Action Area

Jaguar habitat in the U.S. portion of the range is a mix of arid and semi-arid mountain ranges (often called "sky islands") and grasslands that function primarily as a secondary area and corridor for male jaguars dispersing from core populations in Mexico. Critical habitat was designed for jaguar in the United States in 2014 on 764,207 acres in Arizona and New Mexico in six critical habitat units (79 FR 12571). After court ordered revisions to critical habitat there are currently 640,124 acres designated (89 FR 46822) in expansive open spaces in the southwestern US that are at least 100 km^2 in size. There are seven primary constituent elements of jaguar critical habitat, that provide for the needs of the species. These seven elements are: 1) Provide connectivity to Mexico, 2) Contain adequate levels of native prey species, 3) Surface water

sources available within 20 km of each other, 4) Contain from greater than 1 to 50 percent canopy cover within Madrean evergreen woodland, 5) characterized by intermediately, moderately, or highly rugged terrain, 6) below 2,000 m (6,562 feet) in elevation, and 7) are characterized by minimal to no human population density, no major roads, or no stable nighttime lighting over any 1-km^2 (0.4-mi^2) area (USFWS 2018d).

For the critical habitat in the U.S. approximately 68% is federally owned and 65% is managed by the U.S. Forest Service, mainly within the Coronado National Forest. Of jaguar critical habitat on Forest Service lands there are 63,578 acres (9.9%) in the affected environment and 16% of it is operable or operable but complex.

We also conducted an analysis overlapping the entire USFWS range with the proposed action area. Approximately 2.7% of the USFWS range for jaguar is in the proposed action area.

Effects of Roads

No direct mortality of jaguars from vehicle collisions has been documented within the United States. In Mexico and other parts of Latin America, vehicle strikes have been confirmed as a source of jaguar mortality, particularly where major transportation corridors intersect occupied habitat. Studies from Brazil, Mexico, and Central America have reported multiple jaguar fatalities due to vehicle collisions, underscoring the vulnerability of wide-ranging carnivores to road infrastructure. Accordingly, roads within and adjacent to designated critical habitat should be considered a potential stressor, with mitigation measures focused on maintaining connectivity and reducing collision risk.

Roads have been identified as a significant factor contributing to habitat fragmentation and a potential barrier to jaguar (*Panthera onca*) movement and recovery. Habitat loss and fragmentation are listed as primary threats to the species' continued existence, and this fragmentation leads to small, isolated populations that are more susceptible to local extinction. Major roads act as impediments to natural jaguar dispersal, particularly between populations in the United States and Mexico. Furthermore, specific critical habitat criteria exclude areas with major roads, as well as areas with stable nighttime lighting or high human population density, which indicates that roads inherently degrade the quality of suitable habitat. This avoidance behavior is supported by data showing that jaguar activity is negatively correlated with human activity and development, which often accompanies road infrastructure. Consequently, recovery efforts for the species include strategies to mitigate the impediments caused by roads to ensure landscape permeability and connectivity between habitat units.

Roads can also indirectly affect jaguars through illegal hunting and trapping by increasing human entry into otherwise remote habitats; the 2018 Recovery Plan notes that illegal poachers and trappers often use road networks to reach rugged mountain ranges such as the Peloncillo and Santa Rita Mountains in the southwestern U.S., elevating the potential for direct persecution of jaguars and depletion of their prey base (USFWS 2018d). Roads also generate edge effects that alter predator–prey dynamics, with fragmented landscapes with increased edge habitat favoring smaller carnivores like coyotes and bobcats, while reducing prey species such as white-tailed deer and javelina, thereby diminishing habitat quality for jaguars. In addition, human disturbance from noise, light, and recreational activity along road corridors is documented in areas such as

the Coronado National Forest, where increased offroad vehicle use and border related infrastructure reduce the suitability of occupied areas and may displace jaguars from critical movement routes (79 FR 12571).

Effects of Timber/ Vegetation Management

Timber harvest and logging activities can indirectly affect jaguar conservation by altering habitat structure, prey dynamics, and levels of human disturbance. Removal of mature forest cover reduces canopy density and vegetative complexity, diminishing the concealment and stalking opportunities jaguars rely on. Logging operations may also disrupt prey populations such as deer and peccaries by modifying forage availability, thereby reducing prey abundance in occupied areas. Large-scale timber harvest is not currently a primary threat within designated U.S. critical habitat for jaguars but the Recovery Plan emphasizes that these indirect effects can degrade habitat quality and compromise long-term recovery objectives.

Existing Protections

In addition to protections under the ESA, the Coronado National Forest Land and Resource Management Plan (2018) identifies jaguar as a federally listed species requiring special management (available at USFS 2025a). The plan emphasizes maintaining habitat connectivity, limiting disturbance in Sky Island Mountain ranges. Arizona Game and Fish Department recognizes jaguar as endangered under state law with state wildlife action plans that emphasize habitat conservation and coordination with federal agencies.

Critical Habitat Determination

Given that nearly 10% of critical habitat for jaguar in the United States is in inventoried roadless areas, that 16% of that is in operable or operable but complex landscapes, and that the proposed actions could have the potential to affect canopy cover and road density the FS has determined the proposed activity is likely to adversely affect designated critical habitat for jaguar.

Effects Determination

Our exposure data is insufficient at this time and is classified as “unknown” until further analyses can be done at the project level. Based on the small numbers of jaguars in the U.S. and significant potential negative effects of roads on fragmentation of habitat and connectivity, as well other possible indirect effects on human interactions and predator and prey populations, the FS has determined the proposed activity is likely to adversely affect jaguars. Given that nearly 10% of critical habitat for jaguar in the United States is in inventoried roadless areas, that 16% of that is in operable or operable but complex landscapes, and that the proposed actions could have the potential to affect canopy cover and road density the FS has determined the proposed activity is likely to adversely affect designated critical habitat for jaguar.

Fisher, Southern Sierra Nevada DPS

The southern Sierra Nevada (SSN) fisher DPS (*Pekania pennanti*) was listed as Federally Endangered in 2020 with Critical Habitat designated in 2022. At the time of listing both long-term isolation (Knaus et al. 2011, Tucker et al. 2014) and small population size were identified as primary threats to the DPS, with an estimated population size of less than 300 adult fisher

before 2015 (Spencer et al. 2011, Spencer et al. 2015). Since 2015 this DPS has experienced substantial habitat loss and fragmentation due to a historic drought and subsequent large scale tree mortality event (Fettig et al. 2019) followed by a series of megafires in 2020-2021 each exceeding 80,000 acres (Ayars et al. 2023). Cumulatively these disturbances have resulted in large losses of fisher habitat. Hart et al. (2025) found a 48% reduction of female fisher habitat between 2012-2022. Population trend assessed through monitoring of population occupancy was generally stable across the DPS through 2022 (Zielinski et al. 2013a, Tucker et al. in Review, Martin et al. in Review). Mean occupancy dropped from 0.32 (2011-2022) to 0.20 in 2023, which is the lowest occupancy rate recorded since monitoring began with even larger magnitude declines regionally within the DPS (Martin et al. in Review).

Recent habitat losses have also resulted in extreme fragmentation of the remaining habitat. Effects of drought and tree mortality through 2019 (excluding subsequent megafires) resulted in an increase in the number of habitat patches from 74 to 558 and decrease in average patch size from 31,500 acres down to 2,600 acres (Thompson et al. 2020). Additional recent megafires have burned through a number or previously identified linkage areas. This fragmentation has resulted in major concerns for connectivity on multiple scales, both between the remaining patches of reproductive habitat critical for juvenile dispersal as well as in the loss of four major linkage areas that may prevent larger scale movement needed to maintain a viable population (Thompson et al. 2020).

Exposure to the Proposed Action Area and Proposed Actions

In western North America, fisher habitat is primarily associated with late-successional conifer or mixed-conifer hardwood forests. Within these forests fisher habitat is characterized by an abundance of dead and downed wood, dense, multistory canopies, moderate to high canopy cover, and large trees (Lofroth et al. 2010). In the SSN, fishers occur primarily in dense, mature mixed-conifer and ponderosa pine forests at middle elevations that generally do not accumulate persistent deep snow which has been found to limit the movement and distribution of fisher (Zielinski et al. 2017, Pauli et al. 2022). Fisher are habitat specialists with needs that vary depending on age and sex. In the SSN fisher suitable habitat generally categorized as either foraging, resting, or denning habitat (also alternatively modeled as reproductive or female habitat). Reproductive/denning/female habitat categorization represents the specialized habitat needed to support fisher reproductive females that is considered a limiting factor for population viability (Spencer et al. 2015, Thompson et al. 2020).

Foraging Habitat

Fisher's diet in the SSN DPS has been found to include a wide variety of food items suggesting that fishers likely forage diverse vegetation types, though they likely concentrate on foraging in mature forests that are habitat for larger, preferred prey items (e.g., >200 g tree squirrels) (Spencer et al. 2015). However, the widespread tree mortality from 2015-2019 has resulted in dietary shifts finding a reduction in tree squirrels and increase in plant based dietary composition. The post-tree mortality shifts away from medium-sized prey mammalian species that are thought to be the most energetically efficient choice has potential implications for long-term survival SSN fisher and management implications for management of foraging habitat (Pilgrim et al. 2023, Smith et al. 2024).

Resting Habitat

Fishers need a variety of suitable resting habitats and structures (e.g., cavities or platforms in trees or snags) within their home ranges to provide shelter from weather, conserve energy, avoid predators, and safely survey for and consume prey. They use multiple resting sites within their home range to minimize travel time when foraging and rarely reuse the same rest sites. Canopy cover appears to be the most important variable in resting site habitat selection and resting typically occurs in live trees, snags, and logs that are in the largest available diameter classes, and these structures are typically deformed or in a state of decay (Spencer et al. 2015).

Denning Habitat

Denning sites and reproductive habitat that provide sufficient resources for females to bear and raise kits, is considered the most limiting habitat element for the fisher population. Females typically use multiple dens during a single denning season. They give birth care for newborns in natal dens, and later often move kits to a succession of several maternal dens until they are weaned. Therefore, conservation of female reproductive areas requires ensuring clusters of multiple suitable den structures in relatively close proximity persist with their home range. Like resting structures, reproductive dens are usually cavities in large, live or dead trees, in forest stands with dense canopy cover and complex structure. Fishers select den sites in clusters of large, mature trees in stands with variable tree heights and dense canopies, on fairly steep slopes. Suitable denning sites are probably a subset of suitable resting sites, because the requirements are more stringent (Spencer et al. 2015).

Habitat Configuration

Landscape habitat models show that fisher habitat in the southern Sierra Nevada is arranged in a roughly north-south trending collection of narrow habitat patches in a restricted elevation zone that supports forested habitat but is not subject to consistent deep snow throughout the winter. Habitat patches are bisected by a number of major river canyons (e.g., San Joaquin, Kings, and Kern rivers) that have been shown to limit gene flow within the SSN (Tucker et al. 2014, Tucker et al. 2017). This narrow, undulating distribution of habitat makes it susceptible to fragmentation into even smaller patches (Spencer et al. 2015). The tree mortality epidemic and subsequent large scale megafires in the last decade have burned a number of primary connectivity linkages in the SSN DPS (Thompson et al. 2020).

We conducted two analyses to assess the exposure of fisher to the proposed action quantitatively. First, we overlapped the USFWS range map for the Southern Sierra Nevada Fisher, DPS (USFWS 2026a) with the proposed action area and found that approximately 7.6% of the DPS' range was included in IRAs affected by the proposed action. Additionally, we evaluated the overlap of fisher habitat with the action area. Due to the large-scale vegetation changes due to drought and tree mortality in recent year that significantly impacted fisher habitat, we analyzed both pre- and post-drought fisher habitat maps created as part of the SSN fisher Conservation Strategy Interim Guidelines (Thompson et al. 2020, Conservation Biology Institute 2020). We found that approximately 11.1% of pre-drought suitable habitat for fisher overlapped with the proposed action area and 9.2% of post-drought suitable habitat overlapped. Our analysis of post-drought denning habitat showed that approximately 5.9% overlapped with the proposed action

area. We did use occurrence data for this species, because the databases data were drawn from did not distinguish this DPS from other fisher records.

Critical Habitat

The U.S. Fish and Wildlife Service published a revised proposal on November 7, 2022, to designate approximately 595,495 acres of critical habitat across six units (one unit of which is further subdivided into two subunits; 87 FR 66987). This proposed critical habitat has not received a final rule as of the time of publication. Critical habitat was defined as suitable, high-quality denning habitat that includes intermixed foraging and dispersal areas. Such habitat provides structural features for parturition, raising kits, protection from adverse weather conditions, facilitation of safe movement, sites to rest and thermoregulate, foraging opportunities, and cover to reduce predation risk for adults and young. More than 90% of designated critical habitat for SSN fisher is on federal lands (U.S. Forest Service, National Park Service, Bureau of Land Management) with the majority of on Forest Service lands. There are 392,167 acres (65.9%) of critical habitat on Forest Service lands of which 63,835 acres (10.7%) are in the action area.

Due to the recent large scale habitat loss, retaining what habitat remains is highly important to ensure long term species persistence given the small and isolated nature of the SSN fisher DPS. Roadless areas overlap with both critical habitat and broader denning, resting, and foraging habitat and these areas are potentially important to preserving the permeability of remaining foraging habitat that is critical for juvenile fisher dispersal and provide connectivity between blocks of high-quality denning habitat. Additionally, roadless areas have considerable overlap with a number of identified key linkage areas that have already experienced significant cumulative disturbances from tree mortality and wildfire in recent years that may result in the creation of barriers to fisher movement and a loss of connectivity between habitat cores (Spencer et al. 2015, Thompson et al. 2020). In 2017 fisher were north of the Merced River for the first time in 100 years and since then fisher have been increasingly found using and reproducing in this area, representing a potentially important northward expansion. Home ranges in this expansion area overlap extensively with a designated roadless area.

Effects of Roads

Direct Effects

Vehicle collisions are a significant mortality factor for fishers both the SSN DPS and throughout their range. Between December 2007 and January 2013, a research study in the northern portion of the SSN DPS recorded 9 vehicle collision mortalities. Another study reported ten road-killed fishers have been found in Yosemite National Park over the past 2 decades and of the 10 road-killed fishers, 8 were killed during the denning season and of these, 4 were females, 2 were males, and 2 were of unknown sex. Of the 4 females killed during denning season, 2 were lactating (Spencer et al 2015) Another study reported that six percent of necropsied fishers in the SSN between 2007 and 2014 died from vehicular strikes (Gabriel et al. 2015).

A broader review of mortality causes assessed information on 239 radio-collared fishers that died among 7 study sites within the West Coast population segment between 1992 and 2014. Predation was the most common cause of death for radio collared fishers at 5 of the 7 sites,

which was linked to 67% of all known-cause deaths. Vehicle strikes occurred at 4 of 7 study sites and contributed to 8% of all known- cause deaths. Overall, they documented 24 roadkill deaths in the West Coast population segment during the study period, and fishers elsewhere in North America commonly die on highways (Sweitzer et al. 2016).

Indirect Effects

In addition to the risk of vehicular collisions, forest roads and trails may elevate fisher predation by mountain lions, bobcats, and coyote, the most common predators of fisher in the SSN DPS (Wengert et al. 2014), as all these species use trails and forest roads as travel and hunting corridors. Predation sites are closer to roads, on average, than sites where live fishers were detected. Bobcat and fisher interactions are more likely to occur near roads, hard edges (i.e., abrupt changes between closed and open habitat), open areas, and bushy areas (Spencer et al. 2015, Sweitzer et al. 2016).

SSN fisher have high levels of exposure to anticoagulant rodenticides and other toxicants which result in both direct mortality and harmful sub-lethal effects that impact health and behavior and have been linked to increased risk of predation and vehicle caused mortality. This toxicant exposure has been identified as a major threat to the SSS fisher DPS with numerous research studies documenting exposure rates of >80% to one or multiple toxicants based on the results of fisher necropsies (Thompson et al. 2014, Gabriel et al. 2015). The primary source of rodenticide exposure to fishers is from illegal marijuana grow sites on public, private, and Tribal lands in California and these grow sites are frequently found in fisher habitat. Grow sites also tend to be concentrated 1–2 km from roads, which probably facilitates grower access while avoiding places too close to roads (<1km) to minimize risks of detection (Wengert et al. 2021). Consequently, removing roadless protections may lead to increased use previously roadless areas for illegal cultivation potentially further increasing the risk of toxicant exposure for fisher.

In numerous studies fishers have been found to have a negative association with roads. Linden et al. (2017) found a both fisher occupancy and density had a strong negative relationship with roads, while Gompper et al. (2016) found that fisher distribution was negatively associated with paved roads. Ellington et al. (2017) reported a negative correlative trend between increasing road density and fisher patch use, although this relationship was only weakly supported. Ganoe et al. (2025) found that in areas with high human use fisher avoided roads during the day, and that r in areas of lower human use fisher selected areas far away from roads at all times, regardless of the time of day.

Roads have also been documented to negatively impact genetic connectivity for fisher. Garroway et al. (2011) found that road density, river density, and snow depth impeded gene flow during the fisher population expansion in Ontario, Canada. In the SSN DPS it has also been documented that major roads and rivers impeded gene flow of females, and that the influence of landscape features on gene flow is much greater for females than for males (Tucker et al. 2017).

Effects of Timber/ Vegetation Management

Direct Effects

Direct mortality from timber activities for fisher is difficult to detect or quantify due to the species' elusive behavior, large home ranges, low population density and habitat use that makes detection difficult (e.g. use of tree cavities, platforms or broken tops). Mortalities are unlikely to be detected if they occur in cavities or other inaccessible locations or carcasses may be consumed by scavengers, preventing detection. However, as fisher den inside tree cavities there is the potential for direct mortality from timber activities if a den tree is impacted during denning season when fisher kits spend most of the time in tree cavities.

Fisher can also be subjected to noise disturbance from timber activities which have the potential to disrupt denning fisher and potentially impact reproductive success. However, some fisher tolerance of timber activities has been documented even during denning periods. Smith et al (2025) reported a female denning in an active timber area with high levels of noise disturbance that continued to use that den tree for multiple weeks despite high levels of human activities in the vicinity. Yet, they also reported that fishers responded to anthropogenic stressors on an individual basis, where each animal displayed varying tolerance levels with some showing avoidance of treatment areas. Overall fishers retained the same home ranges after treatments occurred focusing activity in other non-disturbed areas of their home ranges. From a management standpoint, treatment size and configuration, management intensity, and management duration may influence observed differences in fisher behavioral responses to treatments (Smith et al. 2025).

Habitat Alteration

Extensive clearcut logging, along with fur trapping, during the late 1800's and early 1900's caused widespread declines in fisher populations across the country and by the early 1900's were extirpated from large portions of their historic range (Powell 1993, Ruggerio et al. 1994). Local extirpations partially attributed to habitat loss from logging and trapping have been documented across the country (e.g. Aubry and Lewis 2003, Williams et al. 2007, Kontos and Bologna 2008).

The Sierra Nevada DPS area experienced extensive habitat changes due to a combination of logging, trapping, and mining associated with the gold rush in the late 1800s. However, settlement at the southern end of the DPS was minimal in comparison due to the absence of gold deposits and steeper topography that restricted human access to forest lands (Beesley 1996). Historical sample locations and genetic analyses indicate that the SSN fisher DPS underwent a significant population decline and bottleneck associated with human development and habitat loss largely attributed to clear cut logging during and immediately after the goldrush and that the area at southern tip of the Sierra Nevada may have acted as a refuge for fisher maintaining a stable population size while fisher in the rest of SSN was in decline (Tucker et al. 2012).

It has been well documented that fishers avoid large open areas and select home ranges area with larger, more contiguous patches of mature forest and avoid high densities of open areas. Studies from both the northern Rockies and California have found that the median amount of open area in a fisher home range was ~5% (Sauder and Rachlow 2014, Raley et al. 2012). Increasing the amount of open area by an additional 5% was estimated to reduce the relative fisher probability

of occurrence by 39% and 50% in different studies (Sauder and Rachlow 2014, Weir and Corbould 2010). Fishers have been reported to avoid clear cut stands but also to continue to use home ranges with a low proportion of clearcuts (<15% of home range area) within them (Zielinski et al. 2013b). Fisher may tolerate some portion of the area treated within their home range, up to a threshold, to gain access to the untreated forest surrounding those areas (Garner 2013).

A number of studies have documented some tolerance of fisher to timber activities, including both non-commercial and commercial logging using forest management strategies that may mitigate timber impacts to fisher habitat. For example, fisher and red squirrel (a potential fisher prey species) habitat use increased with the use of retention harvesting which is the approach of leaving live mature trees behind during forest harvest. Researchers found that higher levels of retention harvest were characterized by greater canopy cover, basal area, and abundance of deadwood were associated with use of forest habitats by late-seral species such as fisher. Many late-seral species showed notable increases in activity between 20% and 50% harvest retention, and use of stands harvested with at least 50% retention was comparable to use of unharvested stands (Franklin et al. 2019).

Sauder and Rachlow (2014) studied fisher used of three comparative management landscapes: roadless and wilderness areas, industrial timberlands managed primarily for timber output, and managed forests with reduced focus on timber and more emphasis on wildlife habitat, recreation and other uses. They found landscapes managed primarily for timber production were not preferred by fisher with capture rates of fishers in industrial landscapes were less than half of those in multiple use forest. While they documented some fisher maintaining entire or significant portions of their annual home ranges in industrial forest, industrial forests as a whole were not in a configuration or pattern preferred by fisher. They also found lands managed as roadless/wilderness were at higher elevation and had significantly more open areas, less mature forest, and reduced proximity of mature forest patches than occupied fisher home ranges.

Zielinski et al. (2013b) investigated the amount of forest management (e.g., timber harvest, vegetation management, prescribed burning) that occurred in areas regularly used by fishers. Most treatment areas were multi-factor with only one area having timber extraction as the dominant activity. Fisher use was lowest in sample areas with above average area treated each year (49.3 treated ha/year) compared to areas with moderate or high fisher use (37.8 and 36.7 treated ha/year, respectively). This suggests that as the area treated increases, fishers may respond by using these areas less frequently. In addition, 3 of the 7 low use areas were dominated by an extractive activity which typically has more intensive effects on habitat elements of importance to fishers. They calculated that fishers consistently occupy at the highest rate of use – places where an average of 2.6% of a home range has been disturbed per year.

From 2009 to 2011 fishers were reintroduced to private timberlands on the Stirling Management Area owned by Sierra Pacific Industries in northern California. The Stirling Management area had a continuous history of selective logging and even aged management dating early 1900s such that by 2008 In 2008, a majority of Stirling (59%) was in a small-tree or moderately dense forest condition with only 23% large tree or moderately dense forest. This reintroduced population was monitored until 2017 and was found to be a well-established and growing population the end of

study providing evidence that fisher can persist in forests managed for timber production. The persistence of fishers on Stirling likely benefited from the retention and recruitment of forest conditions left as a legacy of historical selective management practices under contemporary forest management practices and conservation measures (USFWS 2008b).

Indirect effects

Similar to the potential indirect effects of roads listed previously, timber harvest can impact predator populations as the primary predators of fisher (e.g. bobcat, mountain lion, coyote) are more likely to occur near hard edges (i.e., abrupt changes between closed and open habitat), open areas, and bushy areas that may be created by some types of timber activities (Wengert et al. 2014). Prey populations can also be impacted due to changes in habitat due to logging, but the nature of these changes is variable depending on the type of logging practices uses (Fuller et al. 2004, Escobar and Estades 2021). Timber activities that create reduce forest cover also have the potential to negatively impact genetic connectivity as dense forest has been found to positively influence on genetic connectivity of female fishers (Tucker et al. 2017).

Existing Protections

The impacts of vegetation management, timber, and road or other infrastructure development are also regulated by land management directives. Management direction is guided by an interagency SSN Fisher Conservation Strategy that was first published in 2016. The Strategy was intended to meet the needs for fisher conservation and management for multiple agencies in the Sierra Nevada, including the USDA Forest Service, National Park Service, U.S. Fish and Wildlife Service, California Department of Fish and Wildlife (CDFW), Sierra Nevada Conservancy (SNC), and other local, state, federal, Tribal, and private entities whose actions may affect fishers or their habitat. The SSN fisher distribution is mapped into seven core areas that represent contiguous polygons of modeled suitable fisher habitat that is large enough to support $\geq$5 adult females and considers the results of spatially explicit habitat models and landscape genetic and occupancy patterns. The mapped core areas were designed to provide biologically relevant subdivision for planning and management purposes.

Due to the rapid pace and large scale of habitat changes in the SSN due to drought, tree mortality and wildfire that occurred after initial publication, the Conservation Strategy has undergone several updates with the most recent interim version released in 2020. This 2020 Conservation Strategy publication provides interim recommendations for forest management and restoration projects until a new, comprehensive fisher conservation strategy can be completed with updated habitat information.

Further land management directives are found in the individual land management plans for the Sierra, Sequoia (2023), Stanislaus (1991), and Inyo (2019) National Forests (available at USFS 2025a). Additional management direction for the Stanislaus National Forest, which has does not have a revised land management plan under 2012 Planning Rule, is provided in the 2001 Sierra Nevada Forest Plan Amendment (i.e. Sierra Nevada Framework. Land management plans for the Sequoia, Sierra, and Inyo are tiered the habitat definitions and core areas detailed in the Conservation Strategy, with some plan components applying to each core and or habitat

category, including fisher linkage or connectivity areas, known fisher den clusters, and other categories of fisher habitat.

Critical Habitat Determination

There are 392,167 acres (65.9%) of critical habitat on Forest Service lands of which 63,835 acres (10.7%) are in the action area. Of this 61% of the habitat is defined as operable or operable but complex. Critical habitat was defined as suitable, high-quality denning habitat that includes intermixed foraging and dispersal areas (87 FR 66987). Such habitat provides structural features for parturition, raising kits, protection from adverse weather conditions, facilitation of safe movement, sites to rest and thermoregulate, foraging opportunities, and cover to reduce predation risk for adults and young. Given this, the FS has determined the proposed action is likely to adversely affect designated critical habitat for this species.

Effects Determination

Roadless areas overlap with both critical habitat and broader denning, resting, and foraging habitat. These areas are potentially important to preserving the permeability of remaining foraging habitat that is critical for juvenile fisher dispersal and provide connectivity between blocks of high-quality denning habitat. Given this overlap, the FS has determined the exposure to the proposed action is high and that the proposed action is likely to adversely affect the Southern Sierra Nevada DPS of fisher.

Grizzly Bear

Grizzly bear *(Ursus arctos horribilis)* in the contiguous United States were listed as threatened in 1975 (40 FR 31734). While bears are listed in the contiguous United States, there are several different delineations within that that are important to their recovery. There are 6 different Grizzly Bear Recovery Zones, and these recovery zones are embedded within 6 larger ecosystems: Bitterroot (BE), Cabinet-Yaak (CYE), Greater Yellowstone (GYE), Northern Continental Divide (NCDE), Selkirk (SE), and North Cascades. While the boundaries of these ecosystems have not been officially defined, for the purposes of analyses conducted by the USFWS, they are the area surrounding bear recovery zones in which observed bears are "anticipated to occur as a part of the same population" (USFWS 2024l). Additionally, the bears were listed prior to the 1978 revisions to the Endangered Species Act, which created the concept of distinct population segments (DPS). Most recently, a proposed rule providing a geographically explicit definition of the Grizzly Bear DPS in the contiguous United States was published in the Federal Register in January 2025 (90 FR 4234).

Since listing, populations of bears in the GYE and NCDE ecosystems have tripled (USFWS 2024l). Populations in the SE and CYE are also thought to be increasing (USFWS 2024l). There are no known bear populations in the North Cascades and Bitterroot ecosystems (USFWS 2024l). According to the 2024 SSA, "the primary factors affecting grizzly bears at both the individual and ecosystem levels are excessive human-caused mortality and human activity that reduces the quality and quantity of habitats" (USFWS 2024l).

Exposure to the Proposed Action Area and the Proposed Actions

Grizzly bears use a diverse range of habitats with habitat size, degree of human access, and food availability being primary factors that determine whether an area can support a bear (USFWS 2024l). Today, grizzly bears occur in portions of Idaho, Montana, Washington, and Wyoming (USFWS 2024l). This is a large retraction from their historic distribution, prior to the arrival of European settlers, which included most of the western half of the contiguous U.S. (USFWS 2024l). Critical habitat was proposed for the species in the 1970s (41 FR 48757), but it was never finalized.

IRAs were mentioned in the 2024 SSA as being a conservation mechanism for grizzly bear populations (USFWS 2024l). Specifically, the SSA stated that "lasting" land designations such as Wilderness and IRAs "ensure that large proportions of recovery zones and additional areas outside the recovery zones remain secure for grizzly bears into the future without the development of new roads, extractive industries, or other human structures" (USFWS 2024l).

Currently, large portions of five out of six grizzly bear recovery zones are IRAs. The proposed action does not encompass IRAs in the portions of the Cabinet-Yaak, Bitterroot, and Selkirk recovery zones that are in Idaho (approximately 192 mi^2 in the Cabinet-Yaak and 248 mi^2 in the Selkirk recovery zones). In the Cabinet Yaak-Recovery Zone (total area: 2,589 mi^2), 785 mi^2 is 2001 Roadless Rule IRA. In the portion of the Selkirk Recovery Zone in the continental US (1,116 mi^2), 102 mi^2 is managed as an IRA under the National Roadless Rule (USFWS 2024). A large portion (1,978 mi^2) of the North Cascades Recovery Zone (9,777 mi^2) is also inventoried roadless (USFWS 2024l). Restricted motorized use areas, which include IRAs as well as Tribal Roadless Areas, Tribal Primitive Areas, and National Recreational Areas, protect approximately 19% of the Northern Continental Divide Recovery Zone (USFWS 2024). Additional management delineations for bears in the broader NCDE, just outside of the recovery zone, also include IRAs and additional plan components restricting motorized access in Zone 1 and the Demographic Connectivity Areas (DCAs). While 22% of the 9-mile Demographic Connectivity Area (DCA) is inventoried roadless (USFWS 2024l), plan components restrict motorized access through the entirety of the DCA. Estimates for the contributions of IRAs and other plan components to the GYE focus on "secure" habitat which meet standards for size, distance to motorized access, and other forms of human disturbance (USFWS 2024l). The 2024 SSA estimated that IRAs provide 864 mi^2 of secure habitat within the 9,210 mi^2 Greater Yellowstone Recovery Zone. For the GYE, there is also substantial suitable habitat outside of the recovery zone managed as an inventoried roadless area (2,386 mi^2/ 8,931 mi^2) (USFWS 2024l). We provide a brief overview of existing protections for bears in these areas in the "Existing Protections" section below.

In addition to these values from the literature, we also overlayed the entire UWFWS range of the species with inventoried roadless areas affected by the proposed action and found that approximately 11.8% of the range occurs in such areas. An analysis of grizzly bear occurrence records found that of 243 observations in inventoried roadless areas 153 of them (~63%) occurred within areas that were operable or operable, but complex.

Given that occurrence records are likely to be biased towards observations near existing infrastructure, the overlap of recovery zones and the species' range with inventoried roadless is likely the most informative assessment of exposure.

Effects of Roads

A recent review identified four mechanisms by which roads affect grizzly bears: human caused mortality, habitat displacement, habitat fragmentation, and direct habitat loss (Proctor et al. 2020). In addition to the impacts of roads, we also consider the potential indirect impacts of recreation and anthropogenically caused wildfire on bears, two activities which could increase from expansion of the road network.

Mortality

The 1993 revision to the Grizzly Bear Recovery Plan recommended that road management be given the highest priority in recovery zones as roads "probably" were the biggest threat to bears (USFWS 1993a). Grizzly bear mortality associated with roads can come through multiple mechanisms including direct collisions with vehicles and the negative human interactions that come with increased human use of grizzly bear habitat. Between 2002 and 2023, 57 bears in the demographic monitoring area (DMA) for the GYE and 90 bears in the NCDE DMA died from collisions with vehicles (USFWS 2024l).

"Accidental shooting, poaching, and habituation through direct human contact and/ or food reward all increase with the use of even secondary, unpaved roads by humans" according to the 1993 Recovery Plan (USFWS 1993a). Between 2002 and 2023, 194 bear deaths in the GYE DMA, 69 in the NCDE DMA, 6 in the CYE and 2 in the SE were classified as defense of life (USFWS 2024l). In that same timeframe, 27 bear deaths in the GYE DMA, 18 in the NCDE DMA, 4 in the CYE, and 5 in the SE were classified as cases of "mistaken identity" (USFWS 2024l). Other bears were removed by management after site conflicts with humans or due to concerns for human safety (128 in the GYE DMA and 89 in the NCDE DMA) (USFWS 2024l). One underlying factor influencing human-bear conflict can be food attractants, and many jurisdictions with grizzly bears have implemented food storage requirements to mitigate this risk (USFWS 2024l).

Several studies have shown that grizzly bear survival decreases with increasing road densities (Boulanger et al. 2013; Boulanger and Stenhouse 2014; Ciarniello et al. 2009; Proctor et al. 2017; Schwartz et al. 2010). Boulanger and Stenhouse (2014), showed that both sub-adult and adult female bears had lower survival rates as road densities increased and that adult males suffered similar mortality rates at all road densities. Importantly, of the 19 bear mortalities associated with roads (<500m), 18 occurred in places where the closest road was gravel (Boulanger and Stenhouse 2014). Similarly, bears inhabiting a mountain ecosystem in British Columbia were more likely to die if they were close to secondary or decommissioned roads (Ciarniello et al. 2009). Several other studies show that a large portion of bear mortalities occur within 500m or even 150m of roads (Benn and Herrero 2002; McLellan 2015) or that mortality risk increases as proximity to roads increase (Nielsen et al. 2004a). The impact of roads can have demographic effects- Boulanger and Stenhouse (2014) showed that local population growth decreased when road density exceeded 0.75 km/ km^2.

Displacement

Other studies have shown that bear densities are lower in areas with higher road densities (Mace et al. 1996; Graves et al. 2011; Lamb et al. 2018; Proctor et al. 2023) and higher in areas away from paved roads (Whittington et al. 2025). Densities were higher in areas with restricted vehicle use and highest in areas with high habitat quality and few roads (Lamb et al. 2018). Females in Alberta may select areas of high use within the home range that had lower road densities than the home range overall and select home ranges with larger proportions of land 500km from a road (Proctor et al. 2017). Several other studies suggest that bears select against areas with high road density, against areas of all tested road (Mace et al. 1999), and for roadless areas more than expected (Wakkinen and Kasworm 1997). Others show avoidance of habitat near roads (<500km) in spring and summer (Mattson et al. 1987). Females may remain further distances from roads than males (Chruszcz et al. 2003; Gibeau et al. 2002).

Contextual details about the level of vehicular activity may also affect grizzly bear's use of roads. Some studies suggest bears select against areas near open, as opposed to closed, roads (Kasworm and Manley 1990; Wakkinen and Kasworm 1997; Wielgus et al. 2002). Selection for or against closed roads may be sex-specific, with females showing avoidance even after accounting for habitat (Wielgus et al. 2002). That same study found that neither sex avoided roads used only for forestry after accounting for habitat (Wielgus et al. 2002). Another study found that selection for or against roads may be dependent on road type and road density in the study area, as well as the association of roads and other landscape features of interest (e.g. foraging opportunities) near roads (Ciarniello et al. 2007). Bear use of areas adjacent to roads may be dependent on traffic volume (Chruszcz et al. 2003, Mace et al. 1996; Northrup et al. 2012), with bears avoiding roads with higher volumes. However, in another study in the GYE bears avoided areas near roads regardless of traffic/ road class (McLellan and Shackleton 1988). Habituation also increases how close bears will come to roads (Chruszcz et al. 2003).

A study in Alberta showed that grizzly bears were more likely to cross "narrow unpaved roads" near creeks and open green spaces (Graham et al. 2010). Traffic volume and habitat quality influenced the likelihood of bears to cross roads in another study (Chruszcz et al. 2003). Bears may be more likely to cross roads at night (Waller and Servheen 2005; Northrup et al. 2012), and more likely to cross roads with low (<20 cars per day), traffic volumes (Northrup et al. 2012). While crossing roads, bears may move more quickly (Roever et al. 2010; Waller and Servheen 2005). Sub-adult bears may be more likely to cross roads than adult bears (Waller and Servheen 2005). Another study showed that while all bears avoided crossing highways, female avoidance was stronger than males (Palm et al. 2025).

While traffic is thought to be a deterrent to bears, some of the changes in vegetation that occur near roadsides may be an attractant to bears. For example, one study found that the herbaceous species that are preferred forage by bears in spring were more common near roadsides than in the forest (Roever et al. 2008). One study suggests that the strong association between roads and habitat features (i.e. cutblocks) that provide attractive food resources (Nielsen et al. 2004b) in an industrially developed site may lead bears to be near roads more than otherwise expected (Roever et al. 2010). Another study found that male bears would use areas near high trafficked roads when they were near high quality habitat, but that females would not (Gibeau et al. 2002).

Another study found that bears exhibited a functional response to roads, whereby their use of areas near roads was higher in low productivity areas (Palm et al. 2025). One study suggested that berry food resources may be higher in an area with human infrastructure (roads, towns etc.), but that mortality rates for bears in that same region are higher creating an evolutionary trap (Lamb et al. 2017). Another study found that the positive impacts of huckleberries on bear density and other parameters occurred primarily when food resources were in secure habitat away from roads (Proctor et al. 2023).

Habitat Fragmentation

There is research indicating that roads can fragment grizzly bear habitat, and that this in turn has an effect on genetic and demographic connectivity. Studies in the Southern Canadian Rockies and Northern American Rockies found that highways and areas of higher human development impeded migration of male and female bears, but that female bears were affected more strongly (Proctor et al. 2005; Proctor et al. 2012). There is some evidence that US Highway 2 and an associated railroad cause some fragmentation within the NCDE (Kendall et al. 2009), although bears have been documented crossing this corridor, dependent on traffic volume (Waller and Servheen 2005). Highway 2 also causes fragmentation between the Yaak and Cabinet mountains (Proctor et al. 2018).

Habitat Loss

The 4 ecosystems currently inhabited by grizzly bears have defined standards for core habitat that have a minimum standard for distance from road (USFWS 2024l; see below). By definition core habitat for grizzly bears is lost when a road is constructed, unless it is mitigated by the creation of new secure habitat elsewhere through road decommissioning or other actions.

Indirect Effects of Roads

If road networks are expanded, outdoor recreation in those areas is expected to increase. The primary concern for grizzly bears with recreation is that it will increase the likelihood of bears encountering humans, including encounters that lead to human-caused mortality (USFWS 2024l, see discussion above). Recreation could also displace bears from preferred habitat (USFWS 2024l and below), which could have energetic consequences (White et al. 1999). Typically bears avoid trails with motorized recreation (Graves 2002; Lamb et al. 2018). There is at least one study, however, that suggests bear responses to motorized recreation may be dependent on reproductive state. This study found that male bears and solitary female bears avoided trails with high levels of motorized recreation, but this was not seen in female bears with cubs who responded by moving more quickly (Ladle et al. 2019). The literature on non-motorized recreation and bears suggest that there are many contextual details that influence the way an individual bear may respond to humans recreating nearby. A few studies suggest that bears avoid non-motorized recreationists (Coleman et al. 2013a and Coleman et al. 2013b). Another study suggested that avoidance may be sensitive to the sex of the bear, time of day, and whether bears are near trails or backcountry campsites (Loggers et al. 2025). Additional work suggests avoidance behavior is more likely to be expressed when trails are near low quality habitat (Gibeau et al. 2002). Habituation may also influence the response of bears to non-motorized recreationists (Jope 1985; McLellan and Shackleton 1989; Mace and Waller 1996; Ladle et al. 2019). There is the potential for snowmobiling and other winter sports to disturb bears in dens,

affecting cubs or overwinter energetics (USFWS 2024l). Despite this potential, literature on this is scant (USFWS 2024l), although anecdotal evidence suggests that human activity including recreation could cause den abandonment (Swenson et al. 1997). According to the 2024 SSA, "grizzly bear mortality related to non-motorized recreation is rare and population-level impacts have not been documented."

Increased numbers of anthropogenically-caused fires associated with an expanded road network could affect bears. Despite the species' mobility, there is evidence that wildfires can cause bear mortalities (Blanchard and Knight 1990). However, bears have also been described moving through areas that were still smoldering and feeding on carcasses of prey that succumbed to the fires (Blanchard and Knight 1990). The years immediately post fire may also improve foraging opportunities for bears (USFWS 2024l).

Effects of Timber Harvest/ Vegetation Management

Timber harvest frequently occurs in grizzly bear habitat in the contiguous United States. Timber harvest has been shown to have both positive and negative effects on grizzly bears due to increases in road density and changes to habitat, specifically changes in food availability on the landscape (Colton et al. 2021). Changes in the selection or avoidance of timber harvest depend on the context of the larger landscape as well as human activity. This variability in the effects of timber harvest was recognized in the 2024 SSA which stated, "depending on the type of project, vegetation management can be beneficial, neutral, or harmful to grizzly bears." Research has shown consistently that timber harvest alone has minimal or positive effects on grizzly bears, but only if motorized access is controlled after harvest. The increase of roads from timber harvest and other resource extraction activities can have significant negative effects on grizzly bear mortality and population growth (Proctor et al. 2020).

Mortality Risk from Increases in Road Density from Timber Harvest

Timber harvest requires new roads or the repair and expansion of existing road networks. This increase in road density can lead to increased mortality of grizzly bears through the mechanisms described in the effects of road section above and specific research has shown that road increases from timber harvest can result in higher mortality risk for grizzly bears. Research in British Columbia, Canada compared a population of bears with access to higher quality food but higher road densities because of forestry activities to a population of bears in an area of mountains inaccessible to humans (Ciarniello et al. 2007). The density of bears in the area with better food was less than one quarter than the bears in the mountain area, likely due to a higher mortality risk from humans associated with higher road densities (Ciarniello et al. 2007). Other research in Canada found that bears were illegally killed from roads that were constructed for timber harvest and energy extraction (McLellan 1989) while reviews of grizzly bear mortality across Canada and the contiguous United States have found that human proximity and road access are linked with higher mortality rates across ecosystems (McLellan et al. 1999; Nielsen et al. 2004a).

Grizzly Bear Selection and Avoidance of Timber Harvest and Associated Activities

Bears can change habitat selection both in the short term, in response to logging activities, and in the longer term, in response to the changes in habitat created by timber harvest. The literature on short-term responses to logging activities is small but suggests the potential for negative impacts

to individuals. There is some evidence that bears avoid roads used by trucks hauling logs, and that displacement comes not just from the roads, but from the logging-associated vehicle activity on those roads. A small study of two bears in British Columbia showed that bears avoided the portions of their home range in which log trucks could be heard both during years when trucks were hauling logs and during times of day when trucks were hauling logs (Archibald et al. 1987). A radiotracking study of brown bears in the Greens Creek Drainage in Alaska (n=11) found that while bears remained in the vicinity of an active road construction site (for the mining industry) where blasting was occurring, their proximity to roads decreased during the day while work was ongoing and the number of day beds near the road construction site decreased compared to prior years at the site with no construction activity (Schoen and Beier 1988). In contrast, two observations reported in McLellan and Shackleton (1988) show bears remaining within 1km of heavy machinery use associated with logging. Displacement of bears by machinery is briefly discussed in McLellan and Shackleton 1989, and findings suggest bear responses to heavy machinery may be dependent on surrounding cover.

Grizzly bears show a variety of behavioral responses to areas subject to timber harvest, and these responses depend on the time of year, the amount of other disturbance on the landscape, the amount of bear food in timber harvest areas, and the amount of human activity (Colton et al. 2021). In the Rocky Mountains grizzly bears showed a wide variation in selection, neutral use (use of timber harvest as it was available on the landscape), or avoidance of timber harvest areas depending on the time of year. Two studies found avoidance of timber harvest areas in the spring (McLellan and Hovey 2001; Zager et al. 1983), while five studies found variable selection by individual bears (some mix of selection, avoidance, or neutral use) or neutral use of timber harvest areas (Kearney et al. 2019; Mace et al. 1996; Nielsen et al. 2004b; Stewart et al. 2012, Stewart et al. 2013). Only two studies found selection for timber harvest areas in the spring (Berland et al. 2008; Ciarniello et al. 2015). However, these same studies demonstrated changing habitat use of timber harvest in the summer and fall. Similar to spring, two studies showed avoidance of timber harvest in the summer (McLellan and Hovey 2001; Zager et al. 1983) while five studies showed selection for timber harvest (Ciarniello et al. 2015; Kearney et al. 2019; Mace et al. 1996; Nielsen et al. 2004b; Stewart et al. 2013) and two studies showed neutral or variable selection (Berland et al. 2008; Stewart et al. 2012). In the fall, results were different again with three studies showing avoidance (Ciarniello et al. 2015; McLellan and Hovey 2001; Zager et al. 1983), three studies showing selection (Kearney et al. 2019; Mace et al. 1996; Stewart et al. 2013), and three studies showing variable or neutral selection (Berland et al. 2008; Nielsen et al. 2004b; Stewart et al. 2012).

There are several factors that seem to contribute to this variation in selection, including sex specific selection, varying amounts of bear food in timber harvest units, and the amount of natural disturbance on the landscape (Colton et al. 2021). Several studies found evidence that male and female bears were using timber harvest differently. Stewart et al. (2012) found that male bears used clearcuts more in the spring and summer while females used clearcuts more in the summer and fall. In Alberta, females were also found to use edge habitat created by timber harvest and other human activities more than males, who were more likely to use edges created by natural disturbance (Stewart et al. 2013), while in Montana there was individual variation in selection or avoidance of edge habitat, with some males and females avoiding edges while most

females and some males selected for edge (Sells et al. 2022). Female grizzly bears also showed no shift in their home range due to timber harvest (Sorensen et al. 2015).

Grizzly bear selection of timber harvest is also affected by the amount of grizzly bear food that remains or regrows in logged areas. Bears in British Columbia were more likely to use clearcuts that had higher greenness values (Ciarniello et al. 2015), while researchers in in Alberta found that clearcuts had abundant bear food, although the types of plants that bears foraged were different than forested areas (Nielsen et al. 2004c). However, grizzly bear foods responded to timber harvest differently in different areas, and research southwest British Columbia and northwest Montana found that bears often did not select for timber harvest because of a lack of foraging opportunities and more often used natural disturbances (e.g., avalanche chutes) or moist meadows and riparian areas where preferred plant species were more abundant (McLellan and Hovey 2001; Servheen 1983; Waller and Mace 1997). Several studies have also found that attraction to the food contained in timber harvest areas attracts bears even with the roads associated with timber harvest (Proctor et al. 2020; Roever et al. 2010, Roever et al. 2008), raising concerns that timber harvest with unmanaged motorized access could create sink habitats, particularly for females that were more likely to cross roads than males in at least one study area (Graham et al. 2010). Other studies showed that grizzlies preferred timber harvest areas that had low use roads, fewer roads, or were more remote with less human access (Ciarniello et al. 2015; Mace et al. 1996; McLellan and Shackleton 1988), highlighting the variability in grizzly bear use of timber harvest areas between study areas.

Some of the variation in grizzly bear use of timber harvest between study areas seemed to be a result of landscape context, specifically the amount of natural disturbance on the landscape. In some areas with abundant avalanche chutes or recent wildfires bears preferred to use naturally disturbed areas for foraging (Mace et al. 1996; McLellan and Hovey 2001; Servheen 1983; Waller and Mace 1997), although some of these differences were seasonal. For example, in western Montana grizzly bears used avalanche chutes more than timber harvest in the spring but used timber harvest more in the summer and fall than in the spring (Mace et al. 1996). There is also evidence that bear use of timber harvest was higher in areas with less natural disturbance or in areas with less overall availability of green areas (Ciarniello et al. 2007; Nielsen et al. 2004b).

Effects of Timber Harvest on Grizzly Bear Food

Timber harvest had variable effects on plant species that were foraged by grizzly bears depending on the study area and post-harvest treatments. Some study areas found that plant species fed on by bears were abundant in timber harvest areas, although they typically differed in species composition and identity from forested areas (Apps et al. 2004; Nielsen et al. 2004c; Souliere et al. 2020), while others found that bear food was lower in logged areas (McLellan and Hovey 2001; Servheen 1983; Waller and Mace 1997). In areas where various berry species were important food items reducing forest canopy tended to increase berry production and abundance (Martin 1983; McLellan 2015). However, treatment after timber harvest had a large effect on the persistence of huckleberries and other berry crops, as well as several other plants eaten by grizzly bears. Scarification was found to reduce berries and other preferred plants in multiple other study areas (Larsen et al. 2019; Martin 1983; Nielsen et al. 2004b; Nielsen et al. 2004c). Planting after timber harvest was also found to speed up tree regeneration but led to a rapid decline in grizzly

bear foods because of rapid tree growth (Kearney et al. 2019; Proctor et al. 2018). Overall research supports landscape planning of timber harvest, controlling motorized access after harvest, and consideration of treatments post-harvest to maximize benefits to grizzly bears (Colton et al. 2021).

Existing Protections

There are numerous management directives that guide road use and road construction in grizzly bear habitat. As a result of the 1993 recovery plan (USFWS 1993a), and subsequent litigation, each recovery zone has secure habitat standards that define levels of motorized-access and these are codified as supplements to the 1993 recovery plan and recovery strategies. These standards are briefly summarized for each ecosystem below and they have (or will) be codified in forest plans (USFWS 2024l). These standards remain if the 2001 Roadless Rule is modified or rescinded although the 1993 Recovery Plan states that "the Recovery Plan recognized that grizzly bears would move and reside permanently in areas outside the recovery zones; however, only the area within the recovery zone is managed primarily for grizzly bear habitat" (USFWS 1993a, p. 18). Below we summarize secure habitat standards for each recovery zone. While some of these recovery zones are outside of the affected environment for the proposed action, because bears are wide-ranging species we have provided guidelines for all of them here.

GYE: Secure habitat is "any contiguous area ≥10 acres in size and more than 500 m from an open or gated motorized access route (road or trail), prescribed footprint of a developed site, or recurring low level helicopter line during the non-denning period (March 1 – November 30)." (YES 2024). Gated roads, closed year-round to the public but open to personnel are counted as motorized routes, however decommissioned roads are not (YES 2024). The amount of secure habitat must remain at or above levels in 1998 and can only be changed temporarily (or permanently) under certain conditions (USFWS 2007b; YES 2024). Thirty-seven out of 40 bear management units met these standards as of 2007 (USFWS 2024l). Since then, these standards have continued to be met in those 37 units and conditions are improving in the remaining three units (USFWS 2024l).

NCDE: Secure core habitat in the NCDE is at least 10.12 km^2 in size and more than 500m from a road open to motorized use during the non-denning season or a gated route (USFWS 2018e). Other types of roads, including those "restricted with physical barriers (not gates), decommissioned roads, impassable roads, temporary roads, over-the-snow motorized routes/areas, and non-motorized trails" are permitted in secure core habitat. In the NCDE, the amount of secure habitat, open road densities, and total road densities for bear management subunits and the recovery zone must meet or exceed 2011 baseline levels, and there are conditions that govern any temporary or permanent changes to roads and core habitat (USFWS 2018e). These standards are currently met within the NCDE (USFWS 2024l). Some additional areas outside of the recovery zone have restrictions on motorized use/ secure habitat incorporated in forest plans (USFWS 2024l).

CYE/ SE: Secure habitat is any habitat that is more than 500m from an open motorized road or a high use non-motorized trail (USFWS 2024l). The CYE and SE manage for secure core habitat, open road densities, and total road densities using thresholds that are set for each bear

management unit (USFS 2011). These standards affect a portion of land outside of the recovery zone, as well as within the recovery zone. According to the 2024 SSA, "although motorized access standards have contributed to a positive population trend in the CYE and SE, these standards have not yet been fully implemented."

BE: The overwhelming majority of the BE recovery zone is wilderness and motorized standards have not been established for adjacent areas that might facilitate natural recolonization (USFWS 2024l

North Cascades: In the North Cascades, core habitat is defined by the same 500m standard used in the CYE and SE. Management entities have committed to a "no net loss" of core habitat until further work is done to define seasonal habitat (USFWS 2024l).

In addition to these standards, the 2024 SSA summarizes regulatory mechanisms pertaining to grizzly bears (current as of December 31, 2023) that were considered for the most recent species status assessment (USFWS 2024l). This list, which encompasses everything from Forest Plans, to Conservation Strategies and State Management Plans is an additional resource for evaluating existing protections, although we note that some items (e.g. the 2001 Roadless Rule) would no longer apply under the proposed action. Given that these additional protections are diverse and site-specific they will aid project-level evaluation.

Species Effects Determination

In the 2024 SSA, two of the scenarios evaluated by the Services included loss of protected areas, including loss or partial loss of inventoried roadless protections (Scenario 1 and Scenario 2). The Services' analysis included loss or partial loss of wilderness protections under these scenarios as well. Under both scenarios, resiliency (and subsequently redundancy and representation) of some or all bear populations declined (USFWS 2024l), although the relative contribution of loss of IRAs was not analyzed. Given that there is a high degree of overlap between recovery zones, the full species range, and inventoried roadless areas and the abundant information regarding the impacts of road construction on grizzly bears, the FS has determined the proposed activity is likely to adversely affect this species.

San Joaquin Kit Fox

The San Joaquin kit fox is a subspecies of kit fox that is endemic to the San Joaquin Valley and adjacent valleys and foothills of California. Listed as endangered in 1967 (32 FR 4001), the San Joaquin kit fox is managed according to recovery plans issued in 1983 and revised in 1998 (USFWS 1983a, USFWS 1998a). Critical habitat has not been designated for the San Joaquin kit fox. No range-wide abundance estimates are available; population status is instead monitored using distribution and local monitoring studies (USFWS 2025h, Cypher et al. 2013).

Major threats to the species are land development, predation from other canid species, and disease (USFWS 2025h). Native habitat of San Joaquin kit fox has been reduced dramatically since the 1900s, mainly due to agricultural land conversion but also energy development and urbanization. Today San Joaquin kit foxes occur in fragmented habitat patches of varying quality. They exhibit classic metapopulation dynamics, in which productive patches create a

surplus of animals that disperse and sustain populations in lower quality patches. Habitat connectivity is accordingly deemed essential to long-term viability (USFWS 2020e).

Exposure to the Proposed Action Area and the Proposed Action

The San Joaquin kit fox is associated with arid scrub and grassland vegetation communities of its namesake and nearby valleys, although they can also be found in human altered habitats such as grazed grasslands, petroleum fields, and urban areas (USFWS 2020e). They tend to avoid steep terrain (>15%) and natal dens are limited to gentle slopes (<10%) with sandy-loam soils that facilitate den excavation (USFWS 2020e). Much of the remaining high-quality habitat occur in a narrow band rimming the San Joaquin Valley, with the highest concentration at the southwest edge of the Valley (Cypher et al. 2013). While we could not evaluate the relationship between the Cypher et al. 2013 habitat suitability model and IRAs due to accessibility of spatially explicit model results, we did evaluate the overlay of the UWFWS range map for the species and inventoried roadless. Just over 960 sq km of inventoried roadless areas occur within the range of the kit fox, most of it occurring around the southern rim of the San Joaquin Valley. This represents approximately 1.5% of the species' total range but given that much of the range is comprised of unsuitable habitat (Cypher et al. 2013) the proportion of suitable habitat that is inventoried roadless areas may be much higher. There are no records of the fox in the NRM or GBIF databases, making it impossible to use these records to assess the exposure of the species to the proposed action.

Effects of Roads

Potential direct effects of roads include vehicle strikes, habitat loss and degradation, loss of connectivity, and reduced fitness from disturbance. Potential indirect effects include reduced prey availability and increased competition or predation. Most research in the San Juaquin Valley has focused on the two-lane paved roads (Cypher et al. 2009) or multi-lane highways with medians (Bremner-Harrison et al. 2007) in urban environments. Considerably less data are available for single-lane forest and secondary roads, with past reviews presuming their population level effects to be negligible or highly localized (Bjurlin and Cypher 2003).

New road construction can remove or degrade suitable habitat as well as impede movement between habitat patches. These factors may have significant population-level effects in the case of multi-lane paved roads (Bjurlin and Cypher 2003). Secondary, forest roads are thought to be highly permeable to kit foxes (USFWS 2020e, Bjurlin and Cypher 2003).

A study of 67 radio collared kit foxes in Lokern County found no evidence for detrimental effects of kit fox exposure to 2-lane paved roads on survival, reproductive success, space use or dietary patterns (Cypher et al. 2009). Numerous studies have found that vehicle strikes account for a minor proportion of total kit fox mortality on natural lands, rarely exceeding 10% (Cypher et al. 2000, Bjurlin and Cypher 2003). Kit foxes commonly use disturbed and urban areas and have sustainable populations in Bakersfield and Taft, suggesting they acclimate quickly to human disturbance and can tolerate a high degree of human activity. However, a study of 22 GPS-collared kit foxes from the desert subspecies in Arizona found that individuals avoided areas with higher density off-highway vehicle roads during the winter breeding system, and that off-highway vehicle road networks were the most important predictor of space use (Jones et al.

2017). This suggests that at a certain threshold of off-highway vehicle road density, road effects extend beyond the immediate footprint and can constitute habitat degradation for kit foxes.

Little information is available on the effects of roads on the abundance and distribution of kit fox prey, predators, or competitors. According to one review (Bjurlin and Cypher 2003), disturbed habitats adjacent to roads could possibly favor California ground squirrels over kangaroo rats that are commonly consumed by kit foxes. However, kit foxes in the Lokern Natural Area with different degrees of road exposure exhibited no differences in prey composition detected in scats (Cypher et al. 2009). On the Naval Petroleum Reserves in California, shrub density increased along roads at which could provide cover for coyotes and bobcats that predate kit foxes (Bjurlin and Cypher 2003). Coyotes are a significant cause of mortality for kit foxes (Cypher et al. 2009) but the effect of roads on coyote distribution and abundance in the San Joaquin Valley is not well understood. Finally, roads could serve as travel corridors for the larger-bodied red fox, but a study quantifying niche overlap evaluated the threat from competition with red foxes to be relatively low (Cypher et al. 2022).

Effects of Timber/ Vegetation Management

As kit foxes predominantly use open vegetation communities with short stature, timber activities are unlikely to have detrimental population-level effects.

Existing Protections

There is no mention of the kit fox in the 2023 Sequoia National Forest Plan or the 2005 Los Padres National Forest Plan.

Species Effects Determination

Given that there is overlap of inventoried roadless and the kit fox range in southern portions of the San Joaquin Valley where kit fox habitat suitability is high, we have assigned the exposure level to the proposed action as high, especially considering that remaining habitat for the species is already fragmented. Given this and the potential impacts of roads on the species, the FS has determined the proposed action is likely to adversely affect this species.

Sierra Nevada red fox

The Sierra Nevada red fox is a montane subspecies of red fox that historically occurred throughout the mountainous regions of California and Oregon. It resides today in three geographically distinct regions: the Sierra Nevada, the vicinity of Lassen Peak, and the Oregon Cascades. The population residing in the Sierra Nevada was listed as an endangered DPS under the ESA in 2021 (86 FR 41743). At the time of listing, USFWS estimated the Sierra Nevada DPS to number fewer than 40 individuals (86 FR 41743) and modeled density of foxes inside the DPS region indicated a similarly low abundance (median 42 individuals; 95% CI 32–62; Green et al. 2023). More recent monitoring since 2018 suggests a growing population and expanding range extent, but still an absolute low abundance (Sacks and Quinn 2025; Quinn and Sacks 2022). The DPS is considered a single connected population with little to no internal genetic structuring and long-distance movements documented up to 120 km (Quinn and Sacks 2022; Hatfield et al. 2021).

The greatest identified risk factor to species viability are deleterious effects associated with small population size, including inbreeding depression and hybridization with other subspecies (86 FR 41743). Environmental and ecological threats to the species are not well understood, with few population-specific studies. Habitat was not considered limiting by USFWS and critical habitat has not been designated (86 FR 41743).

Exposure to the Proposed Action Area and Proposed Action

Sierra Nevada red foxes have historically been associated with upper montane, subalpine, and alpine life zones (Grinnell et al. 1937), which in the Sierra Nevada are almost exclusively federally managed. Of the 56 observations of this species in the NRM and GBIF databases, 70% of them have been in portions of IRAs classified as operable or operable but complex. Given the potential for bias in these datasets, we also did a custom analysis based on the probable occupied area, defined as the area encompassed by survey grid cells (4-mi^2) grid cells that contained verified red fox detections from 2010 to 2024. Of the 97 cells with a known red fox detection in the Sierra Nevada (Hatfield et al. 2025), 45 (46%) overlapped with potentially affected IRAs. Approximately 52 mi^2 of IRA environment occurred within these cells, representing 13.4% of the total area of occupied cells. The Dardanelles, Tragedy-Elephants Back, Carson-Iceberg, Iceberg-Mill Creek, and Long Meadow IRAs overlapped detection areas that have supported breeding pairs in the monitored portion of the population (Sacks and Quinn 2025; Quinn and Sacks 2022, Quinn et al. 2019). Additionally, we conducted an analysis overlapping the entire UWFWS range of the Sierra Nevada red fox with the proposed action area and found that approximately 12.2% of these lands were affected.

Red fox habitat in the Sierra Nevada is typified by open whitebark pine and mountain hemlock forest, scrub, patchy meadows, and barren rock and talus fields (Sierra Nevada Red Fox Conservation Advisory Team 2022; USFWS 2018f). Most contemporary detections in the Sierra Nevada are concentrated near or above tree line (mean elevation 3,173 m, range = 2,607–3,651 m; Sierra Nevada Red Fox Conservation Advisory Team 2022), with average tree canopy coverage only 10% of occupied 4-mi cells (Sierra Nevada Red Fox Conservation Advisory Team 2022). The single confirmed denning site in the Sierra Nevada was in an isolated whitebark pine stand at 3,100 meters (Sacks 2025); two verified denning sites in the Lassen population were also located in scattered subalpine stands at approximately 2,800 m in elevation (Sierra Nevada Red Fox Conservation Advisory Team 2022). Critical habitat has not been designated for the species.

Effects of Roads

Seven of 11 verified mortalities of the Sierra Nevada red fox subspecies collected between 2010 and 2020 were caused by vehicle strikes; six in the Oregon Cascade population and one in the Sierra Nevada DPS (Sierra Nevada Red Fox Conservation Advisory Team 2022). However, all seven mortalities occurred on paved roads with two or more lanes. The Sierra Nevada Red Fox Conservation Advisory Team (2022) stated vehicle strikes as "the most prevalent cause of mortality for [Sierra Nevada red fox] identified to date" and recommended minimizing development of new roads that would result in "substantially increased human or motor vehicle presence in areas used by [Sierra Nevada red fox]. Unpaved forest roads are unlikely to pose a barrier to genetic connectivity within the DPS. In Oregon, cameras placed on trails and forest roads had the highest probability of detection (Mahmoud 2025). A landscape genetic study of the

closely related high-elevation Cascade red fox subspecies found no support for roads (paved or unpaved) impeding gene flow (Akins 2017).

Possible indirect effects associated with increased recreation activities could include elevated risk of mortality from pet aggression, disease transmission, vehicle strikes, and other human-wildlife conflicts (Sierra Nevada Red Fox Conservation Advisory Team 2022, Perrine et al. 2010). Red foxes are prone to habituation to humans through food conditioning and have been shown to alter their spatial behavior in areas of concentrated human recreation, presumably to take advantage of anthropogenic food subsidies (Burkholder et al. 2024, Burkholder et al. 2025). Sierra Nevada red foxes have been observed displaying begging behavior toward campers, including within the Sierra Nevada DPS, and human food wrappers have previously been detected inside their scat (Sierra Nevada Red Fox Conservation Advisory Team 2022).

New road construction could facilitate incursion of coyotes, a dominant competitor of Sierra Nevada red fox (USFWS 2018f), into high-elevation Sierra Nevada red fox habitat (Gese et al. 2013). A study in Montana found that coyotes travel on forest roads in snowy winter conditions more than expected based on their availability, even when overlying snow was no more supportive than the surrounding environment (Kolbe et al. 2007). The authors suggested that in addition to compaction, coyotes select for roadbeds as a cleared travel corridor. Similarly, road development has been proposed as a factor that could possibly facilitate incursion of non-native red foxes that are more tolerant of human-disturbed environments into high-elevation Sierra Nevada red fox habitat (Perrine et al. 2010). The USFWS considers genomic integrity vital to the long-term viability of the DPS (USFWS 2018f).

Effects of Timber/ Vegetation Management

Present day habitat associations of Sierra Nevada red fox have minimal overlap with timber zones traditionally targeted for commercial harvest and fuels management (Perrine et al. 2010). However, historical observations in the Sierra Nevada extended lower than they are detected presently (down to 1,370 m; Grinnell et al. 1937), and Sierra Nevada red foxes in the Lassen and Oregon Cascade populations are more frequently detected in mid- and upper-montane forests of red fir, lodgepole, and Jeffrey Pine (Sierra Nevada Red Fox Conservation Advisory Team 2022). This differential use of forest across time and space could mean that current habitat associations in the Sierra Nevada may not represent the full suite of habitats they could potentially use.

Indirect effects of timber harvest and fuels management are unknown but are most likely to be mediated through altered species interactions. Timber practices that shift prey abundance or composition could be beneficial or detrimental, depending if changes have a greater effect on the prey base of red foxes or of generalist competitors like coyotes and gray foxes (Perrine et al. 2010). A recent diet study in the Sierra Nevada DPS suggests that snow-adapted subalpine prey, such as white-tailed jackrabbits, snowshoe hares, and pika may play an outsized role in niche differentiation and long-term coexistence with coyotes (Rosburg-Francot et al. 2025).

Existing Protections

No specific management conditions for the Sierra Nevada red fox have been specified in any existing forest plans.

Species Effects Determination

Given the overlap of verified detections of the Sierra Nevada red fox with inventoried roadless areas and lands within 2 km of inventoried roadless areas, the exposure the species to the proposed action is high. Given this exposure, and the documented impact of roads on the species, the FS has determined the proposed activity is likely to adversely affect this species.

Table 6. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of mammal species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of nest/den trees and snags	Direct mortality; Decreased reproductive output; Habitat fragmentation; Reduction in Connectivity; Loss of habitat Destruction of burrow / den site / roost site; Behavioral avoidance of humans; Increased potential for human conflict; loss of prey
		Tree and Unit Marking	Walking	Human presence	Behavioral avoidance of humans Increased potential for human conflict
			ATV	Noise; Human presence; Trampling/run over	Behavioral avoidance of humans; Increased potential for human conflict; Destruction of burrows
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over	Destruction of burrow: Behavioral avoidance of humans; Loss/fragmentation of habitat, direct mortality
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Loss or gain of habitat; Behavioral avoidance of humans; Increased potential for human conflict; Direct mortality; Increased exposure to toxicants; Loss of forage; loss of prey
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over	Direct mortality; Decreased reproductive output; Habitat fragmentation; Reduction in Connectivity; Loss of habitat; loss of prey; Destruction of burrow / den site / roost site; Behavioral avoidance of humans; Increased potential for human conflict
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans: Increased potential for human conflict
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence	Destruction of burrow / den site / roost site; Behavioral avoidance of humans; Increased potential for human conflict

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans; Increased potential for human conflict
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans; Increased potential for human conflict
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; chemical contamination/nutrients; Drainage Network; Peak/Baseflows	Behavioral avoidance of humans; Increased potential for human conflict; Increased exposure to toxicants; migration impediment
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality Decreased reproductive output Habitat fragmentation resulting in reduction in movement and gene flow Habitat fragmentation - negative effects on individuals/population Loss of habitat Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict Loss of prey species Loss of forage Introduction of competitors Introduction of predators

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site Direct mortality Decreased reproductive output Habitat fragmentation resulting in reduction in movement and gene flow Habitat fragmentation - negative effects on individuals/population Loss of habitat Behavioral avoidance of humans Increased potential for human conflict Loss of prey species Loss of forage Introduction of competitors Introduction of predators
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Behavioral avoidance of humans Increased potential for human conflict
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Behavioral avoidance of humans Increased potential for human conflict
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over	Behavioral avoidance of humans Increased potential for human conflict Increased exposure to toxicants
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Behavioral avoidance of humans Increased potential for human conflict. Exposure to toxicants

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Snow	Plowing	Heavy Equipment	Noise; Soil and snow compaction; Human presence; Trampling/run over	Direct mortality Behavioral avoidance of humans Increased potential for human conflict
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat	Direct mortality Decreased reproductive output Habitat fragmentation resulting in reduction in movement and gene flow Habitat fragmentation - negative effects on individuals/population Loss of habitat Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict Loss of prey species Loss of forage Introduction of competitors Introduction of predators
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence; Loss of nest trees	Loss of habitat Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Loss of nest trees; Soil compaction; Trampling/run over	Loss of habitat Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; Loss of nest trees	Direct Mortality Behavioral avoidance of humans Increased potential for human conflict
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct Mortality Behavioral avoidance of humans Increased potential for human conflict
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust;	Loss of habitat Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal	Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over	Destruction of burrow / den site / roost site Behavioral avoidance of humans Increased potential for human conflict
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict Direct Mortality
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Destruction of burrow / den site / roost site; Behavioral avoidance of humans Increased potential for human conflict Habitat gain

Reptiles

The reptile group includes 16 species that occur within the action area, including 3 Endangered, 9 Threatened, and 4 Proposed Threatened reptile species. The reptile group includes turtles and tortoises (9 species) that are distantly related to snakes and lizards (5 squamate species). The two tortoise species are fully terrestrial species while the 2 alligator snapping turtle species and the flattened musk turtle are fully aquatic, except for females leaving the water to lay eggs terrestrially. The remaining 6 turtle species in this group are considered semi-aquatic to mostly aquatic, spending most of their time in or near water but always nest in terrestrial habitat. This dependency on nesting in terrestrial habitat makes these species vulnerable during their respective nesting seasons. The narrow-headed gartersnake and Northern Mexican gartersnake are highly aquatic while the remaining snake and lizard species are fully terrestrial. Two similarity of appearance (SAT) species, the American alligator and bog turtle are afforded protections but were excluded from the analyses because they are not Threatened, Endangered, or Proposed T&E species. Similarity of appearance listed species are primarily to protect a species that looks similar to the SAT species and meets the definition of threatened or endangered. Our effects analyses do not rely on proper identification of individuals; if relevant and necessary, effects to SAT species will be considered during project-level consultation. The green sea turtle Kemp's ridley sea turtle were also excluded due to lack of overlap with the proposed action.

Table 7. Threatened (T), endangered (E), and proposed threatened (PT) reptile species in the action area. The presence of critical habitat (and its percent overlap with the action area) is also provided. Species and critical effects determinations are included (no effect, NE; not likely to adversely affect, NLAA; likely to adversely affect, LAA; and not applicable, n/a).

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Alligator snapping turtle	*Macrochelys temminckii*	PT	n/a	Low	LAA	n/a
Blunt-nosed leopard lizard	*Gambelia silus*	E	n/a	Moderate	LAA	n/a
Desert tortoise	*Gopherus agassizii*	T	0%	Low	LAA	NE
Eastern indigo snake	*Drymarchon couperi*	T	0%	Low	LAA	NE
Eastern massasauga	*Sistrurus catenatus*	T	0%	Moderate	LAA	NE
Louisiana pinesnake	*Pituophis ruthveni*	T	0%	Low	LAA	NE
Narrow-headed gartersnake	*Thamnophis rufipunctatus*	T	Y (14.18%)	Moderate	LAA	LAA
New Mexican ridge-nosed rattlesnake	*Crotalus willardi obscurus*	T	0%	Unk	LAA	NE
Northern Mexican gartersnake	*Thamnophis eques megalops*	T	0%	Unk	LAA	NE

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Northwestern pond turtle	*Actinemys marmorata*	PT	n/a	High	LAA	n/a
Puerto Rican Boa	*Chilabothrus inornatus*	E	n/a	Unk	LAA	n/a
Sand skink	*Neoseps reynoldsi*	T	n/a	Unk	LAA	n/a
Short-tailed snake	*Stilosoma extenuatum*	PT	n/a	Low	LAA	n/a
Southwestern pond turtle	*Actinemys pallida*	PT	n/a	High	LAA	n/a
Suwannee alligator snapping turtle	*Macrochelys suwanniensis*	T	n/a	Unk	LAA	n/a

Exposure to the proposed action

Exposure of ESA-listed reptile species was estimated from the existing wildlife databases; however, reptile species representation in the database is relatively scarce, resulting in many species having an unknown potential for exposure. Where exposure is unknown the Forest Service assumes some level of exposure is possible.

Response to the proposed action

While habitat fragmentation and habitat degradation affect reptiles similarly to other threatened and endangered species, reptiles have a distinct and much higher likelihood of being struck and killed by vehicles. Individual mortality is recognized as the most adverse response possible. Other effects, such as behavioral responses, will be less impactful to the species as listed. As such, most of the discussion below is focused on the heightened risk of mortality from vehicle strikes.

Fully terrestrial turtles

Desert tortoise (*Gopherus agassizii*)(T)

The desert tortoise is a long-lived species that inhabits the arid southwestern United States. The tortoise occurs on three national forests across the Forest Service Intermountain and Pacific Southwest Regions. Nafus et al. (2013) provide evidence that a reptile with a slow life history (for example, desert tortoises) is susceptible to road presence and that the effect increases with traffic volume. Boarman & Sazaki (2006) suggest that desert tortoise populations are depressed in a zone extending at least 400m from roadways. They speculate that the major cause for this depression zone is road mortality. There is evidence that paved roads and traffic have a detrimental effect on desert tortoise populations, up to 1 km off the road (Bury 1978).

Existing Protections

The Angeles and San Bernadino National Forest Plans states that the Desert tortoise occurs on the districts but does not list any protections. The tortoise is not mentioned in the Humbolt NF Forest Plan.

Semi-aquatic and aquatic turtles

Semi-aquatic and aquatic turtles (7 species) leave water for dispersal and often cross roads for nesting. A model by Gibbs and Shriver (2002) predicted that road networks typical of the northeastern, southeastern, and central regions have the potential to limit land-turtle populations and, to a lesser extent, populations of large-bodied pond turtles. Gibbs and Steen (2005) demonstrated that sex ratios are heavily biased towards male turtles in places with higher road density implying that roads are causing the depletion of breeding females through road mortality. Patrick and Gibbs (2010) showed that freshwater turtle sex ratio (female proportion) and size were negatively correlated with proximity to roads. Their study suggests that roads alter both local- and landscape level turtle populations through a loss of female turtles, and by reducing movement between ponds.

Northwestern pond turtle (*Actinemys marmorata*)(PT)

The northwestern pond turtle can be found in 16 different National Forest districts across the Pacific Southwest and Pacific Northwest Regions of the Forest Service. The species is considered a typical member of the family Emydidae (pond turtles), is a diet generalist and inhabits a variety of permanent and ephemeral aquatic habitats (Holland 1994; Manzo et al. 2021). Road mortality is a risk but ranked lower than many other threats. Male-biased population structure has resulted from road mortality of nesting females, and other threats (Madden-Smith et al. 2005). Nicholson et al. (2020) found museum specimens are male biased over the last century. Northwestern pond turtles can be especially vulnerable to road mortality because they can travel hundreds of meters from aquatic habitat to nest.

Existing Protections

Six Rivers NF Forest Plan states that the district maintains habitat characteristics consistent with their Habitat Capability Model within 300 feet around occupied pond and stream habitats. The Gifford Pinchot NF and the Siuslaw NF Forest Plans list the Northwestern pond turtle as a species as occurring on the districts but do not list any protections. The turtle is not mentioned in any of the other Forest Plans where it occurs.

Southwestern pond turtle (*Actinemys pallida*)(PT)

The southwestern pond turtle occurs on two National Forests in FS Region 5. This species has recently been designated as a separate species from *the northwestern pond turtle* (80 FR 19259). Threats should be similar.

Existing Protections

The Angeles NF Forest Plan states that protection and enhancement of threatened, endangered, proposed, candidate and sensitive species including the western pond turtle will be emphasized in all activities and habitat protection is a priority task. The Los Padres NF Forest Plan prioritizes

the importance of riparian and aquatic species habitat and control of invasive species and habitat restoration for species including the western pond turtle.

Alligator snapping turtle (*Macrochelys temminckii*)(PT)

The alligator snapping turtle is the largest North American freshwater turtle species. It occurs in drainages that flow into the Gulf of America and occurs in 22 National Forest Districts in the Southern and Eastern Regions of the Forest Service. Rosenbaum et al. (2023) provided evidence that the species has a higher occurrence probability in waters of highly forested catchments and are associated with highly turbid water, therefore sediment runoff should not be a concern for this species. Riedle et al. (2006) found that alligator snapping turtles were always associated with submerged cover with a high percentage of overhead canopy cover. Like most turtles, there is a risk of mortality from vehicle collisions.

Existing Protections

The alligator snapping turtle is not mentioned in any of the Forest Plans where it occurs.

Suwannee alligator snapping turtle (*Macrochelys suwanniensis*)(T)

The Suwannee alligator snapping turtle occurs in the Osceola National Forest in the Southern Region of the Forest Service. Males and females have similar survival probabilities suggesting that female are not at greater risk to roads or other factors (Thomas et al. 2022), however Shook et al. (2023) reported that there are documented cases of road mortality in both species of alligator snapping turtles. This species has recently been designated as a separate species from the alligator snapping turtle. Threats should be similar.

Existing Protections

The Suwannee alligator snapping turtle is not mentioned in the Osceola Forest Plan.

Terrestrial snakes

Terrestrial snakes are vulnerable to roads. Roads can be hazardous to snakes and act as barriers to movement. The are interspecific differences in the effects. Some smaller species tend to avoid crossing compared to larger species (Andrews and Gibbons 2005). Wagner et al. (2021) results indicate that high-traffic roads pose a risk to snake populations, but that mortality also occurs on low-traffic roads as well. Large-bodied highly mobile snake species are more likely to encounter roads and are at a higher risk to vehicle mortality.

Eastern indigo snake (*Drymarchon couperi*)(T)

The eastern indigo snake is the largest snake species native to the United States. It occurs in 5 National Forests in the Southern Region of the Forest Service. Hyslop et al. (2009) stated that indigo snakes are frequently killed on roads. Eastern indigo snakes disproportionately use areas near roads and trailways, making them vulnerable (Metcalf 2017).

Existing Protections

The Forest Plan for the National Forests in Florida consider the eastern indigo snake a wildlife species of interest and a target species for reintroduction and augmentation. Guidelines for protection of the eastern indigo snake are found in the Eastern Indigo Snake Recovery Plan,

USFWS. Specific standards and guidelines include, protecting indigo snakes and gopher tortoises encountered by personnel, cooperators, or contractors engaged in activities that may endanger individual specimens. Wildlife biologist should be contacted to safely move these species and collect needed data. The eastern indigo snake is mentioned as occurring in the Conecuh NF Forest Plan but not mentioned in the Desoto NF plan.

Eastern massasauga (*Sistrurus catenatus*)(T)

The eastern massasauga is a small venomous snake species that occurs in two National Forests in the Eastern Region of the Forest Service. Adult massasaugas (are less likely to cross roads in comparison to neonates (Seigel and Pilgrim 2002). Bailey et al. (2011) reported that road mortality does occur in eastern massasaugas. Baker et al. (2016) demonstrated that road kills are a primary source of mortality for massasauga. Shepard et al. (2008) found road mortality was highest in April and October and positively associated with minimum daily temperature. They recorded 42 cases of road mortality of eastern massasauga with the highest number occurring from mid-August to mid-September. Road mortality in eastern massasauga was biased toward adult males, which show an increase in movement in August, coinciding with the peak of the mating season.

Existing Protections

The Huron Manistee NF Forest Plan lists management recommendations for eastern massasauga. Key strategies include avoid mowing, heavy equipment operation, and burning during peak activity periods (spring emergence, late summer birthing, and autumn migration; Restrict burns to early spring before snake emergence, when ambient temperatures remain low (< 10° C) or snakes are inactive, ensuring sufficient unburned refugia remain; and maintain a mosaic of open-canopy basking sites and adjacent dense vegetation, which allows thermoregulation while ensuring adequate cryptic cover. The eastern massasauga is not mentioned in the Midewin National Tallgrass Prairie Plan.

Louisiana pinesnake (*Pituophis ruthveni*)(T)

The Louisiana pinesnake is a large non-venomous snake species that occurs in 3 National Forests in Texas and Louisiana, Southern Region of the Forest Service. Declining habitat quality (fire suppression) is a contributor to decline by indirectly affecting the abundance of prey species, the pocket gopher (Rudolph et al. 2018). Rudolph et al. (2018) also reported that roadkills are a source of mortality for Louisiana pinesnakes. Zappalorti (2016) suggests that among the various reasons for their decline is the construction of new roads through western Louisiana and eastern Texas, coupled with the advent of improved, well-traveled secondary roads that are accessible to all types of vehicles. Himes et al. (2002) suggest that recreational ATV use was the cause of at least one Louisiana pinesnake mortality in their study.

Existing Protections

The Kisatchie NF Forest Plan promotes using fire to achieve desired future conditions; frequently burned old-growth conifer communities that benefit wildlife species including the Louisiana pinesnake. The Texas Forest plan promotes increasing longleaf habitat through restoration and management to provide contiguous habitat for the pinesnake.

Puerto Rican Boa (*Chilabothrus inornatus*)(E)

Puerto Rican boas are the largest native snake species in Puerto Rico and occur in El Yunque National Forest in the Southern Region of the Forest Service (Rivera 2023). The Puerto Rican boa recovery plan (USFWS 1986a) suggests that the species was once more common but deforestation in the 1800s may be the reason for historical declines. Reynolds et al. (2016) documented significant road mortality in the genus (closely related Bahama boa). It is expected that road construction and timber activity will be low on El Yunque but given the lethal impact vehicles have on snake species the Forest Service believes roads could be a threat to the Puerto Rican boa.

Existing Protections

The El Yunque NF Forest Plan mentions providing habitat for the Puerto Rican boa but fails to provide details.

Short-tailed snake (*Stilosoma extenuatum*)(PT)

The Short-tailed snake is a small fossorial species that occurs in the Ocala National Forest in the Southern Region of the Forest Service. This species has a small geographic range (Gibbons and Buhlmann 2001), but there is not much literature on this species. The Listing Rule (59 FR 34270) states that loss and degradation of habitat from urbanization and other historical and ongoing land use changes (such as agriculture and mining) and lack of habitat management (such as lack of prescribed fire in an ecosystem appropriate interval) are potential causes of decline.

Existing Protections

The short-tailed snake is not mentioned in the Ocala NF Forest Plan.

New Mexican ridge-nosed rattlesnake (*Crotalus willardi obscurus*)(T)

The New Mexican ridge-nosed rattlesnake is a small venomous species that occurs in the Coronado National Forest in the Southwestern Region of the Forest Service. Low frequency, high intensity fire events may cause indirect mortality effects on rattlesnakes that extend for some time following the fire event (Davis 2008). Supplemental finding for the New Mexican ridge-nosed rattlesnake recovery plan (USFWS 2019e) states that there is so little information on the species that the threats are not well understood, however populations are so small and isolated that the species is likely in an extinction vortex towards extinction.

Existing Protections

The Coronado NF Forest Plan mentions that the New Mexican ridge-nose rattlesnake occurs in the Madrean pine-oak woodland communities but does not mention any protections.

Aquatic snakes

Narrow-headed gartersnake (*Thamnophis rufipunctatus*)(T)

The narrow-headed gartersnake occurs in 5 National Forests in the Southwestern Region of the Forest Service. It is a highly aquatic species that requires riffle zones to forage for fish (Fleharty 1967). Hibbitts et al. (2009) could not confidently identify the cause of decline at their study site but suggested that changes in prey availability might be responsible. Rosen et al. (2012) stated

that data do not provide support for the hypothesis that post-fire siltation has long-term impacts on narrow-headed gartersnake population status, however direct human killing is a primary source of mortality, implying that increased human presence may me impactful. Holycross and Rosen (2011) also suggested that currently, the main threats come from collecting and killing by humans.

The narrow-headed gartersnake is the only reptile species in the action area with designated critical habitat. More than 14% of designated critical habitat for this species occurs in the action area. The physical and biological features of narrow-headed gartersnake are primarily related to the aquatic environment (water flow and stream structure) which will not be affected by the proposed activity in a significant manner. However, riparian vegetation is also a component of critical habitat for this species, and although impacts to riparian vegetation are likely to be mitigated at the project-level, we cannot discount the potential for effects to occur. As such, the FS has determined designated critical habitat for this species is likely to be adversely affected.

Existing Protections

The Apache NF Forest Plan mentions desired future condition including habitat for the snake, does not mention any no specific protections. The Coconino NF Forest Plan management approach includes collaborating with the Arizona Game and Fish Department to educate residents about Forest Service sensitive species such as the narrow-headed gartersnake. The Gila NF Plan states the forest will Accomplish threatened and endangered species habitat improvements as identified through approved management and recovery plans. The gartersnake is not mentioned in Prescott and Tonto NF Forest Plans.

Northern Mexican gartersnake (*Thamnophis eques megalops*)(T)

The Northern Mexican gartersnake is found in 5 National Forests in the Southwestern Region of the Forest Service. It occurs in riparian areas in central and southeast Arizona and west-central New Mexico (Bauder et al 2024). Primary threats and contributors to the decline of the species include modification of riparian and aquatic habitats from water diversions, habitat loss, loss of native prey, and competition and predation by nonnative species (Rosen and Schwalbe 1988; Holm and Lowe 1995; USFWS 2014a; Boyarski et al. 2019).

Existing Protections

The Coconino and Prescott NF Forest Plans make general statement regarding improving general habitat and occurrence of Northern Mexican gartersnakes. The Apache, Coronado, Gila, and Tonto NF Forest Plans do not mention the gartersnake.

Lizards

Blunt-nosed leopard lizard (*Gambelia silus*)(E)

The blunt-nosed leopard lizard is a relatively large lizard that inhabits the San Joaquin Valley and surrounding foothills. It occurs on the Los Padres National Forest in the Pacific Southwest Region of the Forest Service. Germano and Williams (2005) found three adults (two males, one female) on the dirt roads in their plots that appeared to have been killed by vehicles. Germano et al. (2024) suggest that high-density oil fields likely have a profound negative impact on *blunt-*

nosed leopard lizard because of the great distances moved by individuals that causes them to cross many highly trafficked roads contained in high-density operations. There was a significant positive relationship between habitat patch size and presence of blunt-nosed Leopard Lizards (Bailey and Germano 2015).

Existing Protections

The blunt-nosed leopard lizard is not mentioned in the Los Padres NF Forest Plan.

Sand skink (Neoseps reynoldsi)(T)

The sand skink is a small fossorial lizard that inhabits sandy soils. It can be found in the Ocala National Forest in the Southern Region of the Forest Service. Gianopulos et al. (2001) found that sand skink populations increased over a 5-year period following a clearcut. In general, more sand skinks were found in larger habitat patches, regardless of whether patches were characterized by vegetation or soil (Pike et al. 2007). Richmond et al. (2009) showed that high contrast boundaries created by roads, citrus groves, and other urban development almost certainly constitute low-permeability barriers for a sand-swimming lizard.

Existing Protections

The Ocala NF Forest Plan states that to protect sand skink habitat and aid in the recovery of this species, the Forest Service follows the guidelines found in the Recovery Plan for the Sand Skink and Blue-Tailed Mole Skink, USFWS. Specific guidelines include: Maintain ecotonal areas between longleaf pine-turkey oak and sand pine scrub by prescribed burning every 2-7 years.

Effects Determination

Due to a lack of exposure data for a majority of reptile species, the FS is assuming potential exposure to the proposed action for many of these species. Exposure is assumed to occur if both the species and IRA occur on the same forest. All of the listed reptile species are particularly susceptible to being struck and killed by vehicles in addition to the stressors that also affect other wildlife species (habitat fragmentation, habitat degradation, invasive species, etc.). Based on assumed exposure and the likelihood of mortality from vehicle strikes, the FS has determined the proposed action is likely to adversely affect ESA-listed reptile species that occur in the action area. The FS has determined designated critical habitat for the narrow-headed gartersnake is also likely to be adversely affected.

Table 8. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of reptile species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of trees and snags; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Tree and Unit Marking	Walking	Human presence	Negligible response to human presence.
			ATV	Noise; Human presence; Trampling/run over	Direct mortality, particularly of terrestrial species; Negligible response to noise.
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Direct mortality, particularly of terrestrial species.
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over, vibration	Direct mortality, particularly of terrestrial species., behavioral avoidance

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence; vibration	Direct mortality, particularly of terrestrial species, behavioral avoidance
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients, vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients, vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients, vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over, vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence;	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels.
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Soil compaction; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; behavioral avoidance
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration	Direct mortality, particularly of terrestrial species; loss of fitness/survival in aquatic amphibians in response to changes in water quality; behavioral avoidance
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration; temporary displacement	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; sedimentation; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial species; behavioral avoidance

Birds

Effects of Roads and Timber

The proposed action may result in increased roads in the action area, and those are most likely in places that are potentially operable for timber harvest. Thus, potential effects of both roads and timber on bird species and communities are considered here.

In general, roads can impact birds by fragmenting and decreasing habitat, causing direct fatalities, increasing noise, light, and disturbance, and increasing pollution. Increases in open habitats from road construction can lead to changes in species composition (Kociolek et al. 2015). Species that prefer open areas and are attracted to noise can become more common near roads (Kociolek et al. 2015), with omnivores attracted to increased prey availability that comes with edges (Kroeger et al. 2022). Forest-dependent birds, however, are generally more sensitive to habitat fragmentation than those adapted to more open habitats (Timmers et al. 2022). Consequently, bird richness in habitats with dense tree cover tends to decline with road presence (Kroeger et al. 2022). The presence of roads with higher traffic also tends to be associated with lower forest bird abundance, likely from both noise and mortality (Jack et al. 2015). Birds rely heavily on acoustic communication, and interior forest birds tend to be more sensitive to noise and can even experience reduced reproductive success as a result (Senzaki et al. 2020). Some species have shown increased stress hormones and lower fitness levels when exposed to higher levels of traffic noise (Hayward et al. 2011, Injaian et al. 2018). At a population level, bird species with larger territory sizes may show a slight negative response to roads and/or traffic because of increased exposure to road-related mortality in areas between patches of habitat (Rytwinski and Fahrig 2012). Species with high reproductive rates, small home range sizes, small body sizes, and high flight altitudes, tend to be less vulnerable to road impacts (Kociolek et al. 2015). Direct and short-term adverse impacts to individuals may be partially mitigated by considering timing of road construction, as many bird species are more vulnerable at certain times of year, in particular during breeding and migration periods (Kociolek et al. 2015). In general, decreases in species richness and community integrity are associated with increasing density of roads (Glennon and Porter 2005), though historical road densities seem to play a role, at least for wetland birds (Findlay and Bourdages 2000). Roads (and other types of infrastructure) can have a negative effect on bird abundance up to one kilometer away, on average; bird species vary in terms of their sensitivity to roads (Benitez-Lopez et al. 2010).

Timber harvesting similarly can remove habitat, increase habitat fragmentation, increase noise disturbance, and cause direct mortality to birds. Much of its impact depends on silvicultural prescription and scale. Species that rely on trees for their primary life stages (nesting, roosting, foraging) tend to be negatively impacted by the removal of trees. Loss of large trees, snags, and logs through timber harvest in areas adjacent to roads can have adverse effects on cavity dependent species (Hann et al. 1997 as cited in USFS 2000b). Mature forest species respond most negatively to timber harvest, while shrub and ground foraging species may respond positively, presumably due to the short-term availability of small gaps created by harvesting (Vanderwel et al. 2007). Silvicultural practices that create small gaps that effectively mimic natural disturbance processes can be neutral to forest birds and even enhance habitat (Forsman et

al. 2010). In forests with relative homogeneity such as second-growth forest stands in eastern North America, moderate levels of canopy reduction can increase understory vegetation and subsequently foraging and nesting opportunities for certain birds (Akresh et al. 2023). In general, bird populations are tightly tied to habitat availability and the quality of the habitat. Reductions in stand age, especially through removal of older trees, can decrease bird species abundance even when the total amount of forest cover remains relatively stable (Betts et al. 2022).

The potential activities of timber harvest and road construction/maintenance are deconstructed into their components, with generalized potential impacts and potential responses of birds as shown in Table 9. Additional site-specific analysis would be conducted on all projects with the potential to affect any listed species.

Table 9. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of bird species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of nest/den trees and snags	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Tree and Unit Marking	Walking	Human presence	Disturbance (flushing/relocating)
			ATV	Noise; Human presence; Trampling/run over	Disturbance (flushing/relocating); direct mortality
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Disturbance (flushing/relocating)
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over	Disturbance (flushing/relocating)
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence	Disturbance (flushing/relocating)
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sedimentation	Disturbance (flushing/relocating)
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; chemical contamination/nutrients; Drainage Network; Peak/Baseflows	Disturbance (flushing/relocating)
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence; Loss of nest trees	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Loss of nest trees; Soil compaction; Trampling/run over	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; Loss of nest trees	Direct mortality; habitat loss/fragmentation; decreased nesting success; disturbance (flushing/relocating)
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust;	Disturbance (flushing/relocating)

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal	Disturbance (flushing/relocating)
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over	Disturbance (flushing/relocating)
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; sedimentation	Disturbance (flushing/relocating)
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Disturbance (flushing/relocating)

ESA-listed birds summary

The potential species list from IPaC contained 30 listed or proposed bird species on relevant Forest Service units, of which 4 species have more than one distinct listed entity. Upon further vetting, four species on the original list do not occupy the action area and so are not considered here: Yuma ridgeway's rail *(Rallus obsoletus yumanensis)*, Short-tailed albatross *(Phoebastria albatrus),* Northern aplomado falcon *(Falco femoralis spetentrionalis)*, and Mt. Rainier white-tailed ptarmigan *(Lagopus leucura rainierensis)*. The 26 potentially present species are organized below by their primary habitat association. Each analysis includes brief species information, potential exposure and response to the proposed action, and effects determination. A summary is provided in Table 10.

Table 10. Threatened (T), endangered (E), proposed threatened (PT), and proposed endangered (PE), proposed threatened (PT), and experimental (EXP) bird species in the action area and the units they occur on. The presence of critical habitat (and its percent overlap with the action area) is also provided. Species and critical effects determinations are included (no effect, NE; not likely to adversely affect, NLAA; likely to adversely affect, LAA; not applicable, n/a).

Common Name	Scientific Name	ESA Status	Critical Habitat in Action Area	Exposure	Species Effects Determination	Critical Habitat Effects Determination
California spotted owl	*Strix occidentalis occidentalis*	PE, PT	n/a	High	LAA	n/a
Elfin-woods warbler	*Setophaga angelae*	T	Y (10.6%)	Moderate	NLAA	NLAA
Marbled murrelet	*Brachyramphus marmoratus*	T	Y (13.4%)	Moderate	LAA	LAA
Mexican spotted owl	*Strix occidentalis lucida*	T	Y (8.1%)	Moderate	LAA	LAA
Northern Spotted owl	*Strix occidentalis caurina*	T	Y (15.3%)	High	LAA	LAA
Puerto Rican broad-winged hawk	*Buteo platypterus brunnescens*	E	n/a	Unknown	NLAA	n/a
Puerto Rican parrot	*Amazona vittata*	E	n/a	Low	NLAA	n/a
Puerto Rican sharp-shinned hawk	*Accipiter striatus venator*	E	n/a	Unknown	NLAA	n/a
Red-cockaded woodpecker	*Dryobates borealis*	T	n/a	High	NLAA	n/a
Least Bell's vireo	*Vireo bellii pusillus*	E	Y (6.0%)	High	LAA	LAA
Southwestern willow flycatcher	*Empidonax traillii extimus*	E	Y (<1%)	Moderate	LAA	LAA
Yellow-billed cuckoo	*Coccyzus americanus*	T	Y (<1%)	High	LAA	LAA

Common Name	Scientific Name	ESA Status	Critical Habitat in Action Area	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Cactus ferruginous pygmy-owl	*Glaucidium brasilianum cactorum*	T	n/a	Unknown	NE	n/a
Coastal California gnatcatcher	*Polioptila californica californica*	T	Y (1.0%)	Unknown	NLAA	NLAA
Florida scrub-jay	*Aphelocoma coerulescens*	T	n/a	High	NLAA	n/a
Eastern black rail	*Laterallus jamaicensis* ssp. *jamaicensis*	T	n/a	Unknown	NLAA	n/a
Everglade snail kite	*Rostrhamus sociabilis plumbeus*	E	N	Unknown	NE	n/a
Mississippi sandhill crane	*Antigone canadensis pulla*	E	N	Unknown	NE	n/a
Whooping crane	*Grus americana*	E, Exp	N	Low	NE	n/a
Piping plover	*Charadrius melodus*	T	N	Low	NE	n/a
Rufa red knot	*Calidris canutus rufa*	T	N	Unknown	NE	n/a
Western snowy plover	*Charadrius nivosus nivosus*	T	Y (2.5%)	Unknown	NE	NE
Greater sage-grouse (Bi-state sage-grouse)	*Centrocercus urophasianus*	PT	Y (2.4%)	Moderate	LAA	LAA
Gunnison sage-grouse	*Centrocercus minimus*	T	N	Low	NE	n/a
California condor	*Gymnogyps californianus*	E, Exp	Y (2.2%)	Moderate	NLAA	NLAA

Forest birds

California spotted owl (Strix occidentalis occidentalis)

The California spotted owl (CSO), one of three subspecies of spotted owl, occurs throughout the Sierra Nevada mountain range in California and Nevada, as well as southern and coastal California, with subpopulations distributed among discrete mountain ranges; and in Baja California Norte, Mexico. California spotted owls were proposed for listing in 2023 as two Distinct Population Segments (DPS): threatened in the Sierra Nevada DPS and endangered in the coastal-southern California DPS (88 FR 11600). Spotted owls are long-lived with high adult survival and low reproductive output (USFWS 2022d). They are well-known to prefer mature, structurally complex forests with high canopy cover and large trees. More recently, the importance of habitat heterogeneity has been demonstrated as well as particularly important for foraging and overall fitness (Zulla et al. 2023, Jones et al. 2024a). In the Sierra Nevada, at higher elevations CSO use conifers, with increasing use of hardwoods as elevations decrease. They forage on a variety of small mammals, primarily flying squirrels at higher elevations and

woodrats at lower elevations. In southern California, CSO use platforms and old raptor nests more often, and use smaller trees than in the Sierra Nevada, but will still select for larger trees and high amounts of canopy cover when available (USFWS 2022d). They can be found in riparian and hardwood forests and woodlands, riparian/hardwood forests and woodlands, live oak/big cone fir forests, and redwood/California laurel forests. Coarse woody debris is an important component for food, shelter, and protection for prey species in both DPSs.

The Sierra Nevada DPS is particularly threatened by uncharacteristically large and severe wildfires, tree mortality, drought, and invading barred owls (USFWS 2022d). California spotted owls are widely distributed throughout the Sierra Nevada, with ~2200-2300 estimated owls in 2021 (Kelly et al. 2023). Four years of regional surveys indicated that occupancy has declined slightly between 2021 and 2024 (USFS 2025b). The coastal-southern DPS is impacted by fire and habitat fragmentation. Occupancy analyses on San Bernadino National Forest (historically the largest owl population in southern California and the only with demographic information) indicated a decline by more than 50% from 1991 to 2019 (Tempel et al. 2022). Breeding season surveys in 2022 in the San Gabriel Mountains (San Bernadino and Angeles National Forests), Sierra Pelona Mountains (Angeles National Forest), and the Los Padres National Forest only found 8 currently occupied territories (Barry et al. 2025). In 2024, Forest Service surveys across the four southern California national forests detected owls in 15% of the grids, with none on Cleveland National Forests (USFS 2025b). Forest management remains a controversial topic related to species requiring older forest habitat, but increased frequency and extent of high severity fire in California has caused fuels management to become a key issue. The response of California spotted owls to fuel treatments depends on the type and scale of treatment. In general, with careful application, the long-term benefits of fuels treatments to reduce the risk of severe large-scale wildfire likely outweigh short-term negative impacts (88 FR 11600). A recent study using a dynamic habitat model for the southern-coastal DPS indicated that habitat has declined by more than 50% since 1986, with most of the decline attributed to wildfire and a minimal amount attributed to fuels reduction activities (Barry et al. 2026). See the proposed rule for additional information (88 FR 11600).

Effects

Seventy-five percent of suitable CSO habitat in the Sierra Nevada is on Forest Service-allocated lands (USFWS 2022d). Given protections outlined in local National Forest Plans, roads themselves are unlikely to impact California spotted owls in the Sierra Nevada DPS. Timber harvest, on the other hand, could have negative or positive impacts on owl habitat, depending on specific methods used (USFWS 2022d). The coastal-southern DPS is already highly fragmented, both naturally and resulting from development (USFWS 2022d). New roads there could exacerbate it further. While fuels treatments may be necessary, currently commercial timber harvest does not occur on public lands within the coastal-southern DPS of the California spotted owl (USFWS 2022d).

Dynamic models of suitable nesting/roosting habitat have been developed for CSO, separately for the Sierra population (developed by Ronan Hart) and the coastal-southern population (Barry et al. 2026). From these models, there are about 1,103,997 acres of suitable habitat in the Sierra, 6.2% of which is inside the action area. There are about 82,109 acres of suitable habitat in the

coastal-southern population, 12.8% of which is inside the action area. From the GBIF/NRM dataset, there are a total of 49,662 occurrences on NFS lands, only 5% of which are in the action area, though 78% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations.

Given the direct tie to mature forests, any removal of trees may have short-term adverse impacts on California spotted owls, even when done with long-term benefits as a goal. Adverse effects to CSO could be a consequence of removing suitable habitat (nesting/roosting/foraging) or direct encounters resulting in disturbance (of nesting owls/juveniles in particular). Thus, the proposed action *may affect, and is likely to adversely affect* both the Sierra Nevada DPS and coastal-southern DPS of California spotted owls.

Elfin-woods warbler (Setophaga angelae)

The elfin-woods warbler is a small black and white warbler endemic to Puerto Rico and known in two populations, El Yunque National Forest and Maricao Commonwealth Forest (and adjacent private lands), which are likely isolated from each other (USFWS 2022e). The warbler was listed as threatened in 2016 (81 FR 40534) with critical habitat designated in 2020 (85 FR 39077). It is found on mountain summits in dense stands of short, small, twisted trees and shrubs, such as the dwarf forest type. The essential physical and biological features (PBFs) for its critical habitat include: 1) wet and rain montane forest types (*Podocarpus* forest, dwarf forest, Palo Colorado forest); 2) forested habitat areas that contain active or abandoned shade-grown coffee plantation for forest agricultural lands; and 3) forest habitats not contained within the others described but that have exposed ridge woodlands, timber plantations, or secondary forests within elevations 300-850 m (85 FR 39077). Its threats are primarily habitat modification on private lands and stochastic effects related to small, isolated populations (such as from hurricanes). In 2014, USFWS, USFS, and the Puerto Rico Department of Natural and Environmental Resources signed a Candidate Conservation Agreement (CCA) to work together to conserve the warbler and restore habitat on the two forests (USFWS 2016b). Additional conservation actions have been implemented and agreements developed with private landowners to conserve tropical upland forest and riverbank buffers. In 2010, there were estimated to be about 1,800 mature individuals, but no formal assessment has been made since then, and indications are that occupancy has declined (USFWS 2022e). On El Yunque, the population appears to be declining, and there is not additional continuous montane forested vegetation beyond the forest boundaries (USFWS 2016). Maricao Commonwealth Forest appears stable even after Hurricane Maria in 2017, though it is not uniformly distributed (USFWS 2022e). See the 5-year review for additional information (USFWS 2022e).

Effects

Of the two populations, El Yunque appears to be less stable than that on Maricao (USFWS 2022e). From the GBIF/NRM dataset, there are a total of 27 occurrences on NFS lands, 15% of which are in the action area, though none of those is in a potentially operable area. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations.

There are approximately 2,902 acres of critical habitat within the action area, which is 10.6% of the total critical habitat. None of the critical habitat is within potentially operable areas. Given

the small population size and strictly limited habitat, any additional loss whether to roads or timber would be negative to the species. However, the El Yunque National Forest Land Management Plan Appendix A discusses specific guidelines and management for the Elfin-woods warbler as agreed to as part of the CCA, with a goal of contributing to the recovery of the species. Thus, any future action would need to adhere to those guidelines, which involve ESA consultation (2019 land management plan, available at USFS 2025a).

Given the protections afforded by the forest plan and CCA, and the lack of occurrences or critical habitat within potentially operable areas, road construction and timber harvest are so unlikely to occur where the warbler lives that effects are discountable. Thus, the proposed action *may affect, but is not likely to adversely affect* the Elfin-woods warbler or its critical habitat.

Marbled murrelet (Brachyramphus marmoratus)

Marbled murrelets are small, robin-sized diving seabirds that forage in marine waters, but nest in mature forests. Murrelets live along the west coast, from Alaska to California. The Washington, Oregon, and California DPS were listed as threatened in 1992 (57 FR 45328), with critical habitat designated in 1996 (61 FR 26256) and revised in 2016 (81 FR 51348). Murrelets spend most of their time on the ocean. During the breeding season they rest and feed in near-shore marine environments (1-3 miles offshore) and come inland to nest. Murrelets are dependent upon adequate nesting habitat in mature forest, adequate foraging habitat offshore, and connectivity between the two. They nest in large coniferous trees with large branches or deformities for platform nests, laying a single egg on a mat of moss, lichen, or debris accumulation. Nest stands are old-growth forests with multiple canopy layers and moderate to high canopy closure; though they vary in size from several acres to thousands of acres; larger unfragmented stands tend to be of higher quality (USFWS 2024m). Most murrelets nest within 37 miles of the coast, though they have been found to nest up to 52 miles inland. The physical and biological features essential to critical habitat focus on individual trees with potential platform nests and forested areas within 0.5 mile of such trees, and with a canopy height of at least one-half the site-potential tree height (81 FR 51348). Murrelets are threatened by habitat loss due to wildfire, timber harvest, predation, offshore development, and long-term alterations of weather. The most recent population estimate was 19,000 murrelets in 2022, with indications of a relatively stable population size. Low productivity has likely contributed to the species not recovering within the range. See the Species Report for additional information (USFWS 2024m).

Effects

Timber harvest was historically the main loss of habitat, although it has slowed considerably, losses continue, particularly on nonfederal lands (96% of nesting habitat lost to timber harvest from 1993-2017 was on nonfederal lands, USFWS 2024m). Historically, fire occurred throughout the range of the murrelet but at varying intensities and frequencies. The increase in large-scale, high-severity fire also poses a threat to murrelet habitat (USFWS 2024).

The Northwest Forest Plan Interagency Monitoring Program tracks MAMU habitat and classifies it by quality (Lorenz et al. 2021). As of the 2023 model (developed by Adam Duarte), there was estimated to be 4,378,091 acres of high-suitability nesting habitat, and about 8.2% of that is in the action area. From the GBIF/NRM dataset, there are a total of 3,849 occurrences on NFS

lands, 17% of which are in the action area, and 34% of those are in potentially operable areas. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 493,889 acres of critical habitat within the action area, which is 13.6% of the total critical habitat. About half of that is within potentially operable areas. New roads and opportunities for timber in previously roadless areas could negatively impact the remaining habitat. However, any work would be guided by the Northwest Forest Plan (NWFP, implemented in 1994[24]), which is a multiagency effort to maintain and restore biodiversity, especially within late-successional and old-growth forest systems, while maintaining forest productivity, renewable resources, and rural economies. The NWFP applies to all old-growth forest-associated species within the range of the northern spotted owl, including marbled murrelets.

Given the direct tie to mature forests and large trees, any removal of trees may have short-term adverse impacts on marbled murrelets, even when done with long-term benefits as a goal. Adverse effects to murrelets could be a consequence of removing suitable habitat (nesting/roosting) or direct encounters resulting in disturbance (of nesting murrelets/juveniles in particular). Removal of potential nesting trees and surrounding habitat could have adverse impacts to critical habitat PBFs. Thus, the proposed action *may affect, is likely to adversely affect* the marbled murrelet and its critical habitat.

Mexican spotted owl (Strix occidentalis lucida)

The Mexican spotted owl (MSO), one of three spotted owl subspecies, ranges from Utah, Colorado, Arizona, New Mexico, the western portions of Texas, and south into several states of Mexico. MSO lives in disjunct areas that correspond with isolated mountain ranges and canyon systems. The Mexican spotted owl was listed as threatened in 1993 (58 FR 14248) with critical habitat designated in 2004 (69 FR 53182). Like other spotted owls, this species shows high site fidelity to territories and primarily eats small to medium-sized rodents, though it will eat a variety of prey items that vary with location. In forested habitats, the Mexican spotted owl primarily uses ponderosa pine-Gamble oak, mixed-conifer, and riparian forests. It requires complex canopy cover with habitat heterogeneity for its prey base. Individual MSOs will preferentially forage in forests characterized by a mix of stand ages and stages produced by various types of disturbance but avoid severely burned forest (Jones et al. 2025). In rocky-canyon habitats it inhabits areas with a variety of desert scrub and riparian vegetation communities, and nests in protected caves, as well as rocky ledges or trees (USFWS 2012a). The physical and biological features essential to critical habitat are divided into forest structure, prey base, and canyon habitat. Forest structure requires a range of tree species of varying size and age, but particularly older trees, with shade canopy as well as large dead trees; the prey base requires woody debris and logs, and a range of vegetation conditions including hardwoods and residual plant cover. Canyon habitats need at least one of the following: presence of water, clumps or stringers of vegetation, canyon wall with crevices/ledges/caves, or a high percentage of ground litter/debris (69 FR 53182). The widespread, but fragmented distribution of MSO makes it vulnerable to high-intensity landscape-scale wildfires, though timber harvest was a historical

[24] The 1994 Northwest Forest Plan is available at the Regional Ecosystem Office Northwest Forest Plan website: https://www.fs.usda.gov/r6/reo/overview.php

threat. MSO respond to wildfire consistent with their evolutionary history, occupying areas that experience frequent-fire regimes that are not stand-replacing (Jones et al. 2024b). At the time of listing, there were estimated to be 2,160 MSOs (58 FR 14248). Monitoring from the past 10 years indicates a relatively stable population, though there may be a slight non-significant decline in pair occupancy. Suitable nest-roost habitat has declined across the range to varying degrees since the 1980s (USFWS 2023g). See the 2023 5-year review for additional information (USFWS 2023g).

Effects

Most known MSOs occur on federal lands, the majority of which is managed by the Forest Service in Arizona and New Mexico. To consider potential exposure of the species and habitat, we examined several owl datasets. As part of the conservation strategy, the Recovery Plan recommends that Protected Activity Centers (PACs) around known nest/roost sites of approximately 600 acres be identified (USFWS 2012a). The southwestern forests in AZ and NM currently have about 1,300 PACs totaling 829,906 acres. Of those, 301 PACs intersect the action areas (partially or wholly), totaling 108,819 acres (13%). In the southwestern region, a dynamic model of suitable nest-roost habitat has been developed (Jones et al. 2023, Shirk et al. 2023). As of 2024, there are about 1,070,837 acres of suitable habitat in Region 3, 7% of which is inside the action area. From the GBIF/NRM dataset, there are a total of 13,691 occurrences on NFS lands, 8% of which are in the action area, and 28% of those are in potentially operable areas. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 796,224 acres of critical habitat within the action area, which is 8% of the total critical habitat. Only 1% of total critical habitat is within potentially operable areas. New roads and opportunities for timber could remove MSO habitat, however all Forest Plans in the southwest contain language that requires following listed species Recovery Plans. The Mexican spotted owl Recovery Plan emphasizes that most fuels reduction work can be completed outside of primary nest-roost habitat and PACs (USFWS 2012a).

Given the direct tie to mature forests, any removal of trees may have short-term adverse impacts on Mexican spotted owls, even when done with long-term benefits as a goal. Adverse effects to MSO could be a consequence of removing suitable habitat (nesting/roosting/foraging) or direct encounters resulting in disturbance (of nesting owls/juveniles in particular). Removal of potential nesting trees and surrounding habitat could have adverse impacts to critical habitat PBFs. Thus, the proposed action *may affect, is likely to adversely affect* both the Mexican spotted owl and its critical habitat.

Northern spotted owl (Strix occidentalis caurina)

The Northern spotted owl (NSO), one of three spotted owl subspecies, ranges from southwestern British Columbia, through Washington and Oregon, into northern California. NSO was listed as threatened in 1990 (55 FR 26114). Critical habitat was designated in 1992 (57 FR 1796) and most recently revised in 2021 (86 FR 62606). Like the other subspecies, it is long-lived, territorial, and pair-bonded. Northern spotted owls inhabit structurally complex forests, and primarily eat small mammals, though the focus varies by habitat. They rely on older forest habitats, and the associated structures and characteristics, for nesting, roosting and foraging, including moderate to high canopy over, multi-layer/species canopy with large overstory trees,

with various deformities, fallen trees and woody debris (USFWS 2022f). Foraging habitat is the most variable habitat and is closely tied to prey base. Critical habitat physical and biological features include: 1) forest types that may be in early-, mid-, or late-seral stages and that support the owl across its geographic range; 2) nesting and roosting habitat; 3) foraging habitat; and 4) dispersal habitat; and all of the designated units support at least two of the four PBFs (57 FR 1796).

The Northwest Forest Plan in 1994 was designed in part to arrest the downward trends by providing for late successional/old growth forest over the long term. Reserved lands include late successional reserves, managed reserves, as well as adaptive management areas (USFWS 2022f). The NWFP has reduced habitat loss and allowed for development of new habitat. On non-federal lands, however, habitat has continued to decrease. From 1993 to 2022, habitat decreased by 8.6% across all lands, much of that from wildfires since 2017 (Davis et al. 2022, USFWS 2022f). Recent studies have indicated that interior NSO nest/roost forest tends to burn at lower severity than non-nesting forest and may serve as fire refugia (Lesmeister et al. 2021). Trying to maintain suitable amounts of nest/roost habitat alone, however, is not expected to reverse the trend in occupancy (Franklin et al. 2021). The most recent range-wide northern spotted owl demographic study found a 5.3 percent rate of population decline for all study areas combined, with the rate having increased since the 2011 review. Non-native barred owls are currently the stressor with the largest negative impact on northern spotted owls. Barred owl removal efforts positively impact local spotted owls (USFWS 2022f), and USFWS has developed a Barred Owl Management Strategy to try to help address barred owls at a larger scale in the range of both NSO and CSO (USFWS 2024n). USFWS recently found that NSO meets the definition of endangered, however, it was warranted but precluded from uplisting (USFWS 2022f). See the 2022 Species Form for additional information (USFWS 2022f).

Effects

The Northwest Forest Plan Interagency Monitoring Program tracks NSO habitat; currently, about 55% of suitable NSO habitat is found on Forest Service lands. This dynamic habitat model includes both nest/roost habitat and dispersal habitat (Davis February 18, 2026). Currently, there are about 10,203,105 acres of suitable habitat, 7.8% of which is in the action area. When considering a cover type model that classifies the suitability of nest/roost habitat specifically, there are about 6,004,365 acres of highly suitable nest/roost habitat, 8.8% of which is within the action area. From the GBIF/NRM dataset, there are a total of 29,783 occurrences on NFS lands, 11% of which are in the action area, and 86% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 1,438,583 acres of critical habitat within the action area, which is 15.3% of the total critical habitat. About two-thirds of that is in potentially operable areas. We can assume that habitat will continue to be managed according to the NWFP; however, new roads forest harvest in previously roadless areas could still impact individuals or habitat.

Given the direct tie to mature forests, any removal of trees may have short-term adverse impacts on northern spotted owls, even when done with long-term benefits as a goal. Adverse effects to NSO could be a consequence of removing suitable habitat (nesting/roosting/foraging) or direct

encounters resulting in disturbance (of nesting owls/juveniles in particular). Removal of potential nesting trees and surrounding habitat could have adverse impacts to critical habitat PBFs. Thus, the proposed action *may affect, is likely to adversely affect* both the northern spotted owl and its critical habitat.

Puerto Rican broad-winged hawk (Buteo platypterus brunnescens)

The Puerto Rican broad-winged hawk is an uncommon raptor, endemic to Puerto Rico, and restricted to closed-canopy, montane forests. The species was listed as endangered in 1994 (59 FR 46710). Current sightings of the species have been reported in Río Abajo, Arecibo, Toro Negro, Guilarte and Guajataca Commonwealth forests, as well as El Yunque National Forest and private lands surrounding these forests. The major threats are habitat loss, fragmentation and degradation. In 2017 before Hurricanes Irma and Maria, there were 123 individuals (USFWS 2024o), but surveys in 2022 only detected 43 individuals. The karst region continues to support the largest number of hawks. Since Hurricane Maria, predation and competition from red-talked hawks also appear to be limiting breeding success. While on public lands the species' habitat is mainly protected, most suitable habitat is on private lands with limited regulation (USFWS 2019f). See the 5-year review for additional information (USFWS 2024o).

Effects

The GBIF/NRM dataset does not contain any occurrences for the Puerto Rican broad-winged hawk, so potential exposure as estimated with that dataset is unknown. However, new roads and timber harvest are unlikely where the hawk lives, and the El Yunque National Forest Plan standards include protections for broad-winged hawk nest and roost sites (as well as sharp-shinned hawks and parrots), including no adverse management activities to be conducted within 150 meters of a nest and avoiding potential disturbing activities during the breeding season (November to August) (2019 land management plan, available at USFS 2025a). Given the protections in El Yunque National Forest Plan and that roads and timber operations are unlikely where the hawk lives, the proposed action *may affect, but is not likely to adversely affect* the Puerto Rican broad-winged hawk because effects are discountable.

Puerto Rican parrot (Amazona vittata)

The Puerto Rican parrot is the only native *Amazona* species in the U.S. Though thought to once be abundant and widespread on the Puerto Rican Archipelago's major islands, it was listed as endangered in 1967 with only 24 individuals on El Yunque National Forest (32 FR 4001). By 1912, the island had been more than 80% deforested, likely causing the dramatic decline of the parrot and other forest-associated species (USFWS 2009). A captive breeding program was established in the 1970s to supplement the population. The parrot is a frugivorous cavity nester that prefers mature forests with large cavity forming trees. Secondary forests in the northwestern karst region were identified as suitable for reintroduction because they contain other types of cavities such as cliff pot-holes. In 2023, there were an estimated 32 parrots in El Yunque, more than 210 in Rio Abajo Forest, and 18 in Maricao State Forest, all of which have captive-reared birds as well as recruitment from wild breeding pairs. There are also more than 500 parrots in two captive populations. Populations are recovering after Hurricanes Maria and Irma, however, only the Rio Abajo population appears to be stable. Limited cavities, potential human

disturbance, predation (by red-tailed hawks and pearly-eyed thrasher), small population sizes, and hurricanes continue to be threats to the species. See the 2023 5-year review for additional information (USFWS 2023h).

Effects

Most parrots live in captivity and on non-federal lands. The growth rate for El Yunque from 2018 to 2022 was estimated to be 8.4% (USFWS 2023h). From the GBIF/NRM dataset, there are 12 occurrences on NFS lands, none of which are in the action area. This suggests a low chance of exposure for individuals to potential future roads and/or timber operations. New roads and timber are unlikely in El Yunque, and the El Yunque National Forest Plan standards include protections for parrot nest and roost sites (as well as sharp-shinned hawks and broad-winged hawks), including no adverse management activities to be conducted within 150 meters of a nest and avoiding potential disturbing activities during the breeding season (November to August) (2019 land management plan, available at USFS 2025a). Given the protections in El Yunque National Forest Plan and the lack of occurrences within the action area, roads and timber are unlikely to occur where the parrot lives. Thus, the proposed action *may affect, but is not likely to adversely affect* the Puerto Rican parrot because effects are discountable.

Puerto Rican sharp-shinned hawk (Accipiter striatus venator)

The Puerto Rican sharp-shinned hawk is a subspecies that nests in the subtropical moist forest and the subtropical/lower montane wet forest of Puerto Rico. Deforestation from human expansion and hurricanes have decreased available lands for the hawk, similar to the other Puerto Rican species. The species was listed as endangered in 1967 (32 FR 4001). The overall population is fragmented with isolated populations, primarily restricted to montane forest reserves. Although Puerto Rico has much mature secondary forest, sharp-shinned hawks have not shown an expansion to occupy those (USFWS 2018g). Populations have declined substantially, and there are no recent observations on El Yunque (USFWS 2019f). Surveys in early 2018 after Hurricane Maria only detected 19 individuals. More recent assessments have documented an increase to 22-41 adults (USFWS 2023). See the 5-year review for additional information (USFWS 2023i).

Effects

The GBIF/NRM dataset does not contain any occurrences for the Puerto Rican sharp-shinned hawk, so potential exposure as estimated with that dataset is unknown. However, new roads and timber are unlikely where the hawk lives, and the El Yunque National Forest Plan standards include protections for parrot nest and roost sites (as well as parrots and broad-winged hawks), including no adverse management activities to be conducted within 150 meters of a nest and avoiding potential disturbing activities during the breeding season (November to August) (2019 land management plan, available at USFS 2025a). Given the protections in El Yunque National Forest Plan and that roads and timber operations are unlikely where the hawk lives, the proposed action *may affect, but is not likely to adversely affect* the Puerto Rican sharp-shinned hawk because effects are discountable.

Red-cockaded woodpecker (Dryobates borealis)

The red-cockaded woodpecker is a relatively small woodpecker, known for its unique characteristic of excavating cavities exclusively in living pine trees. Red-cockaded woodpeckers were once common throughout the longleaf pine ecosystem with historical estimates of over 1 million family groups (USFWS 2022g). Much of the longleaf pine ecosystem was eliminated from its original range upon early European settlement followed by widespread timber harvesting. The remaining longleaf pine ecosystem differs greatly from its historical counterpart, with second growth currently just reaching the age (90-100 years) to meet the needs of the woodpecker. Red-cockaded woodpeckers are highly sedentary cooperative breeders with a current patchy distribution in AL, AR, FL, GA, LA, MS, NC, OK, SC, TX, VA. The woodpecker was listed as endangered in 1970 (35 FR 16033) and recently downlisted to threatened in 2024 (89 FR 85294). The woodpeckers require open pine woodlands and savannas with individual large, old pines. A tree must be old and large enough to have sufficient heartwood to completely contain a cavity to prevent pine resin in the sapwood from entering the chamber (89 FR 85294). Suitable foraging habitat consists of mature pines with open canopies, low densities of small pines, minimal hardwood or pine midstory, few or no overstory hardwoods, and abundant native bunchgrass and forb groundcover (USFWS 2003). Frequent fire is necessary to maintain these open pine woodlands. Threats to the red-cockaded woodpecker include historical habitat loss, continued habitat conversion, fragmentation, and stochastic events that can threaten a small population. The woodpecker is conservation-reliant and responds well to active management (89 FR 85294). See the 2022 Species Status Assessment for additional information (USFWS 2022g).

Effects

The Recovery Plan recommends identifying and managing clusters of active and inactive cavity trees within an area that a family group defends (USFWS 2003). The southern region of the USFS maintains a dataset identifying clusters as active, inactive, recruitment, abandoned, and unsuitable. There are currently 4,530 active/inactive/recruitment clusters in that dataset, totaling 68,448 acres. Only 68 of those clusters intersect (partially or wholly) with the action area, totaling 659 acres. From the GBIF/NRM dataset, there are a total of 30,726 occurrences on NFS lands, only 1% of which are in the action area, but 94% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations. However, at a population level, very few would be potentially exposed. Active management is often positive for the species, and all forest plans require implementing protective measures and guidelines in the recovery plan. Given the protections afforded by the forest plans and the positive influence of active management on the species, the proposed action *may affect, but is not likely to adversely affect* the red-cockaded woodpecker.

Riparian birds

Least Bell's vireo (Vireo bellii pusillus)

The least Bell's vireo is a small migratory bird, the westernmost of four subspecies that breeds in southern California and winters in southern Baja California, Mexico. For breeding it is an obligate riparian species, though in winter it uses a greater variety of habitats (USFWS 1998b). The vireo was listed as endangered in 1998 (51 FR 16474) due to habitat loss from agriculture

and urbanization, flood control and grazing, as well as brood parasitism by brown-headed cowbirds. Critical habitat was designated in 1994 (59 FR 4845). The physical and biological features essential to the conservation of the vireo include riparian woodland vegetation with canopy and shrub layers, and some associated upland habitats. At the time of listing there were only 291 territories known in the U.S., but the number of territories has since increased 10-fold (USFWS 2006). Localized improved condition of riparian habitats and cowbird control has likely contributed to this success. However, riparian habitats continue to be increasingly bordered by urbanization and large areas of the historical range remain unoccupied. See the 5-year review (USFWS 2006) and draft Recovery Plan (USFWS 1998b) for additional information.

Effects

From the GBIF/NRM dataset, there are a total of 67 occurrences on NFS lands, only 9% of which are in the action area, but 100% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 2,210 acres of critical habitat within the action area, which is 6% of the total critical habitat. Most of that critical habitat is also within potentially operable areas. Vegetation removal for the purpose of roads and/or timber harvest could decrease and further fragment habitat. Timing is the most important component of protecting the vireo; specifically avoiding work during the breeding season is important and would be a consideration in any future consultation. Most forest plans contain protections regarding riparian areas; however, the vireo requires specifics associated with upland habitats as well. Adverse effects to the vireo could be a consequence of removing suitable habitat and could impact PBFs in critical habitat (particularly upland). Habitat is a major limiting factor, and the proposed action *may affect, is likely to adversely affect* the least bell's vireo and its critical habitat.

Southwestern willow flycatcher (Empidonax traillii extimus)

The southwestern willow flycatcher is a migratory subspecies that breeds in the southwest U.S. and migrates to Central America for the winter. It breeds in dense riparian tree and shrub communities, typically classified as forested or scrub-shrub wetlands (USFWS 2002a). It was listed as endangered in 1995 (60 FR 10694) due to extensive loss and modification of breeding habitat from water diversion and damming, changes in flood and fire regimes, disruption of natural hydrologic cycles, increased grazing, and the spread of non-native plants. In addition, brood parasitism by brown-headed cowbirds has further inhibited reproductive success. The southwestern willow flycatcher's breeding habitats tend to be cottonwood/willow and tamarisk vegetation, although the latter is a non-native species. The spread of the tamarisk leaf beetle is thus a more recent threat to available riparian nesting habitat (USFWS 2025i). Flycatcher critical habitat was designated in 2013 (78 FR 344), with the essential physical and biological features including: riparian habitat along a dynamic river or lakeside that includes dense vegetation with thickets of trees and shrubs, dense ground foliage, nesting sites with dense canopy, and dense patches of riparian forests interspaced with small openings of open water or marsh so it is not uniformly dense. In addition, it requires a variety of insect prey populations within or adjacent to riparian floodplains. See the 5-year review for additional information (USFWS 2025i).

Effects

From the GBIF/NRM dataset, there are a total of 661 occurrences on NFS lands, only 1% of which are in the action area, though 22% of those are in potentially operable areas. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 1,626 acres of critical habitat within the action area, which is 0.8% of the total critical habitat. Only one-third of that critical habitat is within potentially operable areas. Most forest plans contain protections regarding riparian areas. New roads are unlikely in flycatcher habitat, though adjacent roads may degrade habitat and allow more non-native species to invade. Timing is the most important component of protecting southwestern willow flycatchers; specifically, avoiding work during the breeding season is important and would be a consideration in any future consultation. Adverse effects to flycatchers could be a consequence of removing suitable habitat and could impact PBFs in critical habitat. Thus, the proposed action *may affect, is likely to adversely affect* the southwestern willow flycatcher or its critical habitat.

Yellow-billed cuckoo (Coccyzus americanus)

The Distinct Population Segment of the western yellow-billed cuckoo was listed as threatened in 2014 (79 FR 59992). The cuckoo is a migratory species that travels between wintering grounds in Central and South America and breeding grounds in North America, which includes 12 states in the U.S. and 6 in Mexico. The cuckoo breeds in large tracts of dense riparian woodlands along low-gradient streams that particularly include cottonwood and willow. In parts of Arizona and southwest New Mexico, the cuckoo will also breed in more arid woodlands in perennial, intermittent, and ephemeral drainages (USFWS 2019g). The habitat is especially water-limited, and primary threats include habitat loss and degradation from altered watercourse hydrology, land use encroachment, and more recently mining (85 FR 57816). Critical habitat was designated in 2021 (86 FR 20798), with the physical and biological features focused on range-wide breeding habitat, adequate prey base, and hydrologic processes that maintain the habitats. Range-wide breeding habitat is riparian woodlands within floodplains or in upland areas in large blocks; however, the southwest is more variable, and habitat can include a variety of perennial, intermittent, and ephemeral drainages. See the 2020 findings on a delisting petition for additional information (85 FR 57816).

Effects

From the GBIF/NRM dataset, there are a total of 30,323 occurrences on NFS lands, only 2% of which are in the action area, but 86% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 1,993 acres of critical habitat within the action area, which is 0.7% of the total critical habitat. Only one-fifth of that critical habitat is within potentially operable areas. New roads are unlikely in cuckoo habitat, though adjacent roads may degrade habitat and allow more non-native species to invade, as well as access for other activities such as mining. Timing is the most important component of protecting yellow-billed cuckoos; specifically, avoiding work during the breeding season is important and would be a consideration in any future consultation. Adverse effects to cuckoos could be a consequence of removing suitable habitat and could impact PBFs in critical habitat. Thus, the proposed action *may affect, is likely to adversely affect* the yellow-billed cuckoo or its critical habitat.

Scrub birds

Cactus ferruginous pygmy-owl (Glaucidium brasilianum)

The cactus ferruginous pygmy-owl is a small, cryptic subspecies that lives in desert scrub in southern Arizona south through Michoacán, Mexico and in southern Texas south through Nuevo Leon and Tamaulipas, Mexico. The species was listed as threatened in 2023 (88 FR 46910). It is a diurnal non-migratory owl and requires vegetation in fairly dense thickets with saguaros, organ pipe cactus, trees, or other columnar cacti large enough to support cavities for nesting. The pygmy-owl is threatened by habitat fragmentation and loss due to residential and commercial development, as well as long-term changes to weather and conditions (USFWS 2024p). See the Recovery Plan Outline (USFWS 2024p) for additional information.

Effects

Roads could increase habitat fragmentation and remove potential nesting cacti. The GBIF/NRM dataset does not contain any occurrences for the pygmy-owl, so potential exposure as estimated with that dataset is unknown. Furthermore, there are no known occurrences on U.S. Forests (USFWS 2024p). Given the lack of occurrences on National Forest System lands and that new roads would be unlikely to be built in the owl's desert scrub habitat, the proposed action will have *no effect* on the cactus ferruginous pygmy-owl.

Coastal California gnatcatcher (Polioptila californica californica)

The Coastal California gnatcatcher is a small non-migratory songbird that lives in coastal sage scrub in southern California and Baja California, Mexico. They are tightly tied to coastal sage scrub habitat patches within a matrix that supports an insect prey base. The primary threats to the gnatcatcher include urban development, wildland fire, vegetation type conversion, and long-term changes to weather (USFWS 2025j). The species was listed as threatened in 1993 (58 FR 16742) with critical habitat designated in 2007 (72 FR 72010). The physical and biological features considered essential for conservation of the critical habitat include dynamic and successional sage scrub habitats that provide space for all life stages, as well as non-sage scrub habitats such as chaparral grassland and riparian areas in proximity to sage scrub habitats. See the draft Recovery Plan for additional information (USFWS 2025j).

Effects

Roads could increase habitat fragmentation and alter vegetation type, bringing in invasive species. The GBIF/NRM dataset does not contain any occurrences for the gnatcatcher, so potential exposure as estimated with that dataset is unknown. There are approximately 2,024 acres of designated critical habitat within the action area, which is only 1% of the total critical habitat. Less than half of that critical habitat is in potentially operable areas. Because the proposed action is unlikely to occur where the gnatcatcher lives, the proposed action *may affect, but is not likely to adversely affect* the Coastal California gnatcatcher or its critical habitat.

Florida scrub-jay (Aphelocoma coerulescens)

The Florida scrub-jay is endemic to Florida, non-migratory, sedentary, and restricted to scrub and scrubby flatwoods. The scrub-jay historically occupied 39 counties in peninsular Floria but now only occupies conservation lands in 26 of those counties (private land status is unknown)

(USFWS 2025k). The scrub-jay was listed as threatened in 1987 (52 FR 20715). The primary historical cause of declines was loss and fragmentation of habitat from residential and commercial development (USFWS 2019h). The scrub-jay needs large open landscapes with short oak vegetation and open sandy areas with low densities of pine trees. These landscapes were historically maintained with frequent fire, but reductions in fire frequency have led to overgrown habitat conditions. These conditions also increase predator pressure and competition with the blue jay. See the 2025 5-year review for additional information (USFWS 2025k).

Effects

From the GBIF/NRM dataset, there are a total of 2,922 occurrences on NFS lands, only 1% of which are in the action area, but 81% of those are in potentially operable areas. This suggests a high chance of exposure for individuals to potential future roads and/or timber operations. Roads could increase habitat fragmentation and alter vegetation type. However, timber harvest can be used to enhance habitat for the scrub-jay. The species occurs on Ocala National Forest, which has the largest contiguous scrub habitat remaining in peninsular Florida (USFWS 2025k). The Land Management Plan for National Forests in Florida follows guidelines in the scrub-jay recovery plan (1999 land management plan, available at USFS 2025a), to maintain appropriate scrub habitat. Given the protections afforded by the forest plan and the positive influence of active management on the species, the proposed action *may affect, but is not likely to adversely affect* the Florida scrub-jay.

Wetland birds

Eastern black rail (Laterallus jamaicensis ssp. *jamaicensis)*

The eastern black rail is a sparrow-sized secretive marsh bird that was once distributed widely across interior and coastal areas of the eastern U.S., Mexico, Central America, and the Caribbean, but now occupies a small proportion of its former range. Documented or probable breeding locations are found in New Jersey, Delaware, North Carolina, South Carolina, Florida, Mississippi, Louisiana, Texas, and Colorado. It can live in a variety of salt and brackish marsh, as well as freshwater wetlands. The plant structure seems to be most important, with the rail requiring dense overhead perennial herbaceous cover. The rail is primarily threatened by loss and degradation of wetlands due to habitat conversion (USFWS 2025l). The eastern black rail was listed as threatened in 2020 (85 FR 63764). See the draft Recovery Plan for additional information (USFWS 2025l).

Effects

Colorado is not part of the action area in this project, and Colorado has the only known breeding population of interior eastern black rails. The GBIF/NRM dataset does not contain any occurrences for the rail, so potential exposure as estimated with that dataset is unknown. Roads are unlikely to be built through the coastal wetlands, as there are no timber or fuels reduction priorities, however, new roads constructed nearby could cause associated degradation. The proposed action is unlikely to occur where the rail lives (discountable), therefore, the proposed action *may affect, but is not likely to adversely affect* the eastern black rail.

Everglade snail kite (Rostrhamus sociabilis plumbeus)

The Everglade snail kite is a medium-sized hawk that occupies freshwater marshes in Florida and feeds exclusively on apple snails. It was listed as threatened in 1967 (32 FR 4001) and critical habitat was designated in 1977 (42 FR 47840). The snail kite is threatened by habitat fragmentation and loss of apple snails, with documented decreases associated with periods of drought. However, its population has recently expanded northward in Florida, in part due to the invasion of the non-native island apple snail, providing an additional food source. However, year-to-year variability is high, with continued uncertainty regarding water and habitat quality, and food availability (USFWS 2023j). See the 5-year review for additional information (USFWS 2023j).

Effects

The GBIF/NRM dataset does not contain any occurrences for the snail kite, so potential exposure as estimated with that dataset is unknown. There is no critical habitat in or near the action area. Although nomadic, the snail kites are confined to wetland habitats and are not known to occur on Ocala National Forest (which is the only potential forest within their range). Given the lack of known occurrences on National Forest System lands, the proposed action will have *no effect* on the Everglade snail kite.

Mississippi sandhill crane (Antigone canadensis pulla)

The Mississippi sandhill crane is a tall gray crane that occupies wet pine savanna exclusively in Jackson County, Mississippi. The subspecies was listed as endangered in 1973 (38 FR 14678) with critical habitat designated in 1977 (43 FR 39985). Specifically, the Mississippi Sandhill Crane National Wildlife Refuge was established in 1975 to protect habitat for this non-migratory population. The population is not yet self-sustaining and captive-raised juveniles are used to sustain it (USFWS 2024q). Habitat loss and degradation outside of the refuge is an ongoing threat, plus predation being the most common cause of death for the subspecies. See the 5-year review for additional information (USFWS 2024q).

Effects

The only potential habitat on National Forests is on De Soto National Forest, with no known occurrences. The GBIF/NRM dataset does not contain any occurrences for the snail kite, so potential exposure as estimated with that dataset is unknown. There is no critical habitat in or near the action area, and in fact the action area does not include any of the De Soto National Forest. Because the crane and its habitat do not occur in the action area, the proposed action will have *no effect* on the Mississippi sandhill crane.

Whooping crane (Grus americana)

The whooping crane is North America's tallest bird and was listed as endangered in 1967 (32 FR 4001), with critical habitat designated in 1978 (43 FR 20938) and revised in 1997 (62 FR 38932). The only natural self-sustaining population migrates between Aransas National Wildlife Refuge on the Texas coast and Wood Buffalo National Park in Alberta, Canada. All living whooping cranes are descended from 15 that wintered in Texas in 1941. Whooping cranes were historically threatened due to habitat loss and hunting and continue to be threatened by loss of

wetlands to cropland conversion, urbanization, roads/powerlines/wind farms that have all impacted the migratory corridor. Decreased river flows have also degraded migration and wintering habitats (USFWS 2012b). In addition to the endangered population, four areas have been designated as experimental nonessential for the reintroduction of whooping cranes: Florida (58 FR 5647), five Rocky Mountain states (62 FR 38932), nineteen eastern states (66 FR 33903), and Louisiana (76 FR 6066). Reintroduction efforts continue in the eastern migratory population and Louisiana, while the non-migratory Florida population reintroduction program has been discontinued. During winter 2023-2024 there were 162 cranes in reintroduced populations and 144 cranes managed under human care. The Aransas-Wood Buffalo population is considered a conservation success story, with over a 4% long term growth rate, with most recent counts about 536 (McAbee and Conkin 2025). See the Whooping Crane Status Report for additional information (McAbee and Conkin 2025).

Effects

From the GBIF/NRM dataset, there are a total of 21 occurrences on NFS lands, none of which are in the action area. This suggests a low chance of exposure for individuals to potential future roads and/or timber operations. Roads are unlikely to be developed in wetland areas that the crane uses as stopover sites. Given the lack of critical habitat or occurrences within the action area, and that the proposed activities are unlikely to occur in habitat, the proposed action will have *no effect* on the whooping crane.

Shorebirds

Piping plover (Charadrius melodus)

The piping plover is a small migratory shorebird that nests and feeds along coastal sand and gravel beaches in North America. It mostly uses sparsely vegetated sandbars or shorelines on river systems and alkaline lakes. The Atlantic and Northern Great Plains populations were listed as threatened in 1986, and the Great Lakes population was listed as endangered (50 FR 50726). Main threats to these populations include habitat loss (in breeding and wintering areas), damming and water withdrawals, predation, and human disturbance (USFWS 2024r). Critical habitat was designated for the Northern Great Plains breeding population in 2002 (67 FR 57638), the Great Lakes breeding population in 2001 (66 FR 22938), and for all wintering piping plovers in 2001 (66 FR 36038, revised 2009 in 74 FR 23476). See the 2024 5-year Review (USFWS 2024r) and the 2016 Draft Revised Recovery Plan for the Northern Great Plains (USFWS 2016c) for current species information.

Effects

Like most shorebirds, the piping plover stays close to shorelines and beaches without forest. From the GBIF/NRM dataset, there are a total of 144 occurrences on NFS lands, none of which are in the action area. This suggests a low chance of exposure for individuals to potential future roads and/or timber operations. As shoreline habitats do not have timber or fuels reduction needed, new roads are not likely to be proposed in piping plover habitat. In the unlikely event that a new road would be proposed in the future, the project would need to comply with ESA at the time. Critical habitat does not occur in the action area. Because activities related to the

proposed action are highly unlikely to occur in their habitat, there will be *no effect* on the piping plover.

Rufa red knot (Calidris canutus rufa)

The rufa red knot is a robin-sized shorebird known for being an extreme long-distance migrant, requiring spots of habitat from the extreme south of Tierra del Fuego to the far north of the central Canadian Arctic. It nests in dry, slightly elevated tundra locations with sparse vegetation. During migration and winter it uses coastal marine and estuarine habitats with large areas of exposed intertidal sediments. It may also use inland saline lakes in the Northern Great Plains as stopover sites. The rufa red knot is primarily threatened by habitat loss due to coastal development and sea level rise. It was listed as threatened in 2015 (79 FR 73706) and critical habitat was designated in 2023 (88 FR 22530). See the 2021 5-year review for current species information (USFWS 2021c).

Effects

Like most shorebirds, the rufa red knot stays close to shorelines and beaches without forest. The GBIF/NRM dataset does not contain any occurrences for the rufa red knot, so potential exposure as estimated with that dataset is unknown. As shoreline habitats do not have timber or fuels reduction needed, new roads are not likely to be proposed in rufa red knot habitat. In the unlikely event that a new road would be proposed in the future, the project would need to comply with ESA at the time. Critical habitat does not occur in the action area. Because activities related to the proposed action are highly unlikely to occur in their habitat, there will be *no effect* on the rufa red knot.

Western snowy plover (Charadrius nivosus nivosus)

The Pacific Coast Distinct Population Segment of the western snowy plover was listed as threatened in 1993 (58 FR 12864). The western snowy plover subspecies is a small ground nesting shorebird that ranges from Washington, south along the Pacific Coast to southern Baja California, Mexico, with wintering populations as far north as British Columbia, Canada. The western snowy plover needs sparsely vegetated sandy areas for nesting along the coast. Threats include habitat loss and modification from a variety of factors (development, sea level rise), human disturbance (recreation, drone use, etc.), introduced beachgrass, and increased predator populations (USFWS 2024s). Revised critical habitat was designated in 2012 (77 FR 36727) in Washington, Oregon, and California. The primary physical and biological features essential to conservation are sandy beaches, dune systems, salt/mud flats, etc. that include: 1) areas that are below heavily vegetated/developed areas and above high tides, 2) shoreline habitats with little to no vegetation that support invertebrates as prey, 3) organic debris like seaweed that both supports prey and provides cover, and 4) minimal human disturbance. See the 2024 5-year review (USFWS 2024s) and the 2007 Recovery Plan (USFWS 2007c) for current species information.

Effects

Like most shorebirds, western snowy plovers stick closely to shorelines and beaches without forest. The GBIF/NRM dataset does not contain any occurrences for the western snowy plover, so potential exposure as estimated with that dataset is unknown. There are approximately 621

acres of critical habitat in the action area, which is 2.5% of the total critical habitat, almost none of which is in potentially operable areas. As shoreline habitats do not have timber or fuels reduction needed, new roads are not likely to be proposed in snowy plover habitat. In the unlikely event that a new road would be proposed in the future, the project would need to comply with ESA at the time. Because activities related to the proposed action are highly unlikely to occur in their habitat or designated critical habitat, there will be *no effect* on the western snowy plover or its critical habitat.

Grassland birds

Greater sage-grouse (Centrocercus urophasianus)

The greater sage-grouse is the largest North American grouse species, known for elaborate spring mating displays on traditional lekking grounds. The Bi-State greater sage-grouse is a proposed threatened Distinct Population Segment that lives in the California-Nevada border near Mono Lake. The DPS was proposed in 2013 (78 FR 64357) with critical habitat proposed in 2014 (78 FR 64328). The greater sage-grouse is a sagebrush obligate that requires large, interconnected expanses of sagebrush with healthy, native understories. Sage-grouse exhibit strong site fidelity, limiting the degree to which they can adapt when habitat changes. Sagebrush ecosystems are threatened by a range of disturbances including drought, altered wildlife regimes, expansion of invasive species, and human development (Coates et al. 2024). There are multiple impacts to habitat interacting in the Bi-State area including pinyon-juniper encroachment, urbanization and infrastructure development, fire-invasive species cycles, and local increases in predation (USFWS 2020f). However, a 2012 Bi-State Action Plan resulted in planning and implementing conservation actions that have reduced the average annual rate of population decline (Coates et al. 2024). See the 2020 Species Report for additional information (USFWS 2020f).

Effects

From the GBIF/NRM dataset, there are a total of 4,342 occurrences on NFS lands, 25% of which are in the action area, and 11% of those are in potentially operable areas. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations. Approximately 439,330 acres of proposed critical habitat occurs within the action area, which is 23.5% of the total proposed critical habitat. Most of that critical habitat, however, is not within potentially operable areas. Physical and biological features essential to the conservation include sagebrush plant communities and interspersed mesic areas of sufficient size and configuration to provide for foraging, nesting, and rearing habitats (78 FR 64328).

Roads are a known factor contributing to habitat fragmentation in sagebrush ecosystems. In addition, roads can increase human access, spread nonnative plants, provide predator access, and cause direct mortality to sage-grouse (USFWS 2020f). Noise associated with roads and energy development have been shown to decrease lek attendance (Blickley et al. 2012). An extensive road network already occurs throughout the Bi-state area, but all federal lands have made efforts for Bi-state conservation including closing unauthorized routes and restricting new road development (USFWS 2020f). Restrictions on new roads exist in both the Inyo and Humboldt-Toiyabe Forest Plans (USFWS 2020f). As there is already an extensive road network in the area,

new roads are not particularly likely to be developed. However, adverse effects could result from removal of suitable habitat and introducing increased nonnative plants and predators. Removal of sagebrush could have adverse impacts on the critical habitat PBFs. Therefore, the proposed action *may affect, is likely to adversely affect* the proposed Bi-state sage grouse and its proposed critical habitat.

Gunnison sage-grouse (Centrocercus minimus)

The Gunnison sage-grouse lives exclusively in the sagebrush steppe ecosystems of southwestern Colorado and southeastern Utah. The second largest grouse in North America and a sagebrush obligate, Gunnison sage-grouse are found in eight small populations: seven in Colorado and one in Utah. The species was listed in 2014 (79 FR 69191) with critical habitat designated as well (79 FR 69311). The species requires relatively continuous and healthy sagebrush stands with a well-developed understory and uses a mosaic of sagebrush habitats throughout the range (USFWS 2020g). As a sagebrush obligate, it is threatened by loss of sagebrush, particularly due to habitat conversion for agriculture and development, invasive plants, increased pinyon-juniper, recreation, and some grazing practices (USFWS 2024t). See the 2024 5-year review for additional information (USFWS 2024t).

Effects

Because the action area does not include Colorado, only one Gunnison sage-grouse population is within the action area – Monticello in Utah. Monticello is a small satellite population that has steadily decreased since 2009 and may have as few as three individuals remaining, in spite of ongoing efforts to develop and maintain habitat. The population struggles with solar development, cropland and grazing pressures, as well as the effects of increases in ravens (USFWS 2024t). This population does not have designated critical habitat. From the GBIF/NRM dataset, there are a total of 22 occurrences on NFS lands, none of which are in the action area. This suggests a low chance of exposure for individuals to potential future roads and/or timber operations. Given the lack of occurrence overlap with the action area, the proposed action will have *no effect* on the Gunnison sage-grouse.

Other birds

California condor (Gymnogyps californianus)

California condors are the largest landbird in North America and can fly up to 250 miles a day searching for food. They require expansive open spaces such as grasslands for foraging with abundant carrion and well as mountainous cliffs or large broken top trees for nesting. Condors were listed in 1967 (32 FR 4001) and critical habitat was designated in 1974 (41 FR 21914). At the time of listing, condors were restricted to the southern end of the San Joaquin Valley in California. The range continued to contract until a captive breeding program was established in 1982, with the last wild condor captured at Bitter Creek National Wildlife Refuge in 1987. Since 1992, when captive-bred condors were first reintroduced back into the wild, the species has slowly recolonized its former range. Nonessential Experimental Populations were designated in specified portions of Arizona, Nevada, and Utah in 1996 (61 FR 54043) and more recently in specified portions of northern California, northwest Nevada, and Oregon in 2001 (86 FR 15602). The primary threat to species recovery is lead poisoning from spent ammunition. In addition,

powerlines, diseases (specifically highly pathogenic avian influenza, HPAI), intentional shooting, and wind energy contribute to mortalities. The wild populations continue to be supplemented by captive populations, and provided food, as the populations are not yet self-sustaining. See the 2023 5-year review for current species information (USFWS 2023k).

Effects

Condors are primarily threatened by lead poisoning and are not particularly sensitive to habitat loss or fragmentation, as they use such expansive areas. From the GBIF/NRM dataset, there are a total of 1,642 occurrences on NFS lands, 18% of which are in the action area, and 15% of those are in potentially operable areas. This suggests a moderate chance of exposure for individuals to potential future roads and/or timber operations. There are approximately 13,345 acres of critical habitat in the action area, which is 2.2% of total critical habitat. Less than half of that critical habitat is within potentially operable areas. Critical habitat when designated did not provide essential physical and biological features beyond land, water, and airspace up to 3,000 ft in elevation. The proposed action is unlikely to impact the critical habitat PBFs to the degree that they are no longer functional. New road development into previously roadless areas could remove potential nesting trees, however, nesting locations is not a limiting factor for condors.

As mentioned with other species, before any new roads were constructed, or any associated activities occurred in the action area that were previously not allowed, site-specific analyses and ESA compliance would be required. Thus, the proposed action *may affect, but is not likely to adversely affect* the California condor or its critical habitat.

Insects and Arachnids

To assist analyzing effects to a large number of species, ESA-listed species were grouped taxonomically. The insect and arachnid group includes 21 total species including 11 endangered, 7 threatened, 1 proposed endangered, and 2 proposed threatened insects and arachnids (Table 18). Insects and arachnids are widely varied in life history and habitat requirements. Most of the listed insects and arachnids are primarily terrestrial except for larval stages of dragonfly and stonefly species. Most of the insect species are capable of flight and can move around in search of food and nesting sources. Two insect species, the monarch butterfly and Suckley's cuckoo bumble bee, occur on many forests while many of the other species occur on just one or two forests.

Table 11. Threatened (T), endangered (E), proposed threatened (PT), and proposed endangered (PE) insects and arachnid species in the action area. The presence of critical habitat (and its percent overlap with the action area) is also provided. A summary of exposure and the effects determination is also shown.

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
American burying beetle	*Nicrophorus americanus*	T		Low	LAA	
Carson wandering skipper	*Pseudocopaeodes eunus obscurus*	E		Unk	LAA	
Dakota skipper	*Hesperia dacotae*	T	Y (6.6%)	Unk	LAA	LAA

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Franklin's bumble bee	*Bombus franklini*	E		Low	LAA	
Hermes copper butterfly	*Lycaena hermes*	T	Y (1.1%)	High	LAA	NLAA
Hine's emerald dragonfly	*Somatochlora hineana*	E		Low	NLAA	
Karner blue butterfly	*Lycaeides melissa samuelis*	E		Low	NLAA	
Kern primrose sphinx moth	*Euproserpinus euterpe*	T		Moderate	LAA	
Laguna mountains skipper	*Pyrgus ruralis lagunae*	E	Y (3.5%)	Moderate	LAA	NLAA
Mitchell's satyr butterfly	*Neonympha mitchellii mitchellii*	E		Unk	LAA	
Monarch butterfly	*Danaus plexippus*	PT		High	LAA	
Mount Charleston blue butterfly	*Icaricia shasta charlestonensis*	E		Low	LAA	
Oregon silverspot butterfly	*Speyeria zerene hippolyta*	T		Unk	LAA	
Quino checkerspot butterfly	*Euphydryas editha quino*	E	Y (1.3%)	High	LAA	NLAA
Rusty patched bumble bee	*Bombus affinis*	E	Y (<1%)	High	LAA	NLAA
Silverspot	*Speyeria nokomis nokomis*	T		Low	LAA	
Spruce-fir moss spider	*Microhexura montivaga*	E		Unk	NLAA	
Suckley's cuckoo bumble bee	*Bombus suckleyi*	PE		High	LAA	
Taylor's (=whulge) Checkerspot	*Euphydryas editha taylori*	E	Y (1.9%)	Moderate	LAA	NLAA
Valley elderberry longhorn beetle	*Desmocerus californicus dimorphus*	T		Low	NLAA	
Western glacier stonefly	*Zapada glacier*	T		Low	NLAA	
Western regal fritillary	*Argynnis idalia occidentalis*	PT		Unk	LAA	

Road construction, road use, and timber harvest can affect insect and arachnid species by removing habitat and host plants, fragmenting habitat leading to a loss of gene flow between populations, creating edge effects that destabilize environmental conditions that support the species, causing direct mortality through vehicle strikes and right-of-way maintenance equipment, and increasing toxins associated with herbicide and road salt, if used. The primary

effects of the proposed action on insect and arachnid species are discussed below and presented in Table 12. A discussion on species protection afforded by land management plans and other documents beyond the protections of the 2001 Roadless Rule is also included.

Major Stressors

Direct Mortality

Mobile adult insects are expected to avoid activities that might cause direct mortality, except for the use of roads by vehicles. Momeni-Dehaghi et al. (2024), estimate migratory monarchs migrate 3,283 km and cross 3,089 roads during fall migration. Migrating monarch butterflies have an estimated 40% probability of surviving migration that probability drops precipitously with increasing time spent flying at vehicle height. Further, monarch butterflies are not known to avoid roads and vehicles (Momeni-Dehaghi et al. 2024). However, vehicular traffic on NFS roads typically occurs at slow speeds and low volume, minimizing the likelihood of striking flying insects (Munoz et al. 2014). Baxter-Gilbert et al. (2015) found insect mortality rates from vehicle collision to differ among insect families with lepidopterans dying at a rate of 10.1 individuals/km/day and hymenopterans at 26.8 individuals/km/day. Adjacent habitat type appears to affect the impact of roads on pollinating insects. Keilsohn et al. (2018) found that road mortality of insects was greatest when adjacent habitat was meadows or curated lawns and lower when the area was wooded. Road structure also has an impact on insect mortality from strikes as Keilsohn et al. (2018) found lepidopteran mortality was highest when a median was present. NFS roads in remote areas are generally unpaved and single lane, which also minimizes the likelihood of vehicle collision with flying aquatic insects, that often mistake paved roads for water in which to lay their eggs (Munoz et al. 2014). In contrast, insect larvae and pupae are less mobile and can be directly killed by vegetation removal, vehicular strikes, and trampling. Much of the anticipated impacts of roads on insect communities are related to the use of salt to alleviate icy conditions during winter. However, NFS roads do not use salt to remove icy conditions.

Bees and butterflies are important pollinators that may be impacted by road construction and timber harvest. Listed bees can use forested habitat for nesting, overwintering, and foraging. For instance, timber harvests and road construction during overwintering could lead to direct mortality of rusty patched and Suckley's cuckoo bumble bee queens that inhabit leaf litter and shallow soils (89 FR 102074, USFWS 2024u). Similarly, American burying beetles reproduce on buried carcasses just below the soil surface. Soil disturbing activities such as road surfacing and the movement of heavy equipment can crush nesting beetles. However, those same activities can kill small animals that burying beetles can then use for breeding and feeding. In contrast to risks of direct mortality from vehicle strikes for most flying and some ground insects, the spruce-fir moss spider inhabits high elevation moss-covered rocks and is unlikely to be affected by moving vehicles.

Because future timber and road projects may or may not include seasonal restriction for removing trees, prescribed fire, or road construction activities during active flight periods, it is possible an active colony (nest) or hibernating female of insect species could be disturbed, crushed, or burned during project implementation. These effects would result from the crushing

or caving-in of burrows used by colonies, or other similar underground cavities or decaying logs that offer resting and sheltering.

Habitat Fragmentation and Edge Effects

Road construction and subsequent habitat fragmentation pose threats to threatened and endangered insects. The primary effects include outright habitat loss, the creation of barriers to movement, and the degradation of remaining habitat quality through "edge effects" (Munoz et al. 2014). These factors collectively increase the risk of local extinctions and undermine the long-term viability of vulnerable insect populations.

Habitat fragmentation is the physical destruction and division of continuous landscapes into smaller, isolated patches. For threatened and endangered insects, these remaining fragments often become too small to support viable populations, lacking sufficient resources like host plants, breeding grounds, or diverse microhabitats. The reduction in total habitat area is often a key driver of species decline.

Roads act as physical and behavioral barriers to insect movement and are particularly detrimental to species with limited dispersal capabilities. Flightless insects, such as some ground beetles, are highly susceptible, with major roads like highways acting as near-absolute barriers to gene flow (Munoz et al. 2014). Even highly mobile flying insects, like butterflies and bees, often exhibit reluctance to cross roads, even when suitable habitat is nearby. This behavioral avoidance, combined with physical limitations for smaller species, results in the isolation of populations, reduced gene flow, and genetic bottlenecks, which severely compromise a population's ability to adapt to environmental changes.

Habitat degradation around roads further exacerbates the challenges faced by endangered insects. The "edge effect" is a major consequence, where conditions along the road's edge differ drastically from the interior of the habitat patch. These altered conditions, including changes in light, temperature, and humidity, may be unsuitable for species adapted to more stable environments. The disrupted conditions also favor the establishment of non-native and invasive species that outcompete the native flora and fauna the insects rely on.

Finally, roads disrupt essential species interactions, such as predator-prey dynamics and vital pollination processes. Roadsides can attract pollinators with abundant flowering plants but simultaneously expose them to higher risks of mortality. The balance between potential breeding benefits in road rights-of-way and the costs of mortality and exposure to pollutants remains an area of active research.

Loss of Host Plants

The loss and decline of host plants can adversely affect threatened and endangered insects, primarily because many herbivorous insects are specialists that rely exclusively on specific plant species for their survival and reproduction. This specialization means that the disappearance or reduction of a particular host plant directly translates to a loss of food, habitat, and a site for laying eggs, which can lead to a cascade of negative effects that significantly heighten extinction risk for these vulnerable insects (Boggs and Ehrlich 2007).

Most herbivorous insects have evolved to overcome the specific physical and chemical defenses of one or a few closely related plant species. These specialized insects cannot simply switch to another plant when their primary host declines as other plants may be unpalatable, lack the necessary nutrients, or even be toxic (Haan et al. 2018). A classic example is the Monarch butterfly (*Danaus plexippus*), whose caterpillars can only survive by feeding on plants in the milkweed genus (*Asclepias*). The loss of milkweed due to changing land management practices and habitat destruction is directly linked to significant Monarch population declines. Without milkweed, Monarchs cannot complete their life cycle, highlighting how the fate of an insect species is tied directly to the availability of its specific host plant (USFWS 2020h).

There may be a direct loss of floral resources (crushing, uprooting) from heavy equipment use, though this loss is considered temporary and limited to one to two seasons after implementation. Short-term loss of these plants is not expected to cause a wide-spread reduction in host plants or their pollinators. Removing timber will result in varied reductions of canopy cover. The creation of small and large canopy gaps and removal of the smaller understory vegetation is expected to promote improved growing conditions for host plants. Canopy reduction is also expected to increase solar radiation to the ground and support plant and shrub growth, considered favorable to pollinators. On the periphery, tree removal is likely to reduce some of the shade and cover considered important for maintaining overwintering and nesting sites of some insect species (Ulyshen et al. 2024). Roadsides can often act as pollinator breeding habitat if increased light availability results in host plant proliferation (Meinzen et al. 2024). However, insect populations are also negatively affected by roads and traffic. Where roadside habitat is managed for pollinators and traffic is low and slow moving, roadsides could provide a net benefit to pollinator species.

The effects of host plant loss manifest through several mechanisms, the first being reproductive failure (Haan et al. 2018). Adult female insects typically lay their eggs exclusively on their specific host plants. The absence or scarcity of these plants results in fewer places to lay eggs, leading to reduced reproductive success and a smaller next generation. This is particularly true when native plants are displaced by non-native species, which insects often do not recognize as suitable for feeding or oviposition (Tallamy et al. 2020). Furthermore, a decline in host plant abundance means increased competition for food, lower nutritional intake for larvae, slower development, and potentially higher mortality rates. Changes to abiotic factors such as temperature and precipitation can exacerbate this issue by disrupting the synchronized timing of insect life cycles and the availability of their host plants. If a host plant emerges earlier or later than the insect larvae are ready to feed, mass starvation can occur, creating a critical mismatch between the insect's needs and resource availability (Abarca and Spahn 2021).

As host plants become scarce and fragmented, insect populations that rely on them also become isolated in small patches of remaining habitat. This fragmentation prevents insects from moving between populations, leading to reduced gene flow, decreased genetic diversity, and a higher vulnerability to local extinctions.

Insects are a foundational part of the food web, serving as a critical food source for other animals, especially birds and small mammals. The loss of host plants reduces the insect biomass,

which in turn impacts the reproductive success and survival of animals at higher trophic levels, demonstrating a broader ecological impact beyond the initial plant-insect relationship.

Sedimentation and Pollution

Only two ESA-listed insect species have an aquatic lifecycle, the Hine's emerald dragonfly and western glacier stonefly. Aquatic insect larvae are susceptible to being buried by sedimentation, which could lead to direct mortality by clogging their gills. However, Hine's emerald dragonfly larvae primarily inhabit shallow, slow-moving waters of marshes and fens, where sedimentation is not considered a major threat to the species (Environment and Climate Change Canada 2021). In contrast, western glacier stoneflies inhabit cool glacier-fed streams in northern Montana, where sedimentation could be an issue. However, sedimentation is not identified as a stressor to this species as it inhabits remote, undeveloped areas. The major threat to this species is the loss of glacier melt that supports their aquatic life stage. Given the low likelihood these species will be exposed to sedimentation given their habitat and range, and that the other ESA-listed insect species do not have aquatic life stages, the effects of sedimentation on listed insects are not expected to occur.

Existing Protections

The Cleveland, Los Padres, and San Bernardino National Forests are required to avoid activities that result in removal, crushing, burying, burning, or mowing of host plants within critical and occupied habitat for threatened, endangered, and proposed butterfly species, unless guided differently by a species-specific consultation (land management plans available at USFS 2025a). Regardless of the existence of the roadless rule, this design criteria ensures effects to Hermes copper butterfly, Laguna Mountains skipper, Kern primrose sphinx moth, and Quino checkerspot butterfly are minimized or avoided. Similarly, the Huron-Manistee forest plan requires implementation of the Karner Blue Butterfly Recovery Plan and minimization of effects to that species, suggesting in the absence of the Roadless Rule, protections will continue into the foreseeable future.

The USDA Forest Service published Conservation and Management of Monarch Butterflies: A Strategic Framework (USFS 2015) to provide a meaningful approach to monarch conservation on NFS lands. The USDA Forest Service also supports implementation of the North American Monarch Conservation Plan (Commission for Environmental Cooperation 2008).

The Shoshone Forest management plan does not contain species-specific protective measures for the western glacier stonefly.

Critical Habitat

Effects to designated critical habitat are analyzed by reviewing the physical and biological features associated with that habitat and determining the effects the action may have on those features. Most insect critical habitat includes host plants in addition to abiotic characteristics suitable for species growth and reproduction. Only five listed insect species have designated critical habitat within the action area and of those only one, the laguna mountain skipper (3.5%) exceeds more than 2% overlap. In the event activities occur in designated critical habitat the

effects to physical and biological features will be minimized or avoided as dictated by local conditions. Some features of critical habitat, such as open canopy, may be positively affected by future timber and road activities. Most insect species utilize plants that are shrubs and flowering plants that are quicker to establish after disturbance compared to fully mature trees and larger vegetation. Given the small amount of overlap with designated critical habitat, that most vegetation impacts will be temporary, and that some effects may be positive, the proposed action is not likely to adversely affect designated critical habitat for insect species.

Determinations

Refer to Table 11 to see the effects determination for each insect and arachnid species and critical habitat. Where exposure was not discountable due to species range or existing protections, or effects would not be insignificant even if exposed, a likely to adversely affect determination was made.

Table 12. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of insect and arachnid species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of nest/den trees and snags	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Tree and Unit Marking	Walking	Human presence	Negligible response to human presence
			ATV	Noise; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; Negligible response to noise
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Direct mortality, particularly of larvae and less mobile life stages;
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence	Direct mortality, particularly of larvae and less mobile life stages;
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; Loss of host plants leading to reproductive failure

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; Loss of host plants leading to reproductive failure
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sedimentation	Direct mortality, particularly of larvae and less mobile life stages; Loss of host plants leading to reproductive failure; sedimentation leading to mortality of aquatic larval stages
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; chemical contamination/nutrients; Drainage Network; Peak/Baseflows	Direct mortality, particularly of larvae and less mobile life stages; Loss of host plants leading to reproductive failure; sedimentation leading to mortality of aquatic larval stages
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure;
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence; Loss of nest trees	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Loss of nest trees; Soil compaction; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; Loss of nest trees	Direct mortality, particularly of larvae and less mobile life stages;
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust;	Direct mortality, particularly of larvae and less mobile life stages;

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal	Direct mortality, particularly of larvae and less mobile life stages;
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; proliferation of invasive species could negatively affect breeding, feeding, and sheltering
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages; loss of host plants leading to reproductive failure;
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; sedimentation	Direct mortality, particularly of larvae and less mobile life stages; sedimentation leading to mortality of aquatic larval stages
	Revegetation	Seeding	Hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of larvae and less mobile life stages;

Amphibians

The amphibian group includes 19 total species that occur within the affected roadless areas, including 7 Endangered, 8 Threatened, 2 Proposed Endangered and 2 Proposed Threatened amphibian species. The Amphibian group consists of 11 frog and toad species and 10 salamander species. Amphibians are known for their biphasic life histories where most species have an aquatic larval stage and a terrestrial adult stage. For additional discussion of how the aquatic environment may be affected by the proposed action see the effects section for fish, bivalves, and crustaceans. However, there are some exceptions that vary in their life-histories. In this group, the 4 species of salamanders in the family Plethodontidae, the Cheat Mountain salamander, Jemez Mountains salamander, Kern Canyon slender salamander, and relictual slender salamander, do not have an aquatic larval stage, therefore all of the threats to these species will be from impacts to the terrestrial habitat. The Plethodontid species are lungless and are sensitive to changes in microclimate. The 2 hellbender (salamander) species, in this group, are completely aquatic and do not have a terrestrial life-stage. Most of the species are known to occur in several forest districts except 5 species, the Cheat Mountain salamander, Jemez Mountains salamander, Kern Canyon slender salamander, Ozark Hellbender, and relictual slender salamander, have small geographic ranges and only occur in one forest district.

Table 13. Threatened (T), endangered (E), proposed threatened (PT), and proposed endangered (PE) amphibian species in the action area and the units they occur on. The presence of critical habitat (and its percent overlap with the action area) is also provided. Species and critical effects determinations are included (no effect, NE; not likely to adversely affect, NLAA; likely to adversely affect, LAA; and not applicable, n/a).

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Arroyo toad	*Anaxyrus californicus*	E	Y (3.96%)	High	LAA	LAA
California red-legged frog	*Rana draytonii*	T	Y (6.8%)	High	LAA	LAA
California tiger salamander	*Ambystoma californiense*	T	0%	Low	LAA	NE
Cheat Mountain salamander	*Plethodon nettingi*	T	n/a	High	LAA	n/a
Chiricahua leopard frog	*Rana chiricahuensis*	T	Y (1.75%)	Moderate	LAA	LAA
Eastern hellbender	*Cryptobranchus alleganiensis alleganiensis*	PE	n/a	High	LAA	n/a
Foothill yellow-legged frog	*Rana boylii*	E	Y (7.33%)	High	LAA	LAA
Frosted flatwoods salamander	*Ambystoma cingulatum*	T	Y (2.86%)	Low	LAA	LAA
Guajon	*Eleutherodactylus cooki*	T	0%	UnK	LAA	NE
Jemez Mountains salamander	*Plethodon neomexicanus*	E	Y (6.53%)	High	LAA	LAA

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Kern Canyon slender salamander	*Batrachoseps simatus*	PT	Y (19.58%)	High	LAA	LAA
Mountain yellow-legged frog	*Rana muscosa*	E	Y (13.77%)	Moderate	LAA	LAA
Oregon spotted frog	*Rana pretiosa*	T	Y (0.58%)	High	LAA	NLAA
Neuse River waterdog	*Necturus lewisi*	T	0%	UnK	LAA	NE
Ozark hellbender	*Cryptobranchus alleganiensis bishopi*	E	n/a	High	LAA	n/a
Relictual slender salamander	*Batrachoseps relictus*	PE	Y (34.78%)	Moderate	LAA	LAA
Sierra Nevada yellow-legged frog	*Rana sierrae*	E	Y (10.45%)	Moderate	LAA	LAA
Western Spadefoot toad	*Spea hammondii*	PT	n/a	Moderate	LAA	n/a
Yosemite toad	*Anaxyrus canorus*	T	Y (8.09%)	High	LAA	LAA

Exposure to the proposed action

Exposure of amphibian species to the proposed action was calculated using the NRM and GBIF data and categorized as high, moderate, low, or unknown. When unknown, the FS assumes some level of exposure is likely.

Frogs and Toads

All the frog species in this group exhibit a biphasic life history with a terrestrial adult life stage and an aquatic tadpole stage with the exception of the guajon that is fully terrestrial. This is relevant because each species will be exposed to aquatic and terrestrial threats. In the terrestrial stage common threats include habitat loss and direct mortality. Amphibians seem exceptionally vulnerable to death on roads (Beebee 2013). Amphibians rely heavily on vibration detection. Salamanders can sense substrate vibrations (Ross and Smith 1980), and several frog species use ground-borne vibrations as part of their breeding communication (Narins 1990). All amphibian species studied so far demonstrate strong sensitivity to such vibrations (Caorsi et al. 2019). While research on how anthropogenic vibrations influence animal behavior remains limited (Cross et al. 2021), evidence shows that simulated human-made sounds and vibrations can reduce the calling rate of male toads, which may ultimately decrease reproductive success (Caorsi et al. 2019). Anthropogenic disturbances can also have negative physiological effects. For instance, the non-native African clawed frog showed increased stress and mortality when exposed to construction-related noise and vibrations (Felt et al. 2012). In addition, traffic noise can alter calling behavior in frogs that can ultimately affect fitness (Hoskin and Goosem 2010).

In the aquatic stage, amphibians may be exposed to sedimentation caused by runoff after road construction or timber harvest (Jones et al. 2011; Wagner et al. 2013), leading to mortality of these life stages.

Arroyo toad (Anaxyrus californicus)(E)

The arroyo toad occurs in 4 National Forests in the Pacific Southwest Region of the Forest Service. Threats on national forest lands include introduced aquatic predators, road and trail traffic, and tamarisk invasion (Arroyo Toad Recovery Plan, USFWS 1999). Arroyo toad populations are driven by surface water availability (Fisher et al. 2018). Declines of the arroyo toad are attributed to habitat degradation, invasive predators, and altered hydrological regimes (Treglia et al. 2015).

Existing Protections

The Angeles NF Forest Plan has designated Critical Biological Land Use Zones for the protection of species-at-risk and conservation strategies for riparian and aquatic habitats for species of concern, including the arroyo toad. Project planning includes habitat acquisition, vegetation management, invasive species control and fire management to benefit species of concern including the toad. The plan recommends withdrawal from mineral entry to T&E species key habitats in areas of mineral potential where habitat is not protected by any other means and would benefit by withdrawal. Protective measures will be maintained for the period of time needed to provide the necessary protection for sensitive species and key habitats.

The Los Padres, Cleveland and San Bernardino NF Plans have conservation strategies for the arroyo toad including; education on the importance of riparian and aquatic habitat and importance of keeping vehicles on roads; inventory of aquatic species including the arroyo toad; and restricted human access during critical life stages of the toad

California red-legged frog (Rana draytonii)(T)

The California red-legged frog occurs in 3 National Forests in the Forest Service Pacific Southwest Region. Doubledee et al. (2003) suggest that invasive species (bullfrog) are the primary drivers of decline.

Existing Protections

The Angeles NF Forest Plan has a conservation strategy for the California red-legged frog including; education on the importance of riparian and aquatic habitat and importance of keeping vehicles on roads; inventory of aquatic species including the red-legged frog; habitat restoration and control of invasive species. The plan recommends withdrawal from mineral entry to T&E species key habitats in areas of mineral potential where habitat is not protected by any other means and would benefit by withdrawal. Protective measures will be maintained for the period of time needed to provide the necessary protection for sensitive species and key habitats.

The Los Padres NF Plan mentions protecting the California red-legged frog and other sensitive species but does not list specific protections. The California red-legged frog is not mentioned in other Forest Plans where it occurs.

Chiricahua leopard frog (Rana chiricahuensis)(T)

The Chiricahua leopard frog occurs in 4 National Forests in the Southwestern Region of the Forest Service. Increased sediment flows into occupied stock tanks will likely increase the need and frequency of maintenance to remove those sediments, resulting in increased likelihood of mortality of the frogs (USFWS 2007d). Post-fire sediment movement can be especially dramatic, filling in pools and covering stream beds in rocks and cobble. Harris et al. (2022) suggest that the main threats to the Chiricahua leopard frogs are *Batrachochytrium dendrobatidis* (Bd) and invasive bullfrogs.

Existing Protections

The Coronado NF Forest Plan states that efforts are underway to restore a number of species including the Chiricahua leopard frog and new issuance, renewal, modification, and management of grazing permits shall comply with the Coronado National Forest's "Stockpond and Aquatic Habitat Management and Maintenance Guidelines for the Chiricahua Leopard Frog."

The Apache, Cibola, Coconino and Gila NF Plans mention that the leopard frog occurs on the district, but no specific protections are listed. The leopard frog is not mentioned in the Prescott or Tonto NF Forest Plans.

Guajon (Eleutherodactylus cooki)(T)

The guajon occurs on the El Yunque National Forest in Puerto Rico in Region 8. The Listing Rule (62 FR 31757) states that the guajon may require special management considerations or protection due to threats to the primary constituent elements from road construction, agriculture, development, and fishing with chemicals. This species has direct development from egg to froglet, no aquatic larval stage (Townsend and Stewart 1985).

Existing Protections

The Guajon is not mentioned in the El Yunque NF Forest Plan.

Foothill yellow-legged frog (Rana boylii)(E)

The Foothill yellow-legged frog occurs in 7 National Forests in the Pacific Southwest Region of the Forest Service. Lind et al. (2016) suggest that the primary threats to the yellow-legged frog are large human-caused disturbances such as dams and diversions. Ashton et al. (1998) stated that logging and road related mass wasting events have resulted in periodically high levels of siltation. High levels of silt may inhibit the attachment of the egg mass to the substrate. Excessive accumulation of silt on the egg masses may have adverse effects on embryo development, but this needs further study.

Existing Protections

The Foothill yellow-legged frog is not mentioned in any Forest Plans where it occurs.

Mountain yellow-legged frog (Rana muscosa)(E)

The mountain yellow-legged frog occurs on 5 National Forest in the Pacific Southwest Region. Knapp et al. (2007) suggest that disease and invasive fish are major causes of decline. The

Mountain yellow-legged frog Recovery Plan (USFWS 2018h) states that each population is small and highly susceptible to stochastic events. Measures have been taken to reduce the impact of certain threats, including recreation, nonnative trout, and stochastic extinction. However, these threats and other threats to the habitat remain, including illegal activities (cannabis cultivation and suction dredge mining), and legal activities (recreational activities, fire suppression activities, and roadwork construction). Wildfire and long-term alteration of weather patterns both have a high likelihood of affecting southern mountain yellow-legged frog and its habitat; however, the timing and options available to reduce these threats are either limited or unclear.

Existing Protections

The mountain yellow-legged frog is listed as a primary protected species in Angeles NF Forest Plan. There is conservation strategy emphasis on habitat, invasive species control and monitoring. Additional emphasis will be placed on the use by recreationists and urban and national forest infrastructure that is sustainable such that it has minimal effects to species (mountain yellow-legged frog) and their habitat. The San Bernadino NF Plan states the importance of riparian and aquatic species and habitat, value of vegetation management to species at risk, species monitoring, and keeping vehicles on roads for species including the mountain yellow-legged frog.

Inyo, Sequoia and the Sierra NF Plans mentions mountain yellow-legged frog but do not list specific protections.

Sierra Nevada yellow-legged frog (Rana sierrae)(E)

The Sierra Nevada yellow-legged frog occurs in 10 National Forests across the Intermountain and Pacific Southwest Regions of the Forest Service. The declines in populations of Sierra Nevada yellow-legged frogs have been largely attributed to the introduction of trout to historically fishless lakes in high elevations and to the spread of the chytrid fungus (Brown et al. 2019). Threats to all the yellow-legged frog species are similar.

Existing Protections

The Lake Tahoe BMU Land Management Plan has a section for Special Status Species Habitat areas that are defined as areas of quality habitat for Federal Threatened, Endangered, Candidate, and Proposed species, including the Sierra Nevada yellow-legged frog. It calls for work collaboratively with partners to identify and implement additional habitat restoration efforts that expand the range of Sierra Nevada yellow-legged frog within historic habitat throughout the Basin. Specific objectives include; complete restoration of seven high alpine lakes (composed of habitat areas that would support four sub-populations) adjacent to current yellow-legged frog populations in the Desolation wilderness by removing introduced trout species by 2029

The Inyo NF Plan specifies that where pesticide applications are proposed within 500 feet of known occupied sites for Yosemite toad, Sierra Nevada yellow-legged frog, and Mountain yellow-legged frog, applications will be designed to avoid adverse effects to individuals and their habitats. The Sierra Nevada yellow-legged frog is not mentioned in the other Forest Plans where it occurs.

Oregon spotted frog (Rana pretiosa)(T)

The Oregon spotted frog occurs in 2 National Forests in the Pacific Northwest Region of the Forest Service. The Oregon spotted frog Recovery Plan (USFWS 2024v) states that population declines are the results of habitat degradation and fragmentation caused by altered hydrology, invasive vegetation, and non-native predators. Blouin et al. (2010) suggests that there is something about the biology of the Oregon spotted frog that predisposes it to substantial fragmentation, even in relatively undisturbed habitat.

Existing Protections

The Gifford Pichot NF Forest Plan lists the Oregon spotted frog as a sensitive species but does not list protections. The Oregon spotted frog is not mentioned in the Deschutes NF Forest Plan.

Western spadefoot (Spea hammondii)(PT)

The western spadefoot occurs in 7 National Forests in the Pacific Southwest Region of the Forest Service. Shedd (2016) stated where roadways intersect areas inhabited by spadefoot populations, adults and younger terrestrial stage individuals may be observed on road surfaces. Roads increase danger for spadefoots as well, as road-killed western spadefoots have been documented.

Existing Protections

The San Bernardino NF Forest Plan mentions conservation strategies for inventory of riparian and aquatic species, habitat improvement and acquisition, and vegetation treatments for several species of concern, including the western spadefoot. The western spadefoot is not mentioned in any of the other Forest Plans where it occurs.

Yosemite toad (Anaxyrus canorus)(T)

The Yosemite toad occurs in 4 National Forests across the Intermountain and Pacific Southwest Regions of the Forest Service. Sherman and Morton (1993) suggest that chytrid fungus and changing weather (dry winters) are major factors in this species decline.

Existing Protections

The Sierra NF Forest Plan has a species-specific direction intended to minimize adverse impacts of activities on the Yosemite toad including suitable breeding and rearing habitat. This plan direction will help achieve conditions at the meadow scale that enable attainment and maintenance of desired conditions for Yosemite toad and its habitat. Guidelines include; 1) Grazing management practices, such as deferred seasonal grazing, meadow fencing, and modifying grazing period, will be used to avoid negative impacts to Yosemite toad and their habitat to the maximum extent feasible; 2) Yosemite toad breeding dates should be determined locally to avoid co-occurrence of cattle grazing and egg masses, tadpoles, juveniles, and breeding adults to the maximum extent feasible; and 3) monitor if there is sufficient breeding and rearing habitat to support the survival and recovery of local Yosemite toad populations. Specific protections include; 1) Check around equipment for Yosemite toad individuals prior to moving equipment and follow approved handing guidelines if moving individuals is necessary; 2) Check

for Yosemite toad use in open dry lupine covered areas where numerous rodent burrows occur. Reduce impacts to high-quality upland habitat by considering avoiding or minimizing working or walking in or crossing these areas and avoiding vehicle use or storage and equipment storage in these areas; 3) Consider impacts of activities on Yosemite toad dispersal, especially the use of erosion control materials, 4) Reduce impacts on Yosemite toads by implementing a process for monitoring and relocating individual toads from travel ways during motorized special use events or using temporary elevated road or trail structures to avoid direct mortality to toads; and 5) Use burning techniques that allow Yosemite toads to escape burn piles and that leave patches of hiding cover in broadcast burns.

The Inyo NF Plan specifies that where pesticide applications are proposed within 500 feet of known occupied sites for Yosemite toad, Sierra Nevada yellow-legged frog, and Mountain yellow-legged frog, application will be designed to avoid adverse effects to individuals and their habitats. The Yosemite toad is not mentioned in the other Forest Plans where it occurs.

Salamanders

Roads have negative effects on salamanders through direct mortality, damaging habitat, and fragmenting populations. Traffic and wetland proximity increased negative impacts in some studies; abandoned logging roads showed negative effects. Positive effects were limited to habitat creation along roads (Ochs et al. 2024). Semlitsch et al. (2007) found that salamander abundance was significantly lower on old, abandoned logging roads compared with the adjacent upslope sites. These results indicate that forest roads and abandoned logging roads have negative effects on forest-dependent species such as plethodontid salamanders. Further, amphibians rely heavily on vibration detection and salamanders can sense substrate vibrations (Ross and Smith 1980). While all amphibian species studied so far demonstrate strong sensitivity to such vibrations (Caorsi et al. 2019), research on how anthropogenic vibrations influence animal behavior remains limited (Cross et al. 2021). Water siltation and sedimentation are serious threats that can follow logging and road activity.

Fully Aquatic salamanders

Threats are primarily to the aquatic environment.

Neuse River waterdog (*Necturus lewisi*)(T)

The Neuse River waterdog is a fully aquatic salamander species that occurs in the Croatan National Forest in the Southern Region of the Forest Service. There is a strong negative effect of fine sediments on occupancy of *Neuse River waterdog*. Analyses support the hypothesis that observed declines in the occurrence of Neuse River waterdog are a delayed result of sedimentation-induced recruitment limitation (Teitsworth et al. 2024).

Existing Protections

The Neuse River waterdog is not mentioned in the Croatan NF Forest Plan.

Eastern hellbender (Cryptobranchus alleganiensis lleganiensis)(PE)

The eastern hellbender is a fully aquatic salamander species that occurs in 9 National Forests across the Southern and Eastern Regions of the Forest Service. Habitat alterations, including siltation, water impoundment, and water quality degradation are likely the greatest threats to eastern hellbenders. Other threats include incidental take by anglers, collection for the pet trade, and predation by non-native fish (Mayasich et al. 2003).

Existing Protections

The Nantahala and Pisgah FN Forest Plans state that management actions will focus on key species, including eastern hellbenders. The plan calls for maintaining water quality for hellbenders in waters that support them, enhance hellbender and native brook trout populations where habitat is suitable, and restore and/or maintain healthy populations of hellbenders in East Fork and West Fork Pigeon River. Specific protections state that installation of new stream crossings, or their replacement, should avoid fragmenting existing populations of native aquatic species. Replacement of existing crossings should reconnect fragmented populations of native species (including hellbenders) or avoid fragmenting existing populations wherever relevant. Crossings that are barriers should be retained where they protect populations of native species. The eastern hellbender is mentioned in the Daniel Boone NF Plan but no specific protections are listed. The hellbender is not mentioned in any of the other Forest Plans where it occurs.

Ozark hellbender (Cryptobranchus alleganiensis bishopi)(E)

The Ozark hellbender is a fully aquatic salamander species that occurs in the Mark Twain National Forest in the Eastern Region of the Forest Service. The use of interstitial spaces by larval the Ozark hellbender may make them particularly vulnerable to siltation and sedimentation (Pitt et al. 2016). Siltation and sedimentation have been suggested as factors contributing to the decline of the hellbender and its prey due to the degradation and reduction of habitat (Briggler et al. 2007).

Existing Protections

The Ozark hellbender is not mentioned in the Mark Twain NF Forest Plan.

Salamanders with terrestrial adults and aquatic larvae

California tiger salamander (*Ambystoma californiense*)(T)

The California tiger salamander occurs in 3 National Forests in the Pacific Southwest Region of the Forest Service The apparent causes of this decline are habitat loss due to urbanization, adult salamander mortality from automobiles, loss of larvae through habitat drainage, and possibly predation by transient fish populations (Barry and Shaffer 1994). The California tiger salamander recovery plan (USFWS 2017c) states that multiple factors have contributed to population declines of this species, including habitat loss and fragmentation; predation from, and competition with, invasive species; hybridization with nonnative barred tiger salamanders (*Ambystoma tigrinum*); mortality from road crossings; contaminants; and small mammal burrow control efforts (69 FR 47212; USFWS 2014b). Other potential threats include introduction of

diseases such as ranaviruses and chytrid fungi, and long-term alteration of weather patterns (69 FR 47212; USFWS 2014b).

Existing Protections

The California tiger salamander is not mentioned in any of the Forest Plans where it occurs.

Frosted flatwoods salamander (*Ambystoma cingulatum*)(T)

The frosted flatwoods salamander occurs in 4 National Forests in the Southern Region of the Forest Service. The frosted flatwoods salamander recovery plan (USFWS 2021d) states that the improper or overuse of pesticides and herbicides, presence of predatory fish in breeding ponds, lack of regular lightning season fire to maintain suitable habitats, disruptions to hydrology including ditching and addition of impervious surfaces, road construction or increased traffic near breeding ponds, invasive plants, and disease are all threats. Westervelt et al. (2013) Suggest that even modest reductions in annual rainfall (5–10%) would affect the persistence of breeding habitat and result in further population declines and more moderate reductions (20–25%) would result in population extirpation.

Existing Protections

The Florida (Apalachicola and Osceola) NF Forest Plans outline specific protections for the flatwoods salamander. Within the primary buffer zone (600 ft. radius) of breeding ponds: prohibit mechanical site preparation, use only selective timber harvest methods, timber harvest will only occur during dry periods, and no application of pesticides, herbicides, or fertilizers, except directed foliar application of herbicide can be used to control noxious weeds. Injection, frill, girdle, thin-line basal spray or cut stump herbicides may be used to treat undesirable trees if prescribed fire cannot be employed. Within the secondary buffer zone (600-1,500 ft. radius) of breeding ponds: prohibit mechanical site preparation, if clearcutting is used, remove no more than 25 percent of the buffer in a single entry, harvest will only occur during dry periods, and no application of pesticides, herbicides, or fertilizers, except directed foliar application of herbicide can be used to control noxious weeds. Injection, frill, girdle, thin-line basal spray or cut stump herbicides may be used to treat undesirable trees if prescribed fire cannot be employed. The Conecuh NF Forest Plan mentions, in general terms, providing habitat for the salamander and other sensitive species but does not list specific protections.

Lungless terrestrial salamanders

This group is fully terrestrial, sensitive to changes in microclimate, and dependent on woody debris as habitat. Several studies have demonstrated reduced abundances of Plethodontid salamanders following timber harvests (Petranka et al. 1993, 1994).

Cheat Mountain salamander (*Plethodon nettingi*)(T)

The Cheat Mountain salamander occurs in the Monongahela National Forest in the Eastern Region of the Forest Service. Kroschel et al. (2014) suggest that a primary driver of decline is competition with another native salamander species. Dillard et al. (2008) suggest that extensive logging may have contributed to decline by altering the microclimate. Mielke et al. (1986) estimate that when the first European settlers entered what is now West Virginia, there were

nearly 1.5 million acres of red spruce *(Picea nibens),* and that spruce was reduced to approximately 750,000 acres by 1865 due to natural and human-caused factors.

Existing Protections

The Monongahela NF Forest Plan outlines management directions for the Cheat Mountain salamander. The goal is to identify opportunities to reduce fragmentation of populations and habitat. To Achieve this goal, surveys must be conducted and occupied habitat must be delineated prior to proposed vegetation or ground disturbance in known or potential habitat. Ground and vegetation-disturbing activities shall be avoided within occupied habitat and a 300-foot buffer zone around occupied habitat, unless analysis can show that the activities would not have an adverse effect on populations or habitat. Personal use mineral permits shall not authorize removal of material within stream channels or their banks, or within suitable Cheat Mountain salamander habitat.

Jemez Mountains salamander (*Plethodon neomexicanus*)(E)

The Jemez Mountains salamander occurs in the Santa Fe National Forest in the Southwestern Region of the Forest Service. Margolis and Malevich (2016) suggest that fire exclusion has diminished habitat and increases the risk of high severity fire that can cause direct mortality. Bartlow et al. (2022) stated that the lungless Cheat Mountains salamander requires moist surface conditions for activities such as mating and foraging.

Existing Protections

The Santa Fe NF Forest Plan mentions the Jemez Mountains salamander as occurring on the district and makes general comments regarding aquatic habitat management. No specific protection are listed for the salamander.

Kern Canyon slender salamander (*Batrachoseps simatus*)(PT)

The Kern Canyon slender salamander occurs in Sequoia National Forest in the Pacific Southwest Region of the Forest Service. The proposed rule (87 FR 63150) states wildfire and long-term alteration of weather patterns are the primary ongoing threats to the habitat.

Existing Protections

The Sequoia NF Forest Plan lists three watersheds where the Kern Canyon slender salamander in known to occur. The Forest Plan makes general statements regarding sustaining healthy watersheds but does not list specific protections for the salamander.

Relictual slender salamander (*Batrachoseps relictus*)(PE)

The relictual slender salamander occurs in Sequoia National Forest in the Pacific Southwest Region of the Forest Service. The proposed rule (87 FR 63150) states that wildfire and long-term alterations of weather patterns are the primary ongoing threats to the habitat. The maintenance and improvement of Highway 178 through the Kern River Canyon has been implicated in the decline and eventual extirpation of low-elevation populations of *relictual slender salamander* (Hansen 1988; Hansen and Wake 2005). Expansion of the highway involved the destruction of several small spring and seep habitats on the south side of the river that had previously supported populations of this species. Although suitable microhabitat was not entirely destroyed in the

stretch of canyon occupied by *relictual slender salamander*, it is probable that the hydrologic function of roadside waterways was modified significantly. Moss et al. (2025) suggest that removal of downed woody debris may contribute to decline.

Existing Protections

The Sequoia NF Forest Plan mentions the relictual slender salamander as a proposed species for ESA listing but does not list protections.

Response

Although direct mortality from vehicle strikes associated with roads and heavy equipment can occur, most impacts to amphibian species are expected to result from degradation of the aquatic environment and fragmentation of terrestrial habitat as described above. Responses to degradation of aquatic and terrestrial habitats can be as extreme as mortality or lesser effects such as suppression of breeding, feeding, and sheltering.

Effects Determination

Given the potential exposure of most amphibian species and the potential of the proposed activity to result in death or loss of reproduction, feeding, and sheltering the FS has determined the proposed activity is likely to adversely affect all of the ESA-listed amphibians in the action area. Similarly, designated critical in the action area is likely to be adversely affected, primarily through degradation of aquatic habitat by sediment except for critical habitat for the Oregon spotted frog. Critical habitat for this species has less than 1% overlap with the action area and is not likely to be adversely affected.

Table 14. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of amphibian species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of trees and snags; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
		Tree and Unit Marking	Walking	Human presence	Negligible response to human presence.
			ATV	Noise; Human presence; Trampling/run over	Direct mortality, particularly of terrestrial life stages; Noise and disturbance can affect calling activity in some species of frogs.
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over; vibration	Direct mortality, particularly of terrestrial lifestages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Direct mortality, particularly of terrestrial life stages; herbicides are known to decrease amphibian fitness/survival. Noise and disturbance can affect calling activity in some species of frogs.
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels. Noise and disturbance can affect calling activity in some species of frogs; avoidance behavior

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages. Noise and disturbance can affect calling activity in some species of frogs; avoidance behavior
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration	Direct mortality, particularly of terrestrial life stages; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration	Direct mortality; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration; hydrological changes	Direct mortality; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration; hydrological changes	Direct mortality; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages.; avoidance behavior
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence;	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels.
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Soil compaction; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; loss of vegetation leading changes in microclimate, reduction of species dependent on high microclimate moisture levels; avoidance behavior
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; proliferation of invasive species could negatively affect breeding, feeding, and sheltering; avoidance behavior
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration	Direct mortality; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sediment/Turbidity; Chemical Contamination/Nutrients; vibration; hydrological changes	Direct mortality; loss of fitness/survival in aquatic amphibians in response to changes in water quality; avoidance behavior
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; vibration	Direct mortality, particularly of terrestrial life stages; avoidance behavior
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; sedimentation; vibration	Direct mortality, particularly of terrestrial life stages; loss of fitness/survival in aquatic amphibians in response to response to changes in water quality. Noise and disturbance can affect calling activity in some species of frogs; avoidance behavior
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, particularly of terrestrial life stages. Noise and disturbance can affect calling activity in some species of frogs.

Fish

To assist analyzing effects to a large number of species, ESA-listed species were grouped taxonomically. The fish group includes 59 total species including 28 endangered, 27 threatened, 1 proposed endangered, 2 proposed threatened, and 1 similarity of appearance listing (Table 15). Several of these listings also include experimental populations. Unless otherwise discussed below, experimental populations were treated as if they were normal listed entities for the purpose of effects determinations. Fish species are variable, but all rely on sufficient water quality and quantity to support their life history stages.

Table 15. Threatened (T), endangered (E), proposed threatened (PT), proposed endangered (PE), similarity of appearance (SAT), and experimental populations (Exp) of fish in the action area. The presence of critical habitat (and percent overlap with the action area) is also provided. Effects determinations are also shown (NE = No Effect, NLAA = Not Likely to Adversely Affect, LAA = Likely to Adversely Affect, N/A = Not Applicable).

Species Common Name	Scientific Name	Listing Status	CH Present, percent overlap	Species Effects	CH Effects
Gulf sturgeon	*Acipenser oxyrinchus desotoi*	T	No	NLAA	NE
White sturgeon, Kootenai River DPS	*Acipenser transmontanus*	E	No	NLAA	NE
Pallid sturgeon	*Scaphirhynchus albus*	E	N/A	NLAA	N/A
Santa Ana sucker	*Catostomus santaanae*	T	Yes, 0.02	LAA	LAA
Warner sucker	*Catostomus warnerensis*	T	No	NLAA	NE
Shortnose sucker	*Chasmistes brevirostris*	E	Yes, 0.01	NLAA	NLAA
Lost River sucker	*Deltistes luxatus*	E	Yes, 0.01	NLAA	NLAA
Razorback sucker	*Xyrauchen texanus*	E	Yes, 0.0005	NLAA	NLAA
Blue shiner	*Cyprinella caerulea*	T	N/A	LAA	N/A
Spotfin chub	*Erimonax monachus*	T + Exp	No	NLAA	NE
Owens tui chub	*Gila bicolor* ssp. *snyderi*	E	No	NLAA	NE
Humpback chub	*Gila cypha*	T	No	LAA	NE
Sonora chub	*Gila ditaenia*	T	No	LAA	NE
Bonytail	*Gila elegans*	E	No	LAA	NE
Gila chub	*Gila intermedia*	E	Yes, 0.10	NLAA	NLAA
Chihuahua chub	*Gila nigrescens*	T	N/A	LAA	N/A
Yaqui chub	*Gila purpurea*	E	No	LAA	NE
Clear Lake hitch	*Lavinia exilicauda chi*	PT	N/A	LAA	N/A
Little Colorado spinedace	*Lepidomeda vittata*	T	Yes, 0.17	LAA	LAA
Spikedace	*Meda fulgida*	E	Yes, 0.09	LAA	NLAA
Cape Fear shiner	*Notropis mekistocholas*	E	No	NLAA	NE
Colorado pikeminnow	*Ptychocheilus lucius*	E	No	LAA	NE
Santa Ana speckled dace	*Rhinichthys gabrielino*	PT	N/A	LAA	N/A
Long Valley speckled dace	*Rhinichthys nevadensis caldera*	PE	N/A	NLAA	N/A
Kendall warm springs dace	*Rhinichthys osculus thermalis*	E	N/A	NE	N/A
Loach minnow	*Tiaroga cobitis*	E	Yes, 0.10	NLAA	NLAA
Dessert pupfish	*Cyprinodon macularius*	E	No	LAA	NE
Gila topminnow	*Poeciliopsis occidentalis*	E	N/A	LAA	N/A

Species Common Name	Scientific Name	Listing Status	CH Present, percent overlap	Species Effects	CH Effects
Unarmored threespine stickleback	*Gasterosteus aculeatus williamsoni*	E	N/A	LAA	N/A
Etowah darter	*Etheostoma etowahae*	E	N/A	LAA	N/A
Candy darter	*Etheostoma osburni*	E	Yes, 0.07	LAA	LAA
Duskytail darter	*Etheostoma percnurum*	E	N/A	LAA	N/A
Rush darter	*Etheostoma phytophilum*	E	No	LAA	NE
Cherokee darter	*Etheostoma scotti*	T	N/A	LAA	N/A
Kentucky arrow darter	*Etheostoma spilotum*	T	No	LAA	NE
Trispot darter	*Etheostoma trisella*	T	No	LAA	NE
Amber darter	*Percina antesella*	E	No	LAA	NE
Goldline darter	*Percina aurolineata*	T	No	LAA	NE
Peral darter	*Percina aurora*	T	No	LAA	NE
Conasauga logperch	*Percina jenkinsi*	E	No	LAA	NE
Paiute cutthroat trout	*Oncorhynchus clarkii seleniris*	T	N/A	LAA	N/A
Little Kern golden trout	*Oncorhynchus aguabonita whitei*	T	Yes, <0.0001	LAA	NLAA
Lahontan cutthroat trout	*Oncorhynchus clarkii henshawi*	T	N/A	LAA	N/A
Gila trout	*Oncorhynchus gilae*	T	N/A	LAA	N/A
Bull trout	*Salvelinus confluentus*	T + Exp	Yes, 0.009	LAA	LAA
Yaqui catfish	*Ictalurus pricei*	T	No	NE	N/A
Smoky madtom	*Noturus baileyi*	E + Exp	No	LAA	N/A
Yellowfin madtom	*Noturus flavipinnis*	T + Exp	No	LAA	N/A
Frecklebelly madtom	*Noturus munitus*	T	No	LAA	N/A
Carolina madtom	*Noturus furiosus*	E	No	NLAA	N/A

Effects of Roads and Timber

The impact/benefit crosswalk with pathway for fish effects in Table 17 was utilized in the effects analysis for fish when USFWS documents did not align with the described impact/benefits descriptions in Table 16. The impacts/benefits potential effects to fish species and critical habitat from the deconstruction table are summarized in the effects section. When species and/or critical habitat effects are described, duration of short-term effects subside immediately or up to one year, and long-term effects persist over a year. Direct and indirect effects were not distinguished in the analysis because they are hard to differentiate between them in some cases and the effects of each could be close together in timing.

Table 16. Deconstruction of the 2001 Roadless Rule rescission aquatic activities

Activity	*Management Element*	*Task/Method*	*Tool*	*Impact/Benefit*
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment/Motorized Hand Tools; Mules/Horses	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Riparian Resilience; Large Woody Debris
		Tree and Unit Marking	Walking	Chemical Contamination/Nutrients
			ATV	Chemical Contamination/Nutrients
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris; Change in Riparian Vegetation (Temperature); Riparian Resilience
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Riparian Resilience
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Large Woody Debris; Peak/Baseflows; Riparian Resilience
		Timber Haul	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (for example, ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
		Installation (for example, ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
		Replacement or Repair (for example, ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
	Vegetation Management	Brushing, Limbing, Seeding, Mowing, Mulching, Hazard Tree Removal	Heavy Equipment/Hand Tools	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Large Woody Debris
	Surface Repair/Replacement	Road Cut and Fill Repair/Stabilization	Heavy Equipment	Sediment/Turbidity; Spawning Substrate

Activity	*Management Element*	*Task/Method*	*Tool*	*Impact/Benefit*
		Asphalt Repair, Repaving, Chip-Sealing, Skin Patching, and Rocking	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate
		Small Slide Removal (easily handled with maintenance equipment)	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate
		Dust Abatement (for example, Water, Lignin Sulfate, Magnesium Chloride)	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows
	Structure Maintenance, Repair, and Installation	Guardrails, Signs, Gates, etc.	Heavy Equipment, ATV	Sediment/Turbidity; Spawning Substrate
	Bridge Maintenance	Deck Cleaning, Painting, Replacing Decking, and Minor Structural Repairs (outside of the wetted perimeter or above the bankfull)	Heavy Equipment	Chemical Contamination/Nutrients
	Snow	Plowing	Heavy Equipment	Drainage Network
Permanent and Temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris
		Snags - Hand Felling	Motorized Hand Tools	Large Woody Debris
		Snags - Mechanical Felling	Heavy Equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Large Woody Debris
		Tree Removal - Log Decks	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
		Tree Removal - Hauling	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Road Density/Location; Drainage Network
		Soil Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
		Fill Placement	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
		Stream Crossings - Culvert Installation	Heavy Equipment; Pumps; Screens	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network
	Surfacing	Hauling of Surfacing Materials	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients

Activity	*Management Element*	*Task/Method*	*Tool*	*Impact/Benefit*
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients
Temporary Road Removal	Recontouring	Blading/Recontouring	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network
	Scarifying	Material Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate
	Culvert/Drainage Structure Removal	Culvert/Drainage Structure Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network
	Revegetation	Seeding	Hand Tools; Ground Equipment; Aerial Equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris; Riparian Resilience
	Mulching	Weed-free Material (for example, straw)	Hand Tools, ATV	Sediment/Turbidity; Spawning Substrate

Table 17. Impact and/or benefit pathways for effects pathways for fish.

Impact and/or Benefit	Pathways for fish effects
Change in Riparian Vegetation (Temperature)	Water Quality
Chemical Contamination/Nutrients	Habitat Elements
Drainage Network	Channel Conditions and Dynamics; Flow/Hydrology; Habitat Elements
Large Woody Debris	Habitat Elements
Peak/Baseflows	Flow/Hydrology
Riparian Resilience	Habitat Elements
Road Density/Location	Habitat Access
Sediment/Turbidity	Water Quality
Spawning Substrate	Habitat Elements

A generalized effects (impact/benefit) analysis has been completed separately and should be referred to when fish species analysis identifies the impact/benefit as one that will affect the species or critical habitat. When fish species were analyzed and the impact/benefit analysis was general to effects pathways for the species, the crosswalk from Table 17 was used to relate back to the impact/benefit for activities in Table 16. For example, if the USFWS document for a species said water quality degradation was a threat without specifics, Table 17 under column pathway for fish effects has two entries for water quality, one is next to change in riparian vegetation (temperature) and sediment/turbidity; therefore, both change in riparian vegetation (temperature) and sediment/turbidity were analyzed for the species. In some cases, assumptions were made for species and the effects of the activity. If assumptions were made for the effects to individual species or critical habitat, assumptions were spelled out and included in the write-up under the species. Typically, assumptions were made based on identified threats within the referenced USFWS documents.

Change in Riparian Vegetation (Temperature)

Change in riparian vegetation is associated with stream temperature. Tree canopies disrupt the solar radiation from reaching the stream and increasing stream temperatures. Timber felling, landing creation and yarding corridors typically involve the loss of the tree canopy resulting in shade loss. Fish species would be affected differently from any change in riparian vegetation (temperature) based on their specific life history needs and habitat locations.

While stream shade correlates with the width of stream protection buffers, the relationship is quite variable, depending on site-specific factors such as stream size, substrate type, stream discharge, topography (Caissie 2006), channel aspect, and forest structure and species composition. Inputs of cold water from the streambed, seepage areas on the stream bank, and tributaries can help cool the stream on hot summer days if they are sufficiently large relative to the stream discharge (Wondzell 2012). The density of vegetation in riparian areas affects shade and thermal loading to a stream due to the penetration of solar radiation through gaps in the canopy and among the branches and stems (Brazier and Brown 1973; DeWalle 2010). In some instances (such as narrow streams with dense, overhanging streamside vegetation, or stands on the north sides of streams with an east-west orientation), no-harvest buffers as narrow as 30 feet

adjacent to clearcuts can maintain stream shade (Brazier and Brown 1973). Wider buffers, in general, will provide increased protection of stream temperature (Anderson et al. 2007, Leinenbach et al. 2013, Groom et al. 2011a, Groom et al. 2011b).

Timber harvest adjacent to riparian buffers can influence the riparian microclimate adjacent to streams, which can, in turn, influence stream temperatures (Moore et al. 2005a). Gomi et al. (2006) reported that inclusion of a 30m (98') buffer in clear-cut harvesting units increased maximum daily temperatures by <2C in British Columbia. Wilkerson et al. (2006) reported maximum increases of 1.0-1.4 degrees C with 11 m buffers and negligible increases with 23m (75') or partially cut buffers in Maine. DeWalle (2010) theorized that at least 80% shade could be attained with 12m (39') wide buffers on small (<6m [20'] wide) streams. In western Oregon, Zwieniecki and Newton (1999) reported that the presence of near continuous shade from an uncut buffer positioned on the south side of water led to protection from the effect of solar radiation. On the Oregon Coast Range, Groom et al. (2011a) found that on post-harvest lands with a protection buffer (no-cut zone) of 6m (20') and limited harvest in the remainder of the riparian management area increased stream temperature by an average of 0.7 ˚C, but no change to maximum stream temperatures on lands managed to the more stringent state standard of 8m (26') protection buffer and limited harvest in the remainder of the riparian management zone. The high variability in buffer width and shade condition is a function of the many variables that influence the amount of shade produced by riparian vegetation. Low-density stands with limited vertical distribution of branches and foliage will require wider buffer widths to produce the same amount of shade as high-density stands.

Riparian treatments on perennial streams greater than 1,000' from Listed Fish Habitat (LFH) are not expected to increase stream temperatures in LFH when there is hyporheic flow and flow accumulation. Several papers have been published recently indicating that hyporheic flow and stream substrate composition has a significant influence on stream temperature. Janishch et al. (2012) found that that the canopy cover of "buffers" was not a strong predictor for temperature in small (<7 feet wide) headwater streams. Streams with coarse-textured streambeds tended to be thermally unresponsive as compared with fine-textured streambeds or those with small, near-stream wetland areas. Moore et al. (2005b) found that heat exchange associated with hyporheic flow paths was a cooling effect during daytime, with a magnitude up to approximately 25% that of net radiation during the period of maximum daytime warming in clear cut harvest units on streams in British Columbia. Davis et al. (2015) found that in Western Oregon streams, pre-harvest to post-harvest temperature approximately 300m downstream of the harvest will be 56% of the temperature change that occurred within the harvest reach. Rapid cooling has been observed after an extensive exposure to the sun after water re-entered forest cover in Australia (Rutherford et al. 2004), and, in Idaho, warming of headwater streams with no buffers did not lead to warming of fish-bearing streams downstream (Gravelle and Link 2007). Zwieniecki and Newton (1999) indicated that stream temperature increases from harvesting activity were frequently lost within 150–300 m (492-984') beneath downstream cover. This re-emphasizes the important role gravel and large wood play in channels. Therefore, thinning on perennial channels 1,000' or greater from LFH is expected to have only minor negative effects to temperature in LFH.

Accurately predicting the magnitude of stream temperature increases on a highly variable landscape is extremely difficult. However, the research cited above supports the conclusion that while there may be a decrease in effective shade, localized increases in stream temperature are likely to be too small to differentiate from background variability. If localized increases in stream temperature occur, they will likely be less than 1˚C and are unlikely to be measurable in LFH. The tributary streams that flow through the harvest units are small in size (<1-5 cfs in most cases) and research has shown that the magnitude of temperature effect is a function of the amount of water entering (or exchanged with) the stream relative to the stream discharge (Johnson and Jones 2000, Story et al. 2003, Wondzell 2006, Wondzell 2012) and the difference in temperature between the stream and the inflowing water. Therefore, any localized increase in stream temperature from commercial thinning activities would typically have minor negative effect on LFH streams.

Typically, timber harvest activities are associated with an aquatic conservation strategy that provides for a stream protection buffer to limit the solar radiation reaching the stream. It is probable future projects will include some type of minimization measure for stream buffers for LFH areas, but since buffers are unknown for now, there could be projects that will have limited stream protection buffers and therefore have effects on stream temperature. As described above, most of the time timber harvest activities result in insignificant localized stream temperature increases because rapid cooling should occur as the stream flows into the uncut aquatic management zone buffer; therefore, effects should last less than an hour.

Road maintenance typically includes tree removal or brushing near streams and has the potential to create small minor changes to stream temperature. If there are drainage needs for road maintenance, such as culvert replacements, there may be a need to dewater the area and move it to a different location using pumps and screens. Depending on how this is accomplished, it could cause water temperatures to increase. If there are increases in the water temperature, the changes should be minor but could be measurable as future projects are unknown at this time. Past consultations have provided minimization measures that limit effects to temperature fluctuations.

There are a lot of unknowns for future projects in the action area. Even with the unknowns, we know there will be some shade loss from timber harvest, road maintenance and reconstruction, permanent and temporary road construction if implemented with one site potential tree height. Effects are expected to mostly be insignificant, unmeasurable effects, but could result in small localized measurable stream temperatures. Temporary road removal would not create measurable effects to increases in stream water temperature and can regrow shrubs quickly in the area, which would be beneficial.

Fish species react differently to these changes and will be discussed in the species sections below.

Chemical Contamination/Nutrients

Chemical contamination and nutrients for projects in this consultation are associated with equipment utilizing petroleum products and dust abatement chemicals. Timber vegetation management, road maintenance, road construction, and road removal activities all utilize heavy

machinery, some utilize hand tools (e.g., chainsaws), water pumps, and dust abatement chemicals have the potential to create chemical contamination reaching the streams.

Coarse woody debris is an important component of forest ecosystems because it can provide cover for many wildlife species, provide important structures for small mammals, reptiles, and amphibians, and the decomposition of logs on the forest floor that provides nutrients to plants, habitat for wildlife, and food for insects and microorganisms. This process of nutrient input is important for the food web processes for salmonids as they feed on insects and microorganisms during certain life stages. Where timber harvest has been implemented, replanting is important for this process to continue. Harvest areas are typically localized to units and timber harvest must meet certain criteria if it is to be implemented within the aquatic management zones. For many of the forests in the action area, future projects will maintain a no-cut buffer for the aquatic management zone. This will insulate some food-web negative effects.

Any work utilizing chemicals and petroleum products typically requires an approved Spill Prevention Control and Containment Plan (Spill Plan) as required by contract provisions G.3.4.1/BT6.341. The Spill Plan would include a description of the hazardous materials that would be used; and a spill containment kit would be located on-site. All heavy equipment used for refueling would carry a hazardous material recovery kit. All vehicles and machinery would be free of petroleum leaks. Any leaks that occur would be immediately repaired. Power equipment would be refueled at least 150 feet from water bodies or as far away as possible to prevent direct delivery of contaminants into a water body. If local site conditions do not allow for a 150-foot setback, then refueling would be as far away as possible from the water body (USFS 2012a). For all immobile equipment, absorbent pads would be used. All petroleum products being transported or stored would be in approved containers meeting Occupational Safety and Health Administration and appropriate state standards. The Forest Service Contracting Officer would be notified of any spills. Any contaminated soil, vegetation or debris must be removed from National Forest System lands and disposed of in accordance with state laws.

Fuel spills of petroleum products from heavy equipment for all management activities have the potential to result in chemical contamination of streams. Hand tool fuel capacity is small (<1/4 gallon), and thus any potential spills during refueling would be unlikely to reach the stream channel resulting in a neutral effect for hand tools. Water pump fuel capacity is also small, and refueling is completed outside of the stream. It is extremely unlikely that heavy equipment in these activities will spill contaminants due to the contract clause requiring the Spill Plan. It is a standard contract clause that vehicles are checked regularly for leaks by purchasers and timber sale administrators and any leaks are promptly repaired before the vehicle is allowed to continue operating. Based on over 2 decades of experience that encompasses tens of thousands of log truck loads, as well as contractor supplied equipment for instream and road work, there is very low probability of chemical spills, and there have been no known spills from these activities that have impacted LFH. Forest Service best management practices require spill kits to be available if there is a spill and require fueling stations to be located 150 feet from streams when possible, and as far as practical when not possible.

The Forest Service expects limited applications for dust abatement. Dust abatement is typically only implemented during the dry season. Dust abatement can utilize water and chemicals (e.g., lignosulfonate, Magnesium Chloride), depending on the conditions in the work area. The Forest Service has a guide on applications for dust suppression (USFS 1999). The application of dust abatement chemicals would reduce fine sediment delivery to streams typically from dry season haul. Several different chemicals are discussed in the Forest Service guide, with pros and cons of each including an environmental impact section. Most chemicals do have negative environmental impacts, and these will be taken into consideration when utilized.

Lignosulfonate is an organic, nonpetroleum product that has limited ability to migrate since it is a bonding agent that adheres to material in dry conditions. Lignosulfonate is the chemical expected to be utilized near LFH, but there may be other chemicals utilized as well. There is potential for the dust abatement product to directly enter waterways. However, the single application of lignosulfonate during the dry season would reduce the likelihood of it entering a stream channel. There would not be an effect to LFH fish from applying lignosulfonate on road surfaces. Lignosulfonate has been tested to determine its potential impact on water quality and fish species when introduced directly into surface waters and it is rather benign. A 48-hour LC50 of 7,300 parts per million (ppm) was observed by Roald (1977) in rainbow trout (*Oncorhynchus mykiss*). Compounds with values greater than a value of >1,000 are considered to have a low order of toxicity. Due to the limited use and application of lignosulfonate, the timing of application, the use of best management practices for application, and the known non-toxicity to fish at dust abatement concentrations, the probability of affecting listed fish is discountable.

Magnesium chloride can negatively impact aquatic systems by reducing macroinvertebrates (Kotalik et al. 2017) and directly affecting fish (Lewis 1998). If Magnesium chloride is utilized, there will be direct and indirect effects to fish through death (at high enough concentrations) and food web processes. The magnitude of effect is difficult to determine without having project specific information. Future project consultation will determine which chemical is utilized for dust abatement if it is needed but if utilized, we hypothesize the effects will be short-term but not discountable.

Timber harvest, road maintenance and reconstruction, and temporary road removal could cause short-term negative direct effects to the transport of nutrients downstream when implementing the projects, however, when projects utilize a water bypass system, effects would be limited to project implementation. These activities could provide for long-term beneficial effects when maintaining or repairing the current drainage system, installing appropriate sized drainage systems for the stream, and utilizing water bypass systems during the maintenance process.

Road construction could cause short-term negative direct effects to the transport of nutrients downstream when implementing the projects and fuel spills of petroleum products from machinery and equipment used in road work have the potential to result in chemical contamination of streams. Fuel powered equipment (hand tools and heavy machinery) will be used within the Riparian Reserves for road construction, including over stream crossings, where they are needed.

Drainage Network

The natural drainage networks of streams and rivers can be increased through road building and impervious road ditching and channeling that redirects surface flow and increase the active channel network. The primary management effect is associated with the road network, especially road compaction and road drainage systems. An increased drainage network can increase peak volume discharge and cause peak flows to occur earlier than unroaded drainages.

Wemple et al. (1996) studied the effect of channel network extension on two basins in the western Cascades of Oregon and found that 57% of the road network was connected to the stream network (road segments draining to streams, or road segments draining to culvert outlets with gullies incised below them), resulting in an increased drainage density of 21-50%. They concluded that this increase in road drainage density, and enhanced water routing efficiency was a possible explanatory mechanism for the changes in hydrograph shape (i.e., earlier peak, higher volume discharge) as presented in Jones and Grant (1996). Many other watershed factors (e.g., soil percolation rates, soil stability, road construction type) may influence the magnitude or probability that road extension will affect flows.

System road reconstruction, road (system and non-system) maintenance, and permanent (system roads) and temporary (non-system roads) roads will be built, and decommissioning may occur in areas adjacent to streams with LFH as well as in areas that may be several miles upstream from LFH.

Road reconstruction and maintenance have the potential for reducing the distance water travels along roadways, with the implementation of larger cross drain culverts, reduced spacing between culverts, and/or better placement of near-stream cross drains and their outflows. Road Decommissioning may occur near LFH streams, or on tributary streams that feed into LFH streams.

There are no causal effects of timber harvest on the drainage network, and the effects of timber harvest would be neutral to this element.

The only causal mechanism for effects for road maintenance and road re-construction would be through snow plowing. These would create short-term negative effects during the spring when snow melts. These would be localized effects to the area where snow plowing occurs. Snow plowing is not an action that is routinely implemented on Forest Service roads, so the probability of occurrence is low, but now zero. If effects do occur to drainage networks, these effects would be minor and localized.

Permanent and temporary road construction will have negative effects for drainage network lasting as long as the road is in place.

Temporary road removal will have long-term beneficial effects on drainage network. The process would be the opposite of road building.

Large Woody Debris

Large woody debris (LWD) enters streams through chronic and episodic processes (Bisson et al. 1987). Chronic sources include streamside tree mortality and bank undercutting (Murphy and Koski 1989). Episodic sources include windthrow (for example, Harmon et al. 1986; Lienkaemper and Swanson 1986; May and Gresswell 2003), wildfire (Agee 1993), infrequent flooding, landslides/debris flows (Benda et al. 2002, 2003), and insect infestations or disease. The quality, complexity, and function of fish habitat is often affected by the level of instream large woody debris. Large woody debris influences the size and distribution of stream habitat units, provides cover from predators, shades the water surface, and increases the sorting of substrate, providing suitable spawning areas (Megahan 1982; Bilby and Ward 1991; Crispin et al. 1993; Montgomery et al. 1995; May and Gresswell 2003). Large woody debris is instrumental in nutrient retention and provides a substrate for macroinvertebrates (Cederholm and Peterson 1985; Bilby and Ward 1989; Bilby and Bisson 1998; Gregory et al. 1991; Wallace et al. 1995). Large woody debris influences channel morphology (Swanson and Lienkaemper 1978; Heede 1985; Bisson et al. 1987; Swanston 1991). Fish densities are higher in streams with abundant large woody debris (Fausch and Northcote 1992; Roni and Quinn 2001).

The quality, complexity, and function of fish habitat is often affected by the level of instream LWD. LWD influences the size and distribution of stream habitat units, provides cover from predators, shades the water surface, and increases the sorting of substrate, providing suitable spawning areas (Megahan 1982; Bilby and Ward 1991; Crispin et al. 1993; Montgomery et al. 1995; May and Gresswell 2003). LWD is instrumental in nutrient retention and provides a substrate for macroinvertebrates (Cederholm and Peterson 1985; Bilby and Ward 1989; Bilby and Bisson 1998; Gregory et al. 1991; Wallace et al. 1995). LWD influences channel morphology (Swanson and Lienkaemper 1978; Heede 1985; Bisson et al. 1987; Swanston 1991). Fish densities are higher in streams with abundant LWD (Fausch and Northcote 1992; Roni and Quinn 2001).

Timber felling, landings creation, and yarding create pathways for effects to salmonids and critical habitat if they are done within the aquatic management zone. Projects where these activities are completed in aquatic management zones are typically implemented to move the aquatic management zone to a better desired condition, which are better for species and critical habitat. Where large woody debris is lacking, ESA consultations typically contain criteria for large woody debris to be left onsite where it can contribute to future in-stream wood. It is expected that future proposed projects will contain the same requirements. There are direct short-term negative effects from removing large woody debris from aquatic management zone even with minimization measures.

Road maintenance could cause some minor short-term direct negative effects to large woody debris when vegetation management occurs next to the road and is within the aquatic management zone. Large woody debris is important to some fish species and their critical habitat needs. Negative effects would be localized to road sections near LFH. Effects would still be localized and minor even where roads parallel LFH.

Road construction could cause some minor longer-term direct negative effects to large woody debris when roads are built in the aquatic management zone. Large woody debris is important to some fish species and their critical habitat needs. These negative effects would be localized to roads built near LFH. Longer-term effects would still be localized if new roads are built parallel to LFH. Proposed future projects would need to complete ESA consultation prior to any new road building in the affected environment. Past ESA consultations that contain road building as an activity require minimization measures to reduce effects to salmonids and critical habitat. Any future projects would most likely also contain minimization measures as well. The magnitude of effects is difficult to predict. However, road building would be associated with timber harvest and roads for timber harvest are built once to use throughout the harvesting timeframe. This does mean that the effects of large woody debris may be years to re-establish in the area if it is cut. The effects would be localized to the stream segment where the road is built.

Temporary road removal includes removal of culverts, re-establishment of natural drainage patterns, and blocking vehicle access. The most likely point of entry for sediment will be during activities that occur next to streams, typically culvert removal or reducing the fill over culverts that are left in place. It is likely that temp road removal will result in short-term localized sediment delivery to streams. It is also expected that in the long-term, temporary road removal will have a positive effect on sediment delivery by removing roads, and stream crossings, where pulsed sediment delivery can potentially occur.

Peak/Baseflows

Peak/baseflows can be defined as quantity of water and the timing of input of water to an area. Loss of trees and vegetation may change the rate of evapotranspiration, increasing soil moisture and overall water yield. A short term (5-10 years) increase in discharge during the wet and the dry periods could occur from two mechanisms for the thinned stands: increased snow accumulation (wet period) will create small increases in peak flows (Jones and Grant 1996), and reduced canopy (dry periods) will reduce transpiration rates which will account for small increases in summer flows.

Peak (and pulsed) stream flows occur regularly in the some of the action area and carry most of the sediment load with high stream energies that may erode and change channel form (Cooper 2005).

Tree harvest and new permanent and temporary road construction do have potential to change the peak and base flows by decreasing the current canopy cover within the treated stands. Loss of trees and vegetation may change the rate of evapotranspiration, increasing soil moisture and overall water yield. A short term (5-10 years) increase in discharge during the wet and the dry periods could occur from two mechanisms for the thinned stands: increased snow accumulation (wet period) will create small increases in peak flows (Jones and Grant 1996), and reduced canopy (dry periods) will reduce transpiration rates which will account for small increases in summer flows.

Timber harvest has two causal mechanisms for peak/baseflows, these are timber yarding and timber landing construction. The pathway for effects for these two activities is based on

compaction of soil. Soil compaction may cause water runoff to increase when there is heavy rainfall. These effects would be localized and small in magnitude.

The causal effect of road maintenance or reconstruction for peak/baseflows are due to water withdrawals, dewatering of a work site, or water pumping to a different location. These activities create localized short-term direct negative effects that last the duration of the implementation time for these activities but could have long-term beneficial effects if appropriately sized culverts are installed. These activities can last less than an hour for dust abatement to a few days for culvert replacement.

The causal effect of road building for peak/baseflows creates the largest effects to peak/baseflows of any other activity in this consultation. Depending on where roads are built and the associated water diversion systems used for the road, longer-term direct effects are expected since the roadbed can create a dam-like effect on water flow. These effects will last as long as the road is in place.

Temporary road removal will re-establish the natural drainage patterns to the streams resulting in a beneficial effect. Temporary road removal will re-vegetate over time resulting in an intact canopy and a reduction in openings. Temporary roads will be decommissioned at the end of the project. Although this increase in canopy cover on decommissioned roads is positive, at the watershed scale, the road area is small and the creation of temporary roads will offset the canopy closures, thus it is discountable in terms of the potential to affect peak and base flows.

Predicted changes in peak or base stream flows due to the implementation of this action should be minimal.

Water withdrawals have direct effects to peak/base flow. Drafting from streams may occur during spring, summer and fall, which encompasses the base flow period when the impact locally will be greatest, but will be isolated in space and time in that multiple drafting operations is unlikely to occur simultaneously in the action area. Stream flows may be temporarily reduced downstream from the withdrawal point. The amount of flow decrease from water withdrawals would depend on the amount of stream flow, how much water is withdrawn, and the duration of water drafting. Water trucks commonly hold 500 gallons of water, with a withdrawal rate of up to approximately 7.5 gal/sec. This means the maximum withdrawal would last at least 67 sec, and the maximum withdrawal period commonly is less than 5 minutes at lower rates of withdrawal. The magnitude of flow reductions will be insignificant due to the high stream flow volume available and the short period of withdrawal.

Water withdrawal for road work typically occurs during the dry season (generally June to October) in the Pacific Northwest or areas in higher elevation. Timing may change depending on where the work will occur. While water quantity to LFH may be temporarily reduced for short periods of time, the magnitude of flow reductions will be insignificant to affect the species or quality of their critical habitat because the periods of withdrawal will be short and will not affect inflow from adjacent tributary streams and hyporheic flow downstream from the point of withdrawal.

The implementation of projects covered by this programmatic consultation will result mostly in insignificant overall effects to peak and base flows, mainly through drafting of surface water for fuels treatment (during peak flow from fall through spring), and road maintenance (during base flow in summer and early fall). However, future projects may create small significant effects to peak and base flows, it would depend on the specifics of the project.

Vegetation changes will result in short term reduced evapotranspiration, increasing soil water, and therefore increasing yield. Road work will reduce soil water infiltration affecting storage, delivery and timing of flow to stream channels.

Riparian Resilience

Riparian resilience is related to the effects and processes of aquatic management zone to move toward more desirable conditions. Typically, treatments within the aquatic management zone area designed to maintain or improve riparian conditions even though some short-term local impacts are possible. These short-term impacts include a potential slight loss of large wood recruitment from treated riparian stands, potential short-term sediment, and potential minor increase in stream temperature, resulting in a short-term negative effect to this indicator at the local scale. In the long-term the effect will be positive as riparian stand diversity and diameters increase. Utilizing a stream protection buffer determined by the Aquatic Conservation Strategy or the Land Management Plan minimizes short term effects. In the long term, riparian resilience may improve due to benefits expected with future project development.

Timber felling and site restoration can have direct short-term negative effects and long-term beneficial effects, depending on the prescribed treatment within the aquatic management zones. Timber thinning prescriptions within the aquatic management zones can provide long-term beneficial effects by reducing competition between trees allowing trees left to grow and provide future large woody debris. Effects from timber treatments in the aquatic management zone come with short-term negative effects as well just because it takes time for causal mechanisms to improve. Forest thinning remains one of many forest management tools employed to meet ecological, social, and economic objectives on Forests. Thinning within the aquatic management zones is expected to result in a closed canopy within 3-5 years. The temporal boundary for cumulative effects on aquatic species is 5 years. Timber harvest projects within aquatic management zones are expected to be low and the objectives would be to improve the condition of the riparian areas, which would provide for long-term beneficial effects.

Timber yarding and landings have short-term negative effects through compaction of soil, which would take time to rebound. There are measures that are typically used during timber yarding to minimize soil compaction depending on what types of yarding will be utilized. This can range from utilizing slash for heavy equipment to drive over, to skyline yarding. There is no way to predict what type of yarding practices will be utilized for future projects, but we do know any future project will need ESA consultation, and all previous consultations contain project designs that minimize effects to species and critical habitats. The expectation for effects would still be measurable low-level effects to species and critical habitat for this action.

There are no causal effects of road maintenance or reconstruction and permanent and temporary road construction for riparian resilience, and the effects of road maintenance or reconstruction would be neutral to this element.

There are no causal effects of permanent and temporary road construction for riparian resilience, and the effects of permanent and temporary road construction would be neutral to this element.

Temporary road removal will have long-term beneficial effects for riparian resilience by removing the road prism, soil decompaction, and recontouring the landscape within the aquatic management zone.

Road Density/Location

Road density/location is based on where new permanent roads and temporary roads will be built. Almost all forest watersheds in the action area have existing primary and secondary road systems in place. Current road systems need maintenance and repairs over time to maintain safety standards. Any effects to fish and their critical habitats would occur on roads that are hydrologically connected to LFH or where effects can reach LFH. The probability of new roads being built somewhere in the action area is high. The probability of the new road being a permanent road is relatively low, as new permanent roads need funding to build and maintain. Temporary roads are associated with timber harvest needs. If timber is harvested in an area, temporary roads will be built.

There are no causal effects of timber harvest for road density/location, and the effects of timber harvest would be neutral to this element.

There are no causal effects of road maintenance or reconstruction for road density/location, and the effects of road maintenance or reconstruction would be neutral to this element.

New road building would be the largest contributor to increased road density of all actions for the affected environment. Permanent roads are considered system roads, while temporary roads are considered non-system roads. Temporary roads will increase road density for a short time during logging operations, but the increase will be minor over the existing condition as all temporary roads will be fully decommissioned (temporary road removal) after sale contract harvest and yarding operations are complete. Increased road density may affect fish and their critical habitat typically through habitat fragmentation and sedimentation. Legacy roads on Forest Service lands are the largest contributor to habitat fragmentation. New roads would need to meet certain Forest Service standards, which are designed to minimize effects on aquatic systems. Even so, all new roads could have negative effects on fish for as long as they are in place.

Sediment/Turbidity

Sediment and turbidity tend to be the most frequent effects associated with timber harvest tasks, road maintenance and reconstruction and road construction. The potential for surface erosion is directly related to the amount of bare compacted soil exposed to rainfall and runoff and timber road haul; hence timber harvest activities may contribute large quantities of fine sediment to streams (Chamberlin et al. 1991, Swanston and Swanson 1976). Not all hillside erosion reaches

stream channels (Chamberlin et al. 1991, Rice et al. 1972), but the potential exists for negative effects to the sedimentation/turbidity pathway. Heavy equipment can cause compaction and rutting of the ground which aids the movement of sediment- laden runoff. Excessive erosion may occur on poorly designed and placed skid trails, skid trails used during wet conditions, and trails that have connection to stream channels. Soil compaction from ground-based systems can be reduced when this equipment is restricted to dry soil conditions (Sidle and Drlica 1981). Fine sediment deposited in spawning gravels can reduce egg survival and developing alevins (Everest et al. 1987; Hicks et al. 1991) by reducing the availability of dissolved oxygen in the gravel.

The capacity of headwater streams to retain sediment is supported by many studies. The channels contain vegetation and woody debris of varying amounts, much of it small in diameter but in these small streams this material acts as an important obstruction to store sediment (Sullivan et al. 1987). Some studies indicate that annual sediment yields from small watershed make up less than ten percent of the sediment stored in channels (Swanson and Lienkaemper 1978), in other words, headwater stream channels with have a tremendous capability to store large amounts of fine sediment (Benda et al. 2005).

Chase et al. (2019) completed a soil and streamside effects study on 30 harvest units across Western Oregon and Washington on Weyerhaueser ownership on slopes up to 160%; with most 0-60%. Timing of clear-cut harvest was during the wet season (November – March). They observed that streams with riparian buffers >15m (49') wide had little to no stream-adjacent disturbance.

Broderson (1973, cited in Castelle et al. 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. Broderson noted that buffers have little or no effect on sediment removal if sediment-laden water flows across the buffers as channelized flow; buffers can be effective only if they resist channelization and maintain overland flow as sheetflow. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes.

The timing of turbidity delivery to non-LFH stream channels will coincide with the rainstorms (and snowmelt) from fall through spring that will also naturally increase flow volume and turbidity in LFH. Aquatic conservation plans tend to limit the usage of heavy equipment within 150 feet of a stream. Even when following the aquatic conservation strategy and utilizing BMPs, there is the possibility of sediment entering streams and creating direct and indirect effects to species. These effects are typically short-term effects lasting less than one year. The nature and timing of future projects, and inclusion of minimization measures are unknown which means the effects of sediment and turbidity magnitude is uncertain. It is typical to include minimization measures when there is LFH within the action area. Even with minimization measures, sedimentation and turbidity could reach measurable effects for any given project based on the incremental effects of almost every timber harvest activity creating sedimentation/turbidity. Even if measurable effects occur, these effects should be low level effects but would still reach a level of significance to LFH.

Road maintenance and reconstruction may create source areas for sediment input to streams, particularly during heavy rainfall events. Failures of road fill at stream crossings are common

source areas for landslide and debris torrents. Vegetated buffer areas ranging in width from 40 to100 feet were successful in keeping sediment out of streams (Corbett and Lynch 1985, Lynch et al. 1985, Moring 1982, Newbold et al. 1980, and Gomi et al. 2005). Broderson (1973, cited in Castelle et al. 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes. Previous consultations have design measures that minimize sediment effects to streams.

Surfacing roads with rock aggregate or pavement can significantly reduce the amount of erosion from the road surface (Burroughs and King 1989), particularly if used in the wet season. The type of rock used for surfacing can also influence sediment production. Hard, durable rock is more resistant to breakdown while softer rock is more easily crushed into particles small enough to be washed from the road surface (Bilby et al. 1989). Timber hauling may occur year-round, or seasonally depending on the road surface, design, and condition. The Forest Service Roads Standards are described in several different Forest Service publications, including but not limited to the list in Appendix B.

The turbidity generated by restoring flow at a culvert site is likely to affect the channel at the installation and removal sites and for approximately 1,000 feet. Foltz and Yanosek (2005) found that turbidity decreased by an order of magnitude within 328 feet of the disturbance and that turbidity was not expected to be detectable more than 2,400 feet (800 meters) downstream from the project site.

Overall, road maintenance and reconstruction activities are ground disturbing actions and localized increase in sedimentation to streams is expected. However, future projects will most likely include minimization measures in their consultations which will help alleviate effects to species and critical habitats.

In general, modern road construction practices produce less sediment delivered to streams from forest roads than older road construction practices (Copstead et al. 1998). Road construction and maintenance may create source areas for sediment input to streams, particularly during heavy rainfall events. Failures of road fill at stream crossings are common source areas for landslide and debris torrents. Vegetated buffer areas ranging in width from 40 to 100 feet were successful in keeping sediment out of streams (Corbett and Lynch 1985, Lynch et al. 1985, Moring 1982, Newbold et al. 1980, and Gomi et al. 2005). Broderson (1973, cited in Castelle et al. 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes.

Some increases in sedimentation (and resulting increases in turbidity) can be expected during culvert and bridge replacement, as equipment is operating in the stream channel and some trees would need to be removed from the stream bank. However, such increases are likely to be of local extent and of short duration. From experience, typically only 1-2 yards, or less, of sediment enters a stream during culvert projects.

Forest roads are compacted surfaces and can potentially allow for accelerated erosion and potential sediment delivery to stream channels from a variety of sources, including small slumps and slides into the roadway from the cut bank, water channeling from the road or ditches if not properly directed and controlled, and blocked culverts and road fill washouts during floods. Road construction is a ground disturbing action, and the potential exists for some increased sedimentation to streams.

Past consultations include minimization measures to decrease the effects of new road construction to streams. There is reason to believe these minimization measures will be included in any potential future consultation in the action area. Even if minimization measures were included in future consultation, there is reason to believe the effects of sedimentation for new road building would be the largest contributor of sedimentation to stream systems.

Spawning Substrate

Timber harvest activity effects have similar mechanisms between spawning substrate and sediment/turbidity. Spawning substrate can be defined by water quality, water quantity, and substrate. Spawning substrates are necessary for successful reproduction of many fish species. For fish that build redds in gravels, fine sediment deposited in can reduce egg survival and developing alevins (Everest et al. 1987; Hicks et al. 1991) by reducing the availability of dissolved oxygen in the gravel. Sedimentation/turbidity is discussed above as well as temperature. Water quality factors such as dissolved oxygen and water quantity are also important for spawning substrate.

Road reconstruction and new roads construction has the potential to have the largest negative effects to spawning substrate within the affected environment if they are built near the aquatic management zone. Habitat fragmentation would have negative effects on bedload movement and potential become an impediment to habitat connectivity if inappropriate sized culverts were installed. Effects would occur throughout the duration of the road.

Acipenseridae Family - Sturgeon

Gulf sturgeon

Gulf sturgeon are found in the Gulf of America and streams flowing into the Gulf of America. The Gulf sturgeon is an anadromous fish which migrates from salt water into large coastal rivers to spawn and spend the warm months. Most of its life is spent in fresh water. Major population limiting factors are thought to include barriers (dams) to historical spawning habitats, loss of habitat, poor water quality, and overfishing (USFWS 1995b). Historically, the Gulf sturgeon occurred from the Mississippi River east to Tampa Bay. Its present range extends from Lake Pontchartrain and the Pearl River system in Louisiana and Mississippi east to the Suwannee River in Florida. When listed, Gulf sturgeon threats were identified as threats and potential threats identified in the listing rule included modifications to habitat associated with dredged material disposal, desnagging (removal of trees and their roots), and other navigation maintenance activities; incidental take by commercial fishermen; poor water quality associated with contamination by pesticides, heavy metals, and industrial contaminants; aquaculture and incidental or accidental introductions; and the Gulf sturgeon's slow growth and late maturation

(68 FR 13370). Young-of-the-year Gulf sturgeon remain in freshwater feeding on aquatic invertebrates and detritus approximately 10 to 12 months after spawning occurs (68 FR 13370). Additional Gulf sturgeon feeding habits for juvenile and adult life stages can be found in 68 FR 13370.

Gulf sturgeon critical habitat is not located in the action area.

Pallid sturgeon

Pallid sturgeon are found in the Mississippi River system. The Pallid Sturgeon is native to the Missouri and Mississippi rivers and adapted to the pre-development habitat conditions that historically existed in these rivers. These conditions generally can be described as large, free flowing, warm-water, and turbid rivers with a diverse assemblage of dynamic physical habitats. Limiting factors include: 1) activities which affect in-river connectivity and the natural form, function, and hydrologic processes of rivers; 2) illegal harvest; 3) impaired water quality and quantity; 4) entrainment; and 5) life history attributes of the species (i.e., delayed sexual maturity, females not spawning every year, and larval drift requirements). The degree to which these factors affect the species varies among river reaches (USFWS 2014c). Pallid Sturgeon are a bottom-oriented, large river obligate fish inhabiting the Missouri and Mississippi rivers and some tributaries from Montana to Louisiana (USFWS 2014c). Habitat requirements of larval and young-of-year Pallid Sturgeon remain largely undescribed across the species' range, primarily because of low populations of spawning adults and poor recruitment. Scientists think the habitat requirements of larval and young-of-year for pallid sturgeon are similar among all geneses. Pallid Sturgeon have been collected from a variety of turbidity conditions including areas with consistently low turbidities (i.e., 5-100 nephelometric turbidity units (NTU)) to comparatively natural systems with seasonally high turbidity levels (> 1,000 NTU) (USFWS 2014c). Given their physical attributes, the species appears to be highly adapted to low visibility environments (USFWS 2014c).

Pallid sturgeon critical habitat has not been designated.

White sturgeon

White sturgeon are found in the Columbia River system with the Kootenai River Distinct population segment being a landlocked population living their entire life in freshwater. Kootenai River white sturgeon are restricted to approximately 168 miles of the Kootenai River, in Idaho, Montana, and British Columbia, Canada, primarily upstream from Cora Linn Dam at the outflow from Kootenay Lake, British Columbia. Kootenai River white sturgeon critical habitat is contained entirely in Idaho, more than 2 miles downstream of the state border with Montana and will not be considered in this consultation. At the time of ESA listing, threats to the species were decline in the adult population and the almost complete lack of natural recruitment; loss of suitable spawning and rearing habitat because of Libby Dam operations; reduced biological productivity in the basin; and contaminants leading to poor water quality (USFWS 2019i). New information in the revised recovery plan (USFWS 2019i) identified the primary threats stem from the presence and operations of Libby Dam, and fall into three main categories: (1) reductions in peak spring flows; (2) alterations to the annual thermal regime in the Kootenai

River; and (3) reductions to/losses of nutrients and fundamental ecosystem processes (*e.g.*, food web, floodplain interaction, riparian function).

White sturgeon, Kootenai River DPS, critical habitat does not occur in the action area.

Acipenseridae Effects

Generally, sturgeon are long lived species, utilize large river systems for most of their life history needs, and take years to reach sexual maturity, move upstream in large rivers to spawn in freshwater when water flow and temperature conditions they need for spawning are met. All three of the sturgeon species considered in this consultation have been affected by river impoundments affecting spawning, survival of hatchlings, and gene flow (USFWS 2014c; USFWS 1995b; USFWS 2019i). Poor water quality conditions from pollutants, such as heavy metals and agricultural run-off, is believed to negatively affect their survival.

Since all these sturgeon live in large rivers, the pathway for effects would be limited to water quality effects such as increased temperature and sedimentation/turbidity. Activity effects in this consultation would be minor and indistinguishable from background levels since they live in large river systems.

The effects determination is **may affect, not likely to adversely affect** Gulf sturgeon, Pallid sturgeon, and white sturgeon, Kootenai River DPS because effects will be insignificant.

Catostomidae Family - Suckers

Lost River sucker and shortnose sucker

Lost River sucker and shortnose sucker will be discussed together because shortnose suckers significantly with Lost River sucker in the Upper Klamath Lake, its tributaries, or within any of the reservoirs along the Klamath River and in nearby Lost River Basin. Both species have similar life histories, life cycles, and threats to their continued existence. At the time of listing, Lost River sucker and shortnose sucker were known from Upper Klamath Lake and its tributaries and outlet, including a "substantial population" of shortnose sucker in Copco Reservoir, as well as collections of both species from Iron Gate Reservoir and J.C. Boyle Reservoir. Remnants and/or highly hybridized populations were also documented to occur in the Lost River system including both species in Clear Lake Reservoir, but it was apparently presumed that Lost River sucker populations in Sheepy Lake, Lower Klamath Lake, and Tule Lake had been "lost". Although not stated explicitly, shortnose sucker within Gerber Reservoir were likely part of the "highly hybridized populations" in the Lost River Basin referenced in the listing (77 FR 73740). Currently, Clear Lake Reservoir and Upper Klamath Lake and their tributaries support the largest populations of both species. Populations in Klamath River below Keno Dam and in the Lost River drainage below Clear Lake Dam are comprised mostly of adults. These populations are probably functioning as sink populations, as they are not likely self-sustaining because of low recruitment due to the lack of access to spawning habitats. All life stages of these listed suckers have been found in Link River, the outlet of Upper Klamath Lake (USFWS 2013c).

A main factor cited at the time of listing as contributing to population declines was loss or degradation of spawning, rearing, and adult habitats. Current factors limiting species recovery

also include high mortality of larvae and juveniles due to reduced rearing habitat, entrainment in water management structures, poor water quality and negative interactions with introduced species (USFWS 2013c).

Lost River and shortnose suckers have complex life histories that include stream/river, lake, marsh, and shoreline habitats. Both spawn during the spring over gravel substrates in habitats less than 1.3 meters (4 feet) deep in tributary streams and rivers. A smaller but significant number of Lost River sucker also spawn over gravel substrates at shoreline springs along the margins of Upper Klamath Lake. Larvae spend relatively little time in rivers or streams before drifting downstream to the lakes by mid-summer. Larval habitat is generally along the relatively shallow shoreline where emergent vegetation provides cover from predators, protection from currents and turbulence, and abundant food (including zooplankton, macroinvertebrates, and periphyton). Juvenile suckers utilize a wide variety of near and off-shore habitat including emergent wetlands and non-vegetated areas. They increasingly move off-shore into the lake as they grow. Adults occupy open water lake habitats (USFWS 2013c).

Lost River sucker are relatively large fish, up to 0.8 meter long and 4.5 kilograms in weight, distinguished by an elongate body and sub-terminal mouth with a deeply notched lower lip that are relatively more papillose. They have dark backs and sides that fade to yellow or white on the belly. Lost River sucker resiliency for this species is still low and is declining with time as the adult spawning populations reach the end of their expected lifespan. Redundancy is also critically low. There are only three distinct spawning populations: Upper Klamath Lake-springs, Upper Klamath Lake-river, and Clear Lake Reservoir. Two of these spawning populations (Clear Lake Reservoir and Upper Klamath Lake-springs) have very low numbers and are extremely vulnerable to localized catastrophic events (USFWS 2013c, 2024x).

Shortnose sucker is generally distinguished by a smaller head than Lost River sucker also with an oblique, terminal mouth, and thin, but fleshy, lips. The lower lip is deeply notched, giving the appearance of two separate lobes. Coloration is very similar to Lost River sucker, with dark back and sides and a silvery or white belly. Shortnose sucker are generally smaller than Lost River sucker but can still grow to about 0.65 meters. Shortnose sucker are one of three extant species in the genus *Chasmistes*, with the other two being the cui-ui sucker (*Chasmistes cujus*) in Pyramid Lake, Nevada and the June sucker (*Chasmistes liorus*) in Utah Lake, Utah (USFWS 2013). Shortnose sucker suffer from low resiliency as a species. The low resiliency is due to the extremely low numbers in most populations, inadequate access to suitable spawning habitat for most populations, and genetic impurity in most populations (such as impaired representation). Shortnose sucker have more populations across a broader area compared to Lost River sucker, but overall, the species' redundancy remains quite low. There are currently only three known spawning populations (Upper Klamath Lake, Clear Lake Reservoir, and Gerber Reservoir) (USFWS 2013, 2024w).

Water quality conditions in Upper Klamath Lake, one of two lakes supporting the largest populations of the two suckers, are driven by extensive nutrient loading. The lake was highly productive or eutrophic prior to settlement by Europeans in the mid-19th century, but it has become hypereutrophic from increased external loading (pumping of diked wetlands, farm/ranch run-off, and roads) that has enriched the naturally high internal (lake sediments) loads. Much of

the nutrient load entering Upper Klamath Lake comes from non-point sources. Nevertheless, on average, approximately 60 percent of the biologically available phosphorus is derived from internal loads, such as lake sediment, although it has been suggested that these internal loads have been enriched by modern land management practices (USFWS 2013c). Poor water quality in Upper Klamath Lake is particularly associated with high abundance of the blue-green algae *Aphanizomenon flos-aquae*. Core samples of bottom sediments indicate that *Aphanizomenon flos-aquae* was not present in Upper Klamath Lake prior to the 1900s. Its appearance is believed to be associated with increases in productivity of the lake. This alga now dominates the algal community from June to November, and, because of the high phosphorus concentrations and its ability to fix nitrogen, is able to reach seasonally high biomass levels that eventually produce highly degraded water quality. Once the algal bloom subsides, decomposition of the massive amounts of *Aphanizomenon flos-aquae* biomass can lower dissolved oxygen to levels harmful or fatal to fish. The processes that produce these conditions are complex and vary on many time scales: daily, seasonally, and annually. In some years the massive bloom of blue-green algae experiences near catastrophic die-offs (crashes), which provides an abundant source of organic material for bacterial decomposition. During these periods of high levels of decomposition, the bacteria can deplete dissolved oxygen from the water column as they respire, and as a by-product of decomposition nitrogen is also released, which can then convert to toxic forms (un-ionized ammonia) when the pH is basic (greater than 7). There are many other factors, such as wind-driven mixing of the water column, that determine the severity and extent (whether it is localized or lake-wide) of these crashes. Additionally, other cyanobacteria (*Microcystis sp*.) may produce toxins harmful to sucker liver tissue (USFWS 2013c).

Lost River sucker and shortnose sucker critical habitat physical and biological features from (77 FR 73740):

1. Water. Areas with sufficient water quantity and depth within lakes, reservoirs, streams, marshes, springs, groundwater sources, and refugia habitats with minimal physical, biological, or chemical impediments to connectivity. Water must have varied depths to accommodate each life stage: Shallow water (up to 3.28 ft (1.0 m)) for larval life stage, and deeper water (up to 14.8 ft (4.5 m)) for older life stages. The water quality characteristics should include water temperatures of less than 28.0 °C (82.4 °F); pH less than 9.75; dissolved oxygen levels greater than 4.0 mg per L; low levels of microcystin; and un-ionized ammonia (less than 0.5 mg per L). Elements also include natural flow regimes that provide flows during the appropriate time of year or, if flows are controlled, minimal flow departure from a natural hydrograph.

2. Spawning and rearing habitat. Streams and shoreline springs with gravel and cobble substrate at depths typically less than 4.3 ft (1.3 m) with adequate stream velocity to allow spawning to occur. Areas containing emergent vegetation adjacent to open water, provide habitat for rearing and facilitates growth and survival of suckers, as well as protection from predation and protection from currents and turbulence.

3. Food. Areas that contain an abundant forage base, including a broad array of chironomidae, crustacea, and other aquatic macroinvertebrates. With this designation of critical habitat, the USFWS have identified the physical or biological features essential to

the conservation of the species, through the identification of the features' primary constituent elements that support the life-history processes of the species.

Warner sucker

Warner suckers are in the Warner Basin located in southcentral Oregon. Warner suckers mature at 3 to 4 years of age at approximately 5 to 6 inches in length. The species is a moderate-sized member of the family Catostomidae and reaches a maximum length of about 20 inches (50 FR 39117). Habitats of Warner sucker include large natural lakes and associated marshes. Although primarily lacustrine, this species spawns in headwaters of streams, tributary to lakes. Instream barriers and diversion structures have often prohibited the movement of suckers into spawning streams during recent years. During years with high precipitation, enough water flows by the diversion structures so that the suckers can utilize limited reaches of the streams for spawning. However, during periods of low flows, all water is often diverted, thereby eliminating any change for the fish to spawn. If suckers are successful in ascending the barriers, spawned-out fish and progeny are likely to be restricted to small areas of streams because of instream barriers, or sometimes diverted into agricultural fields where they die (50 FR 39117). Connectivity of water is one of the important needs for the species as passage between waterbodies is dependent on connectivity for species success (USFWS 2025n). No threats exist that would likely threaten the survival of the species over a significant portion of its range, although rising temperature is considered to be an upcoming concern since the thermal tolerance of Warner sucker is 23°C and access to cooler water such as springs and deep pools will be important for the continued survival of the species (USFWS 2025n). The three main concerns for the species are human induced stream channel and watershed degradation, irrigation diversion practices, and predation and competition from introduced fishes (USFWS 1998c).

Warner sucker critical habitat is not located in the action area.

Razorback sucker

Razorback sucker is a long lived (40+), large catostomid fish endemic to the Colorado River Basin of the southwestern United States. Historically, razorback suckers were widely distributed in warm-water reaches of larger rivers of the Colorado River Basin from Mexico to Wyoming. Habitats required by adults in rivers include deep runs, eddies, backwaters, and flooded off-channel environments in spring; runs and pools often in shallow water associated with submerged sandbars in summer; and low-velocity runs, pools, and eddies in winter. Spring migrations of adult razorback sucker were associated with spawning in historic accounts, and a variety of local and long-distance movements and habitat-use patterns have been documented. Spawning in rivers occurs over bars of cobble, gravel, and sand substrates during spring runoff at widely ranging flows and water temperatures (typically greater than 14°C). Spawning also occurs in reservoirs over rocky shoals and shorelines. Young razorback suckers require nursery environments with quiet, warm, shallow water such as tributary mouths, backwaters, or inundated floodplain habitats in rivers, and coves or shorelines in reservoirs. Threats to the species include streamflow regulation, habitat modification, competition with and predation by nonnative fish species, and pesticides and pollutants (USFWS 2002b). The razorback sucker is jeopardized by large mainstem dams, water diversions, habitat modification, nonnative fish species, and degraded water quality.

Persistent recruitment failure has depleted and extirpated numerous razorback sucker populations. In the case of lower basin reservoirs (such as, Mead, Mohave, Havasu), the characteristic chronology of extirpation is 40–50 years following dam construction; i.e., the fish that were produced prior to habitat inundation and fragmentation reach maximum longevity with little or no recruitment to replace adult mortality. This occurred because of inundated riverine habitat and poor survival of reproduced young because of predation by nonnative fishes (USFWS 2002b). Razorback suckers are currently found in the Green River, upper Colorado River, and San Juan River subbasins; lower Colorado River between Lake Havasu and Davis Dam; reservoirs of Lakes Mead and Mohave; and in small tributaries of the Gila River subbasin. The razorback suckers in most populations are aged and senile adults with little or no recruitment, except for the middle Green River and Lake Mead, where small numbers of juveniles and young adults indicate low recruitment levels. Currently, the largest concentration of razorback sucker remaining in the entire Colorado River Basin is in Lake Mohave.

Razorback sucker critical habitat physical and biological features include (59 FR 13374):

1. Water - This includes a quantity of water of sufficient quality (such as, temperature, dissolved oxygen, lack of contaminants, nutrients, turbidity, etc.) that is delivered to a specific location in accordance with a hydrologic regime that is required for the particular life stage for each species.

2. Physical Habitat- This includes areas of the Colorado River system that are inhabited or potentially habitable by fish for use in spawning, nursery, feeding, and rearing, or corridors between these areas. In addition to river channels, these areas also include bottom lands, side channels, secondary channels, oxbows, backwaters, and other areas in the 100-year flood plain, which when inundated provide spawning, nursery, feeding and rearing habitats, or access to these habitats.

3. Biological Environment- Food supply, predation, and competition are important elements of the biological environment and are considered components of this constituent element. Food supply is a function of nutrient supply, productivity, and availability to each life stage of the species. Predation and competition, although considered normal components of this environment, are out of balance due to introduced nonnative fish species in many areas.

Special consideration was given to habitats required for reproduction and recruitment. The following selection considerations were used by the Service to help determine areas necessary for survival and recovery of the razorback sucker.

1. Presence of known or suspected wild spawning populations, although recruitment may be limited or nonexistent.

2. Areas where juvenile razorback suckers have been collected or which could provide suitable nursery habitat (backwaters, flooded bottom lands, or coves).

3. Areas presently occupied or that were historically occupied that are considered necessary for recovery and that have the potential for reestablishment of razorback suckers.
4. Areas and water that are required to maintain rangewide fish distribution and diversity under a variety of physical, chemical, and biological conditions.
5. Areas that need special management or protection to ensure razorback survival and recovery.

These areas once met the habitat needs of the razorback sucker and may be recoverable with additional protection and management.

Santa Ana sucker

The Santa Ana sucker is a small, generally less than 6.3 inches, short-lived fish. Santa Ana suckers feed mostly on algae, diatoms, and detritus scraped from rocks and other hard substrates, with aquatic insects making up a very small component of their diet. Larger fish generally feed more on insects than do smaller fish. Santa Ana suckers generally reach sexual maturity in just over 1 year and typically do not live more than 3 years (65 FR 19686). Its range include the Los Angeles River, San Gabriel River, and Santa Ana River basins. The Santa Ana sucker is one of only a few native freshwater species of fish currently extant in southern California. The listed species' range has been reduced in the three watersheds where it occurs and there is no opportunity for natural movement between watersheds. Santa Ana suckers are also patchily distributed within each watershed, with dispersal to other reaches limited by barriers that prevent or severely limit two-way (upstream and downstream) fish passage within the respective watersheds. The primary threat to the Santa Ana sucker includes past and ongoing habitat loss through hydrological modifications throughout the range of the species. The loss and degradation of available habitat (caused by dams, changes in water allocations, and other hydrological modifications) combined with other increasing threats (such as degraded water quality, recreational pressures, and potential effects of nonnative vegetation and predation) have a cumulative effect on Santa Ana sucker and its habitat. Additionally, impassable barriers or areas of unsuitable habitat limit gene flow, thus increasing the vulnerability of small populations to a range of environmental stochastic factors and inbreeding depression (USFWS 2017d).

In streams or rivers that are not restricted by artificial barriers, such as the San Gabriel River (North and East Forks), the upper limit of where fish are likely to occur is determined by the instream gradient and the fish's inability to physically swim upstream when a certain gradient (and thus flow rate) is exceeded. Santa Ana suckers occur in the watersheds draining the San Bernardino and San Gabriel Mountains of southern California. Their historical distribution extended from upper watershed areas to the Pacific Ocean; hence, they can occupy habitats as diverse as mountain streams and rivers in alluvial floodplains (relatively flat landform created by the deposition of sediment over a long period of time from one or many rivers) (USFWS 2017d). The streams that the Santa Ana sucker inhabits are generally perennial streams with water ranging in depth from a few inches to several feet and with currents ranging from slight to swift. Perennial flows with suitable water quality and substrate are needed to support breeding, feeding,

and sheltering. These streams are also naturally subject to periodic, severe flooding and may experience extended periods of low flow as a result of drought conditions that are typical of southern California climate cycles (USFWS 2017d).

Santa Ana sucker critical habitat physical and biological features include a functioning hydrological system that provides sources of water and course sediment necessary to maintain all life stages, including adults, juveniles, larvae, and eggs. Physical and biological features essential to the conservation of Santa Ana sucker are (75 FR 77962):

1. A functioning hydrological system within the historical geographic range of Santa Ana sucker that experiences peaks and ebbs in the water volume (either naturally or regulated) that encompasses areas that provide or contain sources of water and coarse sediment necessary to maintain all life stages of the species, including adults, juveniles, larvae, and eggs, in the riverine environment;

2. Stream channel substrate consisting of a mosaic of loose sand, gravel, cobble, and boulder substrates in a series of riffles, runs, pools, and shallow sandy stream margins necessary to maintain various life stages of the species, including adults, juveniles, larvae, and eggs, in the riverine environment;

3. Water depths greater than 1.2 in (3 cm) and bottom water velocities greater than 0.01 ft per second (0.03 m per second);

4. Clear or only occasionally turbid water;

5. Water temperatures less than 86 °F (30 °C);

6. Instream habitat that includes food sources (such as zooplankton, phytoplankton, and aquatic invertebrates), and associated vegetation such as aquatic emergent vegetation and adjacent riparian vegetation to provide:

    i. Shading to reduce water temperature when ambient temperatures are high,

    ii. shelter during periods of high water velocity, and

    iii. protective cover from predators; and

7. Areas within perennial stream courses that may be periodically dewatered, but that serve as connective corridors between occupied or seasonally occupied habitat and through which the species may move when the habitat is wetted.

All occupied units designated as critical habitat contain the physical and biological features in the appropriate quantity and spatial arrangement essential to the conservation of this species and support multiple life processes for Santa Ana sucker.

Catostomidae Effects

Generally, Lost River sucker, shortnose sucker, and razorback sucker are long lived species, take years to mature, move up rivers to spawn, and have been subjected to habitat modifications

where it is difficult to complete their life history needs. The main threats to the species are water quantity and quality including timing.

Lost River and shortnose suckers live in a system where dissolved oxygen levels become during the summer under certain conditions. They spawn in the spring March – April, when conditions for timber and road building are not appropriate. Sedimentation for this species is typically not a concern for eggs, juveniles, or adults. Although it may create increased nutrient input that could create conditions for algae growth that is tied to dissolved oxygen sinks, the amount of sediment would be insignificant compared to what would be needed to increase algae growth causing dissolved oxygen sinks. Critical habitat in the action area is very small, less than 1 percent. Forest Service best management practices (USFS 2012a) would minimize the amount of sediment reaching connected waters. Any future consultation would most likely contain measures to minimize effects on species. Therefore, the Roadless Rule Rescission **may affect, not likely to adversely affect** Lost River sucker, Lost River sucker critical habitat, shortnose sucker, and shortnose sucker critical habitat because effects are discountable.

Razorback sucker live in highly modified large river systems where dams have created habitats insufficient for recruitment of young into the population. Due to the nature of the large river systems, any minor changes in temperature or sedimentation to the habitat would be indistinguishable from background levels. The amount of critical habitat in the affected environment is minimal, < 0.01%. Forest Service best management practices (USFS 2012a) would minimize the amount of sediment reaching connected waters. Any future consultation would most likely contain measures to minimize effects on species. Therefore, the Roadless Rule Rescission **may affect, not likely to adversely affect** razorback sucker and razorback sucker critical habitat because effects are discountable.

Although in the same family, Santa Ana sucker have a different life history than the others listed under Catostomidae Family in this analysis. Effects of future projects on Santa Ana sucker would most likely be due to sedimentation, potential increase in water temperatures, and habitat fragmentation when projects are implemented. Implementing Forest Service best management practices (USFS 2012a) will help alleviate negative effects of any future project. Low level effects from sedimentation, increase in water temperatures, and habitat fragmentation could occur during future project implementation, the magnitude of which is unknown, but expected to be low. Any future projects that would affect species would still need to complete project level consultation. Due to the unknowns, we might expect some effects on species that are susceptible to reach **may affect, likely to adversely affect on individual** Santa Ana suckers (Table 15).

The amount of critical habitat in the affected environment is minimal, 0.02%. Forest Service best management practices (USFS 2012a) would minimize the amount of sediment. Any future consultation would most likely contain measures to minimize effects on species but there may be some effects that are measurable. Therefore, the Roadless Rule Rescission **may affect, likely to adversely affect** Santa Ana sucker critical habitat.

Leuciscidae Family

The Leuciscidae family has the most species in this consultation and includes all the shiners, chubs, dace, spinedace, loach minnow, spikedace, Colorado pikeminnow, and bonytail identified in Table 15.

Blue shiner

The blue shiner is a medium-sized minnow that grows to about 4 inches in total length. Blue shiner live for 3 years, and most spawning fish are 2 years old. It is represented by approximately six populations occurring in headwater streams of the Coosa River system in Northwest Georgia and Southeast Tennessee, and in smaller tributaries of the Coosa River in Northeast Alabama. Populations are isolated by impoundments, inappropriate habitat, inadequate water quality, and distance (USFWS 1995c). The blue shiner requires water of relatively high quality and appears sensitive to turbidity and siltation. Major threats are habitat alteration and water quality degradation (USFWS 1995c).

Blue shiner primarily occupies second to fourth order, moderate gradient streams within the Ridge and Valley and Piedmont physiographic provinces of Alabama, Georgia, and Tennessee (USFWS 1995c). Most watersheds where it is found are predominately forested, and agriculture and urban development are minimal. The headwaters of Choccolocco and Shoal creeks are in Talladega National Forest. The Weogufka watershed is also primarily forested. The Little River flows through Little River Canyon National Preserve, which is managed by the National Park Service. In Georgia and Tennessee, the blue shiner is found in the upper reaches of streams flowing through two National Forests. The Conasauga River originates within the Chattahoochee National Forest in Georgia and flows into Tennessee where it enters the Cherokee National Forest. Mmnnewauga Creek is within the Cherokee National Forest. Rock Creek, in Georgia is mostly within the Chattahoochee National Forest. Perry and Holly Creek have headwaters in the Chattahoochee National Forest (USFWS 1995c).

The blue shiner can be abundant in the proper habitat and is rare or absent elsewhere, suggesting that it is habitat dependent. It is a fluvial specialist, being found only in flowing water. It prefers sand or sand and gravel substrate sometimes with cobble, low to moderate velocity current, and a depth of about 0.15 to 1 meters (0.5 to 3 feet). Blue shiners are sometimes associated with submerged tree roots and fallen branches. They also occur near water willow (Justicia americana) beds, especially in eddy currents downstream from the beds. In the Little River and Choccolocco Creek, lateral pools away from the main current, and backwaters with sandy substrates were preferred. In Weogufka Creek, a concentration of blue shiner were observed below a leaky beaver dam where flows were moderated. In the Conasauga River and Weogufica Creek, this shiner is also prevalent in long, low-velocity pools (USFWS 1995c).

The diet of the blue shiners is dominated throughout the year by terrestrial insects, indicating that the blue shiner is a visual, drift feeder. The importance of riparian vegetation and a tree canopy along the stream margin of blue shiner habitat is suggested by the importance of terrestrial insects in the diet. Successful visual foraging may be dependent on water clarity (USFWS 1995c).

Blue shiner threats include water quality degradation, for example, nutrification and probable low dissolved oxygen levels. The blue shiner's range has been reduced and fragmented by construction of dams, loss of habitat, and/or water pollution. This has created isolated and relatively small populations. Isolated populations are especially vulnerable to habitat degradation and decreased genetic diversity. Any event that adversely affects an isolated population has the potential to eliminate it. This is especially true for stream-dwelling species since recolonization may be prevented by a variety of barriers, e.g., dams, inappropriate habitat, poor water quality, and distance. Thus, a single, even ephemeral incident or event such as a chemical spill or an extreme drought can permanently reduce the range of an aquatic species with fragmented populations and make it very susceptible to extinction (USFWS 1995c).

The range of the blue shiner within Alabama, Georgia and Tennessee has been reduced and fragmented by geomorphic and hydrologic changes such as reservoirs, bridges, pipelines and roads; aggregate extraction; major pollution events; an increase in turbidity caused by sedimentation; and general declining water quality attributed to non-sustainable urbanization and land use practices. These threats to habitat, primarily poor water quality and fragmentation of populations, are still the major driver in limiting the species distribution, abundance, and viability (USFWS 2021e).

Blue shiner critical habitat has not been designated.

Effects: The pathway for fish effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. The pathway for effect to blue shine is through water quality and habitat access. Activities found in Table 16 under the impact/benefit column containing change in riparian vegetation (temperature), sediment/turbidity, and road density/location may have effects on blue shiner. Habitat access could be affected by temporary or permanent road building, these effects would be in place as long as the temporary road is in place. BMPs would help alleviate negative effects but overall the effects of the action would be **may affect, likely to adversely affect** blue shiner.

Spotfin chub

Spotfin chub is a small cyprinid (maximum standard length 92 mm), with a slightly compressed, elongated body, and live to year 3 (USFWS 1983b; USFWS 2019j). Spotfin chub are more benthic than most other cyprinids and feed primarily on the surfaces of boulders, bedrock, and other substrates that are largely free of vegetation or sediment. Spotfin chub are typically oriented at an angle downward several degrees from horizontal when foraging, move about with quick zigzag motions, remain close to the bottom, and apparently do not opportunistically take drifting insects in the water column like most other shiners. The species is an insectivore, feeding diurnally presumably by both sight and taste in benthic areas of slow to swift current over various substrates with little siltation. The streams may range from 15-60 m width and, where occupied, 0.3-1.0 m depth. Water temperature in their summer habitat usually reaches greater than 20° C, and submerged macrophytes are usually absent, occasionally common. The species has been observed associated with sand, gravel, rubble, boulder, and bedrock substrates (USFWS 1983b; USFWS 2019j).

Spotfin chub is a rare large-stream fish that was widely distributed historically in most of the Tennessee River drainage. A total of 41 streams have records for the species, but at least 16 streams are thought to have lost their population. At present, four population clusters continue to support this species. Spotfin chub is generally sporadic or occasional in distribution and usually occurs at varying levels of abundance from very rare to relatively common in discrete reaches in most streams of occurrence. Spotfin chub typically inhabits moderate runs over bedrock, large boulders, and other substrates in large, clear, upland streams. Spawning and foraging substrates must be swept relatively free of fine sediments (USFWS 2019j).

The construction of major impoundments in much of the range of spotfin chub contributed heavily to habitat alterations for the species and resulted in highly disjunct current populations. Additional losses were likely realized from contaminants, sedimentation, and other habitat alterations (for example, middle North Fork Holston River). Losses from these sources were such that current habitat levels represent but a small percentage of historical habitat available for the species (USFWS 2019j).

Spotfin chub populations continue to be threatened by habitat fragmentation and population isolation, sedimentation and legacy contaminants events, and are susceptible to varying degrees by stochastic events (USFWS 2019j). Anthropogenic stresses such as silt or coal fine sedimentation; pollution; inundation by reservoir; temperature depression of dam tailwater; and channelization are invoked to have exterminated populations. Most of these factors also affect master streams of tributaries, some of whose populations may have been at least partly dependent upon ingress of chubs from the exterminated populations (USFWS 1983b).

The 1983 Recovery Plan listed nine "anthropogenic and natural limiting factors on all known populations" of spotfin chub and divided them into direct and indirect impacts for each extant and extirpated population. These included: 1) impoundment, 2) cold tailwaters, 3) channelization, 4) siltation and/or coal fine sedimentation, 5) pollution (inorganic and/or organic), 6) population renovation, 7) localized collecting, 8) natural cool temperature, and 9) small stream size (USFWS 1983b).

Actions that create sedimentation and contamination to spotfin chub habitat over the long term would be deleterious to the species. BMPs would help alleviate negative effects but overall the effects of the action would be may affect, likely to adversely affect spotfin chub.

Spotfin chub critical habitat is not in the action area.

Effects: The pathway for fish effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. The pathway for effect to spotfin chub is through water quality and habitat access. Activities found in Table 16 under the impact/benefit column containing change in riparian vegetation (temperature), sediment/turbidity, and road density/location may have effects on spotfin chub. Habitat access could be affected by temporary road building; these effects would be in place as long as the temporary or permanent road is in place. The probability of spotfin chub being in the action area are low since they have not been found in over 20 years, but not zero. Considering the species pathway of effects, best management practices, and probability

of presence in the action area, project activities may affect, not likely to adversely affect spotfin chub because effects are discountable.

Owens tui chub

Owens tui chub is dusky-olive above and has blue and gold sides, a gold-colored head, and olive-brown to reddish brown fins. Individuals range from 15 mm to 180 mm in length. It is distinguished from other tui chubs by the presence of lateral radii on the scales with a rounded or shield shaped scale base. This species occurs in low-velocity waters with well-developed beds of aquatic vegetation, rocks, and undercut banks. Vegetation is important for predator avoidance, reproduction, food, and water velocity reduction. The Owens tui chub is an opportunistic omnivore, consuming aquatic insects, vegetation, and detritus. Owens tui chubs feed mainly by gleaning and grazing among submerged vegetation. Its diet varies seasonally; the dominant items in its diet are chironomid larvae and algae in spring, chironomid larvae in summer, hydroptilid caddisflies in fall, and chironomid larvae in winter. Owens tui chub are found in constant water temperature springs and reach sexual maturity at 2 years of age for females and 1 year for males. Males and females will both reach sexual maturity at age 2 in varied temperature waters. Spawning in spring systems is likely triggered by day length and occurs from late winter to early spring. Spawning in riverine and lacustrine or lake-like habitats is triggered by warming water temperatures and occurs in spring and early summer. Spawning typically occurs over gravel substrate or aquatic vegetation, with eggs adhering to these features. Life expectancy is likely several years, with the oldest fish captured estimated at 7 years (USFWS 2026a).

Owens tui chub (populations that have not hybridized with Lahontan tui chub (*Siphateles bicolor pectinifer*)) are restricted to five small, isolated sites within its historical range in the Owens Basin: Hot Creek headwaters (only CD Spring), Little Hot Creek Pond, Upper Owens Gorge, Mule Spring, and White Mountain Research Center, operated by the University of California. The Hot Creek headwaters population site consists of two currently separate springs, AB Spring and CD Spring; however, only the population in CD Spring is not hybridized with Lahontan tui chub (USFWS 2022h).

Threats to Owens tui chub are disease; catastrophic events; demographic, genetic and environmental stochasticity; and long-term alteration of weather patterns (USFWS 2022h).

Although only a few studies have examined Owens tui chub behavior, life history, and habitat use, several aspects of its ecology can be generally surmised from studies of other tui chub subspecies. Tui chubs congregate from late winter to early summer to spawn over aquatic vegetation or gravel substrate. Females may produce many eggs. Tui chubs may reach sexual maturity at 2 years and may live more than 30 years (USFWS 1998d).

Owen's tui chub critical habitat is not in the action area.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing sediment/turbidity may have effects on Owen tui chub. Activities that include water quality pathways associated with impacts/benefits for change in riparian vegetation (temperature), would not be included as an activity causing effects because the species would not

be affected by downstream water temperature fluctuations within the expected effects range. Project activity effects would be limited to connected actions that might cause downstream effects such as timber haul that might cause some sediment/turbidity. Best management practices will minimize sediment/turbidity near riparian areas. After considering the species pathway of effects, and best management practices that would be implemented, project activities would not have measurable effects to species or their critical habitat; therefore, the project **may affect, not likely to adversely affect** Owen tui chub.

Humpback chub

The humpback chub is a large fish endemic to the Colorado River Basin. Adults attain a maximum size of about 480 mm total length and 1.2 kg in weight (USFWS 1990; USFWS 2002c). The greatest numbers of humpback chub have been found in relatively inaccessible canyons. The humpback chub has been referred to as "remarkable" and "bizarre." It is certain that the humpback chub is an unusual and striking fish. It is a medium-sized less than 20 inches total length, freshwater fish, with silvery sides and a brown or o1ivaceous back. It has a narrow, flattened head that may be dorsally concave, a long, fleshy snout with an inferior-subterminal mouth, and small eyes. Adults generally have a pronounced dorsal hump that begins approximately at the dorsal origin of the gill covers, protrudes anteriorly, then ends at the origin of the dorsal fin. The fish's body tapers abruptly to the narrow caudal peduncle, then flares posteriorly into a deeply forked caudal fin. These morphologic adaptations of the humpback chub aid the fish in negotiating turbulent waters in deep canyons of the Colorado River (USFWS 1990; USFWS 2002c).

Six extant wild populations of humpback chub are known: (1) Black Rocks, Colorado River, Colorado; (2) Westwater Canyon, Colorado River, Utah; (3) Yampa Canyon, Yampa River, Colorado; (4) Desolation/Gray Canyons, Green River, Utah; (5) Cataract Canyon, Colorado River, Utah; and (6) the mainstem Colorado River in Marble and Grand Canyons and the Little Colorado River, Arizona. The first five populations are in the Upper Colorado River Basin (i.e., upstream of Glen Canyon Dam, Arizona), and the sixth population is in the Lower Colorado River Basin. Populations of humpback chub are restricted to deep, swift, canyon-bound regions of the mainstem and large tributaries of the Colorado River Basin (USFWS 2002c).

Adults require eddies and sheltered shoreline habitats maintained by high spring flows. These high spring flows maintain channel and habitat diversity, flush sediments from spawning areas, rejuvenate food production, and form gravel and cobble deposits used for spawning. Spawning occurs on the descending limb of the spring hydrograph at water temperatures typically between 16 and 22°C. Young require low-velocity shoreline habitats, including eddies and backwaters, that are more prevalent under base-flow conditions. Threats to the species include streamflow regulation, habitat modification, predation by nonnative fish species, parasitism, hybridization with other native Gila, and pesticides and pollutants (USFWS 1990).

Humpback chub reside in larger river systems where any effect from forest service actions, if it reached the larger river system, would return to background levels quickly but may still have negative impacts to young since they tend to utilize low-velocity shoreline habitats. BMPs would

help alleviate negative effects but overall, the effects of the action would be **may affect, likely to adversely affect** humpback chub.

Humpback chub critical habitat is not in the action area.

Sonora chub

Sonora chub is a member of the Leuciscidae family and is generally less than 5 inches in total length. It is a moderately chubby, dark colored fish, with two prominent black lateral bands on the sides and a dark oval spot at the base of the tail. In breeding males, a red coloration develops at the bases of the lower fins and some orange coloration is present on the belly. The Sonora chub is primarily a pool dweller but is highly secretive and little is known of its behavior and habitat preferences (51 FR 16042).

The Sonora chub is a desert fish adapted to the fluctuations of a desert environment; after drought conditions it has been known to rapidly expand and recolonize California Gulch and newly re-wetted reaches. If habitat conditions along water ways can be maintained, then this ability to respond to favorable water conditions is encouraging for the population to avoid the danger of extinction. Construction of roads or bridges as described above might have temporary adverse effects, but long-term effects can be beneficial to the chub if it reduces off road use. there are two known wild populations within the U.S. and a captive population (USFWS 2019k).

The species occurs in southern Arizona and in northern Sonora, Mexico. The Coronado National Forest monitors and implements conservation projects for the Sonora chub regularly. The Coronado National Forest noted Sonora chub were present in California Gulch in 2012. All Coronado National Forest reports stated that international border security issues make it difficult to monitor populations of this species. The majority of habitat occupied by Sonora chub within the U.S. exists within the Coronado National Forest. Portions of Sycamore Canyon and its tributaries are critical habitat and include the Pajarito Wilderness and Goodding Research Natural Area, each of which lends a high degree of administrative protection for the species. Roadless Rule Rescission could remove the administrative protection for the species. Land ownership patterns in México are variable, and the species has comparatively little administrative protection there (USFWS 2022i).

The three primary threats to the Sonora chub are: 1) non-native species, 2) degradation of habitat, and 3) continued demand for water for human consumption. Current threats to the United States population include the stocking of exotic fishes and their associated parasites, and possible uranium mining activities (51 FR 16042).

Sonora chub critical habitat is not in the action area.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on Sonora chub and Sonora chub critical habitat. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects and indistinguishable from background conditions after a

wildfire. Process based restoration and temporary road removal activities would have short-term negative effects but long-term beneficial effects. Temporary road building would create adverse conditions as long as the road is in place. Best management practices will help minimize adverse effects. Removal of the high degree of administrative protection within the Coronado National Forest Plan; any future project could reach the level of not being discountable and **may affect, likely to adversely affect** Sonora chub.

Bonytail

The **bonytail** is a large cyprinid fish endemic to the Colorado River Basin. Adults attain a maximum size of about 550 mm total length and 1.1 kg in weight. The bonytail is rare in the wild and an unknown, but small number of wild adults exist in Lake Mohave on the mainstem Colorado River of the Lower Colorado River Basin (i.e., downstream of Glen Canyon Dam, Arizona), and there are small numbers of wild individuals in the Green River and upper Colorado River subbasins of the Upper Colorado River Basin (USFWS 2002d).

The bonytail was historically common to abundant in warm-water reaches of larger rivers from Mexico to Wyoming. Little is known about the specific habitat requirements of bonytail because the species was extirpated from most of its historic range prior to extensive fishery surveys. The bonytail is considered adapted to mainstem rivers where it has been observed in pools and eddies. Similar to other closely related *Gila* spp., bonytail in rivers probably spawn in spring over rocky substrates; spawning in reservoirs has been observed over rocky shoals and shorelines. It is hypothesized, based on available distribution data, that flooded bottomland habitats are important growth and conditioning areas for bonytail, particularly as nursery habitats for young. Threats to the species include streamflow regulation, habitat modification, competition with and predation by nonnative fish species, hybridization, and pesticides and pollutants (USFWS 2002d).

Currently no self-sustaining populations of bonytail exist in the wild, and very few individuals have been caught throughout the basin. Releases of hatchery-reared adults into riverine reaches in the upper basin have resulted in low survival, but in 2023 there was evidence of reproduction and recruitment of juveniles (USFWS 2002c; USFWS 2024y).

Bonytail critical habitat is not in the action area.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Although difficult to determine based on the available information for young, we assume activities found in Table 16 under the impact/benefit column containing water quality related to sediment/turbidity may have effects on bonytail young within the pools and eddies as sediment/turbidity could decrease sight feeding. The amount of sediment/turbidity that would accompany timber harvest activities would be short-term negative effects. Temporary and permanent road building would create adverse conditions as long as the road is in place. Best management practices will help minimize adverse effects. Therefore, the project **may affect, likely to adversely** affect bonytail.

Gila chub

In June 2025, the USFWS proposed to delist Gila chub due to findings that it is not a listable entry (90 FR 25552). There has been considerable uncertainty in the scientific literature regarding the taxonomy of Gila chub in the Lower Colorado River basin, specifically relationship between the roundtail chub, headwater chub, and Gila chub (*Gila intermedia*) (from SSS of roundtail chub [USFWS 2025o]). The Gila chub has been listed as endangered with critical habitat since 2002 (70 FR 66664). In the recent Species Status Assessment (USFWS 2025o), information indicated a lack of differentiation between the three species, although there was also contrary information as well. USFWS concluded that that the available information did not support recognizing the Gila chub or headwater chub as distinct taxonomic entities and instead should be classified as roundtail chub. USFWS noted this in their 12-month finding for the roundtail chub and issued an advanced notice of proposed rulemaking to delist the Gila under the justification that it was not a listable entity (87 FR 19657).

Gila chub critical habitat is in the action area and consists of approximately 0.1% of the total designated critical habitat. Gila chub's primary constituent elements are:

1. Perennial pools, areas of higher velocity between pools, and areas of shallow water among plants or eddies all found in headwaters, springs, and cienegas, generally of smaller tributaries;
2. Water temperatures for spawning ranging from 17 to 24 °C (62.6 to 75.2 °F), and seasonally appropriate temperatures for all life stages (varying from approximately 10 °C to 30 °C).
3. Water quality with reduced levels of contaminants, including excessive levels of sediments adverse to Gila chub health, and adequate levels of pH (for example, ranging from 6.5 to 9.5), dissolved oxygen (for example, ranging from 3.0 to 10.0) and conductivity (for example, 100 to 1000 mmhos).
4. Food base consisting of invertebrates (for example, aquatic and terrestrial insects) and aquatic plants (for example, diatoms and filamentous green algae);
5. Sufficient cover consisting of downed logs in the water channel, submerged aquatic vegetation, submerged large tree root wads, undercut banks with sufficient overhanging vegetation, large rocks and boulders with overhangs, a high degree of streambank stability, and a healthy, intact riparian vegetation community;
6. Habitat devoid of nonnative aquatic species detrimental to Gila chub or habitat in which detrimental nonnatives are kept at a level that allows Gila chub to continue to survive and reproduce; and
7. Streams that maintain a natural flow pattern including periodic flooding.

Effects: Gila chub have been found to not be a listable entity by the USFWS, although they have not been removed from the ESA list at the time of writing this biological assessment. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit

column in Table 16. Effects to this species may occur; however, the fact that Gila chub is not a listable entity (87 FR 19657), effects analyses for the species and critical habitat is not necessary. Since this is still a listed entity, the assumption for this document is the project **may affect, not likely to adversely affect** Gila chub and Gila chub critical habitat.

Chihuahua chub

The Chihuahua chub has a fusiform body that resembles some trout. It has a terminal mouth that extends backward to about the margin of the eye. The Chihuahua chub averages 5-6 inches in length at maturity and may reach 12 inches. In individuals over 100 millimeters, the back and sides are brassy green to slate gray, sometimes with two indistinct lateral stripes, and the abdomen is whitish (USFWS 1986b; USFWS 2023l). Chihuahua chub is suggested to be an opportunistic carnivore, feeding on invertebrates and possibly fish. However, no specific studies on food habits have been conducted (USFWS 2023l).

The Chihuahua chub is endemic to the Guzman Basin, where it occurs from the Mimbres River in New Mexico, southward to northwestern Chihuahua, Mexico (USFWS 1986b). The historical range of Chihuahua chub is difficult to determine because of the lack of early collections. Most likely all but the uppermost reaches of permanently watered portions of the Mimbres River, New Mexico, and streams in the Guzman and Laguna Bustillos basins in Mexico were occupied. All streams within the historical range flow into a closed (endorheic) basin; none have a connection with the ocean (USFWS 2023l). Adult Chihuahua chub are considered habitat specialists. They are found in their natural habitats primarily in lateral scour pools, beneath undercut banks, or under other solid objects (e.g., logs, boulders) adjacent to moderate to fast flowing water in small to medium sized streams. Corner and backwater pools containing large woody debris are also used as habitat. Almost all natural habitats occupied by Chihuahua chub have extensive cover composed of organic debris or root wads of large trees. Pools are typically 1-2 m deep with a water velocity of less than 15 cm/second, and substrates are small-grained (sand to pea-size). Juveniles are found in shallower water with or without cover. This habitat type provides both escape cover and a suitable foraging situation. Chubs are rather trout-like in much of their feeding, taking terrestrial insects on the surface as well as aquatic invertebrates and perhaps some fish and vegetation (USFWS 1986b; USFWS 2023l).

The decline of the Chihuahua chub in the Mimbres River appears to be primarily related to loss of habitat. This loss has been due to severe flooding caused by degradation of the watershed and loss of riparian vegetation, and to action taken by local landowners to protect their property from future flooding. The Mimbres River has been channelized and leveed by local landowners in an effort to confine flood waters. As a result, chub habitat has been destroyed. The work has not been effective or enduring, and must be repeated each spring at or near the time the chub spawns, making these activities especially detrimental.

Diversions have reduced the quantity of water in the river, while uneven flows, flooding, and repeated stream modifications by local landowners have combined to eliminate the historical habitat conditions to the point the river terminus is usually dry (USFWS 1986b). Exotic species introductions probably led to predation on and competition with the chub. However, these factors

probably played a minor role in the declining status of the chub compared to the influence of loss of habitat (USFWS 1986b).

Propagation and restocking of the Chihuahua chub has been ongoing since the 1970's. There have been continued stocking efforts. The management direction for the Chihuahua chub is to stock the species as needed to maintain populations and expand their range with the intention of creating a self-sustaining population(s) (USFWS 2023l).

The Chihuahua chubs' distribution and abundance in the US has declined in recent years due to catastrophic effects of wildfires and subsequent ash flow during monsoons. Currently, the US population consists of a few isolated individuals in the Mimbres River basin, found during surveys in late 2022 and early 2023. The species has not been surveyed in Mexico since the 1990s. Given its declining state at that time, lack of regulatory mechanisms, and likely increase of stressors due to alterations of weather patterns, it is presumed the species condition has continued to decline in Mexico. Numerous conservation actions have benefited the species, including obtaining property ownership along the Mimbres River, habitat improvements in stream and riparian areas, and the construction of an off channel refugia pond. Further, the species is maintained in captivity for propagation and release (USFWS 2023l).

Chihuahua chub critical habitat has not been designated.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on Chihuahua chub. However, sediment/turbidity within the range of Chihuahua chub may only be an issue when monsoon rain occurs. Activities that create sediment/turbidity would add to sediment movement into the river. It's difficult to determine whether these sediment inputs from the activities would be measurable or not. Best management practices will help minimize the amount of sediment reaching the river but will not eliminate it. These effects would be short-term effects. Therefore, the project **may affect, likely to adversely affect** Chihuahua chub.

Yaqui chub

Yaqui chub are considered a medium-sized minnow, where adults rarely exceed 6 inches in length. The head and anterior portion of the body are thick and narrows towards the posterior region. Yaqui chub are generally dark colored, but in sunlight can exhibit combinations of green, blues and purples. The most prominent feature of Yaqui chub is a dark, nearly black, triangular caudal spot, which is located at the base of the tail. They live in springs, spring-fed ditches, creeks, (and have adapted favorably to ponds) over substrates of silt, clay, sand, and gravel. Water may be clear or muddy, with associated vegetation including watercress, cattail, sedges, and willows. Depths of capture vary to >1.5 m; and currents are none to moderate. Spawning occurs mostly during spring – early summer and has been recorded in all but midwinter months. The reproductive potential of this fish is very high, with large populations able to quickly develop from only a few adults when habitat conditions are favorable. The species feeds mostly on algae, invertebrates, and detritus (USFWS 2020i). Growth to maturity is rapid, often within the first summer of life. Spawning is protracted throughout the warmer months, with greater activity in spring. Reproductive potential is high and large populations develop quickly from a

few adults (USFWS 1995d). Young fish will spatially separate from larger individuals by occupying areas near shore and in shallow water. Threats to this species include very limited range, making potential loss from catastrophic events more likely; loss, alteration, and degradation of suitable wetland habitat; competition with, and/or depredation by, non-native species; and long-term drought combined with expanding human populations, which are creating increased demand for water for human consumption (USFWS 2020i).

Yaqui chub are restricted to the Rio San Bernardino system in Arizona and Sonora. It was historically and still remains known in Mexico but only from a <3.0 km perennial reach of Rio San Bernardino, immediately south of the international border in Sonora. Most of its range is now restricted within the U.S., and the majority of that is directly protected on San Bernadino National Wildlife Refuge.

Approximately 35 managed populations of Yaqui chub currently occur across the known range with one population occurring in West Turkey Creek on the Coronado National Forest. The 2018 Coronado National Forest Land Management Plan (available at USFS 2025a) contains guidelines and management approaches included in their forest plan that are intended to implement projects that will achieve desired ecosystem conditions for animals and rare plants (Chapter 2 under Animals and Rare plants page 65) where they occur.

Yaqui chub critical habitat is not in the action area.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on Yaqui chub. Sediment/turbidity is associated with activities proposed in the project. Temporary and permanent road building would create adverse conditions as long as the road is in place. Yaqui chub reside in spring-fed areas where sediment/turbidity would dissipate quickly but may still cause adverse effects. Guidelines within the Coronado National Forest would evaluate projects for effects to the species prior to determining to implement in Yaqui chub habitat. Best management practices will help minimize the adverse effects but may not eliminate them. Therefore, project activities **may affect, likely to adversely affect** Yaqui chub.

Clear Lake hitch

The Clear Lake hitch is a subspecies of fish in the North American freshwater fish in the Leuciscidae family. It is restricted to the Clear Lake watershed in Lake County, California. Clear Lake is located approximately 100 miles north of San Francisco in the Coast Ranges and experiences hot, dry summers and cool, wet winters, with most precipitation occurring from November to March. The lake is fed by numerous tributaries, most of which are ephemeral and low gradient as they drain into the lake. Historically, the Clear Lake hitch occurred in numerous lakes and ponds found throughout the Clear Lake watershed, including Clear Lake, Thurston Lake, Upper Blue Lake, Lower Blue Lake, and Lampson Pond. There are no population or abundance estimates for the historical population of Clear Lake hitch. Historical data on the hitch were primarily from observations and incidental catch data from past commercial fishing operations, and local Tribal knowledge and observations (USFWS 2024z).

Adult Clear Lake hitch have a deep and elongated body and can grow to over 350 mm standard length (SL). They have a relatively small conical head, a terminal or slightly upturned mouth, and the morphology of their pharyngeal teeth are evidence of their limnetic lifestyle. Juvenile Clear Lake hitch have been known to have a black spot at the end of the tail that extends to the head region as a gradually fading black stripe; however, as individuals age, their coloring fades and they become a brownish-yellow. Clear Lake hitch are thought to live 4 to 6 years, but it is possible some individuals can live longer (USFWS 2024z).

For most of the year, Clear Lake hitch are only found within their lacustrine environment. However, between February and May, a portion of the reproductive population begins to migrate into the surrounding tributaries to spawn. Eggs are deposited on fine to medium sized gravel within the tributary stream. Fertilized eggs develop and hatch within 7 to 10 days, fry are free-swimming after another 7 to 10 days, and fry stay in the streams anywhere between 11 to 152 days, migrating to the lake once streams dry up. Juvenile Clear Lake hitch are found within the nearshore habitat of the lake where they depend on submerged aquatic vegetation for cover and prey. Juvenile Clear Lake hitch move from the nearshore portion of the lake into open water in early- to late-fall. There is evidence that Clear Lake hitch do not require tributary streams with gravel to spawn but can also spawn successfully in different portions of the lake (such as, along the shore, the mouths of tributaries, and Rodman Slough) that lack a gravel substrate (USFWS 2024z).

Clear Lake hitch critical habitat has not been designated.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on Clear Lake hitch eggs and fry within the tributary streams. Adult Clear Lake hitch are found in open water and project activity causing sediment/turbidity may occur in localized areas depending on environmental conditions. Temporary or permanent road building would create adverse conditions as long as the road is in place. Clear Lake hitch eggs and fry if located in tributaries could be covered with sediment/turbidity causing death to this lifestage. Best management practices will help minimize adverse effects to all life stages but not eliminate the adverse effects. Therefore, project activities **may affect, likely to adversely affect** Clear Lake hitch.

Little Colorado spinedace

Little Colorado spinedace is a small fish, generally less than 100 mm in total length, has a moderately oblique mouth, a strong second “spine” of dorsal fin, moderately high and acute dorsal fin (USFWS 1998e). Little Colorado spinedace is a fish with a limited, highly fragmented distribution and relatively low numbers, making it highly vulnerable to stressors, particularly drought, ground-water and surface water withdrawals, high-severity landscape-scale wildfires, and predation and competition with non-native warm water fishes (USFWS 2019l). Currently, the Little Colorado spinedace occurs in disjunct locations in three subbasins of the Little Colorado River Basin: the Middle Little Colorado, Chevelon Canyon, and Little Colorado Headwaters (USFWS 2019l). Populations are small and the true population size for any occupied

stream is unknown due to yearly fluctuations in their numbers and the difficulty in locating fish (USFWS 2023m).

Little Colorado spinedace live in springs, streams, and rivers with perennial flow, tend to prefer pools, but occurs sporadically throughout the habitat. Little Colorado spinedace has a tolerance for wide temperature fluctuations and habitat types. Little Colorado spinedace' s survival is threatened by habitat loss, habitat modification, competition and predation from non-native fish and introduced parasites (USFWS 2019l).

Current or potential future threats to Little Colorado spinedace include habitat alteration and destruction due to high intensity, landscape scale wildfire, increased groundwater pumping, and long-term drought, and competition and predation with nonnative species. The USFS is implementing thinning and prescribed burning within spinedace watersheds to reduce the threat from high intensity landscape-scale wildfires. Thinning and prescribed burning will also improve water infiltration and increase base flows to occupied, potential, and critical habitat within the project area. Improving water retention in these drainages is vital to maintaining Little Colorado spinedace habitat in a warming and drying climate (USFWS 2023m).

Uncertainties and data gaps that may impede recovery progress include the effects of extended drought and increased groundwater pumping for human water consumption to the persistence of spinedace habitat; the lack of knowledge regarding genetic diversity; and the ability to develop techniques to assist with the control of invasive non-native fishes and other aquatic organisms. The number of areas within the East Clear Creek watershed where spinedace are persisting has increased over the last several years. In addition, occupied sites in the Little Colorado River (upstream of Lyman Reservoir), West Chevelon Creek, Nutrioso Creek, and Rudd Creek continue to support spinedace; although there are long-term concerns regarding habitat in Chevelon Creek. Efforts by Arizona Game and Fish Department, the USFWS, USFS, and Bureau of Land Management have resulted in an increased number of small habitats, in which Little Colorado spinedace occupy (USFWS 2023m).

Little Colorado spinedace critical habitat is in the action area and consists of approximately 0.17% of the total designated critical habitat. Physical and biological features, for all areas of designated critical habitat, include clean, permanent flowing water, with pools and a fine gravel or silt-mud substrate (52 FR 35034).

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on Little Colorado spinedace. Sediment/turbidity is associated with most activities proposed in the project. Temporary and permanent road building would create adverse conditions as long as the road is in place. Little Colorado spinedace reside in streams with clean, flowing water, where sediment/turbidity would dissipate quickly under normal conditions. Best management practices and Apache-Sitgreaves National Forest Plan and Coronado National Forest Plan guidelines would minimize effects. Even so, project activities **may affect, likely to adversely affect** Little Colorado spinedace and Little Colorado spinedace critical habitat.

Spikedace

The spikedace is a small, stream dwelling fish endemic to the Gila River system of Arizona and New Mexico. The body is slender, almost spindle shaped, and lightly compressed laterally. The species also likely occurred in the past in the San Pedro River in Sonora, Mexico. Once widely distributed among moderate sized, intermediate elevation streams in the Gila River system, at least upstream of Phoenix, Arizona, the spikedace is now restricted to scattered populations in relatively short stream reaches (USFWS 1991a). Generally, the species must have been common and likely locally abundant in preferred habitats. Although habitat suitable for spikedace was probably not continuous, it was widespread throughout the species' range. Like most western cyprinids, population abundances and distributions of spikedace probably fluctuated in natural response to local and regional environmental conditions (USFWS 1991a).

Spikedace remain limited in distribution in Aravaipa Creek, the mainstem Gila River, and the West and Middle Forks of the Gila River. Spikedace have not been detected in the Verde River since 1999, and in Eagle Creek since 1989. Translocated populations in Spring Creek, Fossil Creek, the Blue River, and the San Francisco River show promise and, given more time to establish and recover from setbacks, may help in expanding the distribution of spikedace across a great portion of its historical range. Translocation efforts at Muleshoe Canyon, Redfield Canyon, and Bonita Creek have proven unsuccessful. Spikedace are believed to be extirpated from Mangas Creek at this time (USFWS 2023n).

Across their range, spikedace continue to face many threats that were identified in the 2012 reclassification rule (77 FR 10810), including nonnative aquatic species which prey on or compete with them, as well as dewatering of existing habitats, drought, and wildfire. Nonnatives are present in all locations except for the Blue River, Spring Creek, and Fossil Creek, all of which are protected by barriers, and all of which have had to have nonnative removals completed after barrier construction. Nonnative species are present in the Verde River, Eagle Creek, Aravaipa Creek, the Gila River, and the West and Middle Forks of the Gila River. Dewatering also continues at these locations, with intermittent stretches of river now occurring on the Gila and Verde Rivers, Aravaipa Creek, and Eagle Creek (USFWS 2023n).

In addition to the effects outlined in the reclassification rule in 2012, drought, and wildfire have increased since 2012, affecting many spikedace locations including the Verde River, Fossil Creek, Blue River, the Gila River, and the West Fork Gila River. The synergistic effects of drought, and wildfire are likely to result in additional threats to the species in the near future (USFWS 2023n).

Spikedace critical habitat physical and biological features are:

1. Habitat to support all egg, larval, juvenile, and adult spikedace, which includes:
    a. Perennial flows with a stream depth generally less than 1 m (3.3 ft), and with slow to swift flow velocities between 5 and 80 cm per second (1.9 and 31.5 in. per second).

b. Appropriate stream microhabitat types including glides, runs, riffles, the margins of pools and eddies, and backwater components over sand, gravel, and cobble substrates with low or moderate amounts of fine sediment and substrate embeddedness;

c. Appropriate stream habitat with a low gradient of less than approximately 1.0 percent, at elevations below 2,100 m (6,890 ft); and

d. Water temperatures in the general range of 8.0 to 28.0 °C (46.4 to 82.4 °F).

2. An abundant aquatic insect food base consisting of mayflies, true flies, black flies, caddisflies, stoneflies, and dragonflies.

3. Streams with no or no more than low levels of pollutants.

4. Perennial flows, or interrupted stream courses that are periodically dewatered but that serve as connective corridors between occupied or seasonally occupied habitat and through which the species may move when the habitat is wetted.

5. No nonnative aquatic species, or levels of nonnative aquatic species that are sufficiently low as to allow persistence of spikedace.

6. Streams with a natural, unregulated flow regime that allows for periodic flooding or, if flows are modified or regulated, a flow regime that allows for adequate river functions, such as flows capable of transporting sediments.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality may have effects on spikedace. Changes in riparian vegetation (temperature) is associated with removal of trees within the riparian area. Any tree removal within the riparian area would create localized non-measurable increase in water temperature, that would return to background levels a short distance downstream. Sediment/turbidity is associated with most activities proposed in the project. Temporary or permanent road building would create adverse conditions as long as the road is in place. Spikedace critical habitat is in the action area and consists of less than 0.01% of the total designated critical habitat. Physical and biological features for their insect food base may be impacted by sedimentation with USFS actions. Best management practices will help minimize adverse effects. Project activities **may affect, likely to adversely affect** spikedace and spikedace critical habitat.

Cape Fear shiner

The Cape Fear shiner is a small minnow native to the upper Cape Fear River Basin in central North Carolina. The Cape Fear shiner is a minnow native to five counties in the upper Cape Fear River Basin in central North Carolina. It is a narrow endemic, with a single population comprised of three subpopulations based on river subbasins – the Haw, Rocky-Deep, and Cape Fear. The species is most often found in rocky pools and runs adjacent to riffles in wide, shallow segments of rivers with gravel, cobble and/or boulder substrates with forested banks and abundant water willow (*Justicia americana*), riverweed (*Podostemum ceratophyllum*), stream mosses, and filamentous green algae. It is distinguished from all other Notropis minnows by

having a long doubly twisted alimentary tract, has 8 anal fin rays, and its lip coloration includes a small black line on its lower lip margin (aka, “lip liner”). The Cape Fear Shiner feeds mostly on detritus, algae and plant material, and small invertebrates. Spawning begins in early spring and can run through summer, and the species has an assumed lifespan in the wild of two to three years (USFWS 2022j; USFWS 2022k).

It was listed as an endangered species due to its small population size and threats to its habitat from dams and pollution. Over the past three decades, significant conservation actions, particularly successful captive propagation and two dam removals, have benefited the species and improved and connected habitat within the Rocky-Deep River subbasin. Despite these habitat improvements, Cape Fear Shiner resiliency remains very low in the Haw and Cape Fear subbasins, accounting for 62% of its range.

Cape Fear shiner critical habitat is not in the action area.

Effects: The pathway for effects to Cape Fear shiner are through pollution. The assumption made in this document is related activities would be most similar to chemical contamination. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Cape Fear shiner. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. After considering the species pathway of effects, best management practices, that would be implemented, project activities would not have measurable effects on species; therefore, the project **may affect, not likely to adversely affect** Cape Fear shiner.

Colorado pikeminnow

Colorado pikeminnow is a large, long-lived fish and the largest member of the taxonomic family Leuciscidae native to North America. It is endemic to warmwater reaches of large rivers in the Colorado River basin. Prior to the introduction of nonnative, predatory fish species, Colorado pikeminnow was the apex predator in these reaches and adults are mostly piscivorous. Colorado pikeminnow can grow to a large size, historically greater than 5 feet in length and can live over 50 years in the wild. Colorado pikeminnow make long distance migrations (409 miles (mi) [658 kilometers (km)] round trip) to spawn and return to their home range, where they inhabit deep runs, pools, and eddies. Colorado pikeminnow eggs hatch in cobble spawning bars as spring high flows decline, and larvae subsequently emerge and are carried long distances by river flows to low velocity nursery habitats downstream. In these reaches, larvae and juveniles seek low to zero velocity backwaters that provide warm temperatures for growth and abundant macroinvertebrates and small fish prey. Individual Colorado pikeminnow reach sexual maturity between 7 and 10 years of age and may spawn each year during their lifetime (USFWS 2023o).

Historically, Colorado pikeminnow occurred throughout the Colorado River basin, including the Green, Colorado, and San Juan rivers subbasins of Wyoming, Colorado, Utah, and New Mexico; downstream through the Colorado River mainstem in Arizona, Nevada, California, and Mexico; and the Gila River subbasin in Arizona and New Mexico. Colorado pikeminnow inhabit river reaches that historically experienced extremes in both flow and temperature on an annual basis, in addition to high turbidity from sediment inputs as a result of spring snow melt or flash floods. Colorado pikeminnow was extirpated from the Lower Colorado River Basin by the 1970s. The range of Colorado pikeminnow in the Upper Colorado River Basin contracted and populations declined in the Green and upper Colorado River subbasins; the species was functionally extirpated from the San Juan River subbasin in the late 1990s (USFWS 2023o).

Changes in the flow regime as a result of water development throughout the Colorado River basin led to initial declines in Colorado pikeminnow populations. In some cases, this development created barriers to movement in the form of dams and diversions, which have fragmented river reaches and limited access to historical habitats. Diversions can also entrain fish into water delivery systems and lead to direct mortality. Large dams can alter water temperatures through hypolimnetic releases, creating conditions too cold for Colorado pikeminnow growth and reproduction. Water storage and flow management often reduce spring peaks necessary to create and maintain spawning and nursery habitats needed by Colorado pikeminnow, and also reduce base flows, which can limit Colorado pikeminnow range and reduce habitat availability. Predation and competition from invasive, nonnative fishes reduce survival and recruitment of Colorado pikeminnow in all life stages. Contaminants that impact water quality can reduce reproduction and survival of individuals or lead to population reductions in the case of large, toxic spills. Lastly, extended drought reduces streamflow in many river reaches, which strains efforts to manage flows to benefit Colorado pikeminnow (USFWS 2023o). The stressors identified are reductions to natural flow regimes, water temperature depression because of hypolimnetic releases from large dams, physical barriers to movement and the resultant loss of habitat and connectivity, entrainment into water diversion facilities, nonnative fish competition and predation, contaminants, channel simplification, and long-term alteration of weather patterns (USFWS 2025p).

Colorado pikeminnow critical habitat is not in the action area.

Effects: One pathway for effects to Colorado pikeminnow is through toxic spills. The assumption made in this document is related activities would be most similar to chemical contamination. Colorado pikeminnow are also suspectable to degraded water quality, specifically higher sediment/turbidity and lower water temperatures. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination and sediment/turbidity may have effects on Colorado pikeminnow. Changes in riparian vegetation (temperature) effects of proposed activities would create insignificant increases in water temperature, therefore, will not be considered in the analysis. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur

these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Sediment/turbidity is associated with activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects for timber harvest. Temporary or permanent road building would create adverse conditions as long as the road is in place. Colorado pikeminnow reside in large rivers with flowing water, where sediment/turbidity would dissipate quickly under normal conditions. Best management practices will help minimize adverse effects from project activities and project effects would be short-term adverse effects. Therefore, the project **may affect, likely to adversely affect** Colorado pikeminnow.

Santa Ana speckled dace

Santa Ana speckled dace has recently been designated as a distinct species (USFWS 2024aa). Santa Ana speckled dace was historically found throughout river systems at the bases of the San Gabriel, San Bernardino, and San Jacinto Mountain ranges in Los Angeles, San Bernardino and Riverside Counties, California. The northern extent of this species' range was likely the northern San Gabriel Mountains in Los Angeles County and the southernmost extent south of Hemet in Riverside County. The majority of currently occupied habitat is on Federal lands, including Forest Service lands, and the species occurs in four small populations distributed across four isolated river systems. These four populations are composed of 17 extant occurrences found in a highly fragmented environment in the isolated remnants of the headwaters that resulted from historical habitat loss and habitat degradation. Currently Santa Ana speckled dace is found in the headwaters of San Jacinto, Santa Ana, San Gabriel, and Los Angeles rivers. Compared to the historical distribution, the remaining dace populations are relatively small and isolated (USFWS 2024aa).

Adult Santa Ana speckled dace inhabit a variety of stream habitat, commonly inhabiting the same stretches of water as Santa Ana sucker, with a preference for moving water. Although dace have been found in streams with a gradient as high as 10%, gradients above 5% seem to limit their movement due to cascading water from the steep terrain (USFWS 2024aa). Santa Ana speckled dace is a hardy fish where populations can naturally fluctuate through boom-and-bust cycles. To determine the current species needs, it is important to put the current distribution in the context of its historical distribution. Currently the species inhabits only small remnants of its historical range (making up four analysis units), due to habitat loss and habitat degradation. The species now exclusively occupies headwaters in relatively steep mountainous regions that are susceptible to drought, fire, and debris flows during high water years. The species needs adequate flows of low gradient (5% or less) cold, clear water with gravel or cobble substrate for reproduction. The dace also needs adequate numbers of aquatic invertebrates for food and limited or no invasive predators for higher abundances. Dace also require connectivity among remaining occurrences to maintain genetic diversity and allow for recolonization of extirpated occurrences in response to stochastic events like drought, fire, and debris flows (USFWS 2024aa).

Primary threats on Santa Ana speckled dace could include: (1) habitat loss (urban development and agricultural activities); (2) habitat degradation (recreational activities, mining, roadways,

hydrological activities and diversions); (3) habitat fragmentation; (4) increased risk of wildfire; (5) changing climate trends (for example, increased temperatures and longer, more frequent drought periods); (6) nonnative species (increased competition and predation); and (7) small population effects (USFWS 2024aa).

Effects: Santa Ana speckled dace reside in headwater streams of steep mountainous regions and could be in areas where the forest service could implement actions. Actions such as road building could create negative impacts. BMPs would help alleviate negative effects from sedimentation and potentially localized increase in water temperature, road building effects last as long as the road is in place. At this time, there are no plans to implement projects in Santa Ana speckled dace habitat as this consultation is for rule making. Overall, the effects of this action, if implemented within Santa Ana speckled dace habitat could cause adverse effects; however, the National Forest Management Act provides for some protection of the aquatic management zone and forests need to analyze for effects to species for future project planning. The effects of the action **may affect, likely to adversely affect** Santa Ana speckled dace.

Santa Ana speckled dace critical habitat has not been designated since it is still a proposed species.

Long Valley speckled dace

Long Valley speckled dace is a subspecies of the speckled dace (*Rhinichthys osculus*), a small fish that typically measures less than 8 cm but can reach 11 cm standard length (SL). It has a small downfacing mouth, a thick caudal peduncle, small scales, and a pointed snout. Long Valley speckled dace are morphologically adapted to springs and exhibit fewer scales in the lateral line, fewer lateral line pores, deeper bodies, and shorter fins than fluvial speckled dace populations (USFWS 2024ab).

Long Valley speckled dace was historically found throughout seven historical sites, including Hot Creek and various isolated springs and ponds, within the 700,000 year-old Long Valley volcanic caldera in Mono County, CA. This subspecies was likely isolated from other speckled dace populations when Speckled Dace colonized the upper Owens Valley region from the Mono Lake Basin. The formation of steep waterfalls in the Owens River Gorge separated the Owens Valley from Long Valley approximately 100,000 years ago, isolating Long Valley speckled dace from other speckled dace populations. The subspecies generally lives in isolated spring habitats and is genetically distinct from other members of the speckled dace complex (USFWS 2024ab).

Long Valley speckled dace has been extirpated from all historical collection sites except Whitmore Marsh (a 4,000–8,000 m2 (1-2 ac) spring-fed marsh) due primarily to the effects of nonnative species, water diversion, and degraded water quality from recreational activity and cattle grazing. The recent population declines observed at Whitmore Marsh are concerning because multiple threats exist on the landscape, including a recreational swimming pool, grazing, and geothermal development, which can all be characterized as posing a risk to the current viability of the species. Multiple nonnative species and a trout hatchery exist within the historical range and have potentially contributed to the extirpation of populations at historical sites (USFWS 2024ab).

Long Valley speckled dace critical habitat has not been designated since it is still a proposed species.

Effects: Long Valley speckled dace do not occur on National Forest System Lands but are downstream of the forest within a 2-mile buffer. Long Valley speckled dace threats do not coincide with project activity effects; however, there may be a few effects pathways for effects not included in the listing document. Although they are not located on Forest System Lands, if effects occurred, it would be through utilizing the road system. Best management practices will minimize effects that could occur on the road system. Best management practices and geographic distance suggest effects are discountable, thus, project activities **may affect, not likely to adversely affect** Long Valley speckled dace.

Kendall Warm Springs dace

Kendall Warm Springs dace is endemic to one stream (984 feet in length) that originates from a series of thermal springs and seeps. The stream ends in a waterfall and empties into the Green River in Sublette County, Wyoming. The dace's entire habitat occurs on property administered by the U.S. Forest Service, Bridger-Teton National Forest. The number of fish present in the population has never been accurately estimated; however, catch per unit effort data may indicate a possible decline in relative abundance over the last decade. The Kendall Warm Springs dace adults range in size from 0.9 to 2.1 inches (23 to 54 millimeters) (USFWS 2015a).

The Kendall Warm Springs dace is found in only one small thermal spring-fed stream of fast-flowing waters over cobble and gravel substrate associated with emergent aquatic vegetation. The Kendall Warm Springs dace are believed to occupy their entire historic range. The warm springs themselves remain a constant 85°F (29.4°C) year-round. The stream temperature is more variable than the warm springs and has been recorded as low as 78°F (25.6°C) in the winter at the point where it cascades over a waterfall into the Green River (USFWS 2015a; USFWS 2022l).

Primary threats at the time of listing were limited distribution, habitat manipulation, and small population size. Additional threats identified since the time of listing are potential catastrophic habitat loss due to manipulation or pollution of the aquifer that supplies the springs, degradation in habitat quality from potential oil and gas development, and potential non-native species introductions (USFWS 2015a).

Since 1975 the U.S. Forest Service has prohibited bathing, wading, and washing clothes in the Kendall Warm Springs area, but, rarely, illegal activities have been documented over the last several decades. At the time of its listing, its habitat was fragmented into two sections by a road built across the stream prior to 1934. The road culvert bisected the stream at a point approximately two-thirds of the way downstream from the stream's origin. The road culvert has since been removed and replaced with a bridge that spans the stream allowing reconnection of the habitat (USFWS 2015a; USFWS 2022l).

Kendall warm springs dace critical habitat has not been designated.

The Bridger-Teton National Forest Land and Resource Management Plan includes provisions for Kendall Warm Springs dace conservation, such as establishing the spring as a Special Interest Area and establishing protection of Kendall Warm Springs dace as a management objective (1990 land management plan, available at USFS 2025a). Due to this, there would be **no effects** to Kendall Warm Springs dace.

Loach minnow

The loach minnow is a small, secretive fish endemic to the Gila River basin of Arizona and New Mexico, USA, and Sonora, Mexico. The loach minnow is a small, slender fish less than 80 mm (3 in) in length. Loach minnow are found in small to large perennial streams and use shallow, turbulent riffles with primarily cobble substrate and swift currents. The loach minnow uses the spaces between, and in the lee (sheltered) side of, rocks for resting and spawning. It is rare or absent from habitats where fine sediments fill these interstitial spaces (77 FR 10810).

Currently, loach minnow are restricted to: portions of the Gila River and its tributaries, the West, Middle, and East Fork Gila River; the San Francisco and Tularosa rivers and their tributaries, Negrito and Whitewater Creeks (Catron County, New Mexico); the Blue River and its tributaries, Dry Blue, Campbell Blue, Pace, and Frieborn Creeks (Greenlee County, Arizona, and Catron County, New Mexico); Aravaipa Creek and its tributaries, Turkey and Deer Creeks (Graham and Pinal Counties, Arizona); Eagle Creek (Graham and Greenlee Counties, Arizona); the North Fork East Fork Black River (Apache and Greenlee Counties, Arizona); and possibly the White River and its tributaries, the East and North Fork White River (Apache, Gila, and Navajo Counties, Arizona) (77 FR 10810).

The loach minnow is a small, stream dwelling fish with an elongated body, little compressed, and flattened ventrally (USFWS 1991b). Loach minnow are associated only with flowing water in shallow, turbulent riffles where there are swift currents (77 FR 10810).

Two of the primary threats to loach minnow are nonnative fishes and loss of water due to diversions, pumping, drought, or other causes (77 FR 10810).

Loach minnow critical habitat is in the action area and consists of approximately 0.10% of the total designated critical habitat. Loach minnow critical habitat physical and biological features are (77 FR 10810):

1. Habitat to support all egg, larval, juvenile, and adult loach minnow which includes:
    a. Perennial flows with a stream depth of generally less than 1 m (3.3 ft), and with slow to swift flow velocities between 0 and 80 cm per second (0.0 and 31.5 in. per second);
    b. Appropriate microhabitat types including pools, runs, riffles, and rapids over sand, gravel, cobble, and rubble substrates with low or moderate amounts of fine sediment and substrate embeddedness;
    c. Appropriate stream habitats with a low stream gradient of less than 2.5 percent and are at elevations below 2,500 m (8,202 ft); and

d. Water temperatures in the general range of 8.0 to 25.0 °C (46.4 to 77 °F).

2. An abundant aquatic insect food base consisting of mayflies, true flies, black flies, caddisflies, stoneflies, and dragonflies.

3. Streams with no or no more than low levels of pollutants.

4. Perennial flows, or interrupted stream courses that are periodically dewatered but that serve as connective corridors between occupied or seasonally occupied habitat and through which the species may move when the habitat is wetted.

5. No nonnative aquatic species, or levels of nonnative aquatic species that are sufficiently low to allow persistence of loach minnow.

6. Streams with a natural, unregulated flow regime that allows for periodic flooding or, if flows are modified or regulated, a flow regime that allows for adequate river functions, such as flows capable of transporting sediments.

Effects: The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing water quality associated with sediment/turbidity may have effects on loach minnow but any sedimentation would be expected to be a pulse short-term effect returning to background levels quickly as sedimentation may only be an issue when monsoon rains occur. Activities that create sediment/turbidity would add to the amount of sedimentation/turbidity but the amount would be unmeasurable from background conditions during the monsoon rains. Loach minnow critical habitat consists of approximately 0.10% of the total designated critical habitat. Physical and biological features for their insect food base may be impacted by sedimentation with USFS actions but would only occur during the monsoon rains and would be indistinguishable from background conditions. Best management practices will minimize the amount of sediment/turbidity created by activities and is expected to result in discountable effects. Therefore, the project **may affect, not likely to adversely affect** loach minnow and loach minnow critical habitat

Cyprinodontidae Family

Desert pupfish

The desert pupfish is in the Cyprinodontidae family and contains several subspecies that all fall under the ESA. Minimal subspecies information is provided.

At the time of listing in 1986 (51 FR 10842), the historical range of desert pupfish included the Lower Colorado River Basin, the Gila River Basin, Laguna Salada, and the Rio Sonoyta Basin. What was desert pupfish in 1986 is now recognized as three separate species[25], but the ESA listing has not been changed to reflect the taxonomic changes (USFWS 2019m). If a taxonomic name change is made under the ESA, any sub-group split off from the listed entity (*C.*

[25] The USFWS ECOS website (https://ecos.fws.gov/ecp/species/7003) accessed 12/11/2025 (USFWS 2026a), still refers to the desert pupfish as two subspecies Desert Pupfish (*C.m. macularis*) and Quitobaquito Pupfish (*C.m. eremus*) and not three separate species as the 2019 Recovery Plan states (USFWS 2019).

macularius) also would be listed (*C. eremus*). This geographic area includes three species, *C. macularius, C. eremus, and C. arcuatus*. However, the listed entity remains *C. macularius* (USFWS 2019m). In this section all three species will be referred to as desert pupfish since that is what is listed under the ESA and information in USFWS publications and on their websites are inconsistent in the number of species/sub-species they include under the desert pupfish listing. Only two of the species/sub-species are listed under the ESA, *C. macularius* and *C. eremus*.

Desert pupfish is a small, laterally compressed fish with a smoothly rounded body shape that rarely grow larger than 75 millimeters in total length (51 FR 10842). Two subspecies of desert pupfish are recognized: Desert Pupfish (*C.m. macularis*) and Quitobaquito Pupfish (*C.m. eremus*). Desert pupfish was historically widespread, but probably not continuous within its historic range, and abundant in portions of southern Arizona and southeastern California, United States, and northern Baja California and Sonora, Mexico. Historical habitats varied in size, complexity, character and permanence, and included cienegas, springs, streams, and margins of larger lakes and rivers. Although remarkably tolerant of extreme environmental conditions, the species is threatened with extinction throughout its native range primarily because of habitat loss or modification, pollution, and introductions of exotic fishes (USFWS 1993b). Naturally occurring populations of desert pupfish are now restricted in Arizona to Quitobaquito Springs and in California to two streams tributary to, and a few shoreline pools and irrigation drains of the Salton Sea (USFWS 1993b).

Desert pupfish have an extraordinary ability to survive under conditions of high water temperature (to 45˚C), low dissolved oxygen concentration (0.1-0.4 milligrams per liter), and high salinity (salt concentrations twice (68 grams per liter) that of seawater), which exceed tolerances of virtually all other freshwater fishes. They also survive abrupt, absolute changes in both salinity (10-15 grams per liter) and temperature (22-26˚C) that are lethal to most fishes. In less harsh environments where a greater diversity of fishes was found (for example, margins of larger streams and rivers), pupfish typically occupied water shallower than that inhabited by adults of most other species (USFWS 1993b).

Under conditions of abundant food and suitable temperature (mid-to-upper 20s ˚C), desert pupfish may become sexually mature as early as six weeks of age at 1.5 centimeters total length, may breed as early as their first summer but most do not breed until their second summer when they reached a maximum of 7.5 centimeters (USFWS 1993b). Lifespan in the wild is highly variable from less than a year for some populations, two years for others, and up to three years for Quitobaquito pupfish (USFWS 1993b).

Larval desert pupfish in the laboratory begin feeding on tiny invertebrates within a few hours to a day after hatching and presumably do so in the wild as well. As they grow, wild fish become opportunistic omnivores, consuming whatever variety of algae, plants, suitably sized invertebrates, and detritus is available. Adult foods include ostracods, copepods, and other crustaceans and insects, pile worms, mollusks, and bits of aquatic macrophytes torn from available tissues. Detritus or algae are often predominant in their diets. Pupfish at Quitobaquito Spring have been reported to eat their own eggs and young. Foraging is typically a daytime activity, and fish may move in response to daily warming from shallower water during morning to feed in deeper places later in the day (USFWS 1993b).

The two main threats to the desert pupfish are loss and degradation of aquatic habitats, and continual spread and introduction of non-native aquatic species, which prey upon and compete with all pupfish species (USFWS 2019m). More specifically threats include habitat loss (dewatering of springs, some headwaters, and lower portions of major streams and marshlands), habitat modification (stream impoundment, channelization, diversion, and regulation of discharge, plus domestic livestock grazing and other watershed uses such as timber harvest, mining, and road construction), pollution, and interactions with non—native species. These threats of habitat loss, predation, and competition continue to be exacerbated by increasing human development and demand for water, as well as interactions with predicted trends for warmer, drier, and more extreme hydrological conditions associated with long-term alteration of weather patterns (USFWS 1993b, USFWS 2019m).

Desert pupfish critical habitat is not in the action area.

Effects: The pathway for effects to desert pupfish include pollution, timber harvest, and road construction. The assumption made in this document is related activities would be most similar to chemical contamination. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination, timber harvest, and road construction may have effects on desert pupfish. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Timber harvest and road construction are threats to the species. The pathways in Table 16 for these activities could cause effects on species along the different pathways listed. Temporary and permanent road building would have effects for as long as the road is in place and may reach a measurable level. Project implementation may come with short-term and long-term negative effects. After considering the species pathway of effects and best management practices, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** desert pupfish.

Poeciliidae Family

Gila topminnow

The Gila topminnow may be in the action area and is the only species in the Poeciliidae family considered in this consultation Table 15. The Gila topminnow is native to the Gila River Basin of the United States and Mexico, and the Ríos de la Concepción and Sonora of northern Mexico. It was considered one of the most common fishes in the southern part of the Colorado River basin prior to 1940. However, habitat loss and interaction with nonnative fishes, particularly western mosquitofish (*Gambusia affinis*) caused range-wide disappearances and decreases in abundance within the United States. Western mosquitofish is the main reason for decline of the Gila topminnow. The Gila topminnow is the only member of the family Poeciliidae that is native to the Gila River drainage (USFWS 1998f).

Habitat requirements of Gila topminnow are broad. They prefer shallow, warm, fairly quiet waters. However, they can become acclimated to a much wider range of conditions. Both lentic habitats and lotic habitats with moderate current are easily tolerated. Temperatures from near freezing under ice to 37˚C have been reported, with a maximum tolerance of 43˚C for brief periods. Topminnows can live in a wide range of water chemistries, with recorded values of pH from 6.6 to 8.9, dissolved oxygen readings from 2.2 to 11 milligrams/liter, and salinities from very dilute to sea water. The widespread historic distribution of Gila topminnows throughout rivers, streams, marshes, and springs of the Gila River Basin is evidence for their tolerance of these environmental extremes. One reestablished population, Mud Springs, survived for 16 years in a simple cement watering trough before being moved (USFWS 1998f).

Destruction of historically occupied habitats such as the marshes, sloughs, backwaters, and edgewaters of larger rivers and presence of nonnative fishes in such habitats that remain has undoubtedly forced Gila topminnow out of their preferred historic habitats and into the spring-heads and smaller erosive creeks we see them in today. Their tolerance of conditions in these habitats has allowed them to maintain populations with less impact from nonnative fishes (USFWS 1998f).

Gila topminnows are viviparous fish, meaning embryos grow and mature within the female and are born living. Eggs are fertilized internally through deposition of spermatophores (packets of sperm) into the female's genital pore by the male's gonopodium. Female Gila topminnow can store spermatozoa for several months and may produce up to 10 broods after being isolated from males. Sexual maturity can be attained as early as two months or as late as 11 months following birth, dependent upon the season of birth. Breeding occurs primarily during January through August, but in thermally constant springs young may be produced throughout the year. Brood size and the onset of breeding in topminnows can be influenced by several factors including food abundance, photoperiod, temperature, predation upon the population, and female size. Increased food supply and larger female size are believed to contribute to the greater fecundity seen in some topminnow populations (USFWS 1998f).

Gila topminnows are opportunistic omnivorous feeders, having a gut length 1.5 to 2 times SL of the individual. They have weakly spatulate dentition characteristic of an omnivorous diet. Primary food items include detritus, vegetation, amphipods, ostracods, and insect larvae; and rarely, other fishes (USFWS 1998f).

Historic events permanently altered much of the aquatic habitat in the arid southwest, but current and future activities also present a great risk. Land use practices such as livestock grazing, mining, timber cutting, road maintenance, and recreation pose threats through increased erosion, intensified flood events, and decreased groundwater storage to both existing populations and habitats proposed for reestablishment. In addition, continued urban and suburban development and population growth affects potential recovery of the species through increased groundwater pumping and diversions to supply the growing populations, stream and river channelization, and increased water pollution (USFWS 1998f).

Introduction of nonnative pathogens, parasites, plants, invertebrates, amphibians and fish may negatively affect the native fishes of the Southwest. Introduction of the western mosquitofish has

caused the most problems for Gila topminnow. Mosquitofish tolerate similar environmental extremes and occupy similar habitats as Gila topminnow. Mosquitofish prey directly on young topminnows and cause the death of adults due to infection following the shredding or removal of fins. Mosquitofish possess open cephalic canals that improve their ability to detect and find invertebrate and vertebrate prey, a trait lacking in topminnows. Competition for space, resulting in harassment of male and female topminnows by larger, dominant, more aggressive female mosquitofish also seemed instrumental in replacement of Gila topminnow by mosquitofish (USFWS 1998f).

Gila topminnow critical habitat has not been designated.

Effects: Timber harvest and road maintenance are threats to the species. One pathway for effects to Gila topminnow is through water pollution. This is relatively vague when trying to compare effects to project activities. The assumption made in this document is related activities would be most similar to chemical contamination. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. The pathways in Table 16 for these activities could cause effects on species along the different pathways listed. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. USFWS documents did not detail the specific pathway for effects of timber harvest on Gila topminnow so the assumption is any timber harvest would have negative effects to the species. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely** affect Gila topminnow.

Gasterosteidae Family

Unarmored threespine stickleback

The unarmored threespine stickleback could be in the action area and is the only species in the Gasterosteidae family considered in this consultation (Table 15). Unarmored threespine stickleback is a small, scaleless, freshwater fish (approximately 5 centimeters in length), inhabiting slow-moving reaches or quiet-water microhabitats in streams and rivers. Fish typically live one year, with most breeding activity occurring between February and September. Historically, unarmored threespine sticklebacks were found throughout a much larger area including the Los Angeles, San Gabriel, and Santa Ana Rivers, but were extirpated from these areas as a result of habitat loss and degradation due to urbanization. Unarmored threespine sticklebacks were also present in the Santa Maria River drainage in San Luis Obispo County, where they were lost to introgression. Unarmored threespine sticklebacks are currently restricted to a few small, local sites in the Santa Clara River watershed in Los Angeles County, the San Antonio Creek drainage in Santa Barbara County, and isolated occurrences in inland San Bernardino and San Diego counties (USFWS 2021f).

Unarmored threespine sticklebacks prefer rivers or streams with a slow, continuous flow of water with moderate algal cover, which is typical of the conditions in headwaters and tributaries that they tend to occupy. They have been observed in rivers with rapid flows (to 1 meter per second) to static pools with virtually no surface flow. In habitats where water velocity is higher, they reside along the channel edge in areas occupied by aquatic vegetation (watercress, Nasturtium spp.), under the edge of algal (*Cladophora* spp.) mats, or behind obstructions. Unarmored threespine sticklebacks are not distributed uniformly throughout the rivers and streams in which they occur. Sand or gravel substrates, with the presence of detritus and aquatic vegetation, are required for reproduction (USFWS 2021f).

Currently, unarmored threespine sticklebacks occur in the upper Santa Clara River and its tributaries in Los Angeles County, San Antonio Creek in Santa Barbara County, and a cluster of sites in San Bernardino County. Unarmored threespine stickleback was transplanted outside its native range in upper San Felipe Creek in San Diego County, and are presumed extant based on recent surveys. An additional location has been established in Fish Canyon Creek, another tributary of the Santa Clara River. Also, a potential previously unrecognized unarmored threespine stickleback site may occur in Pine Valley Creek, San Diego County (USFWS 2021f).

The unarmored threespine stickleback recovery plan (USFWS 1985) identified the following threats: stream channelization, urbanization, introduction of predators and competitors, introgression, agricultural impacts, oxygen reduction, groundwater removal, transpiration, off-road vehicles, water releases from Drinkwater Reservoir, toxic spills and discharges, addition of water, and impoundment of water. Additionally, the 5-year reviews identified ongoing effects of urbanization, eutrophication, stream channelization, addition of water (water releases), groundwater removal, and water quality were identified as the most critical threats to the species and included other not as critical threats such as non-native predators, disease, introgression, competition, and stochastic extinction (USFWS 2021f).

Unarmored threespine stickleback critical habitat was found to be not prudent in 2022, therefore, has not been designated.

Effects: Unarmored threespine stickleback threats relevant to this consultation are water quality and toxic spills. Toxic spills term is vague, and an assumption was made in this document that it is most similar to chemical contamination. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on unarmored threespine stickleback. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. The pathway for effect to the species is through the water quality pathway. Activities found in Table 16 under the impact/benefit column containing change in riparian vegetation (temperature), sediment/turbidity may have effects on the species but it is unknown if the magnitude of the effect would be measurable. Any future projects that would affect species

would still need to complete project level consultation. Due to the unknowns, the assumption in this document is that some effects on species to reach **may affect, likely to adversely affect** unarmored threespine stickleback.

Percidae Family

Darters and logperch are in the Percidae family. There are 17 species of Percidae that that were considered in this consultation (Table 15). Many were not found in the affected environment and not analyzed. These species fall into two geneses: *Etheostoma* and *Percina*. Generally, these species are small, less than 3 inches, live about 3 years, inhabit specific locations within one or two watersheds, spawn in the spring, and need free flowing, clear, cool water free of contaminants and sedimentation. The main threats to most of these species have been impoundment of their habitat that inhibits their habitat needs and/or fragments their habitat. Other threats tend to be road construction that creates a barrier to fish movement and sediment/turbidity impacts.

Etowah darter

The Etowah darter is a small-sized percid fish, adults being about 45 to 75 mm total length. The body is elongate, moderately compressed, and has a moderately pointed snout. The background body color is medium brown or gray-olive. The lower opercle (gill covers) has a have a pale bluish-green wash which is intensified in breeding males. The side is usually pigmented with 13 or 14 small dark blotches just below the lateral line. The breast in breeding males is dark greenish-blue. The spinous dorsal fin is suffused dusky black olive with a red margin. The soft dorsal fin has four bands: dusky black olive on the basal two-thirds, followed by red, white (sometimes with hint of yellow), and black bands of nearly equal width. The caudal (tail) fin is similarly pigmented except the ventral (on the abdominal side of the body) leading rays have a pale blue wash. The anal fin is suffused with greenish-blue and never has red marks, like greenbreast darters do. The pelvic fins are clear to dusky black with a pale green blue wash; pectoral fins are dusky black. All these color patterns are more vivid in breeding males. Life history is unknown. However, like most darters, the species spawns in spring, and probably lives two to three years.

The Etowah darter is endemic to the Etowah River system in north Georgia. Historically, it may have occurred further downstream in the Etowah River mainstem. However, pre-impoundment (Allatoona Lake) records are not known. The Etowah darter persists in the uppermost Etowah River mainstem and two headwater tributaries, Amicalola and Long Swamp Creeks. The species is known from a total of 19 sites in these streams. Population size at known sites is generally small. However, total population level is not known. Etowah darters inhabit medium to larger creeks and small rivers of moderate to high gradient in swift current areas in riffles with large gravel, cobble, and small boulder substrates. Threats to the species include deteriorating water and benthic habitat quality resulting from siltation and other pollutants (USFWS 2000b).

Etowah darter critical habitat has not been designated.

Effects: The pathway for effects to Etowah darter are through pollution and sediment/turbidity. The assumption made in this document is pollution is most similar to chemical contamination as

an impact. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Etowah darter. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects for timber activities and long-term negative effects for temporary and permanent road building as effects occur as long as the roads are in place. Best management practices will help minimize adverse effects but would still be measurable. After considering the species pathway of effects and best management practices that would be implemented, project activities there could be measurable effects on species; therefore, the project **may affect, likely to adversely affect** Etowah darter.

Candy darter

The candy darter is a small, freshwater fish endemic to 2nd order and larger streams and rivers within portions of the upper Kanawha River basin, which is synonymous with the Gauley and greater New River watersheds in Virginia (VA) and West Virginia (WV). The species is described as a habitat specialist, being most often associated with faster flowing stream segments with coarse bottom substrate (for example, gravel, cobble, rocks, and boulders), that provides shelter for individual darters. Candy darters are intolerant of excessive sedimentation and stream bottom embeddedness (the degree to which gravel, cobble, rocks, and boulders are surrounded by, or covered with, fine sediment particles). The available candy darter occurrence data, all of which were collected after the aquatic habitat in the region was degraded by widespread forest clearing in the late 1800s, indicate the species prefers cool or cold water temperatures, but that warm water conditions may also be tolerated. The fish are opportunistic feeders, eating mostly benthic macroinvertebrates such as mayflies and caddisflies. In streams maintaining favorable habitat conditions, through natural or managed condition, candy darters can be abundant throughout the stream continuum (USFWS 2018i).

Candy darters are sexually mature at 2 years of age and live to a maximum age of 3 years. They are classified as brood-hiding, benthic spawners. It is uncertain whether individual candy darters complete their life cycle within single riffles or riffle complexes spanning just a few hundred meters or are capable of longer, seasonally mediated movements within suitable habitat. Although data are sparse regarding the minimum habitat size and degree of genetic connectivity required for candy darter population viability, the historical distribution of the species and the fundamentals of conservation biology suggest these factors are important to the species. Primary factors needed to support the candy darter include: 1) an absence of nonnative species; 2) unembedded gravel and cobble substrates with minimal sedimentation, 3) adequate water quality (temperatures, physical and chemical parameters), 4) an abundant, diverse benthic macroinvertebrate community; and 5) sufficient water quantity and velocities. Absence or degradation of these features could limit populations of the candy darter (USFWS 2018i).

One of the primary threats that resulted in the listing of the candy darter under the Endangered Species Act is the spread of the introduced variegate darter (*Etheostoma variatum*). This species hybridizes with the candy darter (USFWS 2018). Other ongoing contributing threats include excessive sedimentation, warming water temperatures, habitat fragmentation, changes in water quality and flow, catastrophic events, and competition or predation associated with other introduced species (USFWS 2018i).

Historically, the candy darter occurred in 35 populations distributed across 7 metapopulations located in the Bluestone, Lower New River, Upper Gauley, Lower Gauley, Greenbrier, Upper New, and Middle New watersheds. However, the candy darter has been extirpated from almost half of its historical range; 17 of 35 known populations and 2 of 7 known metapopulations have been extirpated. The species is no longer known to occur in the Bluestone and Lower New River watersheds (USFWS 2018i).

Candy darter critical habitat is located within the action area. The candy darter is the only species in the Percidae family with critical habitat in the action area. There is less than one percent of critical habitat that could be affected by any future action within the affected environment (Table 15). Candy darter physical and biological features are (86 FR 17956):

1. Ratios or densities of nonnative species that allow for maintaining populations of candy darters.

2. A blend of unembedded gravel and cobble that allows for normal breeding, feeding, and sheltering behavior.

3. Adequate water quality characterized by seasonally moderated temperatures and physical and chemical parameters (e.g., pH, dissolved oxygen levels, turbidity) that support normal behavior, growth, and viability of all life stages of the candy darter.

4. An abundant, diverse benthic macroinvertebrate community (e.g., mayfly nymphs, midge larvae, caddisfly larvae) that allows for normal feeding behavior.

5. Sufficient water quantity and velocities that support normal behavior, growth, and viability of all life stages of the candy darter.

Effects: The pathways for effects to candy darter are through degraded water quality, specifically higher sediment/turbidity and higher water temperatures. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column changes in riparian vegetation (temperature) and sediment/turbidity may have effects on candy darter. Changes in riparian vegetation (temperature) is associated with timber harvest activities. Any individual tree removal within the riparian area would create localized increase in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects if activities are outside the riparian area. Temporary or permanent road building would create adverse conditions as long as the road is in place. Best management practices will help minimize adverse effects from activities, but project effects could be measurable. Critical habitat effects pathways

would probably be through sediment and potential impacts to benthic macroinvertebrates through the sediment/turbidity pathway as it would decrease the candy darters ability to find food and through nutrient pathway where there could be less nutrients reaching the stream for growth of candy darter prey. After considering the species pathway of effects and best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** candy darter and candy darter critical habitat.

Duskytail darter

Duskytail darter is a small 2 1/2-inch fish that has a straw to olivaceous colored body, the top of the head is medium to dark gray. It has 10 to 15 long dark vertical bars on the sides of its body, 38 to 48 lateral scales. and 17 to 20 dorsal spines and rays. It is difficult to determine the sex of nonbreeding individuals in the field. during the breeding season. the males become very distinctive. In breeding males, their heads became dark and swollen, and the humeral spot and lateral vertical bars intensified, the first dorsal saddle and vertical bar formed a particularly intense dark yoke, and a dark border of melanophore develops along the pectoral and anal fin margins, and the anal fin whitens proximally, and brilliant gold fleshy knobs develop on the tips of the dorsal fin spines and were conspicuous in contrast to the clear medial and dark basal portions of the fin membrane.

The duskytail is primarily an insectivore. The youngest individuals consume microcrustaceans, chironomid larvae, and sometimes large quantities of heptageniid nymphs. Larger individuals are also mainly benthic insectivores, but they generally feed on larger prey items (chironomid larvae, ephemeropteran nymphs, microcrustaceans. and trichopteran larvae) than the young. Heptageniids were the dominant food item in the three largest duskytail size classes, and large duskytail darter sometimes feed on fish eggs (USFWS 1994, 2021g).

The duskytail darter inhabits the Tennessee and Cumberland River systems. Duskytail darter utilize the edges of gently flowing shallow pools, eddy areas, and slow runs in usually clear water of large creeks and moderately large rivers (10 to 80 meters [33 to 264 feet]) (USFWS 2021g).

Duskytail darters have been and continue to be impacted by water quality deterioration. This deterioration results from siltation (contributed by coal mining and poor land use practices), other water pollution, and impoundments (USFWS 1994).

Duskytail darter critical habitat has not been designated.

Effects: The pathway for effects to duskytail darter are through water pollution, water quality, and sediment/turbidity. The assumption made in this document for water pollution even though it is related to coal mining, which is not in the proposed action, is to relate “water pollution” to activities where chemical contamination is an impact. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on duskytail darter. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for

implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects for timber harvest. Road building would create long-term effects lasting as long as the road is in place. Best management practices will help minimize adverse effects from activities. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** duskytail darter.

Rush darter

The rush darter is endemic to spring fed creeks and spring runs in North Central Alabama. The lifespan is estimated to be 2 to 3 years. The species inhabits tributaries and spring systems of the Clear Creek Drainage, Winston County, Alabama (Wildcat Branch, Doe Branch, Mill Creek); Turkey Creek drainage, Jefferson County, Alabama (Beaver Creek, Tapawingo/Penny Springs, Cunningham Creek, and several unnamed tributaries); and upper Locust Fork drainage, Etowah County, Alabama (Little Cove, Bristow, and Dry Creeks).

When listed, the rush darter was known from localized collection sites within approximately 14.5 km of stream channel. Since listing, rush darters have been collected from 12 sites within approximately 17.9 km of stream or spring runs. All collection sites are separated geographically. Rush darter are found in low densities, and individual rush darters may only be sporadically collected at specific sites within their range. No quantitative population estimates are available for the species; however, where it occurs it is usually uncommon and in low numbers (USFWS 2011c; USFWS 2019n).

All rush darter populations are located above the Fall Line (the inland boundary of the Coastal Plain physiographic region) in the Black Warrior River drainage and in portions of the southern Appalachian Plateau and Valley and Ridge physiographic provinces, Alabama. The species continues to persist in all stream systems it is known to have historically occupied (USFWS 2019). Rush darters prefer springs, spring-fed reaches of relatively low-gradient small streams, and wetland pools which are generally influenced by springs. They have been collected from root masses of vegetation in very shallow, clear, cool, and flowing water, including both small clumps and dense stands of bur reed (*Sparganium* sp.), coontail (*Ceratophyllum* sp.), watercress (*Nasturtium officinale*), and rush (*Juncus* sp.). Substrates have included silt, sand, sand and silt, woody material, muck, and gravel with sand. Water depth at all collection sites ranged from 3.0 cm to 0.5 m, with moderate water velocity in riffles and little to no flow in pools. Rush darters have not been found in higher gradient streams with bedrock substrates and sparse vegetation (USFWS 2019n).

Rush darter populations and habitats remain vulnerable to common land use and water practices despite the protections of regulatory mechanisms afforded by State and Federal laws and regulations. Sedimentation, flow alterations, and non-point source pollution continue to adversely affect the species and its habitats. Information available to us since listing does not indicate that the magnitude or imminence of this threat has changed (USFWS 2019n).

Rush darter populations remain vulnerable to localized extinctions from intentional or accidental toxic chemical spills, rapid urbanization and habitat modification, progressive degradation from runoff (non-point source pollutants), natural catastrophic changes to their habitat (for example, flood scour, drought), other stochastic disturbances, and to decreased fitness from reduced genetic diversity (USFWS 2019n).

Rush darter critical habitat is not located within the action area.

Effects: The pathway for effects to rush darter are through non-point source pollution/accidental toxic chemical spills, water quality, and habitat disturbance. The assumption made in this document for non-point source pollution and accidental toxic chemical spills is most related the pathway for chemical contamination. The pathway for fish effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on rush darter. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) are associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities could result in short-term negative effects. Habitat disturbance is related to habitat elements which correlate to drainage network, large woody debris, riparian resilience, and spawning substrate. There are several activities that have these pathways in the impact/benefit column in Table 16. Discussions of these are found in the Fish Effects of Roads and Timber section under the appropriate headers. Combined, there may be effects on the species and under certain conditions effects could be measurable. Best management practices will help minimize adverse effects from activities and project effects could be adverse. Temporary and permanent road building would create effects from sediment/turbidity lasting as long as the road is in place. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species. Therefore, the project **may affect, likely to adversely affect** rush darter.

Cherokee darter

The Cherokee darter is a small-sized percid fish, adults being about 40 to 65 millimeters total length. The body is elongate, subcylindrical (slightly compressed), with a relatively blunt snout. The side of adults is usually pigmented with eight small dark olive black blotches which develop into vertically elongate, slightly oblique bars in breeding adults, especially in males. The back usually has eight small dark saddles and intervening pale areas. The spinous dorsal (situated on the back) fin of breeding males has a dark olive black band at the base of the fin and nearly uniform russet orange red on the remaining fin, except for a blue margin (USFWS 2000b).

The Cherokee darter is endemic to the Etowah River system in north Georgia. Historically, it was thought to have occurred in most tributaries of the watershed. The Cherokee darter persists in about two dozen tributary streams of the middle and upper Etowah River. Present populations are isolated by Allatoona Reservoir and stretches of degraded habitat in tributary streams (USFWS 2000b).

Cherokee darter inhabit small to medium size creeks of moderate gradient in low current areas with large gravel, cobble, and small boulder substrates. Life history is unknown. However, like most darters, the species spawns in spring, and probably lives two to three years. Threats include impoundments and deteriorating water and benthic habitat quality resulting from siltation and other pollutants (USFWS 2000b).

Cherokee darter critical habitat has not been designated.

Effects: The pathways for effects to Cherokee darter are through pollution, water quality impacts, and siltation. The assumption made in this document for non-point source pollution and accidental toxic chemical spills is most related to the pathway for chemical contamination. The pathways for fish effects are listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Cherokee darter. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Water quality is associated with Changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. Sediment/turbidity created by project activities would have short-term negative effects. Combined, there may be effects on the species and under certain conditions could be measurable. Best management practices will help minimize adverse effects from activities but may not be eliminated. Temporary or permanent road building would create adverse effects lasting as long as the road is in place. After considering the species pathway of effects, best management practices that would be implemented, project activities could have

measurable effects on species. Therefore, the project **may affect, likely to adversely affect** Cherokee darter.

Kentucky arrow darter

The Kentucky arrow darter is a small, compressed fish, with a background color of straw yellow to pale green and a body covered by a variety of stripes and blotches. Breeding males exhibit the most vibrant coloration, including scattered scarlet spots and scarlet to orange vertical bars. Kentucky arrow darters can reach 2 inches in length by the end of the first year and reach a maximum length of 4.7 inches. The species' lifespan is approximately four years. The primary threat to this darter species is habitat modification or destruction from surface coal mining, logging, oil/gas development, land development, agriculture, and inadequate sewage treatment, coupled with ongoing threats associated with inadequate regulatory mechanisms. The species' small, isolated populations also make it vulnerable to natural and catastrophic events related to human activity (for example, drought, resource extraction, pollution spills, etc.) (USFWS 2017e).

Kentucky arrow darters typically inhabit pools or transitional areas between riffles and pools (glides and runs) in moderate- to high-gradient, first- to third-order streams with rocky substrates. The species is most often observed near some type of cover (for example, boulders, bedrock ledges, large cobble, or woody debris) in depths ranging from 10 to 45 centimeters and in streams ranging from 1.5 to 20 meters wide. Kentucky arrow darters typically occupy streams with watersheds of 25.9 square kilometers or less, and many of these habitats, especially in first-order reaches, can be intermittent in nature. During the driest periods, the species appears to survive by retreating into shaded, isolated pools or by dispersing into larger tributaries (USFWS 2017e; USFWS 2022m).

Spawning occurs in riffle habitats from late March to May, with peak activity occurring when water temperatures reach 13°C. Young Kentucky arrow darters can reach 2 inches in length by the end of the first year, when these individuals are generally sexually mature and participate in spawning with older age classes. Juvenile Kentucky arrow darters can be found throughout the channel but are often observed in shallow water along stream margins near root mats, rock ledges, or some other cover. Kentucky arrow darters feed primarily on mayflies (Order Ephemeroptera) and other aquatic insects, but large individuals 2.8 inches often feed on small crayfishes (USFWS 2017e).

The Kentucky arrow darter occurred historically in at least 74 streams in the upper Kentucky River basin of eastern Kentucky. Its distribution spanned portions of 6 smaller sub-basins or watersheds (North Fork Kentucky River, Middle Fork Kentucky River, South Fork Kentucky River, Silver Creek, Sturgeon Creek, and Red River) in 10 Kentucky counties (Breathitt, Clay, Harlan, Jackson, Knott, Lee, Leslie, Owsley, Perry, and Wolfe) (USFWS 2017e).

The primary threat to the Kentucky arrow darter is habitat modification or destruction from resource extraction (surface coal mining, logging, oil/gas development), land development, agriculture, and inadequate sewage treatment, coupled with ongoing threats associated with inadequate regulatory mechanisms. The species' small, isolated populations also make it vulnerable to natural and human-induced catastrophic events (for example, droughts, resource

extraction, pollution spills, etc.). The Kentucky arrow darter's habitat and range have been destroyed, modified, and curtailed due to a variety of anthropogenic activities in the upper Kentucky River drainage. Resource extraction (for example, coal mining, logging, oil/gas well development), land development, agricultural activities, and inadequate sewage treatment have all contributed to the degradation of streams within the range of the species. Another threat to the Kentucky arrow darter is water quality degradation caused by a variety of nonpoint-source pollutants (contaminants from many diffuse and unquantifiable sources), with coal mining being the main contributor to water quality concerns (USFWS 2017e).

The isolated nature of Kentucky arrow darter populations restricts the natural exchange of genetic material between populations and makes natural repopulation following localized extirpations of the species unlikely without human intervention. Populations can be further isolated by anthropogenic barriers, such as dams, perched culverts, and fords, which can limit natural dispersal and restrict or eliminate connectivity among populations. Such dispersal barriers can prevent reestablishment of Kentucky arrow populations in reaches where they suffer localized extinctions due to natural or human-caused events. The isolated nature and small size of many populations also likely make them vulnerable to extirpation from intentional or accidental toxic chemical spills, habitat modification, progressive degradation from runoff (nonpoint-source pollutants), natural catastrophic changes to their habitat (for example, flood scour, drought), and other stochastic disturbances (USFWS 2017e).

Kentucky arrow darter critical habitat is not located within the action area.

Effects: The pathways for effects to Kentucky arrow darter are through toxic chemical spills, water quality, and sediment/turbidity. The assumption made in this document for toxic chemical spills is to relate chemical contamination and will be analyzed in this report as chemical contamination. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Kentucky arrow darter. Chemical contamination from project activity implementation is included in many activities due to the utilization of petroleum products by the equipment needed for implementing projects. As the analysis under the chemical contamination section states, motorized hand tools have small fuel capacities and if spills occur these are not expected to reach the streams. Best management practices will limit fueling near riparian areas, and experience over the years of implementing projects has shown large spills of petroleum products have not occurred. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) are associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would result in short-term negative effects. Best management practices will help minimize adverse effects from activities and project effects. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** Kentucky arrow darter.

Trispot darter

The trispot darter is a small, freshwater fish that occurs in the Coosa River system in the Ridge and Valley ecoregion of Alabama, Georgia, and Tennessee. It is a migratory species that uses distinct breeding and nonbreeding habitats. From approximately April to October, the species occupies its nonbreeding habitat, which consists of small to medium margins of rivers and lower reaches of tributaries with slower velocities. It is associated with detritus, logs, and stands of water willow, and with a substrate that consists of small cobbles, pebbles, gravel, and often a fine layer of silt. During low flow periods, the darters move away from the peripheral zones and toward the main channel; edges of water willow beds, riffles, and pools; and mouths of tributaries. Migration into spawning areas begins in approximately late November or early December, with fish moving from the main channels into tributaries and eventually reaching adjacent seepage areas where they will congregate and remain for the duration of spawning, until approximately late April. Breeding sites are intermittent seepage areas and ditches with little to no flow; shallow depths (12 inches (30 centimeters) or less); moderate leaf litter covering mixed cobble, gravel, sand, and clay; a deep layer of soft silt over clay; and emergent vegetation. Additionally, breeding sites possess channels that maintain base flow throughout the winter and early spring. Trispot darters predominantly feed on mayfly nymphs and midge larvae and pupae (85 FR 61619).

All known records of the trispot darter occur above the fall line in the Ridge and Valley ecoregion of Alabama, Georgia, and Tennessee. Historically, this species occurred throughout the middle to upper Coosa River Basin with records from the mainstem Coosa, Conasauga, and Coosawattee rivers, their tributaries, and tributaries to the Oostanaula River (USFWS 2023p). The long-term viability of the trispot darter is affected by habitat degradation and hydrologic alteration. Specifically, incompatible land-use practices and the associated impacts to habitat connectivity, water quantity, water quality, and linear distribution of occupied habitat influence the darter's viability. To maintain stable and increasing populations of tripsot darter, connectivity of breeding and non-breeding habitat, channel integrity, adequate water quantity and water quality are required (USFWS 2023p).

Trispot darter critical habitat is not located within the action area.

Effects: The pathway for effects on tripsot darter is through water quality. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would result in short-term adverse effects. Best management practices will help minimize adverse effects from activities and project effects could be measurable. After considering the species pathway of effects, best management practices that would be implemented, project activities would have measurable effects on species; therefore, the project **may affect, likely to adversely affect** tripsot darter.

Amber darter

Amber darter occurs in a 33.5- mile (53.9 km) reach of the Conasauga River, Georgia and Tennessee, and a 28-mile (45.1 km) reach of the Etowah River, Georgia; these two populations are isolated from each other by Lake Allatoona on the Etowah River (USFWS 2019o). In both systems amber darter have been extirpated or greatly reduced in abundance in downstream reaches within the historic range. Amber darters generally are found in the rivers' mainstems, preferring shoals with a moveable gravel/small cobble substrate and moderate to swift currents. The presence of clean moveable gravel and high water quality in shoal habitat appears to be highly important in determining distribution. Amber darters forage primarily on snails, limpets, and aquatic insects (USFWS 2019o).

Amber darter current stressors are (1) habitat/water quality degradation associated with agriculture and urbanization that increases levels of sediment, nutrients, estrogens, and herbicides in the river, (2) loss of riparian buffers, (3) decline in Podostemum, an aquatic plant , (4) natural demographic, environmental, and genetic stochastic events that affect small populations with limited geographic range, and (5) climate change (USFWS 2019o).

Long-term alteration of weather patterns is expected to bring more extreme weather events, such as flooding and drought. Hydrologic flows impact the availability and connectivity of shoal habitat, as well as spawning and recruitment behaviors of amber darters. For example, spring floods may be a factor limiting recruitment of amber darters, who benefit from windows of low spring flow for spawning and recruitment success. High flows may damage eggs, wash larvae from nursery areas, prevent juveniles from migrating upstream to suitable shoal habitat, and increase turbidity and sedimentation that degrades habitat. Successive years of high spring flows may limit or eliminate juvenile recruitment at a given site with implications for the population (USFWS 2019o).

Amber darter critical habitat is not located within the action area.

Effects: The pathways for effects to amber darter are through water quality and nutrients; however, the nutrient pathway is through fertilizer and human/animal waste. The proposed actions do not include activities that will increase the nutrient pathway for amber darter and will not be discussed in this analysis. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on amber darter. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) are associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would have short-term negative effects. Best management practices will help minimize adverse effects from activities and project effects could be adverse. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** amber darter.

Goldline darter

The goldline darter is a slender, medium-sized fish, about 75 mm (3 in) long with brownish-red and amber dorsolateral (on the upper side) stripes. It differs from other members of the subgenus Hadropterus in the color pattern of its back, which is pale to dusky (USFWS 2000b).

Goldline darter is known from 43 km of the Cahaba River, historically 79 km, 11 km of the Little Cahaba River, and from Schultz Creek, Cahaba River drainage, Alabama. The species also continues to be found in the Coosawattee River and its tributary Talking Rock Creek; Ellijay River, Mountaintown and Boardtown Creeks; and the Cartecay River, Georgia. Populations appear to be small and localized (USFWS 2000b).

The goldline darter prefers a moderate to swift current and water depths greater than 0.5 meter. It is found over sand or gravel substrate interspersed among cobble and small boulders. Little is known about goldline darter life history (USFWS 2000b). The primary threat to goldline darter is water quality degradation, particularly sedimentation, which has reduced the goldline darters range within the Cahaba River System (USFWS 2000b).

Goldline darter critical habitat has not been designated.

Effects: The pathway for effects to goldline darter are through water quality. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on goldline darter. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects. Best management practices will help minimize adverse effects from activities but may not eliminate them. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** goldline darter.

Pearl darter

The pearl darter is a small bottom dwelling freshwater fish that reaches a maximum standard length of 2.5 inches. The species is short-lived, with a life span of one to three years. The pearl darter historically inhabited the bottom of stream and river channels of the Pearl River drainage and Pascagoula River drainage. Microhabitat features needed to complete the life cycle of the species consist of substrate mixtures of sand, silt, loose clay, gravel, fine particulate organic matter, and coarse particulate organic matter at the steep ends of sandbars and inside river bends where vegetative detritus (for example, decomposing wood and leaves) is deposited. Likely spawning habitats have been described as shallow, swift moving riffles with firm, gravel substrates. Spawning occurs in March to June on gravel riffles in shallow, swift flowing water (USFWS 2023q).

Reduction of the pearl darter to a single surviving population, as well as its rarity, leaves it vulnerable to persistent threats within the Pascagoula River drainage, including geomorphic instability, excessive sedimentation, and water quality degradation from point and non-point sources. It is also vulnerable to random events such as spills and weather events (for example, drought, floods). Pearl darter are not found in impounded waters and are intolerant of lentic habitats that may be formed by gravel mining or other landscape-altering practices. Increased temperatures and decreased precipitation is expected to adversely affect the species due to alteration of hydrologic cycles of headwater reaches, but the extent or magnitude of this threat has not been quantified (USFWS 2023q). Sedimentation and degraded water quality are thought to be major factors in the decline, and ultimately, the extirpation of the pearl darter from the Pearl River drainage (USFWS 2023q).

Peral darter critical habitat is not located within the action area.

Effects: The pathways for effects to Pearl darter are through water quality. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Pearl darter. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Any removal of riparian trees would create localized increases in water temperature. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would be short-term negative effects could be measurable. Best management practices will help minimize adverse effects from activities and project effects but may not eliminate adverse effects. After considering the species pathway of effects, best management practices, and conservation measures that would be implemented, project activities would not have measurable effects on species; therefore, the project **may affect, likely to adversely affect** Pearl darter.

Conasauga logperch

The Conasauga logperch is a larger darter, sometimes exceeding 6 inches in length, and is characterized by having many "tiger-like" vertical dark stripes over a yellow background (50 FR 31597). It most often is encountered in deep gravel runs or pools with small stones and sandy bottoms. Even within suitable habitat, this species is rare, typically observed at low densities of scattered individuals or pairs. Captive mortality in Conasauga logperch suggests maximum lifespan is at least five years. Logperch feed on aquatic insect larvae using its fleshy pig-like snout to flip stones to expose prey. Spawning behavior and details of spawning season conditions are unknown (USFWS 2019). The species occurs only in a 34-mile reach of the Conasauga River mainstem, from near the confluence of the Conasauga and Jacks Rivers, Polk County, Tennessee, downstream to Mitchell Bridge, Murray County, Georgia. No other population, current or historic, is known (USFWS 2019p).

The recovery team determined it was unlikely the Conasauga logperch would ever be removed from the list of endangered and threatened species due to the single population's vulnerability to extinction. Primary threats, at the time of listing, included siltation, proposed reservoir construction on the river's mainstem, catastrophic events, increased silviculture, road and bridge

construction, stream channel modification, and land use changes. Research over the past 30+ years has identified current stressors likely are (1) habitat/water quality degradation associated with agriculture and urbanization that increases levels of sediment, nutrients, estrogens, and herbicides in the river, (2) loss of riparian buffers, (3) decline in *Podostemum*, (4) natural demographic, environmental, and genetic stochastic events that affect small populations with limited geographic range, and (5) climate change (USFWS 2019p).

Conasauga logperch critical habitat is not located within the action area.

Effects: The pathways for effects to Conasauga logperch are through water quality. The pathway for fish and habitat effects listed in Table 17 are crosswalked to the impact/benefit column in Table 16. Activities found in Table 16 under the impact/benefit column containing chemical contamination may have effects on Conasauga logperch. Water quality is associated with changes in riparian vegetation (temperature) and sediment/turbidity (Table 17). Changes in riparian vegetation (temperature) is associated with almost all activities. Sediment/turbidity is associated with most activities proposed in the project. The amount of sediment/turbidity that would accompany proposed project activities would result in short-term negative effects and could be measurable. Best management practices will help minimize adverse effects from activities and project effects but may not eliminate adverse effects. After considering the species pathway of effects, best management practices that would be implemented, project activities could have measurable effects on species; therefore, the project **may affect, likely to adversely affect** Conasauga logperch.

Salmonidae Family

Five trout and two charr in the Salmonidae family may be within the action area (Table 15). Members of the Salmonidae family typically inhabit cool, clean, well-oxygenated freshwater streams and lakes for spawning and juvenile rearing, with many species migrating to the ocean (anadromous) as adults. They require complex, diverse habitats, including gravel beds for spawning, and quiet, deeper, or shaded areas for refuge. Since all members of the Salmonidae family have similar effects pathways, effects determinations will be combined below in the Salmonidae Effects section.

Gila trout

Gila trout is in the Pacific trout clade (a common ancestor and all lineal descendants) along with rainbow trout (*O. mykiss*) and cutthroat trout (*O. clarkii*). Gila trout and rainbow trout are the most closely related of the three species (USFWS 2022n). Gila trout are relatively small salmonids that reach sexual maturity at approximately 6 inches long which would be anywhere from age 2 to age 4 for both females and males (USFWS 2022n). Spawning typically occurs in April when water temperatures reach 8˚C. Redds are constructed in three to four centimeters water depth in fine gravel and coarse sand substrate (USFWS 2022n). Most adult Gila trout live to about age 5 (USFWS 2022n). Gila trout are generally insectivorous feeding on adult Diptera (true flies), Trichoptera (caddisfly) larvae, Ephemeroptera (mayfly) nymphs, and aquatic Coleoptera (beetles) but occasionally eat other fishes (USFWS 2022n).

Current distributions of Gila trout occur in the Gila River drainage basin, while, historical distribution likely included montane, cold-water stream habitats in Sierra, Grant, and Catron counties in New Mexico and Greenlee, Apache, Graham, Gila, Maricopa, and Yavapai counties in Arizona (USFWS 2022n). Habitat of Gila trout currently consists of montane streams ranging from approximately 1,660 m to over 2,800 m elevation (USFWS 2022n). Water-quality impairment in cold-water streams within the historical range of Gila trout falls into two main categories: chemical or physical impairment and water temperature impairment.

Gila trout threats were determined to be habitat degradation from livestock grazing, timber harvest, and wildfire; sport fishing; predation from brown trout (*Salmo trutta*) and disease; inadequate regulatory mechanisms to protect and enhance Gila trout populations and their habitat; and hybridization and competition with nonnative trout, drought, wildfire, and floods.

Gila trout critical habitat has not been designated.

Paiute cutthroat trout

Paiute cutthroat trout is a distinctive member of the cutthroat trout complex, distinguishable from other cutthroat trout by body coloration and the absence, or near absence, of body spots. It is native to Silver King Creek in the East Fork Carson River drainage on the Humboldt-Toiyabe National Forest, Alpine County, California that is approximately 14.7 kilometers (9.1 miles) of stream habitat within a single watershed (USFWS 2004). The life history and habitat requirements for Paiute cutthroat trout are similar to those reported for other western stream-dwelling salmonids. All life stages require cool, well-oxygenated waters. Adult fish prefer stream pool habitat in low gradient meadows with undercut or overhanging banks and abundant riparian vegetation. To spawn successfully, they must have access to flowing waters with clean gravel substrates.

Paiute cutthroat trout reach sexual maturity at the age of 2 years. Paiute cutthroat trout are opportunistic feeders, utilizing whatever aquatic and terrestrial invertebrates occur in the drift. In stream environments Paiute cutthroat trout seldom reach sizes more than 250 millimeters (10 inches) total length. Disease is apparently a significant cause of adult mortality, particularly in the post-spawning period. Few Paiute cutthroat trout apparently live beyond the age of 3 years in a wild stream environment (USFWS 2004).

The Paiute cutthroat trout faces several threats to its existence because of its limited distribution and its susceptibility to displacement by other salmonids. Several events have occurred in the past to imperil its existence, including:

1. The early introduction of rainbow trout and Lahontan cutthroat trout into the Silver King Creek drainage and subsequent introgression,
2. the introduction of beavers into the Silver King Creek drainage,
3. the occurrence of a flood in Silver King Creek that may have eliminated a natural barrier and allowed nonnative salmonids to enter the drainage,

4. the degradation of habitat caused by livestock grazing and off-highway vehicle use in the North Fork of Cottonwood Creek, and

5. excessive angling (USFWS 2004).

Paiute cutthroat trout critical habitat has not been designated for the Paiute cutthroat trout.

Lahontan cutthroat trout

Lahontan cutthroat trout is an inland subspecies of cutthroat trout endemic to the Lahontan basin of northern Nevada, eastern California, and southern Oregon. Lahontan cutthroat trout typically exhibit spots on the top and sides of the head extending to the tip of the snout. Lahontan cutthroat trout inhabit lakes and streams and require spawning and nursery habitat characterized by cool water, pools near cover and velocity breaks, well vegetated and stable stream banks, and relatively silt free rocky substrate in riffle-run areas. Lahontan cutthroat trout mature between 2 and 4 years. Lahontan cutthroat trout can be opportunistic feeders, with diets consisting of drift organisms, typically terrestrial and aquatic insects, zooplankton and fish, but varies depending on which habitat types they reside (USFWS 1995e).

Principal threats to Lahontan cutthroat trout include:

1. habitat loss associated with livestock grazing practices, urban and mining development;

2. water diversions;

3. poor water quality;

4. hybridization with non-native trout; and

5. competition with introduced species of fish (USFWS 1995e).

Lahontan cutthroat trout critical habitat has not been designated.

Little Kern golden trout

The **Little Kern golden trout** is a subspecies of rainbow trout (*Oncorhynchus mykiss*) in the family Salmonidae and endemic only to the Little Kern River drainage in Tulare County, California. The Little Kern River drainage occurs primarily within the Golden Trout Wilderness on the Sequoia National Forest. Smaller areas of the drainage occur outside of designated wilderness in either Sequoia National Park or Sequoia National Forest (USFWS 2023r). The Little Kern golden trout is known only from the Little Kern River in Tulare County, Calif. In the Little Kern River watershed, the water quality of the streams inhabited by this fish is generally good. At the time of listing, the following threats to the subspecies were identified: potential for uncontrolled use of off-road vehicles, improper road construction, careless logging activities, pollution from mining operations or overgrazing, and hybridization with non-native rainbow trout (43 FR 15428). Current threats to the subspecies include: grazing and habitat loss, hybridization and low genetic diversity, limited range and small population size, and long-term alteration of weather patterns. There is no information indicating mining and logging currently threaten the subspecies (USFWS 2023r).

Rainbow trout were introduced into the Little Kern River System in the 1930's. Subsequent hybridization between the Little Kern golden trout and the introduced rainbow trout resulted in a reduction in the number of pure populations of Little Kern golden trout. At present, pure populations of Little Kem golden trout exist only in headwater streams which were not stocked with rainbow trout or have barrier falls which have prevented upstream migration of rainbow trout.

Little Kern golden trout critical habitat include streams in the Little Kem River watershed that are sufficient for individual and population growth and dispersal of the Little Kem golden trout. The pools in stream areas within the designated area are proper habitat for aquatic insects which provide food for the trout. The cobbles and larger rocks provide cover for both juvenile and adult fish. The gravel bottom in pool areas of the Critical Habitat streams provides proper substrate for redds (43 FR 15428).

Dolly Varden

Dolly Varden, a char species, are ESA proposed due to similarity in appearance to bull trout. Prior to 1978, bull trout and Dolly Varden were considered a single species but it had been demonstrated these are two different species. In the coterminous U.S., Dolly Varden only occurs in western Washington, overlapping somewhat with the current bull trout distribution (USFWS 2024ac). Dolly Varden will not be analyzed in this document since they overlap in geography with bull trout and are ESA proposed because of their similarity in appearance to bull trout, an ESA listed species. This project will have the same effects for Dolly Varden as they would for bull trout (Table 15).

Bull trout

Bull trout is a char species and has the largest range of any other salmonid considered in this consultation. Bull trout are distributed from coastal Alaska and western Canada, south to the Pacific Northwest, and east to portions of the middle and northern Rocky Mountains. Within the coterminous United States portion of the range, bull trout are currently known to occur in the Columbia and Snake River basins, the Puget Sound and Olympic Peninsula coastal basins, and the Saint Mary and Upper Klamath River basins. Bull trout are grouped into six recovery units that are biologically important to the rangewide conservation of bull trout and provides recovery criteria for each unit. These six recovery units (Coastal Recovery Unit, Klamath Recovery Unit, Mid-Columbia Recovery Unit, Upper Snake Recovery Unit, Columbia Headwaters Recovery Unit, Saint Mary Recovery Unit; Figure 1) encompass 110 core areas that were occupied at the time of listing, 7 historical core areas, and 1 Research Needs Area (RNA), and comprise over 600 local populations (USFWS 2015b, USFWS 2024ac). When future projects occur in the action area and the action area is within a bull trout core area, analysis at the core area scale for bull trout will be completed at that time. Core area information and maps can be found in the 2015 recovery plan for the coterminous United States population of bull trout (USFWS 2015b) and in the species status assessment for the coterminous distinct population segment of bull trout (USFWS 2024ac).

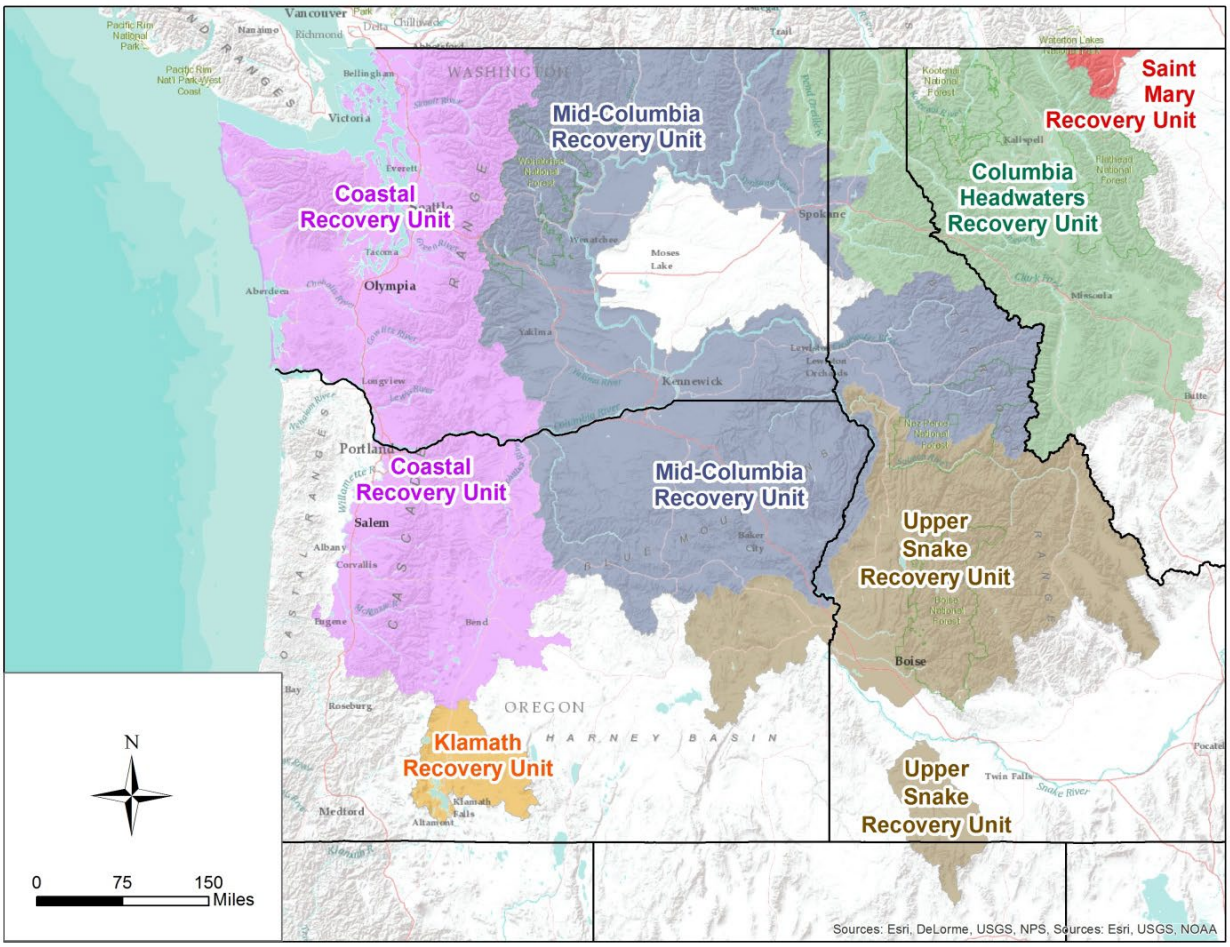


Figure 1. Locations of the six bull trout recovery units in the coterminous United States.

Historical habitat loss and fragmentation, interaction with nonnative species, fish passage issues, and changing climate are the most significant threats affecting bull trout. The magnitude of those threats and their potential synergistic effects vary greatly across recovery units, core areas and among local populations. Historical habitat loss and fragmentation, interaction with nonnative species, and fish passage issues are widely regarded as the most significant primary threat factors affecting bull trout (USFWS 2015b, USFWS 2024ac).

Bull trout have both resident and migratory life history strategies. The size and age of bull trout at maturity depends upon habitat capacity and subsequent life history strategy. Resident fish tend to be smaller at maturity than migratory fish and produce fewer eggs. Bull trout normally reach sexual maturity in 4 to 7 years. They frequently live for 10 years, and occasionally for 20 years or more (USFWS 2015b, USFWS 2024ac).

Of native salmonids in the Pacific Northwest of the United States, bull trout have the most specific habitat requirements, often referred to as “the four Cs”: cold, clean, complex, and connected habitat. Habitat components that influence bull trout distribution and abundance include water temperature, in-stream and stream-bank cover, channel form and stability, valley

form, spawning and rearing substrate, and unobstructed migratory corridors. Juveniles remain in the substrate after hatching; the time from egg deposition to emergence of fry can exceed 200 days. During the relatively long incubation period in the gravel, bull trout eggs are especially vulnerable to fine sediments, streambed scour, and water quality degradation (USFWS 2015b, USFWS 2024ac).

Bull trout need the following core area habitat factors in sufficient condition to successfully complete all stages of their life cycle (USFWS 2015b, USFWS 2024ac):

1. Water Quality – enough clean and cold water to support breeding and feeding, and provide shelter for each life stage to survive.
2. FMO Habitat Access – connectivity for subadults and adults between spawning and rearing habitats and feeding, migrating and overwintering habitats that provide diverse resources.
3. Fish Community Quality – habitat for all life stages with limited interaction with deleterious nonnative species.
4. Instream Quality – in-water habitat with complex substrates, stream morphology, and structure to meet the needs of each life stage.
5. Riparian Quality – intact habitat for all bull trout life stages within the riparian corridor, which provides shade, channel complexities, insect communities, and allochthonous inputs leading to higher quality instream habitat and water quality.
6. Spawning and Rearing Habitat Quantity –sufficient amount and quality of habitat to allow for individual adults to spawn and juveniles to rear.

Most threats described in the recovery plans fall into the category of destruction, modification, or curtailment of habitat. Many of these impacts, such as dewatering, sedimentation, thermal modification, and water quality degradation, are a consequence of specific land and water management activities. There have been increased efforts to mitigate impacts, especially on federal lands where there is a greater conservation emphasis, such as headwater areas where bull trout spawn and rear. For example, improved land management practices, declining timber harvest, and cessation of road building have reduced impacts in riparian habitats. However, legacy effects of past land use continue to degrade bull trout habitat across the range where habitat restoration has not yet occurred, or ecological benefits have not been realized. Loss of connectivity from dams and water diversions continue to isolate and fragment populations, reducing the species' viability today. In the absence of substantially increased conservation effort, these threats are predicted to continue reducing bull trout viability across the range into the future (USFWS 2015b, USFWS 2024ac). Within affected environment, timber harvest and permanent road building in bull trout habitat could create localized threats to the species through water quality impacts and habitat access if roads are built over streams.

Nonnative fish were identified as a significant threat in the original listing of bull trout, and the threat has increased since that time. Today, nonnative fishes are a primary threat in many core

areas across the six recovery units. Nonnative fish of primary concern include both lake trout (*Salvelinus namaycush*) and brook trout (*Salvelinus fontinalis*). Lake trout, a congeneric species whose niche closely overlaps with adfluvial bull trout, can outcompete and prey upon bull trout in lake environments where they co-occur. Brook trout is also a congeneric species that competes with, and can hybridize with, bull trout. Negative effects of brook trout on bull trout appear to vary substantially between watersheds, being relatively severe in small, low-gradient streams with resident or small migratory populations of bull trout. Bull trout populations containing robust numbers of large individuals with migratory life histories can often successfully coexist with brook trout, although the long-term impacts from hybridization are unknown. The presence of either brook trout or lake trout in bull trout waters results in diminished resiliency and ultimately reduced viability. Other piscivorous species that can prey on bull trout can also be found in many bull trout core areas (USFWS 2015b, USFWS 2024ac).

Bull trout require cold water habitat and are vulnerable to the effects of warming climates, changing patterns of precipitation, and dynamic instream hydrologic regimes. Changing climate can also exacerbate other threats. Drought, increased wildfire intensity and frequency, reduction of snowpack, and general warming of waterways all have negative impacts on bull trout. In general, all climate scenarios project some degree of warming, although severity and the effects on amount and seasonality of precipitation are scenario-dependent and the impacts on bull trout populations are expected to vary geographically. Bull trout may not be able to adapt quickly enough to avoid population declines in the future with continued alteration of weather patterns (USFWS 2015b, USFWS 2024ac).

Bull trout critical habitat physical and biological features are essential for the conservation of bull trout critical habitat and may require special management considerations or protection (75 FR 63898):

1. Springs, seeps, groundwater sources, and subsurface water connectivity (hyporheic flows) to contribute to water quality and quantity and provide thermal refugia.
2. Migration habitats with minimal physical, biological, or water quality impediments between spawning, rearing, overwintering, and freshwater and marine foraging habitats, including but not limited to permanent, partial, intermittent, or seasonal barriers.
3. An abundant food base, including terrestrial organisms of riparian origin, aquatic macroinvertebrates, and forage fish.
4. Complex river, stream, lake, reservoir, and marine shoreline aquatic environments, and processes that establish and maintain these aquatic environments, with features such as large wood, side channels, pools, undercut banks and unembedded substrates, to provide a variety of depths, gradients, velocities, and structure.
5. Water temperatures ranging from 2 to 15 °C (36 to 59 °F), with adequate thermal refugia available for temperatures that exceed the upper end of this range. Specific temperatures within this range will depend on bull trout life-history stage and form; geography;

elevation; diurnal and seasonal variation; shading, such as that provided by riparian habitat; streamflow; and local groundwater influence.

6. In spawning and rearing areas, substrate of sufficient amount, size, and composition to ensure success of egg and embryo overwinter survival, fry emergence, and young-of-the-year and juvenile survival. A minimal amount of fine sediment, generally ranging in size from silt to coarse sand, embedded in larger substrates, is characteristic of these conditions. The size and amounts of fine sediment suitable to bull trout will likely vary from system to system.

7. A natural hydrograph, including peak, high, low, and base flows within historic and seasonal ranges or, if flows are controlled, minimal flow departure from a natural hydrograph.

8. Sufficient water quality and quantity such that normal reproduction, growth, and survival are not inhibited.

9. Sufficiently low levels of occurrence of nonnative predatory (e.g., lake trout, walleye, northern pike, smallmouth bass); interbreeding (e.g., brook trout); or competing (e.g., brown trout) species that, if present, are adequately temporally and spatially isolated from bull trout.

Salmonidae Effects

Effects of future projects on members of the Salmonidae family could be varied but would most likely be due to sedimentation, potential increase in water temperatures, and habitat fragmentation when projects are implemented. Impacts would not be the same because of limited distribution of the Paiute cutthroat trout, Little Kern golden trout, Lahontan cutthroat trout, and Gila trout, while bull trout is distributed widely across the affected environment. Implementing Forest Service best management practices (2012) will help alleviate negative effects of any future project. Many National Forests are actively involved in protecting species and protections are expected to continue. Low level effects from sedimentation, increase in water temperatures, and habitat fragmentation could occur during future project implementation, the magnitude of which is unknown, but expected to be low. Any future projects that would affect species would still need to complete project level consultation. Due to the unknowns, we might expect some effects on species that are susceptible to reach **may affect, likely to adversely affect** individual Paiute cutthroat trout, Little Kern golden trout, Lahontan cutthroat trout, Gila trout, bull trout, and Dolly Varden. Individual species effects are listed in Table 15.

Little Kern golden trout and bull trout are the only species with designated critical habitat in the affected environment. The amount of Little Kern golden trout critical habitat in the action area is very minimal, less than 0.0001 percent. Effects to Little Kern golden trout critical habitat are through the same pathways as for the species, so effects analyses will not be duplicated here. Best management practices would minimize effects to Little Kern golden trout critical habitat but may not eliminate adverse effects. The amount of bull trout critical habitat in the action area is minimal, less than 0.10 percent. Land Management Plans within the range of bull trout contain aquatic management strategies that have standards and guidelines for activities that can and

cannot occur in the aquatic management zones. They are the Northwest Forest Plan (NWFP) (USFS and USDI 1994); the interim strategies for managing anadromous fish-producing watersheds in eastern Oregon and Washington, Idaho, and Portions of California (PacFish) (USFS and USDI 1995); and the inland native fish strategy (InFish) (USFS 1995). Within these aquatic strategies, riparian management objectives within the area covered by each strategy need to be met for actions implemented within the aquatic management zones. Aquatic management zones are generally identified within the three strategies as 1 site-potential-tree height or 150 feet, whichever is greater. Combined with the Forest Service best management practices (USFS 2012a), effects to species and critical habitat should be minimized but may not be eliminated. Effects from previous projects have been short-term negative effects for sedimentation/turbidity and water temperature, with long-term beneficial effects from the aquatic conservation strategy guidance. Because of these reasons, future projects in some areas may rise to the level of measurable effects on bull trout critical habitat. The Roadless Rule Rescission **may affect, likely to adversely affect** Little Kern golden trout critical habitat and bull trout critical habitat (Table 15).

Ictaluridae Family

Three species in the Ictaluridae family may be in the action area (Table 15). Since all members of the Ictaluridae family, if affected, have similar effects pathways, effects determinations will be combined below in the Ictaluridae Effects section.

Yaqui catfish

Yaqui catfish[26] are profusely speckled and exhibit a brownish gray coloration on the back, or dorsally, with lighter coloration beneath, ventrally. Their barbels are black, except for the barbels located on the chin, are 15.7 inches average length, cylindrically shaped, lacking scales. weigh 2.2 pounds, making them one of the medium-sized North American catfish species. Yaqui catfish are bottom-dwelling omnivores representing the only known native Ictalurid west of the Continental Divide. Very little is known about the species' biology but believed that it parallels that of channel catfish (*Ictalurus punctatus*). Yaqui catfish are cavity spawners, where males will create or utilize holes in banks, hollow logs, rocks and other submerged material to build a nest. Females are attracted to the nests where spawning occurs, and eggs are laid on the bottom of the nest. Males will stay and protect the nest until the young fish, known as fry, leave the nest. Outside of breeding season, Yaqui catfish live relatively solitary lives in dark shady areas in deeper pools. Catfish will feed during the day or night and tend to feed near the water bottom. Diets can consist of plant and animal material like insects, snails, crawfish, green algae, aquatic plants and small fish. Catfish detect food with their sense of taste, and the barbels around their mouth, which are specialized sensory organs that are covered in tastebuds. Yaqui catfish inhabit moderate to large streams in areas of medium to slow current with sandy or rocky bottoms, at elevations between 4,000 to 5,000 feet.

Currently, the Yaqui catfish is found only in the Rio Yaqui drainage of Sonora, Mexico. It was extirpated from the United States, and in 1997, the Yaqui catfish was reintroduced to San

[26] Much of the species' description and current range was collected from: https://www.fws.gov/species/yaqui-catfish-ictalurus-pricei accessed on 2/13/2026.

Bernardino National Wildlife Refuge and West Turkey Creek in Cochise County, Arizona. However, these introductions did not produce sustained populations and the Yaqui catfish is again considered extirpated from the United States, as is documented in the USFWS 2019 5-Year Review of the species. The USFWS (2019q) 5-year review suggested uplisting the Yaqui catfish.

Many threats to this fish continue or have increased, and these include: loss, alteration, and degradation of suitable habitat through the construction of dams used throughout the Rio Yaqui for agricultural and hydroelectric development; the presence of channel catfish, a common non-native species that hybridizes with Yaqui catfish; and long-term drought combined with expanding human populations, which are creating increased demand for water for human consumption (USFWS 2019r).

Yaqui catfish critical habitat is not in the action area.

Smoky madtom

The Smoky madtom is a small madtom species endemic to the Little Tennessee River basin. The species was thought to be extinct when first described from specimens collected in 1957. In 1980 the species was rediscovered in Citico Creek and was listed as federally endangered in 1984 (49 FR 43065). Currently, there are three known populations, all occur within tributaries of the Little Tennessee River. Two of the populations, Abrams Creek and Citico Creek, are within the known historical range. The USFWS designated a nonessential experimental population for the Smoky madtom in Tellico River that has been stocked from captive propagated individuals from the Citico Creek population. The Abrams Creek and Citico Creek populations appear to be stable. The Tellico River population has documented natural reproduction since 2010 and may also be getting close to being fully self-sustaining. Threats to Smoky madtom habitat have been greatly reduced by management actions by the Forest Service and National Park Service, as most of the range occurs within the Cherokee National Forest and Great Smoky Mountains National Park. While the species continues to be isolated by impoundments, a genetic plan is being implemented to manage gene flow between populations. Source and nonpoint source pollution pose a minor threat to populations due to conservation efforts and watershed management practices of federal land management agencies. Because of the successful establishment of two additional populations (Abrams Creek and Tellico River), active management of the Smoky madtom in the Cherokee National Forest and Great Smoky Mountains National Park, and due to all current populations being stable or increasing (USFWS 2024ad).

The Citico Creek population occupies 14.6 km of habitat almost entirely within the Cherokee National Forest, Monroe County, Tennessee. Protected land surrounding Citico Creek is primarily managed by the Forest Service as part of the Citico Creek Wilderness Area to protect and preserve the natural environment from human influences and allow ecological and biological processes to progress naturally (2004 Cherokee National Forest land management plan, available at USFS 2025a). In the Citico Creek watershed, the USFS (2004 Cherokee National Forest land management plan) prioritizes and implements treatments to improve riparian condition and reduce sediment transport to streams and provide high water quality (USFWS 2024ad).

Most of the species' current habitat lies within lands managed by the Forest Service and the National Park Service, which have restoration, monitoring, and forest management programs with the potential to benefit the species and reduces some impacts from sedimentation and water quality. Sedimentation poses only a minor threat to Smoky madtom populations due to active management by Forest Service and National Park Service. The Cherokee National Forest, for example, is also a vital partner in upgrading and improving road-stream crossings on the Forest to benefit species like the Smoky madtom.

Recreational activities (swimming, wading, horse access to the water) within the Cherokee National Forest can and have resulted in visitors disturbing rocks, impacting nests and dislodging eggs. Smoky madtom populations are also vulnerable to the potential of increased frequency of intense flooding and severity of drought. Alteration of weather patterns is expected to result in increased temperatures, increase in droughts, and increases in extreme weather events, which have the potential to impact species. Reduced water availability may be a significant threat to the Smoky madtom in the long term because an increase in the frequency and duration of dewatering events of suitable habitat may lead to reduced reproduction, nest success, or increase mortality due to the shallow-water habitat occupied by this species (USFWS 2024ad).

Each Smoky madtom population occurs within one watershed. Catastrophic events that could severely impact or extirpate entire Smoky madtom populations include chemical spills, changes in upstream land use that alter stream characteristics and water quality downstream, new impoundments, and potential effects of drought and increases in occurrence of flash flooding events. The likelihood of many of these events are low given the Federal protections afforded to the populations by the National Park Service and Forest Service (USFWS 2024ad).

Smoky madtom critical habitat is not located within the action area.

Yellowfin madtom

The yellowfin madtom is a moderately elongated madtom (maximum known length 92 millimeters slot length). It has a depressed head, large eyes, and a truncate to slightly rounded caudal fin. The pectoral spines are long with highly developed serrae (USFWS 1983c). Yellowfin madtoms are known to occupy small-to-medium size warmwater streams with moderate current, often occurring in pools or backwaters (USFWS 2020j). The yellowfin madtom is vulnerable to loss of genetic diversity and fitness due to small population sizes. Species that are restricted in range and population size are more likely to suffer loss of genetic diversity due to genetic drift, potentially increasing their susceptibility to inbreeding depression and decreasing their ability to adapt to environmental changes (USFWS 2020j).

They probably occurred in many of the lower gradient streams of the Tennessee River basin and has been extirpated from known locations in Chickamauga Creek, Georgia, and Hines Creek, Tennessee. Since the Service listed yellowfin madtom under the ESA, the range of the species has expanded through rediscoveries and reintroduction efforts in the upper Tennessee River drainage. The species was rediscovered in Citico Creek in the Little Tennessee River system, Tennessee, and Copper Creek in the Clinch River system, Virginia. In addition, reintroduction efforts helped bring back the yellowfin madtom to Abrams Creek and the Tellico River in the

Little Tennessee River system, Tennessee, and the North Fork Holston River, Virginia (USFWS 2020j).

Physical habitat destruction resulting from a variety of human-induced impacts such as siltation, disturbance of riparian corridors, and changes in channel morphology continues to plague the Tennessee River watershed. The most significant of these impacts is siltation caused by excessive releases of sediment from activities such as agriculture, resource extraction (for example, coal mining, silviculture), road construction, and urban development. Activities that contribute sediment discharges into a stream system change the erosion or sedimentation pattern, which can lead to the destruction of riparian vegetation, bank collapse, excessive instream sediment deposition, and increased water turbidity and temperatures (USFWS 2020j).

Agriculture continues to threaten the yellowfin madtom in the Clinch and Powell rivers and Copper Creek. Cattle continue to threaten yellowfin madtom habitat in Copper Creek. Agricultural livestock and other unconfined animals often have direct access to waterbodies and can provide a concentrated source of coliform bacteria loads that can directly enter a stream and destroy riparian by trampling vegetation that holds streambanks in place (USFWS 2020j).

Two of seven streams with extant yellowfin madtom populations are located on the Cherokee National Forest and receive management and protection through the National Forest Management Act of 1976 (16 U.S.C. 1600 et seq.) and Cherokee National Forest's 2004 land management plan (available at USFS 2025a). The land management plan is implemented through a series of project-level decisions based on appropriate site-specific analysis and disclosure. It does not contain a commitment to select any specific project; rather, it sets up a framework of desired future conditions with goals, objectives, and standards to guide project proposals. Projects are proposed to solve resource management problems, move the forest environment toward desired future conditions, and supply goods and services to the public (2004 land management plan, available at USFS 2025a). The land management plan contains protective standards that are designed to avoid and minimize potential adverse effects to the Yellowfin Madtom and other sensitive species; however, the Cherokee National Forest continues to consult with the Service when their activities may adversely affect streams supporting the species. The Cherokee National Forest's management under the land management plan contributes substantially to the conservation of the species, and we expect the Cherokee National Forest to continue to implement management actions that benefit the species (USFWS 2020). The populations in Citico Creek and the Tellico River are in the Cherokee National Forest boundaries.

Yellowfin madtom critical habitat is not located within the action area.

Carolina madtom

The Carolina madtom is a small (5 inches), bottom-dwelling member of the catfish family. It has potent venom glands at the spines and is the most strongly armed North American catfish, earning the "furiosus" name. Carolina madtoms typically feed from dusk to dawn and eat aquatic insects, including larval midges, mayflies, caddisflies, dragonflies, and beetles. They nest from mid-May to late-July under natural or artificial cover (for example, mussel shells, bark and other woody debris, glass jars, beverage cans, or other containers). Found only in the Neuse and Tar-

Pamlico River Basins of North Carolina, it typically inhabits medium to large streams and rivers in the Piedmont and Coastal Plain physiographic regions and it requires clean, flowing water habitat and substrates of coarse sand and gravel with leaf litter and abundant cover. Preferred habitats of this species include riffles, runs, and pools of shallow to moderate depth, with leaf packs and sand and gravel substrates, and adequate cover (for example, boulders or woody debris). They are often found in swift waters and will shift to areas of moderate or slow flow during the breeding season (USFWS 2021h).

The Carolina madtom require free-flowing, highly oxygenated streams, water temperatures of 20 – 25°C (68 – 77°F) for breeding, and physical cover for nesting habitat. They rely on uncompacted sand, gravel, and boulder substrates and clean leaf packs that can support insect larvae, which comprise 95% of their diet. Their small body size leaves Carolina madtoms vulnerable to predation. Slow growth delayed reproductive maturity, and a relatively short life span limits the lifetime recruitment potential and their ability to adapt to rapidly changing habitat conditions or respond to punctuated or long-term impacts to populations (USFWS 2021h).

The Carolina madtom was listed as an endangered species with designated critical habitat on June 9, 2021 (86 FR 30688). The primary threat to Carolina madtom is habitat degradation resulting from stressors that affect water quality, water quantity, instream habitat suitability, and habitat connectivity, including watershed-level effects of development on water quality, instream habitat fragmentation, sedimentation from a variety of land uses, and the effects of long-term weather alteration. The remaining small, isolated populations and contracted range of the Carolina madtom further increase their vulnerability to stochastic and catastrophic events (USFWS 2021h).

The greatest threat to the future viability of the Carolina madtom (such as, the ability to sustain populations in the wild over time) is habitat degradation from stressors that influence water quality, water quantity, instream habitat, and habitat connectivity. Sedimentation is induced by variety of land uses (for example, development, agricultural, or forestry activities without adequate riparian protections or proper erosion/sediment control). Carolina madtom are known to avoid areas dominated by fine silts. Sedimentation can result in degraded or unsuitable cover and foraging habitats by increasing substrate embeddedness or rendering leaf packs hypoxic and unable to support insect prey. Some other significant physical and biological stressors include instream barriers, invasive species, and severely diminished populations (USFWS 2021h).

The Croatan National Forest would be the only forest that may have impacts to Carolina madtom within the Trent River system. There are a few smaller tributary streams, Will, Island, and Reedy creeks, that flow into the Trent River near the action area, but Carolina madtom do not inhabit these streams and are found upstream of these creeks in the Trent River system. There may be current roads that could be utilized as haul route for timber and road aggregate, but those would be unknown at this time. If these roads are utilized the effects would be minimal amounts of sediment, depending on the road surface type.

Carolina madtom critical habitat is not located within the action area.

Frecklebelly madtom

The frecklebelly madtom is a small, stout catfish reaching about 4 inches in length and distinctively marked with dark saddles. The species inhabits the main channels and larger tributaries of large river systems in Louisiana, Mississippi, Alabama, Georgia, and Tennessee. The species has a broad but disjunct distribution across the Pearl River watershed and Mobile River Basin, with populations in the Pearl River and Bogue Chitto River in the Pearl River watershed and the Tombigbee, Alabama, Cahaba, Etowah, and Conasauga river systems in the Mobile River Basin (88 FR 13038).

Primary habitat for frecklebelly madtom is associated with fast moving streams often associated with rivers and their tributaries, with substrate consisting of various sizes of gravel. Cover is an important habitat factor for the species, as it provides for concealment against predators, foraging habitat, and nesting habitat. Areas providing firm gravel substrates, such as small pebbles and rocks, are preferred, thus muddy waterway sand still streams are not desirable habitat for this species (88 FR 13038).

The frecklebelly madtom is primarily active at night and has a lifespan of up to five years. There is a lack of information on the breeding behavior and nesting habitat of the species, although it is believed that reproduction occurs between June and July. Nesting sites for madtoms are typically cavities under natural material, such as rocks, logs, empty mussel shells, or human litter, such as inside cans or bottles or under boards. They are opportunistic insectivores and feed on aquatic insect larvae, with the primary food sources being caddisflies, mayflies, blackflies, and midges (USFWS 2023s).

The frecklebelly madtom prefers fast-moving streams with gravel substrate for cover, foraging, and nesting habitats. They avoid muddy waterways and still streams. The presence of coarse and stable gravel substrate is also important for the occurrence of other madtom species. Aquatic vegetation is also important for some parts of their range, where they are often found in or near river weed (*Podostemon*) and under large, flat rocks. They have been collected in moderate to swift currents over boulders, rubble, cobble, and coarse gravel, and around concentrations of river weed in the upper Etowah and Conasauga rivers. Shoal complexity could limit the recovery of the species because stabilizing shoals and recruiting river weed may be unattainable with the surrounding land use and hydrology (USFWS 2023s).

Frecklebelly madtom are sensitive to pollutants from agricultural land use. Hormones, excess nutrients, and herbicide surfactants are postulated to contribute to the species declines observed in the Conasauga watershed. Nonpoint source sediment and agrochemical discharges in predominantly agricultural areas within the frecklebelly madtom's current range could potentially alter its habitat's physical and chemical characteristics, thereby impeding its feeding, sheltering, and reproductive abilities (USFWS 2023s).

Frecklebelly madtom critical habitat is not located within the action area.

Ictaluridae Effects

Yaqui catfish have been extirpated from the U.S.; therefore, there would be **no effect** to Yaqui catfish.

Effects of future projects on all madtoms analyzed in this BA would most likely be from sedimentation and habitat fragmentation. Effects to Smoky and yellowfin madtoms would be low due to protections provided by the Cherokee National Forest Land Management Plan. Implementing Forest Service best management practices (USFS 2012a) will help alleviate negative effects of any future project. National Forests are actively involved in protecting species and protections are expected to continue. Low level effects from sedimentation and habitat fragmentation could occur during future project implementation, the magnitude of which is unknown. Any future projects that would affect species would still need to complete project level consultation. Due to the unknowns, we might expect some effects on individual Smoky madtom, yellowfin madtom, and frecklebelly madtom that are susceptible to reach **may affect, likely to adversely affect** to the species. Carolina madtom could be in adjacent tributaries near the Croatan National Forest. If they are within adjacent tributaries effects would only be from haul which would be dependent on the road surface type. The amount of sedimentation that would reach the stream during haul would not be measurable. We expect Carolina madtom effects, if any, to be discountable; therefore, the determination is **may affect, not likely to adversely affect** Carolina madtom. Individual species effects are listed in Table 15.

Bivalves

To assist analyzing effects to a large number of species, ESA-listed species were grouped taxonomically. The bivalve group includes 45 total species including 25 endangered, 12 threatened, 6 proposed endangered, and 2 proposed threatened freshwater bivalves (Table 18). All the listed bivalve species occur in the southeastern United States and are located in either regions 8 or 9 of the Forest Service. Freshwater bivalves of the United States inhabit a number of freshwater environments, though a majority of those listed under the ESA prefer flowing riverine habitat. Freshwater bivalves are filter feeders that live on or in the substrate. Locomotion is limited and performed with the assistance of a muscular hydraulic foot. Freshwater bivalves of the U.S. have a parasitic larval stage that infect the gills or body of the host fish. Host fish are often species specific to the freshwater bivalve. Threats to native freshwater bivalves of the U.S. are typically associated with decreases in water quality and quantity.

Table 18. Threatened (T), endangered (E), proposed threatened (PT), and proposed endangered (PE) freshwater bivalve species in the action area. The presence of critical habitat (and percent overlap with the action area) is also provided. A summary of exposure and the effects determination is also shown.

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Alabama moccasinshell	*Medionidus acutissimus*	T		Low	LAA	
Appalachian elktoe	*Alasmidonta raveneliana*	E		Low	NLAA	
Arkansas fatmucket	*Lampsilis powellii*	T		Low	LAA	
Atlantic pigtoe	*Fusconaia masoni*	T		Unk	NLAA	
Chipola slabshell	*Elliptio chipolaensis*	T		Unk	LAA	
Clubshell	*Pleurobema clava*	E		Unk	LAA	

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Coosa moccasinshell	*Medionidus parvulus*	E	Y (<1%)	Low	LAA	NLAA
Cumberland mocassinshell	*Medionidus conradicus*	PE		High	LAA	
Fanshell	*Cyprogenia stegaria*	E		Low	NLAA	
Fat threeridge	*Amblema neislerii*	E		Low	LAA	
Finelined pocketbook	*Hamiota altilis*	T	Y (<1%)	Low	LAA	NLAA
Finerayed pigtoe	*Fusconaia cuneolus*	E		Unk	NLAA	
Fluted kidneyshell	*Ptychobranchus subtentus*	E		Low	NLAA	
Georgia pigtoe	*Pleurobema hanleyianum*	E		Low	LAA	
Green floater	*Lasmigona subviridis*	PT	Y (<1%)	Low	LAA	NLAA
James spinymussel	*Parvaspina collina*	E		Unk	NLAA	
Longsolid	*Fusconaia subrotunda*	T	Y (<1%)	Low	LAA	NLAA
Louisiana pearlshell	*Margaritifera hembeli*	T		Low	NLAA	
Louisiana pigote	*Pleurobema riddellii*	PT		Low	LAA	
Northern riffleshel	*Epioblasma rangiana*	E		Unk	LAA	
Ochlockonee moccasinshell	*Medionidus simpsonianus*	E		Low	LAA	
Ouachita rock pocketbook	*Arcidens wheeleri*	E		Unk	LAA	
Ovate clubshell	*Pleurobema perovatum*	E	Y (<1%)	Low	LAA	NLAA
Pink mucket	*Lampsilis abrupta*	E		Low	NLAA	
Purple bankclimber	*Elliptoideus sloatianus*	T		Low	LAA	
Purple bean	*Villosa perpurpurea*	E		Unk	LAA	
Rabbitsfoot	*Quadrula cylindrica cylindrica*	T		Unk	LAA	
Rayed bean	*Villosa fabalis*	E	Y (Proposed) (<1%)	Unk	LAA	NLAA
Round hickorynut	*Obovaria subrotunda*	T		Low	LAA	
Salamander mussel	*Simpsonaias ambigua*	PE		Low	LAA	
Scaleshell mussel	*Leptodea leptodon*	E		Unk	LAA	
Shiny pigtoe	*Fusconaia cor*	E		Unk	NLAA	
Slabside pearlymussel	*Pleuronaia dolabelloides*	E		Unk	NLAA	
Snuffbox mussel	*Epioblasma triquetra*	E		Low	LAA	
Southern clubshell	*Pleurobema decisum*	E	Y (<1%)	Low	LAA	NLAA

Common Name	Scientific Name	ESA Status	Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Southern elktoe	*Alasmidonta triangulata*	PE		Unk	LAA	
Southern pigtoe	*Pleurobema georgianum*	E	Y (<1%)	Low	LAA	NLAA
Speckled pocketbook	*Lampsilis streckeri*	E		Unk	LAA	
Spectaclecase	*Cumberlandia monodonta*	E		Low	NLAA	
Tennessee clubshell	*Pleurobema oviforme*	PE		Low	NLAA	
Tennessee pigtoe	*Pleuronaia barnesiana*	PE		High	LAA	
Texas fawnsfoot	*Truncilla macrodon*	T		Unk	LAA	
Texas heelsplitter	*Potamilus amphichaenus*	PE		Low	LAA	
Triangular kidneyshell	*Ptychobranchus greenii*	E	Y (<1%)	Low	LAA	NLAA
Western fanshell	*Cyprogenia aberti*	T		Low	LAA	

Road construction and timber harvest primarily affects freshwater bivalves through two main pathways: physical and chemical degradation of their habitat and the fragmentation of populations. The most significant impact is often increased sedimentation resulting from soil erosion at construction sites and runoff from new road surfaces. Excess sediment can directly harm bivalves, which are sessile filter feeders, by clogging their gills, interfering with feeding and respiration, and physically burying and smothering individuals, leading to direct mortality or reduced growth rates.

Beyond sedimentation, road construction introduces other stressors. Instream work, like installing culverts, can cause direct mortality and permanent changes to hydrology and streambed composition. This physical alteration may eliminate the stable, clean habitat necessary for bivalve survival. Chemically, runoff from new roads can introduce toxic pollutants such as heavy metals, oils, and de-icing salts, further compromising water quality and the health of the bivalves and their host fish. The FS does not regularly use toxic chemicals to manage the land and does not use salt to clear icy roads. Any activities that use chemicals with toxic properties will be required to analyze their potential effects to the environment, including ESA-listed resources, prior to implementation.

Finally, road-stream crossings like culverts can act as barriers to the movement of fish that serve as essential hosts for the bivalve larval stage (glochidia). Improperly designed culverts can have high water velocities or insufficient water depth, preventing fish from moving upstream and effectively blocking bivalve dispersal and reproduction. This habitat fragmentation can lead to isolated mussel populations with reduced genetic diversity and a higher risk of local extinction (Vaughan 2002).

Major Stressors

Sedimentation

Sedimentation impacts freshwater bivalves by physically degrading their habitat and impairing essential physiological processes. As filter feeders, bivalves rely on relatively clean water and stable substrate. Increased levels of suspended solids directly interfere with feeding and respiration by clogging gills and siphons, which can lead to reduced food intake, energy depletion, and chronic stress. Deposition of fine sediment on the streambed is also detrimental, as it potentially smothers and buries bivalves, leading to suffocation and mortality. Accumulated sediment also alters the natural, stable substrate that mussels require for secure habitation and successful reproduction. Furthermore, increased sedimentation can negatively affect the fish hosts necessary for the larval stage of most bivalves, creating a compounding negative effect that ultimately contributes to population declines and increased extinction risk for many vulnerable species.

Zhu et al. (2024), in a published synthesis on this topic, offer the following on sedimentation effects on bivalves. Freshwater mussels have a complex life cycle that includes fertilization by broadcasting sperm, embryos brooded in marsupial gills, parasitic larvae that attach to fish, and a free-living juvenile stage that typically occupies interstitial spaces in river sediment. In some cases, suspended solids may adversely affect fertilization and brooding of some species. Both juvenile and adult mussels are suspension feeders, and feeding can be inhibited by suspended solids. Because of their small size and rapid metabolism and growth, juvenile mussels are often more sensitive than adults to stressors in laboratory experiments. For example, researchers associated turbidity and sedimentation with recruitment failure in 14 of 24 populations of freshwater pearl mussel (*Margaritifera margaritifera*) that they investigated. Absence of juvenile mussels from a lack of recruitment is considered to be a characteristic feature of many declining mussel populations, which may eventually cause population collapse.

In 2013, Landis et al. linked the reproductive failure of pondmussel (*Ligumia subrostrate*) females to elevated suspended solids. They found that (a) 98-99% of eggs were fertilized regardless of total suspended solids (TSS) levels and (b) total fecundity was not related to TSS levels. Sperm production was not correlated with TSS concentrations. In another study, Landis et al. (2015) assessed the stage-specific disruption of reproduction due to high TSS levels in two mussel species, including a short-term brooding mussel species (Ebonyshell, *Reginaia ebenus*) and a long-term brooding species (Pondmussel, *Ligumia subrostrata*). Although reduced reproductive success was observed in both species at high TSS levels, the effects were different. More female ebonyshell (33-93%) were found fertilized across the TSS gradient (11-92 mg/L) tested, but few glochidia developed when TSS was higher than 20 mg/L. In the case of female pondmussel, similar to the 2013 study, much less female (0- 28%) were fertilized at higher TSS levels, but all those gravid successfully produced fully developed glochidia. The observed differences might be explained by the differences in gill structure (the density of cilia on the gills) and gill function related to habitat use and/or brooding strategies.

Fragmentation

Habitat fragmentation impacts freshwater bivalves by isolating populations and disrupting their complex life cycles, leading to population decline. Improper construction of culverts and other in-stream infrastructure can convert flowing river habitat into non-flowing water features, an unsuitable environment for many river-dwelling mussel species. This not only causes direct habitat loss but also alters environmental conditions such as temperature, flow regimes, and oxygen levels (Lopes-Lima et al. 2018; Pires et al. 2020).

One of the most critical effects of fragmentation is the blockage of native host fish movement, which is essential for the dispersal of bivalve larvae (glochidia). With their host fish unable to pass barriers, mussel populations become physically and genetically isolated. This isolation prevents gene flow between populations, leading to reduced genetic diversity and an increased risk of inbreeding, which can diminish fitness and adaptive potential. Consequently, isolated populations are more vulnerable to environmental stochasticity, disease, pollution events, and local extinction (Osterling and Soderberg 2015).

Further, barriers to natural flow can alter the riverbed dynamics, causing sediment retention upstream and scouring of substrates downstream, which degrades essential microhabitats and negatively affects suspension feeders like bivalves. These impacts, combined with other stressors, have made freshwater bivalves one of the most imperiled animal groups globally (Lopes-Lima et al. 2018; Pires et al. 2020).

Although in-stream work has the potential to fragment riverine habitat, the Forest Service designs and installs culverts in a manner to improve streamflow, not restrict it. Fragmentation can still occur during construction, but those effects are assumed to be temporary and that surface flows will return to normal, or better than normal, when construction has ended. Short-term fragmentation of aquatic habitat is not expected to have population level effects to listed bivalves.

Existing Protections

Land Management Plans or similar planning documents for the following forests provide direction specifically to protect and recover listed bivalve species: Nantahala and Pisgah NF, George Washington and Jefferson NFs, Hoosier NF, Cherokee NF, Kisatchie NF, and Mark Twain NF (available at USFS 2025a). Given land management plan direction on these forests, the Forest Service anticipates a majority of effects to listed bivalve species in these areas to be avoided or minimized. The following species are known to occur on only these forests within the action area: Appalachian elktoe, Atlantic pigtoe, fanshell, finerayed pigtoe, fluted kidneyshell, James spinymussel, Louisiana pearlshell, pink mucket, shiny pigtoe, slabside pearlymussel, spectaclecase, Tennessee clubshell. Where exposure is anticipated to be low or is unknown and planning documents contain protections for ESA-listed bivalves, a determination of NLAA is being made.

Critical Habitat

Effects to designated critical habitat are analyzed by reviewing the physical and biological features associated with that habitat and determining the effects the action may have on those features. Most bivalve critical habitat includes sufficient water quality and quantity to support breeding, feeding, and sheltering. Only nine listed bivalve species have designated critical habitat within the action area and their overlap with the action area are all less than 1%. In the event activities occur in designated critical habitat the effects to physical and biological features will be minimized or avoided as dictated by local conditions. Impacts to critical habitat from in-stream work are expected to be temporary and conditions will return to normal after work is complete. Some features of bivalve critical habitat may be positively affected by future timber and road activities by increasing sunlight on surface waters, which in turn produce higher abundances of algae and zooplankton upon which freshwater bivalves feed. Given the small amount of overlap with designated critical habitat, that most in-stream work effects will be temporary, and that some effects may be positive, the proposed action is not likely to adversely affect designated critical habitat for bivalve species.

Determinations

A majority of listed bivalves are expected to have a low likelihood of exposure to future projects in the affected environment. However, if exposed the response to sedimentation could include mortality and decreased reproductive capability, which could impact individuals and the population as a whole. Only where exposure to the stressors associated with road and timber activities is so low as to be discountable has the Forest Service made a not likely to adversely affect determination (Table 18). If the Forest Service was unable to conclude exposure was unlikely due to location or existing protections, the proposed rescission of the 2001 Roadless Rule is likely to adversely affect these listed bivalves (Table 18). Given the small amount of overlap with designated critical habitat, that most in-stream work effects will be temporary, and that some effects may be positive, the proposed action is not likely to adversely affect designated critical habitat for bivalve species (Table 18).

Table 19. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of bivalve species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Riparian Resilience	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability
		Tree and Unit Marking	Walking	Chemical Contamination/Nutrients	No response anticipated
			ATV	Chemical Contamination/Nutrients	Not a significant source of contaminants; no response anticipated
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Peak/Base Flows; Large Woody Debris; Change in Riparian Vegetation (Temperature); Riparian Resilience	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow and in-channel woody debris not anticipated nor are effects from them.
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Riparian Resilience	Not a significant source of contaminants; no response anticipated
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Large Woody Debris; Peak/Baseflows; Riparian Resilience	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow and in-channel woody debris not anticipated nor are effects from them.
		Timber Haul	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Significant changes to flow and in-channel woody debris not anticipated nor are effects from them.
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Significant changes to flow and in-channel woody debris not anticipated nor are effects from them.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow are expected to be positive in the long term, though short-term impacts may occur; not a significant source of contaminants.
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow are expected to be positive in the long term, though short-term impacts may occur; not a significant source of contaminants.
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow are expected to be positive in the long term, though short-term impacts may occur; not a significant source of contaminants; water pumping, if used, is not expected to affect sessile benthic organisms.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow are expected to be positive in the long term, though short-term impacts may occur; not a significant source of contaminants; water pumping, if used, is not expected to affect sessile benthic organisms. Complete dewatering of occupied T&E habitat will not occur
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Large Woody Debris	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow and in-channel woody debris not anticipated nor are effects from them; not a significant source of contaminants
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; not a significant source of contaminants
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; not a significant source of contaminants

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; not a significant source of contaminants; water pumping, if used, is not expected to affect sessile benthic organisms.
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Sediment/Turbidity; Spawning Substrate	Not expected to be a significant source of sedimentation.
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Chemical Contamination/Nutrients	not a significant source of contaminants, no response anticipated
	Snow	Plowing	Heavy Equipment	Drainage Network	Not expected to significantly affect surface water flow so no response anticipated.
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to flow and in-channel woody debris not anticipated nor are effects from them
		Snags - Hand Felling	Motorized Hand Tools	Large Woody Debris	Significant changes to in-channel woody debris not anticipated nor are effects from them
		Snags - Mechanical Felling	Heavy Equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Large Woody Debris	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Increased growth from increased light resulting in more food availability; Significant changes to in-channel woody debris not anticipated nor are effects from them

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Tree Removal - Log Decks	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
		Tree Removal - Hauling	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Road Density/Location; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Significant changes to flow not anticipated nor are effects from them
		Soil Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
		Fill Placement	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Significant changes to flow is not anticipated nor are effects from them; not a significant source of contamination
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in Riparian Vegetation (Temperature); sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; temporary reduction in fitness while working in the channel, long term benefits after installation; not a significant source of contamination; Increased growth from increased light resulting in more food availability;

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Surfacing	Hauling of surface materials	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; not a significant source of contamination;
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; not a significant source of contamination;
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Significant changes to flow is not anticipated nor are effects from them;
	scarifying	Material removal	Heavy equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation;
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; temporary reduction in fitness while working in the channel, long term benefits after installation
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris; Riparian Resilience	Direct mortality of adults from sedimentation; Reduced reproductive output due to sedimentation; Significant changes to flow or in-channel woody debris are not anticipated nor are effects from them;

Crustaceans

To assist analyzing effects to a large number of species, ESA-listed species were grouped taxonomically. The crustacean group includes 9 total species including 4 endangered and 5 threatened crustaceans (Table 20). These species are aquatic at all life stages, although fairy shrimp eggs can persist through dry conditions until water returns. Two species, the riverside fairy shrimp and Shasta crayfish, potentially occur on two National Forest Service units based on IPAC, while the rest of the listed crustaceans may only occur on one NFS unit (see Appendix A).

Table 20. Threatened (T) and endangered (E) crustacean species in the action area. The presence of critical habitat (and percent overlap with the action area) is also provided. A summary of exposure and the effects determination is also shown.

Common Name	Scientific Name	ESA Status (A)	Critical Habitat in Affected Environment (B)	Exposure (C)	Effects Determinations for Species (D)	Effects Determinations for Critical Habitat (D)
Madison cave isopod	*Antrolana lira*	T	N/A	Low	LAA	N/A
Conservancy fairy shrimp	*Branchinecta conservatio*	E	Yes (15.4%)	High	NLAA	NLAA
Vernal pool fairy shrimp	*Branchinecta lynchi*	T	Yes (4.8%)	High	LAA	LAA
San Diego fairy shrimp	*Branchinecta sandiegonensis*	E	No	Unknown	LAA	N/A
Riverside fairy shrimp	*Streptocephalus woottoni*	E	No	Unknown	LAA	N/A
Big sandy crayfish	*Cambarus callainus*	T	No	Unknown	LAA	N/A
Big Creek crayfish	*Faxonius peruncus*	T	No	Unknown	LAA	N/A
St. Francis River crayfish	*Faxonius quadruncus*	T	Yes (<1%)	Low	LAA	NLAA
Shasta crayfish	*Pacifastacus fortis*	E	N/A	Low	LAA	N/A

Table notes: (A) T = threatened, E = endangered; (B) amount of critical habitat in the affected environment based on spatial analysis (Appendix A) using the USFWS Critical Habitat Shapefile available here for download: https://ecos.fws.gov/ecp/report/table/critical-habitat.html, N/A under critical habitat indicates critical habitat has not been designated; (C) Exposure = Unknown means insufficient data to determine (D) NLAA = may affect, not likely to adversely affect, LAA = may affect, likely to adversely affect, N/A = not applicable;

Four of the ESA-listed crustaceans are fairy shrimp, including the Conservancy fairy shrimp, vernal pool fairy shrimp, riverside fairy shrimp, and San Diego fairy shrimp. Two of these fairy shrimp have critical habitat in the affected environment, the vernal pool fairy shrimp and Conservancy fairy shrimp (71 FR 7118). All four of these fairy shrimp species have short life cycles tied to the wet and dry conditions of vernal pools and other ephemeral freshwater ponds in grasslands, rural areas, and wetlands. Eggs can remain viable as cysts in dry conditions and hatch in response to the return of water in the pool. These species have limited dispersal ability. Dispersal can be facilitated by aquatic birds and mammals that unintentionally distribute eggs or cysts that get caught on their bodies or that they have ingested. All of four of these species have a small range in the United States limited to California, and southern Oregon in the case of the vernal pool fairy shrimp (USFWS 2025q).

Four of the nine ESA-listed crustaceans in the action area are crayfish, including the Big sandy crayfish, Big Creek crayfish, St. Francis River crayfish, and Shasta crayfish. The St. Francis River crayfish has <1% of its total critical habitat overlapping with the action area (88 FR 25512). These crayfish species inhabit streams or rivers with rocky substrate (cobbles, boulders, gravel, volcanic rock) suitable for burrowing, sheltering, and as protective cover. The Big Creek crayfish and St. Francis River crayfish have small ranges limited to the Upper St. Francis River watershed in southeast Missouri. The majority of public land within the Upper St. Francis River watershed is part of the Mark Twain National Forest (USFWS 2023t). The Big Sandy crayfish has a larger range than the other listed crayfish which includes the Appalachian region of Kentucky, Virginia and West Virginia. The Shasta crayfish also has a small range and is restricted to the Pit River drainage system in Shasta County, California (USFWS 2025r).

One of the nine ESA-listed crustaceans in the action area is an isopod, the Madison cave isopod. The Madison cave isopod inhabits deep karst groundwater habitats associated with caves and fissures in West Virginia and Virginia. More specifically, this species occurs in underground lakes or deep karst aquifers (USFWS 1996). The recovery plan for this species describes known populations as extremely small and with low reproductive potential, however it also notes that this species is difficult to study and collect (USFWS 1996). In 1996, the known sites of this species were located on privately-owned property, however the recharge zone of the aquifer at any occupied site is unknown and therefore the zone of potential sources of contamination is also unknown (USFWS 1996).

Road construction, road reconstruction, road use, and timber harvest can affect crustacean species by fragmenting habitat, degrading habitat features and water quality, altering hydrology, and facilitating the spread and introduction of non-native species. Primary threats to the listed fairy shrimp which are relevant to road- and timber-related activities include direct mortality from off-highway vehicles, habitat loss, fragmentation and degradation from ground-disturbing activities, non-native plants, and altered hydrology. Primary threats to the listed crayfish which are relevant to road- and timber-related activities include habitat degradation from ground-disturbing activities that cause erosion and sedimentation, and non-native invasive species. The primary threat to the one listed isopod species in general is thought to be an increasing probability of pollutants entering groundwater due to urban and agricultural development.

The primary effects of the proposed action on crustacean species are discussed below and presented in Table 21. A discussion on species protection afforded by land management plans and other documents beyond the protections of the 2001 Roadless Rule is also included.

Major Stressors

Direct Mortality

Fairy shrimp: Road construction, road reconstruction, and timber harvest activities could result in direct mortality to individuals of listed fairy shrimp species if they are present within project boundaries or within road rights-of-way. Since all four of the listed fairy shrimp species are dispersal limited, the use of heavy equipment and off-road vehicles near or through occupied vernal pools or ephemeral wetland habitat for road construction or timber harvest could result in direct mortality to any life stage via vehicular strikes or trampling. Cysts during the dormant life

stage may be particularly at risk of direct mortality from ground-disturbing activities, since they cannot move on their own and dispersal of cysts is predominantly limited to incidental contact with birds or mammals (USFWS 2025q). Additionally, ground-disturbing activities can result in changes to hydrology and inundation periods which can result in the loss of cysts if pools are lost entirely (no water returns to hatch cysts or sedimentation buries pool and cysts), inundation does not last long enough for cysts to hatch, or less water returns to the pools seasonally (fewer cysts exposed to water needed to hatch) (USFWS 2024ae).

Habitat Fragmentation and Edge Effects

Fairy shrimp: Habitat fragmentation is a major threat to fairy shrimp because it can isolate populations, alter hydrology, and hinder dispersal. Physical barriers like roads can separate vernal pools and other ephemeral wetlands, making it difficult for fairy shrimp to disperse which can lead to reduced genetic diversity, increase susceptibility to environmental disturbances, and reduced reproduction. Altered hydrology caused by nearby development can sometimes result in vernal pools that are no longer suitable as habitat for fairy shrimp species. The surrounding watershed supplies most of the water in vernal pools, up to 60%, so changes to nearby land can indirectly lead to habitat loss even if the pools themselves remain untouched. While direct destruction from development poses a greater threat, altered hydrology is also a growing concern (USFWS 2024ae). Besides direct loss of vernal pools, after a road is constructed, ongoing maintenance of roads or road reconstruction within vernal pool ecosystems may subject the pools to repeated disturbance from off-road vehicles, management activities like vegetation removal or firebreak creation, and the introduction of invasive plants (USFWS 2024ae).

Invasive Species Proliferation

Fairy shrimp: Road construction, road reconstruction, and timber harvest activities can act as disturbances that facilitate the introduction and spread of non-native invasive plants into vernal pool complexes, where they can pose a threat to fairy shrimp species by altering hydrology and water temperature. Invasive grasses at pool margins can block light, retain moisture, and increase thatch (dense layer of dead and living plant material on the soil surface), leading to shorter hydroperiods that may prevent fairy shrimp species from completing their lifecycles (USFWS 2024ae). Additionally, increased vegetative cover due to non-native invasive plant species on the margins of pools can result water temperature changes that are unsuitable thermal conditions for hatching or development. For example, the time to maturity and reproduction for vernal pool fairy shrimp is temperature dependent; they require water temperatures of low enough for cysts to hatch, while immature and adult shrimp may die off if temperatures rise too high (USFWS 2026a, USFWS 2024ae).

Crayfish: Three of the four listed crayfish species are threatened by invasive species proliferation. Invasion by the woodland crayfish (*Faxonius hylas*) has been documented as a primary threat to the viability of the Big Creek crayfish and St. Francis River crayfish, with both of these native species experiencing displacement, hybridization, and range contraction in areas where the woodland crayfish has established (88 FR 25512). The introduction and expansion of the signal crayfish (*Pacifastacus leniusculus*), a generalist species suited to a variety of conditions, is a primary threat to the viability of the Shasta crayfish, a specialist species requiring clean volcanic cobbles and boulders on top of gravel or sand (USFWS 1998g). In areas where

the signal crayfish has rapidly expanded, the Shasta crayfish has shown declines (USFWS 1998g). While the invasive virile crayfish (*Faxonius virilis*) occurs in the same areas and competes with the Big Sandy crayfish in Kentucky, the implications of this competition on Big Sandy crayfish populations are not yet clear (USFWS 2025r). Exposure to invasive species may cause reduced carrying capacity and density of subpopulations, increased predation, increased competition, and displacement.

Sedimentation and Pollution

Fairy shrimp: Ground-disturbing activities in vernal pool watersheds can lead to sedimentation during the wet season, which can have negative impacts on both the hydrology and ecological health of vernal pools and wetlands inhabited by fairy shrimp. Increased sedimentation can alter hydroperiods by changing how water is retained, reduce species richness and abundance of plant and invertebrate communities, bury cysts, clog filter feeding apparatuses, and reduce forage by smothering plant matter or blocking light (USFWS 2024ae).

Crayfish: All four listed crayfish species are threatened by ground-disturbing activities that increase erosion and sedimentation which can lead to smothering of rocky substrates and suitable cavities required by these species. Streams within the Upper St. Francis River watershed, occupied by the Big Creek crayfish and St. Francis River crayfish, may experience increased sedimentation if land uses change or if riparian corridors are cleared (88 FR 25512). Commercial forestry, unpaved roads, gas and road construction were documented as some of the sources of sedimentation within the range of the Big Sandy crayfish (USFWS 2025r). Loss of cover and refugia from sedimentation can lead to reduced carrying capacity and density of subpopulations, increased predation risk, reduced foraging success, and increased competition with non-native crayfish (USFWS 2025r, 2023t, and 1998g).

Madison cave isopod: As a subterranean aquatic obligate species, potential threats to the Madison cave isopod include the loss and modification of habitat (including the surface environment that is their primary source of water and nutrients), groundwater contamination, and groundwater drawdown (USFWS 1996). Groundwater contamination where this species occurs could result in reduced species fitness and reproduction, or depending on the degree and type of contamination it could result in direct mortality to individuals and reduced genetic diversity in subpopulations.

Existing Protections

Federally listed threatened, endangered, and proposed species on NFS Lands, including inventoried roadless areas, would continue to be planned for and managed according to the requirements of law, regulation, and policy. All current Forest Service land management plans, through the National Forest Management Act (NFMA) and planning regulations (1982 Planning Rule or 2012 Planning Rule), are required to maintain and contribute to the recovery of threatened and endangered species and conserve proposed species in the plan area (39 CFR 219.9, 2012 Planning Rule; Sec. 219.19, 1982 Planning Rule). Specifically, under the 2012 Planning Rule, federally listed species are maintained in the plan area by providing the ecological conditions necessary to support the persistence of those species. These ecological conditions may include standards or guidelines related to habitat features, condition, or

connectivity. It is assumed that under all alternatives, future land management plan revisions or amendments would comply with ESA, NFMA, and planning regulations to meet this requirement to the extent feasible within Forest Service authority and consistent with the inherent capability of the plan area (36 CFR 219.9 and FSH 1909.12).

Relevant to the Madison cave isopod, the George Washington National Forest FEIS states there will be no direct or indirect effects to Madison Cave isopod from Forest Service management activities, and only a fraction (less than a tenth of one percent) of potential habitat was determined to be on Forest Service land at the time of the analysis. It was determined that any cumulative effects to the quality or quantity of Madison Cave isopod habitat would be from private land. The strategy of the George Washington National Forest on groundwater issues that cross national forest boundaries and are affected by multiple regionwide impacts, such as increased agricultural use or growing urban development, is to focus on sustaining and improving watershed areas within national forest control while working cooperatively with other agencies and landowners to improve statewide watershed health (USFS 2014).

Relevant to the four listed fairy shrimp, the Final Supplemental Environmental Impact Statement for the Southern California National Forests and the design criteria for the Southern California National Forests, including the Los Padres and Cleveland National Forests, state that implementation of the land management plans will be consistent with Recovery Plans for all ESA-listed species (USFS 2013a, USFS 2005). Further, the known occupied site of the Conservancy fairy shrimp in the Los Padres National Forest is fenced to exclude impacts according to the 2005 land management plan for the Los Padres National Forest (available at USFS 2025a).

Big Creek crayfish and St. Francis River crayfish were federally listed in 2023 and the big sandy crayfish was listed in 2016. The current land management plans for the Mark Twain National Forest and George Washington and Jefferson National Forests (where these species potentially occur based on IPAC) pre-date the listing of these three crayfish species and therefore do not include plan-level analysis or plan components specific to these species (land management plans available at USFS 2025a).

The Shasta crayfish was federally listed in 1988 and the current Northwest Forest Management Plan (which includes the Shasta-Trinity and Lassen National Forests) contains an aquatic conservation strategy that addresses generally managing riparian reserves, key watersheds, watershed analysis, and watershed restoration (FEMAT 1993).

Critical Habitat

Effects to designated critical habitat are analyzed by reviewing the physical and biological features associated with that habitat and determining the effects the action may have on those features. Only three listed crustacean species have designated critical habitat within the action area. The Conservancy fairy shrimp has 15.4% of its total critical habitat within the action area, the vernal pool fairy shrimp has 4.8% of its total critical habitat within the action area, and the St. Francis River crayfish has less than 1% (Table 20). In the event activities occur in designated

critical habitat the effects to physical and biological features will be minimized or avoided as dictated by local conditions.

Critical habitat for the Conservancy fairy shrimp and vernal pool fairy shrimp includes sufficiently connected pools, adequate seasonal water quantity, and suitable structure of inorganic and organic materials within pools to support incubation, maturation, reproduction, sheltering, and feeding (71 FR 7118). For the Conservancy fairy shrimp and vernal pool fairy shrimp, future timber and road activities in occupied critical habitat could result in habitat fragmentation that permanently reduces connectivity between pools and alters hydrology. The designated critical habitat unit that overlaps the action area for the Conservancy fairy shrimp is Unit 8 in Ventura County, California (71 FR 7118). There is a 15.4% overlap of the Conservancy fairy shrimp's critical habitat with the action area and approximately 40% of that overlap area is "likely not operable". The known occupied site of the Conservancy fairy shrimp in the Los Padres National Forest, where the critical habitat overlaps and the only NFS unit this species potentially occurs based on IPAC, is fenced to exclude impacts (2005 land management plan, available at USFS 2025a); therefore, the proposed action may affect, is not likely to adversely affect designated critical habitat for the Conservancy fairy shrimp.

For the vernal pool fairy shrimp, there is a 4.8% overlap of its critical habitat with the action area and approximately 50% of that is "likely not operable". While the Los Padres FEIS states that the vernal pools on the Los Padres are not subject to loss by agriculture or urban development (available at USFS 2025a), the proposed action could result in development of new roads in the long-term which could impact vernal pool critical habitat features. In the short-term, the Final Supplemental Environmental Impact Statement for the Southern California National Forests (USFS 2013a) states that implementation of the land management plans will be consistent with the Vernal Pool Ecosystems of California and southern Oregon Recovery Plan (71 FR 11441). New road construction within occupied critical habitat could potentially result in permanent impacts to the primary constituent elements of this species' critical habitat in the long-term (connectivity, hydrology); therefore, the proposed action may affect, is likely to adversely affect designated critical habitat for the vernal pool fairy shrimp.

Critical habitat for the St. Francis River crayfish includes streams with low flow velocity, shallow depths, appropriate water temperature, connectivity between occupied stream reaches, low stream embeddedness, sufficient forage (invertebrates, periphyton, and plant detritus), and low densities of non-native species (88 FR 25512). For the St. Francis River crayfish, impacts to critical habitat from in-stream work are expected to be temporary and conditions should return to normal after work is complete. Some features of crustacean critical habitat may be positively affected by future timber and road activities by increasing sunlight on surface waters, which in turn produces higher abundances of periphyton upon which crayfish feed. Given the small amount of overlap with designated critical habitat, that most in-stream work effects will be temporary, and that some effects may be positive, the proposed action may affect, is not likely to adversely affect designated critical habitat for the St. Francis River crayfish.

Determinations

For the Madison Cave isopod, of the occurrences available in the GBIF and NRM layers one record was documented on NFS lands, but outside of the action area. The recharge zone of the aquifer at any of the occupied sites is unknown, and therefore the zone of potential sources of contamination is also unknown (USFWS 1996). The recovery plan for this species indicates it may be highly sensitive to disturbances due to its low reproductive potential and small population sizes at some sites, but it is also noted that this species is difficult to study and collect (USFWS 1996). Karst and cave resources can be expected to occur on NFS lands underlain by limestone or marble or areas having exposed basaltic flows. Road construction or reconstruction activities and other developments are sources of sediment, debris, and other pollutants that, when entering karst or cave systems can damage them and their associated resources. While, the analysis done for the George Washington National Forest FEIS states only a fraction (less than a tenth of one percent) of potential habitat was determined to be on NFS lands (USFS 2014), the one occurrence record available is on the Jefferson National Forest, which does not mention this species specifically in the land management plan or FEIS. Because the recharge zone of occupied aquifers is unknown and the amount of potential habitat on the Jefferson National Forest is unknown, the proposed action could lead to groundwater contamination that would be **likely to adversely affect the Madison cave isopod**.

For the Conservancy fairy shrimp, although the exposure is high, it should be noted that this is due to a low number of occurrences overall from GBIF and NRM. This species has been known to occur in inventoried roadless areas on the Los Padres National Forest (2005 land management plan, available at USFS 2025a). Threats to this species include habitat loss, fragmentation, and degradation from development and agriculture. However, the known location of Conservancy fairy shrimp in the Los Padres National Forest is fenced to exclude impacts (2005 land management plan, available at USFS 2025a), therefore the proposed action is **not likely to adversely affect the Conservancy fairy shrimp**.

For the vernal pool fairy shrimp, although the potential exposure to activities within the action area is high, it should be noted that this is due to a low number of occurrences overall from GBIF and NRM. The Los Padres FEIS states that the vernal pools on the Los Padres are not subject to loss by agriculture or urban development (available at USFS 2025a), but the proposed action could result in development of new roads in the long-term which could impact vernal pool habitat. Roads and timber-related activities could impact the vernal pool fairy shrimp resulting in direct mortality, decreased reproduction, decreased gene flow, habitat fragmentation, and habitat alteration through the spread of invasive plant species; therefore, the proposed action is **likely to adversely affect the vernal pool fairy shrimp**.

For the San Diego fairy shrimp and Riverside fairy shrimp, occurrence data was not available for these species from GBIF and NRM and so the exposure to potential activities in the action area is unknown. However, threats to these two species include habitat loss and indirect effects from development and habitat fragmentation. Roads and timber-related activities could impact the San Diego fairy shrimp and Riverside fairy shrimp resulting in direct mortality, decreased reproduction, decreased gene flow, habitat fragmentation, and habitat alteration through the

spread of invasive plant species; therefore, the proposed action is **likely to adversely affect the San Diego fairy shrimp and Riverside fairy shrimp**.

For the big sandy crayfish, the primary impacts to species are from ground-disturbing activities that increase erosion and sedimentation, which degrades the stream habitat required by the species. Under the proposed action, potential in-stream work and sedimentation are expected to be temporary and conditions should return to normal after work is complete. Some features of crustacean habitat may be positively affected by future timber and road activities by increasing sunlight on surface waters, which in turn produce higher abundances of periphyton upon which crayfish feed. However, suitable habitat includes low-silt and unembedded slab boulders; slab boulders and cobble that become highly embedded with sediment deposits are not suitable for this species (USFWS 2025r). Further, females that are reproductively active select slab boulders closer to stream banks in late summer and fall when egg extrusion occurs, typically avoiding higher velocity flows that could dislodge sediments (USFWS 2025r). Depending on the timing and duration, sedimentation from road construction or timber harvest activities could reduce reproductive success of this species. Therefore, even temporary effects from in-stream work due to the proposed action may be **likely to adversely affect the big sandy crayfish**.

For the Big Creek crayfish and St. Francis River crayfish, the impacts to species from potential in-stream work and sedimentation are expected to be temporary and conditions should return to normal after work is complete. Some features of crustacean habitat may be positively affected by future timber and road activities by increasing sunlight on surface waters, which in turn produce higher abundances of periphyton upon which crayfish feed. However, the Big Creek crayfish and St. Francis River crayfish are currently threatened by non-native woodland crayfish species and there is currently no means to slow or stop the spread of this species where they have been introduced (88 FR 25512). The Big Creek crayfish and St. Francis River crayfish have experienced reduced abundances and range contractions where non-native invasive crayfish have established (88 FR 25512). The construction of new roads and potential for increased public access could facilitate the introduction and spread of non-native crayfish into more of the Big Creek crayfish and St. Francis River crayfish ranges and reduce overall viability of these species; therefore, the proposed action is **likely to adversely affect the Big Creek crayfish and St. Francis River crayfish.**

As of 2023, there were only eight isolated populations of Shasta crayfish known to the U.S. Fish and Wildlife Service and the land ownership overlapping these populations was identified as private and state parks (USFWS 2023u). However, of the occurrences available in the GBIF and NRM layers one record was documented on NFS lands, but outside of the action area. The Shasta crayfish has experienced population declines and competition due to the proliferation of the signal crayfish. If the Shasta crayfish is present within the action area or close enough to experience downstream effects, the construction of new roads and potential for increased public access could facilitate the introduction and spread of non-native crayfish into more of the Shasta crayfish's occupied range and reduce overall viability of this species; therefore, the proposed action is **likely to adversely affect the Shasta crayfish.**

Refer to Table 20 to see the effects determination listed for each crustacean species and critical habitat.

Table 21. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of crustacean species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Riparian Resilience	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Timber Harvest	Vegetation Management	Tree and Unit Marking	Walking	Chemical Contamination/Nutrients	Reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Timber Harvest	Vegetation Management	Tree and Unit Marking	ATV	Chemical Contamination/Nutrients	Reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Timber Harvest	Vegetation Management	Landings - Vegetation Clearing and Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Peak/Base Flows; Large Woody Debris; Change in Riparian Vegetation (Temperature); Riparian Resilience	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Timber Harvest	Vegetation Management	Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Riparian Resilience	Reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Timber Harvest	Vegetation Management	Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Large Woody Debris; Peak/Baseflows; Riparian Resilience	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Timber Harvest	Vegetation Management	Timber Haul	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Timber Harvest	Temporary Access Control	Road Gates Installation	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Drainage	Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Drainage	Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Drainage	Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in Riparian Vegetation (Temperature); Sediment/Turbidity; Chemical Contamination/Nutrients; Large Woody Debris	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Roads - Maintenance or Roads - Reconstruction	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Roads - Maintenance or Roads - Reconstruction	Surface Repair/Replacement	Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Surface Repair/Replacement	Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Surface Repair/Replacement	Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Roads - Maintenance or Roads - Reconstruction	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Chemical Contamination/Nutrients	Direct mortality, particularly of less mobile life stages; reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Roads - Maintenance or Roads - Reconstruction	Snow	Plowing	Heavy Equipment	Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Permanent and temporary Road Construction	Vegetation Clearance	Snags - Hand Felling	Motorized Hand Tools	Large Woody Debris	Habitat fragmentation resulting in gene flow reduction and decreased reproduction.
Permanent and temporary Road Construction	Vegetation Clearance	Snags - Mechanical Felling	Heavy Equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Large Woody Debris	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Permanent and temporary Road Construction	Vegetation Clearance	Tree Removal - Log Decks	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Permanent and temporary Road Construction	Vegetation Clearance	Tree Removal - Hauling	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Permanent and temporary Road Construction	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Road Density/Location; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Permanent and temporary Road Construction	Roadway template	Soil Removal	Heavy Equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Permanent and temporary Road Construction	Roadway template	Fill Placement	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Permanent and temporary Road Construction	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Permanent and temporary Road Construction	Drainage	Stream Crossings - Culvert Installation	Heavy Equipment	Change in Riparian Vegetation (Temperature); sediment/Turbidity; Chemical Contamination/Nutrients; Spawning Substrate; Peak/Baseflows; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.
Permanent and temporary Road Construction	Surfacing	Hauling of surface materials	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity
Permanent and temporary Road Construction	Surfacing	Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Sediment/Turbidity; Spawning Substrate; Chemical Contamination/Nutrients	Direct mortality, particularly of less mobile life stages; reduced water quality leading to reduced species fitness and reproduction, or direct mortality and reduced genetic diversity.
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Temporary Road Removal	Scarifying	Material removal	Heavy equipment	Sediment/Turbidity; Spawning Substrate	Direct mortality, particularly of less mobile life stages; sedimentation leading to reduced carrying capacity and density of subpopulations.
Temporary Road Removal	Culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Sediment/Turbidity; Spawning Substrate; Drainage Network	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations.
Temporary Road Removal	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in Vegetation (Temperature); Sediment/Turbidity; Spawning Substrate; Peak/Baseflows; Large Woody Debris; Riparian Resilience	Direct mortality, particularly of less mobile life stages; habitat fragmentation resulting in gene flow reduction and decreased reproduction; sedimentation leading to reduced carrying capacity and density of subpopulations; reduced water quality leading to decreased species fitness and reproduction, or direct mortality and reduced genetic diversity.

Snails

To assist in analyzing effects to a large number of species, ESA-listed species were grouped taxonomically. The snail group includes 4 total species including 3 endangered freshwater snails, and 1 threatened terrestrial snail (Table 22). Two of the listed species have designated critical habitat, however, no critical habitat for these ESA-listed snails overlaps with the action area. Two species, Anthony's riversnail and Alamosa springsnail, potentially occur on two National Forest Service units based on IPAC, while the other two listed snails may only occur on one NFS unit (see Appendix A).

Table 22. Threatened (T) and endangered (E) snail species in the action area. The presence of critical habitat (and percent overlap with the action area) is also provided. A summary of exposure and the effects determination is also shown.

Common Name	Scientific Name	ESA Status (A)	Critical Habitat in Affected Environment (B)	Exposure (C)	Effects Determinations for Species (D)	Effects Determinations for Critical Habitat (D)
Alamosa springsnail	*Tryonia alamosae*	E	N/A	Unknown	NLAA	N/A
Anthony's riversnail	*Athearnia anthonyi*	E	N/A	Unknown	LAA	N/A
Interrupted (=Georgia) Rocksnail	*Leptoxis foremani*	E	No	Unknown	LAA	N/A
Morro shoulderband (=Banded dune) snail	*Helminthoglypta walkeriana*	T	No	Unknown	LAA	N/A

Table notes: (A) T = threatened, E = endangered; (B) amount of critical habitat in the affected environment based on spatial analysis (Appendix A) using the USFWS Critical Habitat Shapefile available here for download: https://ecos.fws.gov/ecp/report/table/critical-habitat.html, N/A under critical habitat indicates critical habitat has not been designated; (C) Exposure = Unknown means insufficient data to determine (D) NLAA = may affect, not likely to adversely affect, LAA = may affect, likely to adversely affect, N/A = not applicable;

Road construction, road reconstruction, road use, and timber harvest can affect snail species by removing habitat, fragmenting habitat leading to a loss of gene flow between populations, creating edge effects that destabilize environmental conditions that support the species, and causing direct mortality through vehicle strikes and right-of-way maintenance equipment. The primary effects of the proposed action on snail species are discussed below and presented in Table 23. A discussion on species protection afforded by land management plans and other documents beyond the protections of the 2001 Roadless Rule is also included.

Three of the ESA-listed snails are freshwater snails, including the Alamosa springsnail, Anthony's riversnail, and interrupted rocksnail. The Alamosa springsnail occurs in one spring complex on private land with the primary threats to this species including collection, vandalism, and effects of droughts and flood events (USFWS 2018j, USFWS 2023v). Anthony's riversnail inhabits medium to large rivers with cobble or boulder substrates (USFWS 2023w). Interrupted rocksnail is endemic to the Coosa River drainage and inhabits bedrock, boulders, cobbles, and gravel in slowly moving water (USFWS 2014d). Both the Anthony's riversnail and interrupted

rocksnail are threatened by the degradation of water quality and habitat, and habitat fragmentation (USFWS 2011d, USFWS 2014d).

One of four ESA-listed snails in the action area is a terrestrial snail, the Morro shoulderband snail. This species is endemic to San Luis Obispo County in the central coastal region of California and is associated with sand or sandy soils, slopes not greater than 10 percent, and native coastal dune scrub vegetation (USFWS 2019s). Threats influencing the viability of Morro shoulderband snail populations at the time of listing included in part off-road vehicle activity and invasive species; however, since the time of listing the USFWS determined that these threats, particularly off-road vehicle activity, were no longer affecting the species (USFWS 2019s). The most common threats currently are those associated with land use practices that eliminate, reduce, fragment, and/or modify habitat used by the species (USFWS 2019s).

Major Stressors

Direct Mortality

Terrestrial snails: Road construction, reconstruction, or maintenance via mowing within the distribution of Morro shoulderband snail can alter habitat characteristics and can result in direct and indirect injury or mortality to snails within both affected and adjacent areas. In addition to direct mechanical injury or mortality, mowing and the construction of roads is likely to increase exposure of this species and therefore increase its vulnerability to desiccation and predation, particularly if these activities occur during the dry season when snails are aestivating (USFWS 2019s).

Habitat Fragmentation and Edge Effects

Freshwater snails: Similar to bivalves, habitat fragmentation impacts freshwater snails by isolating populations, leading to population decline. Improper construction of culverts and other in-stream infrastructure can convert flowing river habitat into non-flowing water features unsuitable for these listed freshwater snails. Additionally, culverts can act as barriers to dispersal of freshwater snails in some cases (Vaughan 2002). This isolation prevents gene flow between populations, leading to reduced genetic diversity, which can diminish fitness and adaptive potential. Consequently, isolated populations are more vulnerable to environmental stochasticity, pollution events, and local extinction (USFWS 2014d). Further, barriers to natural flow can alter the riverbed dynamics, causing sediment retention upstream and scouring of substrates downstream, which degrades essential microhabitats. Although in-stream work has the potential to fragment riverine habitat, the Forest Service designs and installs culverts in a manner to improve streamflow, not restrict it. Fragmentation can still occur during construction but those effects are assumed to be temporary and that surface flows will return to normal, or better than normal, when construction has ended. Short-term fragmentation of aquatic habitat is not expected to have population level effects to listed snails.

Terrestrial snails: Road construction, reconstruction, or maintenance via mowing within the distribution of Morro shoulderband snail can alter required habitat characteristics associated with vegetation structure and humidity. Loss of habitat connectivity and reduced ability to disperse between populations can reduce genetic diversity in the long-term (USFWS 2019s). Mechanical

activities associated with road construction, reconstruction, or mowing could increase vulnerability of the Morro shoulderband snail to desiccation and predation, particularly if mowing occurs during the dry season when snails are aestivating (USFWS 2019s).

Sedimentation and Pollution

Freshwater snails: Freshwater snails can be highly sensitive to water and habitat quality (Vaughan 2002). Anthony's riversnail is threatened by habitat and water quality degradation from activities that cause sedimentation and non-point source pollutants, including timber harvest and road construction (USFWS 2011d). Excess sediment and water pollution from these activities can alter stream channels, reduce riparian vegetation, increase turbidity, degrade water quality, and reduce food availability (USFWS 2011d). However, USFWS noted in the most recent 5-year review for the Anthony's riversnail that implementation of best management practices can be effective in protecting water quality and instream habitats, which can moderate threats in many instances as is the case with forestry (USFWS 2023w). The interrupted rocksnail requires flowing water, stable stream channels with minimal sediment and algae growth, and adequate water quality (USFWS 2014d). Historical causes of water and habitat degradation in the Coosa River and its tributaries where the interrupted rocksnail occurs included drainage from construction runoff USFWS 2014d).

Existing Protections

Federally listed threatened, endangered, and proposed species on NFS Lands, including inventoried roadless areas, would continue to be planned for and managed according to the requirements of law, regulation, and policy. All current Forest Service land management plans, through the National Forest Management Act (NFMA) and planning regulations (1982 Planning Rule or 2012 Planning Rule), are required to maintain and contribute to the recovery of threatened and endangered species and conserve proposed species in the plan area (39 CFR 219.9, 2012 Planning Rule; Sec. 219.19, 1982 Planning Rule). Specifically, under the 2012 Planning Rule, federally listed species are maintained in the plan area by providing the ecological conditions necessary to support the persistence of those species. These ecological conditions may include standards or guidelines related to habitat features, condition, or connectivity. It is assumed that under all alternatives, future land management plan revisions or amendments would comply with ESA, NFMA, and planning regulations to meet this requirement to the extent feasible within Forest Service authority and consistent with the inherent capability of the plan area (36 CFR 219.9 and FSH 1909.12).

The 2022 Cibola National Forest land management plan includes plan components for at-risk species, including the Alamosa springsnail (available at USFS 2025a), although the plan notes this species has only been documented in habitat directly adjacent to the Cibola's boundaries. More specifically, Table 100 of the Cibola National Forest Land Management Plan Final Environmental Impact Statement Volume 2 identifies which plan components alleviate or eliminate key threats to the Alamosa springsnail (USFS 2021). Existing protections identified include, in part, forestwide standards and guidelines to: prevent damage to soil, slope, and watershed conditions from timber harvest; protect stream ecosystems from fragmentation; and locate new roads and trails outside of riparian management zones (USFS 2021). The 2024 Gila

National Forest land management plan and associated documents do not contain species-specific protective measures for the Alamosa springsnail or identify this species as one of the ESA-listed species potentially present on the unit (land management plans available at USFS 2025a).

For the Anthony's riversnail, the 2004 Cherokee National Forest and 2023 Nantahala and Pisgah National Forests (where this species potentially occurs based on IPAC) land management plans and associated documents do not contain species-specific protective measures for this species or identify this species as one of the ESA-listed species potentially present on the unit (land management plans available at USFS 2025a). The Cherokee National Forest land management plan includes forestwide standards for watersheds including a standard that states specifically: "*Resource management activities that may affect soil and/or water quality will implement Tennessee Best Management Practices (BMPs) as a minimum to achieve soil and water quality objectives. when standards exceed BMPs, standards shall take precedence over Tennessee BMPs*". The Nantahala and Pisgah National Forests land management plan includes a forestwide standards for aquatic systems that states: "*Management activities shall be designed to avoid, minimize, or mitigate negative impacts on aquatic habitats and species unless the management objective is to protect a native species from encroachment by a non-native species…*". This land management plan also includes forestwide guidelines for aquatic systems including: "*During forest management activities such as timber harvest or road maintenance, channel-spanning and other passage techniques should be considered to promote passage of aquatic organisms…*" and "*Management activities should follow all applicable North Carolina and National Core Best Management Practices (BMPs) to meet relevant laws, regulations, and policies as described in the streamside zone and water resources sections. Exemptions to regulatory policies may be requested on a case-specific basis where application of the standard policy could result in greater resource damage*".

The interrupted rocksnail was federally listed in 2010 (75 FR 67512). The current land management plan for the Chattahoochee-Oconee National Forest (where these species potentially occur based on IPAC) land management plan is from 2004 and pre-dates the listing of this species and therefore does not include analysis or plan components specific to this species (land management plan available at USFS 2025a). However, this land management plan does include several forestwide standards protective of aquatic habitats.

For the Morrow shoulderband snail, the 2005 Los Padres National Forest (where this species potentially occurs based on IPAC) land management plan and associated documents do not contain species-specific protective measures for this species or identify this species as one of the ESA-listed species potentially present on the unit (land management plans available at USFS 2025a).

Critical Habitat

Final critical habitat was designated for the interrupted rocksnail in 2010 (75 FR 67512) and for the Morro shoulderband snail in 2001 (66 FR 9233). There is no overlap of the designated critical habitat for these two species with the action area.

Determinations

For the Alamosa springsnail, the Cibola National Forest land management plan states that this species has been documented in habitat directly adjacent to the Cibola's boundaries (land management plan available at USFS 2025a) and the USFWS states that this species is endemic to one spring complex on private land in Socorro County, New Mexico (USFWS 2018j, USFWS 2023v). The Gila National Forest land management plan does not identify this species as potentially present on the unit and this species is unlikely to be present based on USFWS documentation (USFS 2025a, USFWS 2018j, USFWS 2023v). The Cibola National Forest land management plan includes plan components to alleviate or eliminate key threats to the Alamosa springsnail, including from threats related from timber harvest and roads (USFS 2025a). Given that the Alamosa springsnail occurs outside of but adjacent to NFS lands and that the closest Forest Service unit to the species includes protective measures for this species related to timber harvest and roads, the proposed action is **not likely to adversely affect the Alamosa springsnail.**

For the Anthony's riversnail, occurrence data was not available for this species from GBIF and NRM and so the exposure to potential activities in the action area is unknown. Documented threats to this species include habitat and water quality degradation due to sedimentation and non-point source pollutants associated with timber harvest and road construction (USFWS 2011d). USFWS noted that implementation of best management practices can be effective in protecting water quality and instream habitats (USFWS 2023w). While the Cherokee National Forest and Nantahala and Pisgah National Forests land management plans do not contain species-specific protective measures for the Anthony's riversnail, they do include forestwide standards and guidelines that require the use of best management practices and protective measures in watersheds and aquatic habitats. However, if this species is exposed activities the response to sedimentation could include mortality and decreased reproductive capability, which could impact individuals and the population as a whole. For this reason, the proposed action is **likely to adversely affect the Anthony's riversnail.**

For the interrupted rocksnail, occurrence data was not available for this species from GBIF and NRM and so the exposure to potential activities in the action area is unknown. This species requires flowing streams with minimal sediment and algae growth, and adequate water quality (USFWS 2014d). Although the Chattahoochee-Oconee National Forest land management plan pre-dates the listing of this species, it does include plan components protective of aquatic habitats. However, if this species is exposed to activities the response to sedimentation could include mortality and decreased reproductive capability, which could impact individuals and the population as a whole. For this reason, the proposed action is **likely to adversely affect the interrupted rocksnail.**

For the Morro shoulderband snail, occurrence data was not available for this species from GBIF and NRM and so the exposure to potential activities in the action area is unknown. However, threats to this species include direct mortality, habitat loss, and indirect effects from habitat fragmentation due to road construction, reconstruction, and maintenance activities. Further, this species prefers areas of less than 10 percent slope where timber operability and roads associated with that activity are more likely. If the species is present in the action area, the proposed action

could impact the Morro shoulderband snail resulting in direct mechanical injury or mortality, increased desiccation and predation, and decreased gene flow; therefore, the proposed action is **likely to adversely affect the Morro shoulderband snail.**

Refer to Table 22 to see the effects determination listed for each snail species and critical habitat.

Table 23. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of snail species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of nest/den trees and snags	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Tree and Unit Marking	Walking	Human presence	Negligible response to human presence
			ATV	Noise; Human presence; Trampling/run over	Direct mortality; negligible response to noise
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil Compaction; Noise; Human presence; Change in vegetation; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence	Direct mortality; reduced food availability
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil Compaction; Noise; Human presence; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Timber Haul	Heavy Equipment	Soil Compaction; Noise; Human presence; Dust; Trampling/run over	Direct mortality
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil compaction; Human presence	Direct mortality
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Sedimentation	Direct mortality; sedimentation leading to mortality or habitat degradation of aquatic species

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; chemical contamination/nutrients; Drainage Network; Peak/Baseflows	Direct mortality; sedimentation leading to mortality or habitat degradation of aquatic species
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence; Loss of nest trees	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Loss of nest trees; Soil compaction; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; Loss of nest trees	Direct mortality
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust;	Direct mortality
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over	Direct mortality; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; sedimentation	Direct mortality; sedimentation leading to mortality or habitat degradation of aquatic species
	Revegetation	Seeding	Hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality

Plants

We have identified 130 flowering plant species, two fern species, one lichen species and one conifer species that occur in the action area (Table 24) which we are referring to as the plant taxa group. The plant taxa group includes 134 total species including 74 endangered and 60 threatened plant species. Of these, 38 species have critical habitat designated and critical habitat is proposed for two species. Plant species ranges in the action area vary from extremely small such as Pedate checker-mallow (*Sidalcea pedata*) encompassing less than 20 acres and occurring on just one National Forest to multi-regional in scale such as whitebark pine (*Pinus albicaulis*) which encompasses over 42.9 million acres on US Forest Service lands (USFWS 2021i) over multiple states and multiple National Forests. Most of the plant species in this analysis are found in upland habitats but 41 of the 134 plant species are found in aquatic, wetland or riparian habitats. The plant species in this assessment are found in a variety of habitat types from hardwood and coniferous forests to swamps and bogs to arid deserts to rock crevices and cliffs. Many plant species are specialized, growing on specific geological parent soil types such as serpentine and limestone soils. Life history of plant species in the action area varies from annual to biennial to very long lived, with some species living multiple centuries (whitebark pine; Arno and Hoff 1989).

Table 24. Threatened (T) and endangered (E) conifer, fern, flowering plant, and lichen species in the action area, the percent overlap of critical habitat with the action area for those species with designated critical habitat, the likely level of exposure to effects, and the effects determinations for the species and critical habitat. Exposure = Unknown means insufficient data to determine. Critical Habitat = N/A means no critical habitat designated.

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Alabama streak-sorus fern	*Thelypteris pilosa var. alabamensis*	T	N/A	Low	LAA	N/A
American Chaffseed	*Schwalbea americana*	E	N/A	Low	LAA	N/A
American Hart's-tongue fern	*Asplenium scolopendrium var. americanum*	T	N/A	Low	LAA	N/A
Arizona cliffrose	*Purshia subintegra*	E	N/A	Moderate	LAA	N/A
Arizona eryngo	*Eryngium sparganophyllum*	E	No	Low	LAA	N/A
Ash-grey paintbrush	*Castilleja cinerea*	T	10.75	Moderate	LAA	LAA
Bakersfield cactus	*Opuntia treleasei*	E	N/A (Proposed CH)	Unknown	LAA	N/A
Bartram's stonecrop	*Graptopetalum bartramii*	T	N/A	Moderate	LAA	N/A
Bear Valley sandwort	*Arenaria ursina*	T	No	High	LAA	N/A
Beardless chinchweed	*Pectis imberbis*	E	0	Low	LAA	N/A
Blowout penstemon	*Penstemon haydenii*	E	N/A	Low	LAA	N/A

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Blue ridge goldenrod	*Solidago spithamaea*	T	N/A	Low	LAA	N/A
Braunton's milk-vetch	*Astragalus brauntonii*	E	No	Unknown	LAA	N/A
Bunched arrowhead	*Sagittaria fasciculata*	E	N/A	Unknown	LAA	N/A
California jewelflower	*Caulanthus californicus*	E	N/A	Moderate	LAA	N/A
California Orcutt grass	*Orcuttia californica*	E	N/A	Low	LAA	N/A
California taraxacum	*Taraxacum californicum*	E	3.07	High	LAA	LAA
Canby's dropwort	*Oxypolis canbyi*	E	N/A	Low	LAA	N/A
Capa rosa	*Callicarpa ampla*	E	N/A	Low	LAA	N/A
Chupacallos	*Pleodendron macranthum*	E	N/A	Low	LAA	N/A
Clay phacelia	*Phacelia argillacea*	E	N/A	Moderate	LAA	N/A
Cook's lomatium	*Lomatium cookii*	E	No	High	LAA	N/A
Cushenbury buckwheat	*Eriogonum ovalifolium var. vineum*	E	2.7	High	LAA	LAA
Cushenbury milk-vetch	*Astragalus albens*	E	3.11	High	LAA	LAA
Cushenbury oxytheca	*Oxytheca parishii var. goodmaniana*	E	0.25	High	LAA	NLAA
Decurrent false aster	*Boltonia decurrens*	T	N/A	Low	LAA	N/A
Dwarf Lake iris	*Iris lacustris*	T	N/A	Low	LAA	N/A
Eastern prairie fringed orchid	*Platanthera leucophaea*	T	N/A	Low	LAA	N/A
Encinitas Baccharis	*Baccharis vanessae*	T	N/A	Low	LAA	N/A
Fassett's locoweed	*Oxytropis campestris var. chartacea*	T	N/A	Low	LAA	N/A
Fleshy-fruit gladecress	*Leavenworthia crassa*	E	No	Low	LAA	N/A
Florida skullcap	*Scutellaria floridana*	T	N/A	Low	LAA	N/A
Gambel's watercress	*Rorippa gambellii*	E	N/A	Low	LAA	N/A
Gentner's Fritillary	*Fritillaria gentneri*	E	N/A	Low	LAA	N/A
Georgia rockcress	*Arabis georgiana*	T	No	Unknown	LAA	N/A
Godfrey's butterwort	*Pinguicula ionantha*	T	N/A	High	LAA	N/A
Greene's tuctoria	*Tuctoria greenei*	E	0.37	Low	LAA	NLAA
Harperella	*Ptilimnium nodosum*	E	N/A	Low	LAA	N/A

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Harper's beauty	*Harperocallis flava*	E	N/A	Low	LAA	N/A
Heliotrope milk-vetch	*Astragalus montii*	T	100	Moderate	LAA	LAA
Heller's blazingstar	*Liatris helleri*	T	N/A	Low	LAA	N/A
Holy ghost ipomopsis	*Ipomopsis sancti-spiritus*	E	N/A	High	LAA	N/A
Houghton's goldenrod	*Solidago houghtonii*	T	N/A	Low	LAA	N/A
Huachuca water-umbel	*Lilaeopsis schaffneriana var. recurva*	E	No	Low	LAA	N/A
Keck's checker-mallow	*Sidalcea keckii*	E	No	Low	LAA	N/A
Kincaid's lupine	*Lupinus sulphureus ssp. kincaidii*	T	No	Unknown	LAA	N/A
Kral's water-plantain	*Sagittaria secundifolia*	T	N/A	Low	LAA	N/A
Kuenzler hedgehog cactus	*Echinocereus fendleri var. kuenzleri*	T	N/A	Unknown	LAA	N/A
Lakeside daisy	*Hymenoxys herbacea*	T	N/A	Moderate	LAA	N/A
Large-flowered skullcap	*Scutellaria montana*	T	N/A	Low	LAA	N/A
Lassics lupine	*Lupinus constancei*	E	8.79	High	LAA	LAA
Last chance townsendia	*Townsendia aprica*	T	N/A	Moderate	LAA	N/A
Layne's butterweed	*Senecio layneae*	T	N/A	Low	LAA	N/A
Leafy prairie-clover	*Dalea foliosa*	E	N/A	Low	LAA	N/A
Lee pincushion cactus	*Coryphantha sneedii var. leei*	T	N/A	Unknown	LAA	N/A
Lewton's polygala	*Polygala lewtonii*	E	N/A	Low	LAA	N/A
Louisiana quillwort	*Isoetes louisianensis*	E	N/A	Low	LAA	N/A
Lyrate bladderpod	*Lesquerella lyrata*	T	N/A	Unknown	LAA	N/A
Macfarlane's four-o'clock	*Mirabilis macfarlanei*	T	N/A	High	LAA	N/A
Maguire primrose	*Primula maguirei*	T	N/A	Moderate	LAA	N/A
McDonald's rock-cress	*Arabis mcdonaldiana*	E	N/A	High	LAA	N/A
Mead's milkweed	*Asclepias meadii*	T	N/A	Low	LAA	N/A
Missouri bladderpod	*Physaria filiformis*	T	N/A	Low	LAA	N/A

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Mohr's Barbara's buttons	*Marshallia mohrii*	T	N/A	Unknown	LAA	N/A
Morefield's Leather flower	*Clematis morefieldii*	E	N/A	Unknown	LAA	N/A
Mountain golden heather	*Hudsonia montana*	T	0.67	High	LAA	NLAA
Munz's onion	*Allium munzii*	E	No	Low	LAA	N/A
Navasota ladies-tresses	*Spiranthes parksii*	E	N/A	Low	LAA	N/A
Neches river rose-mallow	*Hibiscus dasycalyx*	T	No	Low	LAA	N/A
Nevin's barberry	*Berberis nevinii*	E	No	Low	LAA	N/A
No common name	*Ilex sintenisii*	E	N/A	Unknown	LAA	N/A
No common name	*Schoepfia arenaria*	T	N/A	Unknown	LAA	N/A
No common name	*Ternstroemia subsessilis*	E	N/A	Unknown	LAA	N/A
Northeastern bulrush	*Scirpus ancistrochaetus*	E	N/A	High	LAA	N/A
Northern wild monkshood	*Aconitum noveboracense*	T	N/A	Unknown	LAA	N/A
Ocmulgee skullcap	*Scutellaria ocmulgee*	E	No	Low	LAA	N/A
Palo Colorado	*Ternstroemia luquillensis*	E	N/A	Unknown	LAA	N/A
Palo de Jazmin	*Styrax portoricensis*	E	N/A	Low	LAA	N/A
Papery whitlow-wort	*Paronychia chartacea*	T	N/A	Unknown	LAA	N/A
Parish's daisy	*Erigeron parishii*	T	3.07	Moderate	LAA	LAA
Pigeon wings	*Clitoria fragrans*	T	N/A	Low	LAA	N/A
Pima pineapple cactus	*Coryphantha scheeri var. robustispina*	E	N/A	Low	LAA	N/A
Pitcher's thistle	*Cirsium pitcheri*	T	N/A	Low	LAA	N/A
Pondberry	*Lindera melissifolia*	E	N/A	Low	LAA	N/A
Prairie bush-clover	*Lespedeza leptostachya*	T	N/A	Low	LAA	N/A
Price's potato-bean	*Apios priceana*	T	N/A	Low	LAA	N/A
Purple amole	*Chlorogalum purpureum*	T	1.39	Low	LAA	LAA
Pygmy fringe-tree	*Chionanthus pygmaeus*	E	N/A	Unknown	LAA	N/A
Relict trillium	*Trillium reliquum*	E	N/A	Low	LAA	N/A
Roan Mountain bluet	*Hedyotis purpurea var. montana*	E	N/A	Unknown	LAA	N/A
Rock gnome lichen	*Gymnoderma lineare*	E	N/A	Moderate	LAA	N/A

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Rough-leaved loosestrife	*Lysimachia asperulaefolia*	E	N/A	Unknown	LAA	N/A
Ruth's golden aster	*Pityopsis ruthii*	E	N/A	Low	LAA	N/A
Sacramento Mountains thistle	*Cirsium vinaceum*	T	N/A	Moderate	LAA	N/A
Sacramento prickly poppy	*Argemone pleiacantha ssp. pinnatisecta*	E	N/A	Moderate	LAA	N/A
San Bernardino bluegrass	*Poa atropurpurea*	E	6.24	Moderate	LAA	LAA
San Bernardino Mountains bladderpod	*Lesquerella kingii ssp. bernardina*	E	36.34	Moderate	LAA	LAA
San Diego thornmint	*Acanthomintha ilicifolia*	T	No	High	LAA	N/A
San Francisco peaks ragwort	*Packera franciscana*	T	No	Low	LAA	N/A
San Joaquin adobe sunburst	*Pseudobahia peirsonii*	T	N/A	Unknown	LAA	N/A
Santa Ana River woolly-star	*Eriastrum densifolium ssp. sanctorum*	E	N/A	Unknown	LAA	N/A
Schweinitz's sunflower	*Helianthus schweinitzii*	E	N/A	Low	LAA	N/A
Shale barren rock cress	*Boechera serotina*	E	N/A	Unknown	LAA	N/A
Showy stickseed	*Hackelia venusta*	E	N/A	High	LAA	N/A
Slender Orcutt grass	*Orcuttia tenuis*	T	1.44	Moderate	LAA	LAA
Slender-horned spineflower	*Dodecahema leptoceras*	E	N/A	Moderate	LAA	N/A
Small whorled pogonia	*Isotria medeoloides*	T	N/A	High	LAA	N/A
Smooth coneflower	*Echinacea laevigata*	T	N/A	High	LAA	N/A
Southern Mountain wild-buckwheat	*Eriogonum kennedyi var. austromontanum*	T	No	High	LAA	N/A
Spalding's catchfly	*Silene spaldingii*	T	N/A	Moderate	LAA	N/A
Spreading avens	*Geum radiatum*	E	N/A	Moderate	LAA	N/A
Springville clarkia	*Clarkia springvillensis*	T	N/A	High	LAA	N/A
Swamp pink	*Helonias bullata*	T	N/A	High	LAA	N/A
Tennessee yellow-eyed grass	*Xyris tennesseensis*	E	N/A	Unknown	LAA	N/A

Common Name	Scientific Name	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Species Effects Determination	Critical Habitat Effects Determination
Texas golden gladecress	*Leavenworthia texana*	E	No	Low	LAA	N/A
Texas prairie dawn-flower	*Hymenoxys texana*	E	N/A	Low	LAA	N/A
Thread-leaved brodiaea	*Brodiaea filifolia*	T	No	High	LAA	N/A
Todsen's pennyroyal	*Hedeoma todsenii*	E	No	Low	LAA	N/A
Ute ladies'-tresses	*Spiranthes diluvialis*	T	N/A	Moderate	LAA	N/A
Uvillo	*Eugenia haematocarpa*	E	N/A	Moderate	LAA	N/A
Vail Lake ceanothus	*Ceanothus ophiochilus*	T	No	High	LAA	N/A
Virginia sneezeweed	*Helenium virginicum*	T	N/A	High	LAA	N/A
Virginia spiraea	*Spiraea virginiana*	T	N/A	Low	LAA	N/A
Webber's ivesia	*Ivesia webberi*	T	No	High	LAA	N/A
Wenatchee Mountains checkermallow	*Sidalcea oregana var. calva*	E	No	Unknown	LAA	N/A
Western lily	*Lilium occidentale*	E	N/A	High	LAA	N/A
Western prairie fringed orchid	*Platanthera praeclara*	T	N/A	Moderate	LAA	N/A
White birds-in-a-nest	*Macbridea alba*	T	N/A	Low	LAA	N/A
White bladderpod	*Physaria pallida*	E	N/A	Unknown	LAA	N/A
White fringeless orchid	*Platanthera integrilabia*	T	N/A	Low	LAA	N/A
Whitebark pine	*Pinus albicaulis*	T	N/A	High	LAA	N/A
Whorled sunflower	*Helianthus verticillatus*	E	No	Unknown	LAA	N/A
Yreka phlox	*Phlox hirsuta*	E	N/A	Low	LAA	N/A
Zuni fleabane	*Erigeron rhizomatus*	T	N/A	Moderate	LAA	N/A

Threats to these plant species vary and are presented in Table 25. Major stressors applicable to the proposed action and potential effects to threatened and endangered plants are discussed below.

Table 25. Threats to listed plant species occurring or potentially occurring in the action area.

Common Name (ECOS)	Scientific Name (ECOS)	Threats
Alabama streak-sorus fern	*Thelypteris pilosa var. alabamensis*	modification/alteration of habitat, road and dam construction, reservoir management, recreation use, loss of canopy from fire and disease, timber harvest, development, restricted range, small population size
American Chaffseed	*Schwalbea americana*	urban development, land conversion to pine plantations and agriculture, fire suppression or inappropriate fire return interval, roadside maintenance/ditching (direct impacts and change in hydrology), inadequate regulatory mechanisms, herbicide use, small population size, temperature change, extreme weather events, sea level rise, invasive species
American Hart's-tongue fern	*Asplenium scolopendrium var. americanum*	logging, development, quarrying, invasive species, increased drought and decreased winter snowpack, collection, recreation, and observer impacts
Arizona cliffrose	*Purshia subintegra*	urbanization and associated impacts (recreation, road development), mining, grazing, increased aridity, reduced precipitation, invasive species, narrow habitat requirements, a small number of known populations (four), and a restricted, disjunct distribution, herbivory
Arizona eryngo	*Eryngium sparganophyllum*	groundwater extraction; water diversion; livestock grazing; agricultural and urban development; invasive plants; inadequate regulations
Ash-grey paintbrush	*Castilleja cinerea*	urban development, roads and trails, off-road vehicle use, alteration of hydrology, grazing, trampling, invasive nonnative plants, mining, fire suppression,; very limited dispersal range, with most seeds falling within a few meters of the parent plant, making it difficult for the species to recover and colonize new areas after a large-scale disturbance
Bakersfield cactus	*Opuntia treleasei*	habitat loss and fragmentation due to agricultural and urban development, oil, gas, and other mining exploration, invasion of nonnative grasses, off-road vehicle use, and extreme weather events
Bartram's stonecrop	*Graptopetalum bartramii*	roads, bridge construction, culvert establishment/replacement, drying habitat or loss of shade, fire suppression, invasive plants, mining, erosion, trampling, debris and sediment, altered fire regime, recreation activity, and cumulative impacts to small populations.
Bear Valley sandwort	*Arenaria ursina*	urbanization, roads and trails, off-road vehicle traffic, utilities operations, alteration of hydrology, trampling, nonnative invasive plants, mining, fire suppression, creation of fuel breaks. The habitat is extremely fragile and susceptible to long-lasting or irreversible damage from disturbances like off-road vehicle use, trampling, and urbanization.
beardless chinchweed	*Pectis imberbis*	non-native plants; altered fire regimes; drought; grazing by livestock and wildlife; erosion, sedimentation, and burial from road and trail maintenance, recreation, mining, border related activity; post-wildfire runoff; small population size and limited distribution
Blowout penstemon	*Penstemon haydenii*	Energy development, the continual loss of blowout habitat due to stabilization of dunes accompanied with the lack of effective habitat management, herbivory, potential disease, inadequate existing regulatory mechanisms, fire suppression, the direct impact of roads built in an area and the indirect effect of those roads increasing accessibility of the area) of human activities has the potential to threaten the extant populations. These activities include but are not limited to wind energy development, oil and gas exploration and development, quarrying, and recreational use of ORVs. local catastrophes, negative effects to pollinators such as pesticides, drought, higher temperatures, insects, destruction of habitat on private property due to lack of outreach, rangeland vegetation stabilization, fire suppression
Blue ridge goldenrod	*Solidago spithamaea*	Recreation (trampling), woody succession, rockslides, increased temperature and change in precipitation
Braunton's milk-vetch	*Astragalus brauntonii*	urban development, fragmentation of habitat, fragmented land ownership of individual occurrences, alteration of fire cycles, fire suppression, nonnative plants, recreation, grazing, reduction in pollinators
Bunched arrowhead	*Sagittaria fasciculata*	hydrological alteration, land use change, urbanization, development, surface runoff, invasive species, hurricanes. Flooding, limited gene variation and flow

Common Name (ECOS)	Scientific Name (ECOS)	Threats
California jewelflower	*Caulanthus californicus*	habitat loss and fragmentation due to agricultural and urban development, oil, gas, and other mining exploration, competition with nonnative grasses, neonicotinoid pesticides
California Orcutt grass	*Orcuttia californica*	habitat loss from urbanization and agriculture, damage from off-highway vehicles and other recreational activities, competition from invasive plant species. The species is also impacted by fire and fire suppression efforts, and having a naturally small population size
California taraxacum	*Taraxacum californicum*	altered hydrology, urbanization, roads and unauthorized off-highway vehicle (OHV) use, developed and dispersed recreation, mining activities, grazing, competition with other plant species, hybridization, limited numbers of plants, fire suppression measures, and drought.
Canby's dropwort	*Oxypolis canbyi*	habitat loss and degradation caused by wetland drainage for agriculture and forestry, fire suppression, which allows woody shrubs to encroach on its habitat, and hydrological alterations from nearby groundwater withdrawal or construction
Capa rosa	*Callicarpa ampla*	habitat loss from forest management (e.g., establishment and maintenance of plantations, selective cutting), trail and road construction and maintenance, and agriculture, small and localized population
Chupacallos	*Pleodendron macranthum*	forest management practices such as the establishment and maintenance of plantations, selective cutting, trails maintenance, and shelter construction, maintenance of power lines and associated rights-of-way, hurricanes, landslides, flash flooding, extreme weather events, low genetic diversity, and little natural recruitment, small population size
Clay phacelia	*Phacelia argillacea*	transportation and transmission line development, erosion, grazing and herbivory are primary threats. Low level threats include recreational use, loss of genetic diversity, disease, and invasive non-native species.
Cook's lomatium	*Lomatium cookii*	habitat loss and degradation from agricultural and urban development, road construction, mining, and incompatible grazing; competition from non-native plants; habitat alteration due to fire suppression leading to woody species encroachment; road vehicle use; change to WOTUS definitions (vernal pools may no longer be protected under the Clean Water Act)
Cushenbury buckwheat	*Eriogonum ovalifolium var. vineum*	mining, off-highway vehicle use, energy development, road construction, fire suppression activities, and other land use activities that result in trampling or ground disturbance
Cushenbury milk-vetch	*Astragalus albens*	mining, off-highway vehicle use, energy development
Cushenbury oxytheca	*Oxytheca parishii var. goodmaniana*	habitat destruction from limestone mining-direct removal of mined materials, dumping of overburden onto unmined habitats, and road construction associated with the mining operations; off-highway vehicles; invasive species; small population size and limited range; energy development
Decurrent false aster	*Boltonia decurrens*	destruction or conversion of habitat to cropland, prolonged, late season flooding, herbicide use, loss of habitat due to encroachment by woody vegetation
Dwarf Lake iris	*Iris lacustris*	trampling, soil moisture reductions, development, recreational activities, invasive species, woody encroachment and natural succession, drought, high-lake levels and associated flooding
Eastern prairie fringed orchid	*Platanthera leucophaea*	lack of land management of high-quality wetland habitats on which this species depends, including increased development, spread of exotic species, and fire suppression, hydrology alteration
Encinitas Baccharis	*Baccharis vanessae*	urban and recreational development, nonnative plants, fuel modification, fire suppression activities, altered fire regime, trampling, habitat fragmentation
Fassett's locoweed	*Oxytropis campestris var. chartacea*	habitat loss and degradation from development, changes in lake hydrology due to groundwater depletion, damming, and high-capacity wells; competition from invasive plants; agricultural and residential runoff; herbicide drift, trampling from recreational use
Fleshy-fruit gladecress	*Leavenworthia crassa*	grazing, agriculture, development, herbicides, road construction, off-road vehicles, woody encroachment, invasive species, very restricted geographical extent
Florida skullcap	*Scutellaria floridana*	Commercial timber production, clearcutting, selective thinning, soil disturbance, urban development, conversion of timberlands into pasture lands for cattle, and fire management and exclusion, drainage and hydrology alteration, catastrophic events (hurricanes and flooding)

Common Name (ECOS)	Scientific Name (ECOS)	Threats
Gambel's watercress	*Rorippa gambellii*	habitat loss and degradation, competition with both native and nonnative species, development and urbanization, stochastic loss due to small population size and low numbers of individuals, watershed eutrophication, hydrologic alteration, genetic introgression, and effects of increased locally extreme rainfall events, accelerated sea level rise, increased drought, loss of ecosystem services provided by marshes and estuaries, and considerable beach narrowing and erosion.)
Gentner's Fritillary	*Fritillaria gentneri*	high severity fire, herbivory (including grazing), small population size, trampling (recreation), roads, habitat succession
Georgia rockcress	*Arabis georgiana*	timber harvest, road building, invasive species, development, including quarrying, residential and commercial development, recreation, and hydropower dam construction, forest succession
Godfrey's butterwort	*Pinguicula ionantha*	Commercial timber production, urban development, and fire management and exclusion of fire, hydrological alteration, illegal harvest, saltwater inundation by hurricanes and sea level rise, road widening and new roads, off-road vehicles
Greene's tuctoria	*Tuctoria greenei*	habitat loss, competition from introduced and weedy plant species, and improper (too intensive or too little) grazing regimes, small population sizes, unfitting grazing regimes, inadequacy of existing regulatory mechanisms, and pesticides
Harperella	*Ptilimnium nodosum*	Changes to hydrologic regime due to natural and anthropogenic events and development (increase in impervious surfaces), Reduction in water quality, Siltation and erosion from poor land-use practices (logging and agricultural), and road building/maintenance, competition from other plant species, Direct physical disturbance, development, trampling, equipment use, placement of fill, land conversion, or vehicle access points (stream crossings), herbivory, flooding, ice scour and drought
Harper's beauty	*Harperocallis flava*	Industrial forestry practices, fire suppression, and soil and hydrological disturbances; very limited range, small population number, the rarity of this species' habitat (about 85 to 98% of herb bog habitat has been estimated to be lost).
Heliotrope milk-vetch	*Astragalus montii*	oil and gas exploration and development, road construction, off-road vehicle use
Heller's blazingstar	*Liatris helleri*	trampling due to high volume of recreational activities (i.e. hiking and rock climbing); fire suppression; development; small population size and low genetic diversity
Holy ghost ipomopsis	*Ipomopsis sancti-spiritus*	fire suppression (needs open, bare soil), roads (vehicle use, maintenance), invasive species, recreation/trampling, pesticide, extremely limited range and small population size
Houghton's goldenrod	*Solidago houghtonii*	Residential development, dune stabilization projects, recreational vehicle use, and excessive human foot traffic, modification or contamination of ground water, non-native invasive species, increase temperature, decrease in ice cover
Huachuca water-umbel	*Lilaeopsis schaffneriana var. recurva*	habitat degradation due to groundwater pumping and drought, competition from invasive plants, and livestock grazing. Other threats include alteration of fire regimes, increased sedimentation from runoff, and damage from recreation
Keck's checker-mallow	*Sidalcea keckii*	agricultural conversion, development, livestock grazing, altered fire regime, small population size, limited recruitment, habitat fragmentation, and genetic isolation of populations
Kincaid's lupine	*Lupinus sulphureus ssp. kincaidii*	habitat loss from urban and agricultural development; encroachment by woody vegetation due to the absence of natural disturbance (like fire); competition from invasive non-native plants; habitat fragmentation; reduced pollinator movement, and, to a lesser extent, overcollection and insect herbivory
Kral's water-plantain	*Sagittaria secundifolia*	clearing of riparian zones for silviculture, residential expansion, recreational development, surface mining, or agricultural activities, land development, increased recreational use of streams, mining operations, and forestry practices without properly implemented protections and best management practices, water quality, hydrological alteration, impoundments, sedimentation, erosion, scouring floods, flooding from dam breaks, inadequacy of existing regulatory mechanisms, off-road vehicle use, pollutants, garbage dumping and sewage leaks, causing eutrophication and elevated coliform bacteria levels (Whetstone 1988), which foster filamentous algae growth that hinders light availability for the plant's growth and flowering, amplifying vulnerability due to the species' limited population size

Common Name (ECOS)	Scientific Name (ECOS)	Threats
Kuenzler hedgehog cactus	*Echinocereus fendleri var. kuenzleri*	illegal collection; livestock grazing; fire (natural and prescribed can severely impact populations due to slow regeneration); drought
Lakeside daisy	*Hymenoxys herbacea*	quarrying, recreation, fire suppression, construction, invasive species, habitat destruction (powerline maintenance, ATV use, snow plowing), succession of competitive overgrowth by woody species, over-collecting for gardens, inadequacy of existing regulatory mechanisms, small population size and self-incompatibility
Large-flowered skullcap	*Scutellaria montana*	logging, residential development, grazing, wildfire, and timber harvest and forest clearing, mining and quarrying, off-road vehicles, recreation (hiking), construction and maintenance of trails, invasive plants, power line maintenance, wildfire suppression activities (clearing fire lines)
Lassics lupine	*Lupinus constancei*	seed predation by small mammals, drought and high temperatures, woody vegetation encroachment; wildfire, small population size
Last chance townsendia	*Townsendia aprica*	mineral and energy exploration and development, trampling by livestock, wild horses, and burros, off-highway vehicles, inadequacy of regulatory mechanisms, small population sizes
Layne's butterweed	*Senecio layneae*	destruction, modification, and/or curtailment of habitat or range due to urbanization, habitat fragmentation, road construction and maintenance, off road vehicle use, and mining, invasive species, herbicide spraying,
Leafy prairie-clover	*Dalea foliosa*	Habitat degradation due to invasive exotic or native species encroachment, illegal ORV use, incompatible management of utility rights-of-way, overgrazing, fire suppression. Many populations continue to suffer from inadequate management to control encroachment by invasive exotic or native species
Lee pincushion cactus	*Coryphantha sneedii var. leei*	Road maintenance; trail use and maintenance; ungulate trampling; fire suppression; range improvement construction, use, and maintenance; pesticide; severe drought; wildfire
Lewton's polygala	*Polygala lewtonii*	habitat loss, fragmentation, and changes in land use, development, land clearing, habitat modification from inadequate management (lack of fire and suppression of fire), herbivory and feral hogs, Inadequacy of existing regulatory mechanisms, intrinsic factors (pollinator-limited dispersal, small, isolated populations, low recruitment, and restricted range), stochastic events (hurricanes, droughts, wildfires), and human disturbances, off-road vehicles
Louisiana quillwort	*Isoetes louisianensis*	development, gravel mining, off-road vehicle use, sedimentation (including from bridge and culvert construction/replacement), overstory clearing-forestry (need buffer), wild hogs and beavers.
Lyrate bladderpod	*Lesquerella lyrata*	limited range and vulnerability; habitat loss and fragmentation from development (e.g., housing and agriculture); lack of natural or managed disturbance which leads to competitive exclusion by other plants; direct impacts from human activities such as mowing and herbicide application.
Macfarlane's four-o'clock	*Mirabilis macfarlanei*	habitat degradation from invasive plant species like cheatgrass and yellow starthistle, and damage from human activities such as off-road vehicle use, trampling (particularly cattle), and road construction. Other threats include herbicide drift, landslides, small population size, and low seed viability
Maguire primrose	*Primula maguirei*	recreational rock climbing, potential release of mountain goats, road construction was originally considered a threat but not currently
McDonald's rock-cress	*Arabis mcdonaldiana*	mining, road widening and maintenance, woody encroachment in the absence of fire, construction of fire breaks
Mead's milkweed	*Asclepias meadii*	urbanization, agricultural conversion, habitat fragmentation, herbicide/pesticide application, lack of prescribed fire for prairie management, habitat destruction from feral hogs, pollinator declines
Missouri bladderpod	*Physaria filiformis*	lack of preservation and management of glade habitats on which this species depends, residential and commercial development, quarrying, spread of exotic species, encroachment by eastern red cedar (Juniperus virginiana), fire suppression
Mohr's Barbara's buttons	*Marshallia mohrii*	habitat loss from agricultural, residential, and forestry development, woody plant encroachment, and invasive species. Human activities like mowing, herbicide use, and off-road vehicles in rights-of-way are also significant dangers, as are fire suppression, which can lead to overgrowth

Common Name (ECOS)	Scientific Name (ECOS)	Threats
Morefield's Leather Flower	*Clematis morefieldii*	mining, residential development, road maintenance, timber operations/incompatible forestry practices, overgrazing, invasive species, small population size, isolation of populations, habitat fragmentation
Mountain golden heather	*Hudsonia montana*	fire suppression, recreation (trampling), invasives
Munz's onion	*Allium munzii*	urban and agricultural development, clay mining, wildland fire, including pre- and post-burn management (e.g., creation of fire breaks), off-highway vehicle activity, nonnative species
Navasota ladies-tresses	*Spiranthes parksii*	habitat loss and modification, from mines, landfills, pipelines, highway construction projects, private development projects, herbivory, limited federal and state regulatory protection on private lands, lack of long-term habitat management plans and permanent protection mechanisms for occupied habitats to continue
Neches river rose-mallow	*Hibiscus dasycalyx*	development and road construction, conversion of land for residential, commercial, and agriculture use, habitat encroachment (native and non-native plants), hydrological alteration, herbicides, hybridization and genetic swamping with more common, closely related Hibiscus species,
Nevin's barberry	*Berberis nevinii*	urbanization, recreational activities, road widening, fire management practices, altered fire regimes, small population size, low reproductive output
No common name	*Ilex sintenisii*	hurricanes, habitat destruction from development, communication facilities installation, road construction and maintenance, forest management (selective cutting), small population size
No common name	*Schoepfia arenaria*	Deforestation for construction, including urban, industrial and tourist development, Inadequacy of existing regulatory mechanisms, small population, genetic isolation, lack of information on species, hurricanes, landslides, invasive species.
No common name	*Ternstroemia subsessilis*	hurricanes, landslides, genetic variation, forest management, development, road maintenance
Northeastern bulrush	*Scirpus ancistrochaetus*	higher temperatures, drying, water level changes, oil and gas development, development. Road construction, forestry, recreation, and plant competition are factors that may affect the species, but they are not primary factors
Northern wild monkshood	*Aconitum noveboracense*	slow growing, very sensitive to disturbance, and is threatened by various disturbances including dams, reservoirs, road/powerline construction and maintenance, quarrying and logging operations, grazing, development
Ocmulgee skullcap	*Scutellaria ocmulgee*	destruction and modification resulting from development (i.e., urbanization) and deer herbivory, invasion by nonnative invasive plant species, drought, change in rainfall patterns. While silviculture and agriculture are not primary threats, at least one population has been negatively affected.
Palo Colorado	*Ternstroemia luquillensis*	hurricanes, landslides, genetic variation, forest management, development, road maintenance
Palo de Jazmin	*Styrax portoricensis*	forest management, development, hurricanes, landslides, and genetic variation
Papery whitlow-wort	*Paronychia chartacea*	habitat loss and degradation due to agricultural and residential development; changes in its natural fire cycle. Overgrown vegetation in areas that lack periodic wildfires can outcompete this plant, while development fragments and destroys its habitat
Parish's daisy	*Erigeron parishii*	mining, urban development, off-highway vehicle use, energy development, fire suppression activities, and other land use activities that could result in trampling or ground disturbance, grazing
Pigeon wings	*Clitoria fragrans*	habitat loss through development and land conversion, inadequate fire management, non-native plant species, isolated populations within a limited and fragmented geographic range
Pima pineapple cactus	*Coryphantha scheeri var. robustispina*	Habitat loss/alteration from urbanization, commercial development, nonnative plants, alteration of fire regimes, livestock activity, and recreation and border activity. Overutilization (illegal collection) has led to an increase in mortality. Most occur on private and State Trust lands, so they and their habitats are not subject to Federal protection unless there is a Federal nexus to a proposed action. Slow growth, low population numbers, sparsely distributed
Pitcher's thistle	*Cirsium pitcheri*	invasive plants and recreational use (primarily beach use and foot traffic in dunes, but also in some off-road vehicle use); shoreline stabilization projects, high-water

Common Name (ECOS)	Scientific Name (ECOS)	Threats
		levels, alter local dune geomorphic processes affecting the creation and maintenance of PITH habitat; invasive weevil that feeds on the seed heads.
Pondberry	*Lindera melissifolia*	drainage ditching and subsequent habitat conversion; even ditching without later conversion of land use can alter the water regime in a manner that reduces the plant's vigor or eliminates it from the site. Domestic hogs, cattle grazing, and timber harvesting have also impacted the plants at some sites; change to WOTUS definitions
Prairie bush-clover	*Lespedeza leptostachya*	habitat loss from prairie conversion, drought, invasive plants, development, herbicides, vegetation encroachment, and hybridization. Most of its native habitat has been converted to cropland, pasture, or developed sites. Remaining prairies are small and isolated, reducing genetic diversity and resilience.
Price's potato-bean	*Apios priceana*	Heavy clear-cutting, right-of-way maintenance for roads and utilities (e.g., mowing), nonnative, invasive plants, feral hogs, herbicide overspray from agricultural fields
Purple amole	*Chlorogalum purpureum*	displacement from nonnative, invasive species, anthropogenic disturbance from ongoing military activities and/or wildlife species, wildfire (it may have some species benefits, but because of the clumped and limited distribution in a semi-arid environment, wildfire could destroy a substantial portion of the population), including extended drought and increasing temperatures
Pygmy fringe-tree	*Chionanthus pygmaeus*	habitat loss through development and land conversion, inadequate fire management, and non-native plant species
Relict trillium	*Trillium reliquum*	Habitat loss and degradation from development, road construction and maintenance, timber harvest, altered hydrology, White-tailed Deer (*Odocoileus virginianus*) and cutworm caterpillar (*Cerastis tenebrifera*) herbivory, invasive plant species competition and feral hogs.
Roan Mountain bluet	*Hedyotis purpurea var. montana*	habitat destruction from commercial and residential development, trampling from recreation, soil compaction, erosion, air pollution
Rock gnome lichen	*Gymnoderma lineare*	habitat destruction from human recreational activities, development and logging (direct habitat destruction and altering light and humidity requirements), changes to its specialized microclimate due to exotic pests and air pollution, habitat degradation from balsam wooly adelgid (exotic pest kills Fraser fir resulting in loss of forest canopy and subsequent changes in microclimate and desiccation).
Rough-leaved loosestrife	*Lysimachia asperulaefolia*	habitat loss from fire suppression and associated ecological succession; herbicide use and mowing in rights-of-way; low seed production; wetland drainage; hydrological changes
Ruth's golden aster	*Pityopsis ruthii*	hydroelectric dam operations and river management, altered flow (hydrology), woody vegetation encroachment, commercial and private whitewater recreational use and associated trampling of plants., inadequacy of regulatory mechanisms, road and highway construction, water quality, pollutants.
Sacramento Mountains thistle	*Cirsium vinaceum*	water availability and development, grazing, recreation use, invasion from native or non-native insects, invasive or non-native plant species, herbicide application
Sacramento prickly poppy	*Argemone pleiacantha ssp. pinnatisecta*	water development/diversion, livestock grazing and trampling, recreation (off-highway vehicles), surface disturbance, herbicide applications, road and pipeline construction and maintenance, drought
San Bernardino bluegrass	*Poa atropurpurea*	urbanization, land conversion, alteration of hydrological conditions, off-highway vehicle (OHV) activity, road and trail construction, use and maintenance, utilities (telephone trenching) and associated soil removal, and vandalism, mining, trampling by livestock and humans; indirect effects of grazing and browsing; competition with other plant species; non-native, invasive plants, habitat fragmentation; hybridization with the nonnative *P. pratensis* (Kentucky bluegrass); and limited numbers of *P. atropurpurea* individuals, and possible localized distribution, altered hydrology
San Bernardino Mountains bladderpod	*Lesquerella kingii ssp. bernardina*	limestone mining; urban and recreational developments; off-highway vehicle use; road construction; fire suppression activities; loss of pollinators

Common Name (ECOS)	Scientific Name (ECOS)	Threats
San Diego thornmint	*Acanthomintha ilicifolia*	habitat loss, fragmentation, urbanization, development, nonnative species, fire, precipitation changes, mining, agriculture, recreational activities
San Francisco peaks ragwort	*Packera franciscana*	recreational activities like hiker trampling and trail construction, as well as natural events such as avalanches.
San Joaquin adobe sunburst	*Pseudobahia peirsonii*	conversion of habitat to residential development, and to a lesser extent, agricultural land conversion, competition from nonnative plants, incompatible grazing practices (although the species benefits from grazing to remove competition), transmission line maintenance, recreational activities, mining, road construction and maintenance, a flood control project, and other human impacts, flooding, increase in duration of dry season
Santa Ana River woolly-star	*Eriastrum densifolium ssp. sanctorum*	alteration of hydrology (natural fluvial processes) from dam construction, flood control; mining; changes in precipitation and temperature
Schweinitz's sunflower	*Helianthus schweinitzii*	alteration of fire and grazing regimes; residential and industrial development; mining; invasive species competition and woody plant encroachment; highway construction and improvement; utility right of way maintenance; herbicides and herbivory; erosion; incorrect mowing regime; altered fire regime.
Scrub buckwheat	*Eriogonum longifolium var. gnaphalifolium*	habitat loss and degradation from conversion for agriculture and development; fire suppression; invasive species; dumping and off-road vehicle use
Shale barren rock cress	*Boechera serotina*	habitat destruction/damage from construction activities or pedestrian traffic, invasive species, fire suppression, white-tailed deer browse
Showy stickseed	*Hackelia venusta*	single, small population highly vulnerable, fire suppression, trampling, bio-control weevil (*Mogulones cruciger*), competition from nonnative plant species, herbicide and road de-icer use, landslide or mass wasting (downslope earth movement), low rates of viable seed production and seedling establishment
Slender Orcutt grass	*Orcuttia tenuis*	habitat loss and degradation from urban and agricultural development, as well as altered vernal pool hydrology. Other threats include competition from invasive plants, inappropriate grazing, contamination from herbicides and other pollutants, and off-highway vehicle use.
Slender-horned spineflower	*Dodecahema leptoceras*	development (including transportation projects and pipeline construction); mining; altered hydrology/interruption of natural flood regimes (e.g., flood control, reservoir expansion); nonnative invasive plants; dumping and trash
Small whorled pogonia	*Isotria medeoloides*	Conversion of forests for residential or commercial development, including roads, power lines, and clearings, destroys populations and their necessary buffer zones. Many former sites have been lost to development. Improperly managed logging and mechanical clearing can remove canopy trees, drastically changing light levels and disrupting the soil and fungal networks the orchid depends on; vehicles, clearcutting, forest succession to closed canopies, collection, development, pathogen, herbivory (deer), ORVs, small, isolated populations leading to low genetic diversity
Smooth coneflower	*Echinacea laevigata*	habitat loss and degradation from development, fire suppression, and maintenance of rights-of-way like highways; encroachment of invasive species; collecting for commercial and horticultural purposes; damage from herbicides and beetles; particularly drought. relies on open habitats maintained by natural disturbances like fire
Southern Mountain wild-buckwheat	*Eriogonum kennedyi var. austromontanum*	urbanization; recreational activity; roads and trails; fire suppression; alteration of hydrology; nonnative plants; mining; trampling; fuel-wood harvesting; changes in temperature, precipitation, snowpack
Spalding's catchfly	*Silene spaldingii*	invasive nonnative plants, problems associated with small geographically isolated populations, changes in the wildfire regime and wildfire effects, land conversion associated with urban and agricultural development, adverse grazing and trampling by domestic livestock and native herbivores, herbicide spraying, off-road vehicle use, insect damage and disease, impacts from prolonged drought, and inadequacy of existing regulatory mechanisms.
Spreading avens	*Geum radiatum*	high elevation rocky sites, spruce-fir or northern hardwoods forests; logging, roads; recreation (trampling, erosion)

Common Name (ECOS)	Scientific Name (ECOS)	Threats
Springville clarkia	*Clarkia springvillensis*	drought, road maintenance activities, road improvements, competition from nonnative plants, altered fire regimes, genetic risks associated with narrow distribution and small population size
Swamp pink	*Helonias bullata*	wetland habitat loss and degradation caused by development (housing, roads, agriculture), drainage, pollution, sedimentation, groundwater extraction, habitat succession, invasive plants, feral hogs damage to habitat, herbivory (deer), low seedling establishment rate
Telephus spurge	*Euphorbia telephioides*	negatively correlated with density of flatwoods, shading does not tolerate shade or increased groundcover; controlled burns have a positive effect; persists in areas of disturbance such as roadsides; threats- timber production, urban development (real estate, road development and powerline maintenance) and fire suppression (needs fire), cattle grazing, sea level rise, catastrophic events.
Tennessee yellow-eyed grass	*Xyris tennesseensis*	agricultural/silvicultural (timber) production, development, road construction and maintenance, gravel quarries, overcollection, succession (woody plant encroachment), hydrological alteration, invasive plants
Texas golden gladecress	*Leavenworthia texana*	habitat destruction from development projects, woody encroachment and invasive species, extremely limited distribution, and vulnerability to localized disturbances such as pipelines and road construction
Texas prairie dawn-flower	*Hymenoxys texana*	Urbanization, agriculture, infrastructure projects, road construction, invasive species, hydrological alteration, extremely limited distribution
Thread-leaved brodiaea	*Brodiaea filifolia*	urbanization, altered hydrology, discing for fire suppression, nonnative invasive plants, grazing, off-highway vehicle activity, manure dumping, vandalism
Todsen's pennyroyal	*Hedeoma todsenii*	illegal grazing, soil erosion, mineral exploration, wildfire, and its naturally low reproductive success and poor dispersal ability
Ute ladies'-tresses	*Spiranthes diluvialis*	habitat loss from anthropogenic activities (urban development, water management, agriculture, livestock grazing, recreation, and invasive plants), vegetative succession, drought
Uvillo	*Eugenia haematocarpa*	establishment and maintenance of plantations, selective cutting, trails maintenance, and shelter construction; maintenance of power lines and associated rights-of-way, landslides, low genetic diversity, little natural recruitment, hurricanes
Vail Lake ceanothus	*Ceanothus ophiochilus*	Fire suppression and urban development are major threats to this species because they disrupt its natural fire cycle, preventing the necessary conditions for its regeneration (fire-dependent germination). Other threats: invasive nonnative plants, wildland fire and fire management, hybridization/introgression
Virginia sneezeweed	*Helenium virginicum*	wetland habitat destruction and alteration, including residential development, incompatible agriculture, filling and ditching of sinkhole ponds, and hydrological disruption. These activities reduce the seasonally flooded habitats the species depends on. Delisting proposed Aug 5, 2025
Virginia spiraea	*Spiraea virginiana*	competition by both native and non-native species and anthropogenic disturbances to habitat (e.g., development, railroads, roads, clearing, mowing, recreational use of rivers, dam construction), changes in hydrologic flow, herbivory
Webber's ivesia	*Ivesia webberi*	habitat loss and fragmentation from development, invasive species competition, altered fire regimes, livestock grazing, all of which reduce its already limited and specialized range
Wenatchee Mountains checkermallow	*Sidalcea oregana var. calva*	low numbers of individuals in most of the populations; competition from nonnative invasive plants and native overstory and understory plants; landscape level fire suppression; loss of habitat due to human development.
Western lily	*Lilium occidentale*	habitat loss from development (cranberry bogs, housing, roads) and successional encroachment of trees and shrubs in the absence of fire; hydrological alterations; deer and insect herbivory, fungal infections, collection for horticulture, soil compaction, and the small size of many populations leading to a loss of genetic diversity; change to WOTUS definitions
Western prairie fringed orchid	*Platanthera praeclara*	land conversion, hydrologic changes, and encroaching dominant vegetation (native plants and non-native, invasive plant species), Stochastic events (e.g., extreme drought or agricultural practices), genetic diversity, inbreeding depression
White birds-in-a-nest	*Macbridea alba*	habitat loss and degradation from development (i.e. logging, timber production, and coastal development) and poor land management (including fire suppression, fertilization of pine plantations, and off-road vehicle use). Fire suppression is a

Common Name (ECOS)	Scientific Name (ECOS)	Threats
		major threat because this species relies on periodic fires to clear out woody encroachment, and a lack of fire leads to too much shade
White bladderpod	*Physaria pallida*	Oil and gas pipeline, roads and well pad construction, farming and grazing, non-native and shrub species encroachment, surface mining, transportation/road maintenance, herbicide use.
White fringeless orchid	*Platanthera integrilabia*	human development, incompatible silvicultural practices, invasive plant species, disturbance by feral hogs, pond construction, ditching, and incompatible right-of-way maintenance, altered hydrology, herbivory (deer, hogs), changes in populations of pollinators or mycorrhizal fungi, or from intense rainfall events, inadequacy of regulatory mechanisms, low resilience
Whitebark pine	*Pinus albicaulis*	White pine blister rust, mountain pine beetle, altered fire regimes
Whorled sunflower	*Helianthus verticillatus*	habitat loss from agriculture and development, fire suppression, invasive species, small, fragmented populations. These pressures reduce its rare calcareous wet prairie and meadow habitats, making the species critically imperiled.
Yreka phlox	*Phlox hirsuta*	residential development, logging, fire suppression activities, ongoing highway maintenance or construction activities, off-road vehicle use, illegal collection, and vandalism, other threats include competition with exotic plants, herbicide application, grazing by domestic animals, inadequate existing regulatory mechanisms, and potential extirpation as a result of random events
Zuni fleabane	*Erigeron rhizomatus*	mineral exploration and development; prolonged drought; grazing; off-road vehicle use

Road construction, road use, and timber harvest can affect plants and plant habitats by direct physical damage, habitat loss, soil disturbance, changes in microhabitat conditions (e.g., light, humidity), and changes in hydrology that can degrade the fragile ecosystems where rare plants survive. The presence of roads increases human access and vehicle traffic, which can introduce invasive species, spread pollutants, and compact soils, further stressing native vegetation. Timber harvest intensifies these impacts by removing canopy cover, altering microclimates, and disrupting ecological processes essential for plant survival. Collectively, these activities reduce population sizes, isolate plant communities, and can make recovery more difficult under the Endangered Species Act. The primary effects of the proposed action on conifer, fern, flowering plant, and lichen species are discussed below and presented in Table 27. A discussion on species protection afforded by land management plans and other documents beyond the protections of the 2001 Roadless Rule is also included.

Major Stressors

Direct Mortality and Habitat Loss

Road construction and timber harvest physically move or remove vegetation and soil, causing direct mortality of exposed plants in the short term, and reduction, degradation, or elimination of plant populations and habitat, including critical habitat, in the longer term. Even narrow roads can eliminate microhabitats essential for rare species. As seen in the Deconstruction Table (project activities and taxa response, Table 27), trampling is an impact associated with numerous management activities that are part of the proposed action. Severity depends on the intensity, frequency, and duration of disturbance, but trampling can lead to direct mortality or reduced plant size, vigor, and reproductive ability, while more intense disturbances can favor invasive species, decrease species richness, and alter a site's ability to support rare plants.

Heavy machinery and trampling can directly crush plants and seedlings. Activities like trenching and grading can physically move or bury plants. Use of vehicles and heavy machinery in road construction and maintenance, timber harvest, and fire suppression activities, disturb the ground (direct mortality) and lead to soil compaction, which can lead to changes in hydrology, reproduction, and seed dispersal (indirect mortality). Soil disturbance also leads to conditions that favor invasive species (Dixon et al. 2024), which may threaten listed plants and/or habitats.

Roads create favorable conditions for invasive plants by providing light gaps, dispersal corridors, and reduced competition. Vehicles and road maintenance equipment can spread invasive plant seeds, further contributing to their proliferation (Coffin et al. 2021). Off-road or off-highway vehicles (OHVs) are cited as a threat for many threatened and endangered plant species, and OHV use will increase in areas where new roads are built. OHV use leads to continued physical damage to plants, soil disturbance and compaction, and spread of invasive species.

Ground disturbance and soil compaction

Soil compaction from road construction and timber harvest may negatively affect threatened and endangered plants by limiting their access to essential resources (water, air, nutrients) and physically impeding their growth, often giving a competitive advantage to more resilient invasive species. Compacted soil has increased bulk density and soil strength, making it difficult for roots to penetrate and expand. Roots may become stunted, thicker, and shallower, unable to form robust taproots or access deeper soil layers for stable water and nutrient sources. Restricted root systems are less efficient at exploring the soil volume and absorbing nutrients, particularly immobile ones like phosphorus (Frene et al. 2024). This can lead to nutrient deficiencies, even if the nutrients are present in the soil in adequate amounts. In addition to impeding growth, soil compaction drastically reduces the vital pore spaces in the soil that hold air. Roots require oxygen for respiration, and low oxygen levels (anaerobic conditions) in compacted soil can suffocate roots and impair their function. Compacted soil exhibits decreased water infiltration, leading to increased surface runoff and erosion (Pandey and Bennett 2024). This also creates a patchy soil moisture environment, where some areas may be waterlogged (leading to root rot) while others remain too dry and hard for plants to thrive. Beneficial soil organisms, such as earthworms and microbial communities, are highly sensitive to compaction due to reduced pore space and altered aeration. This disruption impairs essential ecosystem services like organic matter decomposition and nutrient cycling, further degrading soil fertility. Soil disturbance or removal also has the potential to degrade or remove the soil seed bank of listed plant species.

Heavy machinery used in salvage operations can compact soil, increase erosion, and disrupt seed banks and seed germination. These changes can be particularly harmful to rare plants with narrow ecological tolerances. Leverkus et al. (2020) conducted a global meta-analysis showing that salvage logging negatively affects ecosystem services which may affect rare plants. Puig-Gironès et al. (2020) observed slower plant regeneration in post-fire salvage logged areas, which may reduce the availability of resources for rare plant species but may benefit other rare plants that depend on more open and disturbed conditions.

Some species will benefit from ground disturbance, (shorter term, recurrent) as they are disturbance dependent. Examples include Price's Potato-bean (*Apios priceana*), which requires

moderate disturbance to maintain open canopies from natural processes and from human created openings including silvicultural activities and road rights-of-way and Brauton's milkvetch (*Astragalus brauntonii*), a post-disturbance species that requires fire or other ground disturbance for seedling germination and to reduce vegetative competition. Of the 134 plant taxa species, approximately 60 (44.8%)-need some level and type of disturbance including fire, ground disturbance, vegetation or canopy removal, and natural hydrological processes (for example, flooding, scouring).

Habitat Fragmentation

Habitat fragmentation is explicitly identified as a threat to 14 plant species in this analysis and likely is a concern for many more listed species within the action area. Road construction and timber harvest break continuous habitats into smaller, isolated patches, which reduces effective population size and increases genetic isolation for rare plants, raising extinction risk through demographic stochasticity and inbreeding (Hulting et al. 2024). Habitat fragmentation can lead to population isolation—plant populations become separated, which restricts the movement of pollinators, limits gene flow, and reduces genetic diversity (leading to inbreeding and lower vigor). This makes populations less resilient and more susceptible to disease and environmental changes. Smaller, isolated populations have a higher risk of local extinction due to random disturbances or demographic stochasticity (chance events).

Fragmentation effects can vary across a species' range but often produce measurable declines in endangered plant performance toward patch edges and range margins (Laurance et al. 2007). At edges, microclimatic shifts (higher light, temperature fluctuations, lower humidity) change germination, seedling survival, and adult physiology for shade-adapted species, while edges also facilitate invasion by nonnative plants and altered herbivore or pathogen dynamics that further depress native recruitment. Fragmentation also alters mutualisms: pollinator and seed-disperser behavior often changes near edges and in small patches, reducing pollination success and seed set for plants that depend on specialized partners, which compounds demographic declines for already threatened taxa (Laurance et al. 2007). Fungal diversity and networks may decline with land use intensity reducing resiliency (Carneiro de Melo Moura et al. 2025) and fragmentation (Grilli et al. 2017). Fungal associations are especially essential to many of the listed orchid species. Research shows reduced reproductive output, poorer offspring performance, and altered functional traits, from habitat fragmentation pushing many vulnerable species closer to extinction (Hulting et al. 2024).

In general, plant species that reproduce asexually (via rhizomes or roots) are highly susceptible to habitat fragmentation by roads because their populations get physically separated and cannot easily expand across the disturbed ground.

Invasive non-native species

Roads and soil disturbance from silvicultural activities can be the main facilitator for the introduction, invasion, spread and dominance of invasive and non-native species (Coffin et al. 2021). Invasive, non-native species are responsible for the decline of the largest number of threatened and endangered species (Prosser and Brain 2024) in the United States, estimated to be 42 percent of listed species (Kurth 2017). Invasive and non-native species causing these declines

include plants, animals, fungi, and pathogens. Of the 134 plant taxa in this analysis 81 (60 percent) are threatened by invasive or non-native species. Invasive and non-native plant species compete with rare plants for light, nutrients and space resulting in mortality, loss of vigor, reduced reproduction and loss of suitable habitat. They can also alter soil conditions (tamarisk) and facilitate more frequent fire (downy brome) thus altering fire regimes and affect forest regeneration (Oswalt et al. 2007). Invasive and non-native animals such as feral hogs affect plants through direct mortality and by altering habitat through ground disturbance. Feral hogs are the most frequently identified non-native animal threatening the plant taxa in this analysis. They can cause ground disturbance especially in wetlands, resulting in plant mortality, altering hydrology and facilitating invasive plant species thus making the habitat less suitable for the short and longer term. Under certain conditions silviculture may be used to help prevent and control invasive species (Muzika 2017). The US Forest Service implements BMPs and has a National Strategic Framework for Invasive Species Management that provides consistent direction across forests to prevent, detect, control, manage, restore and rehabilitate areas with invasive species (USFS 2013b). Forest Service Manual 2900 provides policy and direction for the prevention and control of invasive species on National Forest System lands. This policy and direction will help minimize the effects of invasive and non-native species from the proposed action on the plant taxa identified in this analysis.

Herbicides and Pesticides

Herbicides and other pesticides are used to control invasive, non-native and other non-desirable species along roadsides, disturbed areas and during and after silvicultural activities. Non-target application, overspray, drift, leaching and volatilization can negatively affect plants resulting in mortality, injury, reduced vigor and reproduction and failure to germinate (Wan et al. 2025; Rinella et al. 2009; Fishel 2011). Applications of herbicides and other pesticides may also negatively affect pollinators (Sánchez-Bayo and Wyckhuys 2019; Straw et al. 2021, Straw et al. 2022; Dicks et al. 2021; Linguadoca et al. 2022; Tamburini et al. 2021) resulting in decline of plant pollinated species (Pan et al. 2024; Biesmeijer et al.2006). Herbicides and pesticides were found to be a threat to 25 (18.5 percent) of the 135 plant taxa in this analysis.

The USDA and USDOI developed best management practices to enhance pollinator habitat on federal lands (USDA and USDI 2015). Forest Service Manual Chapter 2150 outlines the Forest Service policy and direction for the safe and effective use of pesticides. Prior to application of pesticides, forests are required to complete a Pesticide Use Proposal (PUP) that is then reviewed for safety and effectiveness. Sensitive areas, including areas where threatened, endangered or proposed plants are located, will be identified during the PUP process. In addition, individual project consultation that includes pesticide use will include design criteria and measures to minimize pesticide impacts to rare plants. As such, the effects of herbicides and pesticides are not considered further in this analysis.

Loss of Pollinators

As discussed elsewhere in this document, the proposed action is likely to adversely affect some insects, some of which serve as essential pollinators for threatened and endangered plant species. Loss of pollinators disrupts specialized plant–pollinator mutualisms, causing reduced fruit and seed set, lower genetic diversity, and heightened extinction risk for already small and fragmented

populations of threatened and endangered plants. Global declines in bees, butterflies, bats, and other pollinators—driven primarily by habitat loss, pesticide exposure, and long-term alteration of weather patterns—undermine reproduction for most angiosperms, as animal pollinators are vital for the reproduction of over 75% of flowering plants. Recent assessments indicate that more than 22% of native North American pollinators face elevated extinction risk (Cornelisse et al. 2025), with native bees particularly imperiled, amplifying reproductive failure and demographic collapse in plant species reliant on these partners. The cascading effects extend beyond individual species, degrading ecosystem functions and resilience as pollinator-dependent plants falter, further threatening biodiversity and ecological stability (Artamendi et al. 2025).

Pollutants

Roads, mining and silvicultural operations produce pollutants that can harm plants both through air deposition on plants and soil and through runoff, vehicle exhaust and spray and through application of dust abatements and snow deicers. Pollutants include dust particulates, nitrogen oxide, sulfur oxides, metals and salts (Neher et al. 2013). Plants can uptake these pollutants through the air and the soil. Roads and some mining and silvicultural activity produce dust, which deposits on leaves resulting in reduced photosynthesis and clogged stomata affecting efficient respiration and transpiration. Pollutants deposited by air, road surface runoff and vehicle spray are found on foliage and in soil resulting in toxicity to plants and change in pH (Mullerova et al. 2011, Neher et al. 2013). Some nutrient deposition from roads can benefit plants by acting as a fertilizer or changing soil pH. Effects on plants can occur more than 30 m from roads (Hao et al. 2025). The spray zone where road pollutants are deposited are 15 ft or greater from roadsides (Gould 2013).

Lichens can be especially sensitive to air pollution along roadsides from vehicle exhaust. Air pollution affects lichen physiology including photosynthesis, respiration, and cell membranes (St. Clair 1987). In addition, the habitat for lichens may also be affected by air pollution from sulfur dioxide emitted by vehicles (USFWS 2020k). Air pollution is considered to affect some lichen species including Rock gnome lichen (*Gymnoderma lineare*) (USFWS 2020k).

Pollutants were found to be a threat to five (3.7 percent) of the 134 plant taxa in this analysis. Pollutants from roads are expected to mostly negatively affect the plant taxa in this analysis although a few species may benefit.

Runoff, Erosion, and Sedimentation

Erosion from roads and other disturbed areas can be from wind or water that carries sediment from roads, roadsides, open pits or other disturbed areas creating sheets, gullies, rills or landslides. Topography, soil type, road type and amount of traffic affect the amount and type of erosion. Erosion and sedimentation occur both constantly and in large pulses during weather and/or use events. Erosion which removes and redeposits sediment can result in burying, bending or uprooting plants, wholly or in part. Depending on the amount of erosion this can affect individual plants or whole occurrences. Effects to rare plants from erosion and runoff include mortality, reduced vigor, damage to all parts of the plant, and reduction of population. Some plants need and benefit from natural erosional and depositional disturbance processes (such as

Ute ladies'-tresses, *Spiranthes diluvialis,* in wetland and riparian areas) whereas others are threatened by this process.

Altered hydrology

Roads and associated transportation support (bridges, culverts, ditches, etc.) can alter hydrology (Coffin et al. 2021; Jones et al. 2000) in rare plant habitat. Of the 134 plant species in this analysis 45 (33.3 percent) have been identified as being threatened by hydrological alteration. Altered hydrology can affect plants by changing the hydroregime such as the timing, duration, frequency and quantity of water (Cassanova and Brock 2000; Raulings et al. 2010; Nielson et al. 2013) in rare plant habitat. Changes in hydroregimes can change habitat conditions and result in habitat conversion that no longer supports rare plants (Cantu de Leija and King 2023, De Jager 2012; Raulings et al. 2010). Changes include flooding, draining, altered hydroperiods both in length of time and seasonality of water retention or drying, and scouring. These changes can affect the physical and chemical soil conditions such as redox potential and oxygen availability and ability for plants to uptake nutrients. The correct amount of water availability is important during life stages such as seedling germination and recruitment and reproduction. Drying conditions can result in plant mortality, reduced vigor and reproduction, and failure to germinate or recruit. Plants can be buried by water or sediment or uprooted either wholly or in part by flooding resulting in mortality. Some plants need flooding or scouring to create and maintain suitable habitat such as the Santa Ana River wooly star (*Eriastrum densifolium* ssp. *sanctorum*) and altering that process reduces suitable habitat. Alterations to hydrology from the proposed action are most likely to result in negative effects to most of the plant species in the action area, especially those that are threatened by altered hydrology although some plants may benefit by increasing water availability.

Light environment and Temperature

Construction and maintenance of roads and mines and silvicultural activities reduce or remove vegetation creating areas with more open canopies. These areas with reduced or removed vegetation and more open canopies create a change in both temperature and light conditions for plants. Surfaces of roads and cleared areas can absorb heat and radiate that heat to the adjacent environment, creating heat islands along roadsides. These higher surface and air temperatures can negatively impact plants by reducing soil water availability, increasing transpiration, creating water stress resulting in plant mortality, reduced growth and vigor. Plants that are adapted to arid conditions may benefit from an increase in heat and aridity.

Likewise open canopies can transmit more light into cleared areas. This can be beneficial to plants adapted to high light conditions but be detrimental to plants that grow in partial or full shade under tree canopies. Old growth forests are important for providing habitat for many species (Franklin et al. 1981). Threatened, endangered and proposed plants whose habitat includes stable closed or partial canopy light environment and are subjected to rapidly changed light and temperature environment are likely to react negatively resulting in mortality and reduced vigor. These openings may make habitat for some threatened, endangered and proposed species unsuitable. Some threatened, endangered and proposed species that are dependent on fire or other disturbances may respond positively to periodic vegetation removal from

silvicultural activities and repeatedly disturbed roadside environments that increase both light availability and temperature.

Fuels management and succession

One of the purposes for the proposed action is to help manage areas with elevated wildfire risk due to years of fire suppression leading to overgrowth or large stands of dead or diseased vegetation. Addressing wildfire risk through prescribed fire or vegetation management may be beneficial to a number of threatened and endangered plant species that require occasional fire or other disturbance to clear away competing vegetation. Lack of fire or succession that occurs due to fire suppression is a threat to 35 (26 percent) of the 134 plant species in this analysis, so the proposed action may benefit these species and their habitats.

Critical Habitat

For critical habitat analyses, we consider the effects of the proposed action on the physical and biological features (PBFs) of the habitat described in the Final critical habitat listing rule to determine the level of impact. Of the 134 total species included in this flowering plant, fern, lichen, and conifer analysis, 38 have critical habitat designated and two have critical habitat proposed. Of the 38 species, 14 have critical habitat that overlaps the action area. Of these, nine species have critical habitat that exceeds 2% overlap with the action area, and of these nine, 5 species exceed 5% critical habitat overlap with the action area. The amount of critical habitat by species overlapping the action area is found in Table 24. The main PBFs for listed threatened and endangered plant species generally fall into categories relating to their essential habitat needs and interactions with their environment. These include:

- **Suitable Plant Community and Habitat:** The specific native plant communities and physical landscapes (e.g., forests, wetlands, specific soil types) required for the plant's survival, growth, and reproduction. This often includes maintaining appropriate plant-plant interactions and overall ecosystem resilience.
- **Appropriate Soil Conditions:** The presence of specific soil types, nutrient levels, and moisture regimes (e.g., adequate water quality and flow regimes) that the plant species requires.
- **Pollination and Seed Dispersal Mechanisms:** The conditions and presence of necessary pollinators or dispersal agents (e.g., insects, birds, specific wind patterns) that allow the plant to reproduce effectively.
- **Protection from Threats:** Features that ensure the plant community is protected from significant threats, such as:
    - **Invasive species:** Mechanisms to resist the introduction and proliferation of non-native species that compete for resources.
    - **Altered fire regimes:** Maintaining a natural fire frequency/intensity or protecting areas where fire regimes have been altered.

- **Overutilization:** Regulatory mechanisms or physical protections against overharvesting for commercial, recreational, or scientific purposes.
- **Disease and Predation:** Defense against specific diseases or predators that pose a significant risk (e.g., fungal diseases).
- **Weather Impacts:** Habitat features that confer resilience to the effects of long-term alteration of weather patterns, such as changes in temperature or precipitation.
- **Genetic Diversity:** The maintenance of sufficient genetic variation within populations to ensure long-term adaptability and reduce risks associated with inbreeding depression.

These features are specific to each species and are detailed in relevant recovery plans and federal register documents.

For one species, Heliotrope milk-vetch (*Astragalus montii*), 100 percent of its critical habitat overlaps with the action area. Likewise, a significant portion of critical habitat for San Bernardino Mountains bladderpod (*Lesquerella kingii* ssp. *bernardina*) occurs in the action areas (36.34 percent). The San Bernadino National Forest Plan includes protections for San Bernardino Mountains bladderpod (Table 26) that will help to reduce or avoid impacts to the species' critical habitat. In the case of Heliotrope milk-vetch, where the proposed action will not benefit the species or its habitat and for the other plant species whose critical habitat overlaps the action area, individual forests will consult on specific projects that may impact critical habitat and develop protective measures to minimize impacts to critical habitat. However, the FS cannot discount some adverse effects may occur, thus the FS has determined the proposed action is likely to adversely affect designated critical habitat for Heliotrope milk-vetch and San Bernardino Mountains bladderpod and any plant species with critical habitat overlap with the action area of more than 1% (Table 24).

Other species with critical habitat overlapping the proposed action area include Cushenbury milk-vetch (*Astragalus albens*), Ash-grey paintbrush (*Castilleja cinerea*), Cushenbury buckwheat (*Eriogonum ovalifolium* var. *vineum*), Parish's daisy (*Erigeron parishii*), Cushenbury oxytheca (*Oxytheca parishii* var. *goodmaniana*), and California taraxacum (*Taraxacum californicum*), all of which are part of the carbonate, montane meadow, or pebble plain habitats of the San Bernadino Mountains. Species and habitat protections are described in the Existing Protections table below (Table 26).

Given the small amount of overlap with designated critical habitat, that most vegetation impacts will be temporary, and that some effects may be positive, the proposed action is not likely to adversely affect designated critical habitat for plant species with overlapping critical habitat of less than 1% (Table 24). In the event activities occur in designated critical habitat, the effects to physical and biological features will be minimized or avoided as dictated by local conditions.

Existing Protections

Due to the large number of plant species, the assessment of species-specific existing protections focuses on species with critical habitat overlapping the action area and/or species that were identified as having a high exposure level based on percent of Inventoried Roadless Area (IRA) occurrences in operable or potentially operable but complex areas. These species are listed in Table 26 with existing protections.

Table 26. Assessment of existing protections for a subset of the 134 conifer, fern, flowering plant, and lichen species analyzed for the roadless rule rescission. The subset consists of species that were identified as having a high exposure level based on percent of Inventoried Roadless Area (IRA) occurrences in operable or potentially operable but complex areas and/or species with critical habitat overlapping the proposed action area. NF = National Forest; LMP = Land Management Plan.

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
Ash-grey paintbrush	*Castilleja cinerea*	T (1998)	10.75	Moderate	Major threat: Intensive recreational use on **pebble plain complexes** habitat. Existing protections in the 2005 San Bernadino NF LMP- Part 2- Strategy (available at USFS 2025a) include Place Specific Standards (p. 99) applicable to *C. cinerea* and *C. cinerea* habitat including: SBNF S1 - "Avoid or minimize any activity that causes long-term damage to ashy-gray paintbrush host plants or host plant habitat in occupied paintbrush habitats" and SBNF S2 - "Avoid or minimize new ground disturbing activities that cause long-term damage to pebble plain habitat." The 2005 San Bernadino NF LMP also includes the following specifications for managing Threatened, Endangered, Proposed, Candidate, and Sensitive Species (p. 124): Emphasize the following practices within **carbonate, montane meadow and pebble plain habitat**: • Develop and implement a transportation plan that results in the reduction in road density and no new roads or motorized trails within carbonate, montane and pebble plain habitat. • Develop and implement a facilities plan for carbonate, montane meadow, and pebble plain habitat that avoids construction of new recreation and administrative facilities within these habitats. • Amend/modify existing special-use authorizations to include provisions for minimizing impacts to carbonate, montane meadow and pebble plain habitat. Avoid new authorizations for special-uses in these habitats where the requested use would adversely affect habitat. In carbonate habitat, mining special-uses will be

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					permitted consistent with the terms of the Carbonate Habitat Management Strategy. • Implement a program of land acquisition and land exchange that will contribute to the carbonate habitat reserve as described in the Carbonate Habitat Management Strategy. • Develop contingency plans that will minimize impacts to carbonate, montane meadow and pebble plain habitat from actions and activities that occur during emergencies. The 2005 LMP also includes a Conservation Strategy (pp. 126-128). The 2013 *C. cinerea* 5-Year Review (USFWS 2013d) notes "unauthorized roads and trails are created each year by users in the heavily used Holcomb Valley area and represent a particular challenge for protection of pebble plains species (USFS 2012b, p. 69). Even with the blocking in 2010 of more than 60 unauthorized routes in this area, including the use of large boulders (USFS 2012b, p. 69), impacts from this threat are expected to continue into the near future due to budget and staffing constraints (USFS 2012b, pp. 69, 74).
Bear Valley sandwort	*Arenaria ursina*	T	0	High	Also found in the **pebble plain habitat** on the San Bernadino NF, so similar protections and ongoing threats apply as discussed above for *Castilleja cinerea.*
California taraxacum	*Taraxacum californicum*	E	3.07	High	Found in montane meadow habitat on the San Bernadino NF. See *C. cinerea* description above for existing protections for carbonate habitat protections in the 2005 San Bernardino NF LMP (available at USFS 2025a).
Cook's lomatium	*Lomatium cookii*	E	0	High	Species was listed in 2002, so there are no specific protections in the 1989 Siskiyou NF LMP, the 1990 Rogue River NF LMP, or the 1994 Northwest Forest Plan. General ESA protections would apply: "avoid adverse impacts to federally listed threatened and endangered species." And "Ground disturbing activities shall not occur within known occupied sites of federally listed plant species unless consistent with approved recovery plans."
Cushenbury buckwheat	*Eriogonum ovalifolium var. vineum*	E	2.7	High	Habitat managed as a Special Interest Area in the San Bernardino NF LMP (2005, Part 2 – Strategy; available at USFS 2025a): Carbonate Habitat with a Carbonate Habitat Management Strategy. The strategy is designed to provide long-term protection for the carbonate endemic plants and also provide for continued mining. Critical habitat not likely to be adversely affected if known locations are part of the Carbonate Habitat Reserve. Also see *C. cinerea* description above for existing protections in the San Bernardino NF LMP (2005, Part 2 – Strategy; available at USFS 2025a).

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
Cushenbury milk-vetch	*Astragalus albens*	E	3.11	High	Same protections for limestone/carbonate habitat on 2005 San Bernadino NF (available at USFS 2025a) as described for *E. ovalifolium* and *C. cinerea*.
Cushenbury oxytheca	*Oxytheca parishii var. goodmaniana*	E	0.25	High	Same protections for limestone/carbonate habitat on San Bernadino NF (2005 LMP available at USFS 2025a) as described for *E. ovalifolium* and *C. cinerea*.
Godfrey's butterwort	*Pinguicula ionantha*	T	N/A	High	From 1999 National Forests of Florida LMP (available at USFS 2025a): "Godfrey's butterwort…inhabit(s) poorly drained coastal pinelands and are found, among other places, on the Apalachicola NF. The Forest Service protects them as guided by *Recovery Plan for Four Plants of the Lower Apalachicola Region, Florida*, USFWS. Specific management practices favoring recovery of these species include prescribed burning on a 3-year average and avoiding mechanical ground disturbance in suitable habitat. In addition, wet savannahs and cypress-dominated wetlands are unsuitable for timber production which precludes timber-related impacts to Godfrey's butterwort and Florida skullcap." "Site-specific evaluations are conducted for any proposed activity that may take place within habitat for these species or near known populations. Measures are taken to avoid adverse effects." "VG-2 – During wildland fire suppression efforts, avoid placing disked or plowed lines in Harper's beauty, Florida skullcap, Godfrey's butterwort, and white birds-in-a-nest habitat. Rehabilitate any lines soon after the fire suppression effort is complete."
Heliotrope milk-vetch	*Astragalus montii* PBFs from CH listing (1987): **Physical:** Limestone shale substrate, steep barren slopes, high elevation, open sunlight **Biological:** Sparse vegetation, native bee pollinators, seedling establishment, low disturbance **Ecological:** Needs space for growth,	T	100	Moderate	No specific protections in the current (1986) Manti-La Sal NF LMP. Plan is currently being revised and, when finalized, is proposed to include protections and conservation measures for *A. montii* and other species by way of standards and guidelines applied to a newly designated Heliotrope Botanical Area. The Manti-La Sal NF has closed some motorized roads and trails in and around *A. montii* habitat and adjusted the timing of grazing for pastures with *A. montii* habitat to later in the summer, after *A. montii* seed has developed and dispersed (pers. communication).

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
	reproduction sites, protection from mining/ grazing, historical distribution				
Holy ghost ipomopsis	*Ipomopsis sancti-spiritus*	E	N/A	High	The Santa Fe NF 2021 LMP includes Desired Conditions for the Pecos River Canyon GA, where *I. sancti-spiritus* is found, but does not include specific protections other than those followed for all threatened and endangered species "Management activities will avoid adverse impacts to federally listed threatened and endangered species." There is a recognition of the threats in the Plan: "Dispersed camping and motorized uses in the lower portion of the GA have resulted in soil compaction and vegetation degradation. Restoration of these areas, such as restrictions and revegetation, is needed in addition to managing popular dispersed camping sites."
Lassics lupine	*Lupinus constancei*	E	8.79	High	The 1995 Six Rivers NF LMP does not contain species-specific protections *for Lupinus constancei* (Lassics lupine), because the plant was listed as federally endangered in October 2023. Instead, the plan provides general protective language for all federally listed threatened and endangered species, which now applies to *L. lupine*. The Forest Service must ensure that management activities avoid adverse impacts to listed species, and coordinate with the U.S. Fish & Wildlife Service on recovery actions.
Macfarlane's four-o'clock	*Mirabilis macfarlanei*	T	N/A	High	Nez Perce-Clearwater NF, Wallowa-Whitman NF. Species occurs within Hells Canyon National Recreation Area. Forests shall administer the objectives set forth in Section 7 of the Hells Canyon National Recreation Act (PL 94-199) including, "preservation, especially in the area generally known as Hells Canyon, of all features and peculiarities believed to be biologically unique including, but not limited to, rare and endemic plant species, rare combinations of aquatic, terrestrial, and atmospheric habitats, and the rare combinations of outstanding and diverse ecosystems and parts of ecosystems associated therewith."

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
McDonald's rock-cress	*Arabis mcdonaldiana*	E	N/A	High	Klamath NF, Rogue River-Siskiyou NF, Klamath National Forest Plan Direction: *Arabis macdonaldiana* is not mentioned in the forest plan but the following applies to sensitive plant species. Sensitive Plant Species 7-1 Enhance Sensitive plant species populations and habitat to maintain reproducing, self-sustaining populations. Conduct an assessment to determine which Sensitive species are at a higher risk. Develop management strategies for higher risk Sensitive plant species first, with the intent of preventing the need for the species to become listed as T&E species. 7-2 Coordinate species maintenance and enhancement goals with other management activities on the site. 7-3 Management activities should imitate the natural ecological processes that created the Sensitive species habitat. Fire, timber management, grazing, or other activities may be used as tools for soil disturbance and removal of competing vegetation in managing the habitat. 7-4 Disturbances to plant populations and occupied habitat should be avoided during critical periods of plant growth. Individual projects shall develop project-level mitigations measures to avoid adverse impacts to Sensitive species. Six -Rivers National Forest Plan Goals Maintain the health and well-being of threatened, endangered and sensitive species and their habitats. Take all steps necessary to ensure that actions authorized, funded, or carried out by the Forest Service are not likely to jeopardize the continued existence of these species. Manage other botanical resources on a sustainable basis. practice the use of native species for all revegetation and erosion control projects. Standards and Guidelines 6-1 Federally listed threatened or endangered plants and their habitats will be managed to achieve recovery plan objectives (FSM 2670.21 and 2672.31). If an approved plan is not available, all known populations and their occupied habitat will be protected from negative impacts associated with forest management activities. The Del Norte population of McDonald's rock-cress is a candidate for federal listing. Determination of whether this species will be. listed is likely to follow the taxonomic treatment now underway. Rogue River-Siskiyou National Forest follows the Northwest Forest Plan which requires a survey and manage approach to listed species.

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
Mountain golden heather	*Hudsonia montana*	T	0.67	High	Nantahala and Pisgah National Forest Land Management Plan Standards applicable to all species groups PAD-S-01 Continue to work with the USFWS to expand known range, increase the population size, and enhance/restore suitable habitat for federally listed species on the Forests and within western North Carolina. PAD-S-02 In areas occupied by federally listed species and species of conservation concern, management shall maintain characteristics required by these species, where necessary for their recovery and/or persistence. Specific to *Hudsonia montana* Objectives: PAD-O-05 Tier 1: Manage and restore *Hudsonia montana* and *Liatris helleri* populations to ensure competing woody plants do not overtop and impact either species. Rocky habitat standards: PAD-S-05 Manage recreational impacts to *Hudsonia montana* and *Liatris helleri* within Linville Gorge Wilderness to protect the species.
Northeastern bulrush	*Scirpus ancistrochaetus*	E	N/A	High	George Washington and Jefferson National Forest Plan Protection and conservation of threatened and endangered (T&E) species drive much of the Plan direction and are a fundamental part of the overall management strategies for species diversity. The T&E species have recovery plans, and the strategy is to follow the finalized plans. The program priorities for T&E species include: · Implement Forest Service actions as recommended in final recovery plans for federally listed species. In the absence of an approved recovery plan, implement and, if necessary, develop interim Forest Service conservation measures. Update interim conservation measures as needed when new science becomes available. · Work with USFWS and other conservation partners to develop recovery plans for new federally listed species and candidate conservation agreements for species proposed for listing. · Coordinate with partners to implement measures to resolve conflicts with threatened and endangered species and their habitats. · Cooperatively monitor trends in population and/or habitat of federally listed species. Amend the Plan as needed when new populations are found. *Scirpus ancistrochaetus* is found within a Special Biological Area with standards that provide additional protections. Monongahela National Forest Plan General Direction Goal TE01 Provide habitat capable of contributing to the survival and recovery of species listed under the ESA. Provide habitat that may help preclude Proposed species from becoming listed.

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					Goal TE02 Integrate TEP habitat management with other resource objectives. Goal TE03 Work with USFWS, WVDNR, and other appropriate personnel to identify and manage habitat for TEP species. Participate in recovery plan development for threatened or endangered species that occur on the Forest, or that may be influenced by Forest management activities. Goal TE04: Within watershed-level planning units, identify TEP species habitat and opportunities to maintain, restore, or enhance habitat conditions. Design and implement management actions at the project level to address opportunities and provide for ecological conditions, population viability, reproductive needs, and habitat components for TEP species. Goal TE05 Collaborate on outreach programs for TEP species and their conservation needs. Standard TE06 When proposed exploration or development of privately owned mineral rights may adversely affect TEP species or habitat, the Forest shall work with state and federal mineral operation permitting agencies to reduce adverse effects.
Parish's daisy	*Erigeron parishii*	T	3.07	Moderate	Same protections for limestone/carbonate habitat on San Bernadino NF (USFS 2005) as described for *E. ovalifolium* and *C. cinerea*.
San Bernardino bluegrass	*Poa atropurpurea*	E	6.24	Moderate	Same Forest Plan protections as listed *for Ceanothus ophiochilus* (below) San Bernadino and Cleveland National Forest Plans (2005). *Poa atropupruea* included in Cleveland NF montane meadow and San Bernadino NF meadow habitat plans.
San Bernardino Mountains bladderpod	*Lesquerella kingii ssp. bernardina*	E	36.34	Moderate	Same Forest Plan protections as listed for *Ceanothus ophiochilus* (below).
San Diego thornmint	*Acanthomintha ilicifolia*	T	0	High	Same Forest Plan protections as listed for *Ceanothus ophiochilus* (below). *Acanthomintha ilicifolia* has a Forest Service management guide and strategy.

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
Showy stickseed	*Hackelia venusta*	E	N/A	High	*Hackelia venusta* is located within the Tumwater Botanical Area on the Wenatchee National Forest in steep, rugged terrain. Wenatchee National Forest Plan: GOAL STATEMENT: Manage areas of significant cultural, geological, botanical, zoological, paleontological, or other special Characteristics so as to protect, preserve, and enhance their intrinsic values DESCRIPTION: Special Interest Areas are classified under 36 CFR 294.1 and managed for recreation use substantially in their natural condition. The purpose of classifying these areas is to protect, and where appropriate, foster public use and enjoyment of the feature or environment. This prescription includes the following: Botanical Area: Lands containing specimens or group exhibits of plants, plant groups, and plant communities which are significant because of form, color occurrence, habitat location, life history, arrangement ecology, environment, rarity and/or other features.
Small whorled pogonia	*Isotria medeoloides*	T	N/A	High	CHATTAHOOCHEE-OCONEE NATIONAL FORESTS Plan GOAL 15 Contribute to conservation and recovery of federally-listed threatened and endangered species through habitat maintenance and/or enhancement and, where possible, for their reintroduction into suitable habitats, and contribute to avoiding the necessity for federal listing of other species under the Endangered Species Act. Objective 15-1 for *Isotria medeloides*: Maintain current number of populations/occurrences through protection and maintenance of existing sites. OBJECTIVE 15.2 Inventory non-native invasive aquatic species encroaching upon T&E habitat within the first 10 years of Plan implementation. GOAL 16 Cooperate with the United States Fish and Wildlife Service (USFWS) and the Georgia Department of Natural Resources in providing habitat for the re-introduction of native wildlife, including threatened, endangered and sensitive species. GOAL 18 Cooperate with the USFWS, Georgia Department of Natural Resources (DNR), and academia in conducting research on National Forest T&E species and their response to management. Standards FW-029 Sites supporting federally-listed threatened and endangered species or individuals needed to maintain viability of species of viability concern are protected from detrimental effects caused by management actions. Site-specific analysis of proposed management actions will identify any protective measures needed in addition to Forest Plan standards, including increasing the width of protective buffers where needed. Management activities occur within these sites

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					only where compatible with recovery of federally-listed species, or maintenance of individuals needed to maintain species viability on the National Forest. FW-030 The USFWS will be given timely notification when an emergency situation such as wildfire, weather-related events, insect outbreak, or disease outbreak has impacted, or appears likely to impact, known locations of Federally-listed species. FW-031 As part of recurrent monitoring and any project inventories, include data collection on existing or potential threats such as nonnative invasive species. FW-032 Non-native invasive species shall be controlled with priority given to areas where they are causing adverse effects to federally-listed species, or to individuals of other species needed to maintain their population viability on the national forest. Nonnative invasive species are not intentionally introduced near these species or individuals, nor will management actions facilitate their inadvertent introduction. George Washington and Jefferson National Forest Plan Protection and conservation of threatened and endangered (T&E) species drive much of the Plan direction and are a fundamental part of the overall management strategies for species diversity. The T&E species have recovery plans, and the strategy is to follow the finalized plans. The program priorities for T&E species include: Implement Forest Service actions as recommended in final recovery plans for federally listed species. In the absence of an approved recovery plan, implement and, if necessary, develop interim Forest Service conservation measures. Update interim conservation measures as needed when new science becomes available. Work with USFWS and other conservation partners to develop recovery plans for new federally listed species and candidate conservation agreements for species proposed for listing. Coordinate with partners to implement measures to resolve conflicts with threatened and endangered species and their habitats. Cooperatively monitor trends in population and/or habitat of federally listed species. Amend the Plan as needed when new populations are found. *Isotria medeloides* is found within a Special Biological Area with standards that provide additional protections. Nantahala and Pisgah National Forests Standards applicable to all species groups PAD-S-01 Continue to work with the USFWS to expand known range, increase the population size, and enhance/restore suitable habitat for

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					federally listed species on the Forests and within western North Carolina. PAD-S-02 In areas occupied by federally listed species and species of conservation concern, management shall maintain characteristics required by these species, where necessary for their recovery and/or persistence. Monongahela National Forest Plan General Direction Goal TE01 Provide habitat capable of contributing to the survival and recovery of species listed under the ESA. Provide habitat that may help preclude Proposed species from becoming listed. Goal TE02 Integrate TEP habitat management with other resource objectives. Goal TE03 Work with USFWS, WVDNR, and other appropriate personnel to identify and manage habitat for TEP species. Participate in recovery plan development for threatened or endangered species that occur on the Forest, or that may be influenced by Forest management activities. Goal TE04: Within watershed-level planning units, identify TEP species habitat and opportunities to maintain, restore, or enhance habitat conditions. Design and implement management actions at the project level to address opportunities and provide for ecological conditions, population viability, reproductive needs, and habitat components for TEP species. Goal TE05 Collaborate on outreach programs for TEP species and their conservation needs. Standard TE06 When proposed exploration or development of privately owned mineral rights may adversely affect TEP species or habitat, the Forest shall work with state and federal mineral operation permitting agencies to reduce adverse effects.
Smooth coneflower	*Echinacea laevigata*	T	N/A	High	CHATTAHOOCHEE-OCONEE NATIONAL FORESTS Plan GOAL 15 Contribute to conservation and recovery of federally-listed threatened and endangered species through habitat maintenance and/or enhancement and, where possible, for their reintroduction into suitable habitats, and contribute to avoiding the necessity for federal listing of other species under the Endangered Species Act. Objective 15-1 for *Echinacea laevigata*: Increase number of populations/occurrences by improving and/or increasing available habitat and relying on natural recruitment rather than reintroduction and propagation; and Increase number of populations/occurrences with the assistance of reintroduction and propagation efforts.

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					OBJECTIVE 15.2 Inventory non-native invasive aquatic species encroaching upon T&E habitat within the first 10 years of Plan implementation. GOAL 16 Cooperate with the United States Fish and Wildlife Service (USFWS) and the Georgia Department of Natural Resources in providing habitat for the reintroduction of native wildlife, including threatened, endangered and sensitive species. GOAL 18 Cooperate with the USFWS, Georgia Department of Natural Resources (DNR), and academia in conducting research on National Forest T&E species and their response to management. Standards FW-029 Sites supporting federally-listed threatened and endangered species or individuals needed to maintain viability of species of viability concern are protected from detrimental effects caused by management actions. Site-specific analysis of proposed management actions will identify any protective measures needed in addition to Forest Plan standards, including increasing the width of protective buffers where needed. Management activities occur within these sites only where compatible with recovery of federally-listed species, or maintenance of individuals needed to maintain species viability on the National Forest. FW-030 The USFWS will be given timely notification when an emergency situation such as wildfire, weather-related events, insect outbreak, or disease outbreak has impacted, or appears likely to impact, known locations of Federally-listed species. FW-031 As part of recurrent monitoring and any project inventories, include data collection on existing or potential threats such as nonnative invasive species. FW-032 Non-native invasive species shall be controlled with priority given to areas where they are causing adverse effects to federally-listed species, or to individuals of other species needed to maintain their population viability on the national forest. Nonnative invasive species are not intentionally introduced near these species or individuals, nor will management actions facilitate their inadvertent introduction. George Washington and Jefferson National Forest Plan Protection and conservation of threatened and endangered (T&E) species drive much of the Plan direction and are a fundamental part of the overall management strategies for species diversity. The T&E species have recovery plans, and the strategy is to follow the finalized plans. The program priorities for T&E species include: · Implement Forest Service actions as recommended in final recovery plans for

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					federally listed species. In the absence of an approved recovery plan, implement and, if necessary develop interim Forest Service conservation measures. Update interim conservation measures as needed when new science becomes available. · Work with USFWS and other conservation partners to develop recovery plans for new federally listed species and candidate conservation agreements for species proposed for listing. · Coordinate with partners to implement measures to resolve conflicts with threatened and endangered species and their habitats. · Cooperatively monitor trends in population and/or habitat of federally listed species. Amend the Plan as needed when new populations are found. *Echinacea laevigata* is found within a Special Biological Area with standards that provide additional protections.
Southern Mountain wild-buckwheat	*Eriogonum kennedyi var. austromontanum*	T	0	High	Same Forest Plan protections as listed for *Ceanothus ophiochilus* (below). *Eriogonum kennedyi var. austromontanum* included in Pebble Plain Habitat Plan.
Springville clarkia	*Clarkia springvillensis*	T	N/A	High	Sequoia National Forest plan: No specific protections for *Clarkia springvillensis* but guidelines for at-risk species including ESA listed species. Guidelines (SPEC-FW-GDL) 01 Design features, mitigation, and project timing considerations should be incorporated into projects that may affect habitat for at-risk species where they occur to minimize impacts to ecological conditions that provide for the persistence of at-risk species.
Swamp pink	*Helonias bullata*	T	N/A	High	CHAT TAHOOCHEE-OCONEE NATIONAL FORESTS Plan: GOAL 15 Contribute to conservation and recovery of federally-listed threatened and endangered species through habitat maintenance and/or enhancement and, where possible, for their reintroduction into suitable habitats, and contribute to avoiding the necessity for federal listing of other species under the Endangered Species Act. Objective 15-1 for *Helonias bullata*: Increase number of populations/occurrences with the assistance of reintroduction and propagation efforts. OBJECTIVE 15.2 Inventory non-native invasive aquatic species encroaching upon T&E habitat within the first 10 years of Plan implementation. GOAL 16 Cooperate with the United States Fish and Wildlife Service (USFWS) and the Georgia Department of Natural Resources in

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					providing habitat for the reintroduction of native wildlife, including threatened, endangered and sensitive species. GOAL 18 Cooperate with the USFWS, Georgia Department of Natural Resources (DNR), and academia in conducting research on National Forest T&E species and their response to management. Standards FW-029 Sites supporting federally-listed threatened and endangered species or individuals needed to maintain viability of species of viability concern are protected from detrimental effects caused by management actions. Site-specific analysis of proposed management actions will identify any protective measures needed in addition to Forest Plan standards, including increasing the width of protective buffers where needed. Management activities occur within these sites only where compatible with recovery of federally-listed species, or maintenance of individuals needed to maintain species viability on the National Forest. FW-030 The USFWS will be given timely notification when an emergency situation such as wildfire, weather-related events, insect outbreak, or disease outbreak has impacted, or appears likely to impact, known locations of Federally-listed species. FW-031 As part of recurrent monitoring and any project inventories, include data collection on existing or potential threats such as nonnative invasive species. FW-032 Non-native invasive species shall be controlled with priority given to areas where they are causing adverse effects to federally-listed species, or to individuals of other species needed to maintain their population viability on the national forest. Nonnative invasive species are not intentionally introduced near these species or individuals, nor will management actions facilitate their inadvertent introduction. George Washington and Jefferson National Forest Plan: Protection and conservation of threatened and endangered (T&E) species drive much of the Plan direction and are a fundamental part of the overall management strategies for species diversity. The T&E species have recovery plans, and the strategy is to follow the finalized plans. The program priorities for T&E species include: Implement Forest Service actions as recommended in final recovery plans for federally listed species. In the absence of an approved recovery plan, implement and, if necessary, develop interim Forest Service conservation measures. Update interim conservation measures as needed when new science becomes available. Work with USFWS and other conservation partners to develop recovery plans for new

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					federally listed species and candidate conservation agreements for species proposed for listing. Coordinate with partners to implement measures to resolve conflicts with threatened and endangered species and their habitats. Cooperatively monitor trends in population and/or habitat of federally listed species. Amend the Plan as needed when new populations are found. *Helonias bullata* is found within riparian areas and Special Biological Area have standards that provide additional protections. Nantahala and Pisgah National Forest Plans: Standards applicable to all species groups PAD-S-01 Continue to work with the USFWS to expand known range, increase the population size, and enhance/restore suitable habitat for federally listed species on the Forests and within western North Carolina. PAD-S-02 In areas occupied by federally listed species and species of conservation concern, management shall maintain characteristics required by these species, where necessary for their recovery and/or persistence. Guidelines applicable to all species groups PAD-G-01 Work with USFWS to ensure that recovery plans and relevant biological opinion guidance for federally listed species are incorporated into project design and implementation. Consider newer information from USFWS five-year reviews when relevant.
Thread-leaved brodiaea	*Brodiaea filifolia*	T	0	High	Same Forest Plan protections as listed *for Ceanothus ophiochilus* (below).
Vail Lake ceanothus	*Ceanothus ophiochilus*	T	0	High	Cleveland National Forest Plan (2005; available at USFS 2025a) Standards: S11: When occupied or suitable habitat for a threatened, endangered, proposed, candidate or sensitive (TEPCS) species is present on an ongoing or proposed project site, consider species guidance documents (see Appendix H) to develop project-specific or activity-specific design criteria. This guidance is intended to provide a range of possible conservation measures that may be selectively applied during site-specific planning to avoid, minimize or mitigate negative long-term effects on threatened, endangered, proposed, candidate or sensitive species and habitat. Involve appropriate resource specialists in the identification of relevant design criteria. Include review of species guidance documents in fire suppression or other emergency actions when and to the extent practicable.

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					S12: When implementing new projects in areas that provide for threatened, endangered, proposed, and candidate species, use design criteria and conservation practices (see Appendix H) so that discretionary uses and facilities promote the conservation and recovery of these species and their habitats. Accept short-term impacts where long-term effects would provide a net benefit for the species and its habitat where needed to achieve multiple-use objectives. S24: Mitigate impacts of on-going uses and management activities on threatened, endangered, proposed, and candidate species. When Implementing Fire Management Activities S37: Design and manage fuel treatments to minimize the risk that treated areas will be used by unauthorized motorized and mechanized vehicles. Mitigate impacts where such use does occur. S38: Avoid establishment of staging areas, helibases, base camps, firelines or other areas of human concentration and equipment use within threatened, endangered and proposed species suitable and occupied habitats and riparian areas to the maximum extent possible when suppression of wildland fire and human safety are not compromised. When Implementing Recreation Activities S34: Where a threatened, endangered, proposed, candidate, or sensitive species occurs in a recreation site or area, take steps to avoid or minimize negative impacts to the threatened, endangered, proposed, candidate or sensitive species and its habitat. Use the least restrictive action that will effectively mitigate adverse impacts to the species and habitat. S35: Manage dispersed recreation activities to ensure that environmental sustainability is maintained by utilizing the following measures: • Discourage camping within 100 feet of sensitive resources and habitats, including meadows and bodies of water (springs, streams, ponds and lakes), or within 1/4 mile of developed recreation facilities. • Discourage camping within 600 feet of any wildlife water source developments, such as guzzlers and water holes, in accordance with state laws.• Motorized and non-motorized vehicle travel is restricted to National Forest System roads and trails and limited areas that are designated for vehicle use. Cleveland National Forest Plan (2005; available at USFS 2025a) Guidance: When planning projects or managing ongoing activities in areas that contain habitat for species of concern (including threatened, endangered, proposed, candidate, and sensitive species and other

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					species identified by biologists as being in danger of population decline or habitat loss) use the information found in various types of species guidance documents to develop project-specific design criteria. Species guidance documents include (as of December 2004): 1. Recovery plans for threatened and endangered species, prepared by U.S. Fish and Wildlife Service. 2. Species management guides and strategies, prepared by or for USDA Forest 3. Species conservation strategies, prepared by or for USDA Forest Service 4. Habitat management guides, plans and strategies, prepared by or for USDA Forest Service 5. Species accounts prepared for this planning effort or subsequent to it (USDA Forest Service).
Virginia sneezeweed	*Helenium virginicum*	T	N/A	High	The George Washington NF LRMP (2014 revision; available at USFS 2025a) includes specific protective language for federally listed plants like *Helenium virginicum* (Virginia sneezeweed), which require management activities avoid adverse impacts to federally listed threatened and endangered species and their habitats, and that sinkhole ponds and other sensitive wetlands are managed to maintain natural hydrology and ecological integrity.
Webber's ivesia	*Ivesia webberi*	T	0	High	Species listed in 2014, so there were no specific protections in the 1986 Humboldt-Toiyabe NF LMP, the 1988 Plumas NF LMP, or the 1994 Northwest Forest Plan. Standard ESA protections would apply.
Western lily	*Lilium occidentale*	E	N/A	High	The Siuslaw National Forest Plan (1990; available at USFS 2025a) does not have specific protections for *Lilium occidentale* (ESA listed 1994) but has the following standards and guidelines for Threatened, Endangered and Sensitive plant species: FW-034 Cooperation - Identify and manage threatened and endangered (T&E) and sensitive species in cooperation with the USDI Fish and Wildlife Service (USFWS), Oregon Department of Fish and Wildlife (ODFW, fish and wildlife), and Oregon Department of Agriculture (plants). FW-035 Conservation - Meet legal and biological requirements for conservation of T&E and sensitive plants and animals Evaluate proposed projects that involve significant ground disturbance or have the potential to alter habitat of T&E or sensitive species to determine if any of these species are present (FSM 2670, T&E and R6 Sensitive Plants and Animals) FW-036 Consultation - Where T&E species are present, make the required determination (a biological assessment for an EIS and a biological evaluation for an environmental assessment) according to the requirements of the Endangered Species Act (Public Law 93-205)

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					Consult with the USFWS and state agencies on each program activity or project that the Forest Service determines may affect T&E species, before any decision is made on the proposed project FW-037 Mitigation - Specify protection or mitigation requirements before carrying out a project [36 CFR 219 27(a)(8)] Conduct management activity and manage habitat for existing Federally classified T&E species so as to achieve objectives of existing recovery plans and not impair recovery of any T&E species FW-038 Biological Evaluation - When T&E and sensitive species and/or their habitat is present, perform a biological (field) evaluation For sensitive species, consult with knowledgeable and interested authorities. Manage habitat for sensitive plants and animals to ensure that viable populations are maintained throughout their existing range Management practices and use of species management guides shall assure that species do not become threatened or endangered because of Forest Service actions FW-039 T & E List - Maintain and update list of T&E and sensitive plants and animals on the Forest periodically as new information is collected. Submit pertinent Forest information to the Regional Office for updating the Regional Forester's Sensitive Species lists and to the Oregon Natural Heritage Program for inclusion in their statewide Data Base FW-040 Disclosure - Do not disclose information on the specific location of T&E or sensitive species to the public.
Whitebark pine	*Pinus albicaulis*	T	N/A	High	The National Whitebark Pine Restoration Plan to which the US Forest Service is a participant is in final stages of development and outlines a restoration approach for whitebark pine rangewide. U.S. Forest Service Range-wide Restoration Strategy for Whitebark Pine (Keane et al. 2012) was developed to promote the conservation of whitebark pine. U.S. Forest Service Restoring Whitebark Pine Ecosystems in the Face of Climate Change (Keane et al. 2017a) is a companion document to the US Forest Service Range-wide Restoration Strategy for Whitebark pine to guide adaptation strategies from climate change, fire and treatments. Guidelines and Best Practices for Managing Fire in Whitebark Pine Stands in the Crown of the Continent (Final Draft, Keane et al. 2017b) outlines strategies and techniques for use of fire to restore whitebark pine. Adaptive Action Plan, Whitebark Pine in the Greater Yellowstone Area (GYCC 2015) outlines conservation plans for whitebark pine in the Greater Yellowstone area including National Forests. Whitebark Pine Restoration Strategy for the Pacific Northwest Region 2009-2013 (Aubry et al. 2008) was developed to restore and

Common Name	Scientific Name and PBFs for CH where applicable	ESA Status	Percent Critical Habitat in Affected Environment*	Exposure	Existing Protections
					conserve populations of whitebark pine. An Interagency Conservation Strategy for *Pinus albicaulis* (Whitebark Pine) in California was developed to outline goals and tasks for whitebark pine conservation (2020). In addition, whitebark pine occurs on 37 national forests. All plans have plan guidance and/or components that conserves threatened plant taxa including whitebark pine.

Determinations

The determinations for each species as shown in Table 24 are based on the amount of known occurrences and critical habitat that overlaps the operable or potentially operable action area, the protections that are in place in individual Forest Plans (Table 26), and that before any project will be able to be implemented it will need to be consulted on separately under ESA. Although we analyzed the extent of the proposed action in the action area, we do not expect that the entire action area will be affected by the proposed action. In addition, some of the proposed actions will benefit certain listed species, such as actions to reduce vegetation succession, which will address a primary threat to 35 of 134 (26.1 percent) species. Refer to Table 24 to see the effects determination for each conifer, fern, flowering plant, and lichen species and critical habitat.

Table 27. Deconstruction of the 2001 Roadless Rule rescission including impacts and potential responses of conifer, fern, flowering plant, and lichen species to these impacts.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment	Change in vegetation; Soil disturbance and Compaction; Noise; Human presence; Trampling/run over; removal of canopy; Loss of nest/den trees and snags	Direct mortality; loss of plant parts (leaves, stems, seeds, roots), reduced vigor; habitat fragmentation resulting in gene flow reduction and decreased reproduction.
		Tree and Unit Marking	Walking	Human presence	Trampling resulting in direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor;
			ATV	Noise; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services
		Landings - Vegetation Clearing and Removal	Heavy Equipment	Soil disturbance and Compaction; Noise; Human presence; Change in vegetation; Trampling/run over; invasive species competition	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), reduced vigor; germination and recruitment failure; habitat fragmentation resulting in gene flow reduction and decreased reproduction; pollinator response to noise resulting in increased or reduced pollination services
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle	Chemical Contamination/Nutrients; Noise; Human Presence; larger patch size of native forbs	Direct mortality, reduced vigor; recruitment failure; chemical contaminants could result in mortality or toxicity to pollinators resulting in decreased pollination and reproductive output; seeding with native forbs could increase pollination service and reproductive output

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment	Change in vegetation; Large woody debris; Soil disturbance and Compaction; Noise; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; pollinator response to noise resulting in increased or reduced pollination services
		Timber Haul	Heavy Equipment	Soil disturbance and Compaction; Noise; Human presence; Dust; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
	Temporary Access Control	Road Gates Installation	Heavy Equipment	Noise; soil disturbance and compaction; Human presence;	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over; altered hydrology	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment; Pumps; Screens	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over; Sedimentation	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment
		Temporary Stream Flow Divergence; Bypass	Heavy Equipment; Pumps; Screens	Sediment/Turbidity; chemical contamination/nutrients; Drainage Network; Peak/Baseflows; altered hydrology	Direct mortality, burial, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment; Hand tools	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and chemicals resulting in increased or reduced pollination services;
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services;
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and chemicals resulting in increased or reduced pollination services; reduced germination and recruitment.
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment; ATV	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services;
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bankfull	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over; altered hydrology	Direct mortality, burial, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services;

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Snow	Plowing	Heavy Equipment	Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; reduced germination and recruitment.
Permanent and temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over; Changes to ground cover and microhabitat	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
		Snags - Hand Felling	Motorized Hand Tools	Large woody debris; Noise; Human presence; Loss of nest trees	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and loss of nesting sites resulting in increased or reduced pollination services.
		Snags - Mechanical Felling	Heavy Equipment	Large woody debris; Noise; Human presence; Lost of nest trees; Soil compaction; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and loss of nest trees resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
		Tree Removal - Log Decks	Heavy Equipment	Noise; Soil disturbance and compaction; Human presence; Trampling/run over; Lost of nest trees	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
		Tree Removal - Hauling	Heavy Equipment	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
	Roadway template	Soil Excavation	Heavy Equipment, Blasting	Noise; Soil disturbance; Human presence; Trampling/run over; Dust;	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment;
		Soil Removal	Heavy Equipment	Noise; soil disturbance; Human presence; Trampling/run over; Dust; Seed bank removal	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment.
		Fill Placement	Heavy Equipment	Noise; Soil disturbance; Human presence; invasive species proliferation; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; habitat fragmentation resulting in gene flow reduction and decreased reproduction; reduced germination and recruitment; proliferation of invasive species could negatively affect recruitment, vigor and reproductive effort.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	Drainage	Ditchlines, Relief Culverts, Waterbars, Riprap	Heavy Equipment	Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.
		Stream Crossings - Culvert Installation	Heavy Equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.
	Surfacing	Hauling of surface materials	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and chemicals resulting in increased or reduced pollination services; reduced germination and recruitment.
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.
	scarifying	Material removal	Heavy equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.

Activity	Management Element	Task/Method	Tool	Impact/Benefit	Response
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment	Change in vegetation; Noise; Soil disturbance and compaction; Human presence; Trampling/run over; sedimentation; altered hydrology	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise resulting in increased or reduced pollination services; reduced germination and recruitment.
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment	Change in vegetation; Noise; Soil compaction; Human presence; Trampling/run over	Direct mortality, loss of plant parts (leaves, stems, seeds, roots), and reduced vigor; pollinator response to noise and revegetation resulting in increased or reduced pollination services;

Cumulative Effects

Under the ESA, cumulative effects are the consequences of future State, Tribal, local, and private activities that are reasonably certain to occur within the action area. The action area for rescission of the Roadless Rule is owned and managed by the US Forest Service. As such, nearly all activities that will occur in the action area will require ESA consultation, with the FS as the lead agency for those consultations. Although private activities may occur within proximity to currently inventoried roadless areas, there are no reasonably certain private activities that overlap the time and scale of this consultation. Cumulative effects will be considered more thoroughly during project level consultation.

PROJECT-LEVEL CONSULTATION

The proposed action described below is a "framework programmatic action" as defined in 50 CFR 402.02, where the framework programmatic action only establishes a framework for the development of specific future action(s) but does not authorize any future action(s). The effects to listed species and designated critical habitat of future actions that are subsequently authorized, funded, or carried out under this program will be addressed in subsequent section 7 consultations, as appropriate. The Forest Service proposes to rescind the 2001 Roadless Rule. This action does not affect state-level roadless rules that may exist, nor does it preclude the development of state-led roadless efforts. Rescinding the Roadless Rule does not authorize, fund, or carry out future forest management activities that could occur in these areas. Rescinding the Roadless Rule does not constrain or alter the statutory requirement of sections 7(a)(1) through section 7(a)(4) of the Endangered Species Act. All future activities in currently inventoried roadless areas will require compliance with the ESA prior to implementation. Rescinding the Roadless Rule does not alter the protections afforded to designated wilderness areas or areas protected by other law, regulation, or policy.

Although rescinding the Roadless Rule does not dictate or prescribe forest management in previously inventoried roadless areas, it is reasonably certain that active forest management and road construction will increase in these areas. However, currently the Forest Service does not have a reliable method to identify, describe, or analyze the likelihood and location of any future projects that may be proposed for currently inventoried roadless areas.

Future projects in current inventoried roadless areas will require compliance environmental laws and regulations, including the Endangered Species Act. As is typical, future project designs in these areas will avoid and minimize adverse effects to ESA-listed resources to the greatest extent feasible. All projects must be consistent with the land management plan that governs management for their geographic location.

Adverse effects to ESA-listed resources from road construction and timber projects in roadless areas may include, but are not limited to: behavioral avoidance, increased energy expenditure, reduction in fitness, abandonment, degradation of unique critical habitat features, and even mortality. However, these effects will be fully explored during project-level compliance efforts, including consultation under section 7(a)(2) of the Endangered Species Act.

The Forest Service has determined that rescinding the Roadless Rule will have no direct effect on ESA-listed resources because it does not authorize, fund, or carry out future forest management activities in areas currently identified as inventoried roadless areas. However, after rescinding the Roadless Rule, future forest management may occur in inventoried roadless areas that result in road construction and timber harvest activities. These future activities may affect ESA-listed resources. Future activities in current inventoried roadless areas will comply with the ESA through project-level consultation, when details about the project are known. The Forest Service shall consult with the Services on any future projects in currently inventoried roadless areas that may affect ESA -listed resources.

CONCLUSION

Rescinding the 2001 Roadless Rule will increase the likelihood of road construction and timber harvests occurring in locations previously considered inventoried roadless areas. In many cases the Forest Service anticipates this increased activity to adversely affect some ESA-listed species and their designated critical habitats. However, all future projects in these areas will be required to comply with environmental law, regulation, and policy, including the Endangered Species Act. Projects that are determined that they "may affect" ESA-listed species or designated critical habitats will undergo consultation under section 7(a)(2) of the ESA. Through consultation with the Services at the project-level, jeopardizing listed species or adversely modifying critical habitat should be avoided.

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APPENDICES

Appendix A: FWS Species and the NFS units on which they occur.

Appendix B: Best Management Practices

Forest Service policy and guidance are described in several different Forest Service publications. The Forest Service Directives System is a series of manuals and handbooks used by Forest Service personnel to guide the agency and manage national forests and grasslands. Forest Service manuals contain legal authorities, objectives, policies, responsibilities, instructions, and guidance regularly used by agency employees to do their work. Forest Service handbooks provide specialized guidance and instruction for carrying out the directions in the manuals. Field issuances are localized supplements to the national manuals and handbooks.

Primary references are available in FS directives such as Forest Service Handbook (FSH) 7709.56 Chapter 60 for typical road maintenance, FSH 2409.15 Chapter 50 for timber related road maintenance, and FSH 7709.56 Chapter 40 for development and construction requirements.

Transportation Facilities - Forest Service Handbook 2409.15 Chapter 50 Transportation Facilities. Web Link: https://www.fs.usda.gov/about-agency/regulations-policies/handbook/240915-50-transportation-facilities

Road Design - Forest Service Handbook 7709.56 - Road Pre-Construction Handbook

Chapter 40 – Design Subchapter 44 – Road Drainage. Web Link: https://www.fs.usda.gov/about-agency/regulations-policies/handbook/770956-40-design

Road Maintenance - Forest Service Handbook 7709.59 Chapter 60. Web Link: https://www.fs.usda.gov/about-agency/regulations-policies/handbook/770959-60-road-maintenance

Guidelines for Road Maintenance Levels. Web Link: https://www.fs.usda.gov/t-d/pubs/pdf/11771811.pdf

Sediment and Chemical Monitoring Protocols - National Best Management Practices for Water Quality Management on National Forest System Lands. Web Link: https://www.fs.usda.gov/sites/default/files/FS_National_Core_BMPs_April2012_sb.pdf

Water/Road Interactions Technology Series Documents (18 total). Web Link: Water-Road Interaction Tech Series Documents. Web Link: https://www.fs.usda.gov/biology/nsaec/products-publications-streamteamdocuments-writsd.html

Storm Damage Risk Reduction Guide for Low Volume Roads. Web Link: https://fsweb.mtdc.wo.fs.fed.us/pubs/pdfpubs/pdf12771814/pdf12771814dpi100.pdf

Stabilization and Rehabilitation Measures for Low Volume Forest Roads. Web Link: https://fcds.org.co/wp-content/uploads/2021/01/road-stabilization-rehab-measures-keller.pdf

Environmentally Sensitive Road Maintenance Practices for Dirt and Gravel Roads. Web Link: https://fsweb.sdtdc.wo.fs.fed.us/pubs/pdf/11771802.pdf

Culvert Scour Assessment. Web Link: https://fsweb.sdtdc.wo.fs.fed.us/pubs/pdf/11771802.pdf